

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**RESPONDENTS' IDENTIFICATION OF EXPERT WITNESSES
FOR ISSUES OTHER THAN CLAIM CONSTRUCTION**

Pursuant to Order No. 6, Setting the Procedural Schedule on June 9, 2026, Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC (collectively "Tower Semiconductor" or "Respondents") hereby identify the following expert witnesses, including their expertise and curriculum vitae, and whom may proffer opinions regarding issues other than claim construction for the asserted claims of the patents-in-suit in this investigation. Respondents may call to offer expert testimony or expert opinions at the hearing in the above-referenced investigation, and as such, Respondents reserve the right to rely upon testimony from any of the experts identified herein. Respondents reserve their rights to amend or supplement these disclosures as this investigation and discovery proceeds.

Subject to the above, Respondents identify the following expert witnesses:

I. Frank Ernst, Ph.D.

Dr. Ernst is an expert in the technical matters of this case. Dr. Ernst's curriculum vitae, attached hereto as Exhibit A, sets forth his qualifications.

Dr. Ernst may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Ernst may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

II. David K. Choi, Ph.D.

Dr. Choi is an expert in the technical matters of this case. Dr. Choi's curriculum vitae, attached hereto as Exhibit B, sets forth his qualifications.

Dr. Choi may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Choi may also offer opinions and testimony in rebuttal to any other

arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

III. Jeffrey Bokor, Ph.D.

Dr. Bokor is an expert in the technical matters of this case. Dr. Bokor's curriculum vitae, attached hereto as Exhibit C, sets forth his qualifications.

Dr. Bokor may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Bokor may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

IV. Sanjay K. Banerjee, Ph.D.

Dr. Banerjee is an expert in the technical matters of this case. Dr. Banerjee's curriculum vitae, attached hereto as Exhibit D, sets forth his qualifications.

Dr. Banerjee may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development,

and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Banerjee may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

V. Suman Datta, Ph.D.

Dr. Datta is an expert in the technical matters of this case. Dr. Datta's curriculum vitae, attached hereto as Exhibit E, sets forth his qualifications.

Dr. Datta may offer opinions and testimony in this Investigation relating to, among other subjects, the technical background and state of the art relevant to the Asserted Patents; the scope and construction of claim terms in the Asserted Patents; the invalidity of the Asserted Patents, including under 35 U.S.C. §§ 102, 103, and 112; the unenforceability of the Asserted Patents; the alleged invention and priority dates of the Asserted Patents; the design, development, and operation of the accused and domestic industry products; noninfringement of the Asserted Patents; non-infringing redesigns or alternatives to the Asserted Patents; and Complainant's failure to satisfy the technical prong of the domestic industry requirement with respect to any claim of the Asserted Patents. Dr. Datta may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

VI. Julia Rowe

Ms. Rowe is an expert in economics and related fields, including financial and economic issues arising in complex business litigation involving intellectual property and other disputes. Ms. Rowe's curriculum vitae, attached hereto as Exhibit F, sets forth her qualifications.

Ms. Rowe may offer opinions and/or testimony in this Investigation relating to, among other subjects, the lack of a domestic industry, Complainant's requested remedies in this Investigation, bonding, and any other related issues pertinent to Respondents' defenses and/or Complainant's contentions for which expert testimony is required or useful. Ms. Rowe may further offer opinions and/or testimony related to patent invalidity, including the lack of any secondary considerations of non-obviousness such as commercial success. Ms. Rowe may also offer opinions and testimony in rebuttal to any other arguments and assertions relating to the Asserted Patents that Complainant or its experts raise in this Investigation.

Date: August 21, 2026

/s/ Stephen R. Smith

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Semiconductor NPB Holdings, Inc., Tower
Semiconductor Newport Beach, Inc., and
Newport Fab LLC*

CERTIFICATE OF SERVICE

I hereby certify that on August 21, 2026, copies of the foregoing were filed with and served upon the following as indicated:

The Honorable Lisa Barton
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EXHIBIT A

Curriculum Vitae

Frank Ernst

Dr. rer. nat. habil.

Leonard Case Jr. Professor of Engineering

Department of Materials Science and Engineering

Case Western Reserve University

orcid.org/0000-0002-4823-2041

August 3, 2026

Academic Degrees

1997	Dr. rer. nat. habil. (Habilitation*)	Materials Science	University of Stuttgart, Germany (Advisors: E. Arzt, F. Aldinger, M. Rühle)
1987	Dr. rer. nat. (Ph. D.)	Physical Metallurgy	University of Göttingen, Germany (Advisor: P. Haasen)
1984	Dipl.-Phys. (Masters)	Physics	University of Göttingen, Germany (Advisor: P. Haasen)

Bio Data

Date of Birth	November 17, 1959
Place of Birth	Göttingen, Germany
Spouse	Sabine Ernst
Children	Meike Ernst (October 20, 1990) Oliver Ernst (September 28, 1992)

*The Habilitation is the highest academic degree in Germany. It is a post-doctoral qualification awarded for demonstrating a special qualification in research and *teaching* at the university level.

**The *venia legendi* is the permission to teach at a German university. It *can* be granted on the basis of a Habilitation.

Appointments

2016–2024	Department Chair Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
2016–2021	Faculty Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2012	Secondary Professor Department of Chemical and Biomolecular Engineering CWRU, Cleveland, (Ohio), USA
2008–2011	Director Case Center for Surface Engineering CWRU, Cleveland, (Ohio), USA
2008–2016	Co-Director Swagelok Center for Surface Analysis of Materials CWRU, Cleveland, (Ohio), USA
Since 2007	Leonard Case Jr. Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
Since 2000	Professor Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA
1997–2000	Lecturer University of Stuttgart, Germany
1989–2000	Senior Scientist Max-Planck-Institut für Metallforschung Stuttgart, Germany
1987–1988	Post-Doctoral Researcher Department of Materials Science and Engineering CWRU, Cleveland, (Ohio), USA

Awards

- 2025 **Nomination for John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2019 **Jessica Melton Perry Award for Disnguished Teaching in Disciplinary & Professional Writing**
CWRU
- 2019 **Graduate Teaching Award**
Case School of Engineering
- 2018 **Nomination for the Graduate Teaching Award**
Case School of Engineering
- 2017 **John S Diekhoff Award for Distinguished Graduate Teaching**
CWRU
- 2016 **Technical Educator Award**
CTSC (Cleveland Technical Societies Council)
- 2010 **Acta Materialia Top Reviewer 2009**
Elsevier
- 2010 **Materials Science and Engineering Outstanding Reviewer 2009**
Elsevier
- 2009 **Materials Science and Engineering Outstanding Reviewer 2008**
Elsevier
- 2006 **Case School of Engineering Outstanding Teacher 2005-2006**
CWRU
- 2005 **Nomination for the**
2005 Carl F. Wittke Award for Excellence in Undergraduate Teaching
CWRU
- 2005 **2004-2005 Case School of Engineering Teaching Leader**
CWRU
- 2004 **Tenure Appointment**
CWRU
- 2001 **Werner Köster Prize 2000**
German Materials Research Society (DGM) and Zeitschrift für Metallkunde
- 1997 **Venia Legendi****
University of Stuttgart
- 1993 **Tenure Appointment**
Max-Planck Society, Max-Planck-Institut für Metallforschung
- 1985 **Graduate Student Research Award**
German Materials Research Society (DGM)
- 1984 **Heineman Scholarship for Graduate Studies**
Minna-James-Heineman-Foundation (for 3 years)

Professional Activities

Professional Memberships

ASM	ASM International
MSNO	Microscopy Society of Northern Ohio
MSA	Microscopy Society of America
ACerS	American Ceramic Society
TMS	The Minerals, Metals & Materials Society
DGE	German Electron Microscopy Society
DGM	German Materials Research Society

Memberships on Boards of Scientific Journals

2010 – today	Metals: Editorial Board
2006 – 2016	International Journal of Materials Research: Advisory Board
2004 - 2013	Materials Science and Engineering A: Editorial Board
1997 – 2006	Zeitschrift für Metallkunde: Advisory Board
1996 – 2002	Journal of Electron Microscopy: Editorial Board

Service as Reviewer

For Scientific Publications

Acta Crystallographica	Acta Materialia
Advanced Materials	Applied Physics Letters
International Journal of Materials Research	Journal of Biomedical Materials Research
Journal of Electron Microscopy	Journal of Physical Chemistry
Journal of Materials Research	Materials Science and Engineering
Materials Science and Technology	Metallurgical and Materials Transactions
Metals	Microscopy & Microanalysis
Nano Letters	Philosophical Magazine
Physica Status Solidi	Physical Review B
Physical Review Letters	Science
Steel Research	Surface Science
Surface Science and Technology	Ultramicroscopy
Wiley & Sons	Zeitschrift für Metallkunde

For Research Agencies

European Science Foundation
German Science Foundation (Research Proposals, Proposals for Research Centers)
Max-Planck-Society (Proposals for International Max-Planck Research Schools)
National Science Foundation
US Department of Defense

Conference Activity

1. Organizer of "Third Symposium on Surface Hardening of Corrosion-Resistant Alloys" *Materials Science & Technology 2014*. Pittsburgh, PA, USA, October 12-16, 2014.
2. Session Chair of Symposium "Grain Boundary Engineering" *XIII International Conference on Intergranular and Interface Boundaries (iib2010)*. Mie, Japan, June 27 to July 2, 2010.
3. Co-Organizer and Session Chair of the "International Symposium on Surface Hardening of Corrosion Resistant Alloys". Cleveland, May 24 to 26, 2010.
4. Co-Organizer and Session Chair of the "International Meeting on Surface Hardening of Stainless Steel". Cleveland, October 22 and 23, 2007.
5. Co-Organizer and Session Chairs of Conference on "Progress and Challenges in Understanding Materials Interfaces." *Material Science and Technology Conference and Exhibition*. Cincinnati, October 15-19, 2006.
6. Organizer and Chair of "Interfaces: A Symposium in Honor of Manfred Rühle." *Annual Meeting of the Microscopy Society of America*. Savannah, Georgia, USA, August 01-05, 2004.
7. Chair of the Symposium "Materials Design." *First Kyoto Workshop on Computational Materials Science*. Kyoto, Japan, March 4, 2003.
8. Organizer and Chair of the *MSNO Winter Symposium on Transmission Electron Microscopy*. Cleveland, Ohio, USA, February 13, 2002.
9. Chair of a Symposium on "Data Analysis and Modeling Across length Scales." *United Engineering Foundation Meeting on Integrating Materials Science into Engineering Structures and Devices*. Lake Arrowhead, California, USA, November 11-15, 2001.
10. Chair of a Symposium on "Quantitative Analysis of Interface Structure by HRTEM." *Annual Meeting of the Microscopy Society of America 2001*. Long Beach, California, USA, August 05-09, 2001.
11. Chair of a Symposium on "Wide Band-Gap Semiconductors." *Microscopy of Semiconducting Materials XII*. Oxford, United Kingdom, March 25-29, 2001.
12. Organizer and Chair of a Symposium on "Quantitative High-Resolution Electron Microscopy." *EUREM 12*. Brno, Czech Republic, July 9-14, 2000.

13. Organizer and Chair of an *International Conference on Quantitative High-Resolution Electron Microscopy*. Berlin, Germany, March 27-29, 2000.
14. Organizer and Chair of a Symposium on "Materials Research." *29th Congress of the German Electron Microscopy Society*. Dortmund, Germany, September 5-10, 1999.
15. Organizer and Chair of a Symposium on "Advances in Quantitative and Coherent High-Resolution Electron Microscopy." *14th International Congress on Electron Microscopy*. Cancun, Mexico, August 31 - September 4, 1998.
16. Organizer and Chair of the International Conference on *Frontiers of Electron Microscopy in Materials Science*. Irsee, Germany, April 19-24, 1998.
17. Organizer and Chair of a Symposium on "Interfaces and Surfaces." *Multinational Congress of the Austrian, German, and Swiss Microscopy Societies*. Regensburg, Germany, September 8-12, 1997.
18. Chair of a Symposium on "Atomic Resolution Microscopy of Surfaces and Interfaces." *Fall Meeting of the Materials Research Society*. Boston, USA, December 2-6, 1996.
19. Chair of a Symposium on "Defects at Hetero-Interfaces." *Annual Meeting of The Materials Society*. San Francisco, USA, February 27-March 3, 1994.
20. Chair of a Symposium on "Defect-Interface Interactions." *Fall Meeting of the Materials Research Society*. Boston, USA, November 29-December 3, 1993.
21. Chair of a Symposium on "Stress Relaxation and Misfit Accommodation in Epitaxial Thin Films." *Annual Meeting of The Materials Society*. San Francisco, USA, February 17-22, 1991.
22. Chair of a Symposium on "HRTEM of Defects in Materials." *Spring Meeting of The Materials Research Society*. San Francisco, USA, April 16-20, 1990.

Publications

Academic Publications

1. F. Ernst: *Struktur von Heterogrenzflächen (Structure of Heterointerfaces)*. Habilitationsschrift (Professorial Thesis), University of Stuttgart, 1997.
2. F. Ernst: *Hochauflösende Elektronenmikroskopie der Entmischung von Al-1 at% Ag at 413 K (Decomposition Kinetics of Al-1 at% Ag at 413 K, Studied by High-Resolution Electron Microscopy)*. Dissertation (Ph. D. Thesis), University of Göttingen, 1987.
3. F. Ernst: *Rekristallisationsexperimente in $\langle 100 \rangle$ -zugverformten Kupfer-Einkristallen mit künstlicher Keimbildung (Recrystallization of $\langle 100 \rangle$ -Strained Copper Single Crystals After Artificial Nucleation)*. Diplomarbeit (Masters Thesis), University of Göttingen, 1984.

Refereed Publications[§]

1. Dernek, Ozan and **Mehdi, Redad** and Yue, Weiqi and Bachman, Jonah A. and Holt, Finley R. and Ponon, Gabriel O. and Tripathi, Pawan K. and Willard, Matthew A. and Ernst, Frank and French, Roger H.: *A Perspective on Synchrotron Data Science*. Interdisciplinary Information Sciences, vol. 31, pp. 83–88, 2025.
2. **Redad Mehdi**, Rounak Chawla, Erika I. Barcelos, Roger H. French, Frank Ernst: *2D-diffractogram Analysis: Kinematic-diffraction Simulator for Neural-network Training-data Generation*. Computational Materials Science, vol. 252, pp. 113777, 2025.
3. **Sameera Nalin Venkat**, Thomas G Ciardi, Mingjian Lu, Preston C DeLeo, Jube Augustino, Adam Goodman, Jayvic Cristian Jimenez, Anirban Mondal, Frank Ernst, Christine A Orme, Yinghui Wu, Roger H French, Laura S Bruckman: *A General Materials Data Science Framework for Quantitative 2D Analysis of Particle Growth from Image Sequences*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 71-82, 2024.
4. **Weiqi Yue**, Pawan K. Tripathi, Gabriel Ponon, Zhuldyz Ualikhankyzy, Donald W. Brown, Bjorn Clausen, Maria Strantza, Darren C. Pagan, Matthew A. Willard, Frank Ernst, Erman Ayday, Vipin Chaudhary, Roger H. French: *Phase Identification in Synchrotron X-ray Diffraction Patterns of Ti-6Al-4V Using Computer Vision and Deep Learning*. Integrating Materials and Manufacturing Innovation, vol. 13, pp. 36-52, 2024.

[§]Author names in **boldface** indicate students or post-doctoral scholars I advised.

5. **WeiQi Yue, Mohommad Redad Mehdi, Pawan K Tripathi Matthew A. Willard, Frank Ernst, Roger H. French:** *Exploring 2d X-ray Diffraction Phase Fraction Analysis With Convolutional Neural Networks: Insights From Kinematic-diffraction Simulations*. MRS Advances, vol. 9, pp. 921-928, 2024.
6. Leveraging Deep Neural Networks for Feature Extraction from 2D XRD Diffractograms Acquired at Synchrotron Beamlines: *Pawan K. Tripathi, WeiQi Yue, Redad Mehdi, Finley Holt, Gabriel Ponon, Vipin Chaudhary, Frank Ernst, Matthew Willard, Bjorn Clausen Donald W. Brown, Daniel Savage, Roger H. French*. Los Alamos National Laboratory (LANL) Report, vol. LA-UR-23-33318, pp. 1, 2023.
7. **C. Illing, Z. Ren, F. Ernst:** *Low-Temperature Carburization: Ex Situ Activation of Austenitic Stainless Steel*. Metals, vol. 2023, pp. 335-359, 13.
8. **C. Illing, M. Bestic, F. Ernst:** *Additive Manufacturing: Corrosion Proofing by Infusion of Interstitial Solute—Exemplified for Alloy 22*. Metals, vol. 2023, pp. 127-143, 13.
9. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the efficacy of concentrated interstitial carbon in enhancing the pitting corrosion resistance of stainless steel; response to Comment by Martin et al.. Scripta Materialia*, vol. 224, pp. 115084, 2023.
10. T. Li, S.C. Chien, **Z. Ren**, W. Windl, F. Ernst, and G.S. Frankel: *Understanding the Efficacy of Concentrated Interstitial Carbon in Enhancing the Pitting Corrosion Resistance of Stainless Steel*. Acta Materialia, vol. 221, pp. 117433-447, 2021.
11. **C. Illing, Z. Ren, A. Agaponova, A. Heuer, F. Ernst:** *Rapid Alloy Surface Engineering Through Closed-Vessel Reagent Pyrolysis*. Metals, vol. 11, pp. 1764-1782, 2021.
12. **Z. Ren, F. Ernst:** *Stress–Corrosion Cracking of AISI 316 L Stainless Steel in Seawater Environments: Effect of Surface Machining*. Metals, vol. 10, pp. 1324-1341, 2020.
13. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *The Passivity of Low-Temperature Carburized Austenitic Stainless Steel AISI-316L in a Simulated Boiling-Water-Reactor Environment*. Journal of Nuclear Materials, vol. 537, pp. 152197, 2020.
14. **Z. Ren, F. Ernst:** *High-Temperature Phase Transformations in AISI 316 Stainless Steel Infused With Concentrated Interstitial Carbon*. Journal of Alloys and Compounds, vol. 819, pp. 153000, 2020.

15. **J.C. Dalton**, F. Ernst, A.H. Heuer: *Low-Temperature Nitridation of 2205 Duplex Stainless Steel*. Metallurgical and Materials Transactions A, vol. 51, pp. 607-617, 2020.
16. **Z. Ren**, F. Ernst: *Electronic Impact of Concentrated Interstitial Carbon on Physical Properties of AISI-316 Austenitic Stainless Steel*. Acta Materialia, vol. 2019, pp. 96-105, 173.
17. **Z. Ren**, A.H. Heuer, F. Ernst: *Ultrahigh-Strength AISI-316 Austenitic Stainless Steel Foils Through Concentrated Interstitial Carbon*. Acta Materialia, vol. 2019, pp. 231-240, 167.
18. **Z. Ren**, S.Eppell, S. Collins, F. Ernst: *Co-Cr-Mo Alloys: Improved Wear Resistance Through Low-Temperature Gas-Phase Nitro-Carburization*. Surface & Coatings Technology, vol. 378, pp. 124943-1-9, 2019.
19. **X. Zhan**, M. Daniil, X. Liu, M.A. Willard, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Magnetic Properties of Amorphous Ni-P*. Journal of Alloys and Compounds, vol. 2019, pp. 742-749, 786.
20. **Z. Li, C. Illing**, A.H. Heuer, F. Ernst: *Low-Temperature Carburization of AL-6XN Enabled by Provisional Passivation Metals* 2018 997-1011
21. **Z. Li, X. Zhan**, T. Paramasivam, W. Niu, A.H. Heuer, S. Maloy, R. Lillard, F. Ernst: *Surface Engineering of IN-718 by Low-Temperature Carburization: Properties and Thermal Stability*. Surface Engineering (2018). 35 281-293
22. **S. Lingala**, D. Jayne, F. Ernst: *Effect of Lubricant Additives on the Tribological Behavior of Aluminum Alloy Against Steel*. International Journal of Materials Research (2018). 789-802
23. W. Niu, **Z. Li**, F. Ernst, R.S. Lillard: *Passivity of Surface-Engineered Alloys in a Simulated Boiling-Water-Reactor Environment*. Corrosion, vol. 74, pp. 635-653, 2018.
24. **X. Zhan**, P. Zhang, P.M. Voyles, X. Liu, R. Akolkar, F. Ernst: *Effect of Tungsten Alloying on Short-to-Medium-Range-Order Evolution and Crystallization Behavior of Near-Eutectic Amorphous Ni-P*. Acta Materialia, vol. 122, pp. 400-411, 2017.
25. **D. Wang**, H. Kahn, F. Ernst, A.H. Heuer: *“Colossal” Interstitial Supersaturation in Delta Ferrite in Stainless Steels: (II) Low-temperature Nitridation of the 17-7 PH Alloy*. Acta Materialia, vol. 124, pp. 237-246, 2017.
26. J. Wang, **Z. Li, D Wang**, S. Qiu, F. Ernst: *Thermal stability of low-temperature-carburized austenitic stainless steel*. Acta Materialia, vol. 128, pp. 235-240, 2017.

27. **A. Zangiabadi, J. Dalton, D. Wang, F. Ernst, A.H. Heuer:** *The Formation of Martensitic Austenite during Nitridation of Stainless Steel.* Metallurgical and Materials Transactions A, vol. 48A, pp. 8-13, 2017.
28. **F. Ernst:** *Background of SAM Atom-Fraction Profiles.* Materials Characterization, vol. 125, pp. 142-151, 2017.
29. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *“Colossal” Interstitial Supersaturation in Delta Ferrite in 17-7 PH Stainless Steels After Low-temperature Nitridation.* Microscopy and Microanalysis (22). 2016 2020-2023
30. **L. Yu, Z. Vashaei, F. Ernst, A. Kumar, R. Akolkar:** *Direct Electroless Deposition of Nano-scale Copper Films on Ruthenium: Role of Ethylene-diamine Additive as Nucleation Promoter.* Journal of the Electrochemical Society, vol. 163, pp. D374-D378, 2015.
31. **X. Zhan and F. Ernst:** *Crystallization Micro-Mechanism of Near-Eutectic Amorphous Ni-P.* Acta Materialia, vol. 104, pp. 274-282, 2016.
32. **H. Kahn, F. Ernst, A.H. Heuer:** *Increasing the Coefficient of Sliding Friction of NiCr at Low Loads by Interstitial Surface Hardening.* Wear, vol. 346-347, pp. 1-5, 2016.
33. **D. Wu, Y. Ge, H. Kahn, A.H. Heuer, F. Ernst:** *Diffusion Profiles after Nitrocarburizing Austenitic Stainless Steel.* Surface Coatings and Technology, vol. 279, pp. 180-185, 2015.
34. **W. Niu, R.S. Lillard, Z. Li, F. Ernst:** *The Mechanism of Passivity in Austenitic Stainless Steel AISI 316L Carburized to Colossal Super Saturation.* Electrochimica Acta (176). 2015 410-419
35. **D. Wang, H. Kahn, F. Ernst, A.H. Heuer:** *NiAl Precipitation in Delta Ferrite Grains of 17-7 Precipitation-hardening Stainless Steel During Low-Temperature Interstitial Hardening.* Scripta Materialia, vol. 108, pp. 136-138, 2015.
36. **D. Wang, C.W. Chen, J.C. Dalton, F. Yang, R. Sharghi-Moshtaghin, H. Kahn, F. Ernst, R.E.A. Williams, D.W. McComb and A.H. Heuer:** *“Colossal” Interstitial Supersaturation of Delta Ferrite in Stainless Steels—I. Low-Temperature Carburization.* Acta Materialia, vol. 86, pp. 193-207, 2015.
37. **X. Zhan and F. Ernst:** *Crystallization of Ni-P Fabricated by Electroless Deposition: Microscopic Mechanism.* Materials Research Society Symposia Proceedings, vol. 1757, pp. mrsf14-1757-uu03-07, 2015.
38. **D. Wang, F. Ernst, H. Kahn, A.H. Heuer:** *Cellular Precipitation at a 17-7 PH Stainless Steel Interphase Interface during Low-Temperature Nitridation.* Metallurgical and Materials Transactions A (45). 2014 1-8

39. X. Gu, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Numerical Simulations of Carbon and Nitrogen Composition-Depth Profiles in Nitrocarburized Austenitic Stainless Steels*. Metallurgical and Materials Transactions, vol. 45A, pp. 4268-4279, 2014.
40. **X. Gu**, G.M. Michal, F. Ernst, H. Kahn, A.H. Heuer: *Concentration Dependent Carbon Diffusivity in Austenite*. Metallurgical and Materials Transactions, vol. 45A, pp. 3790-3799, 2014.
41. Y. Ge, F. Ernst, H. Kahn, A.H. Heuer: *The Effect of Surface Finish on Low-Temperature Acetylene-Based Carburization of 316L Austenitic Stainless Steel*. Metallurgical and Materials Transactions A, vol. 45, pp. 2338-2345, 2014.
42. D. Wu, H. Kahn, G.M. Michal, F. Ernst and A.H. Heuer: *Orientation Dependence of Nitrogen Supersaturation in Austenitic Stainless Steel During Low-Temperature Gas-Phase Nitriding*. Acta Materialia, vol. 2014, pp. 339-350, 79.
43. **Z. Li**, F. Ernst, A.H. Heuer: *Improved Accident Tolerance of Austenitic Stainless Steel Cladding through Colossal Supersaturation with Interstitial Solutes*. Transactions of the American Nuclear Society, vol. 109, pp. 304-306, 2013.
44. **D. Wang**, H. Kahn, J. J. Lewandowski, F. Ernst, G. M. Michal and A. H. Heuer: *Sustained-load Crack Growth of Hydrogen-Charged Surface-Hardened 316 L Stainless Steel*. Materials Science and Engineering A, vol. 556, pp. 43-50, 2012.
45. **J-P. Hsu**, **D. Wang**, H. Kahn, F. Ernst, G.M. Michal, A.H. Heuer: *Fatigue Crack Growth of Interstitially Hardened 316L Stainless Steel in Air and 0.6 M NaCl*. International Journal of Fatigue, vol. 47, pp. 100-105, 2013.
46. Q. He, B. Shyam, **M. Nishijima**, X. Yang, B. Koel, F. Ernst, D. Ramaker, S. Mukerjee: *Highly Stable PtAuRu/C Catalyst Nanoparticles for Methanol Electro-Oxidation*. Journal of Materials Chemistry, vol. 117, pp. 1457-1467, 2013.
47. **N. Matic**, G. S. Chottiner, F. Ernst, D. Scherson: *The Platinum/Titanium-Nitride Interface: X-Ray Photoelectron Spectroscopy Studies*. Electrochemical and Solid State Letters, vol. 15, pp. 79-82, 2012.
48. M.-C. Hsieh, **Y. Ge**, G.M. Michal, F. Ernst, A.H. Heuer: *Volatility Diagrams for Cr-O and Cr-Cl and Their Application to Removing the Passive Film from the Surface of Stainless Steel*. Metallurgical and Materials Transactions, vol. 43, pp. 1187-1201, 2012.

49. **N. Matic**, G. S. Chottiner, F. Ernst, D. Scherson: *The Platinum/Titanium-Nitride Interface: X-Ray Photoelectron Spectroscopy Studies*. Transactions of the Electrochemical Society, vol. 41, pp. 865-874, 2011.
50. H. Kahn, A.H. Heuer, G.M. Michal, F. Ernst, **R. Sharghi-Moshtaghin**, **Y. Ge**, P.M. Natishan, R.J. Rayne, F.J. Martin: *Interstitial Hardening of Duplex 2205 Stainless Steel by Low-Temperature Carburization: Enhanced Mechanical and Electrochemical Performance*. Surface Engineering, vol. Surface Engineering, pp. 2012, 28. 213-219
51. A. H. Heuer, H. Kahn, F. Ernst, G. M. Michal, D. Hovis, R. J. Rayne, F. J. Martin, and P. M. Natishan: *Enhanced Corrosion Resistance of Interstitially Hardened Stainless Steel: Implications of a Critical Passive Layer Thickness for Breakdown*. Acta Materialia, vol. 60, pp. 716-725, 2011.
52. **D. Wu**, H. Kahn, G.M. Michal, F. Ernst, A.H. Heuer: *Ferromagnetism in Interstitially Hardened Austenitic Stainless Steel Induced by Low-Temperature Gas-Phase Nitriding*. Scripta Materialia, vol. 65, pp. 1089-1092, 2011.
53. F. Ernst, **D. Li**, A. H. Heuer, H. Kahn and G. M. Michal: *The Carbide M_7C_3 in Low-Temperature-Carburized Austenitic Stainless Steel*. Acta Materialia, vol. 59, pp. 2268-2276, 2011.
54. **L. O'Donnel**, G.M. Michal, F. Ernst, H. Kahn, and A.H. Heuer: *Wear Maps for Low Temperature Carburized 316L Austenitic Stainless Steel Sliding Against Alumina*. Surface Engineering, vol. 26, pp. 284-292, 2010.
55. L. Zhu, M. Teo, P.C. Wong, K.C. Wong, **I. Narita**, F. Ernst, K.A.R. Mitchell, S.A. Campbell: *Synthesis and Characterization of a $CoSe_2$ Catalyst for the Oxygen Reduction Reaction*. Applied Catalysis A, vol. 386, pp. 157-165, 2010.
56. A. Heuer, H. Kahn, F. Ernst, G. Michal, R. Rayne, F. Martin, P.M. Natishan: *Carbon-Enhanced Passivity of PH13-8 Mo: A Precipitation-Hardened Martensitic Stainless Steel*. Electrochemical and Solid-State Letters C, vol. 13, pp. 37-39, 2010.
57. **R. Sharghi-Moshtaghin**, Y. Ge, H. Kahn, X. Gu, F. Martin, P. Natishan, R.J. Rayne, G.M. Michal, F. Ernst, and A. H. Heuer: *Low-Temperature Carburization of the Ni-base Superalloy IN718: Improvements in Surface Hardness and Crevice Corrosion Resistance*. Metallurgical and Materials Transactions, vol. 41, pp. 2022-2032, 2010.
58. **K. Du**, F. Ernst, **M. Pelsozy**, J. Barthel, K. Tillmann: *Expansion of Interatomic Distances at the Surface of Platinum Catalyst Nanoparticles*. Acta Materialia, vol. 58, pp. 836-845, 2010.

59. **M. Guinel**, A. Bonakdarpour, B. Wang, P. K. Babu, F. Ernst, N. Ramaswamy, S. Mukerjee, A. Wieckowski: *Carbon-Supported, Selenium-Modified Ruthenium-Molybdenum Catalysts for Oxygen Reduction in Acidic Media*. ChemSusChem, vol. 2, pp. 658-664, 2009.
60. H. Kahn, G. M. Michal, F. Ernst, and A. H. Heuer: *Poisson Effects on X-Ray Diffraction Patterns in Low-Temperature-Carburized Austenitic Stainless Steel*. Metallurgical and Materials Transactions, vol. 40, pp. 1799-1804, 2009.
61. F. Martin, P. Natishan, E. Lemieux, T. Newbauer, R. Rayne, R. Bayles, H. Kahn, G.M. Michal, F. Ernst, and A.H. Heuer: *Enhanced Corrosion Resistance of Stainless Steel Carburized at Low Temperature*. Metallurgical and Materials Transactions, vol. 40, pp. 1805-1809, 2009.
62. G.M. Michal, X. Gu, W.D. Jennings, H. Kahn, F. Ernst, and A.H. Heuer: *Paraequilibrium Carburization of Duplex and Ferritic Stainless Steels*. Metallurgical and Materials Transactions, vol. 40, pp. 1781-1790, 2009.
63. F. Ernst, **A. Avishai**, H. Kahn, **X. Gu**, G. M. Michal, A. H. Heuer: *Enhanced Carbon Diffusion in Austenitic Stainless Steel Carburized at Low Temperature*. Metallurgical and Materials Transactions A, vol. 40, pp. 1768-1780, 2009.
64. J. M. Ziegelbauer, D. Gatewood, A. F. Gullá, M. Guinel, F. Ernst, D. E. Ramaker, S. Mukerjee: *Fundamental Investigation of Oxygen Reduction Reaction on Rhodium Sulfide-based Chalcogenides*. Journal of Physical Chemistry C, vol. 113, pp. 6955-6968, 2009.
65. **K. Du**, F. Ernst, M. Garrels, J. Payer: *Formation of Nickel Nanoparticles in Nickel-Ceramic Anodes of Solid-Oxide Fuel Cells*. International Journal of Materials Research, vol. 99, pp. 548-552, 2008.
66. N. Agarwal, H. Kahn, **A. Avishai**, G. Michal, F. Ernst, and A.H. Heuer: *Enhanced Fatigue Resistance in 316L Austenitic Stainless Steel Due to Low-Temperature Paraequilibrium Carburization*. Acta Materialia, vol. 55, pp. 5572-580, 2007.
67. F. J. Martin, E. J. Lemieux, T. M. Newbauer, R. A. Bayles, P. M. Natishan, H. Kahn, G. M. Michal, F. Ernst, A. H. Heuer: *Carburization-Induced Passivity of 316L Austenitic Stainless Steel*. Electrochemical and Solid State Letters, vol. 10, pp. C76-C78, 2007.
68. G. Singh, **Y. Yu**, F. Ernst, R. Raj: *Shear Strength and Sliding at a Metal-Ceramic (Aluminum-Spinel) Interface at Ambient and Elevated Temperature*. Acta Materialia, vol. 55, pp. 3049-57, 2007.
69. F. Ernst, **Y. Cao**, G. M. Michal, A. H. Heuer: *Carbide Precipitation in Austenitic Stainless Steel Carburized at Low-Temperature*. Acta Materialia, vol. 55, pp. 1895-1906, 2007.

70. A. H. Heuer, F. Ernst, H. Kahn, A. Avishai, G. M. Michal, D. J. Pitchure and R. E. Ricker: *Interstitial Defects in an Austenitic Stainless Steel Containing "Colossal" Carbon Concentrations: An Internal Friction Study*. Scripta Materialia, vol. 56, pp. 1067-1070, 2007.
71. Y. Yu, J. Mark, F. Ernst, T. Wagner, R. Raj: *Diffusion Reactions at Al-MgAl₂O₄ Interfaces – and the Effect of Applied Electric Fields*. Journal of Materials Science, vol. 41, pp. 7785-7797, 2006.
72. F. Ernst, G. M. Michal, H. Kahn, A. H. Heuer: *Paraequilibrium Surface Alloying with Interstitial Solutes: A New Concept for Improving the Performance of Medical Devices*, in: *Medical Device Materials III*, Eds. K. Venugopalan and M. Wu. ASM International (Materials Park, Ohio), pp. 27-33, 2006.
73. K. Du, F. Ernst: *Quantitative Assessment of Nanoparticle Size Distributions from HRTEM Images*. International Journal of Materials Research, vol. 97, pp. 928-933, 2006.
74. F. Ernst, G. M. Michal, F. Oba, L. Liu, J. Blush, A. H. Heuer: *Gas-Phase Surface Alloying under "Kinetic Control," A Novel Approach to Improving the Surface Properties of Titanium Alloys*. International Journal of Materials Research, vol. 97, pp. 597-606, 2006.
75. G.M. Michal, F. Ernst, H. Kahn, Y. Cao, F. Oba, N. Agarwal, A.H. Heuer: *Colossal Carbon Supersaturation due to Paraequilibrium Carburization: Stainless Steels with Greatly Improved Mechanical Properties*. Acta Materialia, vol. 54, pp. 1597-1606, 2006.
76. G. M. Michal, F. Ernst, A. H. Heuer: *Carbon Paraequilibrium in Austenitic Stainless Steel*. Metallurgical and Materials Transactions, vol. 37, pp. 1819-1824, 2006.
77. R. Liu, E.A. Kulp, F. Oba, E. W. Bohannon, F. Ernst, J. A. Switzer: *Epitaxial Electrodeposition of High-Aspect-Ratio Cu₂O(110) Nanostructures on InP(111)*. Chemistry of Materials, vol. 17, pp. 725-729, 2005.
78. L. Liu, F. Ernst, G. M. Michal, A. H. Heuer: *Surface Hardening of Ti Alloys by Gas-Phase Nitridation: Kinetic Control of the Nitrogen Surface Activity*. Metallurgical and Materials Transactions, vol. 36, pp. 2429-2434, 2005.
79. F. Oba, F. Ernst, Y. Yu, R. Liu, H. Kothari, J. Switzer: *Epitaxial Growth of Cuprous Oxide Electrodeposited onto Semiconductor and Metal Substrates (invited "Feature Article")*. Journal of the American Ceramic Society, vol. 88, pp. 253-270, 2005.
80. J. Cowen, L. Lucas, F. Ernst, P. Pirouz, A. Hepp, S. Bailey: *Liquid-Phase Deposition of Single-phase Alpha-Copper-Indium-Diselenide*. Materials Science and Engineering B, vol. 116, pp. 311-319, 2005.

81. **R. Adelung**, F. Ernst, **N. Zheng**, U. Landau: *In-situ Nanoscale Observation and Control of Electron-Beam-Induced Cluster Formation*. Journal of Vacuum Science and Technology B, vol. 22, pp. 1797-1802, 2004.
82. K. Lyutovich, M. Oehme, F. Ernst: *Growth of Ultra-Thin and Highly Relaxed SiGe Layers Under In-Situ Introduction of Point Defects*. European Physical Journal of Applied Physics, vol. 27, pp. 341, 2004.
83. F. Ernst, **Y. Cao**, G. M. Michal: *Carbides in Low-Temperature Carburized Stainless Steels*. Acta Materialia, vol. 52, pp. 1469-1477, 2004.
84. R. Liu, **F. Oba**, E. W. Bohannon, F. Ernst, J. A. Switzer: *Shape Control in Epitaxial Electrodeposition: Cu₂O Nanocubes on InP(001)*. Chemistry of Materials, vol. 15, pp. 4882-4885, 2003.
85. R. Liu, **F. Oba**, E. W. Bohannon, F. Ernst, J. A. Switzer: *Epitaxial Electrodeposition of Cu₂O Films onto InP(001)*. Applied Physics Letters, vol. 83, pp. 1944-1946, 2003.
86. **R. Adelung**, R. Kunz, F. Ernst, L. Kipp, M. Skibowski: *Self-Organized Structures on Flat Crystals: Nanowire Networks Formed by Metal Evaporation*. Advances in Solid State Physics, vol. 43, pp. 463-476, 2003.
87. **Y. Cao**, F. Ernst, G. M. Michal: *Colossal Carbon Supersaturation in Austenitic Stainless Steels Carburized at Low Temperature*. Acta Materialia, vol. 51, pp. 4171-4181, 2003.
88. F. Ernst, J. Switzer: *Microstructure of Cu₂O/Si Interfaces, Made by Epitaxial Electrodeposition*. Zeitschrift für Metallkunde, vol. 94, pp. 259-264, 2003.
89. J. Switzer, R. Liu, E. W. Bohannon, F. Ernst: *Epitaxial Electrodeposition of a Crystalline Metal Oxide onto Single-Crystalline Silicon*. The Journal of Physical Chemistry B, vol. 106, pp. 12369-12372, 2002.
90. O. G. Schmidt, U. Denker, S. Christiansen, F. Ernst: *Composition of Self-Assembled Ge/Si Islands in Single and Multiple Layers*. Applied Physics Letters, vol. 14, pp. 2614-2616, 2002.
91. **R. Adelung**, **W. Hartung**, F. Ernst: *Fabrication of Cu-Induced Networks of Linear Nanostructures on Different Length Scales*. Acta Materialia, vol. 50, pp. 4925-4933, 2002.
92. F. Ernst: *Assessing the Mechanical Strength of Interfacial Atomic Bonds by Quantitative HRTEM*. Philosophical Magazine A, vol. 82, pp. 2677-2693, 2002.
93. **U. Klement**, F. Ernst, B. Baretzky, J. M. Plitzko: *Diffusion of Oxygen in CdSe-Photosensor Arrays*. Materials Science and Engineering B, vol. 94, pp. 123-130, 2002.

94. R. Raj, A. Saha, L. An, D. P. H. Hasselman, F. Ernst: *Ion Exchange at a Metal/Ceramic Interface*. Acta Materialia, vol. 50, pp. 1165–1176, 2002.
95. **R. Adelung**, F. Ernst, A. Scott, M. Tabib-Azar, L. Kipp, M. Skibowski, S. Hollensteier, E. Spiecker, W. Jäger, V. Zaporozhchenko, F. Faupel: *Self-Assembled Nanowire Networks by Deposition of Copper onto Layered-Crystal Surfaces*. Advanced Materials, vol. 14, pp. 1056–1061, 2002.
96. O. G. Schmidt, U. Denker, M. Dashiell, N. Y. Jin-Phillipp, K. Eberl, R. Schreiner, H. Gräbeldinger, H. Schweizer, S. Christiansen, F. Ernst: *Laterally Aligned Ge/Si Islands: A New Concept for Faster Field-Effect Transistors*. Materials Science and Engineering B, vol. 89, pp. 101–105, 2002.
97. F. Ernst, **O. Kienzle**, O. G. Schmidt, K. Eberl, J. Zhu, K. Brunner, G. Abstreiter: *Ge-Si Nanostructures for Quantum-Effect Electronic Devices*. Inst. Phys. Conf. Ser., vol. 169, pp. 167–176, 2001.
98. **S. Poulat**, F. Ernst: *Epitaxy of Ge on Sapphire*. Materials Science and Engineering A, vol. 323, pp. 9–16, 2002.
99. F. Ernst, C. Scheu, **R. Schweinfest**, M. Rühle: *Structure, Composition, and Bonding of Grain Boundaries in Metals*, in: *Materials for the Third Millennium*, Eds. R. K. Ray et al.. Oxford & IBH Publishing Co. Pvt. Ltd (New Delhi), pp. 295–323, 2001.
100. G. Wöhl, E. Kasper, T. Hackbarth, H. Kibbel, M. Klose, F. Ernst: *Fully Relaxed Si_{0.7}Ge_{0.3} Buffers Grown on Patterned Silicon Substrates for Hetero-CMOS Transistors*. Journal of Materials Science: Materials in Electronics, vol. 12, pp. 235–240, 2001.
101. F. Ernst, M. L. Mulvihill, **O. Kienzle**, M. Rühle: *Preferred Grain Orientation Relationships in Sintered Perovskite Ceramics*. Journal of the American Ceramic Society, vol. 84, pp. 1885–1890, 2001.
102. **Gallardo-López**, J. Martinez-Fernández, A. Domínguez-Rodríguez, F. Ernst: *Origin of Diffuse Electron Scattering in YCSZ Single Crystals with 24 to 32 mol% Yttria*. Philosophical Magazine A, vol. 81, pp. 1675–1689, 2001.
103. **R. Schweinfest**, S. Köstlmeier, F. Ernst, C. Elsässer, M. W. Finnis: *Atomistic Structure of Al/MgAl₂O₄ and Ag/MgAl₂O₄ Interfaces, Studied by Quantitative HRTEM and ab initio Band Structure Calculations*. Philosophical Magazine A, vol. 81, pp. 927–955, 2001.
104. **S. Hutt**, **O. Kienzle**, F. Ernst, M. Rühle: *Processing and Structure of Grain Boundaries in Strontium Titanate*. Zeitschrift für Metallkunde, vol. 92, pp. 105–109, 2001.

105. K. Brunner, J. Zhu, G. Abstreiter, **O. Kienzle**, F. Ernst: *Self-Ordering of Ge Islands on Si Substrates Mediated by Local Strain Fields*. *Physica Status Solidi (b)*, vol. 224, pp. 531–535, 2001.
106. O. G. Schmidt, U. Denker, K. Eberl, **O. Kienzle**, F. Ernst, R. J. Haug: *Resonant Tunneling Diodes Based on Stacked Self-Assembled Ge/Si Islands*. *Applied Physics Letters*, vol. 77, pp. 4341–4343, 2001.
107. **O. Kienzle**, **S. Hutt**, F. Ernst, M. Rühle: *Grain Boundaries in Strontium Titanate*, in: *Grain Boundary Engineering in Ceramics*, Eds. T. Sakuma, L. M. Sheppard, Y. Ikuhara. The American Ceramic Society (Westerville), pp. 215–222, 2000.
108. O. G. Schmidt, U. Denker, K. Eberl, **O. Kienzle**, F. Ernst: *Effect of Overgrowth Temperature on the Photoluminescence of Ge/Si Islands*. *Applied Physics Letters*, vol. 77, pp. 2509–2511, 2000.
109. **W. Han**, P. Kohler-Redlich, C. Scheu, F. Ernst, M. Rühle, N. Grobert, M. Terrones, H. W. Kroto, D. R. M. Walton: *Carbon Nanotubes as Nanoreactors for Boriding Iron Nanowires*. *Advanced Materials*, vol. 12, pp. 1356–1359, 2000.
110. K. Brunner, J. Zhou, G. Abstreiter, **O. Kienzle**, F. Ernst: *Step Bunching and Correlated SiGe Nanostructures on Si(113)*. *Thin Solid Films*, vol. 369, pp. 39–42, 2000.
111. **W. Han**, P. Redlich, T. Seeger, F. Ernst, M. Rühle, W. Hsu, B. Chang, N. Grobert, M. Terrones, Y. Zhu, H. W. Kroto, D. R. M. Walton: *Aligned CN_x Nanotubes Produced by the Pyrolysis of Ferrocene/ C_{60} Mixtures in Ammonia Atmosphere*. *Applied Physics Letters*, vol. 77, pp. 1807–1809, 2000.
112. K. Eberl, O. G. Schmidt, R. Duschl, **O. Kienzle**, F. Ernst, Y. Rau: *Self-Assembling SiGe and SiGeC Nanostructures for Light Emitters and Tunneling Diodes*. *Thin Solid Films*, vol. 369, pp. 33–38, 2000.
113. **W. Han**, P. Redlich, F. Ernst, M. Rühle: *Growth and Microstructure of Ga_2O_3 Nanorods*. *Solid State Communications*, vol. 115, pp. 527–529, 2000.
114. T. Gödecke, **T. Haalboom**, F. Ernst: *Phase Equilibria of Cu-In-Se I. Stable States and Non-Equilibrium States of the In_2Se_3 - Cu_2Se Subsystem*. *Zeitschrift für Metallkunde*, vol. 91, pp. 622–634, 2000.
115. T. Gödecke, **T. Haalboom**, F. Ernst: *Phase Equilibria of Cu-In-Se II. The Cu - Cu_2Se - In_2Se_3 - In Subsystem*. *Zeitschrift für Metallkunde*, vol. 91, pp. 635–650, 2000.

116. T. Gödecke, **T. Haalboom**, F. Ernst: *Phase Equilibria of Cu-In-Se. III. The In_2Se_3 -Se- Cu_2Se Subsystem*. Zeitschrift für Metallkunde, vol. 91, pp. 651-662, 2000.
117. **W. Han**, P. Redlich, F. Ernst, M. Rühle: *Synthesis of GaN-Carbon Composite Nanotubes and GaN Nanorods by Arc Discharge in Nitrogen Atmosphere*. Applied Physics Letters, vol. 76, pp. 652-654, 2000.
118. O. G. Schmidt, K. Eberl, **O. Kienzle**, F. Ernst, S. Christiansen, H. P. Strunk: *Reduced Critical Thickness and Photoluminescence Line Splitting in Multiple Layers of Self-Assembled Ge/Si Islands*. Materials Science and Engineering B, vol. 74, pp. 248-252, 2000.
119. K. Brunner, J. Zhu, C. Miesner, G. Abstreiter, **O. Kienzle**, F. Ernst: *Self-Organized Periodic Arrays of SiGe Wires and Ge Islands on Miscut Si Substrates*. Physica A, vol. 7, pp. 881-886, 2000.
120. K. Lyutovich, E. Kasper, F. Ernst, M. Bauer, M. Oehme: *Relaxed SiGe Buffer Layer Growth with Point Defect Injection*. Materials Science and Engineering B, vol. 71, pp. 14-19, 2000.
121. K. Eberl, O. G. Schmidt, **O. Kienzle**, F. Ernst: *Preparation and Optical Properties of Ge and C-induced Ge Dots on Si*. Materials Research Society Symposia Proceedings, vol. 571, pp. 355-362, 2000.
122. K. Eberl, O. G. Schmidt, **O. Kienzle**, F. Ernst: *Preparation and Optical Properties of Ge and C-induced Ge Dots on Si*. Thin Solid Films, vol. 373, pp. 164-169, 2000.
123. J. Stangl, T. Roch, G. Bauer, I. Kegel, T. H. Metzger, O. G. Schmidt, K. Eberl, **O. Kienzle**, F. Ernst: *Vertical Correlation of SiGe Islands in SiGe-Si Superlattices: X-ray Diffraction Versus Transmission Electron Microscopy*. Applied Physics Letters, vol. 77, pp. 3953-3955, 2000.
124. M. Bauer, K. Lyutovich, M. Oehme, E. Kasper, H.-J. Herzog, F. Ernst: *Relaxed SiGe Buffers with Thicknesses Below 0.1 μm* . Thin Solid Films, vol. 369, pp. 152-156, 2000.
125. K. Eberl, O. G. Schmidt, **O. Kienzle**, F. Ernst: *Self-Assembling Si/SiGe Nanostructures for Light Emitters*. Diffusion & Defect Data B: Solid State Phenomena, vol. 69-70, pp. 13-21, 1999.
126. J. Stangl, V. Holy, J. Grim, G. Bauer, J. Zhu, K. Brunner, G. Abstreiter, **O. Kienzle**, F. Ernst: *Structural Investigation of Si/SiGe Superlattices on Vicinal (113) Oriented Si*. Thin Solid Films, vol. 357, pp. 71-75, 1999.
127. **W. Han**, P. Redlich, F. Ernst, M. Rühle: *Formation of $(\text{BN})_x\text{C}_y$ and BN Nanotubes Filled with Boron Carbide Nanowires*. Chemistry of Materials, vol. 11, pp. 3620-3623, 1999.

128. **W. Han**, P. Redlich, F. Ernst, M. Rühle: *Synthesizing Boron Nitride Nanotubes Filled with SiC Nanowires by Using Carbon Nanotubes as Templates*. Applied Physics Letters, vol. 75, pp. 1875–1877, 1999.
129. K. Eberl, O. G. Schmidt, **O. Kienzle**, F. Ernst: *Preparation and Optical Properties of Ge and C-induced Ge Dots on Si*. Materials Research Society Symposia Proceedings, vol. 570, pp. 187–195, 1999.
130. J. Zhu, K. Brunner, G. Abstreiter, **O. Kienzle**, F. Ernst, M. Rühle: *Correlated SiGe Wires Shaped by Regular Step Bunches on Mismatched Si(111) Substrates*. Physical Review B, vol. 60, pp. 10935–10940, 1999.
131. F. Ernst, R. Raj, M. Rühle: *Nanomechanical Modeling of Misfit Dislocations at Heterointerfaces*. Zeitschrift für Metallkunde, vol. 90, pp. 961–977, 1999.
132. K. Lyutovich, F. Ernst, E. Kasper, M. Bauer, M. Oehme: *Interaction Between Point Defects and Dislocations in SiGe*. Solid State Phenomena, vol. 69–79, pp. 179–184, 1999.
133. O. G. Schmidt, **O. Kienzle**, Y. Hao, K. Eberl, F. Ernst: *Modified Stranski-Krastanov Growth in Stacked Layers of Self-Assembled Islands*. Applied Physics Letters, vol. 74, pp. 1272–1274, 1999.
134. R. Herberholz, U. Rau, H. W. Schock, **T. Haalboom**, T. Gödecke, F. Ernst, C. Beilharz, K. W. Benz, D. Cahen: *Phase Segregation, Cu Migration, and Junction Formation in Cu(In, Ga)Se₂*. European Physical Journal – Applied Physics, vol. 6, pp. 131–139, 1999.
135. **R. Schweinfest**, F. Ernst, T. Wagner, M. Rühle: *High-Precision Assessment of Interface Lattice Offset*. Journal of Microscopy, vol. 194, pp. 142–151, 1999.
136. **D. Baither**, F. Ernst, T. Wagner, M. Rühle, M. Bartsch, U. Messerschmidt: *Micromechanisms of Fracture in NiAl, Studied by in situ HVEM*. Intermetallics, vol. 9, pp. 479–489, 1999.
137. F. Ernst, **O. Kienzle**, M. Rühle: *Structure and Composition of Grain Boundaries in Ceramics*. Journal of the European Ceramic Society, vol. 19, pp. 665–673, 1999.
138. **O. Kienzle**, F. Ernst, M. Rühle, O. G. Schmidt, K. Eberl: *Germanium “Quantum Dots” Embedded in Silicon: Quantitative TEM Study of Self-Alignment and Coarsening*. Applied Physics Letters, vol. 74, pp. 269–271, 1999.
139. J. Zhu, K. Brunner, G. Abstreiter, **O. Kienzle**, F. Ernst: *Lateral Ordering of Self-Assembled Ge Islands*. Thin Solid Films, vol. 336, pp. 252–255, 1998.

140. O. G. Schmidt, C. Lange, K. Eberl, **O. Kienzle**, F. Ernst: *C-Induced Ge Dots: A Versatile Tool to Fabricate Ultra-Small Ge Nanostructures*. Thin Solid Films, vol. 336, pp. 248–251, 1998.
141. F. Ernst, **A. Recnik**, P. A. Langjahr, P. D. Nellist, M. Rühle: *Atomistic Structure of Misfit Dislocations in SrZrO₃/SrTiO₃ Interfaces*. Acta materialia, vol. 47, pp. 183–198, 1998.
142. J.-F. Guillemoles, **T. Haalboom**, T. Gödecke, F. Ernst, D. Cahen: *Phase and Interface Stability Issues in Chalcopyrite-Based Thin Film Solar Cells*. Materials Research Society Symposia Proceedings, vol. 485, pp. 127–132, 1998.
143. O. G. Schmidt, S. Schieker, K. Eberl, **O. Kienzle**, F. Ernst: *Carbon-Induced Germanium Dots: Kinetically Limited Islanding Process Prevents Coherent Vertical Alignment*. Applied Physics Letters, vol. 73, pp. 659–661, 1998.
144. O. G. Schmidt, C. Lange, K. Eberl, **O. Kienzle**, F. Ernst: *Influence of Pre-Grown Carbon on the Formation of Germanium Dots*. Thin Solid Films, vol. 321, pp. 70–75, 1998.
145. C. Schmidt, M. W. Finnis, F. Ernst, V. Vitek: *Theoretical and Experimental Investigations of Structures and Energies of $\Sigma 3$, [112] Tilt Grain Boundaries in Copper*. Philosophical Magazine A, vol. 77, pp. 1161–1184, 1998.
146. **T. Haalboom**, T. Gödecke, F. Ernst, M. Rühle, R. Herberholz, H. W. Schock, C. Beilharz, K. W. Benz: *Phase Relations and Microstructure of Bulk Material and Thin Films of the Ternary System Cu–In–Se*. Inst. Phys. Conf. Ser., vol. 152 B, pp. 249–252, 1998.
147. P. Pirouz, F. Ernst, Y. Ikuhara: *On Epitaxy and Orientation Relationship in Bicrystals*. Diffusion and Defect Data Part B, vol. 59–60, pp. 51–62, 1998.
148. **T. Gemming**, G. Möbus, M. Exner, F. Ernst, M. Rühle: *Towards **ab-initio** HRTEM Simulations*. Journal of Microscopy, vol. 190, pp. 89–98, 1998.
149. **O. Kienzle**, F. Ernst, G. Möbus: *Reliability of Atom Column Positions in a Ternary System Determined by Quantitative High-Resolution Transmission Electron Microscopy*. Journal of Microscopy, vol. 190, pp. 144–158, 1998.
150. **O. Kienzle**, F. Ernst: *Analysis of Interface Structures by Quantitative HRTEM*, in: *Electron Microscopy and Analysis 1997*, Ed. J. M. Rodenburg. IOP Publishing (Bristol), pp. 457–462, 1998.

151. **O. Kienzle**, M. Exner, F. Ernst: *Atomistic Structure of $\Sigma=3$, (111) Grain Boundaries in Strontium Titanate*. Physica Status Solidi (a), vol. 166, pp. 57–71, 1998.
152. G. Wöhl, C. Schöllhorn, O. G. Schmidt, K. Brunner, K. Eberl, **O. Kienzle**, F. Ernst: *Characterization of Self-Assembled Ge Islands on Si (100) by Atomic Force Microscopy and Transmission Electron Microscopy*. Thin Solid Films, vol. 321, pp. 86–91, 1998.
153. **O. Kienzle**, F. Ernst: *Analysis of Interface Structures by Quantitative HRTEM*. Journal of Computer-Assisted Microscopy, vol. 9, pp. 13–15, 1997.
154. **U. Klement**, F. Ernst: *Microstructure and Opto-Electronic Properties of CdSe Thin Films*. Materials Research Society Symposia Proceedings, vol. 472, pp. 131–136, 1997.
155. F. Ernst, **O. Kienzle**, M. Exner: *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Materials Research Society Symposia Proceedings, vol. 466, pp. 95–106, 1997.
156. K. Lyutovich, F. Ernst, F. Banhart, I. Silier, A. Gutjahr, M. Konuma: *Defect Distribution in Compositionally-Graded Epitaxial SiGe Layers on Si Substrates*. Inst. Phys. Conf. Ser., vol. 157, pp. 131–134, 1997.
157. O. G. Schmidt, C. Lange, K. Eberl, **O. Kienzle**, F. Ernst: *Formation of Carbon-Induced Germanium Dots*. Applied Physics Letters, vol. 71, pp. 2340–2342, 1997.
158. **O. Kienzle**, F. Ernst, M. Rühle: *Atomic-Resolution Transmission Electron Microscopy – Present State and Future Perspectives*. Acta Microscopica, vol. 6, pp. 36–39, 1997.
159. F. Ernst: *Interface Dislocations Forming in GeSi Layers During Epitaxy on {111} Si Substrates at High Temperatures*. Materials Science and Engineering A, vol. 233, pp. 126–138, 1997.
160. A. Mariano, C. Gatts, F. Ernst, M. Rühle: *Determining Periodicities at Grain Boundaries in HRTEM*. Acta Microscopica, vol. 6, pp. 142–145, 1997.
161. **O. Kienzle**, F. Ernst: *Effect of Shear Stress on the Atomistic Structure of a Grain Boundary in Strontium Titanate*. Journal of the American Ceramic Society, vol. 80, pp. 1639–1644, 1997.
162. **U. Klement**, D. Horst, F. Ernst: *Microstructure of Thin Film Photoconductors and its Correlation with Optical and Electronic Properties*. Materials Research Society Symposia Proceedings, vol. 452, pp. 925–930, 1997.

163. **A. Recnik**, P. A. Langjahr, F. Ernst: *Structural Characterization of SrZrO₃/SrTiO₃ Epitaxial Layers by HRTEM*. Journal of Computer-Assisted Microscopy, vol. 9, pp. 35–36, 1997.
164. F. Ernst, M. W. Finnis, W. Gust, A. Koch, C. Schmidt, B. Straumal: *Structure and Energy of Twin Boundaries in Copper*. Zeitschrift für Metallkunde, vol. 87, pp. 911–922, 1996.
165. G. Dehm, F. Ernst, J. Mayer, G. Möbus, H. Mülleians, F. Phillipp, C. Scheu, M. Rühle: *Transmission Electron Microscopy at the Max-Planck-Institut für Metallforschung*. Zeitschrift für Metallkunde, vol. 87, pp. 898–910, 1996.
166. P. Pirouz, Y. Ikuhara, F. Ernst: *On Epitaxy and Orientation Relationship in Physisorption*, in: *Interface Science and Materials Interconnection*. The Japan Institute of Metals (Tokyo), pp. 107–115, 1996.
167. F. Ernst, M. Rühle: *Structure of Twin Boundaries in Copper, Studied by Quantitative High-Resolution Transmission Electron Microscopy*, in: *Interface Science and Materials Interconnection*. The Japan Institute of Metals (Tokyo), pp. 3–10, 1996.
168. **K. Nadarzynski**, F. Ernst: *The Atomistic Structure of a $\Sigma=3$, (111) Grain Boundary in NiAl, Studied by Quantitative High-Resolution Transmission Electron Microscopy*. Philosophical Magazine A, vol. 74, pp. 641–664, 1996.
169. G. Dehm, **K. Nadarzynski**, F. Ernst, M. Rühle: *Quantification of Irradiation Damage Generated During HRTEM with 1250 keV Electrons*. Ultramicroscopy, vol. 63, pp. 49–55, 1996.
170. C. Schmidt, M.W. Finnis, V. Vitek, F. Ernst: *Prediction of the BCC Structure in a Cu $\Sigma 3$ -84°(211) Tilt Grain Boundary and Its Confirmation by HRTEM*, in: *Intergranular and Interphase Boundaries in Materials*, Eds.: A. C. Ferro, E. P. Conde, E. A. Fortes. Transtec Publications Ltd. (Zürich), pp. 337–340, 1996.
171. **K. Nadarzynski** and F. Ernst: *Quantitative HRTEM of the $\Sigma 3(111)$ Grain Boundary in NiAl*, in: *Intergranular and Interphase Boundaries in Materials*, Eds. A. C. Ferro, E. P. Conde, E. A. Fortes. Transtec Publications Ltd. (Zürich), pp. 309–312, 1996.
172. C. Schmidt, F. Ernst, M. W. Finnis, V. Vitek: *Prediction and Observation of the BCC Structure in Pure Copper at a Sigma-3 Grain Boundary*. Physical Review Letters, vol. 75, pp. 2160–2163, 1995.
173. F. Ernst, **D. Hofmann**, **K. Nadarzynski**, **S. Stemmer**, S. K. Streiffer: *Quantitative High-Resolution Electron Microscopy of Interfaces*, in: *Intergranular and Interphase Boundaries in Materials*, Eds. A. C. Ferro, E.

- P. Conde, E. A. Fortes. Transtec Publications Ltd. (Zürich), pp. 23–34, 1996.
174. G. Möbus, F. Ernst: *Determination of Interface Structures by Quantitative Interpretation of HREM Images*, in: *Image Interpretation and Image Processing in Electron Microscopy*, Eds. J. Heydenreich, W. Neumann. Elbe (Wittenberg), pp. 61–70, 1995.
 175. **S. Stemmer**, S. K. Streiffer, F. Ernst, M. Rühle, W.-Y. Hsu, R. Raj: *Domain Configurations in Ferroelectric PbTiO_3 Thin Films: The Influence of Substrate and Film Thickness*. Solid State Ionics, vol. 75, pp. 43–48, 1995.
 176. **S. Stemmer**, S. K. Streiffer, W. Y. Hsu, F. Ernst, R. Raj, M. Rühle: *The Influence of Pt and SrTiO_3 Interlayers on the Microstructure of PbTiO_3 Thin Films Deposited by Laser Ablation on (001) MgO* . Journal of Materials Research, vol. 10, pp. 791–794, 1995.
 177. **S. Stemmer**, S. K. Streiffer, F. Ernst, M. Rühle: *Atomistic Structure of 90° Domain Walls in PbTiO_3 Thin Films*. Philosophical Magazine A, vol. 71, pp. 713–724, 1995.
 178. **S. Stemmer**, S. K. Streiffer, F. Ernst, M. Rühle: *Dislocations in PbTiO_3 Thin Films*. Physica Status Solidi (a), vol. 147, pp. 135–154, 1995.
 179. **D. Hofmann**, F. Ernst, M. Rühle: *Quantitative High-Resolution Transmission Electron Microscopy of a Grain Boundary in Cu*. Transactions of Materials Research Society Japan, vol. 16B, pp. 1331–1338, 1994.
 180. **D. Hofmann**, F. Ernst: *Twin Boundaries with 9R Zone in Cu and Ag Studied by Quantitative HRTEM*. Interface Science, vol. 2, pp. 201–210, 1994.
 181. Ch. Borchers, F. Ernst, F. Haider, R. Hattenhauer, G. Schmitz, P.-J. Wilbrandt, S.-Q. Xiao: *The Investigation of Decomposition Processes by HREM and Unconventional TEM*. Physica Status Solidi (a), vol. 146, pp. 71–87, 1994.
 182. J. Bruley, **S. Stemmer**, F. Ernst, M. Rühle, W. Y. Hsu, R. Raj: *Nanostructure and Chemistry of a (100) MgO /(100) GaAs Interface*. Applied Physics Letters, vol. 65, pp. 564–566, 1994.
 183. F. Ernst: *Dissociation of Misfit Dislocations in $\text{GeSi}/\{111\}\text{Si}$ Interfaces*. Materials Research Society Symposia Proceedings, vol. 319, pp. 165–170, 1994.
 184. F. Ernst, M. Rühle: *Diffusion Reactions at Metal/Ceramic and Ceramic/Ceramic Interfaces*, in: *Diffusion Reactions*, Eds. Y. Limoge, J. L. Bocquet. Transtec Publications Ltd. (Aedermannsdorf), pp. 331–342, 1994.

185. **Y. Tzou**, J. Bruley, F. Ernst, M. Rühle, R. Raj: *Nucleation of Special Orientations During Heteroepitaxial Growth of Diamond on Silicon*. Materials Research Society Symposia Proceedings, vol. 317, pp. 517–522, 1994.
186. **Y. Tzou**, J. Bruley, F. Ernst, M. Rühle, R. Raj: *TEM Study of the Structure and Chemistry of a Diamond/Silicon Interface*. Journal of Materials Research, vol. 9, pp. 1566–1572, 1994.
187. **B. Roos**, F. Ernst: *Thermal Stress-Induced Dislocations in GeSi/Si Heterostructures*. Journal of Crystal Growth, vol. 137, pp. 457–471, 1994.
188. **D. Hofmann**, F. Ernst: *Quantitative High-Resolution Transmission Electron Microscopy of the Incoherent $\Sigma 3$, (211) Boundary in Cu*. Ultramicroscopy, vol. 53, pp. 205–221, 1994.
189. M. Rühle, F. Ernst: *On the Thermal Stability of Metal/Ceramic Interfaces in Composites*, in: *High Temperature Ceramic Matrix Composites*, Eds. R. Naslain, J. Lamon, D. Doumeingts. Woodhead Publishing Ltd. (Bordeaux), 1993.
190. F. Ernst: *Dissociation of Misfit Dislocation Nodes in GeSi/Si-(111)-Interfaces*. Philosophical Magazine A, vol. 68, pp. 1251–1272, 1993.
191. **B. Roos**, F. Ernst: *Thermal Stress-Induced Defects in GeSi/Si Heterostructures*. Inst. Phys. Conf. Ser., vol. 134(6), pp. 337–340, 1993.
192. F. Ernst, M. W. Finnis, **D. Hofmann**, T. Muschik, U. Schönberger, U. Wolf, M. Methfessel: *Theoretical Prediction and Direct Observation of the 9R Structure in Ag*. Physical Review Letters, vol. 69, pp. 620–623, 1992.
193. F. Ernst, P. Pirouz, E. Bauser: *Lattice Mismatch Accommodation at GeSi/(111)Si Interfaces Grown by Liquid Phase Epitaxy*. Physica Status Solidi (a), vol. 131, pp. 651–662, 1992.
194. **A. Trampert**, F. Ernst, C. P. Flynn, H. F. Fischmeister, M. Rühle: *High Resolution Transmission Electron Microscopy Studies of the Ag/MgO Interface*. Acta metallurgica et materialia, vol. 40, pp. S227–S236, 1992.
195. U. Wolf, F. Ernst, T. Muschik, M. W. Finnis, H. F. Fischmeister: *The Influence of Grain Boundary Inclination on the Structure and Energy of $\Sigma 3$ Twin Boundaries in Copper*. Materials Research Society Symposia Proceedings, vol. 238, pp. 177–182, 1992.
196. J. Bruley, F. Ernst, K. Ljutovich: *Germanium Concentration Profiles Across Interfaces and Close to Dislocations in CVD $\text{Si}_{1-x}\text{Ge}_x$ -on-Si Junctions*. Materials Research Society Symposia Proceedings, vol. 238, pp. 469–474, 1992.

197. P. O. Hansson and F. Ernst: *Two-Dimensional Growth of Strained $Ge_{0.85}Si_{0.15}$ on Si (111) by Liquid Phase Epitaxy*. Journal of Applied Physics, vol. 72, pp. 2083–2085, 1992.
198. U. Wolf, F. Ernst, T. Muschik, M. W. Finnis, H. F. Fischmeister: *The Influence of Grain Boundary Inclination on the Structure and Energy of $\Sigma 3$ Grain Boundaries in Copper*. Philosophical Magazine A, vol. 66, pp. 991–1016, 1992.
199. J. X. Chen, F. Ernst, P. O. Hansson, E. Bauser: *Liquid Phase Epitaxy of GeSi on {111}-Si-Substrates: Lattice Defect Structure and Electronic Properties*. Journal of Crystal Growth, vol. 118, pp. 452–460, 1992.
200. K. Weronek, J. Weber, A. Höpner, F. Ernst, R. Buchner, M. Stefaniak, H. Alexander: *Correlation of the D-Band Photoluminescence with Spatial Properties of Dislocations in Silicon*. Materials Science Forum, vol. 83–87, pp. 1315–1320, 1992.
201. F. Ernst, P. Pirouz, A. H. Heuer: *HRTEM Study of a Cu/ Al_2O_3 Interface*. Philosophical Magazine A, vol. 63, pp. 259–277, 1991.
202. P. Pirouz, J. W. Yang, J. A. Powell, F. Ernst: *The Role of Dislocations in the $3C \leftrightarrow 6H$ SiC Polytypic Transformation*. Inst. Phys. Conf. Ser. No., vol. 117(3), pp. 149–154, 1991.
203. J. Mayer, W. Mader, D. Knaus, F. Ernst, M. Rühle: *Structures of Nb/ Al_2O_3 Interfaces Produced by Different Experimental Routes*. Materials Research Society Symposia Proceedings, vol. 183, pp. 55–60, 1990.
204. F. Ernst: *HRTEM of the Cu/MnO Interface in an Internally Oxidized CuMn Alloy*. Materials Research Society Symposia Proceedings, vol. 183, pp. 49–55, 1990.
205. P. Pirouz, F. Ernst: *High Resolution Electron Microscopy: Possibilities and Limitations*, in: *Metal-Ceramic Interfaces*, Eds. M. Rühle et al.. Pergamon Press (New York), pp. 199–221, 1990.
206. F. Ernst, P. Pirouz: *The Formation Mechanism of Planar Defects in Compound Semiconductors Grown Epitaxially on {100} Silicon Substrates*. Journal of Materials Research, vol. 4, pp. 834–842, 1989.
207. F. Ernst, P. Pirouz, A. H. Heuer: *High Resolution Electron Microscopy of an Alumina/Copper Interface*. Materials Research Society Symposia Proceedings, vol. 138, pp. 557–562, 1989.
208. U. Klement, F. Ernst, P.-J. Wilbrandt: *Recrystallization Experiments in Tensile-Deformed $\langle 100 \rangle$ - and $\langle 111 \rangle$ -Oriented Copper Single Crystals*, in: *Textures and Microstructures*, Gordon and Breach. UK (1988), pp. 383–400, 8–9.

209. P. Pirouz, J. Yang, F. Ernst, H.-J. Möller: *Hexagonal Silicon: A New HREM Study*. Materials Research Society Symposia Proceedings, vol. 139, pp. 199–204, 1988.
210. T. T. Cheng, P. Pirouz, F. Ernst: *Inversion Domain Boundary Dislocations in Heteroepitaxial Films*. Materials Research Society Symposia Proceedings, vol. 144, pp. 189–194, 1988.
211. P. Pirouz, F. Ernst, T. T. Cheng: *Heteroepitaxy on (001) Silicon: Growth Mechanisms and Defect Formation*. Materials Research Society Symposia Proceedings, vol. 116, pp. 57–70, 1988.
212. P. Pirouz, S. Farmer, F. Ernst, J. Chung: *Transmission Electron Microscopy of Composites*, in: *Interfaces in Polymer, Ceramic, and Metal Matrix Composites*, Ed. Hsuo Ishida. Elsevier, pp. 141–157, 1988.
213. F. Ernst, P. Pirouz: *Formation of Planar Defects in the Epitaxial Growth of GaP on Si Substrate by MOCVD*. Journal of Applied Physics, vol. 64, pp. 4526–4530, 1988.
214. F. Ernst: *Correction*. Scripta Metallurgica, vol. 22, pp. 305, 1988.
215. F. Ernst, P. Haasen: *The Decomposition Kinetics of Al-1 at% Ag at 413 K Studied by HREM*. Physica Status Solidi (a), vol. 104, pp. 403–416, 1987.
216. F. Ernst, S.Q. Xiao, F. Haider, P.-J. Wilbrandt: *High Resolution Electron Microscopy of Small Precipitates*. Scripta metallurgica, vol. 21, pp. 1189–1194, 1987.
217. F. Ernst, S.Q. Xiao, F. Haider, P.-J. Wilbrandt: *Investigation of the Decomposition of Al-1 at% Ag by HREM of GP-Zones*. European Journal of Cell Biology, vol. 19, pp. 17, 1987.
218. Measurement of Tridentate Astigmatism of a Conventional HRTEM and a high-vacuum HRTEM: R. Schweinfest, T. Gemming, P. Kopold, F. Ernst, M. Rühle. European Journal of Cell Biology, vol. 74, pp. 78–78, 1997.
219. F. Ernst, P.-J. Wilbrandt: *Growth Selection Experiments in Tensile-Deformed $\langle 100 \rangle$ -Oriented Copper Single Crystals*, in: ICOTOM 7, Eds. C. M. Brakman, P. Jongenburger, E. J. Mittemeijer. Netherland Society for Materials Science (Zwijndrecht, The Netherlands), pp. 233–238, 1984.

Refereed Review Articles and Book Chapters

1. S. R. Collins, P. C. Williams, S. V. Marx, A. Heuer, F. Ernst, H. Kahn: *Low-Temperature Carburization of Austenitic Stainless Steels*, in: *ASM Handbook: Vol. 4D, Heat Treating of Irons and Steels*, Eds. J. Dossett,

- G. Totten. ASM International (Materials Park, Ohio, USA), pp. 451-460, 2014.
2. A. H. Heuer, F. Ernst, S. Collins, P. M. Nathishan, H. Kahn: *Corrosion Resistant Alloys: Case Hardening*, in: *Encyclopedia of Iron, Steel, and Their Alloys*, Eds. Rafael Colas, George E. Totten. CRC Press (London, UK), pp. 0-0, 2016.
 3. F. Ernst, W. Sigle: *Microcharacterisation of Materials*, in: *High-Resolution Imaging and Spectrometry of Materials*, Eds. F. Ernst, M. Rühle. Springer Series in Materials Science, Springer Verlag (New York), pp. 1-8, 2002.
 4. F. Ernst, M. Rühle: *Present Developments in High-Resolution Transmission Electron Microscopy*. Current Opinion in Solid State and Materials Science, vol. 2, pp. 469-476, 1997.
 5. F. Ernst: *Metal-Oxide Interfaces*. Materials Science and Engineering R, vol. 147, pp. 97-156, 1995.
 6. A. Berger, P.-J. Wilbrandt, F. Ernst, U. Klement, P. Haasen: *On the Generation of New Orientations During Recrystallization: Recent Results on the Recrystallization of Tensile-Deformed FCC Single Crystals*. Progress in Materials Science, vol. 32, pp. 2-95, 1988.

Refereed Online Publications

1. F. Ernst: *Laue Diffractograms*. Wolfram Research, <http://goo.gl/n8UKdK>, 2014.
2. F. Ernst: *Pole Figures in Crystallography*. Wolfram Research, <http://goo.gl/grspzz>, 2014.

Conference Proceedings[§]

1. **K. Du**, F. Ernst, M. Pelsozy, K. Tillmann, J. Barthel: *Strain analysis on platinum nanoparticles with aberration-corrected transmission electron microscopy*, in: *17th International Microscopy Congress (IMC17)*, Rio de Janeiro, Brazil, September 19 to 24, 2010, pp. CD.
2. **A. Avishai**, D. Isheim, D. N. Seidman, F. Ernst, G. M. Michal, A.H. Heuer: *Local-Electrode Atom Probe (LEAP™) Tomographic Microanalysis of Low-Temperature Gas-Carburized Austenitic Stainless Steel*. Microscopy & Microanalysis, vol. 13 (Supl 2), pp. 1094-1095, 2007.

[§]Author names in **boldface** indicate students or post-doctoral researchers I advised.

3. **J. Gentil**, F. Ernst, G. Michal, A. Heuer: *The Effect of Colossal Carbon Supersaturation on Stainless Steels of the Type PH13-8Mo and AL6XN*, in: *Recent Developments in Steel Processing*, Materials Science & Technology 2006. Editor(s): Matthew Merwin, Brian Nelson (Warrendale, PA), pp. CD, 2006.
4. Michal, G. M. and Ernst F. and **Cao Y.**: *Surface Hardening of Austenitic Steels by Low Temperature Colossal Supersaturation*, in: *Proceedings of Materials Science & Technology*, Volume II. AIST - TMS (Warrendale, PA), pp. 347 - 353, 2004.
5. **K. Du**, F. Ernst, Y. Ikuhara, P. Pirouz: *Preferred Intercrystalline Orientation Relationships and the Near Coincidence of Reciprocal Lattice Points*, in: *Proceedings of Microscopy & Microanalysis*, Savannah, Georgia, USA, August 1-5, 2004.
6. **F. Oba**, R. Liu, **Y. Yu**, E. W. Bohannon, F. Ernst, J. A. Switzer: *TEM Studies of Cu_2O -Si and Cu_2O -InP Interfaces Made by Epitaxial Electrodeposition*, in: *Proceedings of Microscopy & Microanalysis*, Savannah, Georgia, USA, August 1-5, 2004.
7. **J. Cowen**, **L. Lucas**, F. Ernst, P. Pirouz, A. Hepp, S. Bailey: *Liquid-Phase Deposition of α -Cu-In-Se Thin Films*, in: *Proceedings of Space Photovoltaic Research and Technology Conference (SPRAT)*, Ohio Aerospace Institute, September 16-18, 2003.
8. K. Lyutovich, M. Oehme, F. Ernst: *Growth of Ultra-Thin and Highly Relaxed SiGe Layers Under In-Situ Introduction of Point Defects*, in: *Proceedings of 10th International Conference on Defects: Recognition, Imaging and Physics in Semiconductors*, September 29 - October 02, 2003, Batz-sur-Mer, Loire Atlantique, France.
9. M. Tabib-Azar, A. Scott, **R. Adelung**, F. Ernst: *Nano Metallic Solid Electrolyte Switches and Cu Nanowire Interconnects for Molecular Electronics*, in: *Proceedings of 2001 IEEE International Electronic Devices Meeting*, December 3-5, 2001, Washington (DC) USA.
10. **R. Schweinfest**, F. Ernst: *Progress in Structure Analysis of Crystal Defects by Quantitative HRTEM*, in: *Proceedings of the 12th European Congress on Electron Microscopy*, July 9-14, 2000, Brno, Czech Republic, pp. 111-114.
11. G. Wöhl, E. Kasper, T. Hackbarth, H. Kibbel, M. Klose, F. Ernst: *Fully Relaxed $\text{Si}_{0.7}\text{Ge}_{0.3}$ Buffers Grown on Patterned Silicon Substrates for Hetero-CMOS Transistors*, in: *Proceedings of 3rd International Conference on Materials for Microelectronics*, October 16-17, 2000, Dublin, Ireland.

12. **O. Kienzle, S. Hutt, F. Ernst, M. Rühle:** *Grain Boundaries in Strontium Titanate*, in: *Proceedings of 10th JFCC International Workshop on Fine Ceramics 2000*, March 15-17, 2000, Nagoya, Japan.
13. K. Lyutovich, F. Ernst, E. Kasper, M. Bauer, M. Oehme: *Interaction Between Point Defects and Dislocations in SiGe*, in: *Proceedings of GADEST '99*, September 25-28, 1999, Höör, Sweden.
14. F. Ernst, C. Scheu, **R. Schweinfest**, M. Rühle: *Structure, Composition, and Bonding of Grain Boundaries in Metals*, in: *Proceedings of 37th Metallurgists' Day and 53rd Annual Technical Meeting of the Indian Institute of Metals*, November 14-16, 1999, Kanpur, India.
15. M. Bauer, K. Lyutovich, M. Oehme, E. Kasper, H.-J. Herzog, F. Ernst: *Relaxed SiGe Buffers with Thicknesses Below 0.1 Micron* *Proceedings of the International Joint Conference on Silicon Epitaxy and Heterostructures*, in: *September 13-17, 1999, Miyagi, Japan*, , pp.
-
16. Brunner, J. Zhu, G. Abstreiter, **O. Kienzle**, F. Ernst: *Regular Step Bunching and Ordering of Ge Islands on Vicinal Si Surfaces*, in: *Proceedings of the 24th International Conference on the Physics of Semiconductors*, August 2-7, 1998, Jerusalem, Israel.
17. **O. Kienzle**, F. Ernst, G. Möbus: *Error Analysis in HRTEM Structure Retrieval by Multiple Units Analysis*, in: *Electron Microscopy 1998*, vol. III, Eds. H. A. C. Benavides, M. J. Yacaman. Mexican Society of Microscopy (Cancun), pp. 627-628, 1998.
18. **O. Kienzle**, F. Ernst: *Self-Assembled Ge-Si Nanostructures for Quantum-Effect Electronic Devices*, in: *Electron Microscopy 1998*, vol. III, Eds. H. A. C. Benavides, M. J. Yacaman. Mexican Society of Microscopy (Cancun), pp. 365-366, 1998.
19. **R. Schweinfest**, F. Ernst, T. Wagner, M. Rühle: *Quantitative HRTEM at the Al/MgAl₂O₄ Interface: Translation State and Its Reliability*, in: *Electron Microscopy 1998*, vol I, Eds. H. A. C. Benavides, M. J. Yacaman. Mexican Society of Microscopy (Cancun), pp. 635-636, 1998..
20. **R. Schweinfest**, T. Wagner, F. Ernst, M. Rühle: *Atomistic Structure of Al/MgAl₂O₄ Interfaces*, in: *Microscopy and Microanalysis 1997*, Eds. G. W. Bailey, J. M. Corbett, R. V. W. Dimlich, J. R. Michael, N. J. Zaluzec. San Francisco Press Inc. (San Francisco), pp. 326-327, 1997.
21. R. Herberholz, H. W. Schock, U. Rau, J. H. Werner, **T. Haalboom**, F. Ernst, C. Beilharz, K. W. Benz, D. Cahen: *New Aspects of Phase Segregation and Junction Formation in CuInSe₂*, in: *Proceedings of 26th IEEE Photovoltaic Specialists Conference*, September 29-October 3, 1997, Anaheim, USA, pp. 323-326.

22. **D. Baither**, U. Messerschmidt, B. Baufeld, M. Bartsch, F. Ernst: *In-situ Straining Experiments in HVEM to Study Deformation of Zirconia and NiAl*, in: *Proceedings of the 13th National Electron Microscopy Congress with International Participation*, September 1-4, 1997, Ankara, Turkey, pp. 517-530.
23. O. Kienzle, M. Exner, F. Ernst: *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Materials Research Society Symposium Proceedings, vol. 466, pp. 95-106, 1997.
24. **K. Nadarzynski**, **O. Kienzle**, F. Ernst: *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*, in: *Microscopy and Microanalysis 1996*, Eds. G. W. Bailey, J. M. Corbett, R. V. W. Dimlich, J. R. Michael, N. J. Zaluzec. San Francisco Press Inc. (San Francisco), pp. 326-327, 1997.
25. **O. Kienzle**, F. Ernst: *TEM analysis of GeSi quantum point structures*. European Journal of Cell Biology, vol. 74, pp. 52-52, 1997.
26. **T. Gemming**, F. Ernst, M. Rühle: *Atomistische Struktur von Korngrenzen in α -Al₂O₃*, in: *Werkstoffwoche '96, Symposium 7, Materialwissenschaftliche Grundlagen*, Eds. F. Aldinger, H. Mughrabi. DGM-Informationsgesellschaft mbH (Frankfurt), pp. 865-868, 1997.
27. **R. Schweinfest**, F. Ernst, T. Wagner: *Struktur von Al/MgAl₂O₄-Grenzflächen*, in: *Werkstoffwoche '96, Symposium 7, Materialwissenschaftliche Grundlagen*, Eds. F. Aldinger, H. Mughrabi. DGM-Informationsgesellschaft mbH (Frankfurt), pp. 849-852, 1997.
28. **O. Kienzle**, F. Ernst, M. Rühle: *Struktur der $\Sigma 3$ (111) Kippkorngrenze in Strontiumtitanat*, in: *Werkstoffwoche '96, Symposium 7, Materialwissenschaftliche Grundlagen*, Eds. F. Aldinger, H. Mughrabi. DGM-Informationsgesellschaft mbH (Frankfurt), pp. 813-818, 1997.
29. F. Phillipp, F. Ernst: *High-Voltage Atomic Resolution Microscopy with the Stuttgart ARM 1250*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin 1996, CD-ROM.
30. **O. Kienzle**, F. Ernst, M. Rühle: *Quantitative HRTEM of the $\Sigma 3$ (111) Grain Boundary in Strontium Titanate*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin, 1996, CD-ROM.
31. **R. Schweinfest**, F. Ernst, G. Möbus, T. Wagner: *Structure of the Al/MgAl₂O₄ Interface Determined by Quantitative HRTEM*, in: *Proc. 11th European Congress on Electron Microscopy (EUREM 96)*, Ed. U. C. D. Belfield, Dublin, 1996, CD-ROM.

32. **A. Recnik**, M. Ceh, F. Ernst: *Quantitative HRTEM Analysis of Antiphase Boundaries in SrO-Doped Calcium Titanate*, in: *Electron Microscopy 1994, vol. 2A*, Eds. B. Jouffrey and C. Coliex. Les Editions de Physique (Les Ulis), pp. 405–406, 1994.
33. F. Ernst, **D. Hofmann**: *Quantitative HRTEM of Grain Boundaries*, in: *Electron Microscopy 1994, vol. 2A*, Eds. B. Jouffrey and C. Coliex. Les Editions de Physique (Les Ulis), pp. 121–122, 1994.
34. S. K. Streiffer, S. Stemmer, W. Y. Hsu, F. Ernst, R. Raj, M. Rühle: *The Structure And Distribution of 90 Degrees Domains in PbTiO₃ Ferroelectric Thin Films*, in: *Fifty-second Annual Meeting, Microscopy Society of America and Twenty-Ninth Annual Meeting. Microbeam Analysis Society* (Eds. G. W. Bailey and A. J. Garratt-Reed), pp. 564–565, 1994..
35. **D. Hofmann**, G. Möbus, F. Ernst: *Quantitative HRTEM of Incoherent Twin Boundaries in Copper*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al.. Universidad de Granada (Granada), pp. 513–514, 1992.
36. G. Möbus, M. Hohenstein, F. Phillipp, F. Ernst, G. Necker, M. Rühle: *A Procedure to Measure the Chromatic Contrast Transfer Envelope of High-Resolution Electron Microscopes*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al... Universidad de Granada (Granada), pp. 531–532., 1992.
37. W. Mader, F. Ernst, G. Möbus: *HRTEM of Interfaces in Crystalline Solids*, in: *Electron Microscopy 92, vol. 1*, Eds. A. López-Galindo et al... Universidad de Granada (Granada), pp. 477–481, 1992.
38. S. Wen, F. Ernst, H. Jaeger, W. Mader P. Majewski, M. Marie, G. Petzow, M. Rühle: *HRTEM observation of the phase transition from Bi₂Sr₂CaCu₂O₈ to ₂Sr₂Ca₂Cu₃O₁₀*, in: „, Proceedings, annual meeting, Electron Microscopy Society of America. Cambridge University Press (CUP) (1990), pp. 94-95.

Publications as Editor

1. F. Ernst, M. Rühle: *High-Resolution Imaging and Spectrometry of Materials*. Springer Series in Materials Science, Springer Verlag (New York), 2002.
2. F. Ernst, J. Mayer, M. Rühle: *Journal of Microscopy* vol. 194, 1999.
3. W. Mader, F. Ernst: *Physica Status Solidi (a)* vol. 166, 1998.

Other Publications

Patents

1. P. Williams, S. Marx, F. Ernst, **A. Agaponova**: *Enhanced Activation of Self-Passivating Metals*. Patent, US 12371774B2, 2025-07-29.
2. Frank Ernst, Todd Johns, Ronald Edmonson, Cyprian Adair William Illing, Peter C. Williams, Christina Semkow: *Activation of Self-Passivating Metals Using Reagent Coatings for Low-Temperature Nitrocarburization in the Presence of Oxygen-Containing Gas*. Patent Application, filed April 28, 2022. Publication Number: 20220364216..
3. **Z. Ren**, F. Ernst: *Metal Carburization Process To Produce a Uniform, Concentrated Solid Solution of Interstitial Carbon Workpiece and Articles Made From Same*. Patent, US 11,066,735 B2.
4. **Zhe Ren**, Frank Ernst: *Metal Carburization Process to Produce a Uniform Concentrated Solid Solution of Interstitial Carbon Workpiece and Articles Made From Same*. US Patent Application, 56/654,367, filed on October 16, 2019..
5. **Zhen Li**, Frank Ernst, Harold Kahn, Arthur Heuer: *Alloy Surface Activation by Immersion in Aqueous Acid Solution*. Patent, US 10,450,658.
6. P. Williams, S. Marx, F. Ernst, **A. Agaponova**: *Enhanced Activation of Self-Passivating Metals*. Patent, US 2019/10,214,805.
7. **Z. Li**, F. Ernst, X. Zhan, A.H. Heuer: *Inducing Ferromagnetism in Ni-Base Superalloys*. Invention Disclosure, 2017.
8. **A. Agaponova**, P. Williams, S. Marx, F. Ernst: *Enhanced Activation of Self-Passivating Metals*. Patent Application, US 2016/0032442A1.
9. **Z. Li**, F. Ernst: *Alloy Surface Activation by Immersion in Aqueous Acid Solution*. PCT Application, 2015.
10. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. Japan Patent JP 5079963 B2, filed November 21, 2012.
11. **Chi-Hong Liao**, Uziel Landau, Frank Ernst: *A Non-Cyanide Alkaline Electroplating Bath Formulation for Copper Electrodeposition on Metals*. Provisional patent application, 2011.
12. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. US Patent application 10311716, filed February 7, 2003.

13. F. Ernst, G. M. Michal, A. H. Heuer: *Surface Hardening of Ti Alloys by Gas-Phase Nitridation: Kinetic Control of the Nitrogen Activity*. US Patent 2006/0048862 A1.
14. R. W. Bruce, A. H. Heuer, G. M. Michal, F. Ernst: *Titanium Treatment to Minimize Fretting*. US Patent 2006/0289088 A1.
15. T. Gödecke and F. Ernst: *Preparation of Compounds Based on Phase Equilibria of Cu-In-Se*. Patent No. 01947428.7-2119-EP0107461, issued June 29, 2001.

Invited Presentations

1. *Alloy Surface Engineering by Interstitial Carbon or Nitrogen*. Colloquium, Texas A&M University. March 03, 2025.
2. *Alloy Surface Engineering by Infusion of Carbon or Nitrogen*. Colloquium, University of Alabama. November 09, 2023.
3. *Alloy Surface Engineering by Infusion of Interstitial Atoms*. Colloquium, Rice University. November 10, 2022.
4. *A New Perspective on Alloy Surface Engineering by Infusing High Concentrations of Interstitial Atoms*. Colloquium, University of Central Florida. March 12, 2021.
5. *Progress in Surface Engineering by Concentrated Interstitial Solute*. Colloquium, University of California at Davis. October 31, 2017.
6. *Materials Design by Surface Engineering*. Erich Bloch Symposium, University of Buffalo. June 1, 2017.
7. *Surface Engineering by Concentrated Interstitial Solute*. Colloquium, University of Maryland. May 5, 2017.
8. *Basic and Exploratory Research in the Department of Materials Science and Engineering at CWRU*. Army Research Laboratory, Aberdeen, MD, USA. February 14, 2017.
9. *The Department of Materials Science and Engineering*. Colloquium, Chemistry Department, College of Arts and Sciences, CWRU. October 28, 2016.
10. *Metals and Metalworking: DMSE as R&D Partner*. German Metal Working Delegation Meeting, Cleveland, OH, USA. July 12, 2016.
11. *Interdisciplinary Research*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 13, 2016.

12. *The Need to See Small: Structure-Property Relationships*. Atotech Cross-Functional Meeting, Neu-Hardenberg, Germany. April 12, 2016.
13. *CWRU and SCSAM*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
14. *R&D Interests at CWRU*. Atotech Review Meeting, Berlin, Germany. September 28, 2015.
15. *Computational Modeling of Concentration-Dependent Interstitial Diffusion*. Third Symposium on Surface Hardening of Corrosion-Resistant Alloys: Modeling and Structure, held at MS&T-2014, Pittsburg, PA. October 15, 2014.
16. *Low-Temperature Infusion of Interstitial Solutes: A New Concept for Surface Hardening of Steel*. International Symposium on New Trends in Steel Making and Steel Design, Institut für Neue Materialien, Saarbrücken, Germany. November 11, 2013.
17. *Alloy Surface Engineering by Gas-Phase Infusion of Interstitial Solutes*. Colloquium at the University of Birmingham, Birmingham, UK. August 02, 2013.
18. *Alloy Surface Engineering by Colossal Supersaturation with Interstitial Solutes*. Colloquium at the Christian-Albrechts-University, Department of Materials Science, Kiel, Germany. December 17, 2012.
19. *Surface Engineering of Alloys by Colossal Supersaturation with Interstitials*. Colloquium at the Institute of Engineering Innovation School of Engineering, The University of Tokyo, Tokyo, Japan. February 24, 2012.
20. *Colossal Supersaturation with Interstitial Solutes: Physical Principles and Potential for Alloy Surface Engineering*. WPI-AIMR Annual Workshop, Tohoku University, Sendai, Japan. February 2 to 23, 2012.
21. *Colossal Supersaturation: A Powerful Concept for Surface Engineering of Alloys*. ASM Young Member Night, University Club, Pittsburg, USA. February 16, 2012.
22. *Alloy Surface Hardening by Inward Diffusion of Interstitial Solutes*. Spring Seminar Series, Case School of Engineering, Cleveland, Ohio. March 25, 2011.
23. *Colossal Supersaturation with Interstitial Solutes: Perspectives for Surface Engineering of Medical Devices*. Materials Science & Technology 2010, Houston, Texas, USA. October 17 to 21, 2010.
24. *Thermodynamics of Low-Temperature Surface Carburizing*. 18th Congress of the IFHTSE, Rio de Janeiro, Brazil. July 26-30, 2010.

25. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. Kyoto University, Kyoto, Japan. June 27 to July 5, 2010.
26. *Carbide-Metal Interfaces in Low-Temperature-Carburized Structural Alloys*. XIII International Conference on Intergranular and Interface Boundaries (iib2010), Mie, Japan. June 27 to July 2, 2010.
27. *Strain State of Pt Catalyst Nanoparticles*. The 2nd International Symposium on Advanced Microscopy and Theoretical Calculations (AMTC2), Nagoya, Japan. June 24-26, 2010.
28. *Alloy Surface Engineering by Inward Diffusion of Interstitial Solutes*. The Ohio State University, Columbus, Ohio. May 10, 2010.
29. *Applications Transmission Electron Microscopy in Materials Research*. CUNY - The City University of New York, New York, USA. April 29, 2010.
30. *Perspektiven höchstauflösender Mikrocharakterisierung in der Festkörper- und Materialforschung (Perspectives of High-Resolution Microcharacterization in Solid-State and Materials Research..* Institut für Festkörperforschung, Forschungszentrum Jülich, Germany. November 23, 2009.
31. *Surface Engineering of Medical Devices by Low-Temperature Carburization*. ASM Conference on Opportunities for Next Generation Implants. Cleveland. Ohio USA August 5-7, 2008
32. *Impact of Low-Temperature Carburization on the Microstructure of Structural Alloys*. 35th International Conference On Metallurgical Coatings And Thin Films (ICMCTF) 2008. San Diego, USA. April 28 to May 2, 2008
33. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Surface Engineering of Structural Alloys*. Colloquium at the University of Toledo. Toledo, USA. March 20, 2008
34. *Impact of Applied Electric Fields on Diffusion Reactions at Metal-Oxide Interfaces*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
35. *The Carbide-Austenite Interface in Austenitic Stainless Steels with a Colossal Supersaturation of Carbon*. Materials Science & Technology 2006 Conference and Exhibition, Cincinnati, Ohio, USA. October 15-19, 2006.
36. *Diffusion Reactions at Metal-Oxide Interfaces: The Effect of an Applied Electric Field*. 2006 TMS Annual Meeting & Exhibition, San Antonio, Texas, USA. March 12-16, 2006.
37. *The Microscopy and Microanalysis Facility at Case: Enabling Tools for Nanotechnology*. Functional Nanomaterials. Cleveland, Ohio. USA November 17, 2004

38. *TEM Studies of Cu₂O-Si and Cu₂O-InP Interfaces Made by Epitaxial Electrodeposition*. Microscopy & Microanalysis, Savannah, Georgia, USA. August 1-5, 2004.
39. *Nanoscale Characterization of Epitaxial Cu₂O Films Made by Electrodeposition*. Electronic Materials Conference (TMS), Notre Dame. June 23-25, 2004.
40. *Colossal Supersaturation: A New Concept for Surface Engineering*. The International Conference On Metallurgical Coatings And Thin Films ICMCTF, San Diego, California, USA. April 19-23, 2004.
41. *Colossal Supersaturation with Interstitial Atoms: A New Concept for Advanced Structural Alloys*. Colloquium at the Max-Planck-Institut für Eisenforschung, Düsseldorf, Germany. March 23, 2004.
42. *Microstructure and Phases of Cu-In-Se*. National CIS R & D Team Meeting, Orlando, Florida Solar Energy Center, Cocoa. November 13-14, 2003.
43. *Surface Hardening of Austenitic Stainless Steels by Colossal Supersaturation with Interstitial Solutes*. Structure-Property Relationships in High-Performance Ferrous-Based Systems Possessing Composite-Like Structures, Symposium at Materials Science & Technology 2003, Chicago. November 9-1, 2003.
44. *Alloy Surface Hardening by Colossal Supersaturation with Interstitial Solutes*. Colloquium of the Materials Science and Engineering Program, Rochester Institute of Technology. October 28, 2003.
45. *Colossal Supersaturation -A New Concept for Surface Engineering*. Colloquium of the Department of Materials Science and Engineering. The Ohio State University, Columbus. October 22, 2003.
46. *Liquid-Phase Deposition of α -Cu-In-Se Thin Films*. Space Photovoltaic Research and Technology Conference (SPRAT), Ohio Aerospace Institute. September 16-18, 2003.
47. *Characterization of Materials at the Nanometer Scale: Key to Nanotechnology*. NASA Tutorial on Nanotechnology, NASA Glenn Research Center, Cleveland, Ohio. August 23, 2003.
48. *Assessing Mechanical Properties of Interatomic Bonds Across Heterointerfaces*. Frontiers of Electron Microscopy in Materials Science (FEMMS-2003), Claremont, California. October 5-10, 2003.
49. *Structure of Cu₂O/Si Interfaces, Made by Epitaxial Electrodeposition*. Structure at the Limit of Resolution, Symposium at the 105th Annual Meeting of the American Ceramic Society, Nashville. April 27-30, 2003.

50. *Atomistic Structure and Composition of Grain Boundaries and Heterointerfaces*. Colloquium of the Materials Science and Engineering Department, University of Pittsburgh. March 20, 2003.
51. *Quantitative HRTEM and Modeling of Interfaces*. First Kyoto Workshop on Computational Materials Science, Kyoto (Japan). March 01-04, 2003..
52. *Advances in Assessing the Atomistic Structure of Grain Boundaries and Heterointerfaces*. Colloquium of the Engineering Research Institute, The University of Tokyo, Japan. February 28, 2003.
53. *Atomic Resolution and Bonding of Interfaces*. Colloquium of the Materials Engineering Department, University of Colorado in Boulder. January 28, 2003.
54. *High-Resolution Electron Microscopy*. Workshop on Diagnostics, Imaging and Spectroscopy, Cleveland, Ohio (USA). April 26, 2002.
55. *Assessing Interface Structures and Interfacial Atomic Bonds by Quantitative HRTEM*. Microscopy Society of Northern Ohio, Winter Meeting, Cleveland, Ohio (USA). February 13, 2002.
56. *Analysis of Interface Structure by Quantitative HRTEM*. Colloquium of the Department of Physics and Astronomy, Arizona State University, Tempe, Arizona (USA). December 04, 2001.
57. *Bridging the Gap from Atoms to Applications by Advanced Methods of Transmission Electron Microscopy*. United Engineering Foundation Meeting on "Integrating Materials Science into Engineering Structures and Devices," Lake Arrowhead, California (USA). November 11-15, 2001.
58. *Quantitative Analysis of Interface Structure by HRTEM*. Annual Meeting of the Microscopy Society of America, Long Beach, California (USA). August 05-09, 2001.
59. *Structure and Bonding of Metal/Oxide Interfaces*. Gordon Conference on High-Temperature Corrosion, New London, New Hampshire (USA). July 22-27, 2001.
60. *Atomistic Structure of Interfaces: Progress in Experimental Methods and Modeling*. Case School of Engineering, Faculty Seminar Series 2000-2001, CWRU, Cleveland (USA). May 1, 2001.
61. *Ge-Si Nanostructures for Quantum Effect Electronic Devices*. Microscopy of Semiconducting Materials, Oxford, UK. March 25-29, 2001.
62. *Progress in Materials Science by Synergy of High-Resolution, Analytical, and In-Situ TEM*. Workshop on Future Perspectives of Transmission Electron Microscopy, Stuttgart, Germany. January 17-18, 2001

63. *Self-Assembly of Ge-Si Dots by Coherency Strains*. DSRC/DARPA Workshop on Self-Assembly in Manufacturing, Strategic Analysis Conference Center, Arlington, Virginia. December 6-7, 2000.
64. *Progress in Determination of Defect Structures by Iterative Digital HRTEM Image Matching*. EUREM 12, Brno (Czech Republic). July 9-14, 2000.
65. *Grain Boundaries in Strontium Titanate*. Japan Fine Ceramics Center Workshop on Grain Boundaries in Ceramics. Nagoya (Japan), March 16-17, 2000.
66. *Structure, Composition, and Bonding of Grain Boundaries in Metals*. 37th Metallurgists' Day and 53rd Annual Technical Meeting of the Indian Institute of Metals. Kanpur (India) November 14-16, 1999.
67. *Atomistic Structure of Interfaces: Progress in Experimental Techniques and Modeling*. 3rd Slovenian-German Seminar, Bled, Slovenia. November 19-21, 1998.
68. *Ge-Si-Nanostructures for Quantum Effect Electronic Devices*. Technion-Israel Institute of Technology, Haifa, Israel. June 14, 1998.
69. *High Precision Assessment of Interface Structures by HRTEM*. 32nd Annual Meeting of the Israel Society for Microscopy, The Weizmann Institute of Science, Rehovot, Israel. June 11, 1998.
70. *Analysis of Interface Structures by Quantitative HRTEM*. MCEM 97, Portoroz, Slovenia. October 5-8, 1997.
71. *Analysis of Interface Structures by Quantitative HRTEM*. EMAG 97, Cambridge, Great Britain. September 2-5, 1997.
72. *Solving Interface Structures by Quantitative High-Resolution Electron Microscopy*. Physical Colloquium, University of Vienna, Austria. December 11, 1997.
73. *Analysis of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Fall Meeting of the Materials Research Society, Boston. December 1-5, 1996.
74. *Quantitative HRTEM of Interfaces and Structure of Metal/Ceramic Interfaces*. Micromat 1996. Rio de Janeiro, October 13-16, 1996.
75. *Analysis of Grain Boundary Structures by Quantitative HRTEM*. Annual Meeting of the Microscopy Society of America (MSA), Minneapolis. August 11-15, 1996.
76. *Interface Dislocations*. International Workshop on Plastic Deformation of Structural Materials at High Temperatures, Schloss Ringberg, Tegernsee, Germany. July 14-17, 1996.

77. *Quantitative High-Resolution Electron Microscopy of Interfaces*. 7th International Conference on Intergranular and Interphase Boundaries in Materials (iib95), Lissabon, Portugal. June 26-29, 1995.
78. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Modeling*. Colloquium of the Institut für Physikalische Chemie und Elektrochemie, University of Hannover, Germany. May 8, 1995.
79. *Solving Interface Structures by High-Resolution Electron Microscopy and Computer Simulation*. 13th International Congress on Electron Microscopy, Paris, France. July 18-22, 1994.
80. *Quantitative HRTEM of Twin Boundaries*. Annual Congress of the German Physical Society, Münster, Germany. March 21-25, 1994.
81. *Quantitative High-Resolution Electron Microscopy as a Method to Determine the Structure of Grain Boundaries*. German Physical Society, Spring Meeting, Münster. March 21-25, 1994.
82. *Lattice Mismatch Accommodation by Misfit Partial Dislocations*. Annual Meeting of The Materials Society (TMS), San Francisco. February 27-March 3, 1994.
83. *Quantitative HRTEM of Grain Boundaries*. International Symposium on High-Voltage and High-Resolution Electron Microscopy, Stuttgart, Germany. February 21-24, 1994.
84. *Assessment of Interface Structures by Quantitative High-Resolution Transmission Electron Microscopy*. Workshop on Microcharacterization of Materials and Devices, Göttingen, Germany. January 6-7, 1994.
85. *Quantitative Evaluation of HRTEM Images*. Multinational Congress of the Austrian, German, and Swiss Electron Microscopy Societies, Zürich. Switzerland, September 1993..
86. *HREM of Metal/Ceramic Interfaces*. Annual Meeting of the French Society of Electron Microscopy (SFME), Lyon, France. June 29-July 4, 1993..
87. *Misfit Dislocation Networks in 111-GeSi/Si-Interfaces*. Colloquium of the IV. Physikalisches Institut, University of Göttingen. June 24, 1993.
88. *Relaxation of Misfit Stresses in Epitaxial Layers*. Physical-Chemical Colloquium, University of Stuttgart, Germany. June 22, 1993.
89. *Phase Transformations at Metal/Ceramics Interfaces*. International Workshop on Reactive Phase Formation at Interfaces and Diffusion Processes, Aussois, France. May 21-2, 1993..

90. *Misfit Dislocation Structure of GeSi/Si-{111}-Interfaces*. European Science Foundation Conference on Extended Defects in Semiconductors, Holzgau, Germany. August 29-September 3, 1992.
91. *Quantitative Analysis of Intensities in HRTEM Lattice Images*. Annual Congress of the German Electron Microscopy Society (DGE), Darmstadt, Germany. September 8-13, 1991.
92. *Misfit Dislocation Networks in GeSi/Si Interfaces Grown by Liquid Phase Epitaxy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. February 11, 1991.
93. *HRTEM of the Cu/MnO Interface in an Internally Oxidized CuMn Alloy*. Colloquium of the Department of Materials Science and Engineering, CWRU, Cleveland, USA. April 23, 1990.
94. *HRTEM of Metal/Ceramic Interfaces in Internally Oxidized Alloys*. 4th Workshop on Characterization and Properties of Interfaces in Crystals, Delft, Netherlands. June 5-9, 1989.
95. *High-Resolution Electron Microscopy of Semiconductor Heterointerfaces*. Colloquium of the Max-Planck-Institut für Festkörperforschung, Stuttgart. Germany. May 8, 1989.
96. *High-Resolution Electron Microscopy of Cu/Al₂O₃ Interphase Interfaces*. Colloquium of the Institut für Metallphysik, University of Göttingen, Germany. April 18, 1989.

Courses Taught and Developed*

Univ	Code	Course
Stuttgart		Kristallografie und Beugung* (Crystallography and Diffraction)
Stuttgart		Werkstoff-Analytik 1* (Analytical Methods of Materials Science 1*)
Stuttgart		Halbleiter Werkstoffe der Mikorelektronik und Photovoltaik* (Semiconductor Materials for Microelectronics and Photovoltaics*)
CWRU	EMSE-102	Introduction to Materials Science and Engineering*
CWRU	EMSE-228	Mathematical and Computational Methods for Materials Science and Engineering*
CWRU	EMSE-290	Materials Laboratory III
CWRU	EMSE-314	Electrical, Optical, and Magnetic Properties of Materials*
CWRU	EMSE-327	Thermodynamic Stability and Rate Processes*
CWRU	EMSE-368/468	Scientific Writing*
CWRU	EMSE-398-399	Senior Project in Materials Science and Engineering
CWRU	EMSE-465	Surface Engineering of Materials*
CWRU	EMSE-509	Conventional Transmission Electron Microscopy*
CWRU	EMSE-512	Advanced Methods of Transmission Electron Microscopy*
CWRU	EMSE-515	Analytical Methods of Materials Science and Engineering*

Consulting

Period	Client	Activity
Since 07/2026	Cooley LLP	Expert Witness for Tower in: Global-Foundries vs Tower Semiconductor
08/2025- 01/2026	Cooley LLP	Expert Witness for Caihong in: Corning Incorporated vs Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd.
03/2025- 07/2025	Merchant & Gould P. C.	Expert for Caihong in: Inter Partes Review Petitions of Caihong in vs Display Devices Co., Ltd., d/b/a Irico Display Devices Co., Ltd. vs Corning Incorporated
Since 02/2026	Bosch	Consultant for Semiconductor Device Processing
02/2025- 12/2025	Bosch	Consultant for Semiconductor Device Processing
07/2018- 01/2024	Reichman & Jorgensen	Expert Witness for Arconic in: Arconic Corporation et al vs Novelis Inc. et al
05/2018- 08/2018	Voss Aerospace In- dustries Inc.	Mechanical Properties of Aluminum Alloys
01/2017- 12/2024	KFAS	Proposal Reviewer for Kuwait Foundation for the Advancement of Sciences
03/2016- 04/2018	Atotech	Development of a Know-How Management System
03/2014- 08/2021	Skolkovo	Proposal ("Expert") Reviewer
04/2014- 06/2014	Voss Aerospace In- dustries Inc.	Microstructure of Precipitation-Hardened Aluminum Alloys

07/2012	Powdermet	Microscopic Structure of Novel Powders
07/2011– 06/2012	Atotech	Microstructure of Electroplated Cu for Microelectronic Devices
04/2011– 10/2011	AlSher Titania LLC	Structure and Composition of Titania Nanoparticles
01/2011– 12/2017	Pocared	Microstructure of Coatings for Pharmaceutical Containers
09/2008	TRW Automotive Engine Components	Nanohardness and Auger Electron Spectroscopy Studies of Valves
04/2008	DoD	Proposal reviewer for SERDP program
04/2007	DoD	Proposal reviewer for SERDP program
10/2006– 05/2007	Bird Technologies Group	Scanning Electron Microscopy and Analysis of Resistors
11/2004– 04/2005	Patent Litigation	Expert Witness Assistance for Experimental Studies of Coatings
05/2004– 11/2004	OMG Americas	Transmission Electron Microscopy Analysis of Novel Nanoparticles
12/2003– 06/2005	Moen Incorporated, North Olmsted, Ohio	Transmission Electron Microscopy Studies of Coatings
06/2003– 12/2003	IMQ Corporation, Chesterland, Ohio	Transmission Electron Microscopy Studies on the Effect of Boron on the Surface Structure of Ferritic Steels.
02/2003– 05/2003	Inst. of Semiconductor Technology, Stuttgart, Germany	Transmission Electron Microscopy Analysis of Defect Structures in Ultra-thin SiGe Buffer Layers

EXHIBIT B

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Employment History

RF DESIGN CONCEPTS, LTD., EXPERT WITNESS, RF DESIGN, TURN KEY SOLUTIONS 02/2012–Present

Principal & Owner

- Subject Matter Expert in RF/Microwave (Wireless) Hardware: RF Power Amplifiers/Transmitters, RF Front-end Modules/Transceivers, Power Management, Electromagnetics (Signal integrity, antennas, filters, etc.), Wireless Charging, Semiconductors, Digital Electronics, Device Packaging
 - Consultant, Satcom SSPA, C-band RFPA– Trak Microwave
 - Consultant, 24 GHz Radar Front-End – The Ohio State University
 - Consultant, Ultra compact high efficiency load adaptable RFPA– Draper Labs
 - Consultant, High voltage MEMS packaging – Bright Silicon (Aspen Microsystems)
 - Consultant, Ultra high efficiency wireless charging– NuCurrent, Inc.
 - Consultant, Ultra high efficiency RF plasma generation– Advanced Energy
 - Consultant, DARPA MIS superconductor fabrication program– Grid Logic, Corp.
 - Consultant, Quantum devices– ColdQuanta, Inc.
 - Consultant, LTCC “RF” module for optical communications– AvoPhotonics, Inc.
 - Consultant, RF front-end module– Trellisware
 - Consultant, Market/System ecology analysis of nascent technologies– Invensas
 - Product Manager (Consultant)– UX Lab, Samsung Electronics
 - Custom Design of RF power amplifier solutions for commercial wireless (cdma, WCDMA, LTE), defense (radar, EW, SIGINT, etc., medical (MRI, RF ablation), and industrial (RF induction heating, metal sputtering, plasma generation, etc.)
 - Custom Design of DC power management systems, including Switched-mode (Buck, Boost, Buck-boost, etc.) and linear (LDO) power supply designs
- Expert Witness, Patent Litigation & Technical Due Diligence– Fish & Richardson, Wilmer Hale, Kirkland & Ellis, Perkins Coie, Slayden Grubert Beard, Pillsbury Winthrop Shaw Pittman, O’Melveny, Finnegan, Wolf Greenfield, Shelton Coburn, Alston & Bird, Weil, Hoynig Rokh Monegier, Susman Godfrey

MERCURY SYSTEMS, INC., RF POWER AMPLIFIER BUSINESS UNIT

08/2013–03/2014

Technical Director

- Responsible for all aspects of Mercury Systems’ RF Power Amplifier Business Unit
 - Full P&L responsibility for \$2.5M RF Power Amplifier Business Unit
 - Responsible for engineering, sales, manufacturing, and test

- Develop new business opportunities through direct customer contact and proposals (Technical concept, cost models, etc.)
- Lead and manage new product development and sustaining engineering efforts
- Manage development and production schedules, material planning, product testing
- Manage customer relationships

LGS BELL LABS INNOVATIONS, R&D HARDWARE ENGINEERING

05/2011–12/2012

Sr. R&D RF Hardware Engineer (Technical Ladder, Grade 315)

- Development of high power, highly linear and efficient RF transmitter/transceiver solutions supporting major wireless standards (GSM/EDGE, cdma, WCDMA, LTE)
 - First-pass design success of (200W P1dB) 850MHz and (250W P1dB) 900MHz band Doherty RF transmitters
 - First-pass design success of highly linear, thermally compensated TX lineups for 746-960MHz and 1805-2170MHz broadband transmitters
 - Development of high power (46dBm avg), broadband, highly linear RF transmitter
- Technical leadership/project management
 - Coordinate activities of multi-disciplinary teams of RF/electrical engineers, mechanical engineers, drafters, and technicians
 - Insure adherence to proper engineering design practices and manufactureability; Internal design review

AETHERCOMM, INC., ADVANCED DEVELOPMENT GROUP

07/2008–02/2011

Technical Manager/Principal Engineer

- Design Engineering Responsibilities
 - Responsible for complete execution of rapid product development of pulsed/CW, wideband, high power RF transmitters for radar and communications on extremely accelerated schedules per MIL-STD's: Silicon LDMOS, GaAs FET & HBT, and GaN HEMT device technologies
 - Customer specification to top-level design, complete electrical design (RF link budget analysis, RF circuitry, DC power management, control logic, telemetry, fault protection, etc.), bill of materials (BoM), test procedures, and design verification testing
 - Lead, manage, and coordinate activities of multi-disciplinary teams of RF/electrical engineers, mechanical engineers, drafters, and technicians
 - Identify and execute, internal R&D programs, including SBIR's
- Technical Management Responsibilities
 - "Cradle to grave" management of all technical aspects of product development to insure successful, on time delivery of custom RF products
 - Product documentation and design milestone reviews: Compliance matrix, ECN's, test plans (ATP, DVT, TDR), final product approval, and post mortem reports
 - Maintain competent & effective staff through recruitment, mentoring, and performance planning of technical personnel
 - Insure adherence to good engineering design practice and manufactureability; Review designs through all engineering phases

- Support upper management and marketing through customer interaction, speaking in technical forums, and writing technical publications
- Resource & budget planning, allocation, and scheduling of all group activities

SAMSUNG ELECTRONICS, TECHNOLOGY INTELLIGENCE & COLLABORATION

03/2007–07/2008

Section Manager, Technology Strategy Team

- Technical feasibility and business opportunity analysis of bottom-up nanotechnology for next generation digital storage, wearable electronics, and non-lithographic semiconductor technologies
- Market and technology analysis of (low-power wireless) biomedical electronics
- Market and technology analysis of THz imaging for security, healthcare, and industrial applications

RF MICRO DEVICES, INFRASTRUCTURE PRODUCTS GROUP

07/2006–02/2007

Group Leader/Staff Engineer

- Advanced Product Development: GaN HEMT dual carrier UMTS (2.14 GHz WCDMA) Doherty basestation transmitter (240 W max./40 W avg. RF power)
- Led development of scalable GaN HEMT non-linear simulator model
- Project planning, product documentation (PRD, PPA, etc.), and test plan development

NOKIA CORPORATION, NOKIA RESEARCH CENTER

04/2001–03/2006

Senior Research Engineer, Radio Technologies Laboratory

- Advanced RF Research:
 - *RF Circuits*: Complete design, build, test, and system integration of novel LDMOS based Chireix Outphasing (basestation) transmitter for UMTS systems (120W max./20W avg. RF power)
 - *RF Systems*: Development of OFDM RF system simulator (IEEE 802.11a WLAN & ETSI EN-300 744 DVB-T PHY); Baseband PAPR reduction techniques for OFDM radio systems
 - *Power Management*: Complete design, build, and test of high power, high precision, 16 bit digitally-controlled switched-mode power supply
 - *Electromagnetics*: Compact antenna and (metamaterial) filter design
- Project Management:
 - Technical Liaison: Advise on new technologies offered by established and new venture organizations in collaboration with Nokia Ventures and Nokia Mobile Phones New Technology Sourcing
 - Advise and steer corporate funded university research projects
 - Group leader, *Leading Science 8* program, led market research and analysis for “Network and Infrastructure Solutions in Low Average Revenue per User (ARPU) Markets”

WESTERN DIGITAL CORPORATION/ADAPTEC INC.

01/1995–08/1996

Digital Design Engineer, ASIC Design

- Timing Analysis: Static timing analysis of ASIC critical path delays; Custom cell characterization and library generation
- Clock Tree & Pad Cell Drivers: Clock tree timing analysis and design; Custom ASIC pad cell design; Drivers for memory cells
- Power ON Logic: Developed “No re-spin” solution for faulty ASIC ACK/Power Enable
- Meta-stability Analysis: Circuit and system level analysis of meta-stable logic

Academic History

UNIVERSITY OF COLORADO, BOULDER

03/2013–03/2014; 08/2015–08/2016

Adjunct Professor, Department of Electrical, Computer, & Energy Engineering

UNIVERSITY OF CALIFORNIA, SAN DIEGO

08/2004–08/2008

Visiting Scholar (Part-time), Department of Electrical & Computer Engineering

- Established LoA between UC San Diego, Cree Microwave, Inc., and Nokia for collaborative research on the Envelope Tracking (ET) highly linear and efficient transmitter architecture
- Coordination of corporate sponsored research on linear, highly efficient transmitter architectures

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

09/1996–03/2001

Doctor of Philosophy, Electrical & Computer Engineering– Solid-State Electronics (major), Analog/Digital Circuits (minor); Dissertation title: High Efficiency Switched-Mode Power Amplifiers for Wireless Communications

CALIFORNIA STATE UNIVERSITY, FULLERTON

08/1994–01/1996

Master of Science, Electrical & Computer Engineering– Analog/Digital Circuits (major)

UNIVERSITY OF CALIFORNIA, BERKELEY

08/1990–05/1993

Bachelor of Science, Physics

Selected Publications

T. Hung, D. K. Choi, L. E. Larson, P. M. Asbeck, “CMOS Outphasing Class-D Amplifier With Chireix Combiner,” *IEEE Microwave and Wireless Components Letters*, 8:619-621, August, 2007.

I. Hakala, D. K. Choi, L. Gharavi, N. Kajakine, J. Koskela, R. Kaunisto, “A 2.14-GHz Chireix Outphasing Transmitter,” *IEEE MTT Trans*, 4:2129-2138, June, 2005.

D. Qiao, D. Choi, Y. Zhao, D. Kelly, T. Hung, D. Kimball, M. Li, P. Asbeck, “Antenna Impedance Mismatch Measurement and Correction for CDMA Transceivers,” *IEEE MTT-S Digest*, WE3C-4, Long Beach, CA, June, 2005.

D. K. Choi and S. I. Long, “The Effect of Transistor Feedback Capacitance in Class-E Power Amplifiers,” *IEEE Trans. on Circuits and Systems I*, Vol. 50, Issue 12, pp. 1556-1559, December, 2003.

D. K. Choi and S. I. Long, "Finite DC Feed Inductors in Class E Power Amplifiers– A Simplified Approach," *IEEE MTT-S Digest*, TH2D-5, Seattle, WA, June, 2002.

D. K. Choi and S. I. Long, "A Physically Based Analytic Model of FET Class E Power Amplifiers– Designing for Maximum PAE," *IEEE MTT Trans*, 47:1712-1720, September, 1999.

Other Professional/Personal Information

US citizen, US Secret & Top Secret/SCI (In-active) clearance, Senior member IEEE, California State MICRO Fellow (1996), fluent in the Korean language, conversant in the French language.

EXHIBIT C

Jeffrey Bokor, Ph.D.

Paul R. Gray Distinguished Professor Emeritus
EECS Department
University of California
Berkeley, CA 94720

Phone: (510)642-4134
jbokor@berkeley.edu
www.eecs.berkeley.edu/~jbokor

Expertise

- Electronics
- Nanofabrication
- Nanotechnology
- Device Physics
- Nanomagnetism and Spintronics
- Optics and Optoelectronics
- Laser Technology
- Semiconductor Physics
- Semiconductor Process
- VLSI Technology

Professional Summary

Education

<u>Year</u>	<u>College/University</u>	<u>Degree</u>
1980	Stanford University	Ph.D., Electrical Engineering
1976	Stanford University	MS, Electrical Engineering
1975	Massachusetts Institute of Technology	BS, Electrical Engineering

Employment History

From:	1993	University of California, Department of Electrical Engineering and Computer Science, Berkeley CA
To:	2024	Berkeley, CA
Position:		
	2024-present	Emeritus Professor
	1993-2024	Professor
		▪ Department Chair, Department of Electrical Engineering and Computer Sciences, 2019-2021 ▪ Chair, EE Division, Associate Department Chair, Department of Electrical Engineering and Computer Science, 2017-2019 ▪ Associate Dean for Research, College of Engineering, 2012-2017 ▪ Joint Appointment: 2008-2024 , Faculty Senior Scientist, Lawrence Berkeley National Laboratory ▪ Joint Appointment: 2004-2012, Deputy Director for Science, The Molecular Foundry, Lawrence Berkeley National Laboratory ▪ Joint Appointment: 1993-2004, Group Leader for Advanced Lithography, Center for X-ray Optics, Lawrence Berkeley National Laboratory

Curriculum Vitae - Jeffrey Bokor

Current research interests:

- Nanoelectronic devices
- Spintronics
- Carbon electronics
- Ultrafast laser studies of materials

Classroom teaching:

- Undergraduate and graduate courses in semiconductor physics and technology
- Undergraduate and graduate courses in optical physics and engineering
- Freshman seminar introduction to electrical engineering

From:	1980	AT&T, Bell Laboratories
To:	1993	Holmdel and Murray Hill, NJ
Position:	1990-93	Ultra-Large Scale Integrated Circuits (ULSI) Technology Research Department. Department Head. Murray Hill, NJ Leader of group of 15 Ph.D. scientists, 35 total technical personnel. Responsible for directing research to create technologies relevant to silicon IC manufacturing in the year 2000. Also responsible for continuing improvement and operation of 20,000 sq. ft. Class 100 research cleanroom facility. Experience included: ▪ ULSI Process integration ▪ Deep sub-micron silicon MOSFET device physics ▪ Device and circuit design ▪ Advanced process and device modeling software ▪ Advanced lithography
	1987-90	Laser Science Research Department. Department Head, Holmdel, NJ. Leader of group of 10 Ph.D. scientists, 20 total technical personnel. Managed diverse program of basic research in optical physics. Conducted high visibility research program, including: ▪ X-ray lithography ▪ Nonlinear optics ▪ Picosecond optoelectronics ▪ Femtosecond phenomena in electronic materials ▪ Fiber optics ▪ Atomic physics
	1980-87	Quantum Electronics Research Department. Member of Technical Staff, Holmdel, NJ. Independently directed basic research, including: ▪ Nonlinear optics ▪ Picosecond optoelectronics ▪ Semiconductor device physics ▪ Ultraviolet laser technology

Curriculum Vitae - Jeffrey Bokor

Major Research Accomplishments (1980-Present)

- Demonstration of ultrafast (<10 psec) spin-orbit torque switching (2023)
- Demonstration of ultralow contact resistance between semimetal and monolayer semiconductors (2021)
- Demonstration of double-gate graphene nanoribbon transistors (2021)
- Demonstration of transfer-free synthesis of atomically precise graphene nanoribbons on insulating substrates (2021)
- Demonstration of spin-transfer-torque magnetic recording (2019)
- Demonstration of 7 nm magnetic memory using scanned probe magnetic tunnel junction readout (2018)
- Experimental demonstration of ultrafast (<10 psec) magnetic switching by pure charge current pulses (2017)
- High performance bottom-up chemically synthesized graphene nanoribbon transistors (2017)
- Experimental demonstration of the ultimate lowest switching energy set by fundamental physics (Landauer's principle) in single-bit operations on nanomagnetic memory bits (2016)
- Demonstration of spin transfer torque switching in sub-10 nm size magnetic tunnel junctions (2016)
- Demonstration of fluid-suspended magnetic particle manipulation using multiferroic magnetic actuation (2015)
- Direct optical detection of current induced spin accumulation in non-magnetic metals (2015)
- Demonstration of sub-nsec signal propagation in nanomagnetic logic (2015)
- Electric field control of magnetic domain walls for manipulation of magnetic particles in microfluidics (2014)
- High performance nanoscale solution processed carbon nanotube transistors (2013)
- Bottom-up chemically synthesized graphene nanoribbon transistors (2013)
- Nanoscale hyperspectral imaging with tapered plasmonic "Campanile" tip (2012)
- Demonstration of nanomagnetic logic signal propagation (2012)
- Demonstration of fundamental thermodynamic limit of energy dissipation in magnetic memory and logic (2011)
- Demonstration of bandgap and sub-band formation in graphene nanomesh superlattice transistors (2010)
- Self-assembly of gold nanoparticle chain structures for nanophotonics using DNA origami (2010)
- Measurement of Schottky contact junction capacitance, doping profiles and carrier mobility in single semiconductor nanowire and carbon nanotube transistors using ultrasensitive C-V measurements (2009)
- Label-free DNA biosensors using functionalized carbon nanotube transistors (2009)
- Directed assembly of nanophotonic device structure using DNA nanotechnology (2009)
- Theory of spin-dependent scattering in silicon MOSFET transistors (2009)
- Proposed new concepts for nanomagnetic logic technology (2008)
- Study of spin-dependent scattering from impurity donors in silicon MOSFETs for use in quantum computing (2007)

Curriculum Vitae - Jeffrey Bokor

- Measurement of spin coherence of antimony donors in silicon near silicon dioxide interfaces (2006)
- First observation of Stark tuning of donor electron spins in silicon, a key step towards implementation of quantum logic in silicon technology (2006)
- First monolithic integrated circuit with both carbon nanotube and silicon MOS transistors (2004)
- Development of a monolithic inkjet printhead array capable of sub-3 μm droplet generation (2004)
- Study of surface and bulk acoustic phonon excitations in superlattices using picosecond ultrasonics (2003)
- Invention and demonstration of “spacer lithography”, a sub-10 nm patterning technology for nano-catalysts and DNA label-free hybridization detection (2003)
- Development of 10 nm gate-length MOSFET devices (2002)
- Demonstration of extreme ultraviolet lensless Fourier transform holography (2001)
- Demonstration of 30 nm lithography using extreme ultraviolet radiation (2001)
- Demonstration of aberration testing and alignment optimization of a commercial prototype four-mirror ring field extreme ultraviolet optical system using at-wavelength interferometry (2000)
- Development of complementary silicide source/drain technology for sub-20 nm MOSFET devices (2000)
- Development of picosecond ultrasonic characterization of Mo/Si extreme ultraviolet multilayer reflectors (1999)
- Development of 17 nm gate-length FinFET devices (1999)
- First demonstration of the “FinFET”, a manufacturable double-gate MOSFET with 25 nm gate-length (1998)
- Development of at-wavelength defect detection metrology for extreme ultraviolet lithography (1998)
- Co-invention and demonstration of the phase-shifting, point diffraction interferometer, and application to high-accuracy wavefront metrology of extreme ultraviolet lithography optics (1996)
- First demonstration of working MOS devices fabricated with extreme ultraviolet lithography (1996)
- Direct, quantitative measurement of saturation velocity and overshoot velocity in MOS inversion layers (1994)
- Demonstration of interferometry at extreme ultraviolet wavelengths for characterization of lithography optics (1994)
- First direct observation of non-thermal energy distributions and thermalization dynamics in laser-excited metal films (1992)
- Development of 89 GHz f T Si MOSFET transistors with 0.15 μm channel length (1992)
- First feasibility demonstration of diffraction-limited extreme ultraviolet projection lithography (1990)
- First direct observation of surface state kinetics during semiconductor surface recombination (1989)
- Invention and demonstration of the coplanar vacuum photodiode, an ultrafast, high sensitivity photodetector for the visible and ultraviolet (1987)

Curriculum Vitae - Jeffrey Bokor

- First demonstration of time-resolved photoemission for measurement of picosecond electronic signals in integrated circuits (1986)
- First direct detection of band-gap states at GaAs Schottky barrier interfaces (1986)
- Application of time-resolved photoemission to the first measurement of picosecond electron dynamics on InP, GaAs, and Si surfaces (1985)
- Invention and demonstration of picosecond time-resolved ultraviolet photoemission spectroscopy for the study of electron dynamics at semiconductor surfaces (1985)
- Development of a source of 35.5 nm coherent radiation by seventh harmonic generation in a pulsed jet. At the time, this was the shortest wavelength source of coherent radiation yet demonstrated (1983)
- Invention of the pulsed jet harmonic generator for the production of extreme ultraviolet coherent radiation (1983)
- First picosecond pulse krypton-fluoride excimer laser (1982)
- First demonstration of an autoionization pumped laser (1982)
- First third harmonic generation using an excimer laser (1980)
- Development of the first tunable, single-mode excimer laser system (1980)

Technical Consulting

From: 2001 Rodel, Inc.
To: 2002 Newark, DE
Position: Technical advisory board member for extreme ultraviolet lithography

From: 2002 Alpha Golf, Inc.
To: 2002 Kensington, CA
Position: Technical consultant

From: 2002 University Technology Ventures, Inc.
To: present Pleasanton, CA
Position: Technical consultant

From: 2002 Veeco Metrology Group
To: 2006 Santa Barbara, CA
Position: Technical advisory board member

Litigation Support

Consulting and expert witness on patent validity and infringement in 30+ cases since 1998. Equipment inspections. Testimonial experience at deposition and trial. Full list available on request.

Curriculum Vitae - Jeffrey Bokor

Patents

1. Yang Yang, Jon Gorchon, Richard Wilson, Charles-Henri Lambert, Sayeef Salahuddin, Jeffrey Bokor, "Writing of a magnetic memory with electric pulses," US 10,388,349 B2, August 2019.
2. Thomas Schenkel, Cheuk Chi Lo, Christoph Weis, Stephen Lyon, Alexei Tyryshkin, Jeffrey Bokor, "Scalable quantum computer architecture with coupled donor-quantum dot qubits," US 8,816,325 B2, August 2014
3. Brian Lambson, Zheng Gu, David Carlton, Jeffrey Bokor, "Concave nanomagnets with widely tunable anisotropy," US 8,766,754 B2, July 2014
4. Ji Zhu, Jeff Grunes, Yang-Kyu Choi, Jeffrey Bokor, Gabor Somorjai, "Methods for Fabrication of Positional and Compositionally Controlled Nanostructures on Substrate," US 8,486,287 B2, July 2013
5. Jeffrey Bokor, Nathan C. Emley, David Carlton, "Nanomagnetic signal propagation and logic gates," US 8,134,441 B1, March, 2012
6. David Carlton, Nathan C. Emley, Jeffrey Bokor, "Nanomagnetic Register," US 8,138,874 B1, March, 2012
7. Jeffrey Bokor, Patrick Naulleau, "System For Interferometric Distortion Measurements That Define An Optical Path," US 6,559,952 B1, May, 2003
8. Jeffrey Bokor, Yun Lin, "Method And Apparatus For Inspecting Reflection Masks For Defects," US 6,555,828 B1, April, 2003
9. Jeffrey Bokor, Seongtae Jeong, "Multi-Level Scanning Method for Defect Inspection," US 6,484,306 B1, November 2002
10. Chenming Hu, Tsu-Jae King, Vivek Subramanian, Leland Chang, Xuejue Huang, Yang-Kyu Choi, Jakub Tadeusz Kedzierski, Nick Lindert, Jeffrey Bokor, Wen-Chin Lee, "FINFET Transistor Structures having a Double Gate Channel Extending Vertically From a Substrate and Methods of Manufacture," US 6,413,802 B1, July 2002
11. Jeffrey Bokor, Anthony M. Johnson, "High Speed Photodetector Having Dual Transmission Line Geometry," US 4,933,542 A, June 1990
12. Jeffrey Bokor, Anthony M. Johnson, Ralph H. Storz, "High Speed Circuit Measurements Using Photoemission Sampling," US 4,721,910 A, January, 1988

Professional Associations, Service, and Awards

- Fellow, Optical Society of America
- Fellow, American Physical Society
- Fellow, Institute of Electronics and Electrical Engineers
- Senior Scientist, Lawrence Berkeley National Laboratory
- American Association for the Advancement of Science
- Paul R. Gray Distinguished Professorship in Electrical Engineering and Computer Sciences, 2015-2024
- National Semiconductor Distinguished Professorship in Engineering, University of California, 2007-2015
- Deputy Director, Center for Energy Efficient Electronic Science, an NSF Science and Technology Center headquartered at Berkeley, 2010-2020
- Co-Chair, 5th Berkeley Symposium on Energy Efficient Electronic Systems, 2017

Curriculum Vitae - Jeffrey Bokor

- Co-Chair, 4th Berkeley Symposium on Energy Efficient Electronic Systems, 2015
- Co-Chair, 3rd Berkeley Symposium on Energy Efficient Electronic Systems, 2013
- Co-Chair, 2nd Berkeley Symposium on Energy Efficient Electronic Systems, 2011
- Co-Chair, 1st Berkeley Symposium on Energy Efficient Electronic Systems, 2009
- Scientific Advisory Board, Catalan Institute for Nanoscience (ICN), Autonomous University of Barcelona, 2007-2022
- Scientific Advisory Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 2006-2010
- IEEE Paul Rappaport Award for best paper of the year published in Electron Device Society journals, 2002
- DARPA Most Significant Technical Accomplishment Award, 2000
- Program Committee, CMOS devices, International Electron Devices Meeting, 1999-2000
- Scientific Advisory Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 1998-2002
- Technical Advisory Committee, Laser Science and Technology Program, Lawrence Livermore National Laboratory, 1996-1997
- General Co-chair, Quantum Electronics and Laser Science Conference, Baltimore, MD, 1997
- Program Committee, Topical Meeting on Extreme Ultraviolet Lithography, Boston, MA, 1996
- Users Executive Committee, Advanced Light Source, Lawrence Berkeley National Laboratory, 1994-1998
- Program Co-chair, Quantum Electronics and Laser Science Conference, Baltimore, MD, 1995
- Technical Council, Optical Society of America, 1991-1994
- Chairman, Technical Group on X-ray and XUV Physics, Optical Society of America, 1991-1994
- Program Committee, Topical Meeting on Soft X-ray Projection Lithography, Monterey, CA, April, 1992
- Cochairman, Topical Meeting on Soft X-ray Projection Lithography, Monterey, CA, April, 1991
- Cochairman, Topical Meeting on Short Wavelength Coherent Radiation, Generation and Applications, Monterey, CA, March, 1986
- Program Committee, Conference on Lasers and Electro-Optics (CLEO) '84, Anaheim, CA, June, 1984
- Program Committee, Topical Meeting on Laser Techniques in the Extreme Ultraviolet, Boulder, CO, March, 1984
- Program Committee, Topical Meeting on Excimer Lasers, Lake Tahoe, NV, January, 1983
- Program Committee, Topical Meeting on Laser Techniques for Extreme Ultraviolet Spectroscopy, Boulder, CO, March 1982

Curriculum Vitae - Jeffrey Bokor

Publications

1. D. Polley and J. Bokor, "A perspective on ultrafast magnetization switching in ferromagnets using helicity-independent optical pulses," *Journal of Physics D: Applied Physics*, vol. 58, no. 10, p. 102001, (2025).
2. X. Li, H. Singh, J. Lin, S. Zhang, B. Yi, J. Chatterjee, Z. Xiao, S. Mondal, N. Tamura, R. N. Candler, L. You, J. Bokor, and J. Hong, "An energy efficient way for quantitative magnetization switching," *npj Spintronics*, vol. 2, 2024-07-17 2024.
3. M. Goiriena, Z. Xiao, R. Steinhardt, V. Estrada, N. Tamura, R. V. Chopdekar, A. T. N'Diaye, A. Sepúlveda, G. G. Schlom, R. N. Candler, and J. Bokor, "Imaging of voltage-controlled switching of magnetization in highly magnetostrictive epitaxial Fe–Ga microstructures." *Nanoscale* **16**(18): 9021–9028 (2024).
4. D. Polley, A. Pattabi, A. Rastogi, K. Jhuria, E. Diaz, H. Singh, A. Lemaitre, M. Hehn, J. Gorchon, and J. Bokor, "Picosecond spin-orbit torque–induced coherent magnetization switching in a ferromagnet," *Science Advances*, vol. 9, p. eadh5562, 2023.
5. D. Polley, J. Chatterjee, H. Jang, and J. Bokor, "Analysis of ultrafast magnetization switching dynamics in exchange-coupled ferromagnet–ferrimagnet heterostructures," *J. Magn. Magn. Mater.*, vol. 574, 2023.
6. S. Mondal, D. Polley, A. Pattabi, J. Chatterjee, D. Salomoni, L. Aviles-Felix, A. Olivier, M. Rubio-Roy, B. Diény, L. D. B. Prejbeanu, R. Sousa, I. L. Prejbeanu, and J. Bokor, "Single-shot switching in Tb/Co-multilayer based nanoscale magnetic tunnel junctions," *J. Magn. Magn. Mater.*, vol. 581, 2023.
7. Y. C. Lin, Z. Mutlu, G. Borin Barin, Y. Hong, J. P. Llinas, A. Narita, H. Singh, K. Müllen, P. Ruffieux, R. Fasel, and J. Bokor, "Scaling and statistics of bottom-up synthesized armchair graphene nanoribbon transistors," *Carbon*, vol. 205, pp. 519-526, 2023-03-01 2023.
8. J. Engel, K. Jhuria, D. Polley, T. Lühmann, M. Kuhrke, W. Liu, J. Bokor, T. Schenkel, and R. Wunderlich, "Combining femtosecond laser annealing and shallow ion implantation for local color center creation in diamond," *Appl. Phys. Lett.*, vol. 122, 2023.
9. D. Polley, A. Pattabi, J. Chatterjee, S. Mondal, K. Jhuria, H. Singh, J. Gorchon, and J. Bokor, "Progress toward picosecond on-chip magnetic memory," *Applied Physics Letters*, vol. 120, no. 14, p. 140501, 2022.
10. S. Mondal, Y. Lin, D. Polley, C. Su, A. Zettl, S. Salahuddin, and J. Bokor, "Accelerated Ultrafast Magnetization Dynamics at Graphene/CoGd Interfaces," *ACS Nano*, vol. 16, no. 6, pp. 9620-9630, 2022-06-28 2022.
11. G. Borin Barin, Q. Sun, M. Di Giovannantonio, C. Z. Du, X. Y. Wang, J. P. Llinas, Z. Mutlu, Y. Lin, J. Wilhelm, J. Overbeck, C. Daniels, M. Lamparski, H. Sahabudeen, M. L. Perrin, J. I. Urgel, S. Mishra, A. Kinikar, R. Widmer, S. Stolz, M. Bommert, C. Pignedoli, X. Feng, M. Calame, K. Müllen, A. Narita, V. Meunier, J. Bokor, R. Fasel, and P. Ruffieux, "Growth Optimization and Device Integration of Narrow-Bandgap Graphene Nanoribbons," *Small*, p. 2202301, 2022-06-17 2022.
12. M. Guevara De Jesus, Z. Xiao, M. Goiriena-Goikoetxea, R. V. Chopdekar, M. K. Panduranga, P. Shirazi, A. Acosta, J. P. Chang, J. Bokor, G. P. Carman, R. N. Candler, and C. Lynch, "Magnetic state switching in FeGa microstructures," *Smart Materials and Structures*, vol. 31, p. 035005, 2022/01/25 2022.

Curriculum Vitae - Jeffrey Bokor

13. Z. Xiao, C. Lai, R. Zheng, M. Goiriena-Goikoetxea, N. Tamura, C. T. Juarez, C. Perry, H. Singh, J. Bokor, G. P. Carman, and R. N. Candler, "Localized strain profile in surface electrode array for programmable composite multiferroic devices," *Appl. Phys. Lett.*, vol. 118, p. 182901, 2021-05-03 2021.
14. P.-C. Shen, C. Su, Y. Lin, A.-S. Chou, C.-C. Cheng, J.-H. Park, M.-H. Chiu, A.-Y. Lu, H.-L. Tang, M. M. Tavakoli, G. Pitner, X. Ji, Z. Cai, N. Mao, J. Wang, V. Tung, J. Li, J. Bokor, A. Zettl, C.-I. Wu, T. Palacios, L.-J. Li, and J. Kong, "Ultralow contact resistance between semimetal and monolayer semiconductors," *Nature*, vol. 593, pp. 211-217, 2021.
15. Z. Mutlu, J. P. Llinas, P. H. Jacobse, I. Piskun, R. Blackwell, M. F. Crommie, F. R. Fischer, and J. Bokor, "Transfer-Free Synthesis of Atomically Precise Graphene Nanoribbons on Insulating Substrates," *ACS Nano*, vol. 15, p. 2635, Jan 25 2021.
16. Z. Mutlu, P. H. Jacobse, R. D. McCurdy, J. P. Llinas, Y. Lin, G. C. Veber, F. R. Fischer, M. F. Crommie, and J. Bokor, "Bottom-Up Synthesized Nanoporous Graphene Transistors," *Adv Funct Mater*, vol. 31, p. 2103798, 2021.
17. P. B. Meisenheimer, R. A. Steinhardt, S. H. Sung, L. D. Williams, S. Zhuang, M. E. Nowakowski, S. Novakov, M. M. Torunbalci, B. Prasad, C. J. Zollner, Z. Wang, N. M. Dawley, J. Schubert, A. H. Hunter, S. Manipatruni, D. E. Nikonov, I. A. Young, L. Q. Chen, J. Bokor, S. A. Bhave, R. Ramesh, J.-M. Hu, E. Kioupakis, R. Hovden, D. G. Schlom, and J. T. Heron, "Engineering new limits to magnetostriction through metastability in iron-gallium alloys," *Nature Communications*, vol. 12, 2021-12-01 2021.
18. R. D. McCurdy, P. H. Jacobse, I. Piskun, G. C. Veber, D. J. Rizzo, R. Zuzak, Z. Mutlu, J. Bokor, M. F. Crommie, and F. R. Fischer, "Synergetic Bottom-Up Synthesis of Graphene Nanoribbons by Matrix-Assisted Direct Transfer," *J Am Chem Soc*, vol. 143, pp. 4174-4178, 2021.
19. R. Khojah, Z. Xiao, M. K. Panduranga, M. Bogumil, Y. Wang, M. Goiriena-Goikoetxea, R. V. Chopdekar, J. Bokor, G. P. Carman, R. N. Candler, and D. Di Carlo, "Single-Domain Multiferroic Array-Addressable Terfenol-D (SMArT) Micromagnets for Programmable Single-Cell Capture and Release," *Advanced Materials*, vol. 33, p. 2006651, 2021-05-01 2021.
20. F. Jakobs, T. A. Ostler, C. H. Lambert, Y. Yang, S. Salahuddin, R. B. Wilson, J. Gorchon, J. Bokor, and U. Atxitia, "Unifying femtosecond and picosecond single-pulse magnetic switching in Gd-Fe-Co," *Physical Review B*, vol. 103, p. 104422, 2021.
21. M. Goiriena-Goikoetxea, Z. Xiao, A. El-Ghazaly, C. V. Stan, J. Chatterjee, A. Ceballos, A. Pattabi, N. Tamura, R. Lo Conte, F. Hellman, R. Candler, and J. Bokor, "Influence of dislocations and twin walls in BaTiO₃ on the voltage-controlled switching of perpendicular magnetization," *Physical Review Materials*, vol. 5, p. 024401, 2021.
22. S. Das, Z. Hong, V. A. Stoica, M. A. P. Goncalves, Y. T. Shao, E. Parsonnet, E. J. Marksz, S. Saremi, M. R. McCarter, A. Reynoso, C. J. Long, A. M. Hagerstrom, D. Meyers, V. Ravi, B. Prasad, H. Zhou, Z. Zhang, H. Wen, F. Gomez-Ortiz, P. Garcia-Fernandez, J. Bokor, J. Iniguez, J. W. Freeland, N. D. Orloff, J. Junquera, L. Q. Chen, S. Salahuddin, D. A. Muller, L. W. Martin, and R. Ramesh, "Local

Curriculum Vitae - Jeffrey Bokor

- negative permittivity and topological phase transition in polar skyrmions," *Nat Mater*, vol. 20, pp. 194-201, Feb 2021.
23. J. Chatterjee, D. Polley, A. Pattabi, H. Jang, S. Salahuddin, and J. Bokor, "RKKY Exchange Bias Mediated Ultrafast All-Optical Switching of a Ferromagnet," *Adv Funct Mater*, p. 2107490, 2021-11-16 2021.
 24. A. Ceballos, A. Pattabi, A. El-Ghazaly, S. Ruta, C. P. Simon, R. F. L. Evans, T. Ostler, R. W. Chantrell, E. Kennedy, M. Scott, J. Bokor, and F. Hellman, "Role of element-specific damping in ultrafast, helicity-independent, all-optical switching dynamics in amorphous (Gd,Tb)Co thin films," *Physical Review B*, vol. 103, 2021.
 25. Z. Xiao, R. Lo Conte, M. Goiriena-Goikoetxea, R. V. Chopdekar, C. A. Lambert, X. Li, A. T. N'Diaye, P. Shafer, S. Tiwari, A. Barra, A. Chavez, K. P. Mohanchandra, G. P. Carman, K. L. Wang, S. Salahuddin, E. Arenholz, J. Bokor, and R. N. Candler, "Tunable Magnetoelastic Effects in Voltage-Controlled Exchange-Coupled Composite Multiferroic Microstructures," *ACS applied materials & interfaces*, vol. 12, pp. 6752-6760, Feb 5 2020.
 26. H. P. Wong, K. Akarvardar, D. Antoniadis, J. Bokor, C. Hu, T. King-Liu, S. Mitra, J. D. Plummer, and S. Salahuddin, "A Density Metric for Semiconductor Technology," *Proceedings of the IEEE*, vol. 108, pp. 478-482, 2020.
 27. G. Veber, C. S. Diercks, C. Rogers, W. S. Perkins, J. Ciston, K. Lee, J. P. Llinas, A. Liebman-Peláez, C. Zhu, J. Bokor, and F. R. Fischer, "Reticular Growth of Graphene Nanoribbon 2D Covalent Organic Frameworks," *Chem*, vol. 6, p. 1125, 2020.
 28. E. Parsonnet, Y.-L. Huang, T. Gosavi, A. Qualls, D. Nikonov, C.-C. Lin, I. Young, J. Bokor, L. W. Martin, and R. Ramesh, "Toward Intrinsic Ferroelectric Switching in Multiferroic BiFeO₃," *Physical Review Letters*, vol. 125, p. 067601, 08/06/ 2020.
 29. D. Jordán, L. J. Albornoz, J. Gorchon, C. H. Lambert, S. Salahuddin, J. Bokor, J. Curiale, and S. Bustingorry, "Statistically meaningful measure of domain-wall roughness in magnetic thin films," *Physical Review B*, vol. 101, p. 184431, 2020.
 30. K. Jhuria, J. Hohlfeld, A. Pattabi, E. Martin, A. Y. Arriola Córdova, X. Shi, R. Lo Conte, S. Petit-Watelot, J. C. Rojas-Sanchez, G. Malinowski, S. Mangin, A. Lemaître, M. Hehn, J. Bokor, R. B. Wilson, and J. Gorchon, "Spin-orbit torque switching of a ferromagnet with picosecond electrical pulses," *Nature Electronics*, vol. 3, p. 680, 2020.
 31. Y. L. Huang, D. Nikonov, C. Addiego, R. V. Chopdekar, B. Prasad, L. Zhang, J. Chatterjee, H. J. Liu, A. Farhan, Y. H. Chu, M. Yang, M. Ramesh, Z. Q. Qiu, B. D. Huey, C. C. Lin, T. Gosavi, J. Iniguez, J. Bokor, X. Pan, I. Young, L. W. Martin, and R. Ramesh, "Manipulating magnetoelectric energy landscape in multiferroics," *Nat Commun*, vol. 11, p. 2836, Jun 5 2020.
 32. J. Hong, X. Li, N. Xu, H. Chen, S. Cabrini, S. Khizroev, J. Bokor, and L. You, "A Dual Magnetic Tunnel Junction-Based Neuromorphic Device," *Advanced Intelligent Systems*, vol. 2, p. 2000143, 2020.
 33. J. Hong, H.-N. Hwang, Y. Su, S. Li, J. Liang, Y. Park, L. T. Singh, Y. G. Jung, J.-H. Yang, J.-I. Jeong, H. Yang, J. Bokor, C.-C. Hwang, and L. You, "Synthesis and Properties of Monolayer Graphene (MLG)-Covered Fe(111)," *Chemistry of Materials*, vol. 32, pp. 10463-10468, 2020.

Curriculum Vitae - Jeffrey Bokor

34. M. Goiriena-Goikoetxea, D. Muñoz, I. Orue, M. L. Fernández-Gubieda, J. Bokor, A. Muela, and A. García-Arribas, "Disk-shaped magnetic particles for cancer therapy," *Applied Physics Reviews*, vol. 7, p. 011306, 2020.
35. A. El-Ghazaly, J. Gorchon, R. B. Wilson, A. Pattabi, and J. Bokor, "Progress towards ultrafast spintronics applications," *J. Magn. Magn. Mater.*, vol. 502, p. 166478, 2020.
36. G. Pitner, G. Hills, J. P. Llinas, K. M. Persson, R. Park, J. Bokor, S. Mitra, and H. P. Wong, "Low-Temperature Side Contact to Carbon Nanotube Transistors: Resistance Distributions Down to 10 nm Contact Length," *Nano Lett.*, vol. 19, pp. 1083-1089, Feb 13 2019.
37. B. Navarrete, M. Stone, P. Wang, R. Guduru, K. Luongo, A. Hadjikhani, D. Toledo, Y. Emirov, B. Arkook, P. Liang, J. Hong, J. Bokor, and S. Khizroev, "Nanomagnetic Particle-Based Information Processing," *IEEE Trans. Nanotechnol.*, vol. 18, pp. 983-988, 2019.
38. J. Hong, T. Yang, A. N'Diaye, J. Bokor, and L. You, "Effects of Interface Induced Natural Strains on Magnetic Properties of FeRh," *Nanomaterials*, vol. 9, p. 574, 2019.
39. J. Hong, X. Li, O. Lee, W. Tian, S. Khizroev, J. Bokor, and L. You, "Demonstration of spin transfer torque (STT) magnetic recording," *Appl. Phys. Lett.*, vol. 114, p. 243101, 2019.
40. A. El-Ghazaly, B. Tran, A. Ceballos, C.-H. Lambert, A. Pattabi, S. Salahuddin, F. Hellman, and J. Bokor, "Ultrafast magnetization switching in nanoscale magnetic dots," *Applied Physics Letters*, vol. 114, no. 23, p. 232407, 2019.
41. Z. Xiao, R. Lo Conte, C. Chen, C. Y. Liang, A. Sepulveda, J. Bokor, G. P. Carman, and R. N. Candler, "Bi-directional coupling in strain-mediated multiferroic heterostructures with magnetic domains and domain wall motion," *Scientific Reports*, vol. 8, p. 5207, Mar 26 2018.
42. Z. Xiao, R. Khojah, M. Chooljian, R. L. Conte, J. D. Schneider, K. Fitzell, R. V. Chopdekar, Y. Wang, A. Scholl, J. Chang, G. P. Carman, J. Bokor, D. Di Carlo, and R. N. Candler, "Cytocompatible magnetostrictive microstructures for nano- and microparticle manipulation on linear strain response piezoelectrics," *Multifunctional Materials*, vol. 1, p. 014004, 2018.
43. S. Smith, J. P. Llinás, J. Bokor, and S. Salahuddin, "Negative Differential Resistance and Steep Switching in Chevron Graphene Nanoribbon Field-Effect Transistors," *IEEE Electron Device Letters*, vol. 39, pp. 143-146, 2018.
44. J. Hong, K. Dong, J. Bokor, and L. You, "Self-assembled single-digit nanometer memory cells," *Applied Physics Letters*, vol. 113, p. 062404, 2018.
45. Z. Xiao, K. P. Mohanchandra, R. Lo Conte, C. Ty Karaba, J. D. Schneider, A. Chavez, S. Tiwari, H. Sohn, M. E. Nowakowski, A. Scholl, S. H. Tolbert, J. Bokor, G. P. Carman, and R. N. Candler, "Enhanced magnetoelectric coupling in a composite multiferroic system via interposing a thin film polymer," *AIP Advances*, vol. 8, 055907, 2018.
46. R. Lo Conte, Z. Xiao, C. Chen, C. V. Stan, J. Gorchon, A. El-Ghazaly, M. E. Nowakowski, H. Sohn, A. Pattabi, A. Scholl, N. Tamura, A. Sepulveda, G. P. Carman, R. N. Candler, and J. Bokor, "Influence of Nonuniform Micron-Scale Strain Distributions on the Electrical Reorientation of Magnetic Microstructures in

Curriculum Vitae - Jeffrey Bokor

- a Composite Multiferroic Heterostructure," *Nano Lett*, vol. 18, pp. 1952-1961, Mar 14 2018.
47. R. Lo Conte, J. Gorchon, A. Mougin, C. H. A. Lambert, A. El-Ghazaly, A. Scholl, S. Salahuddin, and J. Bokor, "Electrically controlled switching of the magnetization state in multiferroic BaTiO₃/CoFe submicrometer structures," *Physical Review Materials*, vol. 2, 2018.
 48. J. Hong, M. Stone, B. Navarrete, K. Luongo, Q. Zheng, Z. Yuan, K. Xia, N. Xu, J. Bokor, L. You, and S. Khizroev, "3D multilevel spin transfer torque devices," *Applied Physics Letters*, vol. 112, p. 112402, 2018.
 49. Y. Yang, R. B. Wilson, J. Gorchon, C.-H. Lambert, S. Salahuddin, and J. Bokor, "Ultrafast magnetization reversal by picosecond electrical pulses," *Science Advances*, vol. 3, p. E1603117, 2017.
 50. R. B. Wilson, J. Gorchon, Y. Yang, C.-H. Lambert, S. Salahuddin, and J. Bokor, "Ultrafast magnetic switching of GdFeCo with electronic heat currents," *Physical Review B*, vol. 95, 2017.
 51. R. B. Wilson, Y. Yang, J. Gorchon, C. H. Lambert, S. Salahuddin, and J. Bokor, "Electric current induced ultrafast demagnetization," *Physical Review B*, vol. 96, 2017.
 52. H. Sohn, C.-y. Liang, M. E. Nowakowski, Y. Hwang, S. Han, J. Bokor, G. P. Carman, and R. N. Candler, "Deterministic multi-step rotation of magnetic single-domain state in Nickel nanodisks using multiferroic magnetoelastic coupling," *J. Magn. Magn. Mater.*, vol. 439, pp. 196-202, 2017.
 53. J. P. Llinas, A. Fairbrother, G. Borin Barin, W. Shi, K. Lee, S. Wu, B. Yong Choi, R. Braganza, J. Lear, N. Kau, W. Choi, C. Chen, Z. Pedramrazi, T. Dumsloff, A. Narita, X. Feng, K. Mullen, F. Fischer, A. Zettl, P. Ruffieux, E. Yablonovitch, M. Crommie, R. Fasel, and J. Bokor, "Short-channel field-effect transistors with 9-atom and 13-atom wide graphene nanoribbons," *Nat Commun*, vol. 8, p. 633, Sep 21 2017.
 54. D. Chen, Z. Chen, Q. He, J. D. Clarkson, C. R. Serrao, A. K. Yadav, M. E. Nowakowski, Z. Fan, L. You, X. Gao, D. Zeng, L. Chen, A. Y. Borisevich, S. Salahuddin, J.-M. Liu, and J. Bokor, "Interface Engineering of Domain Structures in BiFeO₃ Thin Films," *Nano Letters*, vol. 17, pp. 486-493, 2017/01/11 2017.
 55. J. Gorchon, Y. Yang, and J. Bokor, "Model for multishot all-thermal all-optical switching in ferromagnets," *Physical Review B*, vol. 94, p. 020409, 2016.
 56. J. Gorchon, R. B. Wilson, Y. Yang, A. Pattabi, J. Y. Chen, L. He, J. P. Wang, M. Li, and J. Bokor, "Role of electron and phonon temperatures in the helicity-independent all-optical switching of GdFeCo," *Physical Review B*, vol. 94, 2016.
 57. Y. Kang, J. Bokor, and V. Stojanovic, "Design Requirements for a Spintronic MTJ Logic Device for Pipelined Logic Applications," *IEEE Transactions on Electron Devices*, vol. 63, pp. 1754-1761, 2016.
 58. J. Hong, B. Lambson, S. Dhuey, and J. Bokor, "Experimental test of Landauer's principle in single-bit operations on nanomagnetic memory bits," *Science Advances*, vol. 2, e1501492, 2016.
 59. J. Hong, A. Hadjikhani, M. Stone, P. Liang, F. Allen, V. Safonov, J. Bokor, and S. Khizroev, "The Physics of Spin-transfer Torque Switching in Magnetic Tunneling

Curriculum Vitae - Jeffrey Bokor

- Junctions in Sub-10-nm Size Range," IEEE Transactions on Magnetism, vol. PP, pp. 1-1, 2016.
60. S. B. Desai, S. R. Madhvapathy, A. B. Sachid, J. P. Llinas, Q. Wang, G. H. Ahn, G. Pitner, M. J. Kim, J. Bokor, C. Hu, H.-S. P. Wong, and A. Javey, "MoS₂ transistors with 1-nanometer gate lengths," *Science*, vol. 354, pp. 99-102, 2016.
 61. L. You, O. Lee, D. Bhowmik, D. Labanowski, J. Hong, J. Bokor, and S. Salahuddin, "Switching of perpendicularly polarized nanomagnets with spin orbit torque without an external magnetic field by engineering a tilted anisotropy," *Proceedings of the National Academy of Sciences*, vol. 112, pp. 10310-10315, August 18, 2015.
 62. D. Bhowmik, M. E. Nowakowski, L. You, O. Lee, D. Keating, M. Wong, J. Bokor, and S. Salahuddin, "Deterministic Domain Wall Motion Orthogonal To Current Flow Due To Spin Orbit Torque," *Scientific reports*, vol. 5, p. 11823, 2015.
 63. A. Pattabi, Z. Gu, J. Gorchon, Y. Yang, J. Finley, O. J. Lee, H. A. Raziq, S. Salahuddin, and J. Bokor, "Direct optical detection of current induced spin accumulation in metals by magnetization-induced second harmonic generation," *Applied Physics Letters*, vol. 107, p. 152404, 2015.
 64. Y. Lee, Z. Q. Liu, J. T. Heron, J. D. Clarkson, J. Hong, C. Ko, M. D. Biegalski, U. Aschauer, S. L. Hsu, M. E. Nowakowski, J. Wu, H. M. Christen, S. Salahuddin, J. B. Bokor, N. A. Spaldin, D. G. Schlom, and R. Ramesh, "Large resistivity modulation in mixed-phase metallic systems," *Nat Commun*, vol. 6, 01/07/online 2015.
 65. M. M. Sabry Aly, M. Gao, G. Hills, C.-S. Lee, G. Pitner, M. M. Shulaker, T. F. Wu, M. Asheghi, J. Bokor, F. Franchetti, K. E. Goodson, C. Kozyrakis, I. Markov, K. Olukotun, L. Pileggi, E. Pop, J. Rabaey, C. Re, H. S. P. Wong, and S. Mitra, "Energy-Efficient Abundant-Data Computing: The N3XT 1,000x," *Computer*, vol. 48, pp. 24-33, 2015.
 66. Z. Gu, R. Storz, M. Marcus, A. Doran, A. Young, A. Scholl, W. Chao, D. Carlton, B. Lambson, M. Nowakowski, and J. Bokor, "Time-Resolved Photo-Emission Electron Microscopy of Nanomagnetic Logic Chains," in *Ultrafast Magnetism I*. vol. 159, J. Y. Bigot, W. Hubner, T. Rasing, and R. Chantrell, Eds., Cham: Springer Int Publishing Ag, 2015, pp. 281-283.
 67. Z. Gu, M. E. Nowakowski, D. B. Carlton, R. Storz, M.-Y. Im, J. Hong, W. Chao, B. Lambson, P. Bennett, M. T. Alam, M. A. Marcus, A. Doran, A. Young, A. Scholl, P. Fischer, and J. Bokor, "Sub-nanosecond signal propagation in anisotropy-engineered nanomagnetic logic chains," *Nat Commun*, vol. 6, 03/16/online 2015.
 68. S.-J. Choi, P. Bennett, D. Lee, and J. Bokor, "Highly uniform carbon nanotube nanomesh network transistors," *Nano Research*, vol. 8, pp. 1320-1326, 2015/04/01 2015.
 69. H. Sohn, M. E. Nowakowski, C.-Y. Liang, J. L. Hockel, K. Wetzlar, S. Keller, B. M. McLellan, M. A. Marcus, A. Doran, A. Young, M. Kläui, G. P. Carman, J. Bokor, and R. N. Candler, "Electrically Driven Magnetic Domain Wall Rotation in Multiferroic Heterostructures to Manipulate Suspended On-Chip Magnetic Particles," *ACS Nano*, vol. 9, pp. 4814-4826, 2015.
 70. Y. Lee, Z. Q. Liu, J. T. Heron, J. D. Clarkson, J. Hong, C. Ko, M. D. Biegalski, U. Aschauer, S. L. Hsu, M. E. Nowakowski, J. Wu, H. M. Christen, S. Salahuddin, J.

Curriculum Vitae - Jeffrey Bokor

- B. Bokor, N. A. Spaldin, D. G. Schlom, and R. Ramesh, "Large resistivity modulation in mixed-phase metallic systems," *Nat Commun*, vol. 6, 01/07/online 2015.
71. Z. Gu, R. Storz, M. Marcus, A. Doran, A. Young, A. Scholl, W. Chao, D. Carlton, B. Lambson, and M. Nowakowski, "Time-Resolved Photo-Emission Electron Microscopy of Nanomagnetic Logic Chains," in *Ultrafast Magnetism I*: Springer, 2015, pp. 281-283.
 72. Z. Gu, M. E. Nowakowski, D. B. Carlton, R. Storz, M.-Y. Im, J. Hong, W. Chao, B. Lambson, P. Bennett, M. T. Alam, M. A. Marcus, A. Doran, A. Young, A. Scholl, P. Fischer, and J. Bokor, "Sub-nanosecond signal propagation in anisotropy-engineered nanomagnetic logic chains," *Nat Commun*, vol. 6, 03/16/online 2015.
 73. A. R. Schmidt, E. Henry, C. C. Lo, Y. T. Wang, H. Li, L. Greenman, O. Namaan, T. Schenkel, K. B. Whaley, J. Bokor, E. Yablonovitch, and I. Siddiqi, "A prototype silicon double quantum dot with dispersive microwave readout," *Journal of Applied Physics*, vol. 116, p. 044503, Jul 2014.
 74. C. C. Lo, S. Simmons, R. Lo Nardo, C. D. Weis, A. M. Tyryshkin, J. Meijer, D. Rogalla, S. A. Lyon, J. Bokor, T. Schenkel, and J. J. L. Morton, "Stark shift and field ionization of arsenic donors in Si-28-silicon-on-insulator structures," *Applied Physics Letters*, vol. 104, p. 193502, May 2014.
 75. D. Lee, M. L. Seol, D. I. Moon, P. Bennett, N. Yoder, J. Humes, J. Bokor, Y. K. Choi, and S. J. Choi, "High-performance thin-film transistors produced from highly separated solution-processed carbon nanotubes," *Applied Physics Letters*, vol. 104, p.143508, Apr 2014.
 76. T. Schenkel, C. C. Lo, C. D. Weis, J. Bokor, A. M. Tyryshkin, and S. A. Lyon, "A Spin Quantum Bit Architecture with Coupled Donors and Quantum Dots in Silicon," in *Single-Atom Nanoelectronics*, E. Prati and T. Shinada, Eds., 2013, pp. 255-279.
 77. C. C. Lo, C. D. Weis, J. van Tol, J. Bokor, and T. Schenkel, "All-Electrical Nuclear Spin Polarization of Donors in Silicon," *Physical Review Letters*, vol. 110, p. 57601, Jan 2013.
 78. B. Lambson, Z. Gu, M. Monroe, S. Dhuey, A. Scholl, and J. Bokor, "Concave nanomagnets: investigation of anisotropy properties and applications to nanomagnetic logic," *Applied Physics a-Materials Science & Processing*, vol. 111, pp. 413-421, May 2013.
 79. S. J. Choi, P. Bennett, K. Takei, C. Wang, C. C. Lo, A. Javey, and J. Bokor, "Short-Channel Transistors Constructed with Solution-Processed Carbon Nanotubes," *ACS Nano*, vol. 7, pp. 798-803, Jan 2013.
 80. P. B. Bennett, Z. Pedramrazi, A. Madani, Y.-C. Chen, D. G. de Oteyza, C. Chen, F. R. Fischer, M. F. Crommie, and J. Bokor, "Bottom-up graphene nanoribbon field effect transistors," *Applied Physics Letters*, vol. 103, 253114, 2013.
 81. W. Bao, M. Staffaroni, J. Bokor, M. B. Salmeron, E. Yablonovitch, S. Cabrini, A. Weber-Bargioni, and P. J. Schuck, "Plasmonic near-field probes: a comparison of the campanile geometry with other sharp tips," *Optics Express*, vol. 21, pp. 8166-8176, Apr 2013.
 82. A. T. Zayak, H. Choo, Y. S. Hu, D. J. Gargas, S. Cabrini, J. Bokor, P. J. Schuck, and J. B. Neaton, "Harnessing Chemical Raman Enhancement for Understanding

Curriculum Vitae - Jeffrey Bokor

- Organic Adsorbate Binding on Metal Surfaces," *Journal of Physical Chemistry Letters*, vol. 3, pp. 1357-1362, May 2012.
83. W. Bao, M. Melli, N. Caselli, F. Riboli, D. S. Wiersma, M. Staffaroni, H. Choo, D. F. Ogletree, S. Aloni, J. Bokor, S. Cabrini, F. Intonti, M. B. Salmeron, E. Yablonovitch, P. J. Schuck, and A. Weber-Bargioni, "Mapping Local Charge Recombination Heterogeneity by Multidimensional Nanospectroscopic Imaging," *Science*, vol. 338, pp. 1317-1321, Dec 2012.
 84. D. Carlton, B. Lambson, A. Scholl, A. Young, P. Ashby, S. Dhuey, and J. Bokor, "Investigation of Defects and Errors in Nanomagnetic Logic Circuits," *IEEE Transactions on Nanotechnology*, vol. 11, pp. 760-762, Jul 2012.
 85. S. J. Choi, C. Wang, C. C. Lo, P. Bennett, A. Javey, and J. Bokor, "Comparative study of solution-processed carbon nanotube network transistors," *Applied Physics Letters*, vol. 101, Sep 2012.
 86. H. Choo, M. K. Kim, M. Staffaroni, T. J. Seok, J. Bokor, S. Cabrini, P. J. Schuck, M. C. Wu, and E. Yablonovitch, "Nanofocusing in a metal-insulator-metal gap plasmon waveguide with a three-dimensional linear taper," *Nature Photonics*, vol. 6, pp. 837-843, Dec 2012.
 87. M. Ilg, C. D. Weis, J. Schwartz, A. Persaud, Q. Ji, C. C. Lo, J. Bokor, A. Hegyi, E. Guliyev, I. W. Rangelow, and T. Schenkel, "Improved single ion implantation with scanning probe alignment," *Journal of Vacuum Science & Technology B*, vol. 30, Nov 2012.
 88. B. Lambson, Z. Gu, D. Carlton, S. Dhuey, A. Scholl, A. Doran, A. Young, and J. Bokor, "Cascade-like signal propagation in chains of concave nanomagnets," *Applied Physics Letters*, vol. 100, p. 152406, Apr 2012.
 89. C. C. Lo, F. R. Bradbury, A. M. Tyryshkin, C. D. Weis, J. Bokor, T. Schenkel, and S. A. Lyon, "Suppression of microwave rectification effects in electrically detected magnetic resonance measurements," *Applied Physics Letters*, vol. 100, p. 63510, Feb 2012.
 90. M. T. Martinez, Y. C. Tseng, M. Gonzalez, and J. Bokor, "Streptavidin as CNTs and DNA Linker for the Specific Electronic and Optical Detection of DNA Hybridization," *Journal of Physical Chemistry C*, vol. 116, pp. 22579-22586, Oct 2012.
 91. C. D. Weis, C. C. Lo, V. Lang, A. M. Tyryshkin, R. E. George, K. M. Yu, J. Bokor, S. A. Lyon, J. J. L. Morton, and T. Schenkel, "Electrical activation and electron spin resonance measurements of implanted bismuth in isotopically enriched silicon-28," *Applied Physics Letters*, vol. 100, p. 172104, Apr 2012.
 92. A. T. Zayak, Y. S. Hu, H. Choo, J. Bokor, S. Cabrini, P. J. Schuck, and J. B. Neaton, "Chemical Raman Enhancement of Organic Adsorbates on Metal Surfaces," *Physical Review Letters*, vol. 106, p. 083003, Feb 2011.
 93. J. W. J. Wu, D. Carlton, J. S. Park, Y. Meng, E. Arenholz, A. Doran, A. T. Young, A. Scholl, C. Hwang, H. W. Zhao, J. Bokor, and Z. Q. Qiu, "Direct observation of imprinted antiferromagnetic vortex states in CoO/Fe/Ag(001) discs," *Nature Physics*, vol. 7, pp. 303-306, Apr 2011.
 94. A. Weber-Bargioni, A. Schwartzberg, M. Cornaglia, A. Ismach, J. J. Urban, Y. J. Pang, R. Gordon, J. Bokor, M. B. Salmeron, D. F. Ogletree, P. Ashby, S. Cabrini, and P. J. Schuck, "Hyperspectral Nanoscale Imaging on Dielectric Substrates with

Curriculum Vitae - Jeffrey Bokor

- Coaxial Optical Antenna Scan Probes," *Nano Letters*, vol. 11, pp. 1201-1207, Mar 2011.
95. Z. Liu, R. Brandt, Y. Yahagi, B. Hansen, B. Harteneck, J. Bokor, A. R. Hawkins, and H. Schmidt, "Detecting single nanomagnet dynamics beyond the diffraction limit in varying magnetostatic environments," *Applied Physics Letters*, vol. 98, Jan 2011.
 96. V. L. V. Lang, C. C. Lo, R. E. George, S. A. Lyon, J. Bokor, T. Schenkel, A. Ardavan, and J. J. L. Morton, "Electrically detected magnetic resonance in a W-band microwave cavity," *Review of Scientific Instruments*, vol. 82, p. 034704, Mar 2011.
 97. B. Lambson, D. Carlton, and J. Bokor, "Exploring the Thermodynamic Limits of Computation in Integrated Systems: Magnetic Memory, Nanomagnetic Logic, and the Landauer Limit," *Physical Review Letters*, vol. 107, p. 010604, Jul 2011.
 98. J. Wu, D. Carlton, E. Oelker, J. S. Park, E. Jin, E. Arenholz, A. Scholl, H. Chanyong, J. Bokor, and Z. Q. Qiu, "Switching a magnetic vortex by interlayer coupling in epitaxially grown Co/Cu/Py/Cu(001) trilayer disks," *J. Phys., Condens. Matter.*, vol. 22, pp. 342001 (4 pp.)-342001 (4 pp.), 1 September 2010.
 99. Y. C. Tseng and J. Bokor, "Characterization of the junction capacitance of metal-semiconductor carbon nanotube Schottky contacts," *Applied Physics Letters*, vol. 96, p. 013103, Jan 2010.
 100. M. T. Martinez, Y. C. Tseng, J. P. Salvador, M. P. Marco, N. Ormategui, I. Loinaz, and J. Bokor, "Electronic Anabolic Steroid Recognition with Carbon Nanotube Field-Effect Transistors," *Acs Nano*, vol. 4, pp. 1473-1480, Mar 2010.
 101. X. Liang, Y.-S. Jung, S. Wu, A. Ismach, D. L. Olynick, S. Cabrini, and J. Bokor, "Formation of Bandgap and Subbands in Graphene Nanomeshes with Sub-10 nm Ribbon Width Fabricated via Nanoimprint Lithography," *Nano Letters*, vol. 10, pp. 2454-2460, 2010.
 102. Ismach, C. Druzgalski, S. Penwell, A. Schwartzberg, M. Zheng, A. Javey, J. Bokor, and Y. G. Zhang, "Direct Chemical Vapor Deposition of Graphene on Dielectric Surfaces," *Nano Letters*, vol. 10, pp. 1542-1548, May 2010.
 103. Q. Ding, Z. T. Deng, H. Yan, S. Cabrini, R. N. Zuckermann, and J. Bokor, "Gold Nanoparticle Self-Similar Chain Structure Organized by DNA Origami," *Journal of the American Chemical Society*, vol. 132, pp. 3248+, Mar 2010.
 104. D. Weis, A. Schuh, A. Batra, A. Persaud, I. W. Rangelow, J. Bokor, C. C. Lo, S. Cabrini, D. Olynick, S. Duhey, and T. Schenkel, "Mapping of ion beam induced current changes in FinFETs," *Nuclear Instruments & Methods in Physics Research, Section B-Beam Interactions with Materials and Atoms*, vol. 267, pp. 1222-1225, 2009.
 105. T. Schenkel, C. C. Lo, C. D. Weis, A. Schuh, A. Persaud, and J. Bokor, "Critical issues in the formation of quantum computer test structures by ion implantation," *Nuclear Instruments & Methods in Physics Research Section B-Beam Interactions with Materials and Atoms*, vol. 267, pp. 2563-2566, Aug 2009.
 106. M. T. Martinez, Y. C. Tseng, N. Ormategui, I. Loinaz, R. Eritja, and J. Bokor, "Label-Free DNA Biosensors Based on Functionalized Carbon Nanotube Field Effect Transistors," *Nano Letters*, vol. 9, pp. 530-536, Feb 2009.

Curriculum Vitae - Jeffrey Bokor

107. C. Lo, A. Persaud, S. Dhuey, D. Olynick, F. Borondics, M. C. Martin, H. A. Bechtel, J. Bokor, and T. Schenkel, "Device fabrication and transport measurements of FinFETs built with Si-28 SOI wafers toward donor qubits in silicon," *Semiconductor Science and Technology*, vol. 24, p. 105022, Oct 2009.
108. C. Garnett, Y. C. Tseng, D. R. Khanal, J. Q. Wu, J. Bokor, and P. D. Yang, "Dopant profiling and surface analysis of silicon nanowires using capacitance-voltage measurements," *Nat. Nanotechnol.*, vol. 4, pp. 311-314, May 2009.
109. C. Ford, J. C. Ho, Y. L. Chueh, Y. C. Tseng, Z. Y. Fan, J. Guo, J. Bokor, and A. Javey, "Diameter-Dependent Electron Mobility of InAs Nanowires," *Nano Letters*, vol. 9, pp. 360-365, Jan 2009.
110. Ding, S. Cabrini, R. N. Zuckermann, and J. Bokor, "DNA directed assembly of nanoparticle linear structure for nanophotonics," *Journal of Vacuum Science & Technology B: Microelectronics and Nanometer Structures*, vol. 27, pp. 184-187, 2009.
111. R. de Sousa, C. C. Lo, and J. Bokor, "Spin-dependent scattering in a silicon transistor," *Physical Review B*, vol. 80, p. 9, Jul 2009.
112. D. Weis, A. Schuh, A. Batra, A. Persaud, I. W. Rangelow, J. Bokor, C. C. Lo, S. Cabrini, E. Sideras-Haddad, G. D. Fuchs, R. Hanson, D. D. Awschalom, and T. Schenkel, "Single atom doping for quantum device development in diamond and silicon," *J. Vac. Sci. Technol.*, vol. 26, pp. 2596-2600, 2008.
113. B. Carlton, N. C. Emley, E. Tuchfeld, and J. Bokor, "Simulation Studies of Nanomagnet-Based Logic Architecture," *Nano Letters*, vol. 8, pp. 4173-4178, Dec 2008.
114. Y. Wang and J. Bokor, "Ultra-high-resolution monolithic thermal bubble inkjet print head," *Journal of Micro/Nanolithography, MEMS and MOEMS*, vol. 6, pp. 043009-10, 2007.
115. S. Wang, A. Barman, H. Schmidt, J. D. Maas, A. R. Hawkins, S. Kwon, B. Harteneck, S. Cabrini, and J. Bokor, "Optimization of nano-magneto-optic sensitivity using dual dielectric layer enhancement," *Applied Physics Letters*, vol. 90, p. 252504, Jun 2007.
116. San Paulo, E. Quevy, J. Black, R. T. Howe, R. White, and J. Bokor, "Mode shape imaging of out-of-plane and in-plane vibrating RF micromechanical resonators by atomic force microscopy," *Microelectronic Engineering*, vol. 84, pp. 1354-1357, May-Aug 2007.
117. San Paulo, N. Arellano, J. A. Plaza, R. R. He, C. Carraro, R. Maboudian, R. T. Howe, J. Bokor, and P. D. Yang, "Suspended mechanical structures based on elastic silicon nanowire arrays," *Nano Letters*, vol. 7, pp. 1100-1104, Apr 2007.
118. S. Paulo, J. P. Black, R. M. White, and J. Bokor, "Detection of nanomechanical vibrations by dynamic force microscopy in higher cantilever eigenmodes," *Applied Physics Letters*, vol. 91, p. 053116, Jul 2007.
119. C. Lo, J. Bokor, T. Schenkel, A. M. Tyryshkin, and S. A. Lyon, "Spin-dependent scattering off neutral antimony donors in Si-28 field-effect transistors," *Applied Physics Letters*, vol. 91, p. 242106, Dec 2007.
120. Batra, C. D. Weis, J. Reijonen, A. Persaud, T. Schenkel, S. Cabrini, C. C. Lo, and J. Bokor, "Detection of low energy single ion impacts in micron scale transistors at room temperature," *Applied Physics Letters*, vol. 91, p. 193502, 2007.

Curriculum Vitae - Jeffrey Bokor

121. Barman, S. Wang, J. Maas, A. R. Hawkins, S. Kwon, J. Bokor, A. Liddle, and H. Schmidt, "Size dependent damping in picosecond dynamics of single nanomagnets," *Applied Physics Letters*, vol. 90, pp. 202504-1-3, 2007.
122. M. Tyryshkin, S. A. Lyon, T. Schenkel, J. Bokor, J. Chu, W. Jantsch, F. Schaffler, J. L. Truitt, S. N. Coppersmith, and M. A. Eriksson, "Electron spin coherence in Si," *Physica E-Low-Dimensional Systems & Nanostructures*, vol. 35, pp. 257-263, Dec 2006.
123. Y. C. Tseng, K. Phoa, D. Carlton, and J. Bokor, "Effect of diameter variation in a large set of carbon nanotube transistors," *Nano Letters*, vol. 6, pp. 1364-1368, Jul 2006.
124. T. Schenkel, J. A. Liddle, A. Persaud, A. M. Tyryshkin, S. A. Lyon, R. de Sousa, K. B. Whaley, J. Bokor, J. Shangkuan, and I. Chakarov, "Electrical activation and electron spin coherence of ultralow dose antimony implants in silicon," *Applied Physics Letters*, vol. 88, p. 112101, Mar 2006.
125. T. Schenkel, J. A. Liddle, J. Bokor, A. Persaud, S. J. Park, J. Shangkuan, C. C. Lo, S. Kwon, S. A. Lyon, A. M. Tyryshkin, I. W. Rangelow, Y. Sarov, D. H. Schneider, J. Ager, and R. de Sousa, "Strategies for integration of donor electron spin qubits in silicon," *Microelectronic Engineering*, vol. 83, pp. 1814-1817, Apr-Sep 2006.
126. M. Contreras, X. M. Yan, S. Kwon, J. Bokor, and G. A. Somorjai, "Catalytic CO oxidation reaction studies on lithographically fabricated platinum nanowire arrays with different oxide supports," *Catalysis Letters*, vol. 111, pp. 5-13, Oct 2006.
127. R. Bradbury, A. M. Tyryshkin, G. Sabouret, J. Bokor, T. Schenkel, and S. A. Lyon, "Stark tuning of donor electron spins in silicon," *Physical Review Letters*, vol. 97, p. 176404, Oct 2006.
128. Barman, S. Q. Wang, J. D. Maas, A. R. Hawkins, S. Kwon, A. Liddle, J. Bokor, and H. Schmidt, "Magneto-optical observation of picosecond dynamics of single nanomagnets," *Nano Letters*, vol. 6, pp. 2939-2944, Dec 2006.
129. X. M. Yan, S. Kwon, A. M. Contreras, M. M. Koebel, J. Bokor, and G. A. Somorjai, "Fabrication of dense arrays of platinum nanowires on silica, alumina, zirconia and ceria surfaces as 2-D model catalysts," *Catalysis Letters*, vol. 105, pp. 127-132, Dec 2005.
130. X. M. Yan, S. Kwon, A. M. Contreras, J. Bokor, and G. A. Somorjai, "Fabrication of large number density platinum nanowire arrays by size reduction lithography and nanoimprint lithography," *Nano Letters*, vol. 5, pp. 745-748, Apr 2005.
131. S. Y. Xiong, T. J. King, and J. Bokor, "Study of the extrinsic parasitics in nano-scale transistors," *Semiconductor Science and Technology*, vol. 20, pp. 652-657, Jun 2005.
132. S. Y. Xiong, T. J. King, and J. Bokor, "A comparison study of symmetric ultrathin-body double-gate devices with metal source/drain and doped source/drain," *Ieee Transactions on Electron Devices*, vol. 52, pp. 1859-1867, Aug 2005.
133. S. Y. Xiong and J. Bokor, "Structural optimization of SUTBDG devices for low-power applications," *Ieee Transactions on Electron Devices*, vol. 52, pp. 360-366, Mar 2005.
134. M. D. Shumway, P. Naulleau, K. A. Goldberg, and J. R. Bokor, "Measuring line roughness through aerial image contrast variation using coherent extreme ultraviolet

Curriculum Vitae - Jeffrey Bokor

- spatial filtering technique," *Journal of Vacuum Science & Technology B*, vol. 23, pp. 2844-2847, Nov-Dec 2005.
135. San Paulo, J. Bokor, R. T. Howe, R. He, P. Yang, D. Gao, C. Carraro, and R. Maboudian, "Mechanical elasticity of single and double clamped silicon nanobeams fabricated by the vapor-liquid-solid method," *Applied Physics Letters*, vol. 87, p. 053111, Aug 2005.
 136. A. San Paulo, X. Liu, and J. Bokor, "Scanning acoustic force microscopy characterization of thermal expansion effects on the electromechanical properties of film bulk acoustic resonators," *Applied Physics Letters*, vol. 86, p. 084102, Feb 2005.
 137. N. Qureshi, S. Q. Wang, M. A. Lowther, A. R. Hawkins, S. Kwon, A. Liddle, J. Bokor, and H. Schmidt, "Cavity-enhanced magneto-optical observation of magnetization reversal in individual single-domain nanomagnets," *Nano Letters*, vol. 5, pp. 1413-1417, Jul 2005.
 138. Persaud, S. J. Park, J. A. Liddle, T. Schenkel, J. Bokor, and I. W. Rangelow, "Integration of scanning probes and ion beams," *Nano Letters*, vol. 5, pp. 1087-1091, Jun 2005.
 139. Persaud, J. A. Liddle, T. Schenkel, J. Bokor, T. Ivanov, and I. W. Rangelow, "Ion implantation with scanning probe alignment," *Journal of Vacuum Science & Technology B*, vol. 23, pp. 2798-2800, Nov-Dec 2005.
 140. J. Lin, P. Xuan, and J. Bokor, "Characterization of chemical vapor deposition growth yields of carbon nanotube transistors," *Japanese Journal of Applied Physics Part I-Regular Papers Brief Communications & Review Papers*, vol. 44, pp. 6859-6861, Sep 2005.
 141. S. Kwon, X. M. Yan, A. M. Contreras, J. A. Liddle, G. A. Somorjai, and J. Bokor, "Fabrication of metallic nanodots in large-area arrays by mold-to-mold cross imprinting (MTMCI)," *Nano Letters*, vol. 5, pp. 2557-2562, Dec 2005.
 142. Yagishita, T. J. King, and J. Bokor, "Schottky barrier height reduction and drive current improvement in metal source/drain MOSFET with strained-Si channel," *Japanese Journal of Applied Physics Part I-Regular Papers Short Notes & Review Papers*, vol. 43, pp. 1713-1716, Apr 2004.
 143. S. Y. Xiong, J. Bokor, Q. Xiang, P. Fisher, I. Dudley, P. Rao, H. H. Wang, and B. En, "Is gate line edge roughness a first-order issue in affecting the performance of deep sub-micro bulk MOSFET devices?," *Ieee Transactions on Semiconductor Manufacturing*, vol. 17, pp. 357-361, Aug 2004.
 144. S. Y. Xiong and J. Bokor, "A simulation study of gate line edge roughness. Effects on doping profiles of short-channel MOSFET devices," *Ieee Transactions on Electron Devices*, vol. 51, pp. 228-232, Feb 2004.
 145. Wang, J. Bokor, and A. Lee, "Maskless lithography using drop-on-demand inkjet printing method," *SPIE-Int. Soc. Opt. Eng. Proceedings of the SPIE*, vol. 5374, pp. 628-36, 2004.
 146. C. Tseng, P. Q. Xuan, A. Javey, R. Malloy, Q. Wang, J. Bokor, and H. J. Dai, "Monolithic integration of carbon nanotube devices with silicon MOS technology," *Nano Letters*, vol. 4, pp. 123-127, Jan 2004.
 147. M. D. Shumway, E. L. Snow, K. A. Goldberg, P. Naulleau, H. Cao, M. Chandhok, J. A. Liddle, E. H. Anderson, and J. Bokor, "EUV resist imaging below 50 nm

Curriculum Vitae - Jeffrey Bokor

- using coherent spatial filtering techniques," *Proceedings of the SPIE*, vol. 5374, pp. 454-9, 2004.
148. T. Schenkel, I. W. Rangelow, R. Keller, S. J. Park, J. Nilsson, A. Persaud, V. R. Radmilovic, P. Grabiec, D. H. Schneider, J. A. Liddle, and J. Bokor, "Open questions in electronic sputtering of solids by slow highly charged ions with respect to applications in single ion implantation," *Nuclear Instruments & Methods in Physics Research Section B-Beam Interactions with Materials and Atoms*, vol. 219-20, pp. 200-205, Jun 2004.
 149. N. W. Pu, E. Y. Pan, and J. Bokor, "Sensitive detection of laser damage to Mo/Si multilayers by picosecond ultrasonics," *Applied Physics B-Lasers and Optics*, vol. 79, pp. 107-112, Jul 2004.
 150. Persaud, F. I. Allen, F. Giccluel, S. J. Park, J. A. Liddle, T. Schenkel, T. Ivanov, K. Ivanova, I. W. Rangelow, and J. Bokor, "Single ion implantation with scanning probe alignment," *Journal of Vacuum Science & Technology B*, vol. 22, pp. 2992-2994, Nov-Dec 2004.
 151. S. J. Park, A. Persaud, J. A. Liddle, J. Nilsson, J. Bokor, D. H. Schneider, I. W. Rangelow, and T. Schenkel, "Processing issues in top-down approaches to quantum computer development in silicon," *Microelectronic Engineering*, vol. 73-74, pp. 695-700, Jun 2004.
 152. S. J. Park, J. A. Liddle, A. Persaud, F. I. Allen, T. Schenkel, and J. Bokor, "Formation of 15 nm scale Coulomb blockade structures in silicon by electron beam lithography with a bilayer resist process," *Journal of Vacuum Science & Technology B*, vol. 22, pp. 3115-3118, Nov-Dec 2004.
 153. H. T. Liu, J. K. O. Sin, P. Q. Xuan, and J. Bokor, "Characterization of the ultrathin vertical channel CMOS technology," *Ieee Transactions on Electron Devices*, vol. 51, pp. 106-112, Jan 2004.
 154. Yang-Kyu, J. Grunes, L. Joon Sung, Z. Ji, G. A. Somorjai, L. P. Lee, and J. Bokor, "Sub-lithographic patterning technology for nanowire model catalysts and DNA label-free hybridization detection," *Proceedings of the SPIE - The International Society for Optical Engineering*, vol. 5220, pp. 10-19, 2003.
 155. P. Q. Xuan and J. Bokor, "Investigation of NiSi and TiSi as CMOS gate materials," *Ieee Electron Device Letters*, vol. 24, pp. 634-636, Oct 2003.
 156. S. Y. Xiong and J. Bokor, "Sensitivity of double-gate and FinFET devices to process variations," *Ieee Transactions on Electron Devices*, vol. 50, pp. 2255-2261, Nov 2003.
 157. M. D. Shumway, P. Naulleau, K. A. Goldberg, E. L. Snow, and J. Bokor, "Resist evaluation at 50 nm in the EUV using interferometric spatial-frequency-doubled imaging," *Proceedings of the SPIE*, vol. 5037, pp. 910-16, 2003.
 158. T. Schenkel, A. Persaud, S. J. Park, J. Nilsson, J. Bokor, J. A. Liddle, R. Keller, D. H. Schneider, D. W. Cheng, and D. E. Humphries, "Solid state quantum computer development in silicon with single ion implantation," *Journal of Applied Physics*, vol. 94, pp. 7017-7024, Dec 2003.
 159. N. W. Pu and J. Bokor, "Study of surface and bulk acoustic phonon excitations in superlattices using picosecond ultrasonics," *Physical Review Letters*, vol. 91, p. 76101, Aug 2003.

Curriculum Vitae - Jeffrey Bokor

160. P. P. Naulleau, K. A. Goldberg, P. Batson, J. Bokor, P. Denham, and S. Rekawa, "Fourier-synthesis custom-coherence illuminator for extreme ultraviolet microfield lithography," *Applied Optics*, vol. 42, pp. 820-826, Feb 2003.
161. P. P. Naulleau, K. A. Goldberg, E. H. Anderson, J. Bokor, B. D. Harteneck, K. H. Jackson, D. L. Olynick, F. Salmassi, S. L. Baker, P. B. Mirkarimi, E. A. Spiller, C. C. Walton, D. J. O'Connell, Y. Pei-Yang, and Z. Guojing, "Static EUV micro-exposures using the ETS Set-2 optics," *Proceedings of the SPIE*, vol. 5037, pp. 36-46, 2003.
162. P. Naulleau, K. A. Goldberg, E. H. Anderson, J. Bokor, B. Harteneck, K. Jackson, D. Olynick, F. Salmassi, S. Baker, P. Mirkarimi, E. Spiller, C. Walton, D. O'Connell, P. Y. Yan, and G. J. Zhang, "Printing-based performance analysis of the engineering test stand set-2 optic using a synchrotron exposure station with variable sigma," *Journal of Vacuum Science & Technology B*, vol. 21, pp. 2697-2700, Nov-Dec 2003.
163. P. Naulleau, K. A. Goldberg, E. H. Anderson, J. Bokor, E. Gullikson, B. Harteneck, K. Jackson, D. Olynick, F. Salmassi, S. Baker, P. Mirkarimi, E. Spiller, C. Walton, and G. J. Zhang, "Lithographic characterization of the printability of programmed extreme ultraviolet substrate defects," *Journal of Vacuum Science & Technology B*, vol. 21, pp. 1286-1290, Jul-Aug 2003.
164. J. S. Lee, D. W. Ha, Y. K. Choi, T. J. King, and J. Bokor, "Low-frequency noise characteristics of ultrathin body p-MOSFETs with molybdenum gate," *Ieee Electron Device Letters*, vol. 24, pp. 31-33, Jan 2003.
165. J. S. Lee, Y. K. Choi, D. W. Ha, S. Balasubramanian, T. J. King, and J. Bokor, "Hydrogen annealing effect on DC and low-frequency noise characteristics in CMOS FinFETs," *Ieee Electron Device Letters*, vol. 24, pp. 186-188, Mar 2003.
166. K. A. Goldberg, P. P. Naulleau, P. Denham, S. B. Rekawa, K. H. Jackson, E. H. Anderson, J. A. Liddle, and J. Bokor, "EUV interferometry of the 0.3-NA MET optic," *Proceedings of the SPIE*, vol. 5037, pp. 69-74, 2003.
167. Y. K. Choi, J. Zhu, J. Grunes, J. Bokor, and G. A. Somorjai, "Fabrication of sub-10-nm silicon nanowire arrays by size reduction lithography," *Journal of Physical Chemistry B*, vol. 107, pp. 3340-3343, Apr 2003.
168. Y. K. Choi, J. S. Lee, J. Zhu, G. A. Somorjai, L. P. Lee, and J. Bokor, "Sublithographic nanofabrication technology for nanocatalysts and DNA chips," *Journal of Vacuum Science & Technology B*, vol. 21, pp. 2951-2955, Nov-Dec 2003.
169. Y. K. Choi, D. Ha, T. J. King, and J. Bokor, "Investigation of gate-induced drain leakage (GIDL) current in thin body devices: Single-gate ultra-thin body, symmetrical double-gate, and asymmetrical double-gate MOSFETs," *Japanese Journal of Applied Physics Part I-Regular Papers Short Notes & Review Papers*, vol. 42, pp. 2073-2076, Apr 2003.
170. L. L. Chang, Y. K. Choi, J. Kedzierski, N. Lindert, P. Q. Xuan, J. Bokor, C. M. Hu, and T. J. King, "Moore's law lives on - Ultra-thin body SOI and FinFET CMOS transistors look to continue Moore's law for many years to come," *Ieee Circuits & Devices*, vol. 19, pp. 35-42, Jan 2003.

Curriculum Vitae - Jeffrey Bokor

171. L. L. Chang, Y. K. Choi, D. W. Ha, P. Ranade, S. Y. Xiong, J. Bokor, C. M. Hu, and T. J. King, "Extremely scaled silicon nano-CMOS devices," *Proceedings of the Ieee*, vol. 91, pp. 1860-1873, Nov 2003.
172. M. Yi, T. Haga, C. Walton, C. Larson, and J. Bokor, "'Actinic-only' defects in extreme ultraviolet lithography mask blanks - Native defects at the detection limit of visible-light inspection tools," *Japanese Journal of Applied Physics Part 1- Regular Papers Short Notes & Review Papers*, vol. 41, pp. 4101-4104, Jun 2002.
173. Y. C. Yeo, V. Subramanian, J. Kedzierski, P. Q. Xuan, T. J. King, J. Bokor, and C. M. Hu, "Design and fabrication of 50-nm thin-body p-MOSFETs with a SiGe heterostructure channel," *Ieee Transactions on Electron Devices*, vol. 49, pp. 279-286, Feb 2002.
174. X. Shiying, J. Bokor, X. Qi, P. Fisher, I. M. Dudley, and P. Rao, "Gate line edge roughness effects in 50 nm bulk MOSFET devices," *Proceedings of the SPIE*, vol. 4689, pp. 733-41, 2002.
175. T. Schenkel, A. Persaud, S. J. Park, J. Meijer, J. R. Kingsley, J. W. McDonald, J. P. Holder, J. Bokor, and D. H. Schneider, "Single ion implantation for solid state quantum computer development," *Journal of Vacuum Science & Technology B*, vol. 20, pp. 2819-2823, Nov-Dec 2002.
176. M. Park, M. Yi, P. Mirkarimi, C. Larson, and J. Bokor, "Characterization of extreme ultraviolet lithography mask defects by actinic inspection with broadband extreme ultraviolet illumination," *Journal of Vacuum Science & Technology B*, vol. 20, pp. 3000-3005, Nov-Dec 2002.
177. P. P. Naulleau, K. A. Goldberg, E. H. Anderson, D. T. Attwood, P. J. Batson, J. Bokor, P. Denham, E. M. Gullikson, B. D. Harteneck, B. Hoef, K. H. Jackson, D. L. Olynick, S. Rekawa, F. Salmassi, K. L. Blaedel, H. N. Chapman, L. C. Hale, R. Soufli, E. A. Spiller, D. W. Sweeney, J. R. Taylor, C. C. Walton, A. Ray-Chaudhuri, D. O'Connell, R. Stulen, D. Tichenor, C. W. Gwyn, Y. Pei-Yang, and Z. Guojing, "Static microfield printing at the Advanced Light Source with the ETS Set-2 optic," *Proceedings of the SPIE*, vol. 4688, pp. 64-71, 2002.
178. P. Naulleau, K. A. Goldberg, E. H. Anderson, D. Attwood, P. Batson, J. Bokor, P. Denham, E. Gullikson, B. Harteneck, B. Hoef, K. Jackson, D. Olynick, S. Rekawa, F. Salmassi, K. Blaedel, H. Chapman, L. Hale, P. Mirkarimi, R. Soufli, E. Spiller, D. Sweeney, J. Taylor, C. Walton, D. O'Connell, D. Tichenor, C. W. Gwyn, P. Y. Yan, and G. J. Zhang, "Sub-70 nm extreme ultraviolet lithography at the Advanced Light Source static microfield exposure station using the engineering test stand set-2 optic," *Journal of Vacuum Science & Technology B*, vol. 20, pp. 2829-2833, Nov-Dec 2002.
179. Y. Moonsuk, P. Mincheol, P. B. Mirkarimi, C. C. Larson, and J. Bokor, "At-wavelength inspection of defect smoothing in EUVL masks," *Proceedings of the SPIE*, vol. 4688, pp. 395-400, 2002.
180. J. S. Lee, Y. K. Choi, D. Ha, T. J. King, and J. Bokor, "Low-frequency noise characteristics in p-channel FinFETs," *Ieee Electron Device Letters*, vol. 23, pp. 722-724, Dec 2002.
181. K. A. Goldberg, P. Naulleau, J. Bokor, H. N. Chapman, and A. Barty, "Testing extreme ultraviolet optics with visible-light and extreme ultraviolet interferometry,"

Curriculum Vitae - Jeffrey Bokor

- Journal of Vacuum Science & Technology B*, vol. 20, pp. 2834-2839, Nov-Dec 2002.
182. K. A. Goldberg, P. Naulleau, J. Bokor, and H. N. Chapman, "Honing the accuracy of extreme-ultraviolet optical system testing: at-wavelength and visible-light measurements of the ETS Set-2 projection optic," *Proceedings of the SPIE*, vol. 4688, pp. 329-37, 2002.
 183. K. A. Goldberg, P. Naulleau, and J. Bokor, "Fourier transform interferometer alignment method," *Applied Optics*, vol. 41, pp. 4477-4483, Aug 2002.
 184. M. S. Yi, T. Haga, C. Walton, and J. Bokor, "High sensitivity actinic detection of native defects on extreme ultraviolet lithography mask blanks," *Journal of Vacuum Science & Technology B*, vol. 19, pp. 2401-2405, Nov-Dec 2001.
 185. M. D. Shumway, L. Sang Hun, C. Chang Hyun, P. Naulleau, K. A. Goldberg, and J. Bokor, "Extremely fine-pitch printing with a 10 $\times$ Schwarzschild optic at extreme ultraviolet wavelengths," *Proceedings of the SPIE*, vol. 4343, pp. 357-62, 2001.
 186. P. P. Naulleau, E. H. Anderson, E. M. Gullikson, and J. Bokor, "Fabrication of high-efficiency multilayer-coated binary blazed gratings in the EUV regime," *Optics Communications*, vol. 200, pp. 27-34, Dec 2001.
 187. P. Naulleau, K. A. Goldberg, E. H. Anderson, P. Batson, P. E. Denham, K. H. Jackson, E. M. Gullikson, S. Rekawa, and J. Bokor, "At-wavelength characterization of the extreme ultraviolet Engineering Test Stand Set-2 optic," *Journal of Vacuum Science & Technology B*, vol. 19, pp. 2396-2400, Nov-Dec 2001.
 188. N. Lindert, L. L. Chang, Y. K. Choi, E. H. Anderson, W. C. Lee, T. J. King, J. Bokor, and C. M. Hu, "Sub-60-nm quasi-planar FinFETs fabricated using a simplified process," *Ieee Electron Device Letters*, vol. 22, pp. 487-489, Oct 2001.
 189. S. H. Lee, P. Naulleau, K. A. Goldberg, C. H. Cho, S. Jeong, and J. Bokor, "Extreme-ultraviolet lensless Fourier-transform holography," *Applied Optics*, vol. 40, pp. 2655-2661, Jun 2001.
 190. X. J. Huang, W. C. Lee, C. Kuo, D. Hisamoto, L. L. Chang, J. Kedzierski, E. Anderson, H. Takeuchi, Y. K. Choi, K. Asano, V. Subramanian, T. J. King, J. Bokor, and C. M. Hu, "Sub-50 nm p-channel FinFET," *Ieee Transactions on Electron Devices*, vol. 48, pp. 880-886, May 2001.
 191. K. A. Goldberg and J. Bokor, "Fourier-transform method of phase-shift determination," *Applied Optics*, vol. 40, pp. 2886-2894, Jun 2001.
 192. H. N. Chapman, A. K. Ray-Chaudhuri, D. A. Tichenor, W. C. Replogle, R. H. Stulen, G. D. Kubiak, P. D. Rockett, L. E. Klebanoff, D. O'Connell, A. H. Leung, K. L. Jefferson, J. B. Wronosky, J. S. Taylor, L. C. Hale, K. Blaedel, E. A. Spiller, G. E. Sommargren, J. A. Folta, D. W. Sweeney, E. M. Gullikson, P. Naulleau, K. A. Goldberg, J. Bokor, D. T. Attwood, U. Mickan, R. Hanzen, E. Panning, P. Y. Yan, C. W. Gwyn, and S. H. Lee, "First lithographic results from the extreme ultraviolet Engineering Test Stand," *Journal of Vacuum Science & Technology B*, vol. 19, pp. 2389-2395, Nov-Dec 2001.
 193. M. Yi, S. Jeong, S. Rekawa, and J. Bokor, "Characterization of extreme ultraviolet lithography mask defects from extreme ultraviolet far-field scattering patterns,"

Curriculum Vitae - Jeffrey Bokor

- Journal of Vacuum Science & Technology B*, vol. 18, pp. 2930-2934, Nov-Dec 2000.
194. Y. C. Yeo, V. Subramanian, J. Kedzierski, T. J. King, J. Bokor, C. M. Hu, and P. Q. Xuan, "Nanoscale ultra-thin-body silicon-on-insulator P-MOSFET with a SiGe/Si heterostructure channel," *Ieee Electron Device Letters*, vol. 21, pp. 161-163, Apr 2000.
 195. J. Seongtae, L. Chih-Wei, S. Rekawa, C. C. Walton, and J. Bokor, "Actinic defect counting statistics over 1 cm² area of EUVL mask blank," *Proceedings of the SPIE*, vol. 3997, pp. 431-40, 2000.
 196. L. Sang Hun, F. Piao, P. Naulleau, K. A. Goldberg, W. Oldham, and J. Bokor, "At-wavelength characterization of DUV-radiation-induced damage in fused silica," *Proceedings of the SPIE*, vol. 3998, pp. 724-31, 2000.
 197. L. Sang Hun, P. Naulleau, K. A. Goldberg, C. Chang Hyun, and J. Bokor, "EUV holographic aerial image recording," *Proceedings of the SPIE*, vol. 3997, pp. 823-8., 2000.
 198. P. P. Naulleau, C. H. Cho, E. M. Gullikson, and J. Bokor, "Transmission phase gratings for EUV interferometry," *Journal of Synchrotron Radiation*, vol. 7, pp. 405-410, Nov 2000.
 199. P. Naulleau, K. A. Goldberg, E. M. Gullikson, and J. Bokor, "At-wavelength, system-level flare characterization of extreme-ultraviolet optical systems," *Applied Optics*, vol. 39, pp. 2941-2947, Jun 2000.
 200. P. P. Naulleau, K. A. Goldberg, and J. Bokor, "Extreme ultraviolet carrier-frequency shearing interferometry of a lithographic four-mirror optical system," *Journal of Vacuum Science & Technology B*, vol. 18, pp. 2939-2943, Nov-Dec 2000.
 201. S. H. Lee, J. Bokor, P. Naulleau, S. T. Jeong, and K. A. Goldberg, "Extreme ultraviolet holographic microscopy and its application to extreme ultraviolet mask-blank defect characterization," *Journal of Vacuum Science & Technology B*, vol. 18, pp. 2935-2938, Nov-Dec 2000.
 202. J. Kedzierski, P. Q. Xuan, V. Subramanian, J. Bokor, T. J. King, C. M. Hu, and E. Anderson, "A 20 nm gate-length ultra-thin body p-MOSFET with silicide source/drain," *Superlattices and Microstructures*, vol. 28, pp. 445-452, Nov-Dec 2000.
 203. J. Kedzierski, X. Peiqi, V. Subramanian, J. Bokor, K. Tsu-Jae, H. Chenming, and E. Anderson, "A 20 nm gate-length ultra-thin body p-MOSFET with silicide source/drain," *Superlattices and Microstructures*, vol. 28, pp. 445-52, 2000.
 204. J. Kedzierski, E. Anderson, and J. Bokor, "Calixarene G-line double resist process with 15 nm resolution aid large area exposure capability," *Journal of Vacuum Science & Technology B*, vol. 18, pp. 3428-3430, Nov-Dec 2000.
 205. Hisamoto, W. C. Lee, J. Kedzierski, H. Takeuchi, K. Asano, C. Kuo, E. Anderson, T. J. King, J. Bokor, and C. M. Hu, "FinFET - A self-aligned double-gate MOSFET scalable to 20 nm," *Ieee Transactions on Electron Devices*, vol. 47, pp. 2320-2325, Dec 2000.
 206. K. A. Goldberg, P. Naulleau, P. Batson, P. Denham, E. H. Anderson, H. Chapman, and J. Bokor, "Extreme ultraviolet alignment and testing of a four-mirror ring field

Curriculum Vitae - Jeffrey Bokor

- extreme ultraviolet optical system," *Journal of Vacuum Science & Technology B*, vol. 18, pp. 2911-2915, Nov-Dec 2000.
207. K. A. Golberg, P. Naulleau, P. J. Batson, P. Denham, E. H. Anderson, J. Bokor, and H. N. Chapman, "EUV interferometry of a four-mirror ring field," *Proceedings of the SPIE*, vol. 3997, pp. 867-73, 2000.
208. Y. K. Choi, K. Asano, N. Lindert, V. Subramanian, T. J. King, J. Bokor, and C. M. Hu, "Ultrathin-body SOI MOSFET for deep-sub-tenth micron era," *Ieee Electron Device Letters*, vol. 21, pp. 254-255, May 2000.
209. N. W. Pu, J. Bokor, S. T. Jeong, and R. A. Zhao, "Nondestructive picosecond-ultrasonic characterization of Mo/Si extreme ultraviolet multilayer reflection coatings," *Journal of Vacuum Science & Technology B*, vol. 17, pp. 3014-3018, Nov-Dec 1999.
210. N. W. Pu, J. Bokor, S. Jeong, and R. A. Zhao, "Picosecond ultrasonic study of Mo/Si multilayer structures using an alternating-pump technique," *Applied Physics Letters*, vol. 74, pp. 320-322, Jan 1999.
211. P. Nen-Wen, J. Seongtae, Z. Ri-An, and J. Bokor, "Picosecond ultrasonic characterization of Mo/Si multilayers for extreme ultraviolet lithography," *Proceedings of the SPIE*, vol. 3676, pp. 627-34, 1999.
212. P. Nen-Wen, J. Seongtae, Z. Ri-An, and J. Bokor, "Nondestructive picosecond ultrasonic characterization of Mo/Si multilayers using a novel pump-probe scheme," *Proceedings of the SPIE*, vol. 3749, pp. 196-7, 1999.
213. P. P. Naulleau, K. A. Goldberg, S. H. Lee, C. Chang, D. Attwood, and J. Bokor, "Extreme-ultraviolet phase-shifting point-diffraction interferometer: a wave-front metrology tool with subangstrom reference-wave accuracy," *Applied Optics*, vol. 38, pp. 7252-7263, Dec 1999.
214. P. Naulleau, K. A. Goldberg, S. H. Lee, C. Chang, P. Batson, D. Attwood, and J. Bokor, "Recent advances in EUV phase-shifting point diffraction interferometry," *Proceedings of the SPIE*, vol. 3767, pp. 154-63, 1999.
215. P. Naulleau, K. A. Goldberg, E. M. Gullikson, and J. Bokor, "Interferometric at-wavelength flare characterization of extreme ultraviolet optical systems," *Journal of Vacuum Science & Technology B*, vol. 17, pp. 2987-2991, Nov-Dec 1999.
216. J. Kedzierski, J. Bokor, and E. Anderson, "Novel method for silicon quantum wire transistor fabrication," *Journal of Vacuum Science & Technology B*, vol. 17, pp. 3244-3247, Nov-Dec 1999.
217. S. T. Jeong, L. Johnson, S. Rekawa, C. C. Walton, S. T. Prisbrey, E. Tejn timer, J. H. Underwood, and J. Bokor, "Actinic detection of sub-100 nm defects on extreme ultraviolet lithography mask blanks," *Journal of Vacuum Science & Technology B*, vol. 17, pp. 3009-3013, Nov-Dec 1999.
218. S. Jeong, C. W. Lai, S. Rekawa, C. C. Walton, S. T. Prisbrey, and J. Bokor, "Cross correlation between actinic and visible defect inspection tool for extreme ultraviolet lithography," *Proceedings of the SPIE*, vol. 3873, pp. 814-21, 1999.
219. S. Jeong, L. Johnson, L. Yun, S. Rekawa, Y. Pei-Yang, P. A. Kearney, E. Tejn timer, J. H. Underwood, and J. Bokor, "Actinic EUVL mask blank defect inspection system," *Proceedings of the SPIE*, vol. 3676, pp. 298-308, 1999.
220. S. Jeong and J. Bokor, "Ultrafast carrier dynamics near the Si(100)2x1 surface," *Physical Review B*, vol. 59, pp. 4943-4951, Feb 1999.

Curriculum Vitae - Jeffrey Bokor

221. M. Gullikson, S. Baker, J. E. Bjorkholm, J. Bokor, K. A. Goldberg, J. E. M. Goldsmith, C. Montcalm, P. Naulleau, E. Spiller, D. G. Stearns, J. S. Taylor, and J. H. Underwood, "EUV scattering and flare of 10x projection cameras," *Proceedings of the SPIE*, vol. 3676, pp. 717-23, 1999.
222. K. A. Goldberg, P. Naulleau, L. SangHun, C. Chang, C. Bresloff, R. Gaughan, H. N. Chapman, J. Goldsmith, and J. Bokor, "Direct comparison of EUV and visible-light interferometries," *Proceedings of the SPIE*, vol. 3676, pp. 635-42, 1999.
223. K. A. Goldberg, P. Naulleau, and J. Bokor, "Extreme ultraviolet interferometric measurements of diffraction-limited optics," *Journal of Vacuum Science & Technology B*, vol. 17, pp. 2982-2986, Nov-Dec 1999.
224. T. Attwood, P. Naulleau, K. A. Goldberg, E. Tejnil, C. Chang, R. Beguiristain, P. Batson, J. Bokor, E. M. Gullikson, M. Koike, H. Medeck, and J. H. Underwood, "Tunable coherent radiation in the soft X-ray and extreme ultraviolet spectral regions," *Ieee Journal of Quantum Electronics*, vol. 35, pp. 709-720, May 1999.
225. Tejnil, K. A. Goldberg, and J. Bokor, "Phase effects owing to multilayer coatings in a two-mirror extreme-ultraviolet Schwarzschild objective," *Applied Optics*, vol. 37, pp. 8021-8029, Dec 1998.
226. J. Seongtae, M. Idir, L. Johnson, L. Yun, P. Batson, R. Levesque, P. Kearney, Y. Pei-Yang, E. Gullikson, J. H. Underwood, and J. Bokor, "Actinic detection of EUVL mask blank defects," *Proceedings of the SPIE*, vol. 3546, pp. 524-30, 1998.
227. S. T. Jeong, M. Idir, Y. Lin, L. Johnson, S. Rekawa, M. Jones, P. Denham, P. Batson, R. Levesque, P. Kearney, P. Y. Yan, E. Gullikson, J. H. Underwood, and J. Bokor, "At-wavelength detection of extreme ultraviolet lithography mask blank defects," *Journal of Vacuum Science & Technology B*, vol. 16, pp. 3430-3434, Nov-Dec 1998.
228. K. A. Goldberg, P. Naulleau, S. Lee, C. Bresloff, C. Henderson, D. Attwood, and J. Bokor, "High-accuracy interferometry of extreme ultraviolet lithographic optical systems," *Journal of Vacuum Science & Technology B*, vol. 16, pp. 3435-3439, Nov-Dec 1998.
229. Budiarto, N. W. Pu, S. Jeong, and J. Bokor, "Near-field propagation of terahertz pulses from a large-aperture antenna," *Optics Letters*, vol. 23, pp. 213-215, Feb 1998.
230. Tejnil, K. A. Goldberg, S. H. Lee, H. Medeck, P. J. Batson, P. E. Denham, A. A. MacDowell, J. Bokor, and D. Attwood, "At-wavelength interferometry for extreme ultraviolet lithography," *Journal of Vacuum Science & Technology B*, vol. 15, pp. 2455-2461, Nov-Dec 1997.
231. Sinitsky, R. Tu, C. L. Liang, M. S. Chan, J. Bokor, and C. M. Hu, "AC output conductance of SOI MOSFET's and impact on analog applications," *Ieee Electron Device Letters*, vol. 18, pp. 36-38, Feb 1997.
232. Sinitsky, F. Assaderaghi, M. Orshansky, J. Bokor, and C. Hu, "Velocity overshoot of electrons and holes in Si inversion layers," *Solid-State Electronics*, vol. 41, pp. 1119-1125, Aug 1997.
233. Sinitsky, F. Assaderaghi, C. M. Hu, and J. Bokor, "High field hole velocity and velocity overshoot in silicon inversion layers," *Ieee Electron Device Letters*, vol. 18, pp. 54-56, Feb 1997.

Curriculum Vitae - Jeffrey Bokor

234. J. Orenstein, J. Bokor, E. Budiarto, J. Corson, R. Mallozzi, I. Bozovic, and J. N. Eckstein, "Nonlinear electrodynamics in cuprate superconductors," *Physica C*, vol. 282, pp. 252-255, Aug 1997.
235. Y. Lin and J. Bokor, "Minimum critical defects in extreme-ultraviolet lithography masks," *Journal of Vacuum Science & Technology B*, vol. 15, pp. 2467-2470, Nov-Dec 1997.
236. J. Kedzierski, J. Bokor, and C. Kisielowski, "Fabrication of planar silicon nanowires on silicon-on-insulator using stress limited oxidation," *Journal of Vacuum Science & Technology B*, vol. 15, pp. 2825-2828, Nov-Dec 1997.
237. K. A. Goldberg, E. Tejnil, L. Sang Hun, H. Medeck, D. T. Attwood, K. H. Jackson, and J. Bokor, "Characterization of an EUV Schwarzschild objective using phase-shifting point diffraction interferometry," *Proceedings of the SPIE*, vol. 3048, pp. 264-70, 1997.
238. Assaderaghi, D. Sinitsky, S. A. Parke, J. Bokor, P. K. Ko, and C. M. Hu, "Dynamic threshold-voltage MOSFET (DTMOS) for ultra-low voltage VLSI," *Ieee Transactions on Electron Devices*, vol. 44, pp. 414-422, Mar 1997.
239. Assaderaghi, D. Sinitsky, J. Bokor, P. K. Ko, H. Gaw, and C. M. Hu, "High-field transport of inversion-layer electrons and holes including velocity overshoot," *Ieee Transactions on Electron Devices*, vol. 44, pp. 664-671, Apr 1997.
240. N. C. Tien, S. Jeong, L. M. Phinney, K. Fushinobu, and J. Bokor, "Surface adhesion reduction in silicon microstructures using femtosecond laser pulses," *Applied Physics Letters*, vol. 68, pp. 197-199, Jan 1996.
241. J. H. Son, S. T. Jeong, and J. Bokor, "Noncontact probing of metal-oxide-semiconductor inversion layer mobility," *Applied Physics Letters*, vol. 69, pp. 1779-1780, Sep 1996.
242. K. B. Nguyen, G. F. Cardinale, D. A. Tichenor, G. D. Kubiak, K. Berger, A. K. RayChaudhuri, Y. Perras, S. J. Haney, R. Nissen, K. Krenz, R. H. Stulen, H. Fujioka, C. Hu, J. Bokor, D. M. Tennant, and L. A. Fetter, "Fabrication of metal-oxide-semiconductor devices with extreme ultraviolet lithography," *Journal of Vacuum Science & Technology B*, vol. 14, pp. 4188-4192, Nov-Dec 1996.
243. Medeck, E. Tejnil, K. A. Goldberg, and J. Bokor, "Phase-shifting point diffraction interferometer," *Optics Letters*, vol. 21, pp. 1526-1528, Oct 1996.
244. S. Jeong, H. Zacharias, and J. Bokor, "Ultrafast carrier dynamics on the Si(100)2x1 surface," *Physical Review B*, vol. 54, pp. 17300-17303, Dec 1996.
245. E. Budiarto, J. Margolies, S. Jeong, J. Son, and J. Bokor, "High-intensity terahertz pulses at 1-kHz repetition rate," *Ieee Journal of Quantum Electronics*, vol. 32, pp. 1839-1846, Oct 1996.
246. Bokor, A. R. Neureuther, and W. G. Oldham, "Advanced lithography for ULSI," *Ieee Circuits and Devices Magazine*, vol. 12, pp. 11-15, Jan 1996.
247. R. Beguiristain, K. A. Goldberg, E. Tejnil, J. Bokor, H. Medeck, D. T. Attwood, and K. Jackson, "Interferometry using undulator sources," *Review of Scientific Instruments*, vol. 67, p. 1, 1996.
248. A. Goldberg, R. Beguiristain, J. Bokor, H. Medeck, K. Jackson, D. T. Attwood, G. E. Sommargren, J. P. Spallas, and R. Hostetler, "At-wavelength testing of optics for EUV," *Proceedings of the SPIE*, vol. 2437, pp. 347-54, 1995.

Curriculum Vitae - Jeffrey Bokor

249. A. Goldberg, R. Beguiristain, J. Bokor, H. Medeck, D. T. Attwood, K. Jackson, E. Tejnil, and G. E. Sommargren, "Progress towards lambda/20 extreme ultraviolet interferometry," *Journal of Vacuum Science & Technology B*, vol. 13, pp. 2923-2927, Nov-Dec 1995.
250. G. Park, C. A. King, D. J. Eaglesham, T. W. Sorsch, B. Weir, H. S. Luftman, J. Bokor, and Y. O. Kim, "Ultraslow p⁺-n junctions formed by diffusion from an RTCVD-deposited B:Ge layer," *Proceedings of the SPIE*, vol. 2091, pp. 122-31, 1994.
251. Assaderaghi, S. Parke, D. Sinitsky, J. Bokor, P. K. Ko, and C. M. Hu, "A Dynamic Threshold Voltage Mosfet (Dtmos) for Very-Low Voltage Operation," *Ieee Electron Device Letters*, vol. 15, pp. 510-512, Dec 1994.
252. A. Tichenor, G. D. Kubiak, M. E. Malinowski, R. H. Stulen, S. J. Haney, K. W. Berger, L. A. Brown, W. C. Sweatt, J. E. Bjorkholm, R. R. Freeman, M. D. Himel, A. A. Macdowell, D. M. Tennant, O. R. Wood, J. Bokor, T. E. Jewell, W. M. Mansfield, W. K. Waskiewicz, D. L. White, and D. L. Windt, "Soft-X-Ray Projection Lithography Experiments Using Schwarzschild Imaging Optics," *Applied Optics*, vol. 32, pp. 7068-7071, Dec 1993.
253. H. Sher, U. Mohideen, H. W. K. Tom, O. R. Wood, II, G. D. Aumiller, T. J. McIlrath, J. Bokor, R. R. Freeman, and J. Sugar, "Soft X-ray pulse length measurement by pump-probe absorption spectroscopy," *Proceedings of the SPIE*, vol. 1860, pp. 112-22, 1993.
254. M. Murnane, H. C. Kapteyn, S. P. Gordon, J. Bokor, E. N. Glytsis, and R. W. Falcone, "Efficient Coupling of High-Intensity Subpicosecond Laser-Pulses into Solids," *Applied Physics Letters*, vol. 62, pp. 1068-1070, Mar 1993.
255. U. Mohideen, M. H. Sher, H. W. K. Tom, G. D. Aumiller, O. R. Wood, R. R. Freeman, J. Bokor, and P. H. Bucksbaum, "High-Intensity above-Threshold Ionization of He," *Physical Review Letters*, vol. 71, pp. 509-512, Jul 1993.
256. A. Macdowell, J. E. Bjorkholm, K. Early, R. R. Freeman, M. D. Himel, P. P. Mulgrew, L. H. Szeto, D. W. Taylor, D. M. Tennant, O. R. Wood, J. Bokor, L. Eichner, T. E. Jewell, W. K. Waskiewicz, D. L. White, D. L. Windt, R. M. Dsouza, W. T. Silfvast, and F. Zernike, "Soft-X-Ray Projection Imaging with a 1/1 Ring-Field Optic," *Applied Optics*, vol. 32, pp. 7072-7078, Dec 1993.
257. W. S. Fann, R. Storz, H. W. K. Tom, and J. Bokor, "Direct Measurement of Nonequilibrium Electron-Energy Distributions in Subpicosecond Laser-Heated Gold-Films," *Surface Science*, vol. 283, pp. 221-225, Mar 1993.
258. Attwood, G. Sommargren, R. Beguiristain, K. Nguyen, J. Bokor, N. Ceglio, K. Jackson, M. Koike, and J. Underwood, "Undulator Radiation for at-Wavelength Interferometry of Optics for Extreme-Ultraviolet Lithography," *Applied Optics*, vol. 32, pp. 7022-7031, Dec 1993.
259. R. H. Yan, K. F. Lee, D. Y. Jeon, Y. O. Kim, B. G. Park, M. R. Pinto, C. S. Rafferty, D. M. Tennant, E. H. Westerwick, G. M. Chin, M. D. Morris, K. Early, P. Mulgrew, W. M. Mansfield, R. K. Watts, A. M. Voshchenkov, J. Bokor, R. G. Swartz, and A. Ourmazd, "89-GHz Room-Temperature Silicon Mosfets," *Ieee Electron Device Letters*, vol. 13, pp. 256-258, May 1992.

Curriculum Vitae - Jeffrey Bokor

260. G. Park, J. Bokor, H. S. Luftman, C. S. Rafferty, and M. R. Pinto, "Ultrashallow Junctions for Ulsi Using as-2+ Implantation and Rapid Thermal Anneal," *Ieee Electron Device Letters*, vol. 13, pp. 507-509, Oct 1992.
261. U. Mohideen, H. W. K. Tom, R. R. Freeman, J. Bokor, and P. H. Bucksbaum, "Interaction of Free-Electrons with an Intense Focused Laser-Pulse in Gaussian and Conical Axicon Geometries," *Journal of the Optical Society of America B-Optical Physics*, vol. 9, pp. 2190-2195, Dec 1992.
262. A. Macdowell, J. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, J. Z. Pastalan, L. H. Szeto, D. M. Tennant, O. R. Wood, T. E. Jewell, W. M. Mansfield, W. K. Waskiewicz, D. L. White, and D. L. Windt, "Reduction Imaging with Soft X-Rays for Projection Lithography," *Review of Scientific Instruments*, vol. 63, pp. 737-740, Jan 1992.
263. W. S. Fann, R. Storz, H. W. K. Tom, and J. Bokor, "Direct Measurement of Nonequilibrium Electron-Energy Distributions in Subpicosecond Laser-Heated Gold-Films," *Physical Review Letters*, vol. 68, pp. 2834-2837, May 1992.
264. W. S. Fann, R. Storz, H. W. K. Tom, and J. Bokor, "Electron Thermalization in Gold," *Physical Review B*, vol. 46, pp. 13592-13595, Nov 1992.
265. L. White, J. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, T. E. Jewell, W. M. Mansfield, A. A. Macdowell, L. H. Szeto, D. W. Taylor, D. M. Tennant, W. K. Waskiewicz, D. L. Windt, and O. R. Wood, "Soft-X-Ray Projection Lithography," *Solid State Technology*, vol. 34, pp. 37-42, Jul 1991.
266. A. Tichenor, G. D. Kubiak, M. E. Malinowski, R. H. Stulen, S. J. Haney, K. W. Berger, L. A. Brown, R. R. Freeman, W. M. Mansfield, O. R. Wood, D. M. Tennant, J. E. Bjorkholm, A. A. Macdowell, J. Bokor, T. E. Jewell, D. L. White, D. L. Windt, and W. K. Waskiewicz, "Diffraction-Limited Soft-X-Ray Projection Imaging Using a Laser Plasma Source," *Optics Letters*, vol. 16, pp. 1557-1559, Oct 1991.
267. A. Macdowell, J. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, W. M. Mansfield, J. Pastalan, L. H. Szeto, D. M. Tennant, O. R. Wood, T. E. Jewell, W. K. Waskiewicz, D. L. White, D. L. Windt, W. T. Silfvast, and F. Zernike, "Soft-X-Ray Projection Lithography Using a 1-1 Ring Field Optical-System," *Journal of Vacuum Science & Technology B*, vol. 9, pp. 3193-3197, Nov-Dec 1991.
268. D. Kubiak, D. A. Tichenor, M. E. Malinowski, R. H. Stulen, S. J. Haney, K. W. Berger, L. A. Brown, J. E. Bjorkholm, R. R. Freeman, W. M. Mansfield, D. M. Tennant, O. R. Wood, J. Bokor, T. E. Jewell, D. L. White, D. L. Windt, and W. K. Waskiewicz, "Diffraction-Limited Soft-X-Ray Projection Lithography with a Laser Plasma Source," *Journal of Vacuum Science & Technology B*, vol. 9, pp. 3184-3188, Nov-Dec 1991.
269. W. S. Fann, R. Storz, and J. Bokor, "Observation of above-Threshold Multiphoton Photoelectric-Emission from Image-Potential Surface-States," *Physical Review B*, vol. 44, pp. 10980-10982, Nov 1991.
270. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, W. M. Mansfield, L. Szeto, D. W. Taylor, D. M. Tennant, O. R. Wood, T. E. Jewell, D. L. White, W. K. Waskiewicz, D. L. Windt, and A. A. MacDowell, "Soft X-ray projection lithography," *Optics & Photonics News*, vol. 2, pp. 27-30, 1991.

Curriculum Vitae - Jeffrey Bokor

271. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, J. Gregus, T. E. Jewell, W. M. Mansfield, A. A. Macdowell, M. L. Omalley, E. L. Raab, W. T. Silfvast, L. H. Szeto, D. M. Tennant, W. K. Waskiewicz, D. L. White, D. L. Windt, and O. R. Wood, "Experiments in Projection Lithography Using Soft X-Rays," *Microelectronic Engineering*, vol. 13, pp. 243-250, Mar 1991.
272. T. E. Jewell, M. M. Becker, J. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, W. M. Mansfield, A. A. MacDowell, M. L. O'Malley, E. L. Raab, W. T. Silfvast, L. H. Szeto, D. M. Tennant, W. K. Waskiewicz, D. L. White, D. L. Windt, O. R. Wood, II, and J. H. Bruning, "20:1 projection soft X-ray lithography using tri-level resist," *Proceedings of the SPIE*, vol. vol.1263, pp. 90-8, 1990.
273. E. Bjorkholm, J. Bokor, L. Eichner, R. R. Freeman, J. Gregus, T. E. Jewell, W. M. Mansfield, A. A. Macdowell, E. L. Raab, W. T. Silfvast, L. H. Szeto, D. M. Tennant, W. K. Waskiewicz, D. L. White, D. L. Windt, O. R. Wood, and J. H. Bruning, "Reduction Imaging at 14 Nm Using Multilayer-Coated Optics - Printing of Features Smaller Than 0.1-Mu-M," *Journal of Vacuum Science & Technology B*, vol. 8, pp. 1509-1513, Nov-Dec 1990.
274. J. Halas and J. Bokor, "Surface Recombination on the Si(111)2x1 Surface," *Physical Review Letters*, vol. 62, pp. 1679-1682, Apr 1989.
275. J. Bokor and N. J. Halas, "Time-Resolved Study of Silicon Surface Recombination," *Ieee Journal of Quantum Electronics*, vol. 25, pp. 2550-2555, Dec 1989.
276. J. Bokor, "Ultrafast Dynamics at Semiconductor and Metal-Surfaces," *Science*, vol. 246, pp. 1130-1134, Dec 1989.
277. S. L. Tang, J. Bokor, and R. H. Storz, "Direct Force Measurement in Scanning Tunneling Microscopy," *Applied Physics Letters*, vol. 52, pp. 188-190, Jan 1988.
278. J. Bokor, A. M. Johnson, W. M. Simpson, R. H. Storz, and P. R. Smith, "Coplanar Vacuum Photodiode for Measurement of Short-Wavelength Picosecond Pulses," *Applied Physics Letters*, vol. 53, pp. 2599-2601, Dec 1988.
279. R. Haight, J. Bokor, R. R. Freeman, and P. H. Bucksbaum, "Time and Angle Resolved Ultraviolet Photoemission Spectroscopy Studies of Single-Crystal Surface and Interfaces," *Journal of Vacuum Science & Technology a-Vacuum Surfaces and Films*, vol. 4, pp. 1481-1486, May-Jun 1986.
280. R. Haight and J. Bokor, "Direct Observation of Adsorbate-Induced Band-Gap States on Gaas(110)," *Physical Review Letters*, vol. 56, pp. 2846-2849, Jun 1986.
281. J. Bokor, R. Storz, R. R. Freeman, and P. H. Bucksbaum, "Picosecond Surface Electron Dynamics on Photoexcited Si(111) (2x1) Surfaces," *Physical Review Letters*, vol. 57, pp. 881-884, Aug 1986.
282. J. Bokor, A. M. Johnson, R. H. Storz, and W. M. Simpson, "High-Speed Circuit Measurements Using Photoemission Sampling," *Applied Physics Letters*, vol. 49, pp. 226-228, Jul 1986.
283. J. Bokor, R. R. Freeman, P. H. Bucksbaum, and R. Haight, "Time-Resolved Photoemission from Semiconductor Surfaces and Interfaces," *Journal of the Optical Society of America B-Optical Physics*, vol. 3, pp. P58-P58, Aug 1986.
284. Attwood and J. Bokor, "Coherent Shortwave Sources and Optics Advance toward Exciting Applications," *Laser Focus-Electro-Optics*, vol. 22, pp. 22-&, Jul 1986.

Curriculum Vitae - Jeffrey Bokor

285. R. Haight, J. Bokor, J. Stark, R. H. Storz, R. R. Freeman, and P. H. Bucksbaum, "Picosecond Time-Resolved Photoemission-Study of the Inp(110) Surface," *Physical Review Letters*, vol. 54, pp. 1302-1305, 1985.
286. W. E. Cooke, L. A. Bloomfield, R. R. Freeman, and J. Bokor, "Planetary Atoms," *Lecture Notes in Physics*, vol. 229, pp. 187-198, 1985.
287. J. Bokor, R. Haight, R. H. Storz, J. Stark, R. R. Freeman, and P. H. Bucksbaum, "Time-Resolved and Angle-Resolved Photoemission-Study of Inp(110)," *Physical Review B*, vol. 32, pp. 3669-3675, 1985.
288. J. Bokor, "Review of Coherent Xuv Sources - Lasers and Nonlinear Optics," *Journal of the Optical Society of America a-Optics Image Science and Vision*, vol. 2, pp. P28-P28, 1985.
289. R. M. Jopson, R. R. Freeman, W. E. Cooke, and J. Bokor, "2-Photon Spectroscopy of 7sn'd Autoionizing States of Barium," *Physical Review A*, vol. 29, pp. 3154-3158, 1984.
290. R. Haight, J. Bokor, R. H. Storz, J. B. Stark, P. H. Bucksbaum, and R. R. Freeman, "Photoemission Apparatus Using Xuv Harmonics of a Picosecond Krf Laser," *Proceedings of the Society of Photo-Optical Instrumentation Engineers*, vol. 476, pp. 61-64, 1984.
291. W. E. Cooke, R. M. Jopson, L. A. Bloomfield, R. R. Freeman, and J. Bokor, "Correlations in Highly Excited 2-Electron Atoms - Planetary Behavior," *Aip Conference Proceedings*, pp. 91-100, 1984.
292. H. Bucksbaum and J. Bokor, "Rapid melting and regrowth velocities in silicon heated by ultraviolet picosecond laser pulses," *Physical Review Letters*, vol. 53, pp. 182-5, 1984.
293. J. Bokor, R. Haight, R. H. Storz, J. Stark, R. R. Freeman, and P. H. Bucksbaum, "Photoemission-Studies of Surfaces Using Picosecond Pulses of Coherent Extreme-Ultraviolet Radiation," *Journal of the Optical Society of America B-Optical Physics*, vol. 1, pp. 529-529, 1984.
294. A. Bloomfield, R. R. Freeman, W. E. Cooke, and J. Bokor, "Angular-Momentum Dependence of Autoionization Rates in Doubly Excited Rydberg States of Ba," *Physical Review Letters*, vol. 53, pp. 2234-2237, 1984.
295. R. M. Jopson, R. R. Freeman, W. E. Cooke, and J. Bokor, "Electron Shake-up in 2-Photon Excitation of Core Electrons to Rydberg Autoionizing States," *Physical Review Letters*, vol. 51, pp. 1640-1643, 1983.
296. R. M. Jopson, S. Darack, R. R. Freeman, and J. Bokor, "Laser-Plasma-Induced Extreme-Ultraviolet Radiation from Liquid Mercury," *Optics Letters*, vol. 8, pp. 265-267, 1983.
297. R. W. Falcone and J. Bokor, "Dichroic Beam Splitter for Extreme-Ultraviolet and Visible Radiation," *Optics Letters*, vol. 8, pp. 21-23, 1983.
298. H. Bucksbaum and J. Bokor, "Measurement of fast melting and regrowth velocities in picosecond laser heated silicon," in *Laser-Solid Interactions and Transient Thermal Processing of Materials*: Elsevier, New York, NY, USA., 1983, pp. 51-6.
299. H. Bucksbaum and J. Bokor, "Time Resolved Amorphous-Silicon Formation from Laser Melted Liquid Silicon Films," *Aip Conference Proceedings*, pp. 279-287, 1983.

Curriculum Vitae - Jeffrey Bokor

300. J. Bokor, L. Eichner, R. H. Storz, P. H. Bucksbaum, and R. R. Freeman, "Wavelength Conversion with Excimer Lasers," *Aip Conference Proceedings*, pp. 143-150, 1983.
301. J. Bokor, P. H. Bucksbaum, and R. R. Freeman, "Generation of 35.5-Nm Coherent Radiation," *Optics Letters*, vol. 8, pp. 217-219, 1983.
302. J. Bokor, P. H. Bucksbaum, and D. H. Auston, "Measurement of Picosecond Ultraviolet-Laser Pulse-Widths Using an Electrical Auto-Correlator," *Applied Physics Letters*, vol. 42, pp. 342-344, 1983.
303. J. C. White, J. Bokor, and D. Henderson, "Optically Pumped Atomic Thulium Lasers," *Ieee Journal of Quantum Electronics*, vol. 18, pp. 320-322, 1982.
304. M. Jopson, R. R. Freeman, J. Bokor, and W. E. Cooke, "Planetary Atoms," *Journal of the Optical Society of America*, vol. 72, pp. 1835-1836, 1982.
305. M. Jopson, R. R. Freeman, and J. Bokor, "4-Wave Sum Frequency Generation Below 100 Nm Hg Vapor," *Applied Physics B-Photophysics and Laser Chemistry*, vol. 28, pp. 203-203, 1982.
306. R. M. Jopson, R. R. Freeman, and J. Bokor, "Laser-Produced Plasmas in Hg - a Source of Pulsed Xuv Radiation," *Aip Conference Proceedings*, pp. 303-304, 1982.
307. Heaven, T. A. Miller, R. R. Freeman, J. C. White, and J. Bokor, "2-Photon Absorption, Laser-Induced Fluorescence Detection of Cl Atoms," *Chemical Physics Letters*, vol. 86, pp. 458-462, 1982.
308. R. R. Freeman, R. M. Jopson, J. Bokor, and P. Bucksbaum, "Laser Generation of Light in the Extreme Ultraviolet and Soft-X-Ray Regime," *Aip Conference Proceedings*, pp. 491-494, 1982.
309. R. R. Freeman, R. M. Jopson, and J. Bokor, "Generation of Light Below 100nm in Hg Vapor," *Aip Conference Proceedings*, pp. 422-430, 1982.
310. R. R. Freeman, J. Bokor, and W. E. Cooke, "Four-wave parametric mixing in optically inverted barium ions," *Physical Review A (General Physics)*, vol. 26, pp. 3029-30, 1982.
311. R. R. Freeman, J. Bokor, and W. E. Cooke, "4-Wave Parametric Mixing in Optically Inverted Barium Ions," *Physical Review A*, vol. 26, pp. 3029-3030, 1982.
312. P. H. Bucksbaum, J. Bokor, J. C. White, and R. H. Storz, "A Picosecond, Tunable Krf-Star Excimer Laser Source," *Applied Physics B-Photophysics and Laser Chemistry*, vol. 28, pp. 128-128, 1982.
313. J. Bokor, R. R. Freeman, and W. E. Cooke, "Autoionization-pumped laser," *Physical Review Letters*, vol. 48, pp. 1242-5, 1982.
314. J. Bokor, R. R. Freeman, and W. E. Cooke, "Selective Autoionization - a New Method for Producing Population-Inversion in Atomic Ions," *Applied Physics B-Photophysics and Laser Chemistry*, vol. 28, pp. 200-200, 1982.
315. J. Bokor, R. R. Freeman, and W. E. Cooke, "Photo-Autoionization Pumped Ba Ion Laser," *Aip Conference Proceedings*, pp. 153-162, 1982.
316. J. C. White, J. Bokor, R. R. Freeman, and D. Henderson, "Tunable Arf Excimer-Laser Source," *Optics Letters*, vol. 6, pp. 293-294, 1981.
317. J. C. White, J. Bokor, and R. R. Freeman, "Narrow-Linewidth Tunable Arfstar Excimer-Laser Source," *Ieee Journal of Quantum Electronics*, vol. 17, pp. 80-80, 1981.

Curriculum Vitae - Jeffrey Bokor

318. Rothschild, H. Egger, R. T. Hawkins, J. Bokor, H. Pummer, and C. K. Rhodes, "High-Resolution Spectroscopy of Molecular-Hydrogen in the Extreme Ultraviolet Region," *Physical Review A*, vol. 23, pp. 206-213, 1981.
319. J. Bokor, R. R. Freeman, J. C. White, and R. H. Storz, "2-Photon Excitation of the N=3 Level in H and D Atoms," *Ieee Journal of Quantum Electronics*, vol. 17, pp. 223-223, 1981.
320. J. Bokor, R. R. Freeman, and J. C. White, "Generation of Microjoule-Level Outputs in the Vuv through 4-Wave Parametric Interactions in Mercury," *Journal of the Optical Society of America*, vol. 71, pp. 368-368, 1981.
321. J. Bokor, R. R. Freeman, R. L. Panock, and J. C. White, "Generation of High-Brightness Coherent Radiation in the Vacuum Ultraviolet by 4-Wave Parametric Oscillation in Mercury-Vapor," *Optics Letters*, vol. 6, pp. 182-184, 1981.
322. J. Bokor, R. R. Freeman, R. L. Panock, and J. C. White, "Generation of High Brightness Coherent Radiation in the Vuv Via 4-Wave Parametric Oscillation in Mercury-Vapor," *Ieee Journal of Quantum Electronics*, vol. 17, pp. 10-12, 1981.
323. J. Kligler, J. Bokor, and C. K. Rhodes, "Collisional and Radiative Properties of the H-2e,F1-Sigma-G+ State," *Physical Review A*, vol. 21, pp. 607-617, 1980.
324. R. T. Hawkins, H. Egger, J. Bokor, and C. K. Rhodes, "A Tunable, Ultrahigh Spectral Brightness Krfstar Excimer Laser Source," *Applied Physics Letters*, vol. 36, pp. 391-392, 1980.
325. Egger, R. T. Hawkins, J. Bokor, H. Pummer, M. Rothschild, and C. K. Rhodes, "Tunable, Narrow Bandwidth 83-Nm Light-Source," *Journal of the Optical Society of America*, vol. 70, pp. 648-649, 1980.
326. Egger, R. T. Hawkins, J. Bokor, H. Pummer, M. Rothschild, and C. K. Rhodes, "Generation of High-Spectral-Brightness Tunable Xuv Radiation at 83 Nm," *Optics Letters*, vol. 5, pp. 282-284, 1980.
327. Bokor, J. Y. Zavelovich, and C. K. Rhodes, "Multi-Photon Ultraviolet Spectroscopy of Some 6p Levels in Krypton," *Physical Review A*, vol. 21, pp. 1453-1459, 1980.
328. Bokor, J. Zavelovich, and C. K. Rhodes, "Isotope Effect in Multi-Photon Ultraviolet Photolysis of Co," *Journal of Chemical Physics*, vol. 72, pp. 965-971, 1980.
329. Bokor and C. K. Rhodes, "Energy Splitting between the B-State and C-State of Xenon Chloride," *Journal of Chemical Physics*, vol. 73, pp. 2626-2628, 1980.
330. R. T. Hawkins, H. Egger, J. Bokor, and C. K. Rhodes, "Ultra-Narrow Linewidth Excimer Laser Sources," *Journal of the Optical Society of America*, vol. 69, pp. 1449-1449, 1979.
331. J. Eggleston, J. Dallarosa, W. K. Bischel, J. Bokor, and C. K. Rhodes, "Generation of 16-Mu-M Radiation in (Nh3)-N-14 by 2-Quantum Excitation of the 2-Nu-2-(7,5) State," *Journal of Applied Physics*, vol. 50, pp. 3867-3870, 1979.
332. J. Bokor, W. K. Bischel, and C. K. Rhodes, "Doppler-Free Spectroscopy of the Nu-2-Band in Nh3-N-14 - Application to 16-Mu-M Generation," *Journal of Applied Physics*, vol. 50, pp. 4541-4544, 1979.
333. W. K. Bischel, J. Bokor, D. J. Kligler, and C. K. Rhodes, "Non-Linear Optical Processes in Atoms and Molecules Using Rare-Gas Halide Lasers," *Ieee Journal of Quantum Electronics*, vol. 15, pp. 380-392, 1979.

Curriculum Vitae - Jeffrey Bokor

334. W. K. Bischel, J. Bokor, J. Dallarosa, and C. K. Rhodes, "Arf Laser Photolysis of Ocse .2. Effect of Vibrational-Excitation on Se(1s) Quantum Yields," *Journal of Chemical Physics*, vol. 70, pp. 5593-5597, 1979.
335. J. Bokor, W. K. Bischel, and C. K. Rhodes, "Optical Down Conversion of 10 Mu-M Radiation to 16 Mu-M Range," *Journal of the Optical Society of America*, vol. 68, pp. 671-672, 1978.

Curriculum Vitae - Jeffrey Bokor

Refereed Conference Papers

1. K. Jhuria, V. Ivanov, D. Polley, W. Liu, A. Persaud, Y. Zhiyenbayev, W. Redjem, W. Qarony, P. Parajuli, Q. Ji, A. Gonsalves, J. Bokor, L. Tan, B. Kante, and T. Schenkel, "Programmable quantum emitter formation in silicon," in Quantum Sensing, Imaging, and Precision Metrology II, SPIE 2024.
2. Z. Mutlu and J. Bokor, "Bottom-Up Synthesized Graphene Nanoribbon Transistors," in 2022 6th IEEE Electron Devices Technology & Manufacturing Conference (EDTM), 2022.
3. Z. Mutlu, Y. Lin, G. B. Barin, Z. Zhang, G. Pitner, S. Wang, R. Darawish, M. D. Giovannantonio, H. Wang, J. Cai, M. Passlack, C. H. Diaz, A. Narita, K. Mullen, F. R. Fischer, P. Bandaru, A. C. Kummel, P. Ruffieux, R. Fasel, and J. Bokor, "Short-Channel Double-Gate FETs with Atomically Precise Graphene Nanoribbons," in 2021 IEEE International Electron Devices Meeting (IEDM), 2021.
4. N. McIntyre, C. T. Amelink, and J. Bokor, "Career Development Impacts of a Research Program on Graduate Student and Postdoc Mentors," 2020 ASEE Virtual Annual Conference Virtual On line.
5. N. McIntyre, C. T. Amelink, and J. Bokor, "Positive impacts of a summer research program on graduate students and postdoctoral research mentors," in American Society for Engineering Education Annual Conference & Exposition, Montreal, Canada, 2020.
6. N. McIntyre, C. T. Amelink, and J. Bokor, "Positive Effects of Summer Research Program on Diverse Community College Students," in 2020 IEEE Global Engineering Education Conference (EDUCON), Porto, Portugal, 2020, pp. 465-471.
7. T. Schenkel, C. D. Weis, C. C. Lo, A. Persaud, I. Chakarov, D. H. Schneider, J. Bokor, A. Lapierre, S. Schwarz, and T. Baumann, "Deterministic Doping and the Exploration of Spin Qubits," in Proceedings of the Xii International Symposium on Electron Beam Ion Sources and Traps. vol. 1640, pp. 124-128, 2015.
8. B. Lambson, Z. Gu, J. Bokor, D. Carlton, S. Dhuey, A. Scholl, "Error immunity techniques for nanomagnetic logic," in *2012 IEEE International Electron Devices Meeting (IEDM)*, 2012, p. 4.
9. B. Lambson, J. Bokor, "Temperature dependence of heat dissipation during Landauer erasure of nanomagnets," *2012 12th IEEE Conference on Nanotechnology*, p. 3, 2012.
10. J. Bokor, B. Lambson, D. Carlton, Z. Gu, and J. Hong, "Exploring the thermodynamic limits of computation using nanomagnetic logic," in *Nanoenergy Letters* (<http://www.nanoenergyletters.com/node/189>), 2013, p. 23.
11. J. Bokor, "Prospects for emerging nanoelectronics in mainstream information processing systems," in IEEE/ACM International Conference on Computer Aided Design - Digest of Technical Papers. San Jose, CA, 2006.
12. X. Liu, A. San Paulo, M. Park, and J. Bokor, "Characterization of acoustic vibration modes at GHz frequencies in bulk acoustic wave resonators by combination of scanning laser interferometry and scanning acoustic force microscopy," *MEMS 2005 Technical Digest*, pp. 175-8., 2005.

Curriculum Vitae - Jeffrey Bokor

13. Yang-Kyu Choi, Daewon Ha, E. Snow, J. Bokor, and Tsu-Jae King, "Reliability study of CMOS FinFETs," in *International Electron Devices Meeting* 2003, pp. 7.6.1-4.
14. X. Peiqi, S. Min, B. Harteneck, A. Liddle, J. Bokor, and T. J. King, "FinFET SONOS flash memory for embedded applications," in *IEEE International Electron Devices Meeting* 2003, pp. 26.4.1-4.
15. C. Yang-Kyu, C. Leland, P. Ranade, L. Jeong-Soo, H. Daewon, S. Balasubramanian, A. Agarwal, M. Ameen, K. Tsu-Jae, and J. Bokor, "FinFET process refinements for improved mobility and gate work function engineering," in *International Electron Devices Meeting. Technical Digest*, 2002, pp. 259-62.
16. C. Yang-Kyu, N. Lindert, X. Peiqi, S. Tang, H. Daewon, E. Anderson, K. Tsu-Jae, J. Bokor, and H. Chenming, "Sub-20 nm CMOS FinFET technologies," in *International Electron Devices Meeting. Technical Digest*, 2001, pp. 19.1.1-4.
17. S. H. Tang, L. Chang, N. Lindert, C. Yang-Kyu, L. Wen-Chin, H. Xuejue, V. Subramanian, J. Bokor, K. Tsu-Jae, and H. Chenming, "FinFET-a quasi-planar double-gate MOSFET," in *2001 IEEE International Solid-State Circuits Conference. Digest of Technical Papers*, 2001, pp. 118-19.
18. Y. Moonsuk, T. Haga, C. Walton, and J. Bokor, "'Actinic-only' defects in EUVL mask blanks-native defects, barely detectable by visible-light inspection," in *2001 International Microprocesses and Nanotechnology Conference*, 2001, pp. 88-9.
19. N. Lindert, C. Yang-Kyu, C. Leland, E. Anderson, L. Wenchin, K. Tsu-Jae, J. Bokor, and H. Chenming, "Quasi-planar NMOS FinFETs with sub-100 nm gate lengths," in *Device Research Conference*, 2001, pp. 26-7.
20. N. Lindert, Y. K. Choi, L. Chang, E. Anderson, W. C. Lee, T. J. King, J. Bokor, and C. Hu, "Quasi-planar FinFETs with selectively grown germanium raised source/drain," in *IEEE International SOI Conference*, 2001, pp. 111-12.
21. J. Kedzierski, I. MeiKei, X. Peiqi, J. Bokor, K. Tsu-Jae, and H. Chenming, "Design analysis of thin-body silicide source/drain devices," in *IEEE International SOI Conference*, 2001, pp. 21-2.
22. S. H. Tang, X. Peiqi, J. Bokor, and H. Chenming, "Comparison of short-channel effect and offstate leakage in symmetric vs. asymmetric double gate MOSFETs," in *IEEE International SOI Conference*, 2000, pp. 120-1.
23. X. Peiqi, J. Kedzierski, V. Subramanian, J. Bokor, K. Tsu-Jae, and H. Chenming, "60 nm planarized ultra-thin body solid phase epitaxy MOSFETs," in *58th DRC. Device Research Conference*, 2000, pp. 67-8.
24. J. Kedzierski, P. Xuan, E. H. Anderson, J. Bokor, K. Tsu-Jae, and H. Chenming, "Complementary silicide source/drain thin-body MOSFETs for the 20 nm gate length regime," in *International Electron Devices Meeting*, 2000, pp. 57-60.
25. Y. -K. Choi, J. Yoo-Chan, P. Ranade, H. Takenuchi, Tsu-Jae King, J. Bokor, and H. C. Hu, "30 nm ultra-thin-body SOI MOSFET with selectively deposited Ge raised S/D," in *58th DRC. Device Research Conference*, 2000, pp. 23-4.
26. L. Chang, S. Tang, Tsu-JaeKing, J. Bokor, and C. Hu, "Gate length scaling and threshold voltage control of double-gate MOSFETs," in *International Electron Devices Meeting*, 2000, pp. 719-22.

Curriculum Vitae - Jeffrey Bokor

27. Yang-Kyu Choi, K. Asano, N. Lindert, V. Subramanian, Tsu-Jae King, J. Bokor, and C. Hu, "Ultra-thin body SOI MOSFET for deep-sub-tenth micron era," in *International Electron Devices Meeting* 1999, pp. 919-21.
 28. Xuejue Huang, Wen-Chin Lee, Charles Kuo, D. Hisamoto, Leland Chang, J. Kedzierski, E. Anderson, H. Takeuchi, Yang-Kyu Choi, K. Asano, V. Subramanian, Tsu-Jae King, J. Bokor, and C. Hu, "Sub 50-nm FinFET: PMOS," in *International Electron Devices Meeting*, 1999, pp. 67-70.
 29. V. Subramanian, J. Kedzierski, N. Lindert, H. Tam, Y. Su, J. McHale, K. Cao, T. J. King, J. Bokor, and C. Hu, "A bulk-Si-compatible ultrathin-body SOI technology for sub-100 nm MOSFETs," in *57th Annual Device Research Conference*, 1999, pp. 28-9.
 30. M. S. Krishnan, L. Chang, K. Tsu-Jae, J. Bokor, and C. Hu, "MOSFETs with 9 to 13 Å thick gate oxides," in *International Electron Devices Meeting*, 1999, pp. 241-4.
 31. M. Orshansky, D. Sinitsky, P. Scrobahaci, J. Bokor, and C. Hu, "Impact of velocity overshoot, polysilicon depletion, and inversion layer quantization on NMOSFET scaling," in *56th Annual Device Research Conference*, 1998, pp. 18-19.
 32. M. S. Krishnan, Yee Chia Yeo, Qiang Lu, Tsu-Jae King, J. Bokor, and C. Hu, "Remote charge scattering in MOSFETs with ultra-thin gate dielectrics," in *International Electron Devices Meeting*, 1998, pp. 571-4.
 33. D. Sinitsky, F. Assaderaghi, M. Orshansky, J. Bokor, and C. Hu, "An extension of BSIM3 model incorporating velocity overshoot," in *International Symposium on VLSI Technology, Systems, and Applications*, 1997, pp. 307-10.
 34. H. C. Wann, C. Hu, K. Noda, D. Sinitsky, F. Assaderaghi, and J. Bokor, "Channel doping engineering of MOSFET with adaptable threshold voltage using body effect for low voltage and low power applications," in *International Symposium on VLSI Technology, Systems, and Applications*, 1995, pp. 159-63.
 35. F. Assaderaghi, D. Sinitsky, S. Parke, J. Bokor, P. K. Ko, and C. Hu, "A dynamic threshold voltage MOSFET (DTMOS) for ultra-low voltage operation," in *International Electron Devices Meeting*, 1994, pp. 809-12.
 36. F. Assaderaghi, D. Sinitsky, H. Gaw, J. Bokor, P. K. Ko, and C. Hu, "Saturation velocity and velocity overshoot of inversion layer electrons and holes," in *International Electron Devices Meeting*, 1994, pp. 479-82.
 37. R. H. Yan, K. F. Lee, D. Y. Jeon, Y. O. Kim, B. G. Park, M. R. Pinto, C. S. Rafferty, D. M. Tennant, E. H. Westerwick, G. M. Chin, M. D. Morris, K. Early, P. Mulgrew, W. M. Mansfield, R. K. Watts, A. M. Voshchenkov, J. Bokor, R. G. Swartz, and A. Ourmazd, "High performance 0.1- μm room temperature Si MOSFETs," in *1992 Symposium on VLSI Technology*, 1992, pp. 86-7.
-

EXHIBIT D

Sanjay Banerjee is the Cockrell Family Regents Chair Professor of Electrical and Computer Engineering at the University of Texas at Austin. He received his B.Tech from the Indian Institute of Technology, Kharagpur, and his M.S. and Ph.D. from the University of Illinois at Urbana-Champaign in 1979, 1981 and 1983 respectively, all in electrical engineering. As a Member of the Technical Staff, Corporate Research, Development and Engineering of Texas Instruments Incorporated from 1983-1987, he worked on polysilicon transistors and dynamic random access trench memory cells used by Texas Instruments in the world's first 4Megabit DRAM. He has been Assistant Professor (1987-90), Associate Professor (1990-93), and Professor (1993-) at The University of Texas at Austin. He has over 1000 archival refereed publications/talks, 10 books/chapters, and 35 U.S. patents, and has supervised over 80 Ph.D. and 70 MS students. His students have received 15 Best Papers Awards at various conferences, and he has presented over 150 invited talks. He received the Engineering Foundation Advisory Council Halliburton Award, 1991, the Texas Atomic Energy Fellowship (1990-1997), Cullen Professorship (1997-2001) and the Hocott Research Award from UT Austin (2007). He has won the SIA/SRC University Researcher Award (2017), IEEE Grove Award (2014), Distinguished Alumnus Award, IIT (2005), Industrial R&D 100 Award (2004), ECS Callinan Award (2003), IEEE Millennium Medal (2000), NSF Presidential Young Investigator Award (1988), and several Best Paper Awards and SRC Inventor Recognition Awards. He was a Distinguished Lecturer for IEEE Electron Devices Society, and the General Chair of the IEEE Device Research Conference, 2002. He is a Fellow of IEEE, APS, AAAS and NAI. He is active in the areas of beyond-CMOS nanoelectronic transistors based on 2D materials and spintronics, fabrication and modeling of advanced MOSFETs, and solar cells.

Sanjay Kumar Banerjee

Current Position: Cockrell Family Regents Chair in Electrical and Computer Engineering, 1999-

Education: University of Illinois, PhD (Electrical Engineering), 1983
University of Illinois, MS (Electrical Engineering), 1981
Indian Institute of Technology at Kharagpur, India, B. Tech (Electronics), 1979

Professional Engineer: Texas

Previous Positions: Director, Microelectronics Research Center, 1999-2024
Cullen Trust Endowed Professorship in Engineering, 1997-2001
University of Texas, Assoc. Director, Microelectronics Research Center, 1996-99
University of Texas, Professor, September 1993-
University of Texas, Associate Professor, September 1990- August 1993
University of Texas, Assistant Professor, September 1987- August, 1990
Texas Instruments, Corporate R& D, Member of Technical Staff, 1983-Aug. 1987

Honors and Awards:

Fellow, National Academy of Inventors (2021)
SIA/SRC University Research Award (2017)
IEEE Andrew Grove Award (2014)
Fellow of American Association for Advancement of Science (2007)
Hocott Research Award, Univ. of Texas, 2007
Fellow, American Physical Society, 2006
Distinguished Alumnus Award, IIT, 2005
Industrial R&D 100 Award (with R.Singh) 2004
Electrochemical Society Thomas D. Callinan Award, 2003
IEEE Millennium Medal, 2001
SRC Inventor Recognition Awards, 1994, 2000, 2009
Best Paper Awards (15) @ ISSCC, SRC TECHCON, DARPA, DRC
Who's Who Listings (Marquis)
Cullen Professorship, Univ. of Texas, 1997- 2001
Distinguished Lecturer, IEEE Electron Devices Society (1997-2003), Adcom
Member till 1998
Fellow of IEEE, 1996
Engineering Foundation Advisory Council Halliburton Award, 1991
Texas Atomic Energy Centennial Fellowship, 1990-97
NSF Presidential Young Investigator Award, 1988
Jagadis Bose and National Science Talent Search Scholarships, India, 1974-79
Institute Medal; &Swapan Saha Prize for Highest Ranking UG (ECE), I.I.T., 1979
Phi Kappa Phi

Professional Society and Major Government Committees:

Technical Advisory Board: Applied Novel Devices (current) AstroWatt, DSM Semiconductors, Cambrios, Nanocoolers Inc., BeSang Memories, Organic ID and ITU Ventures; Gerson Lehmann Group, NY; Austin Community College; Asia Pacific IIIT; Rochester Institute of Technology, HSMC Foundry

IEEE Dan Noble Award Committee, 2010-13 (Chair, 2012-13)

Congressional round-table panel member on nanotechnology, Feb. 2008

Member on International Technology Roadmap for Semiconductors

Siemens Westinghouse Science Talent Contest Judge, 2003

Morgan & Claypool Publishers, Lectures in Electronic Materials & Devices, Series Editor

SISPAD, Program Committee, 2005-6

Electrochemical Society Symposium on SiGe, Program Committee, 2004

IRPS, Program Committee, 2005

12th Int. Workshop on Physics of Semiconductor Devices, Int. Advisory Committee

Int. Advisory Committee, Int. Conf. on MEMS and Nanotechnology, IIT, 2005

Program Committee, International SiGe Technology and Device Meeting, 2004-2012

IEEE Device Research Conference Technical Program Chair, 2000-01, General Chair, 2001-02

Editorial Board, Elsevier Science, 2001

IEDM Program Committee, Modeling and Simulation, Session Chair, 2001-03

ECS Session Chair, Toronto, Canada, May 2000

Program Committee, IEEE Int. Conf. Communications, Computers, Devices, Kharagpur, 2000.

IEEE Device Research Conference Program Committee/Local Arrangements Chair, 1999-2000

NSF Workshop Co-Organizer for "Front and Back-end Processes", Austin, TX 1999

Eleventh Int. Ion Implant Tech. Meet. Program Committee and Publications Chair, 1995-1996.

IEEE Symposium on VLSI Technology, Committee Member, 1992-98

NSF Workshop Organizer for "Silicon-Germanium Devices", Austin, TX 1999

IEEE University Government Industry Microelectronics Symp., General Chairman, 1994-1995

IEEE International Electron Devices Meeting, (Device Technology/ Session Chair: 1989-90)

IEEE Conf. on Electromagnetic Field Computation, Chair Comp. in Electron Dev., CA, 1992

Panel Member, SRC Conference on Integration of Novel Processes, 1991

Sponsored Research:

- Grant title:** "Three-Dimensional IC Technology,"
Co-Principal Investigator: S. Banerjee
Other Investigators: D.L. Kwong
Sponsoring Agency: Texas Advanced Technology Program
Duration: June 1988-August 1990.
- Grant title:** "High Speed Devices and VLSI Structures by Laser-Enhanced Epitaxy,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Advanced Technology Program
Duration: June 1988-August 1990.
- Grant title:** "Optoelectronic Devices by Photo-enhanced Chemical Vapor Deposition,"
Principal Investigator: S. Banerjee
Sponsoring Agency: National Science Foundation PYI
Duration: August 1988- July 1993.
- Grant title:** "GaAs-on-Si MESFET Modeling,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Instruments, Inc.
Duration: December 1988- August 1989.
- Grant title:** "Understanding and Modeling of Unit Processes"
Co-Investigator: S. Banerjee
Other Investigators: W. Adcock (PI), A. Tasch (Co-PI), I. Trachtenberg (Co-PI),
D. Kwong, J. Lee, T. Edgar and J. Ekerdt
Sponsoring Agency: SEMATECH and SRC
Duration: December 1988- August 1993.
- Grant title:** "RPCVD Epitaxial Silicon and Insulators for Use in 3-D CMOS Integrated Circuits,"
Co-Investigator: S. Banerjee
Other Investigators: A. Tasch (P.I.), A. Cowley and R. Jones
Sponsoring Agency: Office of Naval Research
Duration: Sept. 1987- March 1990.
- Grant title:** "Ballistic and Quantum Transport in Si Devices at Cryogenic Temperatures"
Principal Investigator: S. Banerjee
Other Investigators: J. Lee
Sponsoring Agency: Texas Advanced Technology Program
Duration: November 1989- November 1991.
- Grant title:** "Polysilicon Transistor Modeling,"
Principal Investigator: S. Banerjee
Sponsoring Agency: Motorola
Duration: September, 1991-August, 1993.

- Grant title:** "Acquisition of High Resolution Transmission Electron Microscope,"
Principal Investigator: L.Rabenberg
Other Investigators: S.Banerjee, J.Goodenough, A.Heller, P.Ho and A.Manthiram
Sponsoring Agency: National Science Foundation
Duration: 10/92-10/93
- Grant title:** "Atomic Layer Epitaxy of Group IV Semiconductors,"
Co-Principal Investigator: S.Banerjee
Other Investigators: A.Tasch (P.I.), A.Cowley, J.Ekerdt and R.Jones
Sponsoring Agency: Office of Naval Research
Duration: February 1991-August 1996.
- Grant title:** "Materials and Bulk Processes"
Co-Investigator: S.Banerjee
Other Investigators: A.Tasch (PI), D.Kwong, J.Lee
Sponsoring Agency: SRC/ SEMATECH
Duration: September 1993- August 1998.
- Grant title:** "Synthesis, Growth and Analysis of Electronic Materials,"
Co-Investigator: S.Banerjee
Other Investigators: J.White (P.I) and 11 others from ECE, Chemistry and Physics
Sponsoring Agency: National Science
Duration: March 1991- March, 1996.
- Grant title:** "Transport in MOSFETs"
Principal Investigator: S.Banerjee
Sponsoring Agency: Motorola
Duration: August, 1993-August, 1994.
- Grant title:** "Flash EEPROMs"
Principal Investigator: S.Banerjee
Sponsoring Agency: AMD
Duration: May, 1993-December, 1996.
- Grant title:** "LDO Thin Film Transistors"
Principal Investigator: S.Banerjee
Sponsoring Agency: Micron
Duration: March, 1993- April, 1995.
- Grant title:** "Ultra Shallow Junction Technology"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: January 1994- December 1996.

- Grant title:** "SIMS Analysis of Polysilicon-on-Silicon"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1994- August 1995.
- Grant title:** "RTP Implant Monitors"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1995- August 1996.
- Grant title:** "Ultra-shallow Junction Formation and 2-D Dopant Profiling"
Principal Investigator: S.Banerjee
Other Investigators: K.Shih
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 1996- December 1997.
- Grant title:** "Analysis of Deep Submicron MOSFETs"
Principal-Investigator: S.Banerjee
Other Investigators: A.Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1998- September 1999.
- Grant title:** "Unrestricted Grant"
Principal Investigator: S.Banerjee
Sponsoring Agency: Various Donors
Duration: No expiration
- Grant title:** "Synthesis, Growth and Analysis of Electronic Materials,"
Co-Investigator: S.Banerjee
Other Investigators: J.White (P.I) and 11 others from ECE, Chemistry and Physics
Sponsoring Agency: National Science Foundation STC
Duration: March 1996- February, 2002.
- Grant title:** "Ultra-shallow Junction Process Integration"
Principal Investigator: S.Banerjee
Sponsoring Agency: SEMATECH
Duration: September 1997- December 2001.
- Grant title:** "Si and Ge Thin Film CVD, Modeling and Control"
Co-Principal Investigator: S.Banerjee
Other Investigators: J.Ekerdt (P.I.), M.Downer, I.Trachtenberg; Univ. of Wisconsin
Sponsoring Agency: Dept. of Defense-MURI
Duration: July 1995-July 2000

- Grant title:** "Ultra-shallow Junction Technology"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 1998- August 2000.
- Grant title:** "Channel Engineering in Si-Ge-C MOSFETs "
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1997- September 2000.
- Grant title:** "Advanced Annealing"
Principal Investigator: S. Banerjee
Sponsoring Agency: Texas Higher Education Coordinating Board
Duration: January 2000- December 2001.
- Grant title:** "Quantum Transport in Heterostructure MOSFETs "
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: October 1999- September 2002.
- Grant title:** "Front End Processing"
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch, D. Kwong, J. Lee
Sponsoring Agency: SRC/ SEMATECH
Duration: April 1998- March 2001.
- Grant title:** "Vertical Si-Ge-C MOSFETs "
Principal-Investigator: S. Banerjee
Sponsoring Agency: Semiconductor Research Corporation
Duration: September 2000- August 2003.
- Grant title:** "Compact Modeling of Gate Current"
Principal-Investigator: S. Banerjee
Other Investigators: F. Register
Sponsoring Agency: Semiconductor Research Corporation
Duration: July 2000- September 2003.
- Grant title:** "Ion Implantation Modeling"
Principal-Investigator: S. Banerjee
Other Investigators: A. Tasch
Sponsoring Agency: Semiconductor Research Corporation
Duration: July 2000- September 2001.

Grant title: "Front End Processing"

Principal-Investigator: S.Banerjee

Other Investigators: D.Kwong, J.Lee, F.Register

Sponsoring Agency: SRC/ SEMATECH

Duration: April 2001- March 2003.

Grant title: "MARCO Focus Center on Device Structures"

PI at UT: S.Banerjee

Other Investigators: D.Kwong (with MIT, Stanford, UC Berkeley)

Sponsoring Agency: DARPA/SRC

Duration: Award announced February 2001 (3 year contract)

Grant title: "SiGe Flash EEPROMS with Quantum Dot Gates"

Principal Investigator: S.Banerjee

Sponsoring Agency: Texas Higher Education Coordinating Board

Duration: January 2002- December 2003.

Grant title: "MARCO Focus Center on Device Structures"

Principal-Investigator: S.Banerjee

Other Investigators: D.Kwong (with MIT, Stanford, UC Berkeley)

Sponsoring Agency: DARPA/SRC

Duration: Sept. 2003 (3 year contract)

Grant title: "High mobility Ge-channel MOSFETs "

Principal-Investigator: S.Banerjee

Sponsoring Agency: Semiconductor Research Corporation

Duration: September 2003-August 2006.

Grant title: "Monte Carlo and Quantum transport "

Principal-Investigator: S.Banerjee

Co-PI: L.F.Register

Sponsoring Agency: Semiconductor Research Corporation

Duration: September 2003-August 2006.

Grant title: "NIRT on Quantum Dot Memories "

Principal-Investigator: S.Banerjee

Other Investigators: J.Ekerdt, F.Register, G.Hwang

Sponsoring Agency: NSF

Duration: September 2003-August 2007.

Grant title: "High mobility Ge-channel MOSFETs "

Principal-Investigator: S.Banerjee

Sponsoring Agency: Texas Higher Education Coordinating Board

Duration: January 2004-Dec. 2005.

- Grant title:** "Advanced Materials Research Center"
Principal-Investigator: S.Banerjee
Other Investigators: 15 others
Sponsoring Agency: Texas
Duration: January 2004- Dec.2005
- Grant title:** "Advanced Processing and Prototyping Center"
Principal-Investigator: S.Banerjee
Other Investigators: 18 others
Sponsoring Agency: DARPA
Duration: 2005- Dec.2006
- Grant title:** "SiGe Nanostructures"
Co-Principal-Investigator: S.Banerjee,
Other Investigators: R.Huang
Sponsoring Agency: DOE
Duration: 2006- Dec.2009
- Grant title:** "Dopant Diffusion Modeling"
Principal-Investigator: S.Banerjee,
Other Investigators: G.Hwang
Sponsoring Agency: SRC
Duration: 2006- Dec.2009
- Grant title:** "NNIN "
Site Director: S.Banerjee
Sponsoring Agency: NSF
Duration: January 2004- August.2015
- Grant title:** "MARCO MSD Focus Center "
UT PI: S.Banerjee
Sponsoring Agency: DARPA/SRC
Duration: Sept. 2007-2012
- Grant title:** "CERA"
Principal-Investigator: S.Banerjee
Co-PIs: F.Register, R.Ruoff, E.Tutuc, A.Macdonald, D.Akinwande
Sponsoring Agency: DARPA/IBM
Duration: Sept. 2007-2012
- Grant title:** "NASCENT ERC "
Principal-Investigator: Bonnecaze, Sreenivasan
Banerjee (Device Thrust co-Leader)
Sponsoring Agency: NSF
Duration: January 2013- Feb.2018

- Grant title:** "Bay Area PV Consortium led by Stanford/Berkeley"
Sponsoring Agency: DOE
Duration: Sept. 2015- August.2017
- Grant title:** "South West Academy of Nanoelectronics"
Director: S.Banerjee,
Other Investigators: F.Register, A.MacDonald and 15 others from 6 schools
Sponsoring Agency: SRC-NRI
Duration: 2006- Dec.2018
- Grant title:** "NSF-NNCI "
Site Director: S.Banerjee
Sponsoring Agency: NSF
Duration: Sept. 2015- August.2025
- Grant title:** "MURI-Room Temperature Polariton Condensates "
Principal-Investigator: Hui Deng (Michigan),
Co-PIs: S.Banerjee and 4 others
Sponsoring Agency: DoD MURI
Duration: Sept. 2017- August.2023
- Grant title:** "Soft-FET "
PI: J.Kulkarni, S.Banerjee (co-PI)
Sponsoring Agency: NSF
Duration: Sept. 2018- August.2022
- Grant title:** "EFRI- sub-contract from UT Dallas"
PI: S.Banerjee
Sponsoring Agency: NSF
Duration: Sept. 2018- August.2022
- Grant title:** "3D CMOS using CVD TMD"
PI: S.Banerjee
Sponsoring Agency: SRC
Duration: Jan. 2021- Dec.2023
- Grant title:** "BEOL Transistor"
PI: S.Banerjee
Sponsoring Agency: TI, \$85k
Duration: June. 2024- Dec.2025
- Grant title:** "TMD FETs"
PI: S.Banerjee
Sponsoring Agency: Samsung, \$720k

Duration: Sept. 2023- Aug.2026

Grant title: "ALE Modeling"

PI: S.Banerjee; co-PI Gyeong Hwang (ChemE)

Sponsoring Agency: Samsung, \$660k

Duration: Sept. 2023- Aug.2026

Grant Title: " Applied Novel Devices- Power FETs"

PI: S.Banerjee. co-PIs, Alex Huang, Alex Hanson

Sponsoring Agency: Army, \$368,753

Duration: Aug. 2023- Aug.2025

Ph.D. supervision:

Keun Park, 1991
Ting Hsu, 1991
Sean Lian, 1991
Shubneesh Batra, 1992
Sittampalam Yoganathan, 1992
Burt Fowler, 1992
David Kinosky, 1993
Surya Bhattacharaya, 1993
Brain Li, 1994
Le-tien Jung, 1994
Avinash Mahajan, 1994
Chung-you Hu, 1995
Indrajit Manna, 1995
Akif Sultan, 1996
Dean Samara, 1997
Soji John, 1998
Jacob Liu, 1999
Rajan Sharma, 1999
Ed Quinones, 1999
David Kencke, 2000
Christine Ouyang (with Tasch), 2000
Xiangdong Chen, 2001
Taehoon Kim, 2001
Siva Mudanai (with Tasch), 2001
Geng Wang (with Tasch), 2001
Yang Chen (with Tasch), 2001
Xin Wang, 2002
Hong-Jyh Li, 2002
Sung –Bo Hwang (with Edgar), 2002
Di Li (with Tasch), 2002
Yang-Yu Fan (with Register) 2002
Zhonghai Shi, 2002
Tat Ngai, 2002
Dong-Won Kim, 2003
Xiao Chen (with Rabenberg), 2003
Puneet Kohli, 2003
David Onsongo, 2003
Kartik Jayanarayan, 2004
Tongsheng Xia, (with Register), 2005
Taras Kirichenko (with Hwang), 2005
James Chen, 2005
Swaroop Ganguly (with MacDonald), 2006
Fei Lei (with Register), 2006
Li Lin (2006)
Sagnik Dey (2006)
Xiangdong Fan (with Register) 2006
Yueran Liu, 2006
David Kelly, 2006
Xiaofeng Fan (with Register) 2006
Bahman Ghosh (with Register) 2007

Sachin Joshi, 2007
Joy Sarkar, 2007
Joseph Donnelly, 2009
Davood Shahrjerdi, 2008
Shan Tang, 2008
Rownak Zaman, 2008
Ning Kong, 2009
Yonghyun Kim, 2010
Hai Liu 2010
Dipanjan Basu (with Register), 2010
Tackhwi Lee, 2010
Se Hoon Lee, 2011
Ferdousi Fahmida, 2011
Jamil Mustafa, 2011
John David (with Register) 2011
Jung Hwan Yum, 2012
Seyoung Kim, (with Tutuc) 2012
Chang, Jiwon, (with Register) 2013
Priyamvada Jadaun, (with Register) 2013
Yujia Zhai, (with Willson) 2013
Michael Ramon (with Akinwande) 2013
Emmanuel Oneyagam, 2014
Sayan Saha, 2015
Urmimala Roy (with Register) 2015
Jason Mantey, 2015
Donghyi Koh (2016)
Chris Corbet (with Tutuc) 2016
Sangwoo Kang (2016)
William Hsu (2016)
Andreas Hsieh (2017)
Tanuj Trivedi (with Neikirk) (2017)
Atresh Sanne (2017)
Tanmoy Pramanik (with Register) (2018)
Jaehyun Ahn (2018)
Hema Movva (with Tutuc) (2018)
Harry Chou (2018)
Amritesh Rai (2019)
Rik Dey (with Register) (2019)
Omar Mohammed (2019)
Aqyan Bhatti (with Register) (2020)
Sayema Chowdhury (2022)
Teja Subrahmanyam (with Kulkarni) 2023
Moonkyu Song (2024)
Matthew Disiena (2025)

M.S.:

D.Bullock, 1990
K.Picone, 1991
J.Shen, 1991
R.Kovelamudi, 1992
S.Krishnan, 1992
M.Lobo, 1992
S.Ngaoram, 1993
A.Khan, 1993
D.Khanderkar, 1993
H.Taufique, 1994
S.Madireddi, 1994
J.Fretwell, 1995
M.Craig, 1995
J.Williamson, 1995
K.Reddy, 1995
J.Damiano, 1995
R.Gupta, 1995
K.Hassan, 1996
A.Lentvorski, 1997
S.Oswal, 1998
C.Seal, 1998
V.Agarwal, 1999
S.Nandan, 1999
S.Ravi, 1999
T.Ngai, 1999
V.Medina, 1999
H.Rahman, 1999
C.Twu, 2000
S. Oak, 2001
G. Shrivastava, 2001
R. Deppensmith, 2002
M. Swaminathan, 2002
L. Lin, 2003
L.Weltzer, 2004
D.Ahmad, 2005
I.Wiedmann, 2005
S. Ramachandran, 2005
A.Nanda 2007
N. Jain, 2008
N.Vora, 2008
K. Varahramyan, 2008
S. Kaur, 2010
Stephen Szczepaniak, 2015
Jessica Depoy, 2020
Alexander Klatt, 2021
Nick Pronin, 2021
Isaac Bodeman, 2022

Postdocs: Samit Ray (Chair Physics, IIT, Dean S N Bose Institute), Mark Loewe (IBM), Amitava Das (startup), Sabrina Grannan (NASA JPL), Freek Prins (Germany), Chuanbin Mao (Chair Prof. Oklahoma), Bhagawan Sahu (Global Foundries), Mathew Gilbert (Assoc. Prof. UIUC), Donwan.Ahn (Korea), Domingo Ferrer (IBM), Aparna Gupta (IIT), Samaresh Gucchait (Asst. Prof., Howard), Sushant Sonde (Argonne), SeHoon Shin (Samsung), Priyamvada Jadaun (Cornell/IMEC), Marylene Palard, Bahniman Ghosh (IIT), Rudresh Ghosh (NovaCentrix), Victor Chi, Sungkyu Kwon, Seung Heon Shin (Samsung), Anupam Roy (Asst. Prof. IIT), Sarmita Majumder (Sheetak), Ansh Gupta (IMEC), Nupur Navlakja, Prakriti Neha, Nilesh Pandey, Rajveer Jha

Current Post-docs: Nilesh Pandey, Rajveer Jha

In progress (Ph.D):

Chris Luth, Ryan Schalip, Siyu Wu, Sunny Bhakta, Hadi Darkhaneh, Jatin Singh, Hongming Zhang, Ashkan Aminian, Luke Sloan, Pranav Rama, John Komoll, Nicholas Watanabe, Blake Broussard, Justin Tullos

Books and Invited Book Chapters:

1. Solid State Electronic Devices, 5th Ed. (2000), 6th Ed. (2005), 7th Ed. (2015), Prentice-Hall by B.Streetman and S.Banerjee
2. Effect of Surface Nitridation on the Electrical Characteristics of Germanium High- κ /Metal Gate Metal-Oxide-Semiconductor Devices, D. Q. Kelly, J. J.-H. Chen, S. Guha, and S. K. Banerjee. Invited Book chapter, Springer, 2007.
3. SiGe HFETs, S.Banerjee, The Silicon Heterostructure Handbook, 2005, Edited by John Cressler.
4. High-k Gate Dielectrics, Y.Fan. S.Mudanai, L. Register and S.Banerjee, 2003
5. Device Miniaturization and Simulation, S.Banerjee and B.Streetman in ULSI Devices, John Wiley, 2000 (C.Chang and S.Sze editors)
6. Dopant Diffusion, S.Banerjee in Handbook of Semiconductor Manufacturing Technology, Marcel Dekker, 2000, 2006 (Y.Nishi, B.Doering and J.Kilby editors).
7. Silicon-germanium Devices, S.Banerjee, Elsevier, 2001.
8. Novel 3D CMOS, S.Dey and S.Banerjee, Solid State Electronics Trends, 2009
9. X. Mou, L. F. Register and S. K. Banerjee, "Ultra-low-power pseudospintronics devices via exciton condensation in coupled two-dimensional material systems," in Nanoscale Materials and Devices for Electronics, Photonics and Solar Energy, Ed. Stephen Goodnick, Anatoli Korkin and Robert Nemanich, Springer, 2015
10. D. Reddy, L. F. Register and S. K. Banerjee, "Bilayer pseudoSpin Field Effect Transistor (BiSFET)" in "Beyond CMOS Logic Switches," T.-J. King and K. Kuhn, Eds., Cambridge: Cambridge Univ. Press, 2015.
11. Devices and defects in two-dimensional materials: outlook and perspectives, A Rai, A Roy, A Valsaraj, S Chowdhury, D Taneja, Y Wang, LF Register, SK Banerjee Defects in Two-Dimensional Materials, 339-401 (Invited Book Chapter) 2021.

1. US20200365464 11/19/2020
Catalyst influenced chemical etching for fabricating three-dimensional sram architectures and optical waveguides, S.Sreenivasan, A.Mallavarapu, J.Kulkarni, M.Watts and S.Banerjee
2. US10,121,962 11/06/2018
Method for fabricating magnetic solid state devices. L.F.Register, B.Ghosh, R.Dey and S.Banerjee
3. US9,825,218 11/21/2017
Transistor that employs collective magnetic effects thereby providing improved energy efficiency, A.MacDonald, L.F. Register, E. Tutuc, I. Sodemann, H. Chen, X. Mou, S. Banerjee
4. US8,709,892 4/29/2014
Nanoparticles in a flash memory using chaperonin proteins, C.Mao, S.Tang and S.Banerjee
5. US8,629,427 1/14/2014
Topological insulator-based field-effect transistor, S.Banerjee, L.Register, A.MacDonald, B.Sahu, P.Jadaun and J.Chang
6. US8,263,967 9/11/2012
Bi-layer pseudo-spin field-effect transistor, S.Banerjee, L.Register, A.MacDonald, D.Reddy, E.Tutuc
7. US8,198,707 6/12/2012
Establishing a uniformly thin dielectric layer on graphene in a semiconductor device without affecting the properties of graphene, L.Colombo, S.Banerjee, S.Kim, E.Tutuc
8. US8,188,460 5/29/2012
Bi-layer pseudo-spin field-effect transistor, S.Banerjee, L.Register, A.MacDonald, D.Reddy, E.Tutuc
9. US8,008,649 8/30/2011
Incorporating gate control over a resonant tunneling structure in CMOS to reduce off-state current leakage, supply voltage and power consumption, L.Register and S.Banerjee
10. US6,744,083 6/1/2004
Submicron MOSFET having asymmetric channel profile, X.Chen and S.Banerjee
11. US6,420,219 7/16/2002
Thin film transistor and method, S.Batra, M.Manning, S.Banerjee and J.Damiano
12. US06320202 11/20/2001
Bottom gated thin film transistors comprising Ge in a channel region, S.Banerjee and S.Batra
13. US06,319,799 11/20/2001
High mobility heterojunction transistor and method, Q.Ouyang, A.Tasch and S.Banerjee
14. US06,313,486, 11/06/2001
Floating gate transistor having silicon germanium channel layer, D.Kencke and S.Banerjee
15. US06,313,487 11/06/2001
Vertical channel floating gate transistor having SiGe channel layer, D.Kencke and S.Banerjee
16. US06214652 4/10/2001
Thin film transistor and method of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
17. US06,200,839 3/13/2001

- Methods of making thin film transistors, S.Batra, M.Manning, S.Banerjee and L.Jung
18. US06166398 12/26/2000
Thin film transistors, S.Batra, M.Manning, S.Banerjee and L.Jung
 19. US06017782 01/25/2000
Thin film transistor and method of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
 20. US05985703 11/16/1999
Method of making thin film transistors, S.Banerjee
 21. US05977560 11/02/1999
Thin film transistor constructions with polycrystalline silicon-germanium alloy doped with carbon in the channel region, S.Banerjee and S.Batra
 22. US05953596 09/14/1999
Methods of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and L.Jung
 23. US05936262 08/10/1999 Thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
 24. US05904513 05/18/1999 Method of forming thin film transistors, S.Batra, M.Manning, S.Banerjee and J.Damiano
 25. US05665981 09/09/1997
Thin film transistors and method of promoting large crystal grain size in the formation of polycrystalline silicon alloy thin films, S.Banerjee and S.Batra
 26. US05548132 08/20/1996
Thin film transistor with large grain size DRW offset region and small grain size source and drain and channel regions, S.Batra, M.Manning, S.Banerjee and J.Damiano
 27. US05546340 08/13/1996
Non-volatile memory array with over-erase correction, C.Hu, R.Richart, S.Garg and S.Banerjee
 28. US05436474 07/25/1995
Modulation doped field effect transistor having built-in drift field, S.Banerjee, A.Tasch and B.Streetman
 29. US05432366 07/11/1995
P-I-N MOSFET for ULSI applications, S.Banerjee, S.Bhattacharya and W.Lynch
 30. US05109259 04/28/1992
Multiple DRAM cells in a trench, S.Banerjee
 31. US05066607 11/19/1991
Method of making a trench DRAM cell with dynamic gain, S.Banerjee
 32. US04999811 03/12/1991
Trench DRAM cell with dynamic gain, S.Banerjee
 33. US04969019 11/06/1990
Three-terminal tunnel device, S.Banerjee
 34. US04864374 09/05/1989
Two-transistor dram cell with high alpha particle immunity, S.Banerjee
 35. US04713678 12/15/1987
dRAM cell and method, R.Womack, H.Shichijo, S.Banerjee and S.Malhi

1. M. Keever, H. Shichijo, K. Hess, S.Banerjee, L. Witkowski, H. Morkoc and B. Streetman, "Measurements of Hot-Electron Conduction and Real Space Transfer in GaAs-AlGaAs Heterojunction Layers," *Appl. Phys. Lett.*, 38(1), 36-38, January 1981.
2. S.Banerjee and B. Streetman, "Experimental Studies of Laterally Seeded Recrystallized Polysilicon on Silicon Dioxide," *Proc. of IEEE UGIM*, v.5, pp.79-83, May 1983.
3. S.Banerjee and B. Streetman, "Theoretical and Experimental Study of Swept Line Electron Beam Annealing of Semiconductors," *J. Appl. Phys.*, 54(6), 1947-1955, June 1983.
4. S.Banerjee and B. Streetman, "Electron and Hole Traps in Silicon-on-Oxide Grown Using Lateral Epitaxy by Seeded Solidification," *J. Phys. D: Appl. Phys.*, November 1983.
5. S.Banerjee, R. DeJule, K. Soda and B. Streetman, "Planar Be-Implanted GaAs Junction Formation Using Swept-Line Electron Beam Annealing," *IEEE Trans. Elec. Dev.*, 30 (12), 1755-1760, December 1983.
6. S.Banerjee, B. Lee, J. Baker, D. Reed and B. Streetman, "Annealing of Ion-Implanted Silicon-on-Insulator Films Using a Scanned Graphite Strip Heater," *Thin Solid Films*, 115, 19-26, 1984.
7. H. Shichijo, S. Malhi, A. Shah, G. Pollack, W. Richardson, M. Elahy, S.Banerjee, R. Womack and P. Chatterjee, "TITE RAM: A New SOI DRAM Gain Cell for MBit DRAMs," *Proc. of ICSSDM*, v. 16, pp. 265-268, June, 1984.
8. S.Banerjee, R. Tong, B. Lee, R. DeJule, B. Streetman and H. Lam, "Implantation and Annealing Studies of Laterally Seeded Recrystallized Silicon on Silicon Dioxide," *J. Electrochem. Society*, 131(6), 1409-1416, June 1984.
9. G. Pollack, W. Richardson, S. Malhi, T. Bonifield, H. Shichijo, S.Banerjee, M. Elahy, A. Shah, R. Womack and P. Chatterjee, "Hydrogen Passivation of Polysilicon MOSFETs From a Plasma Nitride Source," *IEEE Elec. Dev. Letters*, 5(11), 468-470, November 1984.
10. S.Banerjee, M. Elahy, H. Shichijo, G. Pollack, W. Richardson, S. Malhi, A. Shah, P. Chatterjee, H. Lam and R. Womack, "Comparison of Accumulation and Inversion Mode LPCVD Polysilicon MOSFET Characteristics for Memory Applications," *IEEE Trans. Elec. Dev.*, 31(12), 1983, December 1984.
11. M. Elahy, H. Shichijo, P. Chatterjee, A. Shah, S.Banerjee, and R. Womack, "Trench Capacitor Leakage in High Density DRAMs," *IEEE Elec. Dev. Letters*, 5(12), 527-530, December 1984.
12. M. Elahy, H. Shichijo, P. Chatterjee, A. Shah, S.Banerjee and R. Womack, "Trench Capacitor Leakage in MBit DRAMs," *IEEE IEDM Technical Digest*, p. 248-251, December 1984.
13. H. Shichijo, S. Malhi, W. Richardson, G. Pollack, A. Shah, L. Hite, S.Banerjee, M. Elahy, R. Sundaresan, R. Womack, H. Lam and P. Chatterjee, "Polysilicon Transistors in VLSI MOS Memories," **Invited Talk**, *IEEE IEDM Technical Digest*, p. 228-231, December 1984.
14. S.Banerjee, "Reliability Studies on Thin Film Polysilicon MOSFETs," T.I. Semiconductor Reliability Workshop, Dallas, Texas, September 1984.
15. H. Shichijo, S.Banerjee, G. Pollack, W. Richardson, M. Bordelon, A. Shah, S. Malhi, M. Elahy, R. Womack, C. Wang, J. Gallia, H. Davis and P. Chatterjee, "Trench Transistor DRAM Cell," *IEEE Elec. Dev. Letters*, 7(2), 119-121, February 1985.
16. S.Banerjee and B. Streetman, "Planar Junctions in Silicon on Oxide Grown Using Lateral Epitaxy by Seeded Solidification," *IEEE Trans. Elec. Dev.*, 32(4), 850-853, April 1985.
17. S.Banerjee and J. Baker, "Proximity Annealing of Sulfur-Implanted Gallium Arsenide Using a Strip Heater," *Jap. J. Appl. Phys.*, 24(5), L377-L379, May 1985.
18. J. Fossum, A. Ortiz-Conde, H. Shichijo and S.Banerjee, "Anomalous Leakage Current in LPCVD Polysilicon MOSFETs," *IEEE Trans. Elec. Dev.*, 32(9), 1878-1884, September 1985.
19. W. Richardson, D. Bordelon, G. Pollack, A. Shah, S. Malhi, H. Shichijo, S.Banerjee, M. Elahy, R. Womack, C. Wang, J. Gallia, H. Davis and P. Chatterjee, "A Trench Transistor Cross-Point DRAM Cell," *IEDM Technical Digest*, 714-717, December 1985.
20. H. Shichijo, S. Malhi, R. Sundaresan, S.Banerjee and H. Lam, "Process and Device Considerations for Small Grain Polysilicon MOS Transistors," **Invited paper**, *Proc. Mat. Res. Soc.*, v.53, pp.419-428, December 1985.

21. S. Malhi, H. Shichijo, S.Banerjee, R. Sundaresan, M. Elahy, G. Pollack, W. Richardson, A. Shah, L. Hite, R. Womack, P. Chatterjee and H. Lam, "Characteristics and Three-Dimensional Integration of MOSFETs in Small-Grain LPCVD Polycrystalline Silicon," *IEEE Trans. Elec. Dev.*, 32(2), 258-281, February 1985.
22. A.Shah, C. Wang, R. Womack, J. Gallia, H. Shichijo, H. Davis, M. Elahy, S.Banerjee, G. Pollack, W. Richardson, M. Bordelon, S. Malhi, C. Pilch, B. Tran and P. Chatterjee, "A 4MBit DRAM with Cross-Point Trench Transistor Cell," *Proc. IEEE Int. Sol. Stat. Cir.*, February 1986. **(BEST PAPER AWARD)**
23. S.Banerjee, H. Shichijo, A. Nishimura, A. Shah, G. Pollack, W. Richardson, M. Bordelon, S. Malhi, M. Elahy, R. Womack, C. Wang, J. Gallia, H. Davis and P. Chatterjee, "Characterization of Trench Transistors for 3-D Memories," *VLSI Symp. Digest*, pp.79-80, May 1986.
24. A. Shah, C. Wang, R. Womack, J. Gallia, H. Shichijo, H. Davis, M. Elahy, S.Banerjee, G. Pollack, W. Richardson, M. Bordelon, S. Malhi, C. Pilch, B. Tran and P. Chatterjee, "A 4MBit DRAM with Trench Transistor Cell," *IEEE J. Solid State Cir.*, SC-21(5), 618-626, October 1986.
25. K. Rao, M. Elahy, D. Bordelon, S.Banerjee, H. Tsai, W. Richardson and R. Womack, "Trench Capacitor Issues in VLSI DRAM Cells," *IEEE IEDM Tech. Digest*, pp.140-143, December 1986.
26. H. Shichijo, A. Shah, C. Wang, R. Womack, J. Gallia, H. Davis, M. Elahy, S.Banerjee, G. Pollack, W. Richardson, M. Bordelon, S. Malhi, C. Pilch, B. Tran and P. Chatterjee, "A 4MBit DRAM with Cross-Point Trench Transistor Cell," *Nikkei Electronics*, Japan, 7.14, 189-208, 1986.
27. S.Banerjee, J. Coleman, B. Richardson and A. Shah, "A Band-to-Band Tunneling Effect in the Trench Transistor Cell," *VLSI Symp. Digest*, pp. 97-98, May 1987.
28. S.Banerjee, W. Richardson, J. Coleman and A. Chatterjee, "A New Three Terminal Tunnel Device," *IEEE Elec. Dev. Letters*, 8(8), 347-349, August 1987.
29. B. Shen, I. Chen, S.Banerjee, G. Brown, J. Bohlman, P. Chang and R. Doering, "A High Quality Tantalum Oxide Film for Advanced DRAM Application," *IEEE IEDM Tech. Digest*, pp. 582-586, December 1987.
30. S.Banerjee and M. Bordelon, "A Model for the Trench Transistor," *IEEE Trans. Elec. Dev.*, 34 (12), 2485-2493, December 1987.
31. S. Banerjee, "Device Simulation of Trench Transistor Cell," *T.I. Technical Journal*, 4(1), 53-56, January 1987.
32. S.Banerjee, R. Sundaresan, H. Shichijo and S. Malhi, "Hot Electron Degradation of N-Channel Polysilicon MOSFETs," *IEEE Trans. Elec. Dev.*, 35 (2), 152-157, February 1988.
33. S.Banerjee, J. Coleman, W. Richardson and A. Shah, "Leakage Mechanisms in the Trench Transistor Cell," *IEEE Trans. Elec. Dev.*, 35(1), 108-116, January 1988.
34. L.Breaux, B.Anthony, T.Hsu, S.Banerjee and A.Tasch, "Silicon Homoepitaxy at 400 C using Remote Plasma-enhanced Chemical Vapor Deposition," *Proc. Mat. Res. Soc.*, v.2, pp. 47-58, March 1989.
35. K.Park, S.Batra, T.Lian, E.Yun, S.Banerjee, D.Kwong and A.Tasch, "Electronic Devices in Dielectrically Isolated Polycrystalline Silicon Thin Films," *Proc. Mat. Res. Soc.*, v.2, pp. 847-855, March 1989.
36. T.Hsu, L.Breaux, B.Anthony, S.Banerjee and A.Tasch, "RHEED Studies of Silicon Surface Reconstruction after Remote Hydrogen Plasma Cleaning," *Proc. Mat. Res. Soc.*, April, 1989.
37. H. Shin, A.F. Tasch, C.M. Maziar and S.Banerjee, "A New Approach to Verify and Derive a Transverse-Field Dependent Mobility Model for Electrons in MOS Inversion Layers," *IEEE Trans. Elec. Dev.*, 36(6), 1117-1124, June 1989.
38. B.Anthony, L.Breaux, T.Hsu, S.Banerjee and A.Tasch, "In Situ Cleaning of Silicon Substrate Surfaces by Remote-plasma excited Hydrogen," *J.Vac. Sci. Tech. B*, 7(4), 621-626, August 1989.
39. A.Tasch, S.Banerjee, L.Breaux, B.Anthony and T.Hsu, "Low Temperature Epitaxy Using Remote Plasma-enhanced Chemical Vapor Deposition," *Proc. SRC TRC on Silicon Epitaxy*, September 1989.

40. L.Breaux, B.Anthony, T.Hsu, S.Banerjee and A.Tasch, "Homoepitaxial Films Grown on Silicon (100) at 150 C by Remote Plasma-enhanced Chemical Vapor Deposition," *Appl. Phys. Lett.*, 55(18), pp.1885-1887, October, 1989.
41. T.Hsu, B.Anthony, L.Breaux, S.Banerjee and A.Tasch, "Remote Plasma-enhanced Chemical Vapor Deposition of Epitaxial Silicon on Silicon (100) at 150 C," *Proc. Mat. Res. Soc.*, Nov. 1989.
42. B.Anthony, T.Hsu, L.Breaux, S.Banerjee and A.Tasch, "Reaction Kinetics of Epitaxial Silicon Deposition at 200-400 C Using RPCVD," *Proc. Mat. Res. Soc.*, Nov. 1989.
43. S.Batra, K.Park, C.Kyono, S.Bhattacharya, S.Banerjee, C.Maziar, D.Kwong, A.Tasch, M.Rodder and R.Sundaresan, "Rapid Thermal Annealing for H Passivation of Polysilicon MOSFETs from a Si₃N₄ Overcoat," *IEEE IEDM Tech. Dig.*, pp. 455-458, December 1989.
44. S.Banerjee, B.Shen, I.Chen, J.Bohlman, G.Brown and R.Doering, "Conduction Mechanisms in Sputtered Tantalum Pentoxide," *J. Appl. Phys.*, 65(3), 1140-1146, February 1989.
45. T.Hsu, L.Breaux, B.Anthony, S.Banerjee and A.Tasch, "Crystalline Defects in Low Temperature Epitaxial Silicon Growth by RPCVD," *Elect. Mat. Conf.*, Cambridge, Massachusetts, June, 1989.
46. T.Hsu, B.Anthony, L.Breaux, S.Banerjee and A.Tasch, "Low Temperature Silicon In Situ Cleaning and Homoepitaxy by Remote Plasma-enhanced Chemical Vapor Deposition," *Proc. SPIE*, March 1990.
47. K.Park, S.Batra, J.Lin, S.Yoganathan, S.Banerjee, J.Lee, S.Sun, J.Yeagain and G.Lux, "Effects of As Segregation and Electron Trapping on the Capacitance-Voltage Behavior of Polysilicon and Polycide Gates," *Proc. Mat. Res. Soc.*, April 1990.
48. B.Anthony, T.Hsu, L.Breaux, S.Banerjee and A.Tasch, "Experimental Studies of Silicon Growth Kinetics by RPCVD at Low Temperatures," *Proc. SPIE*, March 1990.
49. B.Anthony, T.Hsu, L.Breaux, S.Banerjee and A.Tasch, "Remote Plasma Enhanced CVD: Reaction Kinetics as a Function of Growth Parameters," *Metallurgical. Soc.Proc-AIME*, Anaheim, CA, February 1990.
50. T.Hsu, B.Anthony, L.Breaux, S.Banerjee and A.Tasch, "Defect Microstructure in Single Crystal Silicon Thin Films Grown at 150 C by RPCVD," *Metallurgical. Soc.Proc-AIME*, Anaheim, CA, February 1990.
51. L.Breaux, B.Anthony, T.Hsu, S.Banerjee and A.Tasch, "Defect-Free Remote Hydrogen Plasma Clean of Silicon (100) for Homoepitaxy," *Metallurgical. Soc.Proc-AIME*, Anaheim, CA, February 1990.
52. K.Park, S.Batra, S.Banerjee and G.Lux, "Ultra-shallow junctions in Silicon Using Amorphous and Polycrystalline Silicon Solid Diffusion Sources," *Elec. Mat. Conf.*, Santa Barbara, CA, June 1990.
53. K.Park, S.Batra, S.Banerjee and G.Lux, "Discontinuity of P, As and B profiles at the Interface of Polycrystalline/ single Crystal Silicon," *Elec. Mat. Conf.*, Santa Barbara, CA, June 1990.
54. T.Hsu, B.Anthony, R.Qian, S.Banerjee and A.Tasch, "Hydrogen Passivation of Si(100) Surface by Low Temperature Remote Hydrogen Plasma Clean for Silicon Epitaxy," *Elec. Mat. Conf.*, Santa Barbara, CA, June 1990.
55. B.Anthony, T.Hsu, R.Qian, S.Banerjee and A.Tasch, "The Use of Langmuir Probe Measurements to Investigate the Reaction Mechanisms of Remote Plasma-enhanced Chemical Vapor Deposition," *Elec. Mat. Conf.*, Santa Barbara, CA, June 1990.
56. K.Park, S.Batra and S.Banerjee, "Preferential Grain Growth in Heavily Doped Polysilicon During Rapid Thermal Annealing," *Mat. Res. Soc. Symp.*, San Francisco, CA, April 1990.
57. K.Park, S.Batra, S.Banerjee and G.Lux, "Comparison of Amorphous and Polycrystalline Silicon Films as a Solid Diffusion Source for Advanced VLSI Processes," *Proc. Mat. Res. Soc.*, April 1990.
58. S.Batra, K.Park, S.Banerjee and G.Lux, "Diffusion of As and B in Polysilicon/Single Crystal Silicon Systems," *Proc. Mat. Res. Soc.*, April 1990.
59. S.Batra, K.Park, S.Banerjee and R.Sundaresan, "Comparative Studies of Furnace and Rapid Thermal Passivation for Accumulation and Inversion Mode Polysilicon-on-Oxide MOSFETs," *Proc. Mat. Res. Soc.*, April 1990.

60. S.Banerjee, A.Tasch, B.Anthony, T.Hsu, L.Breaux, and R.Qian "Remote Plasma-enhanced Chemical Vapor Deposition of Homo-Epitaxial Silicon on Silicon (100) at 150 C," *Proc. of Epitaxial Crystal Growth*, v.1, Trans. Tech. Publishers, Budapest, Hungary, April 1990 (**Invited**).
61. T.Hsu, L.Breaux, B.Anthony, S.Banerjee and A.Tasch, "Defect Microstructure in Low Temperature Epitaxial Silicon Grown by RPCVD," *J. Electron. Mat.*, 19 (4), 375-384, April 1990.
62. J.Lin, S.Banerjee, J.Lee and C.Teng, "Soft Breakdown in Titanium-Silicided Shallow Source Drain Junctions," *IEEE Elec. Dev. Lett.*, 11(5), pp.191-193, May, 1990.
63. B.Anthony, T.Hsu, L.Breaux, R.Qian, S.Banerjee and A.Tasch, "Remote Plasma-enhanced CVD of Silicon: Reaction Kinetics as a Function of Growth Parameters," *J. Electron.Mat.*, 19 (10), pp. 1089-1094, May 1990.
64. S.Batra, K.Park, S.Banerjee, D.Kwong, A.Tasch, R.Sundaresan and M.Rodder, "Rapid Thermal Hydrogen Passivation of Polysilicon MOSFETs," *Elec. Dev. Lett.*, 11(5), pp.194-196, May, 1990.
65. T.Hsu, B.Anthony, L.Breaux, R.Qian, S.Banerjee and A.Tasch, "Defect Microstructure in Single Crystal Silicon Thin Films Grown at 150 C by Remote Plasma-enhanced Chemical Vapor Deposition," *J. Electron. Mat.*, 19 (10), pp. 1043-1050, May 1990.
66. B.Anthony, T.Hsu, L.Breaux, R.Qian, S.Banerjee and A.Tasch, "Defect Free Remote Hydrogen Plasma Clean of Si (100) for Homoepitaxy," *J. Electron.Mat.*, 19 (10), pp. 1027-1032, May 1990.
67. K.Park, S.Batra, J.Lin, S.Yoganathan, S.Banerjee, J.Lee, S.Sun, J.Yeagain and G.Lux, "Anomalous Capacitance-Voltage Behavior due to Dopant Segregation and Carrier Trapping in As-implanted Polysilicon and Polycide Gates," *Appl. Phys. Lett.*, 56 (23), 2325-2327, June 1990.
68. S.Bhattacharya, S.Banerjee, J.Lee, A.Tasch and A.Chatterjee, "Design Issues for Achieving Latchup-free, Deep Trench-isolated, Bulk, Non-epitaxial, Submicron CMOS," *IEEE IEDM Tech. Digest*, p. 185-188, December 1990.
69. K.Park, S.Batra and S.Banerjee, "Study of P diffusion from poly into single crystal Si by TEM and SIMS," *Mat. Res. Soc. Symp.*, San Francisco, CA, April 1990.
70. J.Lin, S.Banerjee, J.Lee and C.Teng, "Anomalous Leakage in Titanium Silicided Junctions," *J.Appl. Phys.*, 68 (3), 1082-1087, August 1990.
71. G.Yeric, A. Tasch and S.Banerjee, "A Universal MOSFET Mobility Degradation Model for Circuit Simulation," *IEEE Trans. CAD*, 9(10), pp.1123-1126, October 1990.
72. K.Park, S.Batra, S.Banerjee and G.Lux, "Comparison of Amorphous and Polycrystalline Silicon Films as a Solid Diffusion Source for Advanced VLSI Processes," *Proc. of SRC TECHCON*, v.2, October 1990.
73. B.Anthony, T.Hsu, R.Qian, J.Irby, S.Banerjee and A.Tasch, "In Situ Doping of Low Temperature Epitaxial Silicon Films Grown by Remote Plasma-enhanced Chemical Vapor Deposition," *Proc. Mat. Res. Soc.*, November 1990.
74. B. Fowler, D.Bullock, S.Lian and S.Banerjee, "Silicon Homoepitaxy at 300 C Using ArF Excimer Laser Photolysis of Disilane," *Proc. Mat. Res. Soc.*, November 1990.
75. S.Batra, K.Park, J.Lin, S.Yoganathan, J.Lee, S.Banerjee, S.Sun, J.Yeagain and G.Lux, "Effects of Dopant Redistribution, Segregation and Carrier Trapping in As-implanted MOS Gates," *IEEE Trans. Elec. Dev.*, 37(11), pp.2322-2330, Nov. 1990.
76. S. Yoganathan, S. Banerjee, and S.Chatterjee, "An Efficient Two-dimensional Hydrodynamic Simulator Using a Decoupled Algorithm", *Proc. ISDRS, vol.1*, December 1991.
77. R. Qian, I.Chung, B. Anthony, T. Hsu, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, "Growth and High Resolution TEM Characterization of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Heterostructures by Remote Plasma-enhanced Chemical Vapor Deposition on Si (100)", *Proc. Mat. Res. Soc.*, Dec. 1991.
78. S.Batra, K.Park, S.Banerjee and G.Lux, "Comparison of Arsenic and Boron Diffusion in Polycrystalline/Single Crystal Silicon Systems," *J.Electrochem. Soc.*, 138(2), pp.545-549, Feb. 1991.
79. S.Bhattacharya, S.Banerjee, J.Lee, A.Tasch and A.Chatterjee, "The Impact of Trench Isolation on Latchup Immunity in Bulk, Non-epitaxial CMOS," *IEEE Elec. Dev. Lett.*, 12(2), pp.77-79, Feb. 1991.

80. S.Lian , B. Fowler, D.Bullock, and S.Banerjee, "ArF Excimer Laser-induced Photolytic Growth of Si Homoepitaxial Films from Si₂H₆ at 330 C," *Appl. Phys. Lett.* , 58(5), p.514-516, Feb. 1991.
81. S. Lian, B. Fowler, S. Krishnan, L. Jung, and S. Banerjee, "Silicon Homoepitaxy Using Photo-CVD: A RHEED and TEM Study", invited paper, *Mat. Science & Eng. B*, B10, p.181 Jan., 1991.
82. S. Lian, B. Fowler, S. Krishnan, and S. Banerjee, "Microstructure of Si Epitaxial Films Grown by Photo-Chemical Vapor Decomposition of Si₂H₆", American Physical Society Meeting on Semiconductor Interfaces and Microstructures, Cincinnati, OH, March 1991.
83. R. Qian, B. Anthony, T. Hsu, J. Irby, D. Kinosky, S. Banerjee, and A. Tasch, "Ge_xSi_{1-x}/Si Quantum Well Growth by Remote Plasma-enhanced Chemical Vapor Deposition", American Physical Society Meeting on Semiconductor Interfaces and Microstructures, Cincinnati, OH, March 1991.
84. K.Park, S.Batra, S.Banerjee and G.Lux, "Ultra-shallow junctions in Silicon Using Amorphous and Polycrystalline Silicon Solid Diffusion Sources," *J. Elec. Mat.*, 20 (3), pp. 261-266, March 1991.
85. T.Hsu, B.Anthony, R.Qian, J.Irby, S.Banerjee and A.Tasch, "Cleaning and Passivation of the Si(100) Surface by Low Temperature Remote Hydrogen Plasma Treatment for Si Epitaxy," *J. Electron.Mat.*, 20 (3), pp. 279-287, March 1991.
86. S. Yoganathan, S. Banerjee, T. Itoh, H. Shichijo, and S. El-Ghazaly, "A Highly Efficient Decoupled Transient Simulator Including Energy Balance for GaAs MESFETs", *Proc. NASECODE vol. VII*, April 1991.
87. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee, and G. Lux, "Effects of TiSi₂ Formation on Boron Penetration Through Gate Oxides in MOS Devices Under Rapid Thermal Processing", *Proc. of Mat. Res. Soc.*, April 1991.
88. S. Batra, K. Park, S. Banerjee, T. Smith, B. Mulvaney, and G. Lux, "Lateral Uniformity of Ultra-Shallow Junction Formed by Rapid Thermal Annealing in Polysilicon-on-Silicon Systems", *Proc. of Mat. Res. Soc.*, April 1991.
89. K. Park, S. Batra, and S. Banerjee, "Secondary Grain Growth in Heavily P Doped Polysilicon During Rapid Thermal Annealing", *Proc. of Mat. Res. Soc.*, April 1991.
90. B. Fowler, S. Lian, S. Krishnan, and S. Banerjee, "In situ Multiprocessing for Laser-enhanced Silicon Photo-epitaxy at 300° C", *Proc. of Mat. Res. Soc.*, April 1991.
91. T. Hsu, B. Anthony, R. Qian, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, "Heteroepitaxy of Si/Si_xGe_{1-x} Grown by Remote Plasma-enhanced Chemical Vapor Deposition", *Proc. of Mat. Res. Soc.*, April 1991.
92. B. Anthony, T. Hsu, R. Qian, J. Irby, S. Banerjee, and A. Tasch, "The Use of Langmuir Probe Measurements to Investigate the Reaction Mechanisms of Remote Plasma-enhanced Chemical Vapor Deposition", *J. Electron. Mat.*, 20(4), pp. 309-313, April 1991.
93. S. Lian, B. Fowler, L. Jung, S. Krishnan, and S. Banerjee, "Defect Microstructure of Si Homoepitaxial Films Grown by Laser Enhanced CVD from Si₂H₆", *Proc.of ECS*, May 1991.
94. S. Lian, B. Fowler, S. Krishnan and S. Banerjee, "In situ and Ex situ Characterization of Microstructure of Photo-epitaxial Growth at 300° C for ULSI Applications," SPIE Conference, Dallas, Texas, May 1991.
95. K. Park, S. Batra, and S. Banerjee, "The Role of Negatively Charged Vacancies in Secondary Grain Growth in Polycrystalline Silicon During Rapid Thermal Annealing", *Appl. Phys. Lett.*, 58(21), pp.2414-2416, May 1991.
96. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee, and G. Lux, "Enhancement of Boron Diffusion Through Gate Oxides in MOS Devices Under Rapid Thermal Silicidation", *Appl. Phys. Lett.*, 58(19), p.2123-2125, May 1991.
97. B. Fowler, S. Lian, S. Krishnan, L. Jung, and S. Banerjee, "ArF Laser-enhanced Photo-CVD of Silicon from Disilane", *Proc. of ECS*, May 1991.

98. R. Qian, T. Hsu, B. Anthony, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, "Crystallographic Characterization of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Superlattices Grown by Remote Plasma-enhanced Chemical Vapor Deposition", *J. Appl. Phys.*, 70(6), p. 3324, June 1991.
99. K. Picone, S. Batra, K. Park, J. Lee, and S. Banerjee, "Characterization of Polysilicon-on-Single Crystal Si Junctions Formed by Rapid Thermal Processing for Polysilicon Emitter BJT and Elevated Source/Drain MOSFET Applications," SPIE Conference, Dallas, TX, May 1991.
100. J. Irby, D. Kinosky, R. Qian, B. Anthony, T. Hsu, S. Banerjee, A. Tasch, and C. Magee, "SIMS and Microstructural Analysis of *In situ* B-Doped Si Epitaxial Films Grown at 450° C by Remote Plasma-enhanced Chemical Vapor Deposition", SPIE Conference, Dallas, TX, May 1991.
101. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee, and G. Lux, "Effects of TiSi_2 Formation on Boron Penetration Through Gate Oxides in MOS Devices", SPIE Conference, Dallas, TX, May 1991.
102. B. Fowler, S. Lian, S. Krishnan, and S. Banerjee, "Growth Kinetic Model for ArF Laser-enhanced Si CVD Using a Factorial Design of Experiment", SPIE Conference, Dallas, TX, May 1991.
103. D. Kinosky, R. Qian, J. Irby, T. Hsu, B. Anthony, S. Banerjee, and A. Tasch, "Microstructural Characterization of $\text{Si}/\text{Si}_x\text{Ge}_{1-x}$ Heterostructures Grown by Remote Plasma-enhanced Chemical Vapor Deposition", SPIE Conf., Dallas, TX, May 1991.
104. K. Picone, S. Batra, K. Park, M. Lobo, S. Bhattacharya, J. Lee, and S. Banerjee, "Leakage Characteristics and Lateral Uniformity of Ultra-Shallow Junctions Formed Using Polysilicon and Amorphous Silicon Diffusion Source", Elec. Mat. Conf., Boulder, CO, June 1991.
105. S. Batra, K. Park, K. Picone, M. Lobo, S. Bhattacharya, and S. Banerjee, "Preferential Orientation of Secondary Grains in Heavily Doped Ultra-thin Polysilicon Films", Elec. Mat. Conf., Boulder, CO, June 1991.
106. R. Qian, B. Anthony, T. Hsu, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, "Microstructural Analysis of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Quantum Wells Grown by Remote Plasma-enhanced Chemical Vapor Deposition on Si (100)", Elec. Mat. Conf., Boulder, CO, June 1991.
107. S. Lian, B. Fowler, L. Jung, S. Krishnan, and S. Banerjee, "Characterization of Microstructure of Si Epitaxy Grown by 193 nm ArF Excimer Laser Photolysis of Si_2H_6 ", Elec. Mat. Conf., Boulder, CO, June 1991.
108. B. Fowler, S. Lian, S. Krishnan, L. Jung, and S. Banerjee, "ArF Excimer Laser-Enhanced Photo-CVD of Silicon from Disilane: Growth Kinetic Models", Elec. Mat. Conf., Boulder, CO, June 1991.
109. J. Irby, D. Kinosky, R. Qian, B. Anthony, T. Hsu, S. Banerjee, A. Tasch, and C. Magee, "*In situ* B-Doped Si Epitaxial Films Grown at 450° C by Remote Plasma-enhanced Chemical Vapor Deposition: Physical and Electrical Characterization", Elec. Mat. Conf., Boulder, CO, June 1991.
110. T. Hsu, B. Anthony, R. Qian, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, J. Lin, and H. Marcus, "Electron-beam-induced-oxidation of Hydrogen-passivated Si (100) Surface Prepared by Low Temperature Remote Hydrogen Plasma Clean", Elec. Mat. Conf., Boulder, CO, June 1991.
111. B. Anthony, T. Hsu, R. Qian, S. Banerjee and A. Tasch, "The Use of Langmuir Probe Measurements to Study Reaction Kinetics in Remote Plasma-enhanced Chemical Vapor Deposition of Silicon," *Proc. Mat. Res. Soc.*, v.190, p.267, 1991.
112. T. Hsu, B. Anthony, R. Qian, S. Banerjee and A. Tasch, "Hydrogen Passivation of Silicon (100) by Remote Hydrogen Plasma Treatment," *Proc. Mat. Res. Soc.*, v.190, p.279, 1991.
113. S. Yoganathan, S. Banerjee, T. Itoh, H. Shichijo, and S. El-Ghazaly, "A Numerical Model of GaAs MESFETs Including Energy Balance for Microwave Applications", *IEEE Microwave and Guided Wave Lett.*, 1(7), p. 175, July 1991.
114. K. Park, S. Batra, S. Banerjee, and G. Lux, "Analysis of Ion-implanted Amorphous and Polycrystalline Silicon Films as Diffusion Sources for Ultra-shallow Junctions", *J. Appl. Phys.*, 70(3), pp.1397-1404, August 1991.

115. K. Park, S. Batra, and S. Banerjee, "Analysis of Lateral Uniformity of Ultra-shallow Junctions in Polycrystalline Silicon-on-Single Crystal Silicon Systems", *Appl. Phys. Lett.*, 59(6), p.709-711, August 1991.
116. D. Kinosky, R. Qian, J. Irby, T. Hsu, B. Anthony, S. Banerjee, A. Tasch, C. Magee, and C. Grove, "Low Temperature Growth of $\text{Ge}_x\text{Si}_{1-x}$ /Si Heterostructures on Si (100) by Remote Plasma-enhanced Chemical Vapor Deposition", *Appl. Phys. Lett.*, 59(7), pp.817-819, August 1991.
117. B. Fowler, S. Lian, S. Krishnan, L. Jung, and S. Banerjee, "Modeling of Photo-Chemical Vapor Deposition of Epitaxial Silicon Using an ArF Laser", *Proc. SPIE Microelectronic Processing Integration*, September 1991.
118. S. Lian, B. Fowler, L. Jung, S. Krishnan, and S. Banerjee, "Characterization of Microstructure of Si Films Grown by Laser-enhanced Photo CVD Using Si_2H_6 ", *Proc. SPIE Microelectronic Processing Integration*, September 1991.
- A. Tasch, S. Banerjee, B. Anthony, T. Hsu, R. Qian, J. Irby, and D. Kinosky, "Low Temperature *In situ* Cleaning of Silicon by Remote Plasma Hydrogen", **Invited** Paper, *Proc. ECS*, October 1991.
119. D. Kinosky, B. Anthony, T. Hsu, R. Qian, J. Irby, S. Banerjee, and A. Tasch, "The Hydrogen-terminated Si (100) Surface and its Role in the Successful Achievement of Silicon Epitaxy at Low Temperatures", *Proc. ECS*, October 1991.
120. *S.Banerjee, "The Impact of Impurities on Semiconductor Devices: Electrical Performance and Reliability," **Invited** paper, *Proc. Microcontamination*, v.91, pp.621-624, October 1991.
121. S. Krishnan, S. Lian, B. Fowler, L. Jung, C.Li and S. Banerjee, "Photoepitaxy of Si from Si_2H_6 With and Without Surface Activation of Si by an ArF Excimer Laser," *Proc. of ECS*, October 1991.
122. B. Fowler, S. Lian, S. Krishnan, L. Jung, C.Li and S. Banerjee, "ArF Excimer Laser-enhanced Photo-CVD of Silicon from Disilane:Growth Kinetic Model", *Proc. of ECS*, October 1991.
123. R. Qian, D. Kinosky, T. Hsu, J. Irby, A.Mahajan, S.Thomas, B. Anthony, S. Banerjee, A. Tasch, and C. Magee, "Growth of $\text{Ge}_x\text{Si}_{1-x}$ /Si Heteroepitaxial Films by Remote Plasma-enhanced Chemical Vapor Deposition", 38th AVS Symposium, Seattle, WA, November 1991.
124. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee, and G. Lux, "Effects of TiSi_2 Formation on Boron Penetration Through Gate Oxides in MOS Devices Under RTP", MRS, 1991.
125. A.Sultan, S.Batra, M.Lobo, K.Park and S.Banerjee, "Modeling of Boron Diffusion in Polysilicon-on-Silicon Layers," *Microcryst. Mat. Sci. and Dev. v. 283 (MRS)*, Dec. 1992.
126. T. Hsu, B. Anthony, R. Qian, J. Irby, D. Kinosky, A.Mahajan, S. Banerjee, A. Tasch, and C. Magee, "Remote Plasma-enhanced Chemical Vapor Deposition Process for Low Temperature (<450 C) Epitaxy of Si and $\text{Ge}_x\text{Si}_{1-x}$ ", **invited** paper, *Mat. & Mfg. Proc.*, 7(4), p.593-612, 1992.
127. B. Anthony, T. Hsu, R. Qian, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, "Physical and Electrical Characterization of *In situ* Boron-doped Single Crystal Silicon Films Grown at 450° C Using Remote Plasma-enhanced Chemical Vapor Deposition", *Thin Solid Films*, 207, p.12-14, Jan.1992.
128. T. Hsu, B. Anthony, R. Qian, J. Irby, D. Kinosky, A.Mahajan, S. Banerjee, A. Tasch, and C. Magee, "Advances in Remote Plasma-enhanced Chemical Vapor Deposition for Low Temperature *In Situ* Hydrogen Plasma Clean and Si and $\text{Ge}_x\text{Si}_{1-x}$ Epitaxy", *J. Elec. Mat.*, 21(1), p.65-74, Jan. 1992.
129. S. Batra, K. Park, S. Banerjee, G. Lux, T. Smith, J. Elliot, and B. Mulvaney, "Effect of Grain Microstructure on P Diffusion in Polycrystalline-on-Single Crystal Silicon Systems", *J. Elec. Mat.*, 21(2), p.227-232, Feb.1992.
130. L. Jung, S. Lian, B. Fowler, S. Krishnan, C.Li, D.Samara, I.Manna and S. Banerjee, "A Green's Function Growth Model for Low Temperature Silicon Homoepitaxy by ArF Excimer

- Laser-enhanced Photo Chemical Vapor Deposition,” American Phys. Soc. Meet., San Marcos, TX, March 1992.
131. R. Qian, D. Kinosky, T. Hsu, J. Irby, A. Mahajan, S. Thomas, S. Banerjee and A. Tasch, “Microstructure of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Grown by Remote Plasma-enhanced CVD”, American Phys. Soc. Meet., San Marcos, TX, March 1992.
 132. C. Li, S. Krishnan, L. Jung, D. Samara, I. Manna, S. Lian, B. Fowler and S. Banerjee, “Characteristics of Silicon Growth by Photo-enhanced Chemical Vapor Deposition Using an ArF Excimer Laser,” American Phys. Soc. Meet., San Marcos, TX, March 1992.
 133. S. Thomas, J. Irby, I. Iqbal, R. Qian, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, “Characterization of *In Situ* P-Type and N-Type Doped Si and $\text{Ge}_x\text{Si}_{1-x}$ Films Grown by Remote Plasma-enhanced Chemical Vapor Deposition”, *Proc. Mat. Res. Soc.*, v.268, p.223, 1992.
 134. R. Qian, D. Kinosky, T. Hsu, J. Irby, A. Mahajan, S. Thomas, B. Anthony, S. Banerjee, A. Tasch, and C. Magee, “Growth of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Heteroepitaxial Films by Remote Plasma-enhanced Chemical Vapor Deposition”, *J. Am. Vac. Soc.*, JVST A10(4). p.1920, April 1992.
 135. D. Kinosky, R. Qian, T. Hsu, J. Irby, A. Mahajan, S. Thomas, S. Banerjee, A. Tasch, and C. Magee, “The Dependence of Defect Density on Deposition Parameters in $\text{Ge}_x\text{Si}_{1-x}$ Heterostructures Grown by Remote Plasma-enhanced Chemical Vapor Deposition”, *Proc. Mat. Res. Soc.*, April 1992.
 136. B. Fowler, S. Lian, S. Krishnan, C. Li, L. Jung, D. Samara, I. Manna, and S. Banerjee, “Modeling of Si Deposition Yield at Low Temperature by ArF Excimer Laser Photolysis of Disilane,” *Proc. Mat. Res. Soc.*, April 1992.
 137. *S. Banerjee, “Laser Assisted Silicon Epitaxy,” **Invited** Talk, NATO Workshop on “In Situ Processing,” Viana Do Castelo, Portugal, April 1992.
 138. S. Bhattacharya, S. Banerjee, J. Lee, A. Tasch, and A. Chatterjee, “Parametric Study of Latchup-Immunity of Deep Trench-isolated, Bulk, Non-epitaxial CMOS”, *IEEE Trans. Elec. Dev.*, 39(4), p.921, April 1992.
 139. S. Krishnan, S. Lian, B. Fowler, L. Jung, C. Li, D. Samara, I. Manna and S. Banerjee, “A Parametric Study of the Growth Kinetics of Silicon Homoepitaxial Films by ArF Laser-enhanced Chemical Vapor Deposition,” *Proc. Mat. Res. Soc.*, April 1992.
 140. *S. Banerjee, A. Tasch, B. Anthony, T. Hsu, R. Qian, D. Kinosky, J. Irby, A. Mahajan and S. Thomas, “*In situ* Low Temperature Cleaning and Passivation of Silicon by Remote Hydrogen Plasma for Silicon Based Epitaxy”, **Invited** talk, *Proc. Mat. Res. Soc.*, April 1992.
 141. R. Qian, B. Anthony, T. Hsu, J. Irby, D. Kinosky, S. Banerjee, A. Tasch, and C. Magee, “Structural Analysis of $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Layers Grown by Remote Plasma-enhanced Chemical Vapor Deposition on Si (100)”, *J. Elec. Mat.*, 21(4), p.395-400, April 1992.
 142. W. Chen, J. Lin, S. Banerjee, and J. Lee, “Thermal Stability of Cobalt Disilicide for Self Aligned Silicide Applications,” *Proc. Mat. Res. Soc.*, v.260, p. 163, April 1992.
 143. C. Tsai, K. Li, D. Kinosky, R. Qian, T. Hsu, J. Irby, S. Banerjee, A. Tasch, J. Campbell, B. Hance and J. White, “A Correlation Between Silicon Hydride Species and the Photoluminescence Intensity of Porous Silicon,” *Appl. Phys. Lett.*, v. 60(14), p. 1700, April 1992.
 144. J. Lin, W. Chen, S. Banerjee, J. Lee, and C. Teng, “Study of SITOX (Silicidation Through Oxide) Process and Its Application to Advanced CMOS Devices,” *Proc. Mat. Res. Soc.*, v.260, p.623, April 1992.
 145. S. Krishnan, S. Lian, B. Fowler, L. Jung, C. Li, D. Samara, I. Manna and S. Banerjee, “A Study of the Effect of Deposition Parameters on the Growth Rates and Microstructure of Silicon Homoepitaxial Films Grown by ArF Laser-enhanced Chemical Vapor Deposition,” *Proc. of Electro. Chem. Soc.*, May 1992.
 146. J. Irby, D. Kinosky, R. Qian, B. Anthony, T. Hsu, S. Banerjee, A. Tasch, and C. Magee, “*In situ* B-Doped Si Epitaxial Films Grown at 450° C by Remote Plasma-enhanced Chemical Vapor Deposition: Physical and Electrical Characterization”, *J. Elec. Mat.*, 21(5), p.543-548, May 1992.

147. *S. Banerjee, A. Tasch, A. Mahajan, J. Irby, D. Kinosky, R. Qian and S. Thomas "Silicon ALE Using Disilane with Remote Helium Plasma Bombardment," **Invited** Talk, SRC Topical Research Conference on GeSi, Ithaca, NY, May 1992.
148. S. Lian, B. Fowler, S. Krishnan, L. Jung, C. Li, D. Samara, I. Manna and S. Banerjee, "A Green's Function Approach to a Growth Kinetic Model for Low Temperature Silicon Homoepitaxy by ArF Excimer Laser-enhanced Photo Chemical Vapor Deposition Using Disilane," *Proc. Electro. Chem. Soc.*, May 1992.
149. S. Krishnan, S. Lian, B. Fowler, L. Jung, C. Li and S. Banerjee, "Characterization of Si Homoepitaxial Films Grown with and without Surface Photo-activation by ArF Excimer Laser-induced Photodissociation of Si₂H₆," *J. Elect. Mat.*, 21(6), p. 559, June 1992.
150. J. Irby, S. Thomas, D. Kinosky, R. Qian, A. Mahajan, S. Banerjee, A. Tasch, and C. Magee, "In Situ Doped Si Films Grown by Remote Plasma-enhanced Chemical Vapor Deposition", Elec. Mat. Conf., Boston, MA, June 1992.
151. R. Qian, D. Kinosky, T. Hsu, J. Irby, A. Mahajan, S. Thomas, S. Banerjee and A. Tasch, "Microstructure of Ge_xSi_{1-x} /Si Grown by Remote Plasma-enhanced CVD", Elec. Mat. Conf., Boston, MA, June 1992.
152. S. Thomas, J. Irby, D. Kinosky, R. Qian, A. Mahajan, S. Banerjee, A. Tasch, and C. Magee, "In Situ B and P Doped Si Epitaxial Films Grown by Remote Plasma-enhanced Chemical Vapor Deposition", Am. Vac. Soc. Meeting, Dallas TX, June 1992.
153. A. Mahajan, D. Kinosky, R. Qian, J. Irby, S. Thomas, S. Banerjee and A. Tasch, "Silicon Atomic Layer Epitaxy Using Remote Helium Plasma Assisted Hydrogen Desorption and Disilane as a Precursor", Am. Vac. Soc. Meeting, Dallas TX, June 1992.
154. D. Kinosky, A. Mahajan, J. Irby, R. Qian, S. Thomas, S. Banerjee and A. Tasch, "A Simple Controller for Repetitive Cycles in Atomic Layer Epitaxy," Second International ALE Symposium, Raleigh NC, June 1992.
155. A. Mahajan, J. Irby, D. Kinosky, R. Qian, S. Thomas, S. Banerjee, A. Tasch and T. Picraux, "Silicon Atomic Layer Epitaxy Based on Disilane and Remote Helium Plasma Bombardment," Second International ALE Symposium, Raleigh NC, June 1992.
156. J. Lin, W. Chen, S. Banerjee, and J. Lee, "Cobalt Disilicide as a Dopant Diffusion Source for Polysilicon Gates in MOS Devices," Elec. Mat. Conf., Boston, MA, June 1992.
157. S. Yoganathan, and S. Banerjee, "A New Decoupled Algorithm for Non-Stationary Transient Simulation of GaAs MESFETs", *IEEE Trans. Elec. Dev.* 39(7), p.1578-1587, July 1992.
158. B. Fowler, S. Lian, S. Krishnan, L. Jung, and S. Banerjee, "ArF Excimer Laser-enhanced Photo-Chemical Vapor Deposition of Epitaxial Si from Si₂H₆: A Simple Growth Kinetic Model", *J. Elec. Mat.*, 21(8), p.791-798, August 1992.
159. S. Lian, B. Fowler, S. Krishnan, L. Jung, B. Li and S. Banerjee, "Si Homoepitaxial Films Grown by Photo-enhanced Chemical Vapor Deposition From Si₂H₆", *J. Electrochem. Soc.*, 139(8), p.2273-2277, Aug. 1992.
160. B. Fowler, S. Lian, S. Krishnan, L. Jung, and S. Banerjee, "ArF Laser-enhanced Photo-CVD of Silicon from Disilane", *J. Electrochem. Soc.*, 139(8), p.2314-2318, Aug. 1992.
161. T. Hsu, S. Lin, B. Anthony, R. Qian, J. Irby, D. Kinosky, A. Mahajan, S. Banerjee, A. Tasch, and H. Marcus, "Hydrogen Desorption on Various H-Terminated Surfaces due to Electron Beam Irradiation: Experiments and Modeling", *Appl. Phys. Lett.*, 61(5), p.580-582, Aug. 1992.
162. B. Fowler, S. Lian, S. Krishnan, L. Jung, C. Li, D. Samara, I. Manna, and S. Banerjee, "Epitaxial Silicon Growth Conditions and Kinetics in Low Temperature ArF Excimer Laser Photochemical Vapor Deposition from Disilane," *J. Appl. Phys.*, 72(3), Aug. 1992.
163. S. Yoganathan and S. Banerjee, "An Efficient Two-dimensional Hydrodynamic Simulator for GaAs MESFETs", *Proc. of CEFC*, p. TOD3, August 1992.

164. S.Bhattacharya, R.Kovelamudi, S.Batra, M.Lobo, S.Banerjee, B.Nguyen and P.Tobin, "Hot Carrier Effects in Thin Film, P-channel, H-Passivated Polysilicon-on-Insulator LDD MOSFETs," *Proc. SPIE*, Sept., 1992.
165. S.Bhattacharya, R.Kovelamudi, S.Batra, S.Banerjee, B.Nguyen and P.Tobin, "Parallel Hot Carrier Induced Degradation Mechanisms in H-Passivated, Polysilicon p-MOSFETs," *IEEE Elec. Dev. Lett.*, v.13(9), p.491, Sept. 1992.
166. S.Banerjee, "Laser-enhanced CVD Using Disilane for Low Temperature Si Epitaxy," **Invited** paper, *Proc. SPIE*, Sept. 1992.
167. S.Bhattacharya, S.Banerjee, B.Nguyen and P.Tobin, "Hot Carrier Induced Degradation Mechanisms and Anomalous Leakage Current in H-Passivated, Polysilicon-on-Insulator LDD p-MOSFETs," *IEEE SOI. Conf.*, Pointe Verde, FL, Oct. 1992.
168. B. Fowler, S. Lian, S. Krishnan, L. Jung, C. Li, D. Samara, I. Manna, and S. Banerjee, "Mechanisms of low temperature epitaxial silicon growth using ArF excimer laser photochemical vapor deposition from disilane," *Thin Solid Film*, v. 218, p.48, Nov. 1992.
169. R. Qian, D. Kinosky, A. Mahajan, S. Thomas, S. Banerjee, and A. Tasch, C. Magee, "Growth of Ge-on-Si Structures Using Remote Plasma-enhanced Chemical Vapor Deposition", *Proc. Mat. Res. Soc.*, 1992.
170. D. Kinosky, R. Qian, A.Mahajan, S.Thomas, S. Banerjee and A. Tasch, "Remote Plasma Cleaning and Ion-induced Desorption from the Silicon (100) Surface and Its Application to Si Epitaxy", *Mat.Res. Soc. Symp.*, Dec. 1992.
171. S.Bhattacharaya, M.Lobo, S.Banerjee, and R.Reuss, "Control of Polysilicon Emitter Bipolar Transistor Characteristics by Rapid Thermal or Furnace Anneal of the Polysilicon/Silicon Interface," *Mat.Res. Soc.Symp.*, Dec. 1992,.
172. A.Sultan, S.Batra, M.Lobo, K.Park and S.Banerjee, "Modeling of Boron Diffusion in Polysilicon-on-Silicon Layers," *Mat.Res. Soc.Symp.*, Dec. 1992.
173. S.Bhattacharya, S.Banerjee, B.Nguyen and P.Tobin, "An Analytical Model for the Temperature Dependence of Leakage Current in Polysilicon-on-Insulator (SOI) MOSFETs," *Proc. of Electro. Chem. Soc.*, Oct. 1992.
174. I.Manna, C.Li, L.Jung, S.Lian, D.Samara, B.Fowler and S.Banerjee, "Role of de-ionized water rinse on the surface preparation for low T Si epi film growth by photo CVD," *SISC*, San Diego, CA, Dec. 1992.
175. Wei-Ming Chen, Jengping Lin, Sanjay K. Banerjee, and Jack C. Lee, "Thermal stability and dopant drive-out characteristics of CoSi₂ polycide gates," *J.App. Phys.*, 1993.
176. D. Kinosky, R. Qian, A. Mahajan, S. Thomas, S. Banerjee, A. Tasch, and C. Magee, "Control of Deposition Rate in Remote Plasma-enhanced Chemical Vapor Deposition of Ge_xSi_{1-x}/Si Heteroepitaxial films", *J. Vac. Sci. and Tech.*, B 11(4), Jul/Aug 1993.
177. D. Kinosky, R. Qian, A. Mahajan, S. Thomas, P. Munguia, J. Fretwell, S. Banerjee, A. Tasch, and C. Magee, "HF/Alcohol Preparation of Wafers for the Reduction of Haze in Low Temperature Si Epitaxy by Remote Plasma Chemical Vapor Deposition", *Mat. Res. Soc. Proc.*, 1993.
178. S.Batra, K.Park, S.Banerjee, G.Lux and R.Manukonda "Discontinuity of Boron Diffusion Profiles at the Interface of Polycrystalline/ Single Crystal Silicon," *J. Appl. Phys.*, 73(8), p. 3800, April 1993.
179. A.Sultan, S.Bhattacharaya, S.Batra and S.Banerjee, "A Boltzmann- Matano Analysis Based Model for B Diffusion from Polysilicon into Single Crystal Si," 2nd Int. Workshop on Measurement of Ultra-Shallow Doping Profiles, Raleigh, NC, May 1993.
180. D. Kinosky, R. Qian, A.Mahajan, S.Thomas, S. Banerjee and A. Tasch, "Control of Deposition Rate in Remote Plasma Chemical Vapor Deposition of Ge_xSi_{1-x}/Si Heteroepitaxial Films," *Phys. Chem. Sem. Int.-20*, Williamsburg, VA, Jan. 1993.

181. C.Li, S.Lian, B.Fowler, L.Jung, C.Li, D.Samara, I.Manna, and S.Banerjee, "Characterization of $\text{Si}_{1-x}\text{Ge}_x/\text{Si}$ Heterostructure Grown by ArF Laser-induced Photochemical Vapor Deposition Using Disilane and Digermane," TMS, Denver, CO, Feb. 1993.
182. L.Jung, B.Fowler, C.Li, I.Manna, D.Samara, S.Lian, and S.Banerjee, "Optical Multichannel Analysis of ArF Laser-induced Photochemical Vapor Deposition of Si and $\text{Si}_{1-x}\text{Ge}_x$ Epitaxial Films," TMS, Denver, CO, Feb. 1993.
183. S. Lian, B. Fowler, S. Krishnan, L. Jung, C. Li, I. I. Manna, D. Samara, and S. Banerjee, "Photo-enhanced Chemical Vapor Deposition System Design Considerations," *J. Vac. Sci. Tech.*, v.A11(6), p.2194-2923, Nov/Dec 1993.
184. A.Mahajan, J. Irby, D. Kinosky, R. Qian, S.Thomas, S. Banerjee, A. Tasch and T.Picraux, "Silicon Atomic Layer Epitaxy Based on Disilane and Remote Helium Plasma Bombardment," *Thin Sol. Films*, v.225, p.177, 1993.
185. J.Lin, W.Chen, S.Banerjee, and J.Lee, "Cobalt Disilicide as a Dopant Diffusion Source for Polysilicon Gates in MOS Devices," *J. Elec. Mat.*, v. 22, p. 667, 1993.
186. S. Batra, K. Picone, M. Lobo, S. Bhattacharya, K. Park, J. Lee, and S. Banerjee, "Leakage Characteristics and Lateral Uniformity of Ultra-Shallow Junctions Formed Using Polysilicon and Amorphous Silicon Diffusion Sources", *Sol. State. Elec.*, v.36, p.955, July 1993.
187. S.Batra, N.Jeng, A.Sultan, K.Picone, S.Bhattacharya, K.Park, S.Banerjee, D.Kao, M.Manning and C.Dennison, "Effect of Epitaxial Realignment on the Leakage Behavior of Arsenic Implanted, As-deposited Polycrystalline Si-on-single Crystal Si Diodes," *J.Elec.Mat.*, v.22(5), p.551, 1993.
188. A.Mahajan, B.Kellerman, N.Russell, D.Bonser, J.Ekerdt, A.Campion, S.Banerjee, A.Tasch and J.White, "Adsorption and Desorption of Diethylsilane and Diethylgermane from Si (100): Surface Kinetics for an ALE Approach of Col. IV Epitaxy," 40th Nat. Am. Vac. Soc. Conf. Orlando, FL, Nov. 1993.
189. N.Russell, M.Tiner, J.Ekerdt, D.Bonser, B.Kellerman, A.Campion, J.White, A.Mahajan, S.Banerjee and A.Tasch, " Surface Chemistry Studies of DES and DEG on Si (100)", Ame. Vac. Soc. Meeting, June 1993, Austin, TX.
190. B.Kellerman, A.Mahajan, N.Russell, D.Bonser, J.Ekerdt, A.Campion, S.Banerjee, A.Tasch and J.White, "Adsorption and Desorption of Diethylsilane and Diethylgermane from Si (100): Surface Kinetics for an ALE Approach of Col. IV Epitaxy," American Chemical Society Meeting, Austin, TX, Oct. 1993.
191. I.Manna, C.Li, D.Samara, S.John, A.Lentvorski and S.Banerjee, "Characterization and Modeling of $\text{Si}_x\text{Ge}_{1-x}$ Heterostructure Films Grown by ArF Laser-induced Photochemical Vapor Deposition Using Disilane and Digermane," Elec. Mat. Proc. and Char. Symp., Austin, TX, June 1993.
192. C.Li, S.John, I.Manna, D.Samara, A.Lentvorski, S.Banerjee and M.White, "Photo-enhanced Epitaxy of Si/Ge Layers," American Chemical Society Meeting, Austin, TX, Oct. 1993.
193. J.Fretwell, R.Qian, D.Kinosky, A.Mahajan, P.Munguia, S.Banerjee and A.Tasch, "Effect of Ge Content on Dopant Transition Widths in $\text{Ge}_x\text{Si}_{1-x}$ Films Grown by RPCVD, " American Chemical Society Meeting, Austin, TX, Oct. 1993.
194. D.Kinosky, R.Qian, A.Mahajan, S.Thomas, P.Munguia, J.Fretwell, S.Banerjee and A.Tasch, "HF/Alcohol Preparation of Wafers for the Reduction of Haze in Low Temperature Si Epitaxy by RPCVD," American Chemical Society Meeting, Austin, TX, Oct. 1993.
195. D.Kinosky, R.Qian, A.Mahajan, S.Thomas, P.Munguia, J.Fretwell, S.Banerjee and A.Tasch, "Deposition Rate in RPCVD of $\text{Ge}_x\text{Si}_{1-x}$ Heteroepitaxial Films," American Chemical Society Meeting, Austin, TX, Oct. 1993.
196. L.Jung, S.Bhattacharya, R.Reuss and S.Banerjee, "The Temperature Dependence of the Current Gain of Polysilicon Emitter BJT's," TECHCON, Atlanta, GA, Sept. 1993.

197. A.Sultan, M.Lobo, S.Bhattacharya, S.Banerjee, S.Batra, M.Manning and C.Dennison" A Physically-based Phenomenological Model Using Boltzmann-Matano Analysis for Boron Diffusion from Polysilicon into Single Crystal Silicon," *J.Elec. Mat.*, v.22(9), p.1129-1135, Sept. 1993.
198. I.Manna, S.Bhattacharya and S.Banerjee, "A New Hot Carrier Resistant P-I-N MOSFET Structure," *Proc. SRC TECHCON*, v. 3, p.248-250, October, 1993.
199. S.Murtaza, R.Qian, D.Kinosky, R.Mayer, A.Tasch, S.Banerjee and J.Campbell, "Room Temperature Measurements of Strong Electroabsorption Effects in $\text{Ge}_x\text{Si}_{1-x}/\text{Si}$ Multiple Quantum Wells Grown by Remote Plasma-enhanced Chemical Vapor Deposition," *Appl. Phys. Lett.*, 62(16), p. 1976, 1993.
200. W.Chen, J.Lin, S.Banerjee and J.Lee, "The Impact of Pre-silicidation Heat Treatment and Dopant Effects on the Thermal Stability of CoSi_2 Polycide During Rapid Thermal Annealing," *Proc. Mat. Res. Soc.*, vol. 303, 1993.
201. W.Chen, J.Lin, S.Banerjee and J.Lee, "Using CoSi_2 Polysilicon/Polycide Structure as Gate Diffusion Source in Rapid Thermal Processing," *Proc. Mat. Res. Soc.*, vol. 303, 1993.
202. J.Lin, W.Chen, S.Banerjee and J.Lee, "Enhanced B Diffusion in Si Using CoSi_2 Diffusion Source and Rapid Thermal Annealing," *Proc. Mat. Res. Soc.*, vol. 303, 1993.
203. A.Sultan, S.Bhattacharya, S.Batra and S.Banerjee, "A Boltzmann-Matano Analysis Based Model for B Diffusion from Polysilicon into Single Crystal Si," *Proc. TECHCON*, v.3, p.43-45, October, 1993.
204. D. Kinosky, R. Qian, A. Mahajan, S. Thomas, P. Munguia, J. Fretwell, S. Banerjee, and A. Tasch, "Hydrogen Plasma Cleaning of the Si(100) Surface: Removal of Oxygen and Carbon and the Etching of Si", *Mat. Res. Soc. Proc.*, 1993.
205. N.Russell, S.Asami, J.Dadap, X.Hu, J.Fretwell, M.Downer, S.Banerjee, A.Tasch, J.White and J.Ekerdt, "Surface Chemistry basis for kinetically self-limited epi growth from dsilane/diegermane mixtures," AICHE Meeting, San Francisco, Nov. 1994.
206. *S.Banerjee, "Low Thermal Budget Si Epitaxy," *TMS Laser Materials Processing* , v.4, p.281-294, 1994.
207. A.Sultan, S.Bhattacharya, S.Batra and S.Banerjee, "A Boltzmann-Matano Analysis Based Model for B Diffusion from Polysilicon into Single Crystal Si," *J. Vac.Sci.Tech.*, v. B12, p.391-4, Jan/Feb., 1994.
208. S.Bhattacharya, S.Banerjee, B.Nguyen and P.Tobin, "Temperature Dependence of the Anomalous Leakage Current in Polysilicon-on-Insulator MOSFETs," *IEEE Trans. Elec. Dev.*, v.41(2), p.221-227, Feb. 1994.
209. W.Chen, J.Lin, S.Banerjee, and J.Lee, "A Simultaneous Shallow-junction Formation and Gate Doping P-MOSFET Process Using Cobalt Silicide as a Diffusion Source (SADDs)," *Appl. Phys.Lett.*, v. 64, p. 345-347, Jan. 1994.
210. W.Chen, J.Lin, S.Banerjee and J.Lee, "Degradation Mechanisms and Improvement of Thermal Stability of CoSi_2 Polycide Layers," *Appl. Phys. Lett.*, 64(12), p. 1505-7, March 1994.
211. L.Jung, C.Li, I.Manna, D.Samara, A.Lentvorsi, S.John, B.Fowler, S.Lian and S.Banerjee, "Optical Multichannel Analysis of Laser-induced Fluorescence During ArF Excimer Laser Photolysis of Disilane," *TMS Laser Materials Processing* , v.4, p.267-280, 1994.
212. A.Mahajan, B.Kellerman, N.Russell, S.Banerjee, A.Campion, J.Ekerdt, A.Tasch, J.White and D.Bonser, "Surface Chemistry of Diethylsilane and Diethygermane on Si (100): An ALE Approach," *J. Am. Vac. Soc.* A 12(4), p. 2265-2270, July 1994.
213. I.Manna, S.Bhattacharya and S.Banerjee, "A New Hot Carrier Resistant P-I-N MOSFET Structure," *Elec. Lett.*, v. 30(5), p. 457, 1994.
214. *S.Banerjee, "Laser-Enhanced Epitaxy of Si and Si-Ge," **Invited** talk, IEEE Engineering Foundation, Palm Coast, FL, May 1994.

215. S.Asami, N.Russell, A.Mahajan, P.Steiner, D.Bonser, J.Fretwell, S.Banerjee, A.Tasch, J.White and J.Ekerdt, "Adaptive Temperature Program ALE of Short Period $\text{Si}_{1-x}\text{Ge}_x$ Heterostructures from $\text{Si}_2\text{H}_6/\text{Ge}_2\text{H}_6$," 3rd. Int. Symp. on Atomic Layer Epitaxy and Related Processes (ALE-3), May 25-27, 1994 Sendai, Japan.
216. L.Jung, J.Damiano, S.Banerjee, S.Batra, M.Manning, C.Dennison and N.Saks, "A Leakage Current Model for Sub-Micron Drain-Offset Polysilicon TFT's," Electrochem. Soc. Meet., Oct. 1994, Miami, FL.
217. L.Jung, I.Manna, S.Bhattacharya and S.Banerjee, "A New Hot Carrier Resistant P-I-N MOSFET Structure," SPIE, Oct. 1994, Austin, TX.
218. J.Damiano, L.Jung, S.Banerjee, S.Batra, M.Manning, C.Dennison and N.Saks, "Analysis and Modeling of Sub-Micron Drain-Offset Polysilicon TFT's," SPIE, Oct. 1994, Austin, TX.
219. A.Mahajan, B.Kellerman, N.Russell, S.Banerjee, J.Ekerdt, A.Tasch and J.White, "Surface Chemistry of Potential CVD Precursors, Diethylsilane and Diethylgermane, on Ge(100)," 41st Nat. Am. Vac. Soc. Conf. Denver, CO, Oct. 1994.
220. I.Manna, H.Taufique, A.Pandelides and S.Banerjee, "Si and Si-Ge Epitaxial Layers Grown by Low Temperature Rapid-Thermal Chemical Vapor Deposition from DCS/GERmane/Hydrogen," Mat. Res. Soc.Symp., Nov. 1994, Boston, MA.
221. A.Mahajan, B.Kellerman, N.Russell, J.Heitzinger, S.Banerjee, J.Ekerdt, A.Tasch and J.White, "Surface Chemistry of Diethylsilane and Diethylgermane on Si (100) and Ge(100) Surfaces," Mat. Res. Soc.Symp., Nov. 1994, Boston, MA.
222. S.Murtaza, R.Mayer, M.Rashed, D.Kinosky, C.Maziar, S.Banerjee, A.Tasch, J.Campbell, J.Bean and L.Peticolas, "Room-temperature Electroabsorption in a $\text{Ge}_x\text{Si}_{1-x}$ Photodiode," *IEEE Trans. Elec. Dev.*, 41(12), p. 2297-2300, Dec. 1994.
223. M.Tesaro, A.Campion and S.Banerjee, "Hydrogen Desorption from Si Surfaces Using Ar and He Ion Bombardment," *Surf. Sci.Lett.*, 318, L1171-1174, 1994.
224. S.Asami, N.Russell, A.Mahajan, P.Steiner, D.Bonser, J.Fretwell, S.Banerjee, A.Tasch, J.White and J.Ekerdt, "Adaptive Temperature Program ALE of Short Period $\text{Si}_{1-x}\text{Ge}_x$ Heterostructures from $\text{Si}_2\text{H}_6/\text{Ge}_2\text{H}_6$," *Appl. Surf. Sci.*, v. 82-83, p. 359 (1994).
225. I.Manna, A.Pandelides and S.Banerjee, "Photo-assisted enhancement for growth rate of SiGe Alloys using RTCVD," APS Meeting, Oct. 1994, Austin, TX.
226. C.Hu, S.Banerjee, J.Lee, K.Sadra, B.Streetman and R.Sivan, "Impact of hole quantization in MOS Structures on Effective Oxide Thickness and Threshold Voltage," APS Meeting, Oct. 1994, Austin, TX.
227. J.Fretwell, B.Doris, D.Diaz, C.Cariss, R.Gupta, S.Banerjee and A.Tasch, "Effects of wafer preparation on epi surface morphology and crystallinity as grown by RPCVD," APS Meeting, Oct. 1994, Austin, TX.
228. B.Kellerman, A.Mahajan, N.Russell, J.Ekerdt, S.Banerjee, A.Tasch, A.Campion, J.White and D.Bonser, "Adsorption and Decomposition of DES and DEG on Si(100): Surface Reactions for an ALE Approach to Column IV Epitaxy," *J.Vac.Sci. Tech. A*, 13(4), p. 1819-1825, July 1995.
229. A.Sultan, S.Batra, G.Lux and S.Banerjee, "Modeling of Boron Diffusion in Polysilicon-on-Silicon Layers Using a Rapid Thermal Anneal Step for Ultr-shallow Junction Formation," *Mat. Sci. Eng. B32*, p. 25-32, 1995.
230. S.Thomas, J. Fretwell, D. Kinosky, R. Qian, A. Mahajan, P. Munguia, S. Banerjee, A. Tasch and C.Magee, "In Situ P Doped Si and $\text{Si}_{1-x}\text{Ge}_x$ Epitaxial Films Grown by RPCVD", *J. Elec.Mat.*, 24(3), p. 183-188, March 1995.
231. C.Y.Hu, D.Kencke, S.Banerjee, B.Bandopadhyay, E.Ibok and S.Garg, "Determining Effective Dielectric Thickness of Metal-oxide-semiconductor Structures in Accumulation Mode," *Appl. Phys. Lett.* 66(13), 1638-40, March 1995.

232. A.Sultan, M.Craig, K.Reddy, S.Banerjee, E.Ishida, P.Maillot, T.Neil and L.Larson, "The Dependence of Ultra-shallow Junction Depths on Implant Dose Rates," *App. Phys. Lett.*, 67(9), p. 1223-1225, August 1995.
233. L.Jung, I.Manna and S.Banerjee, "Simulation, Fabrication and Characterization of P-I-N Drain MOSFET Structure for Hot Carrier Suppression," *IEEE Trans. Elec. Dev.*, p. 1591-1599, Sept. 1995.
234. C.Li, S.John and S.Banerjee, "Low Temperature Heteroepitaxial Growth of $\text{Si}_{1-x}\text{Ge}_x$ -on-Si by Photo-enhanced Ultra-High Vacuum Chemical Vapor Deposition Using Si_2H_6 and Ge_2H_6 ," *J.Elec. Mat.* 24(7), p.875-884, July 1995.
235. A.Sultan, M.Craig, K.Reddy, S.Banerjee, E.Ishida, P.Maillot and L.Larson, "The Effect of Implant Dose Rates and Two-Step Anneals on P+-N Ultra-Shallow Junctions," *Proc. of UGIM Symp.*, v. 11, p. 108-112, May 1995.
236. M.Craig, A.Sultan, K.Reddy, E.Ishida, L.Larson, and S.Banerjee, "Enhanced Diffusion Suppression via Two-Step Anneals in Low Energy BF_2 -Implanted Ultra-shallow Junctions," 3rd Int. Workshop on Ultra-shallow Profiles, Raleigh, NC, March 1995.
237. *A.Tasch and S.Banerjee, "Ultra-shallow Junction Formation in Si Using Low Energy Ion Implantation," **invited** talk, European Mat. Res. Soc. Strasbourg, France, May 1995.
238. I.Manna, L.Jung and S.Banerjee, "A Novel Si/SiGe Sandwich Polysilicon TFT Structure for SRAM Applications," IEEE Dev. Res. Conf., Charlottesville, VA, June 1995.
239. K.Pacheo, B.Ferguson, S.Banerjee and B.Mullins, "Homoepitaxial Deposition of Si Employing Supersonic Jets of Disilane," 14th Annual Symp. on Elec. Mat. Proc. and Characterization, Austin, Tx, June 1995.
240. K.Pacheo, B.Ferguson, S.Banerjee and B.Mullins, "Growth and Characterization of Si Thin Films Employing Supersonic Jets," 42nd Ann. Symp. of Am. Vac. Soc., Minneapolis MN, Oct. 1995.
241. A.Mahajan, R.Gupta, J.Ekerdt, A.Tasch and S.Banerjee, "Deposition and Characterization of Delta-doped Layers in Si Using Self-saturating Adsorption of PH_3 and B_2H_6 on Si(100)," Elec. Mat. Conf., Charlottesville VA, June 1995.
242. *S.Banerjee, "Low Energy Ion Implantation for Ultra-shallow Junction Technology in the ULSI Era," SEMICON Southwest Meeting, Austin TX, Oct. 26, 1995.
243. A.Mahajan, B.Kellerman, J.Heitzinger, S.Banerjee, A.Tasch, J.White and J.Ekerdt, "Surface Chemistry of Diethylsilane and Diethylgermane on Ge (100)," *J. Am. Vac. Soc. A* 13, p. 1 (1995).
244. L.Jung, J.Damiano, S.Batra, M.Manning and S.Banerjee, "A Leakage Current Model for Sub-micron Lightly-doped Drain Offset Polysilicon TFTs," *Solid State Elec.*, v.38(12), p. 2069-2073, Dec. 1995.
245. K.Pacheo, B.Ferguson, S.Banerjee and B.Mullins, "Epitaxial Si Growth Employing Supersonic Jets of Disilane: A Model Study of Energetic Jet Deposition," *Appl. Phys.Lett.*, 67(20), p. 2951-2953, November 1995.
246. C. Hu, D.Kencke, S.Banerjee, R.Richart, B.Bandopadhyay, B.Moore, E.Ibok and S.Garg, "Analysis of Substrate-Bias-Enhanced Hot Electron Injection for Self Convergence of Over-Erased Flash Cells," 14th Ann. IEEE NonVol. Semi. Mem. Workshop, Monterey CA, Aug. 1995.
247. J.Fretwell, D.Diaz, S.Banerjee, A.Tasch, and B.Doris, "Study of Effects Resulting from Varied Ex Situ and In Situ Cleaning Steps for Low Temperature Si Deposition by RPCVD," Mat. Res. Soc. Symp. Spring 1995.
248. *A.Tasch and S.Banerjee, "Low Temperature Si-based Epitaxial Growth with RPCVD," Am. Phys. Soc. Meet., **Invited** talk, Austin TX, Oct. 1995.
249. C.Cariss, S.Banerjee and J.Lowell, "Surface Charge Detection on Epitaxial Si Films Grown by RPCVD Using Surface Photovoltage Measurements," Am. Vac. Soc. Symp., Austin TX, June 1995.

250. D.Samara, J.Williamson, C.Shih and S.Banerjee, "STM Induced Deposition of Semiconductor Quantum Dots," STM Conf., Denver CO, June 1995.
251. M. Craig, A. Sultan, K. Reddy, E. Ishida, L. Larson, and S. Banerjee, "Dose Rate and Thermal Budget Optimization for Ultra-shallow Junctions Formed by Sub-5 keV Ion Implantation," 3rd International Workshop on the Measurement & Characterization of Ultra-shallow Doping Profiles in Semiconductors, Research Triangle Park, NC, March, 1995.
252. M. Craig, A. Sultan, K. Reddy, E. Ishida, L. Larson, and S. Banerjee, "Dose Rate and Thermal Budget Optimization for Ultra-shallow Junctions Formed by Sub-5 keV Ion Implantation," *Proc. Workshop on the Measurement & Characterization of Ultra-shallow Doping Profiles in Semiconductors, March, 1995.*
253. I.Manna, K.Liu, and S.Banerjee, "Novel High Performance Polysilicon Heterostructure TFTs Using P-I-N Source/Drains, *Proc. Int. Elec. Dev. Meet., Dec. 1995.*
254. C. Hu, D.Kencke, S.Banerjee, R.Richart, B.Bandopadhyay, B.Moore, E.Ibok and S.Garg, "Substrate-Current-Induced Hot Electron (SCIHE) Injection: A New Convergence Scheme for Flash Memory," *Proc. of Int. Elec. Dev.Meet., Dec. 1995.*
255. S.John, E.Quinones, B.Ferguson, K.Pacheo, B.Mullins and S.Banerjee, "Progression of the Surface Roughness of N+ Epitaxial Films as Analyzed by AFM," *Proc. of Mat. Res. Soc.: Evolution of Thin Film and Surface Microstructure, Nov. 1995.*
256. M. Craig, A. Sultan, K. Reddy, E. Ishida, L. Larson, and S. Banerjee, "Dose Rate and Thermal Budget Optimization for Ultra-shallow Junctions Formed by Sub-5 keV Ion Implantation," *J.Vac.Sci. Tech.B 14(1), p. 255-259, Jan/Feb 1996.*
257. C. Hu, D.Kencke, S.Banerjee, R.Richart, B.Bandopadhyay, B.Moore, E.Ibok and S.Garg, "A Convergence Scheme for Over-erased Flash EEPROMs Using Substrate-bias-enhanced Hot Electron Injection," *Elec. Dev.Lett. 16(11), p. 500, Nov. 1995.*
258. K.Pacheo, B.Ferguson, S.Banerjee and B.Mullins, "Surface Morphology of Homoepitaxial Si Thin Films Grown Using Energetic Supersonic Jets of Disilane," *Appl. Phys.Lett., 69(8), p. 1110-1112, Aug. 1996.*
259. *A.Tasch and S.Banerjee, "Ultra-shallow Junction Formation in Si Using Low Energy Ion Implantation," **invited** paper, *Nuc. Inst. and Methods B, 112, p. 177-183, 1996.*
260. D.Samara, J.Williamson, C.Shih and S.Banerjee, "STM Induced Deposition of Semiconductor Quantum Dots," *J.Vac. Sci. Tech.B, 14(2), p. 1344-1352, Mar/Apr. 1996.*
261. *S.Banerjee, "Laser-Enhanced Epitaxy of Si and Si-Ge," **invited** paper, *Materials Science & Eng. B, 1996.*
262. C.Li, S.John, E.Quinones and S.Banerjee, "Cold-wall UHVCVD of Doped and Undoped Si and Si_{1-x}Ge_x Epitaxial Films Using SiH₄ and Si₂H₆," *J.Vac.Sci. Tech.A 14(1), p. 170-183, Jan/Feb 1996.*
263. C.Hu, S.Banerjee, K.Sadra, B.Streetman and R.Sivan, "Quantization Effects in Inversion Layers of PMOSFETs on Si(100) Substrates," *Elec. Dev. Lett., 17(6), p. 276-278, June 1996.*
264. S. John, E. Quinones, B.Ferguson, C.Mullins and S.Banerjee, "UHV-CVD of Doped and Undoped Epitaxial SiGeC," *Elec. Mat. Conf., June 1996.*
265. R. Sharma, J. Fretwell, B. Doris and S. Banerjee: "Molybdenum Contamination in Low Temperature Epitaxial Silicon Films Grown by Remote Plasma Chemical Vapor Deposition", *Appl. Phys. Lett., 69 (1), pp.109-111, July 1, 1996.*
- A. Sultan, S. Banerjee, S. List, and Mark Rodder, "Role of Silicon Surface in the Removal of Point Defects," *Proc. 11th International Conf. on Ion Implantation Technology, p. 615-618, June 1996.*
266. Sultan, M. Craig, S. Banerjee, S. List, T. Grider, and V. McNeil, "Effect of F on B Penetration Through Gate Oxide for BF(2) Implants Used to Obtain Ultra-shallow Junctions by RTA," *Proc. 11th International Conf. on Ion Implantation Technology, p. 29-32, June 1996.*

267. Sultan, S. Banerjee, S. List, G. Pollack, and H. Hosack, "Evaluation of Si Pre-amorphization for Obtaining Ultra-shallow Junctions," *Proc. 11th International Conf. on Ion Implantation Technology*, p.25-28, June 1996.
268. M. Craig, A. Sultan, S. Banerjee, E. Ishida, and L. Larson, "Carbon Co-implantation for Ultra-shallow p(+)-n Junction Formation," *Proc. 11th International Conf. on Ion Implantation Technology*, p. 665-668, June 1996.
269. Sultan, I. De, S. Banerjee, S. List, and M. Rodder, "Role of Si Surface in the Removal of Point Defects in Ultra-shallow Junctions," *Proc. TECHCON*, 1996.
270. S.R. Nandan, A. Sultan, M. Craig, and S. Banerjee, "Use of Carbon Co-implantation to Reduce TED and Achieve Shallower P-type Junctions," *Proc. TECHCON 1996*.
271. S.Batra, M.Manning, C.Dennison and S.Banerjee, "A New Submicron CMP-based TFT Technology for Future High Density SRAMs," Oct. Electrochem Soc. Meet, San Antonio, TX, 1996.
272. R.Zaman, S.Banerjee, J.Damiano, S.Batra and M.Manning, "Characterization of Polysilicon TFT Gate Dielectrics," Electrochem Soc. Meet, San Antonio, TX, Oct. 1996.
273. R.Zaman, S.Banerjee, S.Batra and M.Manning, "Study of Polysilicon TFTs for High density SRAM Applications," Electrochem Soc. Meet, San Antonio, TX, Oct. 1996.
274. S.Ray, S.John, S.Oswal and S.Banerjee, "Novel SiGeC Channel Heterojunction PMOSFET," *Proc. of Int. Elec. Dev.Meet.*, 1996.
275. A. Sultan, S. Banerjee, S. List, and M. Rodder, "Role of Silicon Surface in the Removal of Point Defects in Ultra-shallow Junctions," *Appl. Phys. Lett.* 69(15), p.2228, Oct. 1996.
276. D.Kencke, R.Richart, S.Garg and S.Banerjee, "A Sixteen Level Scheme for Multilevel Flash/EEPROM's," *Proc. IEDM*, 1996.
277. S. John, E. J. Quinones, B. Ferguson, S. K. Ray, C. B. Mullins, and S. K. Banerjee, "Surface morphology of SiGeC epitaxial films deposited by low temperature UHV-CVD," *Proceedings of the MRS-Thin Films Surfaces: Morphology and Roughening I*, November 1996.
278. *S.Banerjee, "Shallow Junction Implants: As, B, BF₂, C, Ge, In, Si," **Invited** talk, Silicon Valley Ion Implant Users Group Meeting, San Jose, CA, Nov. 19, 1996.
279. *S.Banerjee, "Defect Engineering for Ion Implanted Ultra-shallow Junctions," **Invited** talk, American Vac. Soc. Meet., Austin, TX, June 4, 1997.
280. *S.Banerjee, "Defect Engineering with N and C Co-implants for Ultra-shallow Junctions," **Invited** Talk, National Ion Implant Users Meeting, Austin, TX, Oct. 1997.
281. Doris, J. Fretwell, J. L. Erskine, and S. K. Banerjee, "Effects of *In-situ* doping from B₂H₆ and PH₃ on hydrogen desorption and the Low Temperature Growth Mode of Si on Si(100) by Remote Plasma Enhanced Chemical Vapor Deposition," *Appl. Phys. Lett.*, 70(21), May 1997, pp. 2819-2821.
282. R. Sharma, J. Fretwell, B. Doris and S. Banerjee, "Use of Metal-Oxide-Semiconductor Capacitors in the Analysis of Low-temperature Epitaxial Si Films deposited by Remote Plasma-Enhanced Chemical Vapor Deposition," *J. of Appl. Phys.*, 82(5), Sept. 97, pp. 2684-89.
283. S. John, S. K. Ray, S. K. Oswal, and S. K. Banerjee, "Novel Si_{1-x-y}Ge_xC_y Channel heterojunction PMOSFET," *Proc. SPIE* 1997.
284. S. John, S. K. Ray, S. K. Oswal, and S. K. Banerjee, "Strained Si NMOSFET on relaxed Si_{1-x}Ge_x formed by ion implantation of Ge," *Proc. SPIE* 1997.
285. *S.Banerjee, "Applications of Si-Ge-C in MOS and Bipolar Transistors," **Invited** paper, *Proc. SPIE*, 1997.
286. K.Liu, S.Ray, S.Oswal, N.Chakraborty, R.Chang, D.Kencke and S.Banerjee, "Enhancement of Drain Current in Vertical SiGe/Si PMOS Transistor using Novel CMOS Technology," *Proc. IEEE Dev. Res. Conf*, June 1997.
287. K.Liu, S.Oswal, S.Ray, and S.Banerjee, "SiGe/Si vertical PMOSFET design and Fabrication," *Proc SPIE*, 1997.

288. T.Speranza, J.Nakos, V.Medina, S.Banerjee, G.Williamson, P.Lysaght and D.Lindholm, "Improved Methods for Evaluation of Rapid Thermal Processors," *Proc. Mat. Res. Soc.*, April, 1997.
289. S.Raghu Nandan, V.Agarwal and S.Banerjee, "Study of Low Energy $\text{BF}_2/\text{BCl}_2/\text{BBr}_2$ Implants for Achieving Ultra-shallow P+/N Junctions," *Proc. SPIE* 1997.
290. R.Sharma, J.Fretwell, J.Vaihinger, and S.Banerjee, "Automation of a Remote Plasma CVD System Using LabVIEW," *Proc. SPIE*, v. 3213, 119, 1997.
291. G.Hess, P.Parkinson, B.Gong, Z.Xu, D.Lim, M.Downer, S.John, S.Banerjee and J.Ekerdt. "Evolution of subsurface H from B-doped Si(100)," *Appl. Phys. Lett.*, 71(15), p. 2184, Oct. 1997.
292. C.Seal, D.Samara, and S.Banerjee, "CVD Growth and characterization of undoped and doped Ge and Ge $_{1-x}$ C $_x$ quantum dots on Si," *Appl. Phys. Lett.*, 71(24), p. 3564, 1997.
293. C.Mullins, K.Pacheo and S.Banerjee, "Growth and Characterization of Si Thin Films Employing Supersonic Jets of SiH $_4$ on Polysilicon and Si (100)," *J. Appl. Phys.* 82(1997), 6281-6288.
294. Raghu Srinivasa, Vikas Agarwal, Jinning Liu, Dan Downey, Sanjay Banerjee, "Annealing studies on low-energy As $^+$ and As $_2^+$ implants", presented at the 1998 MRS Spring Meeting, San Francisco.
295. L. Kencke, R. Richart, S. Garg and S. K. Banerjee, A Multilevel Approach Toward Quadrupling the Density of Flash Memory, *IEEE Electron Device Letters*, 19 (3), pp. 86-88, March 1998.
296. L. Kencke, X. Wang, S. Garg, S. K. Banerjee, "Programming and Retention for Quadruple Density Flash EEPROM's," *SRC TECHCON '98*, 1998. **(BEST PAPER AWARD)**
297. Quinones, S. John, S. Ray, S. Oswal, B. Anantharam, S. Banerjee, "Heterojunction PMOSFETS using Si $_{1-x}$ -yGe $_x$ Cy Alloys," *SRC TECHCON* 1998.
298. Q.Ouyang, H.Wang, S.Mudanai, S.Khan, M.Manassian, A.Tasch, C.Maziar and S.Banerjee, "Representative Deep Submicron MOS Device Structures for Device Model and Device Simulator Evaluation," *SRC TECHCON*, 1998.
299. H.Wang, Q.Ouyang, S.Mudanai, S.Khan, C.Maziar, A.Tasch and S.Banerjee, "A Generalized Physically-based Quantum Mechanical Model for Electron and Hole Accumulation Layers," *SRC TECHCON* 1998.
300. K.Liu, S.Ray, S.Oswal, and S.Banerjee, "A Deep Submicron SiGe/Si vertical PMOSFET Fabricated by Ge Ion Implantation," *IEEE Elec. Dev. Lett.*, 19(1), p. 13, Jan. 1998.
301. S.K.Ray, S.John, S.K.Oswal and S.K.Banerjee, "High mobility heterojunction PMOSFET using partially strained SiGeC"—*Proc. International Workshop on Physics of Semiconductor Devices*, 1998.
302. Akif Sultan, Sanjay Banerjee, Scott List, Vincent McNeil, "An approach using a subamorphizing threshold dose silicon implant of optimal energy to achieve shallower junctions", *J. Appl. Phys.*, Vol. 83, No. 12, June 1998, pp 8046 – 8050
303. S.Ray, L.Bera, C.Maiti, S.John and S.Banerjee, "Electrical Characteristics of Plasma Oxidized Si $_{1-x}$ -yGe $_x$ Cy MOS Capacitors," *Appl. Phys. Lett.* 72(8), March 1998.
304. Raghu Srinivasa, Vikas Agarwal, Sanjay Banerjee, "Use of low-energy BF_2^+ , BCl_2^+ and BBr_2^+ implants for the fabrication of ultra-shallow P $^+$ -N junctions", *SRC TECHCON* 1998. **(BEST PAPER AWARD)**
305. Raghu Srinivasa, Vikas Agarwal, Jinning Liu, Dan Downey, Sanjay Banerjee, "Use of screen oxides and As $_2^+$ implants to fabricate ultra shallow N $^+$ -P junctions", *SRC TECHCON* 1998.
306. Vikas Agarwal, Raghu Srinivasa, David Hoglund, Sanjay Banerjee, "Use of nitrogen co-implants to produce shallower source drain extensions", *SRC TECHCON* 1998.

307. D.Kencke, X.Wang, H.Wang, Q.Ouyang, S.Jallepalli, M.Rashed, C.Maziar, A.Tasch and S.Banerjee, "The Origin of Secondary Electron Gate Current: Multiple-stage Monte Carlo Study for Scaled, Low-power Flash Memory," *Proc. IEDM*, p. 889, 1998.
308. S.K.Ray, L.K.Bera. C.K.Maiti, S.John and S.K.Banerjee, "MOS capacitor characteristics of plasma oxide on Partially strained SiGeC films" -Presented at the Int. Conference on Metallurgical Coatings and Thin Films, San Diego, April 26- May 1, 1998.
309. Majeed A. Foad, Roger Webb, Roger Smith, Erin Jones, Amir Al-Bayati, Mark Lee, Vikas Agarwal, Sanjay Banerjee, Jiro Matsuo, and Isao Yamada, "Formation of shallow junctions using decaborane molecular ion implantation; Comparison with molecular dynamics simulation", Ion Implant Technology Conf., May 1998.
310. *S.Banerjee, "Defect Engineering for Ion-implanted Ultra-shallow Junctions," Eaton Conference on Ion Implantation, Napa Valley, CA, Sept. 1998
311. Quinones, S. K. Ray, K. C. Liu, and S. Banerjee, "Enhanced mobility PMOSFETs using tensile-strained Si_{1-y}Cy layers," *IEEE Elec. Dev. Lett.*, 1999, v. 20(7), p. 338.
312. R.Sharma, J.Fretwell, T.Ngai and S.Banerjee, "Remote Plasma-assisted deposition of gate quality oxides without the use of a pre-oxidation step," *J. Electrochem. Soc.* , vol. 146, pp. 2229, 1999.
313. R.Sharma, J.Fretwell, T.Ngai and S.Banerjee, "Properties of gate quality silicon dioxide films deposited on Si-Ge using remote plasma –enhanced chemical vapor deposition," *J. Vac. Sci. Tech. B*, vol. 17, pp. 460, 1999.
314. S.Mudanai, G. Chindalore, W.-K. Shih, H. Wang, Q. Ouyang, A. F. Tasch, C.M. Maziar and S.K. Banerjee, "Models for Electron and Hole Mobilities in MOS Accumulation Layers," *IEEE TED*, vol. 46, no. 8, August 1999.
315. K.-C. Liu, E.J. Quinones, B. Anantharam, X.D. Chen, S.K. Ray, and S.K. Banerjee, "Electrical Characteristics of Deep-Submicron Vertical Si_{1-x}Ge_x n-MOSFETs," *Proc. of Electrochemical Society*, v. 195, May, 1999.
316. S. John, E.Quinones, B.Ferguson, S.Ray, B.Ananthram, S. Middlebrooks, C.Mullins, J.Ekerdt, J.Rawlings, and S.Banerjee, "Properties of UHVCVD Grown Si_{1-x}yGe_xCy Epitaxial Films," *J. Electrochem. Soc.* Dec. 1999.
317. S. John, S. K. Ray, E. Quinones, S. K. Oswal, and S. K. Banerjee, "Heterostructure P-channel metal-oxide-semiconductor transistor utilizing a Si_{1-x}Ge_xCy channel," *App. Phys. Lett.* v. 74, No.6, pp. 847., 8 Feb, 1999.
318. S. John, S. K. Ray, E. Quinones, and S. K. Banerjee, "Strained Si n-channel metal-oxide-semiconductor transistor on relaxed Si_{1-x}Ge_x formed by ion implantation of Ge," *Appl. Phys. Lett.* v. 74, No. 14, pp. , 5 April 1999.
319. E. Quinones, S.K. Ray, K.C. Liu and S.K. Banerjee "Enhanced Mobility PMOSFET's Using Tensile-Strained Si_{1-y}Cy Layers," *Elec.Dev.Lett*, July 1999.
320. D.Kencke, W.Chen, H.Wang, S.Mudanai, Q.Ouyang, A.Tasch and S.Banerjee, "Source-side barrier effects with very high-K dielectrics in 50 nm Si MOSFETs," *Dev. Res. Conf.*, 1999.
321. *S.Banerjee, "Tinkering with the well-tempered MOSFET: Engineering the source-channel barrier," SRC-NASA Ames Third Device Modeling Workshop, Aug., 1999.
322. K. C. Liu*, X. Wang, E. Quinones, X. Chen, X. D. Chen, D. Kencke, B. Anantharam, R. D. Chang, S. K. Ray, S. K. Oswal, C. Y. Tu, and S. K. Banerjee, "A Novel Sidewall Strained-Si Channel nMOSFET", *DRC*, 1999.
323. M.Downer, L.Mantese, J.Ekerdt and S.Banerjee, "In situ real time monitoring of SiGe CVD by spectroscopic ellipsometry and SHG," *European MRS*, Strasbourg France, June 1999.
324. K. C. Liu*, X. Wang, E. Quinones, X. Chen, X. D. Chen, D. Kencke, B. Anantharam, R. D. Chang, S. K. Ray, S. K. Oswal, C. Y. Tu, and S. K. Banerjee, "A Novel Sidewall Strained-Si Channel nMOSFET", *IEDM*, 1999.
325. W.Qi, R.Nieh, T.Ngai, S.Banerjee and J.Lee, "MOSCAP and MOSFET characteristics using ZrO₂ gate dielectric deposited directly on Si," *IEDM* 1999.

326. Di Li, Ganesh Balamurugan, Borna Obradovic, Geng Wang, Yang Chen, and Al F. Tasch, "Computationally Efficient Model for 2-D Ion Implantation", Fifth International Symposium on Process Physics and modeling in Semiconductor Device Manufacturing, Electrochemical Society, 195th Meeting, May 2-6, 1999, Seattle, WA.
327. T.Kim, X.Chen and S.Banerjee, "Growth of relaxed SiGe structures by RTCVD," ECS Meeting, Austin, Texas, Feb. 2000.
328. Xiangdong Chen, Xiangdong Wang, Kou-chen Liu, Dong-won Kim and Sanjay Banerjee " Scanning tunneling spectroscopy investigation of strained $\text{Si}_{1-x}\text{Ge}_x$ band structure" Journal of Material Research, (16)6,p. 1257, June, 2000.
329. *S.Banerjee, "High ramp rate rapid thermal annealing for ultra-shallow junctions", with P.Kohli, H.Li, T.Kirichenko, S.Ganguly, **Invited** talk and paper, Electrochem. Soc. Proc. , Toronto, 2000.
330. Xiangdong Chen, Xiangdong Wang, Kou-chen Liu, Dong-won Kim and Sanjay Banerjee " Scanning tunneling spectroscopy investigation of strained $\text{Si}_{1-x}\text{Ge}_x$ -on Si band Offsets" presented at MRS, Spring 2000, San Francisco.
331. Qiqing Ouyang, Xiangdong Chen, S. Mudanai, Xin Wang, David Kencke, Al. F. Tasch, Leonard F. Register, and Sanjay Banerjee, " A Novel Si/SiGe Heterojunction pMOSFET with Reduced Short-Channel Effects and Enhanced Drive Current," IEEE Trans. on Electron Devices, Vol. 47, No 10, Oct., 2000.
332. T.Ngai, J.Lee, S.Banerjee, "Electrical Properties of ZrO_2 Gate Dielectric on SiGe," Appl. Phys. Lett., 76(4), p. 502, Jan. 2000.
333. Zhonghai Shi, Xiangdong Chen, David Onsongo, Eduardo Quinones, and Sanjay K. Banerjee "Simulation and optimization of strained $\text{Si}_{1-x}\text{Ge}_x$ buried channel p-MOSFETs," Solid State Electronics, 44 (7): 1223-1228 JUL 2000.
334. E. Quinones, S. John, S.K. Ray and S.K. Banerjee "Design, Fabrication, and Analysis of SiGeC Heterojunction PMOSFETs," Trans.Elec.Dev., Sept. 2000.
335. H.J.Li, P.Kohli, S.Ganguly, T.Kirichenko, S.Banerjee, P.Zeitsoff, "B Diffusion in Si in presence of other species," Appl. Phys. Lett., 77(17), p. 2683, 2000.
336. T.Ngai, J.Lee, S.Banerjee, "Electrical Properties of ZrO_2 Gate Dielectric on SiGe," DRC, 2000.
337. Q.Ouyang, X.Chen, ..A.Tasch, S.Banerjee, "Bandgap Engineering in Deep Submicron Vertical PMOSFETs," Dev. Res. Conf., 2000.Xiangdong Chen, QiqingOuyang, Kou-Chen Liu, Zhonghai Shi, Al Tasch and Sanjay Banerjee "Vertical P-MOSFETS with heterojunction between source/drain and channel," Dev.Res. Conf. 2000, Denver.
339. S. Mudanai, Y-Y Fan, Q. Ouyang, A. F. Tasch, F. Register, D. L. Kwong, and S. Banerjee, "Modeling of Direct Tunneling Current Through Gate Dielectric Stacks", SISPAD 2000, September, 2000.
340. "Modeling High-K Gate Current from p-type Si Inversion Layers", Y-Y Fan, S. Mudanai, Q.Li, J.Lee, A. F. Tasch, F. Register, and S. Banerjee, DRC June, 2000.
341. "Two-Dimensional Bandgap Engineering in a Novel Si/SiGe pMOSFET with Enhanced Device Performance and Scalability," Q. Ouyang, X. D. Chen, S. Mudanai, D. L. Kencke, X. Wang, A. F. Tasch, and L. F. Register, and S. K. Banerjee, SISPAD, September 2000.
342. "Two-Dimensional Bandgap Engineering in a Novel Si/SiGe pMOSFET," Q. Ouyang, X. D. Chen, S. Mudanai, D. L. Kencke, X. Wang, A. F. Tasch, and L. F. Register, and S. K. Banerjee, TECHCON, September 2000.
343. "Electron Transport Properties in Novel Orthorhombically-strained Si Material Explored by the Monte Carlo Method," Xin wang, D. L. Kencke, K. C. Liu, A. F. Tasch,Jr., L. F. Register and S. K. Banerjee, SISPAD 2000
344. "Monte Carlo study on electron transport properties in novel orthorhombically-strained Si material" Xin Wang, D. L. Kencke, K. C. Liu, A. F. Tasch,Jr. L. F. Register and S. K. Banerjee, SRC-TECHCON 2000

345. Y.Chen, G.Weng,D.Li, S.Oak, G.Srivasta, A.Tasch and S.Banerjee, "Universal model for implantation of any species into Si," SRC TECHCON, 2000.
346. D.Kencke, X.Wang, Q.Ouyang, S.Mudanai, A.Tasch and S.Banerjee, "Enhanced Secondary Electron Injection in Novel SiGe Flash Memory Devices," Int. Elec. Dev. Meet., 2000.
347. W.Chen, Q.Ouyang, F.Register and S.Banerjee, "Quantum effects Along the Channel of Ultra-scaled Si MOSFETs?", Int. Elec. Dev. Meet., 2000.
348. Hong-Jyh Li, Puneet Kohli, Swaroop Ganguly, Taras A. Kirichenko, Peter Zeitzoff, Kenneth Torres and Sanjay Banerjee, "Boron Diffusion and Activation in the Presence of Other Species" *IEDM 2000 Technical Digest pp.515-518, Dec. 2000, San Francisco*
349. S. Mudanai, Y-Y. Fan, Q. Ouyang, A. F. Tasch, D. L. Kwong, F. Register and S. Banerjee; Modeling of Direct Tunneling Current Through Gate Dielectric Stacks; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
350. X. Chen, Q. Ouyang, K-C. Liu, Z. Shi, A. Tasch and S. Banerjee; Sub-100nm Vertical P-MOSFETs with Heterojunction Between Source and Channel; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
351. Z. Shi, X. Chen, D. Onsongo, E. J. Quinones and S. K. Banerjee; Simulation and Optimization of Strained Si(1-x)Ge(x) Buried Channel p-MOSFETs; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
352. X.Chen, K.Liu, S.Ray and S.Banerjee, "Bandgap Engineering in Vertical PMOSFETs," IEEE Int. Conf.on, Communications, Computers and Devices, Kharagpur, 2000.
353. S.Ray, S.Maikap, S.Samanta, S.Banerjee and C.Maiti, "Charge Trapping Characteristics of Ultrathin Oxynitrides on Si/SiGeC/Si Heterolayers, IEEE Int. Conf.on, Communications, Computers and Devices, Kharagpur, 2000.
354. D. Onsongo, E. Quinones, Z. Shi, P. Williamson and S. Banerjee; Tensile-Strained Si(1-y)C(y) Alloy pMOSFETs on UHVCVD Epitaxial Films; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
355. S. K. Jayanarayanan, X. Chen, K-C. Liu and S. Banerjee; Hole and Electron Mobility Enhancement in Strained SiGe Vertical MOSFETs; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
356. Y-Y. Fan, S. Mudanai, W. Qi, J. C. Lee, A. F. Tasch, L. F. Register and S. K. Banerjee; Modeling Gate Current Through High K Stacked Dielectrics: Electron Tunneling from Inversion Layers; TECHCON 2000, Semiconductor Research Corporation, Phoenix, Arizona, September 21-23, 2000.
357. S. Maikap, L. K. Bera, S. K. Ray, S. John, S. K. Banerjee and C. K.Maiti, "Electrical characterization of Si/Si_{1-x}Ge_x/Si quantum well heterostuctures using a MOS capacitor", Solid-State. Electron., Vol. 44, pp. 1029-1034, 2000.
358. S. Maikap, S. K. Ray, S. John, S. K. Banerjee and C. K. Maiti, "Electrical characterization of ultra-thin gate oxides on Si/Si_{1-x-y}Ge_xC_y/Si quantum well heterostructures", Semicond. Sci. Technol., 2000.
359. G.Kar, A.Dhar, S.ray, S.John and S.Banerjee, "Hall mobilities in B-doped strained SiGe and SiGeC layers grown by UHVCVD," J. Appl. Phys. 88(4), p. 2039, August 2000.
360. D. L. Kencke, Q. Ouyang, W. Chen, H. Wang, S. Mudanai, , A. Tasch, and S. K. Banerjee, "Tinkering with the well-tempered MOSFET: source-channel barrier modulation with high permittivity dielectrics", Superlattices and Microstructures, 27 (2/3), 2000.
361. S.Mudanai, G. Chindalore, W.-K. Shih, H. Wang, Q. Ouyang, A. F. Tasch, C.M. Maziar and S.K. Banerjee "Models for Electron and Hole Mobilities in MOS Accumulation Layers," IEEE TED, vol. 46, no. 8, p. 1749, 2000.
362. S. Mudanai, Y.-Y. Fan, Q. Ouyang, A. F. Tasch, and S. K. Banerjee, "Modeling of Direct Tunneling Current Through Gate Dielectric Stacks," *IEEE TED, Computational Electronics: New Challenges and Directions*, pp. 1851-1857, Vol. 47, No. 10, October, 2000

363. Xiangdong Chen, Qiqing Ouyang, David M. Onsongo, S. K. Jayanarayanan, Al Tasch and Sanjay Banerjee "SiGe heterojunction vertical p-type metal-oxide-semiconductor field-effect transistor with Si cap," *Applied Physics Letters*, Vol 77, No. 11, Sep. 11, 2000
364. Xin Wang, D. L. Kencke, K. C. Liu, A. F. Tasch, Jr., L. F. Register, and S. K. Banerjee, "Monte Carlo Simulation on Electron Transport in simple Orthorhombically-strained Silicon", *J. Appl. Phys.* 88, 4717 (2000).
365. S. Mudanai, L. F. Register, A. F. Tasch, and S. K. Banerjee, "Understanding the effects of Wave Function Penetration on the Inversion Layer Capacitance of NMOSFETs," *IEEE Electron Device Letters*, pp. 145-147, Vol. 22, No. 3 March 2001.
366. S. Mudanai, L. F. Register, A. F. Tasch, and S. K. Banerjee, "A New and Accurate Quantum Mechanical Compact Model for NMOS Gate Capacitance," the 59th *Annual Device Research Conference*, June 2001.
367. Qiqing (Christine) Ouyang, Xiangdong Chen, Al F. Tasch, Leonard F. Register, Sanjay K. Banerjee, Jack O Chu, and John A. Ott, "Fabrication of a Novel Vertical pMOSFET with Enhanced Drive Current and Reduced Short-Channel Effects and Floating Body Effects", *VLSI Symp. Proc.* July 2001.
368. T. Ngai, R. Sharma, J. Fretwell, X. Chen, J. Chen, W. Brookover, and S. Banerjee, "Remote plasma deposited gate dielectrics on Si and SiGe MOSFETs", *Electrochemical Society Meeting*, March 2001
369. T. Ngai, R. Sharma, J. Fretwell, X. Chen, J. Chen, and S. Banerjee, "Improvement of SiO₂/SiGe Interface of SiGe pMOSFETs Using Water Vapor Annealing", 2001 *Electronic Materials Conference*
370. T. Ngai, K. Onishi, R. Choi, C. Kang, J. Fretwell, X. Chen, J. Chen, J.C. Lee, and S.K. Banerjee, "Electrical Properties of HfO₂ Gate Dielectric on SiGe", 2001 *Electronic Materials Conference*
371. S. Oak, G. Shrivastava, Y. Chen, G. Wang, D. Li, A. F. Tasch, S. Banerjee, "Modeling of implant into photoresist," *IWCE*, Urbana, IL 2001.
372. Wang, B. Obradovic, Y. Chen, D. Li, S. Oak, G. Shrivastav, S. Banerjee and A. Tasch, "A Computationally Efficient Target Search Algorithm for A Monte Carlo Ion Implantation Simulator", *Journal of Technology Computer Aided Design*, <http://www.ieee.org/products/online/journal/tcad/Wang-May01/search.pdf>
373. G. Shrivastav, G. Wang, Y. Chen, D. Li, S. Oak, A. Tasch and S. Banerjee, *A Universal Analytical 1-D Model for Ion Implantation into Single-Crystal Silicon Based on Legendre Polynomials*, *IWCE*, 2001.
374. Shrivastav, G, Wang, Y. chen, D Li, S. Oak, A. Tasch and S. Banerjee, *Monte Carlo modeling of ion implantation into SiGe*, Presented at EMC, June 2001.
375. Di Li, Geng Wang, Yang Chen, Gaurav Shrivastav, Stimit Oak, Al Tasch and Sanjay Banerjee, "A Computationally Efficient Model for Three-dimensional Monte Carlo Simulation of Ion Implantation into Complex Structures", *SISPAD 2001 Athens, Greece*.
376. Di Li, Geng Wang, Yang Chen, Gaurav Shrivastav, Stimit Oak, Al Tasch and Sanjay Banerjee, "A Computationally Efficient Model for Three-dimensional Monte Carlo Simulation of Ion Implantation into Complex Structures", *IWCE 2001, Urbana, IL*.
377. Hong-Jyh Li, Puneet Kohli, Swaroop Ganguly, Taras A. Kirichenko, Peter Zeitzoff, Kenneth Torres and Sanjay Banerjee, "Boron Diffusion and Activation in the Presence of Other Species" *International Conference on Computational Nanoscience, South Carolina March, 2001 (ICCN 2001.)*
378. Hong-Jyh Li, Xiao Chen, Peter Zeitzoff, Kenneth Torres, and Sanjay Banerjee, "Empirical Parameter-Free Recipe for Implantation and Annealing Modeling of N⁺ Source and Drain in the 180nm and 100nm Technology Node", 199th *Electrochemical Society Conference, Washington, D.C., March 2001*

379. Hong-Jyh Li, Taras A. Kirichenko, Swaroop Ganguly, Puneet Kohli, Sanjay Banerjee, David Sing, Peter Zeitzoff, Kenneth Torres, Steve McCoy, Kiefer Elliott "A Study of Rapid Thermal Annealing of Shallow BF₂ Implants", *Ultra Shallow Junctions-2001, Napa, California, April, 2001*
380. Hong-Jyh Li, Taras A. Kirichenko, Swaroop Ganguly, Puneet Kohli, Sanjay Banerjee, David Sing, Peter Zeitzoff, Kenneth Torres, Steve McCoy, Kiefer Elliott "A Study of Ultra-High Ramp Rate Thermal Annealing", *The ECS International Semiconductor Technology Conference (ISTC-2001), Shanghai, China, May, 2001*
381. Hong-Jyh Li, Sanjay Banerjee, Peter Zeitzoff, and Kenneth Torres, "TCAD Validation and Process Optimization Prediction Using Taguchi Method", *2001 Quality and Productivity Research Conference*
382. B.Choi, M.Meissl, M.Coburn, T.Bailey, P.Ruchoeft, S.Sreenivasan, F.Prins, S.Banerjee, J.Ekerdt and G.Willson, "Layer to layer alignment for step and flash lithography," SPIE, March 2001.
383. *S.Banerjee, "RPCVD oxides on SiGe," **Invited** talk, Electrochem. Soc. Proc. 2001. **Invited** paper with Tat Ngai, X.Chen and J. Chen.
384. P.Kohli, S.Ganguly, T.Kirichenko, H-J. Li, S.Banerjee, E.Graetz, M.Shevelev, "Microwave Annealing of P/N Ultra Shallow Junctions," USJ Workshop, 2001.
385. G. Wang, Y. Chen, D. Li, S. Oak, G. Shrivastav, S. Banerjee, A. Tasch, P. Merril*, R. Bleiler*, "Oxygen Recoil Effect During Ion Implantation Through Oxide", J APPL PHYS 89 (11): 5997-6001 Part 1 JUN 1 2001
386. Y. Chen, G. Wang, D. Li, Stimit Oak, Gaurav Shrivastav, A.F. Tasch and Sanjay Banerjee, "Monte Carlo Simulation Models for Ion Implantation into Silicon Nitride and Titanium Silicide," EMC, 2001.
387. Ray SK, Maikap S, Samanta SK, S. Banerjee, Charge trapping characteristics of ultrathin oxynitrides on Si/Si_{1-x-y}GexCy/Si heterolayers, SOLID STATE ELECTRON 45 (11): 1951-1955 NOV 2001
388. Ray SK, Kar GS, Banerjee SK, Characteristics of UHVCVD grown Si/Si_{1-x-y}GexCy/Si quantum well heterostructure, APPL SURF SCI 182 (3-4): 361-365 Sp. Iss. SI OCT 22 2001
389. Chen XD, Liu KC, Ouyang QC, Banerjee SK, Hole and electron mobility enhancement in strained SiGe vertical MOSFETs, IEEE T ELECTRON DEV 48 (9): 1975-1980 SEP 2001
390. Kar GS, Ray SK, Kim T, Banerjee SK, Estimation of hole mobility in strained Si_{1-x}Gex buried channel heterostructure PMOSFET, SOLID STATE ELECTRON 45 (5): 669-676 MAY 2001
391. Ouyang QQC, Chen XD, Tasch AF, Banerjee SK, Built-in longitudinal field effects in sub-100-nm graded Si_{1-x}Gex channel PMOSFETs, IEEE T ELECTRON DEV 48 (6): 1245-1250 JUN 2001
392. Ngai T, Qi WJ, Sharma R, Banerjee SK, Transconductance improvement in surface-channel SiGe p-metal-oxide-silicon field-effect transistors using a ZrO₂ gate dielectric, APPL PHYS LETT 78 (20): 3085-3087 MAY 14 2001
393. Maikap S, Ray SK, Banerjee SK, Electrical properties of O₂/NO-plasma grown oxynitride films on partially strain compensated Si/Si_{1-x-y}GexCy/Si heterolayers, SEMICONDUCTOR SCI TECH 16 (3): 160-163 MAR 2001
394. Li D, Wang G, Chen Y, Tasch AF, Banerjee SK, A computationally efficient simulator for three-dimensional Monte Carlo simulation of ion implantation into complex structures, NUCL INSTRUM METH B 184 (4): 500-508 DEC 2001
395. Chen XD, Liu KC, Ray S, Banerjee SK, Bandgap engineering in vertical P-MOSFETs, SOLID STATE ELECTRON 45 (11): 1939-1943 NOV 2001
396. Kim DW, Kim YH, Chen XD, Prins F, KWONG DL, Banerjee SK, Growth of germanium quantum dots on different dielectric substrates by chemical-vapor deposition, J VAC SCI TECHNOL B 19 (4): 1104-1108 JUL-AUG 2001

397. Chen XD, Ouyang QQ, Jayanarayanan SK, Prins F and Banerjee S, Vertical p-type high-mobility heterojunction metal-oxide-semiconductor field-effect transistors, APPL PHYS LETT 78 (21): 3334-3336 MAY 21 2001
398. Chen XD, Ouyang Q, Jayanarayanan SK, Banerjee SK, An asymmetric Si/Si_{1-x}Ge_x channel vertical p-type metal-oxide-semiconductor field-effect transistor, SOLID STATE ELECTRON 45 (2): 281-285 FEB 2001
399. Xiangdong Chen, Kou-Chen Liu, Sankaran Kartik Jayanarayanan, and Sanjay Banerjee, "Electron mobility enhancement in strained SiGe vertical n-type metal-oxide semiconductor field-effect transistors" Appl. Phys. Lett., 78(3), P.377, 2001.
400. Dong-Won Kim, Young-Hee Kim, Freek E. Prins, Dim-Lee Kwong, and Sanjay Banerjee, "Fabrication of Germanium Dots on Different Dielectric Substrates", 2001 Electrical Materials Conference, June 2001.
401. Dong-Won Kim, Freek Prins, Kil-Soo Ko, C. H. Lee, Dim-Lee Kwong, and Sanjay Banerjee, "The Characterization of Initial Growth of Polycrystalline Silicon Germanium Films on Zirconium Oxide." 2001 MRS Fall, Nov. 2001.
402. Sankaran Jayanarayanan, Freek Prins, Xiangdong Chen, Sanjay Banerjee, "Enhanced Mobility in 100 nm strained SiGe vertical PMOSFETs fabricated by UHVCVD," Materials Research Symposium, November 2001.
403. Di Li, Li Lin, Geng Wang, Yang Chen, Gaurav Shrivastav, Stimit Oak, Al F. Tasch, Sanjay K. Banerjee, "Monte Carlo Simulation of Ion Implantation (3-dimensional) and Defect Modeling During Implantation Process", International Electron Device Meeting, Technical Digest, December, 2001, Washington, DC.
404. L. F. Register, S. Mudanai and S. K. Banerjee, "A New and Accurate Quantum Mechanical Compact Model for NMOS Gate Capacitance," Compact Modeling Council Meeting, Texas Instruments, Dallas, TX, 2001.
405. S. K. Ray, G. S. Kar, S. Maikap and S. K. Banerjee, " Comparison of temperature dependent device parameters for strained-Si_{1-x}Ge_x and partially strain compensated Si_{1-x-y}Ge_xC_y heterostructure PMOSFETs", "First Int'l Workshop on New Group IV(Si-Ge-C) Semiconductors, 2001", Miyagi, Japan.
406. Qiqing C. Ouyang, Xiangdong Chen, S. K. Jayanarayanan, Freek Prins, Sanjay Banerjee "Performance enhancement in vertical sub-100nm nmosfets with graded doped channels", "Fourth IEEE International Caracas conference on devices, circuits and systems," April 2002, Aruba.
407. Di Li, Shyh Horng, Chuck Machala, Al F. Tasch and Sanjay K. Banerjee, "Sb implant Modeling into Si ", Simulation of Semiconductor Processes and Devices, 2002, Japan.
408. *S.Banerjee, "Two Ways to Mess up the Si MOSFET: SiGe Channels and High-k gate Dielectrics," Micron Distinguished Lecture Series, Boise, April 2002.
409. *S.Banerjee, "Two Ways to Mess up the Si MOSFET: SiGe Channels and High-k gate Dielectrics," IEEE EDS Distinguished Lecture Series, Portland, Maine, 2002.
410. *S.Banerjee, "Bandgap and Strain Engineered Planar and Vertical Si-Ge-C MOSFETs," Int. Symp. On Nano and Giga Microelectronics, Moscow, Russia, Sept. 2002.
411. *S.Banerjee, "SiGe MOSFETs with high dose Ge implantation," CAARI, Nov. 2002, Denton, TX.
412. *L.F.Register, W.Chen and S.Banerjee, "Phonon scattering in nanostructures: functions and challenges to modeling," IEEE Nano 2002, Arlington, WVA.
413. Y.Chen, D. Li, L.Lin, A.Tasch and S.Banerjee, "Quantum mechanical model of electronic stopping power for ions in a free electron gas over the entire energy range," 14th IIT, Taos, NM, Sept. 2002.
414. H.Li, T.Kirichenko, P.Kohli, E.Graetz, P.Zeitsoff, R.Tichy, L.Larson and S.Banerjee, "B Retarded Diffusion in the presence of In and Ge," 14th IIT, Taos, NM, Sept. 2002.
415. W.Chen, L.Register and S.Banerjee, "Simulation of quantum effects along the channel of Ultra-scaled Si-based MOSFETs," DRC , Santa Barbara, 2002.

416. D.Kim, F.Prins, T.Kim, D.Kwong and S.Banerjee, "Charge retention Characteristics of SiGe Quantum Dot Flash Memories," DRC , Santa Barbara, 2002.
417. Z.Shi, D.Onsongo, X.Chen, R.Nieh and S.Banerjee, "Ni Silicidation Techniques for Strained SiGe/SiGeC/SiC Alloys," EMC , Santa Barbara, 2002.
418. D.Kim, S.Hwang, T.Edgar and S.Banerjee, "Characterization of initial nucleation of SiGe Dots on SiO₂ and HfO₂ substrates for nanoelectronic devices," EMC , Santa Barbara, 2002.
419. Kar GS, Dhar A, Bera LK, Banerjee SK, Effect of carbon on lattice strain and hole mobility in Si_{1-x}Ge_x alloys, J MATER SCI-MATER EL 13 (1): 49-55 2002
420. Kar GS, Maikap S, Ray SK, et al., Effective mobility and alloy scattering in the strain compensated SiGeC inversion layer SEMICOND SCI TECH 17 (5): 471-475 MAY 2002
421. Chen WQ, Register LF, Banerjee SK, Simulation of quantum effects along the channel of ultrascaled Si-based MOSFETs, IEEE T ELECTRON DEV 49 (4): 652-657 APR 2002
422. Ngai T, Chen X, Chen J, et al., Improving SiO₂/SiGe interface of SiGe p-metal-oxide-silicon field-effect transistors using water vapor annealing, APPL PHYS LETT 80 (10): 1773-1775 MAR 11 2002
423. Kar GS, Maikap S, Banerjee SK, et al., Hole velocity overshoot in partially strain compensated Si_{0.793}Ge_{0.207} inversion layers, ELECTRON LETT 38 (3): 141-142 JAN 31 2002
424. Kohli P, Ganguly S, Kirichenko T, et al. Microwave annealing for ultra-shallow junction formation, J ELECTRON MATER 31 (3): 214-219 MAR 2002
425. Ganguly S, Lin L, Kohli P, et al. Comparison of low energy BF₂⁺, BCl₂⁺, and BBr₂⁺ implants for the fabrication of ultrashallow P⁺-N junctions, J APPL PHYS 91 (4): 2023-2027 FEB 15 2002
426. Li D, Shrivastav G, Wang G, Tasch A, Banerjee S, Accurate and computationally efficient analytical 1-D and 2-D ion implantation models based on Legendre polynomials, IEEE T ELECTRON DEV 49 (7): 1172-1182 JUL 2002.
427. Sanjay Banerjee, "Silicon-Germanium -Carbon Self Assembled Quantum Dot Growth and Applications in Electronic Memory Device." International Conference on Science and Technology of Interfaces, TMS, Seattle, Feb. 2002. **Invited** paper with Dong Kim.
428. Chen Y, Wang G, Li D, Banerjee SK et al. "A universal ion implantation model for all species into single-crystal silicon," IEEE T ELECTRON DEV 49 (9): 1519-1525 SEP 2002.
429. Kar GS, Maikap S, Banerjee SK, et al., "Series resistance and mobility degradation factor in C-incorporated SiGe heterostructure p-type metal-oxide semiconductor field-effect transistors," SEMICOND SCI TECH 17 (9): 938-941 SEP 2002.
430. Kar GS, Dhar A, Bera LK, Banerjee SK, "Effect of carbon on lattice strain and hole mobility in Si_{1-x}Ge_x alloys," J MATER SCI-MATER EL 13 (1): 49-55 2002.
431. Li HJ, Kirichenko TA, Kohli P, Banerjee SK et al., "Boron retarded diffusion in the presence of indium or germanium," IEEE ELECTR DEVICE L 23 (11): 646-648 NOV 2002.
432. Wang X, Kencke DL, Liu KC, et al., "Band alignments in sidewall strained Si/strained SiGe heterostructures," SOLID STATE ELECTRON 46 (12): 2021-2025 DEC 2002.
433. Fan YY, Nieh RE, Lee JC, Banerjee SK et al., "Voltage- and temperature-dependent gate capacitance and current model: Application to ZrO₂ n-channel MOS capacitor," IEEE T ELECTRON DEV 49 (11): 1969-1978 NOV 2002
434. Chen XD, Ouyang QC, Wang G, Banerjee SK et al., "Improved hot-carrier and short-channel performance in vertical nMOSFETs with graded channel doping," IEEE T ELECTRON DEV 49 (11): 1962-1968 NOV 2002
435. Mudanai S, Li F, Samavedam SB, Banerjee SK et al., "Interfacial defect states in HfO₂ and ZrO₂ nMOS capacitors," IEEE ELECTR DEVICE L 23 (12): 728-730 DEC 2002
436. Y.Chen, D. Li, L.Lin, A.Tasch and S.Banerjee, "Quantum mechanical model of electronic stopping power for ions in a free electron gas," 14th Int.Conf. on Ion Implant Technology, Taos, NM, Sept. 2002.

437. D.Kim, F.Prins, K.Ko, C.Lee, D.Kwong and S.Banerjee, "The Characteriation of initial growth of polycrystalline SiGe fims on ZrO₂," MRS, v.686, 2002.
438. Li HJ, Bennett J, Zeitzoff P, Banerjee SK et al., "Indium out-diffusion from silicon during rapid thermal annealing," IEEE ELECTR DEVICE L 24 (4): 221-223 APR 2003
439. Fan YY, Xiang Q, An J, Banerjee SK et al., "Impact of interfacial layer and transition region on gate current performance for high-K gate dielectric stack: Its tradeoff with gate capacitance," IEEE T ELECTRON DEV 50 (2): 433-439 FEB 2003
440. Kim DW, Prins FE, Kim T, Banerjee SK et al., "Reduction of charge-transport characteristics of SiGe dot floating gate memory device with ZrO₂ tunneling oxide," IEEE T ELECTRON DEV 50 (2): 510-513 FEB 2003
441. Kim DW, Hwang SB, Edgar TF, Banerjee SK et al., "Characterization of SiGe quantum dots on SiO₂ and HfO₂ grown by rapid thermal chemical deposition for nanoelectronic devices," J ELECTROCHEM SOC 150 (4): G240-G243 APR 2003
442. Shi ZH, Onsongo D, Onishi K, Banerjee SK et al., "Mobility enhancement in surface channel SiGePMOSFETs with HfO₂ gate dielectrics," IEEE ELECTR DEVICE L 24 (1): 34-36 JAN 2003
443. Shi ZH, Onsongo D, Chen X, Banerjee SK et al., "Nickel silicidation techniques for strained Si_{1-x}Gex, Si_{1-x}yGexCy, and Si_{1-y}Cy alloys material-device applications," J ELECTRON MATER 32 (3): 184-190 MAR 2003
444. Banerjee SK, "Bandgap and Strain engineered SiGeC vertical and planar," MOSFETs, Microelectronic Eng. 69 (2-4): 106-117 SEP 2003. **(Invited paper)**
445. Xia TS, Register LE, Banerjee SK, "Quantum transport in double-gate MOSFETs with complex band structure," IEEE T ELECTRON DEV 50 (6): 1511-1516 JUN 2003
446. S.Banerjee, "Quantum mechanical modeling of CV and IV behavior of SiO₂ and high-k dielectrics," ECS Callinan Address, Paris, April 2003. (with L. Register, Y.Fan and S.Mudanai) **(INVITED)**
447. X.Chen, Z.Shi, J.Zhou, L.Rabenberg and S.Banerjee, "High resolution TEM of silicide formation and stability of Ni/Si and Ni/SiGe," TMS, Seattle, Feb. 2003
448. Z.Shi, D.Onsongo, and S.Banerjee, "Performance Enhancement in 70 nm channel length strained SiGe PMOSFETs," 1st SiGe workshop, Nagoya, Japan, Jan. 2003.
449. T.Kirichenko, D.Yu, S.Banerjee and G.Hwang, "Dynamics of intrinsic defects in Si (001) surfaces and Si/SiO₂ interfaces: A first principles study," APS March Meeting, 2003, Austin, TX.
450. D.Kim, T.Kim, Y.Liu, L.Weltzer and S.Banerjee, "SiGe quantum dot memory devices with HfO₂ tunneling oxide," DRC, Salt Lake, UT, June 2003.
451. F.Li, S.Mudanai, Y.Fan, L.Register and S.Banerjee, "Compact model of MOSFET electron tunneling current through ultra-thin SiO₂ and high-k stacks," DRC, Salt Lake, UT, June 2003.
452. P.Kohli, A.Jain, H.Bu, S.Chakravarthi, CMachala, S.Dunham and S.Banerjee, "Effect of sidewall nitride spacer process of B dose loss in ultra-shallow junction formation," USJ 2003.
453. H.Li, P.Zeitzoff, R.Tichy, L.Larson and S.Banerjee, "B diffusion in Si with pre-amorphization of different species," USJ 2003.
454. D.Onsongo, Z.Shi and S.Banerjee, "Enhanced performance in strained SiGe surface channel and buried channel PMOSFETs," SRC TECHCON, Dallas, Aug. 2003.
455. F.Li, S.Mudanai, Y.Fan, L.Register and S.Banerjee, "Compact gate capacitance and gate current modeling of MOSFETs with high-k gate dielectrics," SRC TECHCON, Dallas, Aug. 2003.
456. H.Li, T.Rhoad, P.Zeitzoff, R.Tichy, L.Larson and S.Banerjee, "B diffusion in low energy B/BF₂ implants with pre-amorphization of different species," MRS, April 2003.
457. F.Li, S.Mudanai, Y.Fan, L.Register and S.Banerjee, "A simulated annealing approach for automatic extraction of device and material parameters of MOS with SiO₂/high-k gate stacks" UGIM, Bose, ID, June 2003.
458. J.Oh, S.Banerjee and J.Campbell, "Metal-Ge-metal photodetectors on heteroepitaxial Ge-on-Si with amorphous Ge Schottky barrier enhancement layers," LEOS 2003.

459. S.Banerjee, "SiGe MOS Technology," NSF Workshop on New Materials and Devices, Stanford Univ., Aug. 21, 2003 (**INVITED**)
460. B.Smith, N.Stacey, J.Donnely, D.Onsongo, T.Bailey, C.Mackay, D.Resnick, W.Dauksher, D.Mancini, K.Nordquist, S.Sreenivasan, S.Banerjee, J.Ekerdt and C.Willson, "Employing Step and Flash Imprint Lithography for Gate Level Patterning of a MOSFET Device," ML 5037-124, SPIE 2003.
461. W. Chen, L. F. Register and S.K. Banerjee, "Two Dimensional Quantum Mechanical Simulation of Electron Transport in Nano-Scale Si-Based MOSFETs, Physica E 19, 28-32 (2003) (Proceedings of the Fourth International Symposium on Nanostructures and Mesoscopic Systems [NanoMES] 2003).
462. L. F. Register, W. Chen and Sanjay K. Banerjee, "Bridging the Gap Between Classical and Semiclassical Transport in Nanoscale MOSFETs: Schrödinger Equation Monte Carlo-2D," presented on The 15th Biennial IEEE University/Government/Industry Microelectronics Symposium (UGIM'03), Boise, Idaho, June 30-July 2, 2003 (**INVITED**).
463. W. Chen, L. F. Register, and S. K. Banerjee, "Scattering in a Nano-Scale MOSFET: A Quantum Transport Analysis," presented on IEEE Nano2003, San Francisco, CA, August 2003.
464. Chen X, Shi Z, Banerjee SK, Zhou JP, Rabenberg LK, "High-resolution transmission electron microscopy of silicide formation and morphology development of Ni/Si and Ni/Si_{1-x}Ge_x," J ELECTRON MATER 32 (11): 1171-1181 NOV 2003
465. Kim DW, Kim T, Banerjee SK, Memory characterization of SiGe quantum dot flash memories with HfO₂ and SiO₂ tunneling dielectrics," ELECTRON DEV 50 (9): 1823-1829 SEP 2003
466. Deen J, Marinov O, Onsongo D, Dey S, and Banerjee SK, "Low frequency noise in SiGeC-based pMOSFETs," Fluctuation and Noise Symposium in Maspalomas, Gran Canaria, Spain, May 2004
467. Register LF, Chen W and Banerjee SK, "SEMC-2D for simulation fo quantum transport and scattering in nanoscale non-classical CMOS," 5th Motorola Workshop on Computational Materials and Electronics, Austin, TX, 2004.
468. Kirichenko TA, Yu D, Banerjee SK and Hwang G, "Structure and dynamics of Si interstitials at Si (100) and Si(100)/SiO₂, ECS Symp. Proc., San Antonio , 2004.
469. Kelly D, Onsongo D, Dey S, Wise R, Cleavelin R, and Banerjee SK, "Enhanced hot-electron performance of strained Si NMOS over unstrained Si, Proc. IRPS, 2004.
470. Banerjee SK and Register LF, "Quantum mechanical modeling of CV and IV behavior of SiO₂ and high-k dielectrics," 5th Motorola Workshop on Computational Materials and Electronics, Austin, TX, 2004.
471. Kirichenko TA, Yu D, Hwang G, and Banerjee SK, "Behavior of vacancies, interstitials, and B-interstitail pairs at Si_{1-x}Ge_x/SiO₂ interface," MRS Symp. Proc., San Francisco , 2004.
472. Banerjee SK, Kim D, Kim T, Weltzer L, Liu Y, Tang S and Palard M, "Nanoparticle floating gate flash memories, DRC 2004, **Invited Talk**.
473. Banerjee SK, "SiGeC HFETs," International Conference on Physics of Semiconductors, Phoenix, July 2004, **Invited Tutorial**.
474. Tongsheng Xia, Leonard F. Register, and Sanjay K. Banerjee, "Complex band structure-based non-equilibrium Green's function (NEGF) transport studies for Ultra-scaled Carbon nanotube (CNT) transistors," DRC, 2004.
475. S.Jayanarayan, W.Bai, D.Kwong, S.Banerjee, "Si and SiGe Vertical MOSFETs with CVD HFO₂ gate dielectric", EMC 2004.
476. S.Joshi, X.Chen, D.Kelly, T.Ngai, J.Chen and S.Banerjee, "MOS Capacitors on epitaxial Ge/SiGe with high-k dielectrics, EMC 2004.
477. S.Banerjee, Nanoelectronics **invited** short course and organizer, SEMI, Austin, 2004.
478. S.Banerjee, "Microelectronics: The End the Beginning or Beginning of the End," **Invited** talk to SEMATECH Board of Directors, San Jose, 2004

479. L.Weltzer and S.Banerjee, "Enhanced CHISEL programming in flash memory devices with SiGe buried layer," NVM Workshop, Nov. 2004.
480. S.Banerjee, "Nanocrystal Floating Gate Memory," *Invited talk* at SRC Forum on Non-volatile Memory, Stanford Univ., Nov. 2004.
481. Shi ZH, Onsongo D, Banerjee SK, "Mobility and performance enhancement in compressively strained SiGe channel PMOSFETs," APPL SURF SCI 224 (1-4): 248-253 MAR 15 2004
482. Oh J, Banerjee SK, Campbell JC, "Metal-germanium-metal photodetectors on heteroepitaxial Ge-On-Si with amorphous Ge Schottky barrier enhancement layers," IEEE PHOTONIC TECH L 16 (2): 581-583 FEB 2004
483. Quinones E, Onsongo D, Shi Z, Banerjee SK, "Evaluation of heterojunction MOSFETs using UHVCVD deposited tensile-strained Si_{1-y}Cy alloy layers," SOLID STATE ELECTRON 48 (3): 379-387 MAR 2004.
484. Xia TS, Register LF, Banerjee SK, "Quantum transport in carbon nanotube transistors: Complex band structure effects," J APPL PHYS 95 (3): 1597-1599 FEB 1 2004
485. Kohli P, Jain A, Bu H, Chakravarthi S, Machala C, Dunham S and Banerjee SK, "Effect of nitride sidewall spacer process on B dose loss in ultrashallow junction formation, J. Vac. Sci. Technol. B 22(1), 471-476, Jan/Feb. 2004.
486. Xia TS, Register LF, Banerjee SK, Calculations and applications of the complex band structure for carbon nanotube field-effect transistors, PHYSICAL REVIEW B 70 (4): Art. No. 045322 JUL 2004
487. Kirichenko TA, Hwang GS, Banerjee SK "Interaction of neutral vacancies and interstitials with the Si(001) surface," Phys. Rev. B, 70 (4): Art. No. 045321 JUL 2004.
488. Kirichenko TA, Hwang GS, Banerjee SK, "Mechanisms of monovacancy annihilation and type-A defect creation on Si(001)-2x1," Surf. Sci. 555 (1-3): 187-192 APR 20 2004.
489. Fan X , Wang X, Winstead B, Register L, Ravaioli U, Banerjee SK, "Monte Carlo simulation of strained Si MOSFET," IEEE Trans. Elec. Dev. 51 (6): 962-970 JUN 2004.
490. Kirichenko TA, Banerjee SK, Hwang GS , "Surface chemistry effects on vacancy and interstitial annihilation on Si(001)," PHYSICA STATUS SOLIDI B-BASIC RESEARCH 241 (10): 2303-2312 AUG 2004
491. Chen JJH, Bojarczuk NA, Shang HL, Copel M, Hannon JB, Karasinski J, Preisler E, Banerjee SK, Guha S, "Ultrathin Al₂O₃ and HfO₂ gate dielectrics on surface-nitrided Ge," IEEE TRANSACTIONS ON ELECTRON DEVICES 51 (9): 1441-1447 SEP 2004
492. Chen X, Joshi S, Chen J, Ngai T, Banerjee SK , "MOS capacitors on epitaxial Ge-Si_{1-x}Ge_x with high-kappa dielectrics using RPCVD," IEEE TRANSACTIONS ON ELECTRON DEVICES 51 (9): 1532-1534 SEP 2004
493. Harrison SA, Yu D, Edgar TF, Hwang GS, Kirichenko TA, Banerjee SK, "Origin of vacancy and interstitial stabilization at the amorphous-crystalline Si interface," JOURNAL OF APPLIED PHYSICS 96 (6): 3334-3338 SEP 15 2004
494. Shi ZH, Onsongo D, Rai R, Samavedam SB, Banerjee SK, "Hole mobility enhancement and Si cap optimization in nanoscale strained Si_{1-x}Ge_xPMOSFETs," SOLID-STATE ELECTRONICS 48 (12): 2299-2306 DEC 2004
495. Lin L, Kirichenko T, Banerjee SK, Hwang GS, Boron diffusion in strained Si: A first-principles study," JOURNAL OF APPLIED PHYSICS 96 (10): 5543-5547 NOV 15 2004
496. Li H-J, Zeitzoff P, Larson L and Banerjee S, "B diffusion in Si with pre-amorphization of different species," J. Vac. Sci. and Tech. B 22(5), 2380-2383, Sept. 2004.
497. S.Jayanarayan, W.Bai, D.Kwong and S.Banerjee, "Si and SiGe vertical MOSFETs with CVD HFO₂ Gate Dielectric," TMS Letters, 2004.
498. Onsongo D, Kelly DQ, Dey S, Wise RL, Cleavelin CR, Banerjee SK, "Improved hot-electron reliability in strained-Si nMOS," IEEE TRANSACTIONS ON ELECTRON DEVICES 51 (12): 2193-2199 DEC 2004

499. Kohli P, Chakravarthi S, Jain A, Bu H, Mehrotra M, Dunham ST, Banerjee SK, "Fundamental characterization of the effect of nitride sidewall spacer process on boron dose loss in ultra-shallow junction formation," MATERIALS SCIENCE AND ENGINEERING B-SOLID STATE MATERIALS FOR ADVANCED TECHNOLOGY 114: 390-396 DEC 15 2004
500. Gate Oxide Reliability Studies for Strained Si on Relaxed SiGe MOS Devices: Sachin Joshi, Doreen Ahmad, Marylene Palard, David Q.Kelly, David Onsongo, Sagnik Dey, L. Fei, T. Torack, Mike Seacrist, Bruce Kellerman, Sanjay K. Banerjee, EMC, Santa Barbara, 2005.
501. $\text{Ge}_{1-y}\text{C}_y$ MOS Capacitors with HfO_2 Gate Dielectric and TaN Gate: David Quest Kelly; Joseph Patrick Donnelly; Sagnik Dey; Sachin Vineet Joshi; Sanjay K. Banerjee; EMC, Santa Barbara, 2005.
502. Theoretical Study of Boron Diffusion in Strained Si: Li Lin; Taras Kirichenko; Bhagawan Sahu; Joo Hwan Yoo; Sanjay Banerjee; EMC, Santa Barbara, 2005.
503. S.Guha, J.Chen, S.Banerjee, et. al., "Ge and high-k oxides for CMOS-materials challenges," MRS, 2005, **Invited** paper.
504. Carrier Confinement in Almost Pure Ge Channels Grown on Si Substrates by Rapidly Graded $\text{Si}_{1-x}\text{Ge}_x$ Growth: Sachin Joshi; Sagnik Dey; Kevin Jones; Michelle Chaumont; Alan Campion; David Q. Kelly; Joseph Donnelly; Sanjay K. Banerjee; EMC, Santa Barbara, 2005.
505. Rownak Jyoti Zaman, Weize Xiong, Andrea Franke, Rudy Quintilla, Nirmal Chaudhury, Thomas Schultz, Rinn Cleavelin, Rick Wise, Mike Pas, Shaofeng Yu, Klaus Schrufer and Sanjay K. Banerjee; "A Detailed Study of Hydrogen Annealing Process Effects on Silicon (011) Nano Structures., Spring MRS 2005.
506. S.Banerjee, "Microelectronics: The End the Beginning or Beginning of the End," **Invited** talk to Nanosummit, Houston, 2005
507. Electrical Transport Modeling of CNTFETs, T.Xia, L.Register and S.Banerjee, EMC 2005
508. Tunnel Oxide Thickness Dependence of Activation Energy for SiGe Quantum Dot Flash Memory, Y. Liu, S. Tang, D. Yu, G. Hwang, and S. Banerjee, DRC, Santa Barbara, 2005.
509. Monte Carlo Study of Germanium N- and P- MOSFETs, B. Ghosh, X.-F. Fan, L. F. Register, and S. K. Banerjee, DRC, Santa Barbara, 2005.
510. Tongsheng Xia, Leonard F. Register, and Sanjay K. Banerjee, "Transmission of the band gap states in Schottky barrier carbon nanotube transistors", 2nd conference of NDSI (nanoscale devices & system integration), (April, 2005).
511. S.Ganguly, L.Register, S.Banerjee and A.MacDonald, "Bias-controlled magnetization switch in a magnetic semiconductor RTD," 2nd conference of NDSI (nanoscale devices & system integration), 2005. (April, 2005)
512. S.Stanley, S.Joshi, S.Banerjee and J.Ekerdt, "Directed Self-assembly of Si and Ge Nanocrystals on HfO_2 through Kinetically Driven Patterning," AiChE Annual Meeting-Engineering Sciences and Fundamentals, 2005.
513. S.Banerjee, J.Donnelly, J.Chen, S.Joshi, D.Kelly, D.Ahmad, S.Dey, S.Guha, **Invited** Talk, and ECS Proc. Paper, "Ge-on-Si MOSFETs with HfO_2 gate dielectric," ECS, Quebec, May 2005
514. S.Banerjee, S.Joshi, D.Ahmad, M.Palard, D.Kelly, D.Onsongo, S.Dey, L.Fei, T.Toprack, M.Seacrist, B.Kellerman, and S.Banerjee, **Invited** Talk, and ECS Proc. paper "Oxide and hot carrier reliability studies for strained Si on relaxed SiGe MOS devices," ECS, Quebec, May 2005.
515. Tongsheng Xia, Leonard F. Register, and Sanjay K. Banerjee, "Simulation study of the carbon nanotube field effect transistors beyond the complex band structure region", Solid-State Electronics, Vol. 49, p860, 2005. (May, 2005)
516. Kohli P, Jain A, Chakravarthi S, Bu H, Dunham ST, Banerjee S, "Interactions of B dopant atoms and Si interstitials with SiO_2 films during annealing for ultra-shallow junction formation," JOURNAL OF APPLIED PHYSICS 97 (7): Art. No. 073520 APR 1 2005
517. Ghosh B, Wang X, Fan XF, Register LF, Banerjee SK, "Monte Carlo study of germanium n- and pMOSFETs," IEEE TRANSACTIONS ON ELECTRON DEVICES 52 (4): 547-553 APR 2005

518. Xia TS, Register LF, Banerjee SK, "Simulation study of the carbon nanotube field effect transistors beyond the complex band structure effect," SOLID-STATE ELECTRONICS 49 (5): 860-864 MAY 2005
519. Wiemer JC, Spencer TA, Banerjee SK, "Magnetic-field effects on Coulomb blockade conductance oscillations in a normal metal-insulator-normal metal double-barrier tunnel junction formed using atomic force microscope nanolithography," JOURNAL OF APPLIED PHYSICS 97 (8): Art. No. 084319 APR 15 2005
520. Li F, Mudanai S, Register LF, Banerjee SK, A physically based compact gate C-V model for ultrathin (EOT similar to 1 nm and below) gate dielectric MOS devices, IEEE TRANSACTIONS ON ELECTRON DEVICES 52 (6): 1148-1158 JUN 2005
521. Kirichenko TA, Yu D, Banerjee SK, Hwang GS, Silicon interstitials at Si/SiO₂ interfaces: Density functional calculations, PHYSICAL REVIEW B 72 (3): Art. No. 035345 JUL 2005
522. Dey S, Joshi S, Banerjee SK, Current-crowding effect in multiple cantilever channel MOSFET, SOLID-STATE ELECTRONICS 49 (7): 1248-1250 JUL 2005
523. Kelly DQ, Dey S, Onsongo D, Banerjee SK, Considerations for evaluating hot-electron reliability of strained Si n-channel MOSFETs, MICROELECTRONICS RELIABILITY 45 (7-8): 1033-1040 JUL-AUG 2005, **INVITED PAPER**
524. Ganguly S, Register LF, Banerjee S, MacDonald AH, Bias-voltage-controlled magnetization switch in ferromagnetic semiconductor resonant tunneling diodes, PHYSICAL REVIEW B 71 (24): 245306 JUN 2005
525. Lin L, Kirichenko T, Sahu BR, Hwang GS, Banerjee SK, Theoretical study of B diffusion with charged defects in strained Si, PHYSICAL REVIEW B 72 (20): Art. No. 205206 NOV 2005
526. Yu D, Hwang GS, Kirichenko TA, Banerjee SK, Structure and diffusion of excess Si atoms in SiO₂, PHYSICAL REVIEW B 72 (20): Art. No. 205204 NOV 2005
527. David Q. Kelly, Joseph P. Donnelly, Sachin V. Joshi, Sagnik Dey, Domingo I. García Gutiérrez, Miguel José Yacamán, and Sanjay K. Banerjee: Drive Current Enhancement in High- κ /Metal Gate Germanium-Carbide pMOSFETs Fabricated Directly on Si Substrates, IEDM 2005.
528. David Kelly and Sanjay K. Banerjee, Ge_{1-y}C_y MOS Devices with High- κ Gate Dielectrics and Metal Gates, SRC TECHCON Portland, Oct. 2005.
529. M.Oye, J.Hurst, C.Shih, S.Banerjee, J.Campbell, A.Holmes and T.Mattord, AFM study of sapphire surfaces annealed with H₂O flux from a baffled MBE effusion cell loaded with Al(OH)₃, NAMBE 2005.
530. S. Joshi, S. Dey, D. Garcia-Gutierrez, M. Chaumont, M. Yacaman, A. Campion, D.Q. Kelly, J. Donnelly and S.K. Banerjee: Ge channel MOSFETs fabricated using thin Ge on strained SiGe epitaxial layers using bulk Si substrates, HfO₂ gate dielectric and TaN metal gate electrode, SRC TECHCON Portland, Oct. 2005.
531. S.Banerjee, High mobility Strained Ge MOSFETs with high-k gate dielectrics, ISDRS, Washington D.C., Dec. 2005. **Invited** talk (with J.Donnely, D.Kelly, S.Joshi, S.Dey, D.Shahrjerdi, I.Wiedeman and D.Ahmad)
532. Wanqiang Chen, Leonard F. Register, and Sanjay K. Banerjee, "Schroedinger Equation Monte Carlo"-based simulation of nanoscale double gate MOSFETs, SRC TECHCON Portland, Oct. 2005.
533. S.Tang, C.Mao, ... S.Banerjee, Nanocrystal flash memory made with protein mediated assembly, Washington D.C. IEDM 2005.
534. Bahniman Ghosh, Xiao-Feng Fan, Leonard F. Register and Sanjay K. Banerjee, Monte Carlo simulation study of III-V semiconductor PMOSFETs, TECHCON Portland, Oct. 2005.
535. Stanley SK, Joshi SV, Banerjee SK, Ekerdt JG, Ge interactions on HfO₂ surfaces and kinetically driven patterning of Ge nanocrystals on HfO₂, JOURNAL OF VACUUM SCIENCE & TECHNOLOGY A 24 (1): 78-83 JAN-FEB 2006

536. Ghosh B, Fan XF, Register LF, Banerjee SK, Monte Carlo study of strained germanium nanoscale bulk pMOSFETs, IEEE TRANSACTIONS ON ELECTRON DEVICES 53 (3): 533-537 MAR 2006
537. Ghosh B, Chen JH, Fan XF, Register LF, Banerjee SK, Monte Carlo study of remote Coulomb and remote surface roughness scattering in nanoscale GePMOSFETs with ultrathin high-kappa dielectrics, SOLID-STATE ELECTRONICS 50 (2): 248-253 FEB 2006
538. Ganguly S, Register LF, MacDonald AH, Banerjee SK, Two-level voltage-controlled magnetization switch using a ferromagnetic semiconductor resonant-tunneling diode, IEEE TRANSACTIONS ON NANOTECHNOLOGY 5 (1): 30-36 JAN 2006
539. Ganguly S, MacDonald AH, Register LF, Banerjee S, Intrinsic Curie temperature bistability in ferromagnetic semiconductor resonant tunneling diodes, PHYSICAL REVIEW B 73 (3): Art. No. 033310 JAN 2006
540. Xia TS, Register LF, Banerjee SK, Transmission through the band-gap states in Schottky-Barrier carbon nanotube transistors, IEEE TRANSACTIONS ON NANOTECHNOLOGY 5 (2): 80-83 MAR 2006
541. Stanley SK, Joshi SV, Banerjee SK, Ekerdt JG, Surface reactions and kinetically-driven patterning scheme for selective deposition of Si and Ge nanoparticle arrays on HfO₂, SURFACE SCIENCE 600 (5): L54-L57 MAR 1 2006
542. Kelly DQ, Donnelly JP, Dey S, Joshi SV, Gutierrez DIG, Yacaman MJ, Banerjee SK, BC high-kappa/metal gate Ge/C alloy pMOSFETs fabricated directly on Si (100) substrates, IEEE ELECTRON DEVICE LETTERS 27 (4): 265-268 APR 2006
543. Schricker AD, Joshi SV, Hanrath T, Banerjee SK, Korgel BA, Temperature dependence of the field effect mobility of solution-grown germanium nanowires, JOURNAL OF PHYSICAL CHEMISTRY B 110 (13): 6816-6823 APR 6 2006
544. Kelly DQ, Wiedmann I, Donnelly JP, Joshi SV, Dey S, Banerjee SK, Garcia-Gutierrez DI, Jose-Yacaman M, Thin germanium-carbon alloy layers grown directly on silicon for metal-oxide-semiconductor device applications, APPLIED PHYSICS LETTERS 88 (15): Art. No. 152101 APR 10 2006
545. Jayanarayanan SK, Dey S, Donnelly JP, Banerjee SK, A novel 50 nm vertical MOSFET with a dielectric pocket, SOLID-STATE ELECTRONICS 50 (5): 897-900 MAY 2006
546. Oye MM, Hurst JB, Shahrjerdi D, Kulkarni NN, Muller A, Beck AL, Sidhu R, Shih CK, Banerjee SK, Campbell JC, Holmes AL, Mattord TJ, Reifsnider JM, Atomic force microscopy study of sapphire surfaces annealed with a H₂O flux from a baffled molecular-beam epitaxy effusion cell loaded with Al(OH)(3), JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B 24 (3): 1572-1576 MAY-JUN 2006
547. Liu YR, Tang S, Banerjee SK, Tunnel oxide thickness dependence of activation energy for retention time in SiGe quantum dot flash memory, APPLIED PHYSICS LETTERS 88 (21): Art. No. 213504 MAY 22 2006
548. Li F, Mudanai SP, Fan YY, Register LF, Banerjee SK, Physically based quantum-mechanical compact model of MOS devices substrate-injected tunneling current through ultrathin (EOT similar to 1 nm) SiO₂ and high-k gate stacks, IEEE TRANSACTIONS ON ELECTRON DEVICES 53 (5): 1096-1106 MAY 2006
549. Shahrjerdi D, Oye MM, Holmes AL, Banerjee SK, Unpinned metal gate/high-kappa GaAs capacitors: Fabrication and characterization, APPLIED PHYSICS LETTERS 89 (4): Art. No. 043501 JUL 24 2006
550. Li F, Tseng HH, Register LF, Tobin PJ, Banerjee SK, Asymmetry in gate capacitance-voltage (C-V) behavior of ultrathin metal gate MOSFETs with HfO₂ gate dielectrics, IEEE TRANSACTIONS ON ELECTRON DEVICES 53 (8): 1943-1946 AUG 2006
551. Dey S, Joshi S, Garcia-Gutierrez D, Chaumont M, Champion A, Jose-Yacaman M, Banerjee SK, Pure germanium epitaxial growth on thin strained silicon-germanium graded layers on bulk

- silicon substrate for high-mobility channel metal-oxide-semiconductor field-effect transistors, JOURNAL OF ELECTRONIC MATERIALS 35 (8): 1607-1612 AUG 2006
552. Y.Kim, T.Kirichenko, G.Hwang and S.Banerjee, "Ab initio study of B diffusion retardation in SiGe," MRS San Francisco, Spring, 2006.
 553. Scott K. Stanley, Yueran Liu, Sanjay K. Banerjee, John G. Ekerdt, Core-shell Ge nanoparticles for enhanced interface stability, MRS Spring 2006.
 554. Coffee SS, Stanley SK, Winkenwerder WA, Shahrjerdi D, Banerjee SK, Ekerdt JG, Using self-assembly and selective chemical vapor deposition for precise positioning of individual germanium nanoparticles on hafnia, MRS Spring 2006.
 555. B.Sahu, L.Kleinman and S.Banerjee, Density functional investigation of magnetism in Mn doped Si, MRS, Fall, 2005.
 556. Z.Huang, N.Kong, J.Oh, S.Banerjee and J.Campbell, Effectiveness of SiGe buffer layers in reducing dark current in Ge-on-Si photodetectors, Univ. Park, EMC, 2006.
 557. M.Oye, D.Shahrjerdi, .. J.Harris, A.Holmes and S.Banerjee, GaAs growth on Si substrates using thin (~80 nm) SiGe step-graded buffer layer for high-k III-V MOSFET applications, Univ. Park, EMC 2006.
 558. D.Kelly, S.Banerjee, "Thin Germanium-Carbon Layers on Silicon for Metal-Oxide-Semiconductor Devices" 3rd International Silicon Germanium Technology and Device Meeting, Princeton, NJ, May 2006.
 559. D.Basu, M.Gilbert, and S.Banerjee, "Effect of Elastic Processes and Ballistic Recovery in Si Nanowire transistors," Int. Workshop on Computational Electronics, Vienna, May 2006
 560. S.Banerjee, "Strained channel Si-Ge-C MOSFETs," SEMICON, Seoul Korea, Feb. 2006.
- Invited talk**
561. S.Banerjee, "An overview of alternative charge and magnetic moment based devices," AVS ICMI conference, Austin, March 2006, **Invited** talk.
 562. M. Gilbert and S.Banerjee, Ballistic recovery in III-V nanowire transistors," AVS ICMI conference, Austin, March 2006.
 563. D.Basu, M.Gilbert and S.Banerjee, "Surface roughness exacerbated performance degradation in Si nanowire transistors," AVS- ICMI conference, Austin, March 2006.
 564. D.Kelly, ...M.Yacaman and S.Banerjee, "Thin Ge-C alloy layers on Si for improved hole mobility in buried channel pMOSFETs," AVS ICMI conference, Austin, March 2006.
 565. D. Shahrjerdi, J. Sarkar, X. Gao, D. Q. Kelly, S. K. Banerjee, Fabrication of Self-Assembled Ni Nanocrystal Flash Memories Using a Polymeric Template, Univ. Park DRC 2006.
 566. J.Sarkar, S.Dey, Y.Liu, D.Shahrjerdi, D.Kelly and S.Banerjee, Vertical (3_D) Flash Memory with SiGe nanocrystal Floating Gate, DRC 2006
 567. Yueran Liu, Shan Tang, Chuanbin Mao and Sanjay Banerjee, SiC Nanocrystal Flash Memory Fabricated with Protein-mediated Assembly, DRC 2006.
 568. D. Shahrjerdi, J. Sarkar, S. K. Banerjee, Fabrication of Dense Ordered Arrays of Metal Dots for Flash Memory Application, MRS 2006.
 569. Bahniman Ghosh, Xiao-Feng Fan, Leonard F. Register and Sanjay K. Banerjee, Monte Carlo simulation study of III-V semiconductor PMOSFETs, SRC Symp. 2006
 570. Bahniman Ghosh, Xiao-Feng Fan, Leonard F. Register and Sanjay K. Banerjee, Monte Carlo study of remote Coulomb and remote surface roughness scattering in nanoscale Ge PMOSFETs with ultrathin high-k dielectrics, SISPAD 2006
 571. Xiao-Feng Fan, Bahniman Ghosh, Leonard F. Register and Sanjay K. Banerjee, Scalability of Biaxially Strained Si NMOS on Technology Roadmap, SISPAD 2006
 572. T.Kirichenko, D.Yu, S.Banerjee and G.Hwang, Vacancy at Si-SiO₂ Interface: Ab-Initio Study, SISPAD 2006
 573. X.Fan, F.Register and S.Banerjee, Hole mobility enhancement with <110> uniaxial stress in Si up to 2GPa, SRC Symp. 2006.

574. Y.Kim, S.Harrison, S.Lee, S.Banerjee and G.Hwang, Ab-initio calculation of As-vacancy deactivation and interstitial-mediated As diffusion in strained Si, SISPAD 2006
575. David Kelly and Sanjay K. Banerjee, Thin $\text{Ge}_{1-y}\text{C}_y$ layers grown directly on Si for MOS Device Applications, SRC Symp. 2006. **(BEST PAPER AWARD.)**
576. S. Joshi, S.Dey, .. and S.K. Banerjee, Strained Si and Ge MOSFETs fabricated using multiple thin SiGe dislocation blocking layers, SRC Symp. 2006.
577. S.Banerjee, Ge-C MOSFETs, ECS Meeting, Cancun Mexico, Oct. 2006. **Invited** talk (with D.Kelly, D.Garcia and M.Yacaman)
578. Z.Huang, Kong, J.Oh, S.K. Banerjee and J.Campbell, Interdigitated photodiode fabricated in high quality Ge with thin SiGe buffer layers, IEEE LEOS Summer, 2006.
579. J. Na, C. Krug, S. Joshi, D. Heh, P. Sivasubramani, P. D. Kirsch, R. Choi, B. H. Lee, R. Jammy, S. K. Banerjee, and J. C. Lee, Improved passivation and characterization of the Ge/HfSiO interface enabling surface channel Ge pMOSFETs, SISC, 2006.
580. Sachin Joshi, Sagnik Dey, Michelle Chaumont, Alan Campion and Sanjay K. Banerjee, Ultra-thin Si_{1-x}Ge_x dislocation blocking layers for Ge /strained Si CMOS devices;±, International Symposium on Advanced Gate Stack Technology, September 2006
581. Swaroop Ganguly, L.F. Register, A.H. MacDonald, and S.K. Banerjee, “Scattering Dependence of Magnetization Switching in Ferromagnetic Resonant Tunneling Diodes”, *European Materials Research Society Fall Meeting*, Warsaw, September 2006.
582. Garcia-Gutierrez DI, Jose-Yacaman M, Lu SF, Kelly DQ, Banerjee SK , Carbon segregation as a strain relaxation mechanism in thin germanium-carbon layers deposited directly on silicon, JOURNAL OF APPLIED PHYSICS 100 (4): Art. No. 044323 AUG 15 2006
583. Li F, Register LF, Hasan MM, Banerjee SK A program for device model parameter extraction from gate capacitance and current of ultrathin SiO₂ and high-k gate stacks, IEEE TRANSACTIONS ON ELECTRON DEVICES 53 (9): 2118-2127 SEP 2006
584. Z.Huang, Kong, Duan, S.K. Banerjee and J.Campbell, 21GHz Bandwidth Ge on Si photodiode using thin SiGe buffers, Sel. Topics J. Quantm Elec. 12 (6): 1450-1454 Part 2, NOV-DEC 2006
585. Ganguly S, MacDonald AH, Register LF, Banerjee SK, Scattering dependence of bias-controlled magnetization switching in ferromagnetic resonant tunneling diodes PHYSICAL REVIEW B 74 (15): OCT 2006
586. Basu D, Gilbert MJ, Banerjee SK, Surface roughness exacerbated performance degradation in silicon nanowire transistors, JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B 24 (5): 2424-2428 SEP-OCT 2006
587. Liu YR, Dey S, Tang S, Kelly DQ, Sarkar J, Banerjee SK, Improved performance of SiGe nanocrystal memory with VARIOT tunnel barrier, IEEE TRANSACTIONS ON ELECTRON DEVICES 53 (10): 2598-2602 OCT 2006
588. Schrodinger Equation Monte Carlo-3D for Simulation of Nanoscale MOSFETs, Liu KM, Chen WQ, Register LF, Banerjee SK, 7th IEEE Conference on Nanotechnology, AUG 02-05, 2007 Hong Kong, PEOPLES R CHINA, **2007 7TH IEEE CONFERENCE ON NANOTECHNOLOGY, VOL 1-3 Pages: 1128-1133 Published: 2007**
589. High performance pMOSFETs using Si/Si_{1-x}Ge_x/Si quantum wells with high-k/metal gate stacks and additive uniaxial strain for 22 nm technology node, Suthram S, Majhi P, Sun G, Kalra P, Harris HR, Choi KJ, Heh D, Oh J, Kelly D, , Cho BJ, Hussain MM, Smith C, Banerjee S, Tsai W, Thompson SE, Tseng HH, Jammy R, IEEE International Electron Devices Meeting, DEC 10-12, 2007 Washington, DC
590. Protein-mediated assembly of nanocrystal floating gate in a vertical flash cell Sarkar J, Tang S, Garcia D, Banerjee SK, 22nd IEEE Non-Volatile Semiconductor Memory Workshop, Monterey, CA Pages: 34-35 Aug. 2007

591. Joshi S, Sahu B, Banerjee SK, Ciucivara A, Kleinman L, Wise R, Cleavelin R, Pinto A, Seacrist M, Ries M, Theoretical and experimental investigation of valence band offsets for direct silicon bond hybrid orientation technology, *APPLIED PHYSICS LETTERS* 90 (4): Art. No. 043503 JAN 22 2007
592. Fan XF, Register LF, Winstead B, Foisy MC, Chen WQ, Zheng X, Ghosh B, Banerjee SK, Hole mobility and thermal velocity enhancement for uniaxial stress in Si up to 4 GPa, *IEEE TRANSACTIONS ON ELECTRON DEVICES* 54 (2): 291-296 FEB 2007
593. Kelly DQ, Wiedmann I, Garcia-Gutierrez DI, Jose-Yacamán M, Banerjee SK, Thin germanium-carbon layers deposited directly on silicon for metal-oxide-semiconductor devices, *SEMICONDUCTOR SCIENCE AND TECHNOLOGY* 22 (1): S204-S207 Sp. Iss. SI, JAN 2007
594. Kong N, Banerjee SK, Kirichenko TA, Anderson SGH, Foisy MC, Enhanced and retarded diffusion of arsenic in silicon by point defect engineering, *APPLIED PHYSICS LETTERS* 90 (6): Art. No. 062107 FEB 5 2007
595. Gilbert MJ, Banerjee SK, Ballistic recovery in III-V nanowire transistors, *JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B* 25 (1): 189-193 JAN-FEB 2007
596. Tang S, Mao CB, Liu YR, Kelly DQ, Banerjee SK, Protein-mediated nanocrystal assembly for flash memory fabrication, *IEEE TRANSACTIONS ON ELECTRON DEVICES* 54 (3): 433-438 MAR 2007
597. Gilbert M and Banerjee SK, Ballistic to Diffusive Crossover in III-V Nanowire Transistors, *IEEE TRANSACTIONS ON ELECTRON DEVICES* 54 (4): 645-653, APRIL 2007
598. Hongki Min, B.R. Sahu, Sanjay K. Banerjee, A.H. MacDonald, Ab Initio Theory of Gate Induced Gaps in Graphene Bilayers, *PHYSICAL REVIEW B* 75 (15): Art. No. 155115 APR 2007
599. Adrian Ciucivara, B. R. Sahu, Sachin Joshi, Sanjay K. Banerjee, and Leonard Kleinman, Density functional study of Si(001)/Si(110) and Si(100)/Si(110) interfaces, *PHYSICAL REVIEW B* 75 (11): Art. No. 113309 MAR 2007
600. Huang ZH, Oh J, Banerjee SK, Campbell JC, Effectiveness of SiGe buffer layers in reducing dark currents of Ge-on-Si photodetectors, *IEEE JOURNAL OF QUANTUM ELECTRONICS* 43 (3-4): 238-242 MAR-APR 2007
601. Sarkar J, Dey S, Shahrjerdi D, Banerjee SK, Vertical Flash memory cell with nanocrystal floating gate for ultradense integration and good retention *IEEE ELECTRON DEVICE LETTERS* 28 (5): 449-451 MAY 2007
602. Joshi S, Krug C, Heh D, Na HJ, Harris HR, Oh JW, Kirsch PD, Majhi P, Lee BH, Tseng HH, Jammy R, Lee JC, Banerjee SK, Improved Ge surface passivation with ultrathin SiO_x enabling high-mobility surface channel pMOSFETs featuring a HfSiO/WN gate stack, *IEEE ELECTRON DEVICE LETTERS* 28 (4): 308-311 APR 2007
603. Joshi S, Dey S, Chaumont M, Champion A, Banerjee SK, Ultra-thin Si_{1-x}Ge_x dislocation blocking layers for Ge/strained SiCMOS devices, *JOURNAL OF ELECTRONIC MATERIALS* 36 (6): 641-647 JUN 2007
604. Oye MM, Shahrjerdi D, Ok I, Hurst JB, Lewis SD, Dey S, Kelly DQ, Joshi S, Mattord TJ, Yu X, Wistey MA, Harris JS, Holmes AL, Lee JC, Banerjee SK, Molecular-beam epitaxy growth of device-compatible GaAs on silicon substrates with thin (similar to 80 nm) Si_{1-x}Ge_x step-graded buffer layers for high-kappa III-V metal-oxide-semiconductor field effect transistor applications, *JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B* 25 (3): 1098-1102 MAY-JUN 2007
605. Sarkar, Joy, Tang, Shan, Shahrjerdi, Davood, Banerjee, Sanjay K., Vertical flash memory with protein-mediated assembly of nanocrystal floating gate, *APPLIED PHYSICS LETTERS* 90 (10): Art. No. 103512 MAR 5 2007
606. S. Dey, Sachin Joshi, P.Majhi and S.K. Banerjee, Gate-all-around (GAA) fully depleted (FD) cantilever channel MOSFET with high-k and metal Gate, *Materials Research Society Spring Meeting*, 2007

607. S.-H. Lee, S. Dey, Sachin Joshi , P. Majhi and S. K. Banerjee, SiGe cantilever channel gate-all-around (GAA) fully depleted (FD) PMOSFET with high-k and metal gate, Device Research Conference, 2007
608. Doreen Ahmad, Sagnik Dey, Marylene Palard, Fahmida Ferdousi, Mustafa Jamil, Emanuel Tutuc and Sanjay K. Banerjee, Nano-Imprint Lithography to fabricate Nanowire Field Effect Transistor (NWFET), EMC 2007.
609. M. Liu, W. Chen, L. F. Register and S. Banerjee, "Schrödinger Equation Monte Carlo-3D for Simulation of Nanoscale MOSFETs," IEEE Review of Advances on Micro, Nano, and Molecular Systems, IEEE-NANO 2007, Hong Kong Aug. 2-5, 2007.
610. Yonghyun Kim, Taras A. Kirichenko, Ning Kong, Larry Larson, and Sanjay K. Banerjee, Stability and Diffusion of Small Arsenic Interstitial Complexes in Crystalline Si: First-principle Studies, Int. Conf. Defects in Semicond. 2007.
611. Ning Kong, Taras A. Kirichenko, Gyeong S. Hwang, Mark C. Foisy, Steven G. H. Anderson, Sanjay K. Banerjee, "An experimental and simulation study of arsenic diffusion behavior in point defect engineered silicon" 2007 MRS Proceeding 0994-F10-02.
612. Sanjay Banerjee, Strain and Bandgap Engineered High Channel Mobility MOSFETs, Keynote talk, ASME, Austin, June 2007.
613. *Sanjay Banerjee, Shan Tang, Joy Sarkar, Davood Shahrjerdi, Chang Lee, Flash Memory with Nanoparticle Floating Gate, Toronto, Particles Aug. 2007, **Invited**
614. Sanjay Banerjee, Materials, devices, and heterogeneous integration for new functions, GOMAC, Orlando, FL, Jan. 2007, **Invited**.
615. S.Banerjee, Retrospect and Prospects of Nanocrystal Floating Gate Memories, MARCO MSD e-workshop, July 2007.
616. Shan Tang, C. H. Lee, X. Gao and Sanjay Banerjee, "Flash Memory Fabricated with Protein-Mediated PbSe Nanocrystal Assembly as Floating Gate," IEEE Device Research Conference (DRC) 2007, p93-94.
617. Sarkar, Shan Tang, D. Garcia and Sanjay Banerjee, "Protein-mediated assembly of nanocrystal floating gate in a vertical flash cell," 22nd IEEE Non-Volatile Semiconductor Memory Workshop 2007.
618. S.Banerjee, Future of Microelectronics: The beginning of the end or the end of the beginning? Plenary talk, April 2007, TAMEST Conference, Austin.
619. S.Banerjee, New Materials and Structures for Transistors based on Spin, Charge and Wavefunction Phase Control, NIST, Gaithersburg, MD, March, 2007, **Invited**.
620. S.Banerjee, Computational Aspects of Nanowire FETs, NCN NRI Meeting, Indianapolis, Feb. 2007, **Invited**.
621. S.Banerjee, Beyond CMOS, Brussels, April 2007, **Invited**.
622. S.Banerjee (with N.Peppas and K.Roy), How Microelectronics will Impact and Improve Drug Discovery and Delivery, Plenary Talk, Pharma Conf. Austin, Jan. 2007.
623. Sachin Joshi, C. Krug, D. Heh, H. J. Na, R. Harris, J. Oh, P. D. Kirsch, P. Majhi, B. H. Lee, R. Jammy and S. K. Banerjee, Improved Ge Interface Passivation with Ultrathin SiO₂ Enabling High Mobility Surface Channel PMOSFETs Featuring a HfSiO₂ / WN gate stack, 34th Conference on the Physics and Chemistry of Semiconductor Interfaces, Salt Lake City, Jan 2007, (Young Scientist Award at PCSI 2007.)
624. Sachin Joshi, Adrian Ciucivara, Bhagwan Sahu, Leonard Kleinman, Rick Wise, Mike Seacrist, Rinn Cleavelin, Angelo Pinto, J-T. Huang and Sanjay Banerjee, "Theoretical and experimental investigations of valence band offsets in a Direct Silicon Bond hybrid orientation technology", 34th Conference on the Physics and Chemistry of Semiconductor Interfaces, Jan 2007 (Young Scientist Award at PCSI 2007).
625. Angelo Pinto, Sachin Joshi, Y.-T. Huang, Rick Wise, Rinn Cleavelin, Mike Seacrist, Mike Ries, Manfred Ramin, Melissa Freeman, Billy Nguyen, Kenneth Matthews, Bruce Wilks, Charles

- Stager, Mike Ma, C.-T. Lin and Sanjay K. Banerjee, Bonded layer thickness optimization for a DSB-HOT Technology, Materials Research Society meeting, Spring 2007.
626. Sachin Joshi, S Dey, C Krug, S Lee, P Kirsch, W Wang, A Campion, D Heh, H Harris, P Majhi, B H Lee, H H Tseng, R Jammy, J Lee, S Banerjee, 3X hole mobility enhancement in epitaxially grown SiGe PMOSFETs on (110) Si substrates for hybrid orientation technology, Device Research Conference, 2007.
 627. M.Oye,... S.Banerjee, et. al. Effects of different plasma species on optical properties of dilute nitrides grown by plasma MBE, EMC 2007
 628. D.Sharjedi, T.Aykol, D. Shahrjerdi, T. Akyol, M. Oye, G. Balakrishnan, A. Khoshakhlagh, E. Tutuc, D. Huffaker and S. K. Banerjee, "Atomic layer deposition of Al₂O₃ and HfO₂ gate dielectrics on GaAs for MOSFET applications", presented at EMC 2007.
 629. H.Min, B.Sahu, S.Banerjee, A.MacDonald, Bilayer Graphene, APS March meeting, 2007. (A28.00009)
 630. B.Sahu and S.Banerjee, Computational study of local meta-magnetic states in Mn doped Si, APS March meeting, 2007. (L14.00001)
 631. A.Ciucivara, B.Sahu, S.Banerjee and L.Kleinman, Expt. and theoretical study of Si (001)/(110) junctions, APS March meeting, 2007. (N39.00007)
 632. *S.Banerjee, SWAN overview, Int. Assoc. Nano, San Francisco, Oct. 2006, **Invited**
 633. Kelly DQ, Lee S, Kalra P, Harris R, Oh J, Kirsch P, Banerjee SK, Majhi P, Tseng H, Jammy R, Interrelationship between electrical and physical properties of subcritical Si-Ge layers grown directly on silicon for short channel high-performance pMOSFETs, MICROELECTRONIC ENGINEERING 84 (9-10): 2054-2057 SEP-OCT 2007
 634. D. Shahrjerdi, E. Tutuc and S. K. Banerjee, "Impact of surface chemical treatment on capacitance-voltage characteristics of GaAs metal-oxide-semiconductor capacitors with Al₂O₃ gate dielectric", APPLIED PHYSICS LETTERS 91, 1 (2007)
 635. Sanjay K. Banerjee, Emanuel Tutuc, Joseph Donnelly and Davood Shahrjerdi, Enhanced Channel Mobility Materials for MOSFETs on Si Substrates, The Electro-Chemical Society (ECS) Transactions, October 2007. (**Invited**)
 636. S. Suthram, P. Majhi, G. Sun, P. Kalra, R. Harris, K. J. Choi, D. Heh, J. Oh, D. Kelly, R. Choi, B.J. Cho, M. M. Hussain, C. Smith, S. Banerjee, W. Tsai, S. E. Thompson, H. H. Tseng, R. Jammy Demonstration of High Performance PMOSFETs Using Si/Si_xGe_{1-x}/Si Quantum Wells with High-k/Metal Gate Stacks and Uniaxial Strain Additivity for 22 nm Technology and Beyond, IEDM 2007
 637. L.F. Register, M.M. Hasan, F. Li, and S.K. Banerjee, "Efficient Gate Capacitance and Current Modeling of High-k Gate Stacks," The Electro-Chemical Society (ECS) Transactions, Physics and Technology of High-k Gate Dielectrics 5, Oct. 2007. (**Invited talk by Banerjee**)
 638. D. Basu, M. J. Gilbert, L. F. Register, and S. K. Banerjee, Effect of Elastic Processes in Silicon Nanowire Transistors, SRC TECHCON, Austin, Sept, 2007.
 639. Sanjay Banerjee, "How might the nanoscale transistors of the future be made?" NanoTEX, Dallas, Oct. 2007, **Invited**. Also two panel discussion talks on memory and beyond CMOS.
 640. Ning Kong, Taras A. Kirichenko, Gyeong S. Hwang, Mark C. Foisy, Steven G. H. Anderson, Sanjay K. Banerjee, "Arsenic diffusion in point defect engineered silicon" SRC TECHCON, Austin, 2007.
 641. Shan Tang, C. H. Lee, X. Gao and Sanjay Banerjee, "Bio-Nano Fabrication of Flash Memories with PbSe Nanocrystal Floating Gates," SRC TECHCON. Austin 2007.
 642. H.Chen, M.J. Gilbert, L.F. Register, and S.K. Banerjee, Rashba Effect in a Gate-controlled Aharonov-Bohm Device, SRC TECHCON, Austin, Oct. 2007
 643. Y.Kim, Taras A. Kirichenko, Ning Kong, Larry Larson, S.Banerjee, Stability and Diffusion of Small Arsenic Interstitial Complexes in Crystalline Silicon: First-principle Studies, SRC TECHCON, Austin, Oct. 2007

644. Sanjay Banerjee, Shan Tang, C. B. Mao, J. Sarkar, H. Liu, D. Shahrjerdi, C. H. Lee and J. D. Trent, "Bio-Nano Approaches to Fabrication of Quantum Dot Floating Gate Flash Memories," International Conference on Solid State Devices and Materials (SSDM) 2007, **invited** paper; also S.Banerjee, Bio-nano Fabrication of Nanoparticle floating gate Memory, Panasonic Research Center, Sept. 2007, Nara, Japan, **Invited**.
645. Tackhwi Lee, Jack C. Lee and Sanjay K. Banerjee "Characterization of Dysprosium oxide (Dy₂O₃) incorporated HfO₂ gate oxide devices, " SISC 2007.
646. Hai Liu, Wyatt Winkenwerder, Yueran Liu, Scott K. Stanley, John G. Ekerdt, Sanjay K. Banerjee, "Core-shell Germanium-Silicon Nanoparticle Structure for High K Nonvolatile Memory Applications," International Semiconductor Device Research Symposium 2007.
647. D. Shahrjerdi*, N. Nuntawong, T. Akyol, G. Balakrishnan, A. Khoshakhlagh, S. Dey, S. R. Bank, E. Tutuc, D. Huffaker and S. K. Banerjee, "Fabrication of Unpinned Metal Gate/ High-k GaAs MOS capacitors on Ge/ Si/ _{1-x} /Ge/ _x // Si Substrates", /North American Conference on Molecular Beam Epitaxy/, 2007
648. Improved Electrical Characteristics of Ge-on-Si Field-Effect Transistors With Controlled Ge Epitaxial Layer Thickness on Si Substrates, Jungwoo Oh; Majhi, P.; Lee, H.; Oooksang Yoo; Banerjee, S.; Chang Yong Kang; Ji-Woon Yang; Harris, R.; Hsing-Huang Tseng; Jammy,R.; Electron Device Letters, IEEE Volume 28 (11), Nov. 2007 Pp:1044 - 1046
649. Junction Passivation for Direct Silicon Bond Hybrid Orientation Technology, Joshi, S.; Pinto, A.; Huang, Y.-T.; Wise, R.; Cleavelin, R.; Seacrist, M.; Ries, M.; Ramin, M.; Freeman, M.; Nguyen, B.; Matthews, K.; Wilks, B.; Denning, L.; Johnson, C.; Bennet, J.; Ma, M.; Lin, C.-T.; Banerjee, S.K., Electron Devices, IEEE Transactions on Volume 54, Issue 8, Aug. 2007 Page(s):2045 – 2050.
650. Fabrication of Ni nanocrystal flash memories using a polymeric self-assembly approach, Shahrjerdi D, Garcia-Gutierrez DI, Banerjee SK, IEEE ELECTRON DEVICE LETTERS Volume: 28 Issue: 9 Pages: 793-796 SEP 2007
651. Trigate FET Device Characteristics Improvement Using a Hydrogen Anneal Process With a Novel Hard Mask Approach, Zaman, R.J.; Mathews, K.; Xiong Weize; Banerjee, S.K.; Electron Device Letters, IEEE, Volume 28, Issue 10, Oct. 2007 Page(s):916 – 918
652. GaAs metal-oxide-semiconductor capacitors using atomic layer deposition of HfO₂ gate dielectric: Fabrication and characterization, Shahrjerdi D, Garcia-Gutierrez DI, Akyol T., S. R. Bank, E. Tutuc, J. C. Lee, Banerjee S., APPLIED PHYSICS LETTERS Volume: 19 Article Number: 193503, NOV 5 2007
653. Selective silicon nanoparticle growth on high-density arrays of silicon nitride Author(s): Coffee SS, Shahrjerdi D, Banerjee SK, Ekerdt J., JOURNAL OF CRYSTAL GROWTH Volume: 308 Issue: 2 Pages: 269-277 Published: OCT 15 2007
654. First-principles studies of di-arsenic interstitial and its implications for arsenic-interstitial diffusion in crystalline silicon, Kim Y, Kirichenko TA, Kong N, Larson L, Banerjee SK, PHYSICA B-CONDENSED MATTER Volume: 401 Pages: 144-147, DEC 15 2007
655. Effect of edge roughness on electronic transport in graphene nanoribbon channel metal-oxide-semiconductor field-effect transistors, Basu D, Gilbert MJ, Register LF, Macdonald A, Banerjee SK, APPLIED PHYSICS LETTERS Volume: 92 Issue: 4 Article Number: 042114, JAN 28 2008
656. High mobility strained Ge PMOSFETs with high-kappa gate dielectric and metal gate on Si substrate, Donnelly JP, Kelly DQ, Garcia-Gutierrez DI, Jose-Yacamán M, Banerjee SK, ELECTRONICS LETTERS Volume: 44 Issue: 3 Pages: 240-U24, JAN 31 2008
657. Schrodinger equation Monte Carlo in two dimensions for simulation of nanoscale metal-oxide-semiconductor field effect transistors, Chen WQ, Register LF, Banerjee SK, JOURNAL OF APPLIED PHYSICS Volume: 103 Issue: 2 Article Number: 024508 JAN 15 2008
658. Demonstration of high-performance PMOSFETs using Si-SixGe_{1-x}-Si quantum wells with high-kappa/metal-gate stacks, Majhi P, Kalra P, Harris R, Choi KJ, Heh D, Oh J, Kelly D,

- Choi R, Cho BJ, Banerjee S, Tsai W, Tseng H, Jammy R, IEEE ELECTRON DEVICE LETTERS, Vol: 29 (1), pp. 99-101, JAN 2008.
659. Effects of Non-Ideal Edges in Graphene Nanoribbons, D. BASU, M.J. GILBERT, L.F. REGISTER, S.K. BANERJEE, APS March Meeting 2008, New Orleans.
 660. Evidence of Real-Space Transfer in Buried-Channel $\text{Ge}_{1-x}\text{C}_x$ Devices EN-SHAO LIU, DAVID KELLY, JOSEPH DONNELLY, EMANUEL TUTUC, SANJAY BANERJEE, APS March Meeting 2008, New Orleans.
 661. Ferromagnetism in Mn-implanted Ge and epitaxial GeC, SAMARESH GUCHHAIT, JOHN MARKERT, MUSTAFA JAMIL, SANJAY BANERJEE, APS March Meeting 2008, New Orleans.
 662. Electronic properties of one-dimensional graphene bi-layer ribbons, BHAGAWAN SAHU, HONGKI MIN, ALLAN MACDONALD, SANJAY BANERJEE, APS March Meeting 2008, New Orleans.
 663. Tackhwi Lee, Yonghyun Kim, and Sanjay K. Banerjee, "Potential Approach to grow Epitaxial Graphene by using Carbon Implant on the Si(111) Substrate" MRS Spring 2008.
 664. A Comprehensive Study of Growth Techniques & Characterization of Epitaxial $\text{Ge}_{1-x}\text{C}_x$ (111) Layers Grown Directly on Si (111) for MOS Applications, Mustafa Jamil, Joseph P. Donnelly, Se-Hoon Lee, Davood Shahrjerdi, Emanuel Tutuc and Sanjay K. Banerjee, MRS Spring 2008.
 665. S.Banerjee, "Microelectronics: The End the Beginning or Beginning of the End?" **Invited** talk to SRC Board of Directors, March 2008, Washington DC. Also Congressional panel on Nanotechnology.
 666. N.Jain, E.Tutuc, S.Banerjee, L.Register, Performance Analysis of Ge Tunneling Field Effect Transistors, DRC Santa Barbara 2008
 667. E-S.Liu, N.Jain, K.Varahramyan, J.Nah, S.Banerjee, E.Tutuc, Impact of metal contact depth on device performance in back gated semiconductor nanowire FET, DRC 2008
 668. F.Ferdousi, J.Sarkar, T.Aykol, D.Shahredji, J.Donnely, E.Tutuc and S.Banerjee, Vertical Flash Memory with Protein Assembled Nanoparticle Floating Gate and Al_2O_3 Dielectric, DRC 2008.
 669. H.Zhao, D.Shahrjerdi, F.Zhu, I.Ok, H.Kim, S.Banerjee, J.Lee, ALD Al_2O_3 Gate Dielectrics on InP using S Passivation, EMC 2008
 670. J.Mustafa, S.Guchhait, J.Markert, E.Tutuc and S.Banerjee, Ferromagnetism in Epi Grown GeC layers on Si (100), EMC 2008.
 671. D. Shahrjerdi, N. Nuntawong, G. Balakrishnan, D. I. Garcia-Gutierrez, A. Khoshakhlagh, E. Tutuc, D. Huffaker, J. C. Lee and S. K. Banerjee, "Fabrication and characterization of metal-oxide-semiconductor GaAs capacitors on $\text{Ge}/\text{Si}_{1-x}\text{Ge}_x/\text{Si}$ substrates with Al_2O_3 gate dielectric", J. Vacuum Science Tech B. v. 26(3), May 2008
 672. Oh, Jungwoo, Majhi, Prashant, Lee, Banerjee, Sanjay, Harris, Rusty, Tseng, Hsing-Huang, Jammy, Raj Combined effects of an epitaxial Ge channel and Si substrate on Ge-on-Si metal-oxide-semiconductor capacitors and field effect transistors JAPANESE JOURNAL OF APPLIED PHYSICS Volume: 47 Issue: 4 Pages: 2656-2659 Part: Part 2 APR 2008
 673. Oh, Jungwoo, Majhi, Prashant, Tseng, Hsing-Huang, Jammy, Raj, Kelly, David Q., Banerjee, Sanjay K., Campbell, Joe C., Interface characteristics of ZrO_2 high-k gate dielectrics on epitaxial Ge capacitor layers after thermal desorption of Ge native oxide and Ge nitridation, THIN SOLID FILMS Vol: 516 (12), p. 4107-4110 APR 2008
 674. Bhagawan Sahu, Sanjay K. Banerjee and Leonard Kleinman, Density functional study of bulk silicon lightly doped with manganese, PHYSICAL REVIEW B Volume: 77 Issue: 15 Article 155202 APR 2008
 675. Shahrjerdi, D., Aykol, T., Ramon, M., Garcia-Gutierrez, D. I., Tutuc, E., Banerjee, S. K. Self-aligned inversion-type enhancement-mode GaAs metal-oxide-semiconductor field-effect

- transistor with Al₂O₃ gate dielectric, APPLIED PHYSICS LETTERS Volume: 92 Issue: 20 Article Number: 203505 MAY 19 2008
676. Shahrjerdi, Davood, Rotter, Thomas, Balakrishnan, Ganish, Huffaker, Diana, Tutuc, Emanuel, Banerjee, Sanjay K. Fabrication of self-aligned enhancement-mode In_{0.53}Ga_{0.47}As MOSFETs with TaN/HfO₂/AlN gate stack, IEEE ELECTRON DEVICE LETTERS Volume: 29 Issue: 6 Pages: 557-560 JUN 2008
 677. Zhao, Han, Shahrjerdi, Davood, Zhu, Feng, Zhang, Manhong, Kim, Hyoungh-Sub, Ok, Injo, Yum, Jung Hwan, Park, Sung Il, Banerjee, Sanjay K., Lee, Jack C. Gate-first inversion-type InP metal-oxide-semiconductor field-effect transistors with atomic-layer-deposited Al₂O₃ gate dielectric, APPLIED PHYSICS LETTERS Volume: 92 Issue: 23 Article Number: 233508 JUN 9 2008
 678. Shahrjerdi, D., Garcia-Gutierrez, D. I., Tutuc, E., Banerjee, S. K. Chemical and physical interface studies of the atomic-layer-deposited Al₂O₃ on GaAs substrates, APPLIED PHYSICS LETTERS Volume: 92 Issue: 22 Article Number: 223501 JUN 2 2008
 679. Zhao, Han, Shahrjerdi, Davood, Zhu, Feng, Kim, Hyoungh-Sub, Ok, Injo, Zhang, Manhong, Yum, Jung Hwan, Banerjee, Sanjay K., Lee, Jack C. Inversion-type InP MOSFETs with EOT of 21 angstrom using atomic layer deposited Al₂O₃ gate dielectric, ELECTROCHEMICAL AND SOLID STATE LETTERS Volume: 11 Issue: 8 Pages: H233-H235 2008
 680. Zhao, Han, Shahrjerdi, Davood, Zhu, Feng, Kim, Hyoungh-Sub, Ok, Injo, Zhang, Manhong, Yum, Jung Hwan, Banerjee, Sanjay K., Lee, Jack C. Inversion-type indium phosphide metal-oxide-semiconductor field-effect transistors with equivalent oxide thickness of 12 angstrom using stacked HfAlO_x/HfO₂ gate dielectric, APPLIED PHYSICS LETTERS Volume: 92 Issue: 25 Article Number: 253506 JUN 23 2008
 681. Kong, Ning, Kirichenko, Taras A., Kim, Yonghyun, Foisy, Mark C., Banerjee, Sanjay K. Physically based kinetic Monte Carlo modeling of arsenic-interstitial interaction and arsenic uphill diffusion during ultrashallow junction formation, JOURNAL OF APPLIED PHYSICS Volume: 104 Issue: 1 Article Number: 013514 JUL 1 2008
 682. Sahu, Bhagawan, Min, Hongki, MacDonald, A. H., Banerjee, Sanjay K. Energy gaps, magnetism, and electric-field effects in bilayer graphene nanoribbons, PHYSICAL REVIEW B Volume: 78 Issue: 4 Article Number: 045404 JUL 2008
 683. Ning Kong, Taras A. Kirichenko, Gyeong S. Hwang, and Sanjay K. Banerjee Interstitial-based boron diffusion dynamics in amorphous silicon, Appl. Phys. Lett. 93, 082109 (2008)
 684. D. Shahrjerdi, D. I. Garcia-Gutierrez, and S. K. Banerjee, "Fabrication of Self-aligned Enhancement-mode n-channel GaAs MOSFETs Employing a Wet Clean Process for GaAs Substrates", 214th Int. Electrochemical Society meeting, Hawaii, 2008
 685. S. Kim, J. Nah, I. Jo, D. Shahrjerdi, L. Colombo, Z. Yao, E. Tutuc, S. Banerjee, CARRIER MOBILITY IN GRAPHENE DEVICES WITH HIGH- κ DIELECTRICS, SRC TECHCON, 2008
 686. J. Mustafa, S. Guchhait, D. Ferrer, E. Tutuc, J. Markert, L. Colombo and S. Banerjee, Ferromagnetism in Mn-implanted Epitaxially Grown GeC on (100) Si, SRC TECHCON, 2008
 687. H. Chen, L. Register, M. Gilbert, S. Banerjee, Resonant Tunneling enhanced FET for low voltage switching: concept and quantum transport simulation, SRC TECHCON, 2008
 688. F. Ferdousi, J. Sarkar, S. Tang, D. Shahrjerdi, E. Tutuc, S. Banerjee, Improved Memory Operation of Vertical Flash Devices with Al₂O₃ control oxide and nanocrystal floating gate, SRC TECHCON, 2008
 689. H. Liu, .., S. Banerjee, Fabrication of Ni Nanocrystal Flash Memory using Protein-Induced Assembly, SRC TECHCON, 2008
 690. N. Shi, F. Register, S. Banerjee, A Full-Band Monte Carlo Simulator with Quantum-Confinement-Enhanced Scattering, SRC TECHCON, Austin, TX, 2008
 691. D. Shahrjerdi, E. Tutuc, and S. K. Banerjee, "Self-aligned Enhancement-mode GaAs MOSFETs with Al₂O₃ gate dielectric", SRC TECHCON, Austin, TX, 2008

692. Dipanjan Basu, L.Register, M.Gilbert, A.Macdonald and S.Banerjee, Effect of Non-Ideal Edges on Electronic Transport in Graphene Nanoribbon Channel Metal Oxide Semiconductor Field-Effect Transistors, SRC TECHCON, 2008
693. E.Tutuc, S.Banerjee, J.Nah, K.Vahramyan, N.Jain and D.Ferrer, Opportunities for Group IV nanowire devices in Si CMOS technology, **Invited**, Transactions of ECS, Oct. 2008.
694. M.Ramon, D. Shahrjerdi, C.Young, D. Garcia-Gutierrez, T. Akyol, S. Banerjee, Ultra short pulsed IV Characterization of GaAs FET with Al₂O₃ Gate dielectric, SISC, Dec. 08.
695. T.Lee and S.Banerjee, Device Performance of HfON charge trap layer NAND flash memory, SISC, 2008.
696. T.Lee, S.Park, J.Lee and S.Banerjee, Breakdown mechanism for thin EOT Dy₂O₃/HfO₂ dielectric, IEEE IRW Conf. Oct. 2008.
697. N.Shi, L.Register and S.Banerjee, On strain and scattering in deeply-scaled NMOSFETs: a quantum corrected semiclassical Monte Carlo analysis, IEDM 2008.
698. S.H. Lee, P.Mahji,...S.K.Banerjee and R.Jammy, "Demonstration of 55 nm pMOSFETs with Si/Si_{0.25}Ge_{0.75} channels, high ON/OFF and controlled SCE," IEEE Elec. Dev. Lett. 29(9), p. 1017, Sept. 2008.
699. "Doping of Ge-Si_xGe_{1-x} core-shell nanowires using low energy ion implantation", Junghyo Nah*, K. Varahramyan, E.-S. Liu, S.K. Banerjee, E. Tutuc, Appl. Phys. Lett. 93, 203108 (2008).
700. Core-Shell Germanium-Silicon Nanocrystal Floating Gate for Nonvolatile Memory Applications, Liu H, Winkenwerder W, Liu YR, Ferrer D, Shahrjerdi D, Stanley SK, Ekerdt JG, S.Banerjee, IEEE TRANSACTIONS ON ELECTRON DEVICES Vol: 55 (12) Pg: 3610-3614, DEC 2008
701. Schrodinger equation Monte Carlo in three dimensions for simulation of carrier transport in three-dimensional nanoscale metal oxide semiconductor field-effect transistors, Liu KM, Chen WQ, Register LF, S.K.Banerjee, JOURNAL OF APPLIED PHYSICS v. (11): 114515, DEC 1 2008
702. S.Banerjee, F.Register, E.Tutuc, A.Macdonald, D.Reddy and D.Basu, Microelectronics: the Beginning of the end or the end of the beginning?", *Invited Keynote*, ACM TAU workshop, Austin, TX, Feb. 2009.
703. Vth Variation and Strain Control of High Ge% Thin SiGe Channels by Millisecond Anneal Realizing High Performance pMOSFET beyond 16nm node, S.Lee, ...S.Banerjee, R.Jammy, VLSI Symp. Kyoto, 2009.
704. High-Mobility Dual-gated Graphene Field-Effect Transistors with Al₂O₃ Dielectric, Seyoung Kim, Junghyo Nah, Insun Jo, Davood Shahrjerdi, Luigi Colombo, Zhen Yao, Emanuel Tutuc and Sanjay K. Banerjee, APS March Meeting 2009.
705. Epitaxial Growth and Characterization of Void-Free 3C-SiC Films on Germanium-Modified Si Substrates using RTCVD, DOMINGO FERRER, SHAGANDEEP KAUR, SAYAN SAHA, SEYOUNG KIM, EMANUEL TUTUC, SANJAY BANERJEE, APS March Meeting 2009.
706. Top-gate Ge-Si_xGe_{1-x} core-shell nanowire field effect transistors with highly doped source and drain, JUNGHYO NAH, E.-S. LIU, D. SHAHRJERDI, K. M. VARAHRAMYAN, S. K. BANERJEE, E. TUTUC, APS March Meeting 2009.
707. Epitaxial growth of Ge-Si_xGe_{1-x} core-shell nanowireheterostructures with tunable shell content, KAMRAN VARAHRAMYAN, DOMINGO FERRER, EMANUEL TUTUC, SANJAY BANERJEE, APS March Meeting 2009.
708. Prospects of Spin Injection in Germanium Nanowires, EN-SHAO LIU, KAMRAN VARAHRAMYAN, JUNGHYO NAH, SANJAY BANERJEE, EMANUEL TUTUC, APS March Meeting 2009.
709. Mn-implanted GeC: An Amorphous Ferromagnetic Material, SAMARESH GUCHHAIT, M. JAMIL, D. FERRER, E.TUTUC, J. MARKERT, S. BANERJEE,A. LI-FATOU, L. COLOMBO, APS March Meeting 2009.

710. Negative Differential Resistance in Buried-Channel GexC1-x pMOSFETs, Liu ES, Kelly DQ, Donnelly JP, Tutuc E, Banerjee, Sanjay, IEEE ELECTRON DEVICE LETTERS Vol: 30 (2), pg: 136-138, FEB 2009
711. S.K.Banerjee, L.Register, E.Tutuc, D.Reddy and A.Macdonald, "Bilayer pseudoSpin Field Effect Transistor (BiSFET): a proposed new logic device," IEEE Elec. Dev. Lett. P. 157, 30(2), Feb. 2009.
712. Protein-Assembled Nanocrystal-Based Vertical Flash Memory Devices with Al₂O₃ Integration Ferdousi F, Sarkar J, Tang S, E.Tutuc and S.K.Banerjee, JOURNAL OF ELECTRONIC MATERIALS Vol. 38 (3) Pg: 438-442, MAR 2009
713. Realization of dual-gated Ge-SixGe1-x core-shell nanowire field effect transistors with highly doped source and drain, Nah J, Liu ES, Shahrjerdi D, Varahramyan, K. M., Banerjee, S. K., Tutuc, E., APPLIED PHYSICS LETTERS Vol: 94 (6) Article Number: 063117, FEB 9 2009
714. Realization of a high mobility dual-gated graphene field-effect transistor with Al₂O₃ dielectric, Kim S, Nah J, Jo I, Shahrjerdi, Davood, Colombo, Luigi, Yao, Zhen, Tutuc, Emanuel, Banerjee, Sanjay K., APPLIED PHYSICS LETTERS Volume: 94 (6) Article Number: 062107, FEB 9 2009
715. First-principles studies of small arsenic interstitial complexes in crystalline silicon, Yonghyun Kim, Taras A. Kirichenko, Ning Kong, Graeme Henkelman, and Sanjay K. Banerjee, Phys. Rev. B, Volume: 79 (7), FEB 2009
716. Rownak Jyoti Zaman, Kenneth Matthews, Mohammad Mehedi Hasan, Weize Xiong, Leonard Franklin Register, and Sanjay K. Banerjee, "A Novel Low-Cost Trigate Process Suitable for Embedded CMOS 1T-1C Pseudo-SRAM Application, IEEE TRANSACTIONS ON ELECTRON DEVICES, VOL. 56, NO. 3, MARCH 2009
717. Shape- and Size-Controlled Fabrication of Lead Chalcogenide Nanocrystals for Information Storage Devices, Domingo Ferrer, Hai Liu, Erik Taylor and Sanjay Banerjee, MRS Spring 2009.
718. Sun, H. C. Floresca, J. G. Wang, J. Mustafa, S. Guchhait, D. Ferrer, S. K. Banerjee, G. Lian, L. Colombo and M. J. Kim, "Amorphous Structure and Stability of Mn Implanted GeC Ferromagnetic Semiconductor," Microscopy and Microanalysis, 2009 (in press).
719. Large-Area Synthesis of High-Quality and Uniform Graphene Films on Copper Foils, Li XS, Cai WW, An JH, Kim S, Nah J, Yang DX, Piner R, Velamakanni A, Jung I, Tutuc E, Banerjee, SK, Colombo L, Ruoff RS, Science, Vol: 324 Issue: 5932 Pages: 1312-1314, JUN 5 2009.
720. Band engineered epitaxial Ge-SixGe1-x core-shell nanowire heterostructures, Varahramyan KM, Ferrer D, Tutuc E, Banerjee, SK, APPLIED PHYSICS LETTERS Volume: 95(3) , JUL 20 2009
721. Shahrjerdi, J. Nah, T. Akyol, M. Ramon, E. Tutuc, and S. K. Banerjee, Accurate Inversion Charge and Mobility Measurements in Enhancement-mode GaAs Field-Effect Transistors with High-k Gate Dielectrics, DRC, 2009
722. J.Nah, E.Liu,..., SBanerjee, E.Tutuc, Top Gated Ge-SixGe1-x core shell NWFETs with highly doped S/D, DRC, 2009.
723. Bilayer pseudoSpin Field Effect Transistor (BiSFET)- a proposed logic device and circuits, Dharmendar Reddy, Leonard F. Register, Emanuel Tutuc, Allan MacDonald and Sanjay K. Banerjee, DRC, 2009.
724. Jamil, E. Liu, F. Ferdousi, J. Donnelly, E. Tutuc, L. Colombo, and S. Banerjee, Effect of Si-Cap Thickness on Device Performance of Buried Channel Si/Ge1-xCx/Si Devices, Electronic Materials Conference, 2009.
725. J. Oh, I. Ok, C. Kang, M. Jamil, S. Lee, W. Loh, J. Huang, B. Sassman, L. Smith, S. Parthasarathy, B. E. Coss, W. Choi, H. Lee, M. Cho, S. Banerjee, P. Majhi, P. Kirsch, H. Tseng and R. Jammy, Mechanisms for Low On-State Current of Ge (SiGe) nMOSFETs: A Comparative Study on Gate Stack, Resistance, and Orientation-Dependent Effective Masses, Symposium on VLSI Technology, 2009 .

726. Effects of Extension profile Engineering to suppress B TED on reliability of high-k metal gate SiGe PMOSFETs, M.Park,... S.Banerjee, Ra.Jammy, Int. Gate Stack Meeting, CA, 2009.
727. Growth and Electronic Properties of Ge-SiGe core shell NW heterostructures, E.Tutuc, K.Varahamyan, J.Nah, E.Liu, D.Shahredji, S.Banerjee, **invited**, SPIE, 2009.
728. Comprehensive Design Methodology of Extension Profile to Suppress Boron TED in High Performance High-k/ Metal SiGe pMOSFETs, C. Y. Kang, Y. H. Kim, M. S. Park, J. W. Oh, B. G. Min, K. S. Lee, S. K. Banerjee, P. Majhi, H. H. Tseng and R. Jammy, *SSDM*, Japan, 2009.
729. S.Banerjee, L.Colombo and E.Tutuc, High-k materials on alternative substrates: Ge, GaAs and graphene, **Invited** talk, ECS Fall meeting, Vienna, Austria Oct. 2009.
730. S.Banerjee, "Nanocrystal Floating Gate Memories," **Invited** talk, Chip on the Dunes, Brazil, Sept. 2009.
731. Resonant Injection Enhanced Field Effect Transistor for Low Voltage Switching: Concept and Quantum Transport Simulation, Hui Chen, L.F. Register, S.K. Banerjee, *SISPAD* Sept. 2009
732. Semiclassical Monte Carlo with quantum-confinement enhanced scattering: Quantum correction and application to short-channel device performance vs. mobility for biaxial-tensile-strained silicon nMOSFETs, Ningyu Shi, Leonard F. Register and Sanjay K. Banerjee, *SISPAD* 2009.
733. Atomistic simulation of band-to-band tunneling in III-V nanowire field-effect transistors, Dipanjan Basu, Leonard F. Register, Matthew J. Gilbert and Sanjay K. Banerjee, *SISPAD*, 2009.
734. Atomistic Kinetic Monte Carlo (KMC) Modeling for Self-Interstitial Clusters and Boron-Interstitial Clusters in SiGe, Yonghyun Kim, T.Kirichenko, N.King, S.Banerjee, *SRC TECHCON*, 2009,
735. Interlayer Tunneling Across Coupled Graphene Nanoribbons Dipanjan Basu, L.Register, A.MacDonald, S.Banerjee, *SRC TECHCON*, Austin, 2009.
736. Low-Temperature Rapid Thermal Chemical Vapor Deposition of 3C-SiC on Si using Thin Ge Buffer Layer Shagandeep Kaur, .. E.Tutuc, S.Banerjee, *SRC TECHCON*, Austin, 2009.
737. Arsenic-Defect Complexes at SiO₂/Si Interfaces, Ning Kong, Taras. A. Kirichenko, Gyeong S. Hwang, Sanjay. K. Banerjee, *SRC TECHCON*, Austin, 2009.
738. Bilayer pseudoSpin Field Effect Transistor (BiSFET)- a proposed logic device and circuits, Dharmendar Reddy, Leonard F. Register, Emanuel Tutuc, Allan MacDonald, Sanjay K. Banerjee, *SRC TECHCON*, Austin, 2009.
739. Jamil, D. Ferrer, S. Kaur, E. Tutuc, and S. K. Banerjee, "Electrical Characteristics and Thermal Stability Study of TaN/Al₂O₃/GeO₂/Ge (111) MOS Capacitors," *MRS Fall Meeting* 2009.
740. Arsenic defect complexes at SiO₂/Si interfaces: A density functional theory study , Kong N, Kirichenko TA, Hwang GS, Banerjee SK, *PHYSICAL REVIEW B* Volume: 80 Issue: 20 Article Number: 205328, NOV 2009
741. Physical and electrical characterizations of metal-oxide-semiconductor capacitors fabricated on GaAs substrates with different surface chemical treatments and Al₂O₃ gate dielectric, Garcia-Gutierrez DI, Shahrjerdi D, Kaushik V, Banerjee SK, *JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B* Volume: 27 Issue: 6 Pages: 2390-2395, NOV 2009
742. Flicker-Noise Improvement in 100-nm L-g Si_{0.50}Ge_{0.50} Strained Quantum-Well Transistors Using Ultrathin Si Cap Layer, Li F, Lee SH, Fang Z, Banerjee SK, Datta S, *IEEE ELECTRON DEVICE LETTERS* Volume: 31 Issue: 1 Pages: 47-49, JAN 2010
743. Scaling Properties of Ge-SixGe_{1-x} Core-Shell Nanowire Field-Effect Transistors, Nah J, Liu ES, Varahramyan KM, Banerjee SK, Tutuc E, *IEEE TRANSACTIONS ON ELECTRON DEVICES* Volume: 57 Issue: 2 Pages: 491-495 Published: FEB 2010
744. Effects of magnetism and electric field on the energy gap of bilayer graphene nanoflakes, Sahu B, Min HK, Banerjee SK, *PHYSICAL REVIEW B* Volume: 81 Issue: 4 Article Number: 045414, JAN 2010

745. Role of Metal-Semiconductor Contact in Nanowire Field-Effect Transistors, Liu ES, Jain N, Varahramyan KM, Banerjee SK, Tutuc E, IEEE TRANSACTIONS ON NANOTECHNOLOGY Volume: 9 Issue: 2 Pages: 237-242, MAR 2010
746. Bilayer Pseudospin Field-Effect Transistor: Applications to Boolean Logic, Reddy D, Register LF, Tutuc E, Banerjee SK, IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 57 Issue: 4 Pages: 755-764, APR 2010
747. Effects of Si-cap thickness and temperature on device performance of Si/Ge_{1-x}C_x/Si p-MOSFETs, Jamil M, Liu ES, Ferdousi F, Tutuc E, Banerjee SK, SEMICONDUCTOR SCIENCE AND TECHNOLOGY Volume: 25 Issue: 4 Article No: 045005, APR 2010
748. Analytical Model of Short-Channel Double-Gate JFETs, Chang, J.; Kapoor, A. K.; Register, L. F, Banerjee, S. K.; IEEE Transactions on Elec. Dev, Volume: 57, Issue: 8, Aug. 2010.
749. Tight-binding study of electron-hole pair condensation in graphene bilayers: Gate control and system-parameter dependence, Basu D, Register LF, Reddy, Macdonald AH, Banerjee SK, PHYSICAL REVIEW B Volume: 82 Issue: 7 Article Number: 075409 : AUG 10 2010
750. S. Kim, I. Jo, J. Nah, S. K. Banerjee and E. Tutuc, "Independently-contacted Graphene Bilayers: Towards pseudospin-based graphene logic devices," The Sixth International Nanotechnology Conference on Communications and Cooperation, May 2010, Grenoble, France
751. Effects of InP barrier layer thicknesses and different ALD oxides on device performance of In_{0.7}Ga_{0.3}As MOSFETs, Han Zhao, Ning Kong, Yen-Ting Chen, Yanzhen Wang, Fei Xue, Fei Zhou, Sanjay K. Banerjee and Jack C. Lee, DRC 2010
752. Ge-Si_xGe_{1-x} Core-Shell Nanowire Tunneling Field-Effect Transistors, J. Nah, Y. Kim, E.–S. Liu, K. M. Varahramyan, S. K. Banerjee, and E. Tutuc, DRC 2010
753. Characterization and Modeling of Exfoliated and CVD Graphene Field Effect Transistors, S. Banerjee, E. Tutuc and L.F.Register, GOMAC Inv. Talk, 2010.
754. Changing Front-End Dielectric Requirements for End-of-the-Roadmap CMOS and Beyond, Leonard. F. Register, Dipanjan Basu, Mohammad M. Hasan, Dhamendar Reddy, Ningyu Shi, and Sanjay K. Banerjee, ECS Inv. Talk, 2010.
755. Characterization and Modeling of Exfoliated and CVD Graphene FETs, S. Banerjee, E. Tutuc, L.F.Register, R.Ruoff, S.Kim, D.Basu and L.Colombo, MRS Inv. Talk, April 2010.
756. Epitaxial GeSiGe core Shell NW heterostructures and high performance FETs, E.Tutuc, J.Nah, K. Varahramyan, E.Liu, D.Ferrer and S.Banerjee, **Invited** MRS talk, April 2010.
757. Bandgap engineering by magnetism and external electric fields in multilayer graphene sheets, B.Sahu, H.Min, S.Banerjee and A.Macdoanld, MRS April 2010.
758. Retention modeling of nanocrystalline flash memories: A Monte Carlo approach, Bahniman Ghosh and Sanjay K. Banerjee, Singapore, 2010.
759. Shape dependency of magnetic properties of FePt nanostructures, S.Guchhait, D.Ferrer, H.Liu, F.Ferdousi, C.Corbet and S.K. Banerjee, APS March Meeting, 2010.
760. Lateral spin injection in Ge nanowires, E.Liu, J.Nah, K. Varahramyana, S.Banerjee and E.Tutuc, APS March Meeting, 2010.
761. Graphene and its Applications, S.Banerjee, F. Register, E.Tutuc, Nikkei Electronics Symp. **Invited** talk, Tokyo, 2010.
762. Hole band anisotropy effect on ON state performance of biaxial compressive strained SiGe short channel QW pMOSFETs, S.H.Lee,S.Banerjee and R.Jammy, VLSI TSA, Taiwan, 2010.
763. Coulomb Drag in Independently Contacted Graphene Bilayers, Seyoung Kim, Insun Jo, Junghyo Nah, Zhen Yao, Sanjay K. Banerjee, Emanuel Tutuc SRC TECHCON, Austin, **Best paper**, Sept. 2010
764. Growth of Epitaxial Ge:C on Si(111) by Ultra High Vacuum Chemical Vapor Deposition for CMOS Applications, M. Jamil, D. Ferrer, M. Ramon, J. Mantey E. Tutuc, L. Colombo, and S. K. Banerjee, SRC TECHCON, Austin, Sept. 2010

765. Stepped broken-gap Hetero-barrier Tunnel Field Effect Transistor (HetTFET) for ultra-low power and high speed, Mohammad M. Hasan, Leonard F. Register and Sanjay K. Banerjee, SRC TECHCON, Austin, Sept. 2010
766. Nanomagnets for Spin Torque Memory Devices, D.Ferrer,S.K.Banerjee, SRC TECHCON, Austin, Sept. 2010
767. Graphene FETs and Beyond, Sanjay Banerjee, L.Register, E.Tutuc, R.Ruoff, A.MacDonald, L.Colombo*, G.Carpenter#, S.Kim. D.Basu, D.Reddy, **Invited** talk, Carbon Electronics Workshop, Albany, NY, Sept. 2010
768. Electron transport in dual-gated mono and bilayer graphene devices with high-k, E.Tutuc, S.Kim, B.Fallah, K.Lee, J.Nah, S.K.Banerjee, AVS Oct. 2010, **Invited**.
769. Graphene for Beyond Scaled CMOS, Luigi Colombo, S. Kim, E. Tutuc, R. Ruoff, L. Register, S.K. Banerjee, SEMATECH Int. Gate Stack Workshop, Sept.2010, **Invited**.
770. Effects of edge magnetism and external electric field on energy gaps in multilayer graphene nanoribbons, Sahu B, Min HK, Banerjee SK, PHYSICAL REVIEW B Volume: 82 Issue: 11 Article Number: 115426, SEP 14 2010
771. Retention modeling of nanocrystalline flash memories: A Monte Carlo approach, Ghosh B, Liu H, Winstead B, Foisy, M, Banerjee SK, SOLID-STATE ELECTRONICS Volume: 54 Issue: 11 Pages: 1295-1299, NOV 2010
772. Graphene for CMOS and Beyond CMOS Applications, S. K. Banerjee, L.F.Register, E. Tutuc, D.Basu, S.Kim, D.Reddy and A.H. MacDonald, Proceedings of IEEE, Dec. 2010, vol. 98(12), pg. 2032, **Invited** paper.
773. "The Future of the Semiconductor Industry," S.K.Banerjee, Keynote speech to Semiconductor Industry Association and Federal Reserve, Austin TX, Dec. 2010.
774. "Nanoelectronics Research Initiative," S.Banerjee, Keynote, NSF, Dec. 2010.
775. Magnetism in bulk and finite size graphene multilayers and its effect on the band gaps, APS March Meeting, Dallas, 2011
776. Coulomb Drag in Independently Contacted Graphene Bilayers, APS March Meeting, Dallas, 2011
777. Dependence of Condensate Formation in Graphene Bilayers on Relative Layer Orientation, APS March Meeting, Dallas, 2011
778. First-principles study of polarization in graphene, APS March Meeting, Dallas, 2011
779. Tunneling between two independently contacted graphene layers Effect of Dielectric Materials on the Topological Insulator Bi₂Se₃ Surface States, APS March Meeting, Dallas, 2011
780. Numerical simulation of time-dependent transport in graphene, APS March Meeting, Dallas, 2011
781. Ferromagnetism in Mn-implanted HOPG, APS March Meeting, Dallas, 2011
782. Device characteristics of HfON charge-trap layer nonvolatile memory, Lee T, Banerjee SK, JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B Volume: 28 Issue: 5 Pages: 1005-1010, SEP-OCT 2010
783. High-Mobility TaN/Al₂O₃/Ge(111) n-MOSFETs With RTO-Grown Passivation Layer, Jamil M, Oh J, Ramon M, Kaur S, Majhi P, Tutuc E, Banerjee SK, IEEE ELECTRON DEVICE LETTERS Volume: 31 Issue: 11 Pages: 1208-1210, NOV 2010
784. Hall mobility measurements in enhancement-mode GaAs field-effect transistors with Al₂O₃ gate dielectric, Shahrjerdi D, Nah J, Hekmatshoar B, Akyol T, Ramon M, Tutuc E, Banerjee SK, APPLIED PHYSICS LETTERS Volume: 97 Issue: 21 Article Number: 213506, NOV 22 2010
785. Reduced Gate-Leakage Current and Charge Trapping Characteristics of Dysprosium-Incorporated HfO₂ Gate-Oxide n-MOS Devices, Lee T, Banerjee SK, IEEE TRANSACTIONS ON ELECTRON DEVICES , Volume: 58 Issue: 2 Pages: 562-566, FEB 2011

786. Low-Frequency Acoustic Phonon Temperature Distribution in Electrically Biased Graphene, Jo I, Hsu IK, Lee YJ, Sadeghi MM, Kim S, Cronin S, Tutuc E, Banerjee SK, Yao Z, Shi L, NANO LETTERS Volume: 11 Issue: 1 Pages: 85-90 , JAN 2011
787. Quantum Transport Simulation of Strain and Orientation Effects in Sub-20 nm Silicon-on-Insulator FinFETs, Liu KM, Register LF, Banerjee SK, IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 58 Issue: 1 Pages: 4-10, JAN 2011
788. ON-State Performance Enhancement and Channel-Direction-Dependent Performance of a Biaxial Compressive Strained Si_{0.5}Ge_{0.5} Quantum-Well pMOSFET Along $\langle 110 \rangle$ and $\langle 100 \rangle$ Channel Directions, Lee SH, Nainani A, Oh J, Jeon K, Kirsch PD, , Register LF, Banerjee SK, Jammy R, IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 58 Issue: 4 Pages: 985-995, APR 2011
789. Atomic layer deposited beryllium oxide: Effective passivation layer for III-V metal/oxide/semiconductor devices, Yum JH, Akyol T, Lei M, Hudnall T, Bersuker G, Downer M, Bielawski CW, Lee JC, Banerjee SK, JOURNAL OF APPLIED PHYSICS Volume: 109 Issue: 6 Article Number: 064101,MAR 15 2011
790. Mechanism of V-FB/V-TH shift in Dysprosium incorporated HfO₂ gate dielectric n-Type Metal-Oxide-Semiconductor devices, Lee T, Choi K, Ando T, Park DG, Gribelyuk MA, Kwon U, Banerjee SK, JOURNAL OF VACUUM SCIENCE & TECHNOLOGY, Volume: 29 Issue: 2, Number: 021209, MAR 2011
791. "Role of Metal-Semiconductor Contact in Nanowire Field-Effect Transistors", J. Nah, E.-S. Liu, N. Jain, K. M. Varahramyan, J. Nah, S. K. Banerjee, E. Tutuc, IEEE Transactions on Nanotechnology 9, 237 (2010).
792. "Graphene Field-Effect Transistors using Large-Area Monolayer Graphene Grown by Chemical Vapor Deposition on Co Thin Films," Michael E. Ramon,...S.Banerjee, SRC Techcon, Austin, TX, September 13, 2011.
793. "Tunneling Resistance Between Independently Contacted Graphene Layers" Chris M Corbet,... E.Tutuc, S.Banerjee, SRC Techcon, Austin, TX, September 13, 2011.
794. "Spin Torque Switching of a Tilted Spin Valve Structure with Co/Pt/Co/Ni Multilayer," Urmimala Roy, F.Ferdousi ..., S.Banerjee, SRC Techcon, Austin, TX, September 13, 2011
795. Effect of Dielectric Materials on the Topological Insulator Bi(2)Se(3) Surface States," Jiwon Chang, B.Sahu, F.Register, S.Banerjee, SRC Techcon, Austin, TX, September 13, 2011
796. "Graphene FETs Using Large-Area Monolayer Graphene Grown by CVD on Co Thin Films", M.Ramon, A.Gupta, C. Corbet, D. A. Ferrer, H.C.P. Movva, G. Carpenter, L. Colombo, G. Bourianoff, M. Doczy, D. Akinwande,E.Tutuc, S.Banerjee, DRC Santa Barbara, 2011.
797. Introduction of ALD Beryllium Oxide Gate Dielectric for III-V MOS Devices, T.Akyol, J.H.Yum, D. A. Ferrer, M. Lei, M. Downer, C. W. Bielawski, T. W. Hudnall, G. Bersuker,J.Lee and S.Banerjee, DRC Santa Barbara, 2011.
798. Device and Circuit Performance Evaluation and Improvement of SiGe Tunnel FETs Mishra R, Ghosh B, Banerjee SK, Conference Information: International Conference on Enabling Science and Nanotechnology, DEC 01-03, 2010 Kuala Lumpur City Ctr, Kuala Lumpur, MALAYSIA ENABLING SCIENCE AND NANOTECHNOLOGY Book Series: AIP Conference Proceedings Volume: 1341 Pages: 185-192 Published: 2011
799. Neural Network Modeling of Degradation of Solar Cells Gupta H, Ghosh B, Banerjee SK, Conference Information: International Conference on Enabling Science and Nanotechnology, DEC 01-03, 2010 Kuala Lumpur City Ctr, Kuala Lumpur, MALAYSIA ENABLING SCIENCE AND NANOTECHNOLOGY Book Series: AIP Conference Proceedings Volume: 1341 Pages: 249-253 Published: 2011
800. Coulomb drag of massless fermions in graphene Kim Seyoung; Jo Insun; Nah Junghyo; Yao, Z ; Banerjee, S. K.; Tutuc, E, PHYSICAL REVIEW B Volume: 83 Issue: 16 PhysRevB.83.161401 APR 8 2011

801. Fullerene-Based Hybrid Devices for High-Density Nonvolatile Memory
Ferdousi F.; Jamil M.; Liu H.; Kaur, S.; Ferrer, D ; Colombo, L ; Banerjee SK, IEEE
TRANSACTIONS ON NANOTECHNOLOGY Volume: 10 Issue: 3 Pages: 572-
575 TNANO.2010.2053215 MAY 2011
802. Stepped Broken-Gap Heterobarrier Tunneling Field-Effect Transistor for Ultralow Power
and High Speed, Register Leonard F.; Hasan Mohammad M.; Banerjee Sanjay K.
IEEE ELECTRON DEVICE LETTERS Volume: 32 Issue: 6 Pages: 743-
745 LED.2011.2126038 JUN 2011
803. Density functional study of ternary topological insulator thin films, Chang Jiwon; Register
Leonard F.; Banerjee Sanjay K.; Sahu B.
PHYSICAL REVIEW B Volume: 83 Issue: 23 PhysRevB. 83.235108 JUN 6 2011
804. Inversion type InP metal oxide semiconductor field effect transistor using novel atomic
layer deposited BeO gate dielectric, Yum, JH ; Akyol, T ; Lei, M ; Ferrer, DA ; Hudnall, TW ;
Downer, M ; Bielawski, CW ; Bersuker, G ; Lee, JC ; Banerjee, S. K., APPLIED PHYSICS
LETTERS Volume: 99 Issue: 3 Article Number: 033502 DOI:
10.1063/1.3614446 Published: JUL 18 2011
805. 3D-Monte Carlo study of short channel tri-gate nanowire MOSFETs
David J. K.; Register L. F.; Banerjee S. K.; SOLID-STATE ELECTRONICS Volume: 61 Issue:
1 Pages: 7-12 DOI: 10.1016/j.sse.2010.12.013 JUL 2011
806. Origin of shape anisotropy effects in solution-phase synthesized FePt nanomagnets Ferrer
D. A.; Guchhait S.; Liu H.; Ferdousi, F ; Corbet, C ; Xu, H ; Doczy, M ; Bourianoff, G ; Mathew,
L ; Rao, R ; Saha, S ; Ramon, M ; Ganguly, S ; Markert, JT ;Banerjee, S. K.
JOURNAL OF APPLIED PHYSICS Volume: 110 Issue: 1 Article Number: 014316 DOI:
10.1063/1.3608109 JUL 1 2011
807. Effect of interlayer bare tunneling on electron-hole coherence in graphene bilayers
Author(s): Basu D.; Register L. F.; MacDonald A. H.;Banerjee SK,
PHYSICAL REVIEW B Volume: 84 Issue: 3 Article Number: 035449 DOI:
10.1103/PhysRevB.84.035449 JUL 28 2011
808. Ferromagnetism in Mn-implanted epitaxially grown Ge on Si(100)
Author(s): Guchhait S.; Jamil M.; Ohldag H.; Mehta, A ; Arenholz, E ; Lian, G ; LiFatou, A ;
Ferrer, DA ; Markert, JT ; Colombo, L ; Banerjee, S. K.
PHYSICAL REVIEW B Volume: 84 Issue: 2 Article Number: 024432 DOI:
10.1103/PhysRevB.84.024432 JUL 29 2011
809. Graphene field-effect transistors, Reddy Dharmendar; Register Leonard F.; Carpenter Gary
D.; Banerjee SK, JOURNAL OF PHYSICS D-APPLIED PHYSICS Volume: 44 Issue:
31 Article Number: 313001 DOI: 10.1088/0022-3727/44/31/313001 AUG 10 2011
810. Edge saturation effects on the magnetism and band gaps in multilayer graphene ribbons
and flakes, Sahu Bhagawan; Min Hongki; Banerjee Sanjay K.
PHYSICAL REVIEW B Volume: 84 Issue: 7 Article Number: 075481 DOI:
10.1103/PhysRevB.84.075481 AUG 16 2011
811. Tackhwi Lee, and Sanjay K. Banerjee, "*V_{TH} shift Mechanism in Dysprosium
(Dy)incorporated HfO₂ gate nMOS devices*", IEEE IRPS 2011. (International Reliability and
Physics Symposium)
812. Metal-Ge-metal photodetectors integrated with dielectric waveguides, J.Cervantes-
Gonzalez, D.Ahn, A Torres-Jacome, S.K.Banerjee and I. Zaldivar-Huerta, Int. Commision for
Optics, Puebla, Mexico, Aug. 2011.
813. Graphene Nanoelectronics, S.Banerjee, L.F.Register, E.Tutuc, D.Akinwande, L.Colombo
and G.Carpenter, TechConnect World Conf, Boston, June 2011, **Invited** talk.
814. Graphene nanoelectronics: a simulation perspective, L.F. Register, D.Basu, X.Mou,
D.Reddy, G.Carpenter, A.Hassibi.A.Macdonald, and S.Banerjee, TechConnect World Conf,
Boston, June 2011, **Invited** talk.

815. Tunnel FETs, S.Banerjee, L.F.Register, E.Tutuc, D.Reddy, D.Basu, M.Hasan and H.Chen, SEMATECH TFET Workshop, Austin, **Invited** talk.
816. Dielectric deposition and electron transport in graphene devices, E.Tutuc, B.Fallahazad, S.Kim, K.Lee, M.Ramon, S.Banerjee and L.Colombo, ECS May, 2011, **Invited**.
817. Monocrystalline Si solar cell technology, R.Rao, L.Mathew, S.Saha, ... E.Oneyagam, D.Jawarani, J.Fossum and S.Banerjee, 37th IEEE Photovoltaic Specialists Conf. Washington DC. July 2011
818. Exfoliated 40 micron Ge cost effective photovoltaic cell, E.Oneyagam, J.Mantey, R.Rao, L.Mathew, M.Hilali, S.Saha, D.Jawarani, S.Sreenivasan and S.Banerjee, 37th IEEE Photovoltaic Specialists Conf. Washington DC., July 2011
819. Novel low cost 25 micron thin exfoliated monocrystalline Si solar cell technology, R.Rao, L.Mathew, S.Saha,...D.Jawarani, E.Oneyagam, S.Banerjee and J.Fossum, European PVSC, Hamburg, Germany, Sept. 2011.
820. A Proposed Novel Graphene Switch: Bilayer Pseudospin Field Effect Transistor, Sanjay Banerjee, Frank Register and Emanuel Tutuc, ICCAD, Nov. 2011, San Jose, **Invited** talk
821. ALD dielectrics for electronic devices, S.Banerjee, Cambridge Nanotech Fall ALD Symposium, Atlanata, GA, Nov. 2011, **Invited**.
822. ALD Beryllium Oxide as a High-k Gate Dielectric for III-V MOS Devices, J. H. Yum, T. Akyol, M. Lei, D. A. Ferrer, Todd. W. Hudnall, M. Downer, C. W. Bielawski, G. Bersuker, J. C. Lee, and S. K. Banerjee, AVS-ALD Conf. 2011
823. ALD Beryllium Oxide: Novel Barrier Layer for High Performance Gate Stacks on Si and High Mobility Substrates, J. H. Yum, T. Akyol, M. Lei, D. A. Ferrer, Todd. W. Hudnall, M. Downer, C. W. Bielawski, G. Bersuker, J. C. Lee, and S. K. Banerjee, IEDM 2011
824. Direct Measurement of the Fermi Energy in Graphene Using a Double Layer Structure, SEYOUNG KIM, INSUN JO, DAVID DILLEN, DOMINGO FERRER, BABAK FALLAHAZAD, ZHEN YAO, SANJAY BANERJEE, EMANUEL TUTUC, APS March Meeting, 2012
825. Ab-initio study of dilute nitride substitutional and split-interstitial impurities in gallium antimonide (N-GaSb), PRIYAMVADA JADAUN, HARI P. NAIR, SETH R. BANK, SANJAY BANERJEE, APS March Meeting, 2012.
826. Spinglass Dynamics of Amorphous Ferromagnetic Ge:Mn, Samarsh Guchhait, Sanjay Banerjee, Raymond Orbach, APS March Meeting, 2012.
827. Impact of Millisecond Flash-Assisted Rapid Thermal Annealing on SiGe Heterostructure Channel pMOSFETs With a High-k/Metal Gate, Lee Se-Hoon; Majhi Prashant; Ferrer Domingo A., Hung, Pui-Yee, Huang, Jeff, Oh, Jungwoo, Loh, Wei-Yip, Sassman, Barry, Byoung-Gi, Tseng, Hsing-Huang, Harris, Rusty, Gennadi Bersuker, Kirsch, Paul D., Jammy, Raj Banerjee, Sanjay K., IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 58 Issue: 9 Pages: 2917-2923, SEP 2011
828. High-Performance Ge nMOSFETs With n(+)-p Junctions Formed by "Spin-On Dopant" Jamil M.; Mantey J.; Onyegam E. U.; Carpenter, G. D., Tutuc, E., Banerjee, S. K, IEEE ELECTRON DEVICE LETTERS Volume: 32 Issue: 9 Pages: 1203-1205, SEP 2011
829. CMOS-Compatible Synthesis of Large-Area, High-Mobility Graphene by Chemical Vapor Deposition of Acetylene on Cobalt Thin Films Ramon Michael E.; Gupta Aparna; Corbet Chris; Ferrer, Domingo A, Movva, Hema C. P., Carpenter, Gary, Colombo, Luigi, Bourianoff, George, Doczy, Mark Akinwande, Deji, Tutuc, Emanuel Banerjee, Sanjay K., ACS NANO Volume: 5 Issue: 9 Pages: 7198-7204, SEP 2011
830. Dielectric capping effects on binary and ternary topological insulator surface states Author(s): Chang Jiwon; Jadaun Priyamvada; Register Leonard F.; Banerjee, Sanjay K., Sahu, Bhagawan, PHYSICAL REVIEW B Volume: 84 Issue: 15 Article Number: 155105, Published: OCT 10 2011

831. A study of highly crystalline novel beryllium oxide film using atomic layer deposition
Author(s): Yum J. H.; Akyol T.; Lei M.; Ferrer, D. A. Hudnall, Todd W., Downer, M, Bielawski, C. W. Bersuker, G. Lee, J. C.; Banerjee, S. K. Source: JOURNAL OF CRYSTAL GROWTH Volume: 334 Issue: 1 Pages: 126-133 Published: NOV 1 2011
832. Comparison of the self-cleaning effects and electrical characteristics of BeO and Al(2)O(3) deposited as an interface passivation layer on GaAs MOS devices
Author(s): Yum J. H.; Akyol T.; Ferrer D. A.; Lee, J. C., Banerjee, S. K., Lei, M., Downer, M. Hudnall, Todd W., C. W. Bersuker, G., Source: JOURNAL OF VACUUM SCIENCE & TECHNOLOGY A Volume: 29 Issue: 6 Article Number: 061501 Published: NOV 2011
833. Epitaxial ALD BeO: Efficient Oxygen Diffusion Barrier for EOT Scaling and Reliability Improvement, Yum Jung Hwan; Bersuker Gennadi; Akyol Tarik; (Ferrer, D. A, Lei, Ming, Park, Keun Woo; Hudnall, Todd W., Downer, Mike C., Bielawski, Christopher W, Yu, Edward T, Price, Jimmy, Lee, Jack C.; Banerjee, Sanjay K., IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 58 Issue: 12 Pages: 4384-4392 Published: DEC 2011
834. Density functional theory based study of graphene and dielectric oxide interfaces , Jadaun Priyamvada; Banerjee Sanjay K.; Register Leonard F.; Banerjee, SK, Sahu, B, Source: JOURNAL OF PHYSICS-CONDENSED MATTER Volume: 23 Issue: 50 Article Number: 505503 Published: DEC 21 2011
835. Role of Confinement on Carrier Transport in Ge-Si(x)Ge(1-x) Core-Shell Nanowires, Nah Junghyo; Dillen David C.; Varahramyan Kamran M.; Banerjee, SK, Tutuc E, Source: NANO LETTERS Volume: 12 Issue: 1 Pages: 108-112 Published: JAN 2012
836. Direct Measurement of the Fermi Energy in Graphene Using a Double Layer Structure, Seyoung Kim, Insun Jo, D. C. Dillen, D. A. Ferrer, B. Fallahazad, Z. Yao, S. K. Banerjee, E. Tutuc, Phys Rev. Lett, 2012.
837. U. Onyegam, D. Sarkar, M. Hilali, S. Saha, R.A. Rao, L. Mathew, D. Jawarani, S. Smith, S.K. Banerjee, "Exfoliated Thin, Flexible Monocrystalline Germanium Heterojunction Solar Cells." 38th IEEE PVSC 2012, Austin, TX
838. Sarkar, E.U. Onyegam, S. Saha, A. Rao, L. Mathew, R.S. Smith, D. Xu, D. Jawarani, R. Garcia, S.K. Banerjee, "Remote Plasma Chemical Vapor Depositon for High Efficiency Ultra Thin ~25 Microns Crystalline Si Solar Cells." Presented at 38th IEEE PVSC 2012, Austin, TX.
839. R. Rao, L. Mathew, D. Sarkar, S. Smith, S. Saha, R. Garcia, R. Stout, A. Gurmu, M. Ainom, E. Onyegam, D. Xu, D. Jawarani, U. Das, S. Banerjee, J. Fossum, "A low-Cost Kerfless Thin Exfoliated Si Solar Cell Technology." Presented at 38th IEEE PVSC 2012, Austin, TX.
840. L. Mathew, R. Rao, D. Sarkar, S. Banerjee, D. Jawarani, J. Fossum, R. Garcia, S. Smith, D. Xu, M. Ainom, E. Onyegam, R. Stout, S. Saha, A. Gurmu, A Novel Low-Cost ~25µm-Thin Monocrystalline Silicon Bifacial Solar Cell Technology with Flexible and Rigid Form-Factor and Electroplated Contacts." Presented at 38th IEEE PVSC 2012, Austin, TX.
841. Integration and Reliability of Ultrathin Silicon Solar Cells and Modules Fabricated using SOM Technology, D. Jawarani, D. Xu, R. S. Smith, R. S. A. Rao, L. Mathew, S. Saha, D. Sarkar, C. Vass, S. K. Banerjee, P. S. Ho, 38th IEEE PVSC, June, 2012.
842. Integration and Reliability of Thin Silicon Solar Cells and Modules Fabricated using SOM® Technology, D. Jawarani, D. Xu, R. S. Smith, S. Saha, R. S. A. Rao, L. Mathew, D. Sarkar, E. U. Onyegam, M. Ainom, R. S. Garcia, A. Gurmu, R. S. Stout, C. Vass, S. K. Banerjee, P. S. Ho, J. G. Fossum, 27th EU PVSEC, September, 2012
843. Electrical and physical characteristics for crystalline atomic layer deposited beryllium oxide thin film on Si and GaAs substrates, Yum J. H.; Akyol T.; Lei M.; et al. Banerjee SK, THIN SOLID FILMS Volume: 520 Issue: 7 Pages: 3091-3095, JAN 31 2012
844. Theoretical approach to evaluating beryllium oxide as a gate dielectric considering electromagnetics and thermal stability, Yum J. H.; Bersuker G.; Oh J.; et al. Banerjee SK, APPLIED PHYSICS LETTERS Volume: 100 Issue: 5 Article Number: 053501, JAN 30 2012

845. A path-sum Monte Carlo approach for many-electron systems within a tight-binding basis, David J. K.; Register L. F.; Banerjee S. K. Source: JOURNAL OF COMPUTATIONAL ELECTRONICS Volume: 11 Issue: 2 Pages: 172-181, JUN 2012
846. Band offsets of atomic layer deposited Al₂O₃ and HfO₂ on Si measured by linear and nonlinear internal photoemission, Lei M.; Yum J. H.; Banerjee S. K.; et al. Source: PHYSICA STATUS SOLIDI B-BASIC SOLID STATE PHYSICS Volume: 249 Issue: 6 Pages: 1160-1165, JUN 2012
847. Spin-transfer-torque switching in spin valve structures with perpendicular, canted, and in-plane magnetic anisotropies, Roy U.; Seinige H.; Ferdousi F.; et al, Banerjee SK, JOURNAL OF APPLIED PHYSICS Volume: 111 Issue: 7 Article Number: 07C913, APR 1 2012
848. Semiclassical Monte Carlo Analysis of Graphene FETs, David J. K.; Register L. F.; Banerjee S. K., IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 59 Issue: 4 Pages: 976-982, APR 2012
849. Spectroscopic evaluation of band alignment of atomic layer deposited BeO on Si(100), Lei Ming; Yum J. H.; Price J.; et al., Banerjee SK, APPLIED PHYSICS LETTERS Volume: 100 Issue: 12 Article Number: 122906, MAR 19 2012
850. Density functional theory studies of interactions of graphene with its environment: Substrate, gate dielectric and edge effects, Jadaun, Priyamvada, Sahu, Bhagawan R, Register, Leonard F. Banerjee, Sanjay K, SOLID STATE COMMUNICATIONS Volume: 152 Issue: 15 Special Issue: SI Pages: 1497-1502, AUG 2012
851. D. Akinwande, K. N. Parrish, M. E. Ramon, L. Tao, S. Banerjee, "Wafer-scale graphene technology and GHz nanoelectronics," International Semiconductor Device Research Symposium, 2011. **(Invited)**
852. Ramon, K. N. Parrish, J. Lee, C. W. Magnuson, L. Tao, R. S. Ruoff, S. K. Banerjee, D. Akinwande, "Graphene Frequency Doubler with Record 3GHz Bandwidth and the Maximum Conversion Gain Prospects," IEEE MTT-S International Microwave Symposium, June 2012
853. M. Ramon, K. N. Parrish, Sk. F. Chowdhury, C. W. Magnuson, H. C. P. Movva, R. S. Ruoff, S. K. Banerjee, D. Akinwande, "3GHz Graphene Frequency Doubler on Quartz Operating Beyond the Transit Frequency," IEEE Transactions on Nanotechnology, 10.1109/TNANO.2012.2203826.
854. M. Ramon, K. N. Parrish, Sk. F. Chowdhury, C. W. Magnuson, H. C. P. Movva, R. S. Ruoff, S. K. Banerjee, D. Akinwande, "3GHz Graphene Frequency Doubler on Quartz Operating Beyond the Transit Frequency," Semiconductor Research Corporation TECHCON, 2012.
855. H. C. P. Movva, M. E. Ramon, C. M. Corbet, Sk. F. Chowdhury, G. Carpenter, E. Tutuc, S. K. Banerjee, "Self-aligned graphene field-effect transistors with chemically doped source/drain access regions," Semiconductor Research Corporation TECHCON, 2012.
856. H. C. P. Movva, M. E. Ramon, C. M. Corbet, Sk. F. Chowdhury, G. Carpenter, E. Tutuc, S. K. Banerjee, "Graphene field-effect transistors with self-aligned spin-on-doping of source/drain access regions," 70th Device Research Conference, June 2012.
857. K. N. Parrish, M. E. Ramon, S. K. Banerjee, D. Akinwande, "A Compact Model for Graphene FETs for Linear and Non-linear Circuits," 17th International Conference on Simulation of Semiconductor Processes and Devices, September 2012.
858. J. Chang, L. F. Register, S. K. Banerjee, "Atomistic Quantum Transport Simulation of Topological Insulator Bi₂Se₃Tunnel FETs", SISPAD, Sep 5-7 , 2012, Denver, CO, U.S.A.
859. J. Chang, L. F. Register, S. K. Banerjee, "Possible Applications of Topological Insulator for Tunnel FETs", DRC, Jun 18-20 , 2012, University Park, PA, U.S.A.
860. J. Chang, L. F. Register, S. K. Banerjee, "Computational Study of Tunnel FETs Based on Topological Insulator Bi₂Se₃", Techcon, Sep 10-11, 2012, Austin, TX. U.S.A.
861. Self-aligned graphene field-effect transistors with chemically doped source/drain access regions, Hema C.P. Movva, Michael E. Ramón, Chris M. Corbet, Sk. Fahad Chowdhury, Gary Carpenter, Emanuel Tutuc and Sanjay K. Banerjee, Techcon, Sep 10-11, 2012, Austin, TX. U.S.A.

862. D.Reddy, F.Register, S.Banerjee, Bilayer Graphene Vertical TFET, Techcon, Sep 10-11, 2012, Austin, TX. U.S.A.
863. D. Reddy, P. Jadaun, A. Valsaraj, L. F. Register, S. K. Banerjee, "Time Dependent Quantum Transport in Graphene," SISPAD 2012, Sept. 5 - Sept. 7, 2012.
864. Jadaun, Hari P. Nair, Seth R. Bank, S. K. Banerjee, "Ab-initio study of dilute nitride substitutional and split-interstitial impurities in gallium antimonide (N-GaSb)" APS March Meeting, Feb 27 - Mar 2, 2012
865. S. Kim, I. Jo, D. Dillen, B. Fallahazad, D. Ferrer, Z. Yao, S. K. Banerjee, E. Tutuc, "'Direct Measurement of the Fermi Energy in Graphene Using a Double Layer Structure'", APS March Meeting Conference, Boston (2012).
866. S.Gucchait, S. K. Banerjee, R.Orbach, "Spinglass Dynamics in FM Ge:Mn", APS March Meeting Conference, Boston (2012)
867. J.. Mantey, W. Hsu, M. Jamil, E. U. Onyegam, E Tutuc, and S. K. Banerjee, "Germanium nMOSFETs with GeO₂ Passivation and n+/p Junctions Formed by Spin-on Dopants", Inter. SiGe Tech. and Dev. Meeting (ISTDM) Session 3.5, June 4-6, 2012 (2012).
868. D.Reddy, F.Register, S.Banerjee, Bilayer Graphene Vertical TFET, DRC, 2012.
869. D. Koh, J. H. Yum, T. Akyol, D. A. Ferrer, M. Lei, Todd. W. Hudnall, M. C. Downer, C. W. Bielawski, R.Hill, G. Bersuker and S. K. Banerjee, Novel atomic layer deposited thin film beryllium oxide for InGaAs MOS Devices, IRPM, 2012.
870. S. K. Banerjee, L. F. Register, E. Tutuc, D. Reddy, S. Kim, D. Basu, C. Corbet, L. Colombo, G. Carpenter, A. H. MacDonald, "Novel Double Layer Graphene Transistors-Bilayer Pseudospin FETs and 2D-2D Tunnel FETs", 70th Device Research Conference (2012) (**invited**).
871. Leonard F. Register, D. Reddy, X. Mau, W. Jung, I. Sodemann, D. Pesin, A. Hassibi, A. H. MacDonald, and S. K. Banerjee, "Bilayer PseudoSpin Field Effect Transistor (BiSFET): Concepts and Critical Issues for Realization," 221st ECS Meeting - Seattle, Washington, May 6 - May 10, 2012; ECS Transactions Vol. 45, No. 4, pp 3-14, **Invited**
872. Leonard F. Register and Sanjay K. Banerjee, "Southwest Academy of Nanotechnology (SWAN)," The Eight International Nanotechnology Conference on Communication and Cooperation, Tsukuba, Japan, May 8-11, 2012, **Invited**
873. Sanjay. Banerjee, Leonard F. Register, Emanuel Tutuc, Dharmendar Reddy, Dipanjan Basu, Mohammad Mehedi Hasan, and Hui Chen, "Tunnel FETs," SEMATECH TFET Workshop, Austin Texas, **Invited**
874. S.Banerjee, SWAN, GOMAC Las Vegas, March 2012, **Invited**
875. Novel Transistor Concepts Based on 2D Systems- Graphene and Topological Insulators, Sanjay Banerjee and Frank Register, NSF/AFOSR 2D Workshop, Washington DC, May 2012, **Invited**.
- A. H. MacDonald, D. Pesin, I. Sodemann, L. F. Register, and S. K. Banerjee, "Physical Properties of Bilayer Exciton Condensates," 221st ECS Meeting - Seattle, Washington, May 6 - May 10, 2012, **Invited**
876. B.Doering and S.Banerjee, Beyond CMOS, SPIE Lithography Meeting, Williamsburg, VA, June 2012, **Invited**.
877. Electron Transport and Strain Mapping in Ge-Si_xGe_{1-x} Core-Shell Nanowire Heterostructures, D. C. Dillen, J. Nah, K. M. Varahramyan, S. K. Banerjee, and E. Tutuc, ECS Meeting, 2012, **Invited**
878. ETB-QW InAs MOSFET with scaled body for improved electrostatics, T.Kim, et. al., IEDM, 2012
879. S.Banerjee,, BiSFET, SISC, Dec. 2012, **Invited**.
880. D. Jawarani, R.S. Smith, S. Saha, R.A. Rao, L. Mathew, M. Ainom, R. Garcia, A. Gurmu, R. Stout & C. Vass, D. Xu, E.U. Onyegam, S. Banerjee & P.S. Ho, , D. Sarkar & J.G. Fossum, Integration and Reliability of Thin Silicon Solar Cells and Modules Fabricated Using SOM® Technology, EU PVSC, 2012, Germany.

881. L. Mathew, R.-A. Rao, S. Saha, S. Smith, R. Garcia, R. Stout, A. Gurmu, M. Ainom, D. Xu & D. Jawarani, D. Sarkar & J. Fossum, E. Onyegam & S. Banerjee, A Novel Low Cost ~25 μ m Thin Monocrystalline Silicon Bi-Facial Solar Cell Technology with Flexible and Rigid Form-Factors and Electroplated Contacts, EU PVSC, 2012, Germany.
882. Yujia Zhai, Marylene Palard, Leo Mathew, Muhammad Mustafa Hussain, C. Grant Willson, Emanuel Tutuc and Sanjay K. Banerjee, Fabrication of Three-Dimensional MIS Nano-Capacitor Based on Nanoimprinted Single Crystal Silicon Nanowire Arrays, Micro and Nanosystems, 2012, 4
883. Self-aligned graphene field-effect transistors with polyethyleneimine doped source/drain access regions, Movva, HCP ; Ramon, ME ; Corbet, CM ; Sonde, S ; Chowdhury, SF ; Carpenter, G ; Tutuc, E ; Banerjee, SK , APPLIED PHYSICS LETTERS Volume: 101 Issue: 18, OCT 29 2012
884. Zhai, Yujia; Mathew, Leo; Rao, Rajesh; Xu, Dewei; Banerjee, Sanjay, "High Performance Flexible Thin-Film Transistors Exfoliated from Bulk Wafer," Nano Letters, 2012. Also **Best Paper** at DARPA conference.
885. Mohamed M Hilali, Shuqiang Yang, Mike Miller, Frank Xu, Sanjay Banerjee and S V Sreenivasan, Enhanced photocurrent in thin-film amorphous silicon solar cells via shape controlled three-dimensional nanostructures, Nanotechnology 23 405203, Sept. 2012.
886. First principles study of Stage-1 graphene intercalates, IBr and ICl, PRIYAMVADA JADAUN, LEONARD F. REGISTER, SANJAY K. BANERJEE, APS March Meet, 2013.
887. Topological Classification of Crystalline Insulators with Point Group Symmetry, DI XIAO, PRIYAMVADA JADAUN, QIAN NIU, SANJAY BANERJEE, APS March Meet, 2013.
888. Growth of topological insulators on Si(111)-(7 $\times$ 7) surfaces by molecular beam epitaxy, ANUPAM ROY, SUSHANT SONDE, SAMARESH, GUCHHAIT, SANJAY BANERJEE, APS March Meet, 2013.
889. A study of capping layers for sulfur monolayer doping on III-V junctions, Yum, JH ; Shin, HS ; Hill, R ; Oh, J ; Lee, HD ; Mushinski, RM ; Hudnall, TW ; Bielawski, CW ; Banerjee, S. K.; Loh, WY ; Wang, WE ; Kirsch, P, APPLIED PHYSICS LETTERS Volume: 101 Issue: 25: DEC 17 2012
890. Germanium metal-semiconductor-metal photodetectors evanescently coupled with upper-level silicon oxynitride dielectric waveguides, Cervantes-Gonzalez, JC ; Ahn, D ; Zheng, XG ; Banerjee, Sanjay K.; Jacome, AT ; Campbell, JC ; Zaldivar-Huerta, IE, APPLIED PHYSICS LETTERS Volume: 101 Issue: 26: DEC 24 2012
891. Topological insulator Bi₂Se₃ thin films as an alternative channel material in metal-oxide-semiconductor field-effect transistors, Chang, Jiwon; Register, Leonard F.; Banerjee, Sanjay K, JOURNAL OF APPLIED PHYSICS Volume: 112 Issue: 12: DEC 15 2012
892. Uniform Wafer-Scale Chemical Vapor Deposition of Graphene on Evaporated Cu (111) Film with Quality Comparable to Exfoliated Monolayer, Tao, L; Lee, J ; Holt, M ; Chou, H ; McDonnell, SJ ; Ferrer, DA ; Babenco, MG ; Wallace, RM¹; Banerjee, Sanjay K.; Ruoff, RS ; Akinwande, D, JOURNAL OF PHYSICAL CHEMISTRY C Volume: 116 Issue: 45 Pages: 24068-24074: NOV 15 2012.
893. Fast and slow transient charging in various III-V field-effect transistors with atomic-layer-deposited-Al₂O₃ gate dielectric, Ramon, ME ; Akyol, T; Shahrjerdi, D; Young, CD; Cheng, JL ; Register, LF; Banerjee, Sanjay K, APPLIED PHYSICS LETTERS Volume:102 Issue:2 Article Number:022104, JAN 14 2013
894. Exfoliated, thin, flexible germanium heterojunction solar cell with record FF=58.1%, Onyegam, EU; Sarkar, D; Hilali, M; Saha, S; Rao, RA; Mathew, L; Jawarani, D; Mantey, J; Ainom, M; Garcia, R; James, W; Banerjee, S. K., SOLAR ENERGY MATERIALS AND SOLAR CELLS Volume:111 Pages:206-211 APR 2013

895. Single heterojunction solar cells on exfoliated flexible similar to 25 μm thick monocrystalline silicon substrates, Saha, S; Hilali, MM; Onyegam, EU; Sarkar, D; Jawarani, D; Rao, RA; Mathew, L; Smith, RS; Xu, DW; Das, UK; Sopori, B; Banerjee, Sanjay K, APL, 2013
896. Two-dimensional weak anti-localization in Bi₂Te₃ thin film grown on Si(111)-(7x7) surface by molecular beam epitaxy, Roy, A; Guchhait, S; Sonde, S; Dey, R; Pramanik, T; Rai, A; Movva, HCP; Colombo, L; Banerjee, Sanjay K, APPLIED PHYSICS LETTERS Volume:102 Issue:16 Article Number:163118 APR 22 2013
897. Ultra-smooth epitaxial Ge grown on Si(001) utilizing a thin C-doped Ge buffer layer, Mantey, J; Hsu, W; James, J; Onyegam, EU; Guchhait, S; Banerjee, S. K., APPLIED PHYSICS LETTERS Volume:102 Issue:19 Article Number:192111 MAY 13 2013
898. Micromagnetic study of spin-transfer-torque switching of a ferromagnetic cross towards multi-state spin-transfer-torque based random access memory, Roy, U ; Pramanik, T; Tsoi, M; Register, LF; Banerjee, Sanjay K., JOURNAL OF APPLIED PHYSICS Volume:113 Issue:22 Article Number:223904 JUN 14 2013
899. Novel atomic layer deposited thin film beryllium oxide for InGaAs MOS Devices, Koh, D; Yum, JH; Akyol, T; Ferrer, DA; Lei, M Hudnall, TW; Downer, M; Bielawski, CW; Hill, R; Bersuker, G; Banerjee, S. K, 2012 INTERNATIONAL CONFERENCE ON INDIUM PHOSPHIDE AND RELATED MATERIALS (IPRM) Book Series:International Conference on Indium Phosphide and Related Materials Pages:163-166 2013
900. A Comparative Study of Gate First and Last Processes Using ALD Beryllium Oxide as an Interface Passivation Layer for Si MOSFETs, J. H. Yum, H. S. Shin, Ryan M. Mushinski, Todd. W. Hudnall, J. Oh, P. Kirsch, R. Jammy, R. J. W. Hill, W. Y. Loh, C. W. Bielawski, G. Bersuker S. K. Banerjee, and W. Wang, VLSI TSA, 2013
901. A Novel Low-Cost Method for Fabricating Bifacial Solar Cells, S. Saha, R. A. Rao, L. Mathew, M. Ainom and S. K. Banerjee, PVSC, 2013
902. Amorphous/Crystalline Silicon Heterojunction Cells Realized via Remote Plasma Chemical Vapor Deposition: Influence of Hydrogen Dilution, RF power, and Sample Z-height Position, E. U. Onyegam, S. Saha, R. Rao, L. Mathew, M. Hilali, William James and S.K. Banerjee, PVSC 2013
903. ALD Gate Dielectrics for High Mobility and Tunnel Transistors J.Mantey, C.Corbet, F.Chowdhury, D.Koh, J.Yum, H.Movva, M.Ramon, D.Reddy, L.F.Register and S.K. Banerjee, MRS **Invited** talk, 2013
904. Interactions between two contacted graphene layers, C. Corbet, E.Tutuc, S.Banerjee, FCMN NIST 2013.
905. Intercalated bilayer graphene heterostructures towards the BISFET, H.Movva, P.Jadaun, F.Register and S.Banerjee, INC9, Germany, 2013
906. Design and verification of PECVD multilayer nanoscale PV device, S.Hajimirza, J. Howell, M.Holt, S.Saha, D. Akinwande and S.Banerjee, ASME HT 2013
907. Full band quantum transport simulation of MoS₂ transistors, J.Chang, F.Register, S.Banerjee, DRC 2013
908. Novel Graphene Devices, C. Corbet, M. Ramon, H. Movva, D. Reddy, S.Kang, F.Chowdhury, D.Akinwande, E. Tutuc, F.Register and S. K. Banerjee, ECS **Invited**, Oct. 2013
909. Quantum Transport Simulation of Bilayer Pseudospin Field-Effect Transistor (BiSFET) on Tightbinding Hartree-Fock Model, Xuehao Mou, Leonard F. Register, and Sanjay K. Banerjee, SISPAD, Scotland, 2013.
910. Comparison of Ballistic Transport Characteristics of Monolayer Transition Metal Dichalcogenides (TMDs) MX₂ (M = Mo, W; X = S, Se, Te) n-MOSFETs, Jiwon Chang, Leonard F. Register, and Sanjay K. Banerjee, SISPAD 2013
911. Xuehao Mou, Leonard F. Register and Sanjay K. Banerjee, "Quantum Transport Simulations on the Feasibility of the Bilayer PseudoSpin Field Effect Transistor (BiSFET)", IEDM 2013, Washington, D. C., Dec. 2013

912. Sub-100 nm InGaAs Quantum-Well (QW) Tri-Gate MOSFETs with Al₂O₃/HfO₂ (EOT < 1 nm) for Low-Power Logic Applications, T.-W. Kim, D.-H. Kim, D.H. Koh, H.M. Kwon, R.H. Baek, D. Veksler, C. Huffman, K. Matthews, S. Oktyabrsky, A. Greene, Y. Ohsawa, A. Ko, H. Nakajima, M. Takahashi, T. Nishizuka, H. Ohtake, S. K. Banerjee, S.H. Shin, D.-H. Ko, C. Kang, D. Gilmer, R.J.W. Hill, W. Maszara, C. Hobbs and P.D. Kirsch, IEDM 2013
913. Xuehao Mou, Dax M. Crum, Dipanjan Basu, Leonard F. Register, Sanjay K. Banerjee, "On the Possibility of Sub- $k_B T/q$ Switching in BiSFETs: Quantum Transport Simulations in Graphene Bilayers with Many-body Exchange Interactions", Techcon 2013, Austin, Texas, Sep. 2013
914. Tanmoy Pramanik, U.Roy, M.Tsoi, Leonard F. Register, Sanjay K. Banerjee, "Micromagnetic study of spin-transfer-torque switching of a ferromagnetic cross towards multi-state spin-transfer-torque based random access memory", Techcon 2013, Austin, Texas, Sep. 2013
915. Chris Corbet, S. McDonnell, S. Sonde, B. Fallahazad, S. Larentis, D. Hinojos, R. Wallace, E. Tutuc, S.K. Banerjee, Fabrication and characterization of large arrays of graphene tunneling field effect transistors (TFETs), Techcon 2013, Austin, Texas, Sep. 2013
916. Topological classification of crystalline insulators with space group symmetry, Jadaun, P, Xiao, D ; Niu, Q; Banerjee, Sanjay K., PHYSICAL REVIEW B Volume:88 Issue:8 Article Number:085110, :AUG 8 2013
917. Strained-Si/strained-Ge type-II staggered heterojunction gate-normal-tunneling field-effect transistor, William Hsu, Jason Mantey, Leonard F. Register, and Sanjay K. Banerjee, APL, 27 August 2013
918. Theory and synthesis of bilayer graphene intercalated with ICl and IBr for low power device applications, Priyamvada Jadaun, Hema C. P. Movva, Leonard F. Register, and Sanjay K. Banerjee, J. Appl. Phys. 114, 063702 (2013); Aug. 2013.
919. Atomistic full-band simulations of monolayer MoS₂ transistors, Chang, Jiwon; Register, Leonard F.; Banerjee, Sanjay K, APPLIED PHYSICS LETTERS Volume: 103 Issue: 22, NOV 25 2013
920. Low interface defect density of atomic layer deposition BeO with self-cleaning reaction for InGaAs metal oxide semiconductor field effect transistors H. S. Shin, J. H. Yum, D. W. Johnson, H. R. Harris, Todd. W. Hudnall, J. Oh, P. Kirsch, W.-E. Wang, C. W. Bielawski, S. K. Banerjee, J. C. Lee, and H. D. Lee, Applied Physics Letters 103, 223504 (2013)
921. Reliability study of methods to suppress boron transient enhanced diffusion in high-k/metal gate Si/SiGe channel pMOSFETs, Park, MS, Kim, Y; Lee, KT Kang, CY; Min, BG; Oh, J; Majhi, P; Tseng, HH;; Banerjee, Sanjay K.; Lee, JS; Jammy, R; Jeong, YH, MICROELECTRONIC ENGINEERING Volume:112 Special Issue:SI Pages:80-83, DEC 2013
922. Magnetic ordering of implanted Mn in HOPG substrates, Guchhait, S; Ohldag, H; Arenholz, E; Ferrer, DA; Mehta, A; Banerjee, Sanjay K., PHYSICAL REVIEW B Volume:88 Issue:17 NOV 27 2013
923. Evidence for hydrogen two-level systems in atomic layer deposition oxides, Khalil, MS; Stoutimore, MJA; Gladchenko, S ; Holder, AM; Musgrave, CB; Kozen, AC; Rubloff, G; Liu, YQ;Gordon, RG; Yum, JH; Banerjee, S. K.; Lobb, CJ; Osborn, KD, APPLIED PHYSICS LETTERS Volume:103 Issue:16, OCT 14 2013
924. The Bilayer Pseudospin Field-Effect Transistor: Overview and Quantum Transport Simulation, Leonard F. Register, Xuehao Mou, Sanjay K. Banerjee, ECS Orlando, 2014, **Invited**
925. Improving the electrical characteristics of graphene field effect transistors by hexamethyldisilazane interaction SK. CHOWDHURY, SOMAYYEH RAHIMI, SUSHANT SONDE, LI TAO, SANJAY BANERJEE, DEJI AKINWANDE, APS March Meeting 2014
926. Electronic and optical properties of GaSb:N from first principles, PRIYAMVADA JADAUN, HARI NAIR, VINCENZO LORDI, SETH BANK, SANJAY BANERJEE, APS March Meeting 2014

927. Transfer-free growth of atomically thin hexagonal boron nitride, SUSHANT SONDE, NING LU, MOON KIM, LUIGI COLOMBO, SANJAY K. BANERJEE, APS March Meeting 2014
928. Novel Low Power Transistors in 2D Dirac Materials: Graphene and Topological Insulators, Sanjay Banerjee and Leonard Frank Register, NanoGiga, ASU, **Plenary talk**, March 2014
929. Novel Low Power Transistors in 2D Dirac Materials: Graphene and Topological Insulators, Sanjay Banerjee and Leonard Frank Register, ISVLSI, **Florida, Plenary talk**, July 2014
930. Graphene Transfer Using Sacrificial PIB Layer onto 1nm Al₂O₃/TiOPc/Graphene Gate Stacks, Jun Hong Park, Hema Chandra Prakash Movva, Erich W. Kinder, Hao Lu, Deji Akinwande, Iljo Kwak, Susan K. Fullerton, Sanjay K. Banerjee, Andrew Kummel, MRS 2014
931. Characterization of ALD Beryllium Oxide as a Potential High-k Gate Dielectric for Low-Leakage AlGaIn/GaN MOSHEMTs, Johnson, Derek W.; Yum, Jung Hwan; Hudnall, Todd W.; ... SK. Banerjee, R. Harris, JOURNAL OF ELECTRONIC MATERIALS Volume: 43 Issue: 1 Pages: 151-154, JAN 2014
932. Improved Cleaning Process for Textured similar to 25 μ m Flexible Mono-Crystalline Silicon Heterojunction Solar Cells with Metal Backing, Saha, Sayan; Hilali, Mohamed M.; Onyegam, Emmanuel U.; ... S. K. Banerjee, ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY Volume: 3 Issue: 7 Pages: Q142-Q145 Published: 2014
933. Atomic layer etching of BeO using BCl₃/Ar for the interface passivation layer of III-V MOS devices, Min, K. S.; Kang, S. H.; Kim, J. K.; ... Bielawski, CW; Banerjee, S. K.; Bersuker, G; Jhon, MS, MICROELECTRONIC ENGINEERING, Volume: 114, Pages: 121-125, FEB 2014
934. Impact of contact and access resistances in graphene field-effect transistors on quartz substrates for radio frequency applications, Ramon, ME; Movva, HCP; Chowdhury, SF; Parrish, KN; Rai, A; Magnuson, CW; Ruoff, RS; Akinwande, D; Banerjee, SK, APPLIED PHYSICS LETTERS, Volume: 104 (7), FEB 17 2014
935. Poly(methyl methacrylate) as a self-assembled gate dielectric for graphene field-effect transistors, Sanne, A.; Movva, H. C. P.; Kang, S.; ... SK Banerjee, APPLIED PHYSICS LETTERS Volume: 104 Issue: 8 FEB 24 2014
936. Ballistic performance comparison of monolayer transition metal dichalcogenide MX₂ (M = Mo, W; X = S, Se, Te) metal-oxide-semiconductor field effect transistors, Chang, Jiwon; Register, Leonard F.; Banerjee, Sanjay K. JOURNAL OF APPLIED PHYSICS Volume: 115 Issue: 8 Article Number: 084506 FEB 28 2014
937. Atomistic simulation of the electronic states of adatoms in monolayer MoS₂, Chang, Jiwon; Larentis, Stefano; Tutuc, Emanuel; L. Register, SK. Banerjee, APPLIED PHYSICS LETTERS Volume: 104 Issue: 14 Article Number: 141603 APR 7 2014
938. Realization of dual-heterojunction solar cells on ultra-thin similar to 25 μ m, flexible silicon substrates, Onyegam, Emmanuel U.; Sarkar, Dabraj; Hilali, Mohamed M.; ... SK Banerjee, APPLIED PHYSICS LETTERS Volume: 104 Issue: 15 Article Number: 153902 APR 14 2014
939. L-g=100 nm In_{0.7}Ga_{0.3}As quantum well metal-oxide semiconductor field-effect transistors with atomic layer deposited beryllium oxide as interfacial layer, Koh, D.; Kwon, H. M.; Kim, T-W; ... SK. Banerjee, APPLIED PHYSICS LETTERS Volume: 104 Issue: 16 Article Number: 163502 APR 21 2014
940. Investigation of atomic layer deposited beryllium oxide material properties for high-k dielectric applications, Koh, Donghyi; Yum, Jung-Hwan; Banerjee, Sanjay K.; et al, JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B Volume: 32 Issue: 3 Article Number: 03D117 MAY 2014
941. Micromagnetic simulations of spin-wave normal modes and the spin-transfer-torque driven magnetization dynamics of a ferromagnetic cross, Pramanik, Tanmoy; Roy, Urmimala; Tsoi, Maxim; L. Register, SK. Banerjee, JOURNAL OF APPLIED PHYSICS Volume: 115 Issue: 17 Article Number: 17D123 MAY 7 2014

942. Strong spin-orbit coupling and Zeeman spin splitting in angle dependent magnetoresistance of Bi₂Te₃, Dey, Rik; Pramanik, Tanmoy; Roy, Anupam; L.Colombo, F.Register, SK.Banerjee, APPLIED PHYSICS LETTERS Volume: 104 Issue: 22 Article Number: 223111, JUN 2 2014
943. Improvement of graphene field-effect transistors by hexamethyldisilazane surface treatment, Sk. Fahad Chowdhury, Sushant Sonde, Somayyeh Rahimi, Li Tao, Sanjay Banerjee and Deji Akinwande, Appl. Phys. Lett. 105, 033117, July 2014.
944. The Bilayer Pseudospin Field-Effect Transistor: Overview and Quantum Transport Simulation, Leonard F. Register, Xuehao Mou, and Sanjay K. Banerjee, ECS 2014 **Invited**.
945. Effect of HfO₂ and Al₂O₃ on monolayer MoS₂ electronic structure, A. Valsaraj, J. Chang, L. F. Register, and S. K. Banerjee, Dev. Res. Conf. 2014.
946. Intra- and Inter-layer Current Flow within Exciton Condensate in Bilayer Graphene Systems, Xuehao Mou, Leonard F. Register and Sanjay K. Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
947. Proposal of a multi-state memory using voltage controlled magnetic anisotropy of a cross-shaped ferromagnet, Tanmoy Pramanik, Urmimala Roy, Nima Asoudegi, Leonard F. Register and Sanjay K. Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
948. Characteristics of Cerium Oxide Based Resistive Random Access Memories, Cheng-Chih Hsieh, Anupam Roy, Amritesh Rai, Sanjay Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
949. Oxidized titanium as a graphene gate dielectric and its conduction mechanisms, Chris Corbet, Connor McClellan, Kyoungwhan Kim, Sushant Sonde, Emanuel Tutuc, Sanjay Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
950. DFT Study of Effect of HfO₂ on Monolayer MoS₂, Amithraj Valsaraj, Jiwon Changb, Leonard F. Registera, and Sanjay K. Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
951. Evidence of Strong Spin-Orbit Coupling & Effects of Zeeman Spin Splitting in Angle-Dependent Anisotropic Magnetoresistance of Topological Insulator Bi₂Te₃ Thin Film, Rik Dey, Tanmoy Pramanik, Anupam Roy, Amritesh Rai, Samaresh Guchhait, Sushant S Sonde, Hema CP Movva, Luigi Colombo, Leonard F Register and Sanjay K Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
952. Numeric Technique for Pauli-Exclusion Principle in Monte Carlo Simulation, Dax M. Crum, A. Valsaraj, J.K. David, L. F. Register, and S. K. Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
953. Quantum-Corrected Semi-Classical Ensemble Monte Carlo Simulator for Nano-Scale III-V In Ga As Tri-Gate FinFET, Dax M. Crum, A. Valsaraj, J.K. David, B. Sahu, ^{0.53 0.47} Z. Krivokapic, L. F. Register, and S. K. Banerjee, Int.Conf. Phys. Semi. Austin, TX, Aug. 2014.
954. Oxidized Titanium as a Graphene Gate Dielectric and Its Conduction Mechanisms, C.Corbet, E.Tutuc, S.Banerjee, SRC TECHCON, Sept. 2014 (**Best paper**)
955. Improving the Electrical Characteristics of Graphene Field-Effect Transistors by Hexamethyldisilazane Interaction, S.F. Chowdhury, S.Banerjee, D.Akinwande, SRC TECHCON, Sept. 2014
956. Device Applications of Exciton Condensates in Bilayer Graphene Systems, X.Mou, X.Wu, F.Register and S.Banerjee, SRC TECHCON, Sept. 2014
957. Self Assembled Ordered Phthalocyanine Monolayers on 2D Semiconductors for Subnanometer dielectric ALD Nucleation, Jun Hong Park, Iljo Kwak, Pabitra Choudhury, Kasra Sardashti, Hema Chandra Prakash, Sanjay Banerjee, Susan Fullerton§ Andrew C. Kummel, SRC TECHCON, Sept. 2014
958. Density-Functional-Theory-Based Study of Monolayer MoS₂ on Oxide”A. Valsaraj, L.F. Register, S.K. Banerjee and J. Chang, SISPAD 2014, Japan.

959. Semi-Classical Ensemble Monte Carlo Simulator Using Innovative Quantum Corrections for Nano-Scale n-Channel FinFETs” D.M. Crum, A. Valsaraj, L.F. Register, and S.K. Banerjee, SISPAD 2014, Japan.
960. Interplay among Bilayer PseudoSpin Field-Effect Transistor (BiSFET) Performance, BiSFET Scaling and Condensate Strength”, X. Mou, L.F. Register, and S.K. Banerjee, SISPAD 2014, Japan.
961. Leonard F. Register, Urmimala Roy, Rik Dey, Tanmoy Pramanik, Jiwon Chang* and Sanjay K. Banerjee, “Topological insulators in electronic and spintronic applications,” Workshop on Innovative Nanoscale Devices and Systems (WINDS) 2014, Big Island, Hawaii, 12/1/2014-12/5/2014. (**Invited talk**)
962. Light trapping in ultrathin 25 μ m exfoliated Si solar cells , Hilali, Mohamed M.; Saha, Sayan; Onyegam, Emmanuel; Rao, Rajesh; Mathew, Leo; Banerjee, SK, APPLIED OPTICS Volume: 53 Issue: 27 Pages: 6140-6147 SEP 20 2014
963. Electronics based on two-dimensional materials, Fiori, Gianluca; Bonaccorso, Francesco; Iannaccone, Giuseppe; Palacios, TomasNeumaier, Daniel; Seabaugh, Alan; Banerjee, Sanjay K.; Colombo, L, NATURE NANOTECHNOLOGY Volume: 9 Issue: 10 Pages: 768-779 OCT 2014
964. Oxidized Titanium as a Gate Dielectric for Graphene Field Effect Transistors and Its Tunneling Mechanisms, Corbet, Chris M.; McClellan, Connor; Kim, Kyoungwhan; Sonde, Sushant; Tutuc, Emanuel; Banerjee, SK ACS NANO Volume: 8 Issue: 10 Pages: 10480-10485 OCT 2014
965. Thin, relaxed Si_{1-x}Ge_x virtual substrates on Si grown using C-doped Ge buffers, Hsu, William; Mantey, Jason; Hsieh, Cheng-Chih; Roy, Anupam); Banerjee, SK, APPLIED PHYSICS LETTERS Volume: 105 Issue: 15 Article Number: 152107 OCT 13 2014
966. Vertical finFET with Salicide Contact for Potential Power Applications, Zhai, YJ ; Mathew, L ; Rao, R; Sreenivasan, SV; Willson, CG; Banerjee, SK ECS JOURNAL OF SOLID STATE SCIENCE AND TECHNOLOGY, Volume: 3 , Issue: 10 ,Pages: Q203-Q206; Published: 2014
967. High-Performance Vertical Gate-All-Around Silicon Nanowire FET With High-kappa/Metal Gate, Zhai, YJ; Mathew, L ; Rao, R ; Palard, M; Chopra, S; Ekerdt, JG; Register, LF; Banerjee, Sanjay K., IEEE TRANSACTIONS ON ELECTRON DEVICES, Volume: 61, Issue: 11, Pages: 3896-3900, DOI: 10.1109/TED.2014.2353658, NOV 2014
968. Magnetic and magneto-transport studies of MBE grown Cr₂Te₃ thin films with perpendicular magnetic anisotropy, Anupam Roy, Samaresh Guchhait, Rik Dey, Tanmoy Pramanik, Cheng-Chih Hsieh, Amrithesh Rai, Sanjay Banerjee, Bulletin of the American Physical Society 60 (2015), APS March Meeting 2015 at San Antonio, Texas, USA (March 2-6, 2015)
969. Bilayer Graphene-Hexagonal Boron Nitride Heterostructure Negative Differential Resistance Interlayer Tunnel FETs, S.Kang...,F.Register, E.Tutuc, SK. Banerjee, APS Meeting, San Antonio, March 2015.
970. Resonant Tunneling in Double Bilayer Graphene Heterostructures, B.Fallahazad,... F.Register,SK. Banerjee, E.Tutuc, APS Meeting, San Antonio, March 2015.
971. Low temperature magnetoresistance studies in MBE grown topological insulator thin films, Rik Dey, Anupam Roy, Tanmoy Pramanik, Samaresh Guchhait, Sushant Sonde, Amrithesh Rai, Sarmita Majumder, Bahaniman Ghosh, Leonard Register, Sanjay Banerjee, Bulletin of the American Physical Society 60 (2015), APS March Meeting 2015 at San Antonio, Texas, USA (March 2-6, 2015)
972. Air stable doping of MoS₂ FETs using TiO_x sol-gel, Amrithesh Rai, Rudresh Ghosh, Anupam Roy, Amithraj Valsaraj, Hema C P Movva, Sangwoo Kang, Emanuel Tutuc, Leonard Register, Sanjay Banerjee, Bulletin of the American Physical Society 60 (2015), APS March Meeting 2015 at San Antonio, Texas, USA (March 2-6, 2015)

973. Van der Waals epitaxy of Bismuth Telluro-Sulfide nanosheets and magnetotransport in devices, T.Trivedi... SK. Banerjee, APS Meeting, San Antonio, March 2015.
974. Ambipolar conduction in MoS₂/WSe₂ hetero-bilayers, H.Movva... SK. Banerjee, APS Meeting, San Antonio, March 2015.
975. Rhenium Disulfide Depletion-Load Inverter, C.McClellan... E.Tutuc, SK. Banerjee, APS Meeting, San Antonio, March 2015.
976. Ultra Low Energy Switching of Ferromagnet with Perpendicular Anisotropy on Topological Insulator by Voltage Controlled Magnetic Anisotropy, B.Ghosh... F.Register, SK. Banerjee, APS Meeting, San Antonio, March 2015.
977. Magnetization switching of a nanomagnet by spin polarized surface states of a topological insulator, U.Roy...F.Register, SK. Banerjee, APS Meeting, San Antonio, March 2015.
978. Semiconducting Behavior, Schottky Barriers and Field Effect Transistors in Ultrathin Rhenium DiSulfide, C.Corbet... E.Tutuc, SK. Banerjee, APS Meeting, San Antonio, March 2015.
979. Gate-Tunable Resonant Tunneling in Double Bilayer Graphene Heterostructures, B. Fallahazad ;K. Lee; S. Kang; ..LF Regsiter, SK Banerjee and E.Tutuc, NANO LETTERS Volume: 15 Issue: 1 Pages: 428-433 Published: JAN 2015
980. Field Effect Transistors with Current Saturation and Voltage Gain in Ultrathin ReS₂, Corbett, Chris M.; McClellan, Connor; Rai, Amrithesh; E.Tutuc; SK Banerjee, ACS NANO Volume: 9 Issue: 1 Pages: 363-370 Published: JAN 2015
981. Comment on "Assessment of field-induced quantum confinement in heterogate germanium electron-hole bilayer tunnel field-effect transistor" [Appl. Phys. Lett. 105, 082108 (2014)], Hsu, William; Mantey, Jason; Register, Leonard F.; SK Banerjee, APPLIED PHYSICS LETTERS Volume: 106 Issue: 2 Article Number: 026102 Published: JAN 12 2015
982. Top-gated chemical vapor deposited MoS₂ field-effect transistors on Si₃N₄ substrates, Sanne, A.; Ghosh, R.; Rai, A.; SK Banerjee, APPLIED PHYSICS LETTERS Volume: 106 Issue: 6 Article Number: 062101 Published: FEB 9 2015
983. Air Stable Doping and Intrinsic Mobility Enhancement in Monolayer Molybdenum Disulfide by Amorphous Titanium Suboxide Encapsulation, Rai, Amrithesh; Valsaraj, Amithraj; Movva, Hema C. P.; ... LF Register, E. Tutuc, SK Banerjee, NANO LETTERS Volume: 15 Issue: 7 Pages: 4329-4336 Published: JUL 2015
984. On the Electrostatic Control of Gate-Normal-Tunneling Field-Effect Transistors, Hsu, William; Mantey, Jason; Register, Leonard F.; SK. Banerjee, IEEE TRANSACTIONS ON ELECTRON DEVICES Volume: 62 Issue: 7 Pages: 2292-2299 Published: JUL 2015
985. Perpendicular Magnetic Anisotropy and Spin Glass-like Behavior in Molecular Beam Epitaxy Grown Chromium Telluride Thin Films, Roy, Anupam; Guchhait, Samareesh; Dey, Rik; ... SK. Banerjee, ACS NANO Volume: 9 Issue: 4 Pages: 3772-3779 Published: APR 2015
986. Bilayer Graphene-Hexagonal Boron Nitride Heterostructure Negative Differential Resistance Interlayer Tunnel FET, Kang, Sangwoo; Fallahazad, Babak; Lee, Kayoung; et al. Movva, Hema, Kim, Kyoungwan, Corbet, Chris M, Taniguchi, Takashi, Watanabe, Kenji, Colombo, Luigi, Register, Leonard F, Tutuc, Emanuel, Banerjee, Sanjay K., IEEE ELECTRON DEVICE LETTERS Volume: 36 Issue: 4 Pages: 405-407 Published: APR 2015
987. Magnetization switching of a metallic nanomagnet via current-induced surface spin-polarization of an underlying topological insulator, Roy, Urmimala; Dey, Rik; Pramanik, Tanmoy; LF. Register, SK. Banerjee, JOURNAL OF APPLIED PHYSICS Volume: 117 Issue: 16 Article Number: 163906 Published: APR 28 2015
988. Characteristics and mechanism study of cerium oxide based random access memories, Hsieh, Cheng-Chih; Roy, Anupam; Rai, Amrithesh;... SK. Banerjee, APPLIED PHYSICS LETTERS Volume: 106 Issue: 17 Article Number: 173108 Published: APR 27 2015
- 989.

990. T. Pramanik, U Roy, L. F. Register, S. K. Banerjee, "Proposal of a multi-state memory using voltage controlled magnetic anisotropy of a cross-shaped ferromagnet," IEEE Trans. Nanotech. **14**, 883-888 (July 2015).
991. Radio Frequency Transistors and Circuits Based on CVD MoS₂, Sanne, Atresh; Ghosh, Rudresh; Rai, Amritesh; Yogeesh, Maruthi Nagavalli; Shin, Seung Heon; Sharma, Ankit; Jarvis, Karalee; Mathew, Leo; Rao, Rajesh; Akinwande, Deji; Banerjee, Sanjay, Nano letters, Volume:15 Issue:8, Pages:5039-45, DOI:10.1021/acs.nanolett.5b01080 Published:2015-Aug-12 (Epub 2015 Jul 08)
992. Write error rate in spin-transfer-torque random access memory including micromagnetic effects, Urmimala Roy, David L. Kencke*, Tanmoy Pramanik, Leonard F. Register and Sanjay K. Banerjee, DRC Ohio, 2015
993. Cerium oxide based bipolar resistive switching memory with low operation voltage and high resistance ratio, Cheng-Chih Hsieh, Anupam Roy, Yao-Feng Chang, Amritesh Rai and Sanjay Banerjee, Proceedings of 73rd IEEE Device Research Conference at Ohio State University, Columbus, Ohio (June 21-24, 2015) Page 101-102.
994. MOD-TFET, W.Hsu... L.Register, SK.Banerjee, DRC Ohio, 2015
995. Gate Tunable Resonant Tunneling in Graphene Heterostructures, E.Tutuc, ... L.Register, S.Banerjee, **Invited** Talk, DRC Ohio, 2015
996. Self. Assembled Ordered Phthalocyanine Monolayers on 2D, J.Park, .. SK Banerjee, S.Fullerton, A.Kummel, **Invited**, MRS 2015
997. Monolayer Organic Films for Nucleation of ALD, J.Park, ... S.Banerjee,... A.Kummel, **Invited** SISC 2015
998. Radio Frequency Transistors and Circuits Based On CVD MoS₂, A.Sanne,... R,Rao, L.Mathew, D.Akinwande, SK.Banerjee, DRC Ohio, 2015
999. Interfacial-oxygen-vacancy mediated doping of MoS₂ by high-K dielectrics, A Rai, A Valsaraj, H C P Movva, A Roy, E Tutuc, L F Register and S K Banerjee, Proceedings of 73rd IEEE Device Research Conference at Ohio State University, Columbus, Ohio (June 21-24, 2015) Page 189-190.
1000. Top-gated WSe₂ field-effect transistors with Pt contacts, Hema C. P. Movva, Amritesh Rai, Sangwoo Kang, Kyoungwhan Kim, Samareesh Guchhait, Takashi Taniguchi, Kenji Watanabe, Emanuel Tutuc, and Sanjay K. Banerjee, **Best Paper**, DRC Ohio, 2015
1001. Interlayer Tunnel FETs and High Frequency FETs in Graphene and TMDs, **Invited talk**, (with H.Movva, S.Kang, A.Sanne, R.Ghosh, B.Fallahzad, A.Rai, E.Tutuc, L.F.Register), IEEE Photonics Conf. 2D Materials, July 2015.
1002. Interlayer Tunnel FETs and High Frequency FETs in Graphene and TMDs, **Invited talk**, (with H.Movva, S.Kang, A.Sanne, R.Ghosh, B.Fallahzad, A.Rai, E.Tutuc, L.F.Register), Int. Mat. Res. Conf. Cancun, Aug. 2015.
1003. Contact Engineering for Novel Transistors in 2D Materials, **Invited Talk** (with H.Movva, S.Kang, A.Sanne, R.Ghosh, B.Fallahzad, A.Rai, L.Colombo, D. Akinwande, E.Tutuc, L.F.Register), Metallization Conf. Austin, Sept. 2015.
1004. Interlayer Tunnel FETs, **Invited Talk**, (with Kang, Sangwoo; Fallahzad, Babak; Lee, Kayoung; et al. Movva, Hema, Kim, Kyoungwhan, Corbet, Chris M, Taniguchi, Takashi, Watanabe, Kenji, Colombo, Luigi, Register, Leonard F, Tutuc, Emanuel, AVS, San Jose, Oct. 2015
1005. High Frequency Prospects of 2D Materials in Flex Nano, S.Park,...SK.Banerjee and D. Akinwande, IEDM 2015
1006. Spin transfer torque switching of ferromagnets on topological insulators, S. Majumder, S. Guchhait, R. Dey, L. F. Register, and S. K. Banerjee, SRC TECHCON, Austin, 2015
1007. Voltage Induced Switching of Nanomagnetism Topological Insulator Magnetoelectric Devices through Ruderman-Kittel-Kasuya-Yosida Interactions, B.Ghosh, R.Dey, LF. Register, SK.Banerjee, SRC TECHCON, Austin, 2015

1008. Synthesis and characterization of h-BN, graphene, and other 2D materials, Harry Chou, Rudresh Ghosh, Carl Magnuson, Ariel Ismach, Andrei Dolocan, Rodney Ruoff, and Sanjay Banerjee, SRC TECHCON, Austin, 2015
1009. Exploring nanoscale InGaAs and Si FinFETs using quantum-corrected semi-classical Monte Carlo, Dax M. Crum, A. Valsaraj, L. F. Register, S. K. Banerjee, B. Sahu, and Z. Krivokopic, SRC TECHCON, Austin, 2015, **BEST PAPER**.
1010. High Mobility WSe₂ FETs with Platinum Contacts, Hema C.P. Movva, Amritesh Rai, Sangwoo Kang, Kyoungwan Kim, Emanuel Tutuc, and Sanjay K. Banerjee, SRC TECHCON, Austin, 2015
1011. Low Temperature Raman and Photoluminescence Measurements of MoS₂ Layer Grown by Chemical Vapor Deposition, Barbara Nichols, R. Ghosh, S.K. Banerjee, AVS, Oct. 2015, San Jose.
1012. Defects Compensation and Refining Optical Luminescence in Organic/Transition Metal Dichalcogenide Heterostructure, J.H. Park, , A.M. Sanne, H.C.P. Movva, S. Vishwanath, Il Jo Kwak, UC San Diego, H. Xing, J. Robertson, S.K. Banerjee, A.C. Kummel, AVS, Oct. 2015, San Jose.
1013. Dielectrics Layer Deposition on 2D Materials by Functionalization with Polar Titanyl Phthalocyanine, JunHong Park, S. Fathipour, I.J. Kwak, UC San Diego, H.C.P. Movva, UT-Austin, S. Vishwanath, H. Xing, S.K. Banerjee, A.C. Seabaugh, A.C. Kummel, AVS, Oct. 2015, San Jose.
1014. Substitutional Doping of Metal Contact for Monolayer Transition Metal Dichalcogenides: a Density Functional Theory Based Study, Amithraj Valsaraj, Leonard F. Register, J.Chang, and Sanjay K. Banerjee, SISPAD 2015.
1015. Theoretical Study of the Spontaneous Electron-Hole Exciton Condensates between n and p-type MoS₂ Monolayers, toward beyond CMOS Applications, Xian Wu, Xuehao Mou, Leonard F. Register, and Sanjay K. Banerjee, SISPAD, 2015.
1016. Impact of Gate Oxide Complex Band Structure on n-Channel III-V FinFETs, Dax M. Crum, Amithraj Valsaraj, Leonard F. Register, Sanjay K. Banerjee, Bhagawan Sahu, Zoran Krivakopic, Srinivasa Banna, Deepak Nayak, SISPAD 2015.
1017. Two Dimensional Electronics, **Invited Colloquium**, UT San Antonio, 2015.
1018. Two dimensional Transistor, **Invited Colloquium**, CREOL, Univ. Central Florida, 2015.
1019. Double Bilayer Graphene Negative Differential Resistance Vertical Interlayer Tunnel FET, Sangwoo Kang, Babak Fallahazad, Kayoung Lee, Hema C. P. Movva, Chris M. Corbet, Kyoungwan Kim, Takashi Taniguchi, Kenji Watanabe, Luigi Colombo, Leonard F. Register, Emanuel Tutuc, Sanjay K. Banerjee, SRC TECHCON, Austin, 2015, **BEST PAPER**.
1020. Semiconducting Behavior and Logic in Ultrathin Rhenium Disulfide Field Effect Transistors, Chris M. Corbet, Connor McClellan, Amritesh Rai, Sushant Sudam Sonde, Emanuel Tutuc, and Sanjay K. Banerjee, SRC TECHCON, Austin, 2015, **BEST PAPER**.
1021. Quantum transport simulation of exciton condensate transport physics in a double-layer graphene system, X Mou, LF Register, AH MacDonald, SK Banerjee, Physical Review B 92 (23), 235413, 2015.
1022. Large-Area Monolayer MoS₂ for Flexible Low-Power RF Nanoelectronics in the GHz Regime, HY Chang, MN Yogeesh, R Ghosh, A Rai, A Sanne, S Yang, N Lu, S.K. Banerjee, D.Akinwande, Advanced Materials, 2015.
1023. Theoretical and experimental investigation of vacancy-based doping of monolayer MoS₂ on oxide, A Valsaraj, J Chang, A Rai, LF Register, SK Banerjee, 2D Materials 2 (4), 045009, 2015.
1024. Damage free Ar ion plasma surface treatment on In_{0.53}Ga_{0.47}As-on-silicon metal-oxide-semiconductor device, D Koh, SH Shin, J Ahn, S Sonde, HM Kwon, T Orzali, DH Kim, TW Kim, ...SK Banerjee, Applied Physics Letters 107 (18), 183509, 2015.
1025. In Situ Observation of Initial Stage in Dielectric Growth and Deposition of Ultrahigh Nucleation Density Dielectric on Two-Dimensional Surfaces, JH Park, HCP Movva, E Chagarov,

- K Sardashti, H Chou, I Kwak, KT Hu, ...SK Banerjee, AC Kummel, Nano letters 15 (10), 6626-6633, 2015.
1026. High-Mobility Holes in Dual-Gated WSe₂ Field-Effect Transistors, HCP Movva, A Rai, S Kang, K Kim, B Fallahazad, T Taniguchi, ...E.Tutuc, SK. Banerjee, ACS Nano 9 (10), 10402-104101, 2015.
 1027. High-mobility surface states and conductance fluctuations in Bismuth Telluro-Sulfide topological insulator devices TANUJ TRIVEDI, SUSHANT SONDE, HEMA C. P. MOVVA, SANJAY K. BANERJEE, APS March Meeting, Baltimore, 2016.
 1028. Shubnikov-de Haas oscillations of high mobility holes in monolayer and bilayer WSe₂: spin-valley locking, effective mass, and inter-layer coupling, BABAK FALLAHAZAD, HEMA CHANDRA PRAKASH MOVVA, KYOUNGHWAN KIM, STEFANO LARENTIS, TAKASHI TANIGUCHI, KENJI WATANABE, SANJAY K. BANERJEE, EMANUEL TUTUC, APS March Meeting, Baltimore, 2016.
 1029. Effects of High-Energy X-Ray Radiation on MoS₂ FETs, AMRITESH RAI, LAXMAN THOUTAM, WEI ZHANG, KIRAN KOVI, SANJAY BANERJEE, SAPTARSHI DAS, APS March Meeting, Baltimore, 2016.
 1030. Effects of Strain on CVD-Grown Few-Layered Terrace Structures of MoS₂, AMBER MCCREARY, R. GHOSH, M. AMANI, J. WANG, K.-A. DUERLOO, A. SHARMA, K. JARVIS, E. REED, A. DONGARE, S.K. BANERJEE, Univ. of Texas at Austin, M. TERRONES, R. NAMBURU, M. DUBEY, APS March Meeting, Baltimore, 2016.
 1031. Beyond CMOS Devices in 2D Materials, **Invited Talk**, (with H.Movva, S.Kang, A.Sanne, R.Ghosh, B.Fallahzad, A.Rai, L.Colombo, D. Akinwande, E.Tutuc, L.F.Register), Mat. Res. Soc. Conf. Phoenix, April 2016.
 1032. Electronics and Spintronics in Flatland, **Invited**, 2D Workshop Brookhaven National Lab, April 2016.
 1033. Enhancing the performance of Ge pFETs using novel BF₂⁺ implantation, W.Hsu...S.K.Banerjee, DRC 2016.
 1034. Electronics and Spintronics in Flatland, **Invited**, 2D Workshop Argonne National Lab, June, 2016.
 1035. Enhancing the performance of Ge pFETs using novel BF₂⁺ implantation, W.Hsu...S.K.Banerjee, ISTDM Japan 2016.
 1036. Insights into ITFET performance improvement, S.Kang,...LF Register, E.Tutuc, SK Banerjee, DRC 2016.
 1037. RT gate tunable NDR in MoS₂/hBN/WSe₂ heterostructure, H.Movva...E.Tutuc, SK Banerjee. DRC 2016.
 1038. Towards wafer scale monolayer MoS₂ flexible low power RF for IoT, M.Yogeesh,...SK Banerjee, D.Akinwande, DRC 2016.
 1039. Performance Factors for Interlayer Tunnel FETs Based on Layered Graphene-Hexagonal Boron Nitride-Graphene Heterostructures, Sangwoo Kang, Nitin Prasad, Hema C. P. Movva, Amritesh Rai, Kyoungwan Kim, Takashi Taniguchi, Kenji Watanabe, Leonard F. Register, Emanuel Tutuc, Sanjay K. Banerjee, SRC Techcon 2016.
 1040. Weak antilocalization, electron-electron interaction and universal conductance fluctuations in van der Waals epitaxially grown Bismuth Telluro-Sulfide topological insulators, Tanuj Trivedi, Sushant Sonde, Hema C. P. Movva, Anupam Roy, and Sanjay K. Banerjee, SRC Techcon **BEST PAPER**, 2016.
 1041. Improved Contact Resistance and Current Saturation in ReSe₂ Field Effect Transistors, Omar Mohammed, Amritesh Rai, Chris M. Corbet, Emanuel Tutuc, and Sanjay K. Banerjee, SRC Techcon 2016.
 1042. 2D Materials Synthesis and Graphene Thinned Silicon Solar Cell, Harry Chou, Jae Hyun Ahn, Rudresh Ghosh, Ariel Ismach, Xiaohan Wang, Yufeng Hao, Luigi Colombo, Rodney Ruoff, and Sanjay K. Banerjee, SRC Techcon 2016.

1043. Full-Band Simulations of Single-Particle Resonant Tunneling in Transition Metal Dichalcogenide-Based Interlayer Tunneling Field-Effect Transistors, Xian Wu, Xuehao Mou, Leonard F. Register, and Sanjay K. Banerjee, SRC Techcon 2016.
1044. Write error rate of spin-transfer-torque random access memory including micromagnetic effects using rare event enhancement, Tanmoy Pramanik, Urmimala Roy, Leonard F. Register, and Sanjay K. Banerjee, SRC Techcon 2016.
1045. Air-stable ambipolar hBN encapsulated monolayer MoTe₂ field effect transistors, S. Larentis, A. Rai, K. Kim, T. Taniguchi, K. Watanabe, S.K. Banerjee, and E. Tutuc, SRC Techcon 2016.
1046. Effect of Rotational Misalignment on Interlayer Coupling in a Graphene/hBN/Graphene van der Waal's heterostructure, Amithraj Valsaraj, Leonard F. Register and Sanjay K. Banerjee, SISPAD 2016.
1047. Full-Band Simulations of Single-Particle Resonant Tunneling in Transition Metal Dichalcogenide-Based Interlayer Tunneling Field-Effect Transistors, Xian Wu, Xuehao Mou, Leonard F. Register, and Sanjay K. Banerjee, SISPAD 2016.
1048. Phthalocyanine Monolayer Nucleation of Gate Oxide ALD on Single Layer Graphene and TMD Surfaces, Andrew Kummel, Jun Hong Park, Iljo Kwak, Evgeniy Chagarov, Hema Movva, Harry Chou, Sanjay K Banerjee, Sara Fathipour, Alan Seabaugh, Susan Fullerton, Suresh Vishwanath, Huili Grace Xing and Pabitra Choudhury, ECS Invited talk, 2016
1049. Performance and carrier transport analysis of In_{0.7}Ga_{0.3}As quantum-well MOSFETs with Al₂O₃/HfO₂ gate stack, SW Son, JH Park, JM Baek, JS Kim, DK Kim, SH Shin, SK Banerjee, ...Solid-State Electronics 123, 63-67, 2016
1050. High Phosphorous Dopant Activation in Germanium Using Laser Spike Annealing, W Hsu, X Wang, F Wen, Y Wang, A Dolocan, T Kim, E Tutuc, S Banerjee, IEEE Electron Device Letters, vol. 9, 2016.
1051. First-principles simulation of oxygen vacancy migration in HfO_x, CeO_x, and at their interfaces for applications in resistive random-access memories, AA Bhatti, CC Hsieh, A Roy, LF Register, SK Banerjee, Journal of Computational Electronics 15 (3), 741-748, 2016
1052. The Positive Effects of Hydrophobic Fluoropolymers on the Electrical Properties of MoS₂ Transistors, S Rahimi, R Ghosh, S Kim, A Dodabalapur, S Banerjee, D Akinwande, Applied Sciences 6 (9), 236, 2016
1053. Novel BF Implantation for High Performance Ge pMOSFETs, W Hsu, T Kim, H Chou, A Rai, SK Banerjee, IEEE Electron Device Lett., vol. 37, no. 8, p. 954, 2016
1054. Effects of Electrode Layer Band Structure on the Performance of Multilayer Graphene-hBN-Graphene Interlayer Tunnel Field Effect Transistors, S Kang, N Prasad, HCP Movva, A Rai, K Kim, X Mou, T Taniguchi, ...LF Register, E. Tutuc, SK Banerjee, Nano Letters 16 (8), 4975-4981, 2016
1055. Diffusion and recrystallization of B implanted in crystalline and pre-amorphized Ge in the presence of F, W Hsu, T Kim, A Benítez-Lara, H Chou, A Dolocan, A Rai, ...SK Banerjee, Journal of Applied Physics 120 (1), 015701, 2016
1056. High-frequency characteristics of L_g= 60 nm InGaAs MOS high-electron-mobility-transistor (MOS-HEMT) with Al₂O₃ gate insulator, TW Kim, JS Kim, DK Kim, SH Shin, WS Park, S Banerjee, DH Kim, Electronics Letters 52 (10), 870-872, 2016
1057. Improved contact resistance in ReSe₂ thin film field-effect transistors, CM Corbet, SS Sonde, E Tutuc, SK Banerjee, Applied Physics Letters 108 (16), 162104, 2016.
1058. Large area chemical vapor deposition growth of monolayer MoSe₂ and its controlled sulfurization to MoS₂, R Ghosh, JS Kim, A Roy, H Chou, M Vu, SK Banerjee, D Akinwande, Journal of Materials Research 31 (07), 917-922, 2016
1059. Black Phosphorous Thin-Film Transistor and RF Circuit Applications, SF Chowdhury, MN Yogeesh, SK Banerjee, D Akinwande, IEEE Electron Device Letters 37 (4), 449-451, 2016

1060. Write error rate of spin-transfer-torque random access memory including micromagnetic effects using rare event enhancement, U Roy, T Pramanik, LF Register, SK Banerjee, JAP, 2016
1061. Influence of electron-beam lithography exposure current level on the transport characteristics of graphene field effect transistors, S Kang, HCP Movva, A Sanne, A Rai, SK Banerjee, Journal of Applied Physics 119 (12), 124502, 2016
1062. Large magnetoresistance at room temperature in ferromagnet/topological insulator contacts, S Majumder, S Guchhait, R Dey, L Register, SK Banerjee, IEEE Trans. Nanotech, 2016
1063. Nanoscale doping of compound semiconductors by solid phase dopant diffusion, J Ahn, H Chou, D Koh, T Kim, A Roy, J Song, SK Banerjee, Applied Physics Letters 108 (12), 122107, 2016
1064. Voltage-Controlled Low-Energy Switching of Nanomagnets through Ruderman-Kittel-Kasuya-Yosida Interactions for Magnetoelectric Device Applications, B Ghosh, R Dey, LF Register, SK Banerjee, J. Appl. Phys. July 2016
1065. Nuclear reaction analysis for H, Li, Be, B, C, N, O and F with an RBS check, WA Lanford, M Parenti, BJ Nordell, MM Paquette, AN Caruso, ...SK,..Banerjee, Nuclear Instruments and Methods in Physics Research Section B: Beam, 2016
1066. Structural and Electrical Properties of MoTe₂ and MoSe₂ Grown by Molecular Beam Epitaxy, A Roy, HCP Movva, B Satpati, K Kim, R Dey, A Rai, T Pramanik, ...SK.Banerjee, ACS applied materials & interfaces 8 (11), 7396-7402, 2016
1067. Shubnikov–de Haas Oscillations of High-Mobility Holes in Monolayer and Bilayer WSe₂: Landau Level Degeneracy, Effective Mass, and Negative Compressibility, B Fallahazad, HCP Movva, K Kim, S Larentis, T Taniguchi, K Watanabe, ...SK Banerjee, E.Tutuc, Physical review letters 116 (8), 086601, 2016
1068. Effects of Uniaxial and Biaxial Strain on Few-Layered Terrace Structures of MoS₂ Grown by Vapor Transport, A McCreary, R Ghosh, M Amani, J Wang, KAN Duerloo, A Sharma, ...SK Banerjee.. M.Dubey, ACS nano 10 (3), 3186-3197, 2016
1069. Van der Waals heterostructures with high accuracy rotational alignment, K Kim, M Yankowitz, B Fallahazad, S Kang, HCP Movva, S Huang, ..SK.Banerjee, E.Tutuc, Nano letters 16 (3), 1989-1995, 2016
1070. Weak antilocalization and universal conductance fluctuations in bismuth telluro-sulfide topological insulators, T Trivedi, S Sonde, HCP Movva, SK Banerjee, Journal of Applied Physics 119 (5), 055706, 2016.
1071. Novel Characterization Techniques for 2D Materials: Visualizing Inherent and External Defects, Rudresh Ghosh, S.K. Banerjee, D. Akinwande, University of Texas at Austin, Nashville, AVS, Nov. 2016
1072. Passivation of Transition Metal Chalcogenide Surface via Sulfur Layer to Enhanced Metal Contact, JunHong Park, A. Rai, I.J. Kwak, S. Bhattacharjee, K. Ganapathi, N. Bhat, S.K. Banerjee, A.C. Kummel, Nashville, AVS, Nov. 2016
1073. X. Mou, X. Wu, L. F. Register and S. K. Banerjee, “Device application of exciton condensates in layered materials,” *EMN meeting on quantum 2016*, Phuket, Thailand, April 2016. (Invited talk)
1074. Methods for modeling non-equilibrium degenerate statistics and quantum-confined scattering in 3D ensemble Monte Carlo transport simulations, DM Crum, A Valsaraj, JK David, LF Register, SK Banerjee, Journal of Applied Physics 120 (22), 224301, 2016
1075. Multi-barrier inter-layer tunnel field-effect transistor, N Prasad, X Mou, LF Register, SK Banerjee, Electron Devices Meeting (IEDM), 2016
1076. A sub-1-volt analog metal oxide memristive-based synaptic device with large conductance change for energy-efficient spike-based computing systems, CC Hsieh, A Roy, YF Chang, D Shahrjerdi, SK Banerjee, Applied Physics Letters 109 (22), 223501, 2016

1077. Localization and interaction effects of epitaxial Bi₂Se₃ bulk states in two-dimensional limit, R Dey, A Roy, T Pramanik, S Guchhait, S Sonde, A Rai, LF Register, SK Banerjee, Journal of Applied Physics 120 (16), 164301, 2016
1078. DFT simulations of inter-graphene-layer coupling with rotationally misaligned hBN tunnel barriers in graphene/hBN/graphene tunnel FETs, A Valsaraj, LF Register, E Tutuc, SK Banerjee, Journal of Applied Physics 120 (13), 134310, 2016
1079. Write error rate of spin-transfer-torque random access memory including micromagnetic effects using rare event enhancement, U Roy, T Pramanik, LF Register, SK Banerjee, IEEE Transactions on Magnetics 52 (10), 1-6, 2016
1080. Performance and carrier transport analysis of In 0.7 Ga 0.3 As quantum-well MOSFETs with Al₂O₃/HfO₂ gate stack, SW Son, JH Park, JM Baek, JS Kim, DK Kim, SH Shin, SK Banerjee,..., Solid-State Electronics 123, 63-67, 2016
1081. Effect of rotational misalignment on interlayer coupling in a graphene/hBN/graphene van der Waal's heterostructure, A Valsaraj, LF Register, SK Banerjee, Simulation of Semiconductor Processes and Devices (SISPAD), 2016 .
1082. Full-band simulations of single-particle resonant tunneling in transition metal dichalcogenide-based interlayer tunneling field-effect transistors, X Wu, X Mou, LF Register, SK Banerjee, Simulation of Semiconductor Processes and Devices (SISPAD), 2016
1083. Probing Anisotropy in Transition-Metal Dichalcogenides using Polarized Raman Spectroscopy, J. HARDING, J. R. SIMPSON, J.-A. YAN, A. MCCREARY, M. TERRONES, D. RHODES, L. BALICAS, R. GHOSH, S. BANERJEE, A. R. HIGHT WALKER, APS March Meeting, 2017
1084. Nano-imaging of Electrical Properties of MoSe₂/WSe₂ Vertical Heterostructures, Di Wu, Wei Li, Maruthi Yogeesh, Amrithesh Rai, Sanjay Banerjee, Deji Akinwande, Keji Lai, APS March Meeting, 2017
1085. Large Area CVD MoS₂ RF transistors with GHz performance, MARUTHI NAGAVALLI YOGESH, ATRESH SANNE, SAUNGEUN PARK, DEJI AKINWADE, SANJAY BANERJEE, APS March Meeting, 2017
1086. Dual-Gated MoTe₂/MoS₂ van der Waals Heterojunction p-n Diode, AMRITESH RAI, HEMA C. P. MOVVA, SANGWOO KANG, STEFANO LARENTIS, ANUPAM ROY, EMANUEL TUTUC, SANJAY K. BANERJEE, APS March Meeting, 2017
1087. Carrier Density Dependent Quantum Hall States Sequence of Holes in WSe₂, HEMA C. P. MOVVA, BABAK FALLAHAZAD, KYOUNGHWAN KIM, STEFANO LARENTIS, TAKASHI TANIGUCHI, KENJI WATANABE, SANJAY K. BANERJEE, EMANUEL TUTUC, APS March Meeting, 2017
1088. Spin Hall Effect in Monolayer and Bilayer WSe₂, BABAK FALLAHAZAD, HEMA C. P. MOVVA, XIAO LI, TAKASHI TANIGUCHI, KENJI WATANABE, QIAN NIU, SANJAY K. BANERJEE, EMANUEL TUTUC, APS March Meeting, 2017
1089. Extraction of interlayer coupling and hopping potentials with misaligned hBN tunnel barriers in graphene/hBN/graphene tunnel FETs, AMITHRAJ VALSARAJ, LEONARD F. REGISTER, SANJAY K. BANERJEE, APS March Meeting, 2017
1090. Ultrathin, wafer-scale hexagonal boron nitride on dielectric surfaces by diffusion and segregation mechanism, S Sonde, A Dolocan, N Lu, C Corbet, MJ Kim, E Tutuc, SK Banerjee, L.Colombo, 2D Materials 4 (2), 025052, 2017
1091. Detection of current induced spin polarization in epitaxial Bi₂Te₃ thin film, R Dey, A Roy, T Pramanik, A Rai, S Heon Shin, S Majumder, LF Register, SK Banerjee, Applied Physics Letters 110 (12), 122403, 2017
1092. Laser Spike Annealing for Shallow Junctions in Ge CMOS, W Hsu, F Wen, X Wang, Y Wang, A Dolocan, A Roy, T Kim, E Tutuc, SK Banerjee, IEEE Transactions on Electron Devices 64 (2), 346-352, 2017

1093. A simulation study of voltage-assisted low-energy switching of a perpendicular anisotropy ferromagnet on a topological insulator, B Ghosh, R Dey, LF Register, SK Banerjee, Journal of Computational Electronics, 1-7, 2017
1094. Novel Memory and Logic Devices in Graphene, S. Banerjee, F.Register, E.Tutuc, D.Akinwande and L.Colombo, European Graphene Workshop, Barcelona Spain, March 2017 **(Invited)**
1095. S.K. Banerjee, Electronics in Flatland, ECS, New Orleans may 2017 **(Invited)**
1096. Intra-Domain Periodic Defects in Monolayer MoS₂, A Roy, R Ghosh, A Rai, A Sanne, K Kim, HCP Movva, R Dey, T Pramanik, ...E.Tutuc, S.K.Banerjee Appl. Phys. Lett. (2017)
1097. Graphene-Al₂O₃-silicon heterojunction solar cells on flexible silicon substrates, J Ahn, H Chou, SK Banerjee, Journal of Applied Physics 121 (16), 163105, 2017
1098. Reconfigurable Complementary Monolayer MoTe₂ Field-Effect Transistors for Integrated Circuits, S Larentis, B Fallahazad, HCP Movva, K Kim, A Rai, T Taniguchi, S.Banerjee, E.Tutuc, ACS Nano 2017.
1099. Embedded gate CVD MoS₂ microwave FETs, Atresh Sanne, Saungeun Park, Rudresh Ghosh, Maruthi Nagavalli Yogeesh, Chison Liu, Leo Mathew, Rajesh Rao, Deji Akinwande, Sanjay Kumar Banerjee. npj 2D Materials and Applications, Nature Publishing Group, Volume 1, Issue 1, Pages 26, 2017/8/10
1100. Out-of-Plane Electromechanical Response of Monolayer Molybdenum Disulfide Measured by Piezoresponse Force Microscopy, Christopher J. Brennan, Rudresh Ghosh, Kalhan Koul, Sanjay K. Banerjee, Nanshu Lu, and Edward T. Yu, *Nano Letters* **Article ASAP**, DOI: 10.1021/acs.nanolett.7b02123, 2017
1101. Versatile Large-Area Custom-Feature van der Waals Epitaxy of Topological Insulators, Tanuj Trivedi, Anupam Roy, Hema C. P. Movva, Emily S. Walker, Seth R. Bank, Dean P. Neikirk, and Sanjay K. Banerjee, *ACS Nano* **2017** 11 (7), 7457-7467, DOI: 10.1021/acsnano.7b03894, July 2017
1102. Interfacial reactions at Fe/topological insulator spin contacts, Sarmita Majumder, Karalee Jarvis, Sanjay K Banerjee, Karen L Kavanagh, Journal of Vacuum Science & Technology B, Nanotechnology and Microelectronics: Materials, Processing, Measurement, and Phenomen, Volume 35(4), 2017
1103. Experimental evidence of exciton capture by mid-gap defects in CVD grown monolayer MoSe₂, Ke Chen, Rudresh Ghosh, Xianghai Meng, Anupam Roy, Joon-Seok Kim, Feng He, Sarah C Mason, Xiaochuan Xu, Jung-Fu Lin, Deji Akinwande, Sanjay K Banerjee, Yaguo Wang, npj 2D Materials and Applications, Vol. 1(1), 2017.
1104. A.Sanne, Maruthi Yogesh, D.Akinwande, SK Banerjee, Sanjay Banerjee, E-mode RF transistors and circuit model using CVD MoS₂, Device Research Conf 2017
1105. Machine learning for variability aware statistical device design: The case of perpendicular spin-transfer-torque random access memory, Urmimala Roy, Tanmoy Pramanik, Subhendu Roy, Leonard F Register, Sanjay K Banerjee, Device Research Conference, 2017
1106. A universal model for interface-type threshold switching phenomena by comprehensive study of Vanadium oxide-based selector, Chih-Yang Lin, Ying-Chen Chen, Meiqi Guo, Chih-Hung Pan, Fu-Yuan Jin, Yi-Ting Tseng, Cheng Chih Hsieh, Xiaohan Wu, Min-Chen Chen, Yao-Feng Chang, Fei Zhou, Burt Fowler, Kuan-Chang Chang, Tsung-Ming Tsai, Ting-Chang Chang, Yonggang Zhao, Simon M Sze, Sanjay Banerjee, Jack C Lee, International Symposium on VLSI Technology, Systems and Application (VLSI-TSA), 2017
1107. Tip-enhanced Raman spectroscopy of semiconductor nanostructures, Zhongjian Zhang, David Dillen, Christopher J Brennan, Alex De Palma, Gabriel Cossio, Rudresh Ghosh, Sanjay Banerjee, Emanuel Tutuc, T Yu Edward, Nanoimaging and Nanospectroscopy V, Proc. SPIE, Vol.10350, Pages 103500A

1108. Enhancement-Mode RF Transistors and Circuit 1 Model using CVD MoS₂, Atresh Sanne, Saungeun Park, Rudresh Ghosh, Maruthi Nagavalli Yogeesh, Chison Liu, Leo Mathew, Rajesh Rao, Deji Akinwande, and Sanjay Kumar Banerjee, SRC Techcon 2017
1109. MoTe₂/MoS₂ Van der Waals Heterostructures with Diode-like Behavior and Negative Differential Transconductance, Amritesh Rai, Hema C.P. Movva, Sangwoo Kang, Stefano Larentis, Anupam Roy, Emanuel Tutuc, and Sanjay K. Banerjee, SRC Techcon 2017
1110. Micromagnetic effects on the write error rates of in-plane and perpendicular spin-transfer-torque random access memory via rare-event-enhanced simulation, Tanmoy Pramanik, Leonard F. Register, and Sanjay K. Banerjee, SRC Techcon 2017
1111. Interlayer Tunnel Field-Effect Transistors (ITFETs): Physics, Fabrication and Applications, Sangwoo Kang, Xuehao Mou, Babak Fallahazad, Nitin Prasad, Xian Wu, Amithraj Valsaraj, Hema CP Movva, Kyoungwhan Kim, Emanuel Tutuc, Leonard F Register, Sanjay K Banerjee, Journal of Physics D: Applied Physics, Vol. 50(38), 2017
1112. Angular dependence of magnetization reversal in epitaxial chromium telluride thin films with perpendicular magnetic anisotropy, T Pramanik, A Roy, R Dey, A Rai, S Guchhait, HCP Movva, CC Hsieh, S.K.Banerjee, Journal of Magnetism and Magnetic Materials, 2017
1113. ReS₂-based interlayer tunnel field effect transistor, OB Mohammed, HCP Movva, N Prasad, A Valsaraj, S Kang, CM Corbet, L.F.Register, E.Tutuc, S.K. Banerjee, Journal of Applied Physics 122 (24), 245701, 2017
1114. Bilayer Pseudospin Junction Transistor (BiSJT) for “Beyond-CMOS” Logic, X Mou, LF Register, AH MacDonald, SK Banerjee, IEEE Transactions on Electron Devices 64 (11), 4759-4762, 2017
1115. Carrier Trapping by Oxygen Impurities in Molybdenum Diselenide, K Chen, A Roy, A Rai, A Valsaraj, X Meng, F He, X Xu, LF Register, SK Banerjee, Y.Wang, ACS applied materials & interfaces, 2017
1116. Short-Term Relaxation in HfO_x/CeO_x Resistive Random Access Memory With Selector, Cheng-Chih Hsieh, Yao-Feng Chang, Yoocham Jeon, Anupam Roy, Davood Shahrjerdi, and Sanjay K. Banerjee, EDL v. 38(7) 2017.
1117. Defect passivation of transition metal dichalcogenides via a charge transfer van der Waals interface, JH Park, A Sanne, Y Guo, M Amani, K Zhang, HCP Movva, JA Robinson, Ali Javey, John Robertson, Sanjay K Banerjee, Andrew C Kummel Science Advances 3 (10), e1701661, 2017
1118. Resonant Interlayer Tunneling in 2D Van Der Waals-Materials-Based Channel-Dielectric-Channel Systems and Possible Device and Circuit Applications, Leonard F Register, G William Burg, Chris M Corbet, Babak Fallahazad, Sangwoo Kang, Kyoungwhan Kim, Stefano Larentis, Kayoung Lee, Omar Mohammed, Xuehao Mou, Hema CP Movva, Nitin Prasad, Dharmendar Reddy, Amithraj Valsaraj, Xian Wu, Sanjay K Banerjee, Emanuel Tutuc, Nishtha Sharma, Qingxiao Wang, Moon Kim, Andrew Marshall, Jiamin Xue, Takashi Taniguchi, Kenji Watanabe, Luigi Colombo, ECS **Invited talk**, 2017
1119. Electronics in Flatland, SK. Banerjee (with L.F. Register, E.Tutuc, D. Akinwande), APS March Meeting 2018 **Invited Talk**.
1120. Tunable Γ -K Valley Populations in Hole-Doped Trilayer WSe₂, H.Movva, .. SBanerjee, A.MacDonald, E.Tutuc, APS March Meeting 2018
1121. Shubnikov-de Haas Oscillations in bilayer MoSe₂: effective mass and carrier density dependent quantum Hall states sequence, S.Larentis, ... S.Banerjee, E.Tutuc, APS March Meeting 2018
1122. Carrier Trapping by Oxygen Impurities in Molybdenum Diselenide, A.Rai, K.Chen, .. S.Banerjee, Y.Wang, APS March Meeting 2018
1123. Electronics in Flatland, SK. Banerjee (with L.F. Register, E.Tutuc, D. Akinwande), Korean Semiconductor Society, **Plenary Talk (2018)**.

1124. Towards band structure and band offset engineering of monolayer Mo (1-x) W (x) S₂ via Strain, Joon-Seok Kim, Rafia Ahmad, Tribhuwan Pandey, Amritesh Rai, Simin Feng, Jing Yang, Zhong Lin, Mauricio Terrones, Sanjay K Banerjee, Abhishek K Singh, Deji Akinwande, Jung-Fu Lin, 2D Materials 5 (1), 015008, 2018.
1125. Probing nanoscale variations in strain and band structure of Mo S₂ on Au nanopylramids using tip-enhanced Raman spectroscopy, Z Zhang, AC De Palma, CJ Brennan, G Cossio, R Ghosh, SK Banerjee, E.Yu, Physical Review B 97 (8), 085305, 2018
1126. Conversion of spin current into charge current in a topological insulator: Role of the interface, R Dey, N Prasad, LF Register, SK Banerjee, Physical Review B 97 (17), 174406, 2018
1127. Accelerated carrier recombination by grain boundary/edge defects in MBE grown transition metal dichalcogenides, K Chen, A Roy, A Rai, HCP Movva, X Meng, F He, SK Banerjee, Y Wang, APL Materials 6 (5), 056103, 2018
1128. Highly improved passivation of c-Si surfaces using a gradient i-a-Si: H layer, S Lee, J Ahn, L Mathew, R Rao, Z Zhang, JH Kim, SK Banerjee, ET Yu, Journal of Applied Physics 123 (16), 163101, 2018
1129. Tunable Gamma-K Valley Populations in Hole-Doped Trilayer WSe₂, Hema CP Movva, Timothy Lovorn, Babak Fallahazad, Stefano Larentis, Kyoungwan Kim, Takashi Taniguchi, Kenji Watanabe, Sanjay K Banerjee, Allan H MacDonald, Emanuel Tutuc, PHYSICAL REVIEW LETTERS, Vol: 120 (10) MAR 2018
1130. Modeling all-electrical detection of the inverse Edelstein effect by spin-polarized tunneling in a topological-insulator/ferromagnetic-metal heterostructure, R Dey, LF Register, SK Banerjee, Physical Review B 97 (14), 144417, 2018
1131. Scaling Challenges in Solid State Memory, S.Banerjee, SPIE Workshop, San Jose 2018, **Invited.**
1132. Large effective mass and interaction-enhanced Zeeman splitting of K-valley electrons in MoSe₂, S Larentis, HCP Movva, B Fallahazad, K Kim, A Behroozi, T Taniguchi, Sanjay K Banerjee, Emanuel Tutuc, Physical Review B 97 (20), 201407, 2018.
1133. Towards mm-wave nanoelectronics and RF switches using MoS₂ 2D Semiconductor, M Kim, S Park, A Sanne, SK Banerjee, D Akinwande, 2018 IEEE/MTT-S International Microwave Symposium-IMS, 352-354
1134. Enhanced P-Type Behavior in 2D WSe₂ via Chemical Defect Engineering, A Rai, JH Park, C Zhang, I Kwak, S Wolf, S Vishwanath, X Lin, J Furdyna, A.Kummel, S.K.Banerjee, 2018 76th Device Research Conference (DRC), 1-2.
1135. Progress in Contact, Doping and Mobility Engineering of MoS₂: An Atomically Thin 2D Semiconductor, A Rai, H Movva, A Roy, D Taneja, S Chowdhury, S Banerjee, Crystals 8 (8), 316 (2018).
1136. Write error rates of in-plane spin-transfer-torque random access memory calculated from rare-event enhanced micromagnetic simulations, T Pramanik, U Roy, P Jadaun, LF Register, SK Banerjee, Journal of Magnetism and Magnetic Materials 467, 96-107 (2018).
1137. Enhancement of Resonance by the Use of Multiple Tunnel Barriers in Bilayer Graphene-Based Interlayer Tunnel Field Effect Transistors, N Prasad, SK Banerjee, LF Register, SISPAD 2018.
1138. Memory and Logic soft error improvement using phase transition material assisted transistors, N.Teja, A.Rai, SK Banerjee, J.Kulkarni, ICEE India 2018.
1139. Growth of lateral graphene/h-BN heterostructure on copper foils by chemical vapor deposition, Pingping Zhuang, Weiye Lin, Harry Chou, Anupam Roy, Weiwei Cai, Sanjay K Banerjee, Nanotechnology, Vol. 30(3),Pg. 03LT01, (2018).
1140. Dependence of h-BN Film Thickness as Grown on Nickel Single-Crystal Substrates of Different Orientations, Harry Chou, Sarmita Majumder, Anupam Roy, Massimo Catalano, Pingping Zhuang, Manuel Quevedo-Lopez, Luigi Colombo, Sanjay K Banerjee, ACS applied materials & interfaces, vol.10(51), pg. 44862-44870 (2018)

1141. Moir'e Excitons in Van der Waals Heterostructures, Kha Tran, Galan Moody, Fengcheng Wu, Xiaobo Lu, Junho Choi, Akshay Singh, Jacob Embley, André Zepeda, Marshall Campbell, Kyoungwan Kim, Amritesh Rai, Travis Autry, Daniel A Sanchez, Takashi Taniguchi, Kenji Watanabe, Nanshu Lu, Sanjay K Banerjee, Emanuel Tutuc, Li Yang, Allan H MacDonald, Kevin L Silverman, Xiaoqin Li, APS March Meeting 2019.
1142. Two dimensional SrTiO₃ membranes, W Guo, A Posadas, A Demkov, A Roy, A Rai, S Banerjee, K Olsson, XE Li, Bulletin of the American Physical Society, 2019.
1143. Atomic layer deposition of cobalt oxide on oxide substrates and low temperature reduction to form ultrathin cobalt metal films, Zizhuo Zhang, Himamshu C Nallan, Brennan M Coffey, Thong Q Ngo, Tanmoy Pramanik, Sanjay K Banerjee, John G EkerdtJournal of Vacuum Science & Technology A **37**, 010903 (2019)
1144. Stress-induced bandgap renormalization in atomic crystals, Zheng Sun, Jonathan Beaumariage, Hema CP Movva, Sayema Chowdhury, Anupam Roy, Sanjay K Banerjee, David W Snoke, Solid State Communications, Vol 288, pg. 18-21, Feb. 2019.
1145. Visualization of Local Conductance in MoS₂/WSe₂ Heterostructure Transistors, Di Wu, Wei Li, Amritesh Rai, Xiaoyu Wu, Hema CP Movva, Maruthi Nagavalli Yogeesh, Zhaodong Chu, Sanjay K Banerjee, Deji Akinwande, Keji Lai, Nano letters, Feb. 2019.
1146. Evidence for moiré excitons in van der Waals heterostructures, Kha Tran, Galan Moody, Fengcheng Wu, Xiaobo Lu, Junho Choi, Kyoungwan Kim, Amritesh Rai, Daniel A Sanchez, Jiamin Quan, Akshay Singh, Jacob Embley, André Zepeda, Marshall Campbell, Travis Autry, Takashi Taniguchi, Kenji Watanabe, Nanshu Lu, Sanjay K Banerjee, Kevin L Silverman, Suenne Kim, Emanuel Tutuc, Li Yang, Allan H MacDonald, Xiaoqin Li, Nature (Feb. 2019)
1147. Neural Network Assisted Compact Model for Accurate Characterization of Cycle-to-cycle Variations in 2-D h-BN based RRAM devices, Jacob N. Rohan, Pingping Zhuang, SS Teja Nibhanupudi, Sanjay K. Banerjee, Jaydeep P. Kulkarni, Dev. Res. Conf. 2019.
1148. Tunnel Barrier Thickness, Interlayer Rotational Alignment, and Top Gating Effects on ReS₂/hBN/ReS₂ Resonant Interlayer Tunnel Field Effect Transistors, Omar B. Mohammed, Leonard F. Register, and Sanjay K. Banerjee, Dev Res. Conf, 2019.
1149. Fractal Growth of WSe₂, Using Chemical Vapor Deposition Sayema Chowdhury, Anupam Roy and Sanjay Banerjee, Elec. Mat. Conf., 2019.
1150. Epitaxial Growth and Characterization of Chromium Selenide Thin-Films on c-Al₂O₃(0001) Anupam Roy, Tanmoy Pramanik, Rik Dey, Amritesh Rai and Sanjay Banerjee, Elec. Mat. Conf., 2019.
1151. Two Terminal Devices Based on Two-Dimensional Materials Omar B, Mohammed and Sanjay Banerjee, Elec. Mat. Conf., 2019
1152. Tunable Long-Term and Short-Term Memories in a Single Magnetic Tunnel Junction Based Synapse, Nitin Prasad, Tanmoy Pramanik, Sanjay K. Banerjee and Leonard F. Register, MMM Conf. , 2019.
1153. Simulation of exciton condensate-mediated quantum transport in a double-monolayer transition metal dichalcogenide system, X Wu, X Mou, LF Register, SK Banerjee, Physical Review B **99** (3), 035113, 2019
1154. Electron redistribution and energy transfer in graphene/MoS₂ heterostructure, W Lin, P Zhuang, H Chou, Y Gu, R Roberts, W Li, SK Banerjee, W Cai, D.Akinwande, Applied Physics Letters **114** (11), 113103, 2019.
1155. Thinnest Nonvolatile Memory Based on Monolayer h-BN, X Wu, R Ge, PA Chen, H Chou, Z Zhang, Y Zhang, S Banerjee, JC Lee, D. Akinwande, Advanced Materials **31** (15), 1806790, 2019
1156. Band Structure Engineering of Layered WSe₂ via One-Step Chemical Functionalization, JH Park, A Rai, J Hwang, C Zhang, I Kwak, SF Wolf, S Vishwanath, X Liu, SK. Banerjee, KJ. Cho, AC. Kummel, ACS Nano, 1377545-7555, 2019.

1157. Valence and conduction band offsets at beryllium oxide interfaces with silicon carbide and III-V nitrides, Donghyi Koh, Sanjay K Banerjee, Chris Locke, Stephen E Sadow, Justin Brockman, Markus Kuhn, Sean W King, *Journal of Vacuum Science & Technology B* **37**, 041206 (2019).
1158. Electronics in Flatland, (with Register, Tutuc, Akinwande), Graphene Week, Helsinki, Sept. 2019. **(Invited)**
1159. Electronics in Flatland, (with Register, Tutuc, Akinwande), American Vac. Society, Ohio, Sept. 2019. **(Invited)**
1160. Electronics in Flatland, (with Register, Tutuc, Akinwande), Int. Workshop on Physics of Semiconductors, Kolkata, India, Dec. 2019. **(Invited Plenary)**
1161. Tunable Long-Term and Short-Term Memories in a Single Magnetic Tunnel Junction Based Synapse, Nitin Prasad, Tanmoy Pramanik, Sanjay K. Banerjee and Leonard F. Register, MMM Conf. , 2019.
1162. Transition from 3D to 2D and fractal to compact domains of CVD Grown MoSe₂, Sayema Chowdhury, Anupam Roy, Sanjay K. Banerjee, Materials Res. Society Symposium, 2020.
1163. Solid Electrolytic Substrates for High Performance TMD Transistors and Circuits, Md. Hasibul Alam, Zifan Xu, Sayema Chowdhury, Zhanzhi Jiang, Deepyanti Taneja, Sanjay K. Banerjee, Keji Lai, Maria Helena Braga and Deji Akinwande, Device Research Conference, June 2020.
1164. Large-Area Metal Organic Chemical Vapor Deposition Growth of Few Layer MoS₂ and its Controlled Sulfurization to Monolayer MoS₂, Sayema Chowdhury, Anupam Roy, Chison Liu, Rudresh Ghosh, Harry Chou and Sanjay K. Banerjee, Electronics Materials Conference, June 2020.
1165. Electronics in Flatland, (with Register, Tutuc, Akinwande), Vebleo Conf. on Nanomaterials, Sweden, Aug. 2020 (Invited Keynote)
1166. Design of giant spin Hall effect in transition metal oxides, P. Jadaun, L. F. Register, S. K. Banerjee, *Materials Research Society (MRS)*, online, Nov 2020
1167. Design of giant DMI and its microscopic origin in magnetic bilayers, Priyamvada Jadaun, Leonard F Register, and Sanjay K Banerjee, MMM Conference, 2020.
1168. Stacking-Order-Driven Optical Properties and Carrier Dynamics in ReS₂, Y Zhou, N Maity, A Rai, R Juneja, X Meng, A Roy, Y Zhang, X Xu, JF Lin, S.K.Banerjee, A.Singh, Y.Wang, *Advanced Materials* 32 (22), 1908311, 2020.
1169. Contact Engineering of Layered MoS₂ via Chemically Dipping Treatments, S Bang, S Lee, A Rai, NT Duong, I Kawk, S Wolf, CH Chung, SK Banerjee, A.Kummel, M.Jeong, *Advanced Functional Materials* 30 (16), 2000250, 2020.
1170. Two-Dimensional to Three-Dimensional Growth of Transition Metal Diselenides by Chemical Vapor Deposition: Interplay between Fractal, Dendritic, and Compact Morphologies, S Chowdhury, A.Roy, I Bodemann, SK Banerjee. *ACS Applied Materials & Interfaces* 12 (13), 15885-15892, 2020
1171. Realizing both short-and long-term memory within a single magnetic tunnel junction based synapse, N Prasad, T Pramanik, SK Banerjee, LF Register, *Journal of Applied Physics* 127 (9), 093904, 2020.
1172. Structural and magnetic properties of molecular beam epitaxy grown chromium selenide thin films, A Roy, R Dey, T Pramanik, A Rai, R Schalip, S Majumder, S Guchhait, S.K.Banerjee, *Physical Review Materials* 4 (2), 025001, 2020.
1173. X-ray photoelectron spectroscopy investigation of the valence band offset at beryllium oxide-diamond interfaces, D Koh, T.Hudnall, C.Bielawski, SK Banerjee, J Brockman, M Kuhn, SW King, *Diamond and Related Materials* 101, 107647, 2020.
1174. Nonpolar Resistive Switching of Multilayer-hBN-Based Memories, P Zhuang, W Lin, J Ahn, M Catalano, H Chou, A Roy, M Quevedo-Lopez, L.Colombo, S.K.Banerjee, *Advanced Electronic Materials* 6 (1), 1900979, 2020

1175. Progress in Materials Development for the Rapid Efficiency Advancement of Perovskite Solar Cells, W.Chi and S.K. Banerjee, Small, May 2020.
1176. Lithium-Ion Electrolytic Substrates for sub-1V High-Performance TMD Transistors and Amplifiers, Md Hasibul Alam, Sayema Chowdhury, Zifan Zhu, Deepyanti Taneja, Keji Lai, Sanjay K. Banerjee, Maria Helena Braga, and Deji Akinwande, Nature Communications, 2020.
1177. The microscopic origin of DMI in magnetic bilayers and prediction of giant DMI in new bilayers, Priyamvada Jadaun, Leonard F Register, Sanjay K Banerjee, Nature Partner Journal Computation, 2020.
1178. Rational design principles for giant spin Hall effect in 5d-transition metal oxides, P Jadaun, LF Register, SK Banerjee, Proceedings of the National Academy of Sciences 117 (22), 11878-11886, 2020
1179. Quantum-Corrected Semiclassical Monte Carlo Study of Channel Scaling in Nanoscale Si, Ge, and $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ n-channel FinFETs, Aqyan A. Bhatti, Nupur Navlakha, Dax M. Crum, Sanjay K. Banerjee, and Leonard F. Register, IEEE Nanotechnology magazine, Dec. 2020.
1180. A Library of Atomically Thin 2D Materials Featuring the Conductive-Point Resistive Switching Phenomenon, R Ge, X Wu, L Liang, SM Hus, Y Gu, E Okogbue, H Chou, J Shi, Y Zhang, Sanjay K Banerjee, Yeonwoong Jung, Jack C Lee, Deji Akinwande, Advanced Materials, 2007792, 2020
1181. Two-dimensional transport model of spin-polarized tunneling in a topological-insulator/tunnel-barrier/ferromagnetic-metal heterostructure, R Dey, LF Register, SK Banerjee, Physical Review B 102 (14), 144414, 2020
1182. Optoelectronic mixing with high-frequency graphene transistors , MH Alam, Z Xu, S Chowdhury, Z Jiang, D Taneja, SK Banerjee, K Lai, D.Akinwande, NATURE COMMUNICATIONS, vol 11, 3203, 2020.
1183. Machine Learning Assisted Compact Modeling of Cycle-to-cycle Variations in 2-D h-BN based RRAM devices, Jacob N. Rohan, P. Zhuang, S. S. Teja Nibhanupudi, Sanjay K. Banerjee, and Jaydeep P. Kulkarni, Government Microelectronic Conference (GOMACTech), March 2020
1184. Stability Improvement of Perovskite Solar Cells by Compositional and Interfacial Engineering, W Chi, SK Banerjee, Chemistry of Materials, v.33 (2021).
1185. Machine Learning for Statistical Modeling: The Case of Perpendicular Spin-Transfer-Torque Random Access Memory, U Roy, T Pramanik, S Roy, A Chatterjee, LF Register, SK Banerjee, ACM Transactions on Design Automation of Electronic Systems (TODAES) 26 (2021).
1186. Two-Step Growth of Uniform Monolayer MoS_2 Nanosheets by Metal–Organic Chemical Vapor Deposition, S Chowdhury, A Roy, C Liu, MH Alam, R Ghosh, H Chou, D Akinwande, S.K. Banerjee, ACS Omega (2021).
1187. Centimeter-Scale MoS_2 on Solid Electrolyte Substrate by Sulfurization of Molybdenum Thin Film, MH Alam, SST Nibhanupudi, SK Banerjee, D Akinwande, IEEE 21st International Conference on Nanotechnology (NANO), 421-424, 2021.
1188. X-Ray Photoemission Investigation of the Beryllium Oxide Band Alignment with Magnesium Oxide and Estimates for Other Insulating and Conducting Oxides, D Koh, TW Hudnall, CW Bielawski, S Banerjee, J Brockman, M Kuhn, S.King, ECS Transactions 102 (3), 127, 2021.
1189. Development of perovskite solar cells by incorporating quantum dots, W Chi, SK Banerjee, Chemical Engineering Journal, 131588, 2021.
1190. Direct growth of MoS_2 on electrolytic substrate and realization of high-mobility transistors, MH Alam, S Chowdhury, A Roy, MH Braga, SK Banerjee, D Akinwande, Physical Review Materials 5 (5), 054003, 2021.
1191. Method to enhance resonant interlayer tunneling in bilayer-graphene systems, N Prasad, X Wu, SK Banerjee, LF Register, Journal of Computational Electronics, 1-6, 2021.
1192. Achieving Resistance against Moisture and Oxygen for Perovskite Solar Cells with High Efficiency and Stability, W Chi, SK Banerjee, Chemistry of Materials, 2021.

1193. Recent progress on measurement of spin–charge interconversion in topological insulators using ferromagnetic resonance, R Dey, A Roy, LF Register, SK Banerjee, APL Materials 9 (6), 060702, **Invited**, 2021.
1194. Method to enhance resonant interlayer tunneling in bilayer-graphene systems, N Prasad, X Wu, SK Banerjee, LF Register, Journal of Computational Electronics, 1-6, 2021.
1195. Development of perovskite solar cells by incorporating quantum dots, W Chi, SK Banerjee, Chemical Engineering Journal 426, 131588.
1196. Electronics and Spintronics in 2D, UC Berkeley invited talk, Sept, 2021
1197. Electronics in Flatland, IEEE EDS invited talk, Oct., 2021
1198. Electronics and Spintronics in 2D, Indian National Academy of Sciences, invited talk, Dec, 2021.
1199. Development of perovskite solar cells by incorporating quantum dots, W Chi, SK Banerjee, Chemical Engineering Journal 426, 131588.
1200. Room-Temperature Processed Lateral Trench-Metal–Insulator–Semiconductor Schottky Barrier Diodes with Amorphous Gallium Oxide (α -Ga₂O₃) Thin Films on Si, H Ebrahimi-Darkhaneh, Z Shamsi, MGR Banda, M Quevedo-Lopez, SK Banerjee, physica status solidi (a), 2022.
1201. Performance Improvement of Perovskite Solar Cells by Interactions between Nano-Sized Quantum Dots and Perovskite, W Chi, SK Banerjee, Advanced Functional Materials, 2022.
1202. Wafer-Scalable Single-Layer Amorphous Molybdenum Trioxide, MH Alam, S Chowdhury, A Roy, X Wu, R Ge, MA Rodder, J Chen, Y Lu, J.Lee, SK Banerjee, J. Warner, D.Akinwande, ACS nano 16 (3), 3756-3767, 2022.
1203. Application of perovskite quantum dots as an absorber in perovskite solar cells, W Chi, SK Banerjee, Angewandte Chemie 134 (9), 2022.
1204. High-quality crystal Mg-doped p-type Ga₂O₃ thin film by pulse laser deposition, Hadi Ebrahimi-Darkhaneh, Mahsa Shekarnoush, Josefina Arellano-Jimenez, Rodolfo Rodriguez, Luigi Colombo, Manuel Quevedo-Lopez, Sanjay K. Banerjee, J. Mat. Science 2022
1205. Emerging 2D Materials for Tunneling Field Effect Transistors, N.Navlkha, F.Register, S.Banerjee, IEEE Latin American Devices Conf. 2022
1206. Experimental demonstration of sub-nanosecond switching in 2D hexagonal Boron Nitride resistive memory devices, SS Teja Nibhanupudi, Dmitry Veksler, Anupam Roy, Matthew Coupin, Kevin C. Matthews, Jamie Warner, Gennadi Bersuker, Jaydeep P. Kulkarni and Sanjay K. Banerjee, IEEE Dev. Res. Conf, 2022.
1207. Millimeter-wave switch based on monolayer hexagonal boron nitride, Sung Jin Yang, Frank Makal, Paul Peterson, Jason Alikpala, Christopher J. Luth, Sanjay K. Banerjee, Andreas Roessler, and Deji Akinwande, IEEE Dev. Res. Conf, 2022.
1208. Molecular Beam Epitaxy Growth of Iron Selenide Thin Films on *c*-Al₂O₃(0001), Ryan Schalip, Anupam Roy, and Sanjay K. Banerjee, Elec. Mat. Conf. 2022.
1209. Role of Hydrogen in Suppressing Secondary Nucleation in Chemical Vapor-Deposited MoS₂, S. Chowdhury, A. Roy, ... F.Mangolini, D.Akinwande, SK. Banerjee, ACS APPLIED ELECTRONIC MATERIALS 4 (12) , pp.6133-6141, Dec. 2022.
1210. Phase-Field Modeling of Chemical Vapor-Deposited 2D MoSe₂ Domains with Varying Morphology for Electronic Devices and Catalytic Applications, A.Roy,

- T.Pramanik, S.Chowdhury, SK. Banerjee, ACS APPLIED NANO MATERIALS 5 (10) , pp.15488-15497, 2022.
1211. Engineering strategies for two-dimensional perovskite solar cells, W.Chi and SK. Banerjee, TRENDS IN CHEMISTRY 4 (11), Oct. 2022.
 1212. Plasmons in 2D Materials, R.Xue, ...S.Chowdhury, SK. Banerjee, D.Forrest, H.Deng, D.Snoke, APS March Meeting 2023.
 1213. Band Alignment in BP/MoS₂ heterostructure: Role of Charge Redistribution, Electric Field, Biaxial Strain, and Layer Engineering, N Navlakha, P Jadaun, LF Register, SK Banerjee, J. Elec. Materials, v.52, p.1474, 2023.
 1214. Comparison and integration of CuInGaSe and perovskite solar cells, W.Chi and SK. Banerjee, JOURNAL OF ENERGY CHEMISTRY 78 , pp.463-475, 2023
 1215. Reconfigurable Low-Voltage Hexagonal Boron Nitride Nonvolatile Switches for Millimeter-Wave Wireless Communications, S. Yang, M. Dahan, S.Teja, C.Luth, SK. Banerjee, M.Kim, A.Roessler, E.Yalon, D. Akinwande, NANO LETTERS , pp.1152-1158, Jan. 2023
 1216. Perovskite/Silicon Tandem Solar Cells: Choice of Bottom Devices and Recombination Layers, W.Chi, SK. Banerjee, K.Jayawardena, R.Silva and S.Seok, ACS Energy Lett. 8, 1535–1550, 2023.
 1217. Proximity effects of 2D antiferromagnets on superconductivity in exfoliated niobium disulfide, Matthew N. Disiena, Christopher Luth, S. S. Teja Nibhanupudi, Jatin V. Singh, A. Ansh, Sarmita Majumder, and Sanjay K. Banerjee, J. Appl. Phys, 134(2), July 2023.
 1218. Self-Compliant Threshold Switching Devices with High On/Off ratio by Control of Quantized Conductance in Ag Filaments, Moonkyu Song, Sangheon Lee, S. S. Teja Nibhanupudi, Jatin Vikram Singh, Matthew Disiena, Christopher J. Luth, Siyu Wu, Matthew J. Coupin, Jamie H. Warner, and Sanjay K. Banerjee, Nano Lett. 23(7), July 2023
 1219. Semi-classical Monte Carlo study of the impact of tensile strain on the performance limits of monolayer MoS₂ n-channel MOSFETs, A.Bhatti, AA; B.Archer, BT; N.Navlakha, L.Register, S. Banerjee , Nov 28 2023, JOURNAL OF APPLIED PHYSICS 134 (20).
 1220. Simultaneous Determination of Thermal Conductivity and Heat Capacity in Thin Films with Picosecond Transient Thermorefectance and Picosecond Laser Flash, Ye, ZF; Park, JH; .. S.K. Banerjee, Wang, YG, NANOSCALE AND MICROSCALE THERMOPHYSICAL ENGINEERING 27 (3-4) , pp.182-194, Oct. 2023.
 1221. Characteristics of the thermodynamics of Novel Ag-cation Controllable Diffusive Memristors for Artificial Neuron Application, Moonkyu Song, Sangheon Lee, and Sanjay K. Banerjee, Mat. Res. Symp, Nov. 2023.
 1222. Comparative Study of Steep Switching Devices for 1T Dynamic Memory, Nupur Navlakha Md Hasan Raza Ansari Leonard F. Register, Sanjay Banerjee, Latin America Device Research Conf., 2023.
 1223. Improvement of Power Consumption and Linearity of Integrate/Fire Characteristic using Diffusive Memristors with Defective Graphene for Artificial Neuron Application, Moonkyu Song, Sangheon Lee, S. S. Teja Nibhanupudi, Siyu Wu, and Sanjay K. Banerjee, Solid State Electronics, May 2024.
 1224. Ultra Fast Switching Memristors Based on Two-dimensional Materials , SS Teja Nibhanupudi, Anupam Roy, Dmitry Veksler, Matthew Coupin, Kevin C. Matthews, Matthew Disiena, Ansh, Jatin V Singh, Ioana R Gearba-Dolocan, Jamie Warner, Jaydeep

- P. Kulkarni, Gennadi Bersuker, Sanjay K. Banerjee, Nature Communications, March 2024.
1225. Microelectronics: The Beginning of the End or the End of the Beginning?" S.Banerjee, Plenary talk, Device Research Conference, June 2024.
 1226. 2-D Analytical Modeling of the Magnetic Tunnel Junctions Including Multidomain Effects: Predictive Insights and Design Optimization, Pandey, N; Chauhan, YS; Register LF; Banerjee, SK, IEEE TRANSACTIONS ON ELECTRON DEVICES, 71 (7) , pp.4347-4354, July 2024
 1227. Low-Temperature Synthesis of WSe₂ by the Selenization Process under Ultrahigh Vacuum for BEOL Compatible Reconfigurable Neurons, Nibhanupudi, SST; Roy, A; (...); Banerjee, SK, ACS APPLIED MATERIALS & INTERFACES, 16 (17) , pp.22326-22333, Apr 18 2024.
 1228. Ferroelectric proximity effects in two-dimensional FeSeTe. Matthew N. Disiena Christopher Luth, Reid Shattuck, Jatin V. Singh, and Sanjay K. Banerjee, Journal of Applied Physics, 136(3), 2024
 1229. Dynamics of Domains and its Impact on Gate Tunneling in CMOS-Compatible FeFETs, Nilesh Pandey, Yogesh Singh Chauhan, Leonard F. Register, and Sanjay K. Banerjee, IEEE Elec. Dev, Letters, VOL. 45, NO. 7, JULY 2024.
 1230. Multi-Domain Dynamics and Ultimate Scalability of CMOS-Compatible FeFETs, Nilesh Pandey, Yogesh Singh Chauhan, Leonard F. Register, and Sanjay K. Banerjee, IEEE Tran. Elec. Dev, VOL. 71, NO. 7, JULY 2024.
 1231. Estimating Heterogeneous|Homogeneous Junction Semiconductor Device Internal Electrical Properties (Charge Density, Potential, Electric Field ...) with In-house Technology Computed Aided Design (TCAD) Tools, A.Banerjee and S. Banerjee, International Conference on Nanostructures and Nanomaterials (ICN May 2024), Kottayam Kerala, India.
 1232. Impact of Multi-Domain on Ferroelectric Tunnel Junction Design Metrics, Nilesh Pandey, Yogesh S. Chauhan, Leonard F. Register, and Sanjay K. Banerjee, Device Res. Conf. June 2024.
 1233. Impact of Multi-Domain Microscopic Interactions on Magnetic Tunnel Junction's Static and Transient Characteristics, Nilesh Pandey, Yogesh S. Chauhan, Leonard F. Register, and Sanjay K. Banerjee, Device Res. Conf. June 2024
 1234. Multi-Domain Dynamics and Ultimate Scalability of CMOS-Compatible FeFETs, N.Pandey, Y.Chauhan, L.Register, SK Banerjee, IEEE ELECTRON DEVICE LETTERS, Volume 45 (11), Nov. 2024
 1235. Comparative Study of Steep Switching Devices for 1T Dynamic Memory, Navlakha, N; Ansari, HR; Register LF, Banerjee, SK, Jun 2024 TECNOLOGIA EN MARCHA37 , pp.110-117
 1236. Nanoelectronics Using 2D Materials, Sk Banerjee, IEEE IndCon, IIT Kharagpur, December 2024 (Invited).
 1237. Increased nonreciprocal current in iron superconductor antiferromagnet interface, Christopher Luth, Rajveer Jha, Matthew Disiena, Luke Sloan, Sanjay K Banerjee, CEC-ICMC Conference, May 2025.
 1238. Comprehensive TCAD Analysis of BEOL-Compatible In₂O₃ Thin-Film Transistors, Nilesh Pandey, Ujwal Radhakrishna, Jian-Yu Lin, Chang Niu, Orlando Lazaro, Gianluca Boselli, Luigi Colombo², Baher Haroun, Peide D. Ye, Leonard F. Register, and Sanjay K. Banerjee, Device Research Conference, June 2025.
 1239. Gate Controlled Nonreciprocal Critical Current in a Superconductor Oxide Interface, C. Luth, R. Schalip, L. Sloan, J. Singh, M. Disiena, and S. Banerjee, Device Research Conference,

June 2025

1240. Diode Effect from Increased Flux Pinning of Iron Based Superconductor Proximity Effect, C. Luth, R. Jha, M. Disiena, R. Schalip, L. Sloan, and S. Banerjee, Electronic Materials Conference, June 2025.
1241. Improved Accuracy and Efficiency in Full-Band Semi-Classical Monte Carlo Carrier Transport Simulations via Smearing and Marching Tetrahedra Methods, Donghyeok Lee, Seong Woo Kang, Leonard F. Register, Sanjay K. Banerjee, Jiwon Chang, SISPAD Conference, Grenoble 2025.
1242. Giant memory window performance and low power consumption of hexagonal boron nitride monolayer atomistor, Sung Jin Yang, Yu-Rim Jeon, Dongyoon Kim, Sivasakthya Mohan, Shanmukh Kutagulla, Matthew Disiena, Sanjay K. Banerjee and Deji Akinwande, *npj 2D Materials and Applications* volume 9, Article number: 9 (2025).
1243. Proximity Effect of Antiferromagnetic Material on FeSeTe Superconductor Hongming Zhang,* Christopher Luth, Rajveer Jha, Luke Sloan, Matthew Disiena, Jatin Singh, and Sanjay K. Banerjee Cite This: ACS Appl. Mater. Interfaces 2025, 17, 12797–12804
1244. Enhancing chemical vapor deposition growth and fabrication techniques to maximize hole conduction in tungsten diselenide for monolithic CMOS integration □ Special Collection: Honoring John Ekerdt's Contributions to Chemical Reaction Engineering and Surface Chemistry for Electronic Materials Jatin Vikram Singh ; Matthew N. Disiena ; S. S. Teja Nibhanupudi ; Nicholas T. Watanabe ; JaeHyun Ahn ; Dong-Won Kim; Anupam Roy ; Sanjay K. Banerjee J. Vac. Sci. Technol. A 43, 022201 (2025)
1245. Analytical Modeling of Short-Channel Effects in BEOL-Compatible Thin-Film Transistors Nilesh Pandey, Ujwal Radhakrishna, Jian-Yu Lin, Chang Niu, Orlando Lazaro, Gianluca Boselli, Luigi Colombo, Bahar Haroun, Peide D. Ye, Leonard F. Register and Sanjay K. Banerjee, IEEE TRANSACTIONS ON ELECTRON DEVICES, VOL. 72, NO. 5, MAY 2025.
1246. Fabrication and Characterization of Ga₂O₃ FinFETs on Patterned Silicon Substrate Hadi Ebrahimi-Darkhaneh, Leunam Fernandez-Izquierdo, Josefina Arellano-Jimenez, Manuel Quevedo-Lopez, and Sanjay K. Banerjee, Advanced Electronic Materials, 2025.
1247. Engineering Si-Qubit MOSFET: Quantum-Electrostatic Integration at Cryogenic Temperatures Nilesh Pandey, Dipanjan Basu, Yogesh Singh Chauhan, Leonard F. Register, and Sanjay K. Banerjee, IEEE TRANSACTIONS ON ELECTRON DEVICES, VOL. 72, NO. 7, JULY 2025.
1248. CBRAM based on CVD-grown MoSe₂ with the coexistence of volatile and non-volatile resistive switching, Wu, Siyu, Singh, JV, Schalip, R, Nibhanupudi, SST, Roy, A, Banerjee, SK, JOURNAL OF APPLIED PHYSICS, Volume, 138, Issue 3, 2025.
1249. Effect of Mo Doping on the Superconducting Properties of Nb_{1-x}Mo_xSe₂, Matthew N. Disiena, Rajveer Jha, Luke Sloan, and Sanjay K. Banerjee, Oct 28 2025, ACS APPLIED ELECTRONIC MATERIALS, 7(20), pp.9526-9533
1250. Signature of quantum phase slips in a layered quasi-one-dimensional Nb₂PdS₅ superconductor, Jha, R., Sloan, L., Zhang, H., Luth, C., Disiena, M., Singh, J., Banerjee, S. K., ACS NANOSCIENCE AU, Oct. 2025.
1251. Progress in the scale-up of single-junction and tandem perovskite solar devices, Weiguang Chi, Sanjay K. Banerjee, Solar Cells, CHEMICAL ENGINEERING JOURNAL, Nov. 2025
1252. Semi-Classical Monte Carlo Simulation of Quantum Confinement Effects in Si Nanosheet and Fin FETs in Industry Standard Orientation, Uiho Lee, Jaeyeon Kim,

Byoung Hak Hong, Wookhyun Kwon, Branch T. Archer, Leonard F. Register, IEEE,
Sanjay K. Banerjee, and Jiwon Chang, IEEE EDL Dec. 2025.

EXHIBIT E

Suman Datta

Joseph M Pettit Endowed Chair Professor
Georgia Research Alliance (GRA) Eminent Scholar
School of Electrical and Computer Engineering
Georgia Institute of Technology

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Biography

Suman Datta received the B.Tech degree in Electrical Engineering from the Indian Institute of Technology, Kanpur, India and the Ph.D degree in Electrical and Computer Engineering from University of Cincinnati, Cincinnati, Ohio. He joined Georgia Tech's School of Electrical and Computer Engineering (ECE) as the Joseph M. Pettit Chair and Georgia Research Alliance (GRA) Eminent Scholar in the Fall of 2022. From 2016 till 2022, he was the Stinson Endowed Chair Professor of Nanotechnology in the Department of Electrical Engineering at the University of Notre Dame, Notre Dame, Indiana, where he directed research in the Nanoelectronic Devices and Circuits Laboratory. He was the Director of the multi-university \$40.4M SRC and DARPA sponsored Applications and Systems-driven Center for Energy-efficient integrated NanoTechnologies (ASCENT) till Dec 2022. In addition, he was the Director of the SRC and NSF sponsored Center for Extremely Energy Efficient Collective Electronics (EXCEL). His research involves heterogeneous computing, brain-inspired computing, in-memory/near-memory computing and cryogenic computing using advanced CMOS and beyond CMOS devices. Prior to Notre Dame, he was a Professor of Electrical Engineering at The Pennsylvania State University, University Park, from 2007 to 2015. From 1999 till 2007, he was in the Advanced Transistor Group at Intel Corporation, Hillsboro, where he pioneered several generations of high-performance logic transistor technologies including high-k/metal gate, Tri-gate (FinFET) and non-silicon channel CMOS transistors that have been deployed in Intel microprocessors for high-performance cloud and client computing applications.

Prof. Datta has co-authored 12 book chapters and over 510 refereed journal and conference papers, holds 185 United States patents and has presented numerous invited and keynote talks. His work has won 3 best paper awards (DRC 2010 and 2011), Test-of-time award (VLSI Symposium 2020), best paper nominations (IEDM 2016, 2018) and received over 43,475 citations (*h-index* = 106). He is a recipient of the Intel Achievement Award (2003), the Intel Logic Technology Quality Award (2002), the Penn State Engineering Alumni Association (PSEAS) Outstanding Research Award (2012), the SEMI Award for North America (2012), the PSEAS Premier Research Award (2015), and the SIA-SRC University Researcher Award (2024). Datta recently received the 2026 IEEE Andrew S. Grove Award *for contributions to and leadership in high-performance and energy-efficient logic transistor research* – the IEEE Technical Field Award presented by the IEEE Board of Directors for outstanding contributions to solid-state devices and technology. Datta served as the General Chair of premier IEEE conferences (DRC and IEDM) and served on the Technical Program Committee of top-tier IEEE conferences (VLSI Technology Symposium, IRPS, VLSI-TSA etc.). In 2013, he was named Fellow of the Institute of Electrical and Electronics Engineers (IEEE) *for his contributions to high-performance advanced silicon and compound semiconductor transistor technologies*. In 2016, he was named Fellow of the National Academy of Inventors (NAI) *in recognition of his inventions that have made a tangible impact on quality of life, economic development and the welfare of society*.

Education

PhD, Electrical and Computer Engineering Dept., University of Cincinnati, OH, September 1999

Bachelors, Electrical Engineering Dept., Indian Institute of Technology, Kanpur, India, June 1995

Experience

Last updated, Mar 2026

Joseph M Pettit Endowed Chair in Advanced Computing and Professor (2022 – Now)

School of Electrical and Computer Engineering, Georgia Institute of Technology, GA

Stinson Endowed Chair Professor (2015 – 2022)

Department of Electrical Engineering, University of Notre Dame, IN

Director, Applications and Systems Driven Center for Energy-Efficient NanoTechnology (2018-2023)

DARPA/ SRC supported multi-university \$40.4M center with 29 PIs, 14 schools, 120 researchers

Director, Center for Extremely Energy-Efficient Collective Electronics (2016 - 2021)

NSF/ SRC supported multi-university \$4.49M center with 10 PIs, 5 schools, 15 researchers

Professor (2011 – 2015)

Department of Electrical Engineering, Penn State University, PA

Monkowski Associate Professor (2007-2011)

Department of Electrical Engineering, Penn State University, PA

Principal Engineer (2005-2007)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Senior Staff Engineer (2003-2005)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Staff Engineer (2002-2003)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Senior Process Engineer (1999-2002)

Logic Technology Development, Intel Corporation, Hillsboro, OR

Technology CAD (TCAD) Engineer (1999)

Avanti Corporation (now Synopsis), Fremont, CA

Technology Consultant (2008 – 2022)

TSMC (Taiwan Semiconductor Manufacturing Corp)

UMC (United Microelectronics Corp)

Samsung Electronics

Micron

Subject Matter Expert (2020-2023)

Expert Witness for Samsung in the KIPB LLC versus Samsung Electronics Co. (2020)

Expert Witness for Samsung in the Daedalus LLC versus Samsung Electronics Co ITC Case
(2023)

Research Career

Funded Projects

NRI

Last updated, Mar 2026

- Energy Efficient Transistor and Architectures, (06/08-05/11)- Phase 1, **Nanoelectronics Research Institute/Semiconductor Research Corporation (NRI/SRC)**, \$525,920 (Total: \$873,920 including PSU match of \$348,000), (PI with T. Mayer, V. Narayanan, D. Schlom)(40% share)
- Energy Efficient Transistor and Architectures, (06/11 – 12/12) - Phase 1.5, **Nanoelectronics Research Institute/Semiconductor Research Corporation (NRI/SRC) and National Institute of Standards and Technology (NIST)**, \$280,000 (PI with T. Mayer)(50% share)
- E2CDA: Extremely Energy Efficient Collective Electronics, (10/16 – 8/31), **National Science Foundation (NSF) and Semiconductor Research Corporation (SRC)**, \$4,290,000 (Center Director and PI) (Candidate's share \$430,000)

NSF

- Ultra-sensitive Magnetic Sensors Integrating the Giant Magnetoelectric Effect with MEMs and Advanced Microelectronics, (10/08-9/11), **National Science Foundation/Division of Electrical, Communications and Cyber Systems (NSF/ECCS)**, \$352,273 (Co-PI with Q. Zhang (PI), EE, PSU and Q. Yang, Radiology, PSU Hershey Medical Center)(50% share)
- EMT/NANO:Co-Exploration of Device and System Architecture for Quantum NanoElectronics,” (09/08-08/11), **National Science Foundation/Division of Computing and Communication Foundation (NSF/CCF)**, \$200,020 (Co-PI with V. Narayanan (PI), EE, PSU)(50% share)
- Collaborative: Mixed Anion and Cation Based Transistor Architecture for Ultra-Low Power Complementary Logic Applications, (10/10-09/13), **National Science Foundation/ Division of Electrical, Communications and Cyber Systems (NSF/ECCS)**, \$472,753 (PI with M. Hudait, EE, Virginia Tech)(50% share)
- MRSEC: Center for Nanoscale Science Supplement titled “Very Low Energy Dissipation Computing using Inter-band Tunneling Injected Non-equilibrium Ballistic Carriers,” (10/10-09/13), **National Science Foundation/Nanotechnology Research Initiative (NSF/NRI)**, \$300,000, (Co-PI with T. Mayer(PI), EE, PSU)(50% share)
- ERC: NSF Nanosystems Engineering Research Center (NERC) for “Advanced Self-Powered Systems of Integrated Sensors and Technologies (ASSIST),” (10/12 – 09/19), **National Science Foundation**, \$18,500,000 (Co-PI and Low Power Nanoelectronics Theme Leader)(Candidate's share \$600,000 to date)
- EXPEDITION: NSF Expedition in Computing for “Visual Cortex on Silicon,” (10/13 – 09/18), **National Science Foundation**, \$10,000,000 (Low Power Nanoelectronics Theme Leader)(Candidate's share \$500,000)

- EFRI 2DARE: Ultra-Low Power, Collective-State Device Technology Based on Electron Correlation in Two-Dimensional Atomic Layers, (9/13 – 12/17), **National Science Foundation**, \$2,000,000 (Co-PI)(Candidate's share \$500,000)
- MRSEC: Center for Nanoscale Science,” (10/13-09/16), **National Science Foundation (NSF)**, \$18,300,000, (Senior Personnel) (Candidate's share \$250,000)
- SemiSynBio: Cardiac Muscle-Cell-Based Coupled Oscillator Networks for Collective Computing (07/18 - 06/21), **National Science Foundation (NSF)**, \$1,125,000 (Co-PI) (Candidate's share \$300,000)
- 2D Defect Zoology, Functionalization, and Application, **National Science Foundation (NSF)** (04/18 - 9/19), \$75,000 (Candidate's share 100%)
- SemiSynBio: Cardiac Muscle-Cell-Based Coupled Oscillator Networks for Collective Computing (07/18 - 06/21), **National Science Foundation (NSF)**, \$1,125,000 (Co-PI) (Candidate's share \$300,000)
- Collaborative Research: DMREF: Accelerated Design, Discovery, and Deployment of Electronic Phase Transitions (ADEPT) (1/24 - 12/27), **National Science Foundation (NSF)**, \$2,000,000 (Co-PI) (Candidate's share \$400,000)
- FuSe: A Reconfigurable Ferroelectronics Platform for Collective Computing (FALCON) (10/23 - 9/26), **National Science Foundation (NSF)**, \$1,290,000 (Co-PI) (Candidate's share \$300,000)
- “EFRI BEGIN OI: Implementation of Dense Associative Memory through Cardiac muscle cell-based Reprogrammable Bio-Oscillatory Neural Networks”, **National Science Foundation**, 09/2024-08/2028, \$400,000

DARPA, NSA, ONR, NIST, DTRA, SRC

- Mixed Anion Arsenide-Antimonide Channel Transistors with High-k Gate Stack, (11/09-10/12) **Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA)**, \$226,000 (PI)
- Architecture-Device Co-Design for Ultra-Low Power High Performance Design, (10/09-09/11), **National Security Agency (NSA)**, \$600,000 (Co-PI with V. Narayanan, EE, PSU)(50% share)
- Correlated Electron Switching Based Tunnel Transistors, (7/11-6/15), **Office of Naval Research (ONR)**, \$1,923,700 (PI with V. Gopalan, R. Engel-Herbert, MSE, PSU, D. Schlom, MSE, Cornell, K. Rabe, Physics, Rutgers)(25% share)
- Development and Demonstration of Next Generation Electronic Warfare Components based on Graphene Technologies, (01/02/12 – 12/31/14), **Office of Naval Research (ONR)**, \$1,280,030 (Co-PI with J. Robinson, Electro-optic Center, PSU)(33% share)

- Ultrafast Spectroscopy in Heterojunction Tunnel Transistors, (10/11 – 9/13), **National Institute of Standards and Technology (NIST)**, \$120,000 (PI)
- Basic Single-Event and Total-Ionizing Dose Mechanisms in Antimony (Sb)-based CMOS Transistors with High-K Dielectric, (4/01/14-3/31/17), **Defense Threat Reduction Agency (DTRA)**, \$1,745,560 (PI with D. McMorro, NRL, K. Saraswat, EE, Stanford U.)(33% share)
- Architecture-Device Co-Design for Ultra-Low Power High Performance Design- Phase 2, (10/12-09/14), **National Security Agency (NSA)**, \$600,000 (Co-PI with V. Narayanan, EE, PSU)(50% share)
- Center for Low Energy Systems (LEAST) FCRP with Notre Dame Univ. (01/13-12/17)
Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA) (total center funding is \$ 30,000,000), Candidate is PI and Theme Leader for the “Quantum Engineered Steep Slope Transistors” \$ 4,000,000 (over 5 years) (25% share)
- Oxide-based Reconfigurable Single-Electron Logic for Beyond CMOS, (10/13 - 9/14) **(Semiconductor Research Corporation/Sematech)** \$62,125 (PI)
- Basic Single-Event and Total-Ionizing Dose Mechanisms in GE/InGaAs-based CMOS Transistors with ALD High-k Dielectric, (09/14 – 08/17) **Defense Threat Reduction Agency (DTRA)**, \$1,045,560 (PI with C. Cress, NRL, K. Saraswat, EE, Stanford U.)(33% share)
- Landau FET Using Mott Hubbard Phase Transition, (1/14 - 12/17) **(Semiconductor Research Corporation)** \$325,000 (PI)
- Ferroelectric Field Effect Transistor with Steep Switching Slope and Non-Volatile Functionality, (1/16 – 12/18) **(Semiconductor Research Corporation)** \$300,000 (PI)
- Orbital Ordering Driven Threshold Switches for Select Devices in 3D X-Point Memories (11/15 – 10/17) **Semiconductor Research Corporation and Defense Advanced Research Projects Agency (SRC/DARPA)** \$ 500,000 (PI with S. Gupta, Penn St, and S. Guha, University of Chicago)(50% share)
- ASCENT - Applications and Systems driven Center for Energy-Efficient Integrated NanoTechnologies (01/18 - 12/22) **DARPA / Semiconductor Research Corporation** \$ 38,386,478 (Director and Center Leader) (Candidate’s share: \$1,200,000)
- IMPACT - Innovative Materials and Processes for Accelerated Compute Technologies (IMPACT) Research Center (01/20 – 12/22) **DARPA / Semiconductor Research Corporation** \$ 6,000,000 (PI) (Candidate’s share: \$1,000,000)
- Multi-Bit-per-Cell (MBC) Ferroelectric FET Memory Using Ferroelectric (FE) Superlattice with Anti-Ferroelectric (AFE) Interfacial Coupling, (1/20 – 12/22) **(Semiconductor Research Corporation)** \$255,000 (PI)

- Multi-Component Semiconducting Oxide FETs: Materials-Device Co-Design, Synthesis, NanoFabrication, Characterization and Benchmarking, (1/20 – 12/22) (**Semiconductor Research Corporation**) \$255,000 (PI)
- T-MUSIC: Negative Capacitance Enabled Scaling to Achieve 1 THz Cut-off Frequency Transistors on a CMOS Platform, (0/19 – 3/21) (**DARPA**) \$444,302 (PI)
- IIRM-URA: Interaction with Ionizing Radiation with Matter, University Research Alliance (IIRM-URA) (07/20 – 12/24) (**DTRA**) \$1,872,523 (PI)
- FALCON: Ferroelectric Analog Compute in Memory (11/23 – 4/28) (**DARPA**) \$9,000,000 (PI)
- 3DFEM: Center for 3D Ferroelectric Microelectronics (08/22 – 7/24) (**DOE**) \$240,000 (PI)
- Center for Heterogenous Integration of Micro Electronic Systems (CHIMES) (01/23 – 12/27) (**SRC/DARPA**) \$1,094,092 (PI)
- Center for Processing with Intelligent Storage and Memory (PRISM) (01/23 – 12/27) (**SRC/DARPA**) \$1,089,581 (PI)
- Zero Schottky Barrier Ohmic Contact to Silicon Using Metal Interlayer (01/23 – 12/25) (**SRC**) \$405,000 (PI)
- Advanced Technology Boosters to Enable Low Temperature Logic Technology (LTLT) on a CMOS Platform (01/22 – 10/24) (**DARPA**) \$2,741,301 (PI)
- Atoms-to-Systems Co-Design: Transforming Data Flow to Accelerate Scientific Discovery (09/21 – 10/24) (**DOE**) \$525,000 (Co-PI)
- Machine Learning Enabled High Precision and Fast Thermal Model of Nanoscale Transistor Arrays (9/23-8/24) (**DARPA**) \$500,000 (Co-PI)
- FALCON: Ferroelectric Analog Compute-in-Memory, **DARPA**, 12/2023-8/2026, \$3,000,000 (PI ~ 20% share)

Industry

- Compound Semiconductor Based Heterojunction Tunnel Transistors for Ultra Low Power Logic Applications-Phase 2, (09/01/09-08/31/12), **Intel Corporation**, \$255,000 (PI)
- Ultra-Low Resistance Ohmic Contacts for III-V Digital Logic, (04/01/09-05/01/11), **Intel Corporation**, \$250,000 (Co-PI with S. Mohny, Mat. Sc., PSU)(50% share)
- Post CMOS circuits and architecture, (10/01/10-09/30/13), **Academic Research Office (ARO), Intel Corporation**, \$170,000 (PI)
- Sub-0.4V Logic Circuits with Steep Sub-threshold Slope Inter-band Tunnel FETs-Phase 2, (06/09-06/11) **Intel Corporation**, \$70,000 (PI)

- Supply voltage scalability of III-V based heterojunction tunnel transistors -Phase 1, (09/08-08/09), **Intel Corporation**, \$85,000 (PI)
- Multi-Gate III-V QWFET , (3/1/11-2/28/14), **Global Foundries**, \$165,000 (PI)
- Germanium and III-V Devices, (08/11 – 07/12), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Ultra Low Resistivity Metal Insulator Semiconductor (MIS) Contacts, (10/12 – 09/13), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Reliability Assessment of Highly Scaled High-k Gate Stacks, (10/12 – 09/13), **Applied Materials (AMAT)**, \$ 60,000 (PI)
- Variation Study of 3D Transistors, (10/11 – 09/14), **Lam Research**, \$ 75,000 (PI)
- III-V-based Nanowire MOSFET and NanoPillar Tunnel FET for Ultra Low-Power Nanoelectronics, (2/12 – 07/15), **Samsung GRO**, \$ 240,000 (PI)
- Ultra-scaled III-V FinFETs for Next Generation Nanoelectronics (11/15 – 1/17), **Samsung Electronics**, \$ 120,000 (PI)
- 5nm Node Logic Transistor Option for Mobile System on a Chip (“SOC”) (2/16 – 1/17), **Qualcomm**, \$ 50,000 (PI)
- Low Voltage, High Non-Linearity (High ION & Low IOFF) Select Devices for Cross-point Memory Applications (06/17 - 05/20), **Intel Corporation**, \$300,000 (PI)
- Functional Oxide based Cross-point Resistive Devices for Deep Neural Networks (04/17 - 04/19) **IBM Corporation**, \$60,000 (PI)
- Investigation of Oxygen Insertion Channel Field Effect Transistor Architecture for Enhanced Transport and Improved Positive Bias Temperature Instability (4/17 – 2/19) **ATOMERA**, \$117,393 (PI)
- Transport in Gate-All-Around (GAA) Nanowire (NW) Transistors: Impact of NW Shape and Diameter (11/16 - 10/18), **Silicon Valley Community Foundation** \$25,000 (PI)
- Ferroelectric Transistor Technologies for Neuromorphic Computing (03/23 – 3/25) **Versum Materials**, \$600,000 (PI)
- A scalable physical compute engine for energy-based learning and inference (09/21 – 9/24) **Intel Corporation**, \$255,000 (PI)
- Technology Boosters for DRAM Logic Transistors, (5/2024 - 04/2026), **Micron Technology**, \$227,272 (PI)
- Performance and reliability boosters for amorphous oxide semiconductor (AOS) TFT", (5/2024 - 04/2027), **Samsung Electronics**, \$450,000 (PI ~ 50%)

Total number of publications to date:

– **490 total (240 journal, 250 refereed conferences, 186 patents)**

Last updated, Mar 2026

– h-index = **106** with **>43,475** citations (Google Scholar)

Total number of graduate students to date:

Postdoctoral scholars: **11 completed, 4 current**

PhD's: **29 graduated** (including 2 IBM PhD Fellows, 1 Lam research PhD Fellow), **8 current**;

MS's: **9 graduated** (with thesis option)

Total number of issued United States patents: **186**

Graduated Students (29 PhDs, 11 Postdoctoral Researchers, 7 Masters)

1. Tanmoy Maiti (Postdoctoral Associate) (08/09-08/10) (Currently, Assistant Professor at Indian Institute of Technology, Kanpur, India)
2. Ramakrishnan Krishnan, PhD, 12/2009: Reliability Effects Of Soft Errors and NBTI in Current and Emerging Digital Circuits (Currently, Senior Staff Engineer, Advanced Technology Platforms Group, Taiwan Semiconductor and Manufacturing Corp (TSMC), Hsinchu, Taiwan)
3. Saurabh Mookerjee, PhD, 08/2010: Simulation, Design and Fabrication of Tunnel Transistors with steep sub-threshold slopes (Currently, Senior Device Engineer, Logic Technology Development, Intel Corporation, Hillsboro, Oregon)
4. Wei-Chieh Kao, MS (Thesis), 05/2010: Impact of Non-ideal Interfaces on Transistor Performance (Currently, PhD student at Arizona State University)
5. Vikram Sampat Kumar, MS (Thesis), 04/15/2010: An FPGA-based Real Time Tracking For Indoor Environment
6. Srijith Rajamohan, MS (Thesis), 04/2010: A Neural Network based Classifier on the Cell Broadband Engine
7. Ashkar Ali, MS (Thesis), 03/2009: Transport in Silicon Quantum Dots Embedded in a Rare Earth Oxide
8. Chad Ostrowski, BS (Honor's Thesis) 12/2009: Analytical Modeling of Tunnel Diodes
9. Vinay Saripalli, PhD, 10/31/11: Device Architecture Co-Design for Ultra Low Power Logic Using Emerging Tunneling Based Devices (Currently Senior CAD Engineer, Intel Corporation, Santa Clara)
10. Zhao Feng, PhD, 08/31/2011 : Ultra Sensitive Magnetic Sensors Integrating the Giant Magnetoelectric Effect with MEMS and Advanced CMOS (Currently Design Engineer, Texas Instruments, Dallas)
11. Salil Mujumdar, MS (Thesis), 05/2011: Strain Engineering in Nanoscale Transistors (Thesis option) (Current Device Engineer, Inter Molecular Foundry, San Jose)

12. Ashish Agrawal, MS (Thesis), 05/2011: Noise measurement and modeling of nanoscale devices (Thesis option) (Currently Ph.D. candidate at Penn State)
13. Ashkar Ali, PhD, 06/2012: Design and Fabrication of Ultra-low power and High Performance Quantum-well Transistors (IBM PhD Fellow 2010-2011, Currently Senior Device Engineer at Intel Corporation)
14. Feng Li, PhD, 08/2012: Ultra-sensitive Chip-Scale Magnetometers (Currently Design Engineer, Freescale Semiconductors)
15. Srinidhi Kestur, PhD, 01/2012: Accelerating computationally intensive applications using Reconfigurable systems (Currently Senior Design Engineer, Intel Corporation)
16. Euichul Hwang, PhD, 09/2012: Multi-gate III-V Metal Oxide Semiconductor FETs (Currently Device Engineer, Samsung Advanced Institute of Technology, SAIT)
17. Dheeraj Mohata, PhD, 01/2013: Arsenide-Antimonide Hetero-Junction Transistors for Low Power Logic Applications (Currently Integration Engineer, RF Micro Devices)
18. Ayan Kar, Post Doctoral Researcher, 02/13: (Currently Senior Device Engineer, Intel Corporation)
19. Eugene Freeman, MS (Thesis), 11/13: Correlated Electron Based Switches (Currently PhD student at Penn State)
20. Bijesh Rajamohanan: PhD, 05/2014: Fabrication, Characterization and Physics of III-V Tunneling Field Effect Transistors for Low Power Logic and RF Applications (Currently Senior Device Engineer, Sandisk Corporation)
21. Lu Liu: PhD, 05/2014: Classical and Coulomb Blockade III-V Multi-Gate Quantum Well Field Effect Transistors for Ultra Low Power Logic Applications (Currently Senior Device Engineer, Intel Corporation)
22. Ashish Agrawal: PhD, 12/2014: Physics and Technology of Strained Germanium Quantum Well FinFET for Low Power P-Channel Application (Currently Senior Device Engineer, Intel Corporation)
23. Huichu Liu: PhD, 5/2014: Circuit-Device Interaction for Steep Switching Slope Devices (Currently Senior Architecture Engineer, Intel Corporation)
24. Matt Hollander: PhD, 11/2015: Two-Dimensional Materials for Novel Electronic Applications: The Graphene Mixer and TaS₂ Hyper FET (Currently Senior Device Engineer, Micron Corporation)

25. Arun VT: PhD, 7/2015: Physics and technology of nanoscale III-V field effect transistors for low power electronics (Currently Senior Device Engineer, Intel Corporation)
26. Nidhi Agrawal: PhD, 7/2015: Numerical Simulation of Variation in 3D NonSilicon Transistors (Currently Senior Reliability Engineer, Micron Corporation)
27. Ali Razavieh (Post Doctoral Associate, PhD, Purdue University, West Lafayette, Indiana)(Currently, Senior Device Engineer, Global Foundries)
28. Bikas Das (Post Doctoral Associate, PhD, Indian Association of Cultivation of Science, Kolkata, India)(Currently Assistant Professor, IISER Calcutta)
29. Sandeepan Das Gupta (Post Doctoral Associate, PhD. Vanderbilt University)(Currently Senior Device Engineer, Micron Corporation)
30. Himanshu Madan: PhD, 12/2015: RF Electronics based on Emerging Devices (Currently Senior Device Engineer, Intel Corporation)
31. Mike Barth: Ph.D., 08/01/2016: Antimonide based Low Power Nanoelectronics (Currently Senior Process Engineer, Intel Corporation)
32. Rahul Pandey: Ph.D., 08/01/2016: Electrical Noise in Emerging Devices (Currently Senior Device Engineer, Intel Corporation)
33. Ramkrishna Ghosh (Post Doctoral Associate, PhD, Indian Institute of Science, IISc, Bangalore, India) (Assistant Professor, Jawaharlal Nehru University (JNU), Delhi, India)
34. Pankaj Sharma (Post Doctoral Associate, PhD, EPFL, Lausanne, Switzerland)(Currently Senior Process Integration Engineer, Micron Corporation)
35. Nikhil Shukla: PhD, 08/01/2017: Computational Device and Circuit Concepts using Electronic Phase Transition (Assistant Professor, ECE, University of Virginia, Charlottesville)
37. Xueqing Li (Post Doctoral Associate, PhD, Tsinghua University)(Assistant Professor, Tsinghua University, Beijing, China)
38. Matt Jerry: Ph.D.,10/01/2018: Collective Phenomena Based Solid State Devices and Their Natural Computing Applications (Senior Memory Array Engineer, Micron)

39. Kai Ni (Post Doctoral Associate, PhD, Vanderbilt University)(Assistant Professor, University of Notre Dame, Notre Dame, Indiana, USA)
40. Benjamin Grisafe: Ph.D., 07/23/2020: Threshold Switching Phenomena and Their Application to Electronic Devices (Process Integration Engineer, Northrop Grumman)
41. Jeff Smith: Ph.D., 11/20/2020: Novel Silicon and Non-Silicon Transistors For Low Power Logic Applications (Senior Device Engineer, Wolfspeed)
42. Abhishek Khanna: Ph.D., 4/6/2023: Phase Transition Oscillators Based Ising Machines for Combinatorial Optimization Problem Solvers (Senior Device Integration Engineer, Micron)
43. Wriddhi Chakrobarty: Ph.D. 4/4/2022: Cryogenic CMOS Technology and Monolithic 3D Integration for high Performance Computing (Senior Process Integration Engineer, Intel Corporation)
44. Jorge Gomez: Ph.D., 06/2021: Functional Oxide-Based Emerging Nanoelectronic Devices for Low Power Logi, Memory and High Frequency Applications (Research Scientist, Meta)
45. Huacheng Ye: Ph.D., 08/01/2022: Sequential Monolithic Three Dimensional Circuits (Senior Device Engineer, Western Digital)
46. Sourav Dutta (Post Doctoral Associate, PhD, Georgia Tech) (Currently, Assistant Professor, University of Texas, Dallas)
47. Navnidhi Upadhyay (Post Doctoral Associate, PhD, University of Massachusetts, Amherst) (Currently, R&D Engineer, EMD Electronics)
48. Khandker Akif Aabrar, PhD, 08/2024 Ferroelectric and Monolithic 3D (M3D) Memory Technologies for High-Performance, Energy-Efficient Computing First employment (Senior Process Integration Engineer, Micron Technology)

Current Research Interests

- Energy Efficient CMOS (e.g. next-gen gate stacks and source-drain contacts)
- Cryogenic FinFET CMOS (e.g. multi-state gain-cell memory)
- Monolithic 3D integration (e.g. low temperature processed oxide semiconductor transistors)
- In-memory computing (e.g. artificial neural networks using analog ferroelectric memory)
- Neuromorphic computing (e.g. spiking neural networks)
- Physics Inspired computing (e.g. Potts and Ising solvers using coupled oscillators)

- Biology Inspired computing (e.g. coupled oscillators using living heart muscle cells)

Awards and Honors

- SIA (Semiconductor Industry Association)-SRC (Semiconductor Research Corporation) University Researcher Award (2024)
- VLSI Technology and Circuits Test of Time Award (2023)
- Facebook Faculty Award (2020)
- Stinson Endowed Professorship (2019)
- Fellow of National Academy of Inventors (NAI) (2017) for “outstanding inventions that have made a tangible impact on quality of life, economic development and the welfare of society”
- Penn State Engineering Alumni Society (PSEAS) Premier Research Award (2015)
- IEEE Fellow for “contributions to for contributions to high-performance advanced silicon and compound semiconductor transistor technologies” (2013)
- SEMI Award for North America “in recognition of their pioneering work in the development, integration and introduction of a successful high-k dielectric and metal electrode gate stack for 45 nm CMOS IC production” (2012)
- IBM Faculty Award (2012)
- Penn State Engineering Alumni Society (PSEAS) Outstanding Research Award (2012)
- Distinguished Lecturer of IEEE Electron Devices Society (2011)
- Joseph Monkowsky Professorship for Faculty Early Career Development, The Pennsylvania State University (2007)
- Intel Achievement Award (the highest technical honor at Intel) for “developing the world’s first high-K/metal gate CMOS transistors with record-setting performance” (2003)
- Divisional Achievement Award from Intel Logic Technology Development Group for “invention and successful demonstration of high performance Tri-gate CMOS transistors” (2002)
- All India Rank of 124 among 300,000 students who appeared for Indian Institute of Technology Joint Entrance Examination (IIT-JEE) (1995)

Research Supervision (current)

Postdoctoral Researcher (5)

1. Sunbin Deng, PhD, Hong Kong University of Science & Technology (HKUST)
2. Hyeonwoo Park, PhD, Seoul National University (SNU)
3. Hyun Jae Lee, PhD, Seoul National University (SNU)
4. Dr. Songhyeon Kuk, PhD, Korea Advanced Institute of Technology (KAIST)
5. Dr. Junsu Yu, PhD, Seoul National University (SNU)

Doctoral Students (9)

1. Dyutimoy Chakraborty: Analog Compute in Memory (Start Date: 09/2023)
2. Rinku Dutta: Electro-thermal Transport in Extremely Scaled Transistors (Start Date: 09/2023)
3. Sharadindugopal Kirtania: Reliability Physics of Oxide Transistors (Start Date: 08/2022)
4. Eknath Sarkar: Gate-all-around Transistors (Start Date: 09/2023)
5. Jaewon Shin: Ferroelectric Non-volatile transistors (Start Date: 09/2023)
6. Chengyang Zhang: Zero Barrier Schottky Contact to Semiconductors (Start Date: 09/2022)
7. Emmanuel Quezada: BEOL Oxide Transistors (Start Date: 08/2024)
8. Ishaan Joshi: State Space Model Implementation on Ferroelectric Hardware (Start Date: 08/2025)
9. Wan-Hsuan Lo: DTCO for CFET (Start Date: 08/2025)

MS Students (0)

Publications

Book Chapters

- [1] V. Saripalli, V. Narayanan and S. Datta, “Ultra Low Energy Binary Diagram Circuits Using Few Electron Transistors”, *Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering*, Springer Berlin Heidelberg, October 2009
- [2] V. Eachempati, R. Das, V. Narayanan, Y. Xie, S. Datta and C. Das, “HeTERO: Hybrid Topology Exploration for RF based On Chip Networks”, *Communication Architectures for System-on-Chip (SoC)*, CRC Press, September 2010
- [3] S. Datta, D. Schlom, “Gate Oxides beyond SiO₂”, *Multifunctional Oxide Heterostructures*, Oxford University Press, September 2010
- [4] S. Datta, “III-V MOSFETs”, *Future Intelligent Integrated Systems: New Paths to Augmented Silicon CMOS Technologies*, WSPC-Pan Stanford (Singapore), January 2013
- [5] Nikhil Shukla, S. Datta, A. Parihar, A. Raychowdhury, “Computing with Relaxation Oscillators”, *Future Trends in Microelectronics*, Wiley, March 2016
- [6] J. Robinson, M. Hollander, S. Datta, “Epitaxial Graphene: Progress on Synthesis and Device Integration”, *2D Materials for Nanoelectronics*, CRC Press, May 2016
- [7] Suman Datta, “Tri-Gate Transistors”, *Nanoscale Silicon Devices*, CRC Press, Jan 2016

- [8] Ahmedullah Aziz, Sumitha George, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, "Sensing in Ferroelectric Memories and Flip-Flops", *Sensing of Non-Volatile Memory Demystified*, Springer, Aug 2018
- [9] Xueqing Li, Moon Seok Kim, Ahmedullah Aziz, Matthew Jerry, Nikhil Shukla, John Sampson, Sumeet Gupta, Suman Datta, Vijaykrishnan Narayanan, "Emerging Steep-Slope Devices and Circuits: Opportunities and Challenges", *Beyond CMOS Technologies for Next Generation Computer Design*, Springer, Aug 2018
- [10] Ahmedullah Aziz, Sandeep Krishna Thirumala, Danni Wang, Sumitha George, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, "Sensing in Ferroelectric Memories and Flip-Flops, *Sensing of Non-Volatile Memory Demystified*, Springer, 2019
- [11] Robert Chau, Suman Datta, "Field-Effect Transistor R&D in the United States: Past, Present, and Future", *The FET Centennial*, Wiley, 2026

Journal Articles

- [258] Madhav Vadlamani, Dyutimoy Chakraborty, Jianwei Jia, Halid Mulaosmanovic, S. Duenkel, S. Beyer, Suman Datta, S. Yu, "Cryogenic Characterization of Ferroelectric Nonvolatile Capacitors" in IEEE Transactions on Electron Devices, Mar 2026
- [257] J. Sonawane, S. Deng, C. Zhang; Omkar Phadke; Faaiq Waqar; Suman Datta "Asymmetric Underlap–Overlap 2T Gain Cell Enabling Voltage-Sensing Readout With Minimized Capacitive Coupling," in IEEE Electron Device Letters, vol. 47, no. 3, pp. 522-525, March 2026
- [256] X. Niu, S. Deng, F. X. Liang, K. A. Aabrar, Y. Qin, J. Duan, Suman Datta, V. Narayanan, X. Hu, K. Ni, "Forming Process in BEOL Oxide Channel Ferroelectric FET," in IEEE Transactions on Electron Devices, vol. 72, no. 12, pp. 6963-6969, Dec. 2025.
- [255] K. Toprasertpong, T. Mikolajick, Suman Datta, Q. Huang, Shinichi Takagi, "Emerging applications: Neuromorphic computing and reservoir computing," MRS Bulletin, vol. 50, no. 9, pp. 1032-1042, 2025.
- [254] J. Kwak, S. Deng, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, J. Lee, Suman Datta, S. Yu, "Machine learning-assisted compact modeling of AC stress-induced bias temperature instability in top- and bottom-gate W-doped In₂O₃ channel semiconductor transistors," IEEE Transactions on Electron Devices, vol. 72, no. 8, pp. 4163-4169, 2025.

- [253] J. Kwak, S. Deng, J. Lee, Suman Datta, S. Yu, "Backside active power delivery with hybrid DC-DC converter enabled by amorphous oxide semiconductor transistors," *IEEE Transactions on VLSI Systems*, vol. 33, no. 8, pp. 2153-2162, 2025.
- [252] O. Phadke, P. V. Ravindran, H. Mulaosmanovic, S. Dunkel, S. Beyer, A. I. Khan, Suman Datta, S. Yu, "Impact of the endurance measurement methodology on ferroelectric field effect transistor under the capacitive non-destructive read," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2312-2318, 2025.
- [251] O. Phadke, K. A. Aabrar, G. Choe, Y.-C. Luo, S. G. Kirtania, A. I. Khan, Suman Datta, S. Yu, "Effect of DC stress on low-frequency noise characteristics of W-doped In₂O₃ BEOL transistors," *IEEE Transactions on Electron Devices*, vol. 72, no. 3, pp. 1489-1493, 2025.
- [250] H. Park, S. G. Kirtania, E. Sarkar, D. Chakraborty, C. Zhang, H. J. Lee, J. Shin, S. Yu, A. I. Khan, H. Kim, C. Im, M. J. Hong, D. Ha, Suman Datta, "Extreme EOT scaling in tungsten-doped In₂O₃ MOSFETs for enhanced stability and drive current," *IEEE Transactions on Electron Devices*, vol. 72, no. 12, pp. 7136-7144, 2026.
- [249] S. G. Kirtania, H. Park, O. Phadke, E. Sarkar, D. Chakraborty, F. G. Waqar, J. Shin, A. I. Khan, S. Yu, Suman Datta, "Amorphous indium oxide channel FeFETs with write voltage of 0.9 V and endurance > 10¹² for refresh-free embedded memory," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2691-2699, 2025.
- [248] E. Sarkar, C. Zhang, D. Chakraborty, S. G. Kirtania, K. A. Aabrar, H. Park, J. Shin, H. J. Lee, M. Tian, A. I. Khan, S. Yu, Suman Datta, "First demonstration of high-performance and extremely stable W-doped In₂O₃ gate-all-around (GAA) nanosheet FET," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2662-2669, 2025.
- [247] S. Woo, G. Choe, A. I. Khan, Suman Datta, S. Yu, "Design of superlattice ferroelectric-metal field-effect transistor for triple-level cell 3D NAND Flash," *Microelectronic Engineering*, vol. 295, pp. 112276, 2025.
- [246] L. Fernandes, P. V. Ravindran, J. Chen, M. Tian, D. Das, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, Suman Datta, A. I. Khan, "Optimizing memory window for ferroelectric NAND applications: An experimental study on dielectric material selection and layer positioning," *IEEE Transactions on Electron Devices*, vol. 72, no. 1, pp. 234-239, 2025.

- [245] P. Venkatesan, L. Fernandes, P. Ravikumar, C. Park, H. Tran, Z. Wang, H. Jayasankar, A. Garlapati, T. Song, M. Tian, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, S. Yu, Suman Datta, W. Kim, D. Ha, A. I. Khan, "Demonstration of robust retention in band engineered FeFETs for NAND storage applications using tunnel dielectric layer," IEEE Electron Device Letters, vol. 46, no. 3, pp. 397-400, 2025, editor's pick.
- [244] P. Ravikumar, P. Venkatesan, C. Park, N. Afroze, M. Tian, W. Chern, Suman Datta, S. Yu, S. Mahapatra, A. I. Khan, "The first switch effect in ferroelectric field-effect transistors," IEEE Transactions on Device and Materials Reliability, vol. 25, no. 3, pp. 365-370, 2025.
- [243] Tanvir H Pantha, A. Khanna, H. Ye, S. Niazi, M. P Kamal, B. Chakraborty, S. Alam, E. Weinstock, N. Babu, S. Mukhopadhyay, Y. Chiu, Suman Datta, K. Y Camsari, S. Dutta, "Addressing the Connectivity Bottleneck With BEOL FeFETs for 3-D CMOS + X Ising Machines," in IEEE Transactions on Electron Devices, vol. 72, no. 6, pp. 3335-3342, June 2025
- [242] S. G. Kirtania, H. Park, O. Phadke, E. Sarkar, D. Chakraborty, F. G Waqar, J. Shin, A. Khan, S. Yu, Suman Datta, "Amorphous Indium Oxide Channel FEFETs With Write Voltage of 0.9 V and Endurance >10¹² for Refresh-Free Embedded Memory," in IEEE Transactions on Electron Devices, vol. 72, no. 5, pp. 2691-2699, May 2025
- [241] FVE Hensling, P Vogt, J Park, SL Shang, H Ye, YM Wu, Suman Datta, D. Schlom, "Fully Transparent Epitaxial Oxide Thin-Film Transistor Fabricated at Back-End-of-Line Temperature by Suboxide Molecular-Beam Epitaxy", Advanced Electronic Materials, vol. 11, no. 3, pp. 24004992025, 2025
- [240] P Venkatesan, L Fernandes, P Ravikumar, C Park, H Tran, Z Wang, S Lim, K. Kim, S. Yu, S. Datta, W. Kim, D. Ha, A. Khan, Demonstration of Robust Retention in Band engineered FEFETs for NAND Storage Applications using Tunnel Dielectric Layer, IEEE Electron Device Letters, 2024.
- [239] L Fernandes, PV Ravindran, J Chen, M Tian, D Das, H Chen, W Chern, ... S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, S. Datta, A. Khan, Optimizing Memory Window for Ferroelectric Nand Applications: An Experimental Study on Dielectric Material Selection and Layer Positioning, IEEE Transactions on Electron Devices, vol. 72, no. 1, pp. 234-239, Jan. 2025
- [238] DB Durham, M Noor, KA Aabrar, Y Liu, S. Datta, K Cho, S Guha, C Phatak Direct Imaging of Asymmetric Interfaces and Electrostatic Potentials inside a Hafnia–Zirconia Ferroelectric Nanocapacitor, ACS Applied Materials & Interfaces 16 (49), 68562-68568, 2024

- [237] S. Deng, J. Kwak, J. Lee, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K.A. Aabrar, S. Yu, S. Datta, "Monolithic Switched-Capacitor DC-DC Converters Using BEOL-Compatible Oxide Power Transistors and Superlattice MIM Capacitors," in IEEE Transactions on Electron Devices, vol. 71, no. 12, pp. 7999-8006, Dec. 2024
- [236] Z. Zhao, S. Woo, K. A. Aabrar, S. G. Kirtania, Z. Jiang, S. Deng, Y. Xiao, H. Mulaosmanovic, S. Duenkel, D. Kleimaier, S. Soss, S. Beyer, R. Joshi, S. Meninger, M. Mohamed, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, V. Narayanan, S. Datta, S. Yu, K. Ni, Paving the Way for Pass Disturb-Free Vertical NAND Storage via a Dedicated and String-Compatible Pass Gate, ACS Applied Materials & Interfaces, vol 16, No 41, pp 55619-55626, 2024
- [234] O. Phadke, S. Gopal Kirtania, D. Chakraborty, S. Datta and S. Yu, "Suppressed Capacitive Coupling in 2 Transistor Gain Cell with Oxide Channel and Split Gate," in IEEE Transactions on Electron Devices, vol. 71, no. 11, pp. 6749-6755, Nov. 2024
- [233] P. Venkatesan, C. Park, T. Song, L. Fernandes, D. Das, N. Afroze, P. G. Ravikumar, M. Tian, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Mahapatra, S. Yu, S. Datta, A. Khan, "Disturb and its Mitigation in Ferroelectric Field-Effect Transistors With Large Memory Window for NAND Flash Applications," in IEEE Electron Device Letters, vol. 45, no. 12, pp. 2367-2370, Dec. 2024
- [232] R. Ramesh, S. Salahuddin, S. Datta, C H Diaz, D E Nikonov, I A Young, D. Ham, M. F. Chang, W. S. Khwa, A. S. Lele, C. Binck, .. L. Bellaiche, P.Wu, J. Cai, J. Appenzeller, Supriyo Datta, K. Y Camsari, J.Kwon, J A C Incorvia, I. Asselberghs, F. Ciubotaru, S. Couet, Christoph Adelman, Y.Zheng, A. M Lindenberg, P. Evans, P. Ercius, I. P Radu, Roadmap on low-power electronics, Applied Physics Letters Materials, vol 12, no 9, 2024
- [231] FVE Hensling, P Vogt, J Park, SL Shang, H Ye, YM Wu , Yu-Mi Wu, K. Smith, V. Show, K. Azizie, H. Paik, D. Jena, H. G Xing, Y E. Suyolcu, P. A van Aken, S. Datta, Z. K. Liu, D. G Schlom Fully Transparent Epitaxial Oxide Thin-Film Transistor Fabricated at Back-End-of-Line Temperature by Suboxide Molecular-Beam Epitaxy, Advanced Electronic Materials, pp 2400499, 2024
- [230] M. M. Islam, S. Alam, M. A. Jahangir, G. S Rose, S. Datta, V. Narayanan, S. K. Gupta, A. Aziz "Reimagining Sense Amplifiers: Harnessing Phase Transition Materials for Current and Voltage Sensing," in IEEE Transactions on Nanotechnology, vol. 23, pp. 606-614, 2024

- [229] L. Fernandes, P. Ravindran, T. Song, D. Das, C. Park, N. Afroze, M. Tian, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, S. Datta, A. Khan, "Material Choices for Tunnel Dielectric Layer and Gate Blocking Layer for Ferroelectric NAND Applications," in *IEEE Electron Device Letters*, vol. 45, no. 10, pp. 1776-1779, Oct. 2024
- [228] F. -X. Liang, S. Kumar, K. Ni, W. Chakraborty, Y. Singh Chauhan, H. Amrouch, S. Datta, M. Niemier, X. S. Hu, "A Physics-Based Model for Oxide–Semiconductor-Based Ferroelectric Field-Effect Transistors," in *IEEE Transactions on Electron Devices*, vol. 71, no. 7, pp. 4397-4402, July 2024
- [227] G. Stone, Y. Shi, M. Jerry, V. Stoica, H. Paik, Z. Cai, D. G. Schlom, R. Engel-Herbert, S. Datta, H. Wen, L. Q. Chen, V. Gopalan, "In-Operando Spatiotemporal Imaging of Coupled Film-Substrate Elastodynamics During an Insulator-to-Metal Transition", *Advanced Materials*, vol 36, no 24, June 2024
- [226] R. Saligram, A. Raychowdhury, S. Datta, "The future is frozen: cryogenic CMOS for high-performance computing", *Chip*, vol 3, no 1, March 224
- [225] Yaoqiao Hu, Darrell Schlom, Suman Datta, and Kyeongjae Cho, Ab Initio Design of Pb²⁺-based Ternary Oxides for Phase Stable and Hole Dopable p-type Oxides, *Chemistry of Materials* (2024)
- [224] Rakshith Saligram, Arijit Raychowdhury, Suman Datta, The Future is Frozen: Cryogenic CMOS for High-Performance Computing (Invited), *Chip* (2023)
- [223] S. Woo, K. A. Aabrar, S. Datta and S. Yu, "Analyzing Total-Ionizing-Dose Induced Memory Window Degradation in Ferroelectric FinFET," in *IEEE Transactions on Device and Materials Reliability*, (2023)
- [222] Y. -C. Luo et al., "Correlated Oxide Selector for Cross-Point Embedded Non-Volatile Memory," in *IEEE Transactions on Electron Devices*, vol. 71, no. 1, pp. 916-921, (Jan. 2024)
- [221] A. D. Gaidhane, R. Saligram, W. Chakraborty, S. Datta, A. Raychowdhury and Y. Cao, "Design Exploration of 14 nm FinFET for Energy-Efficient Cryogenic Computing," in *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, vol. 9, no. 2, pp. 108-115 (Dec. 2023)
- [220] T. -H. Kim et al., "Tunable Non-Volatile Gate-to-Source/Drain Capacitance of FeFET for Capacitive Synapse," in *IEEE Electron Device Letters*, vol. 44, no. 10, pp. 1628-1631 (Oct. 2023)

- [219] Daniel B Durham et al, Quantitative Electrostatic Potential Mapping in Dense Polycrystalline Functional Materials and Devices, *Microscopy and Microanalysis*, Volume 29 (August 2023)
- [218] X. Ma et al., "A 2-Transistor-2-Capacitor Ferroelectric Edge Compute-in-Memory Scheme With Disturb-Free Inference and High Endurance," in *IEEE Electron Device Letters*, vol. 44, no. 7, pp. 1088-1091 (July 2023)
- [217] Jiaying Ji et al, "Large-Scale Cardiac Muscle Cell-Based Coupled Oscillator Network for Vertex Coloring Problem" *Advanced Intelligent Systems* (February 2023)
- [216] Yaoqiao Hu, Darrell Schlom, Suman Datta, Kyeongjae Cho, Amorphous Ta₂SnO₆: A hole-dopable p-type oxide, *Applied Surface Science*, Volume 613 (2023)
- [215] Yaoqiao Hu, Darrell Schlom, Suman Datta, Kyeongjae Cho, Ilmenite and amorphous SnTiO₃ as p-type oxide semiconductors, *Journal of Materials Chemistry C* (2023#)
- [214] Y Hu, D Schlom, S Datta, K Cho "Interlayer Engineering of Band Gap and Hole Mobility in p-Type Oxide SnO" *ACS Applied Materials & Interfaces* (May 2022)
- [213] Y Luo, S Dutta, A Kaul, S Kyu Lim, M Bakir, S Datta, S Yu "A Compute-in-Memory Hardware Accelerator Design with Back-end-of-line (BEOL) Transistor based Reconfigurable Interconnect" *IEEE Journal on Emerging and Selected Topics in Circuits and Systems* (May 2022)
- [212] S Dutta, G Detorakis, A Khanna, B Grisafe, E Neftci, S Datta "Neural sampling machine with stochastic synapse allows brain-like learning and inference" *Nature Communications* (May 2022)
- [211] S Cheema, N Shanker, L Wang, C Hsu, S Hsu, Y Liao.... S Salahuddin "Ultrathin ferroic HfO₂–ZrO₂ superlattice gate stack for advanced transistors" *Nature* (April 2022)
- [210] Y Lee, Y Hu, D Kim, S Datta, K Cho "First-principles mobility prediction for amorphous semiconductors" *Physical Review B* (Feb 2022)
- [209] S Dutta, H Ye, AA Khandker, SG Kirtania, A Khanna, K Ni, S Datta "Logic Compatible High-Performance Ferroelectric Transistor Memory" *IEEE Electron Device Letters* (Feb 2022)
- [208] KA Aabrar, SG Kirtania, FX Liang, J Gomez, M San Jose, Y Luo, H Ye, S Dutta, P Ravikumar, P Ravindran, A Khan, S Yu, S Datta "BEOL-Compatible Superlattice FEFET Analog Synapse With Improved Linearity and Symmetry of Weight Update" *IEEE Transactions on Electron Devices* (Jan 2022)

- [207] DV Christensen, R Dittmann, B Linares-Barranco, A Sebastian, M Le Gallo, A Redaelli, S Slesazeck, T Mikolajick, S Spiga, S Menzel, I Valov, G Milano, C Ricciardi, S Liang, F Miao, M Lanza, T J Quill, S T Keene, A Salleo, ... N Pryds "2022 roadmap on neuromorphic computing and engineering" *Neuromorphic Computing and Engineering* (Jan 2022)
- [206] Y Liao, K Akif Aabrar, W Chakraborty, W Li, S Datta, and S Salahuddin "Large Injection Velocities in Highly Scaled, Fully Depleted Silicon on Insulator Transistors" *IEEE Electron Device Letters* (Dec 2021)
- [205] R Saligram, W Chakraborty, N Cao, Y Cao, S Datta, A Raychowdhury "Power Performance Analysis of Digital Standard Cells for 28nm Bulk CMOS at Cryogenic Temperature using BSIM models" *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits* (Nov 2021)
- [204] W Chakraborty, K Akif Aabrar, J Gomez, R Saligram, A Raychowdhury, P Fay, and S Datta "Characterization and Modeling of 22nm FDSOI Cryogenic RF CMOS" *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits* (Nov 2021)
- [203] J A Smith, K Ni, H Takeuchi, RJ Stephenson, Y Chen, M Hytha, S Li, P E Nicollian, R J Mears, and S Datta "Intermixing reduction in ultra-thin titanium nitride/hafnium oxide film stacks grown on oxygen-inserted silicon and associated reduction of the interface charge dipole," *Journal of Applied Physics* (Nov 2021)
- [202] R Saligram, S Datta, A Raychowdhury "CryoMem: A 4–300-K 1.3-GHz Hybrid 2T-Gain-Cell-Based eDRAM Macro in 28-nm Logic Process for Cryogenic Applications" *IEEE Solid-State Circuits Letters* (Oct 2021)
- [201] R Saligram, S Datta, and A Raychowdhury "Scaled Back End of Line Interconnects at Cryogenic Temperatures" *IEEE Electron Device Letters* 42 (11), 1674-1677 (Oct 2021)
- [200] S Dutta, A Khanna, AS Assoa, H Paik, DG Schlom, Z Toroczkai, A Raychowdhury, and S Datta "An Ising Hamiltonian solver based on coupled stochastic phase-transition nano-oscillators" *Nature Electronics*, (July 2021)
- [199] M Si, A Murray, Z Lin, J Andler, J Li, J Noh, S Alajlouni, C Niu, X Lyu, D Zheng, K Maize, A Shakouri, S Datta, R Agrawal, and PD Ye "BEOL Compatible Indium-Tin-Oxide Transistors: Switching of Ultrahigh-Density 2-D Electron Gas Over $0.8 \times 10^{14}/\text{cm}^2$ at Oxide/Oxide Interface by the Change of Ferroelectric Polarization" *IEEE Transactions on Electron Devices* (July 2021)

- [198] S Dutta, H Ye, A Khanna, Y Luo, L Pentecost, A A Khandker, W Chakraborty, G Wei, D Brooks, M Niemier, X Sharon Hu, S Yu, K Ni, and S Datta "Logic Compatible High-Performance Ferroelectric Transistor Memory" *IEEE Electron Device Letters* (May 2021)
- [197] D V Christensen ... N Pryds "2021 roadmap on neuromorphic computing and engineering" arXiv preprint arXiv:2105.05956 (May 2021)
- [196] S Cheema, N Shanker, L Wang, C Hsu, S Hsu, Y Liao, M San Jose, J Gomez, W Li, J Bae, S Volkman, D Kwon, Y Rho, G Pinelli, Ra Rastogi, D Pipitone, C Stull, M Cook, B Tyrrell, V Stoica, Z Zhang, J Freeland, C Tassone, A Mehta, G Soheli, D Thompson, D Ik Suh, W Koo, K Nam, D Jung, W Song, C Lin, S Nam, J Heo, C Grigoropoulos, P Shafer, P Fay, R Ramesh, Jim Ciston, Suman Datta, Mohamed Mohamed, C Hu, S Salahuddin "Atomic-scale ferroic HfO₂-ZrO₂ superlattice gate stack for advanced transistors" *Nature* (April 2021)
- [195] X Ren, J Gomez, M K Bashar, J Ji, U Can, H Chang, N Shukla, S Datta, and P Zorlutuna "Cardiac Muscle Cell-Based Coupled Oscillator Network for Collective Computing" *Advanced Intelligent Systems* (April 2021)
- [194] N Tasneem, P Venkatesan Ravindran, Z Wang, J Gomez, J Hur, S Yu, S Datta, and A Khan "Differential charge boost in hysteretic ferroelectric–dielectric heterostructure capacitors at steady state" *Applied Physics Lett.* 118, 122901 (March 2021)
- [193] M Si, A Murray, Z Lin, J Andler, J Li, J Noh, S Alajlouni, C Niu, X Lyu, D Zheng, K Maize, A Shakouri, S Datta, R Agrawal, and P Ye "BEOL Compatible Indium-Tin-Oxide Transistors: Switching of Ultrahigh-Density 2-D Electron Gas Over $0.8 \times 10^{14}/\text{cm}^2$ at Oxide/Oxide Interface by the Change of Ferroelectric Polarization" *IEEE Transactions on Electron Devices* (March 2021)
- [192] B Chakrabarti, H Chan, K Alam, A Koneru, T E Gage, L E Ocola, R Divan, D Rosenmann, A Khanna, B Grisafe, T Sanders, S Datta, I Arslan, S KRS Sankaranarayan, and S Guha "Nanoporous Dielectric Resistive Memories Using Sequential Infiltration Synthesis" *ACS Nano* March 2021
- [191] S Dutta, G Detorakis, A Khanna, B Grisafe, E Neftci, and S Datta "Neural Sampling Machine with Stochastic Synapse allows Brain-like Learning and Inference" arXiv preprint arXiv:2102.10477
- [190] S Dutta, B Grisafe, C Frentzel, Z Enciso, M San Jose, J Smith, K Ni, S Joshi, and S Datta "Experimental Demonstration of Gate-Level Logic Camouflaging and Run-Time Reconfigurability

Using Ferroelectric FET for Hardware Security" *IEEE Transactions on Electron Devices* 68 (2), 516-522 Jan 2021

[189] Y Hu, X Yao, DG Schlom, S Datta, K Cho "First Principles Design of High Hole Mobility p-Type Sn–O–X Ternary Oxides: Valence Orbital Engineering of Sn²⁺ in Sn²⁺–O–X by Selection of Appropriate Elements X" *Chemistry of Materials* Dec 2020

[188] W Chakraborty, H Ye, B Grisafe, I Lightcap, and S Datta "Low Thermal Budget (<250 °C) Dual-Gate Amorphous Indium Tungsten Oxide (IWO) Thin-Film Transistor for Monolithic 3-D Integration" *IEEE Transactions on Electron Devices* Vol, 67. No 12, pp 5336-5342 Nov 2020

[187] K Berggren, Q Xia, K K Likharev, D B Strukov,....., J Yang, K Roy, S Datta, and A Raychowdhury "Roadmap on Emerging Hardware and Technology for Machine Learning" *Nanotechnology* Vol 32 Issue 1 pp 012002 Oct 2020

[186] A Khan, A Keshavarzi, and S Datta "The Future of Ferroelectric Field-effect Transistor Technology" *Nature Electronics* Oct 2020

[185] A Keshavarzi, K Ni, W Van Den Hoek, S Datta, and A Raychowdhury "FerroElectronics for Edge Intelligence" *IEEE Micro* Sept 2020

[184] M Si, J Andler, X Lyu, C Niu, S Datta, R Agrawal, and P Ye "Indium-Tin-Oxide Transistors with One Nanometer Thick Channel and Ferroelectric Gating" *ACS Nano* Aug 2020

[183] P Wang, Z Wang, X Sun, J Hur, S Datta, A Khan, and S Yu "Investigating Ferroelectric Minor Loop Dynamics and History Effect--Part II: Physical Modeling and Impact on Neural Network Training" *IEEE Transactions on Electron Devices*, DOI: 10.1109/TED.2020.3009956 Volume: 67 , Issue: 9 , Sept. 2020

[182] S Dutta, A Khanna, H Paik, D Schlom, A Raychowdhury, Z Toroczkai, and S Datta "Ising Hamiltonian Solver using Stochastic Phase-Transition Nano-Oscillators" arXiv preprint arXiv:2007.12331 Jul 24, 2020

[181] C Wu, H Ye, B Grisafe, S Datta and P Fay "Ferroelectric Polarization Switching Behavior of Hf_{0.5}Zr_{0.5}O₂ Gate Dielectrics on Gallium Nitride High-Electron-Mobility-Transistor Heterostructures" *Physica Status Solidi (a)* <https://doi.org/10.1002/pssa.201900717> April 2020

[180] S Dutta, C Shafer, J Gomez, K Ni, S Joshi, S Datta "Supervised Learning in All FeFET-Based Spiking Neural Network: Opportunities and Challenges" *Frontiers in Neuroscience* Jun 2020 DOI:10.3389/fnins.2020.00634

- [179] A Kazemi, R Rajaei, K Ni, S Datta, M Niemier, XS Hu "A Hybrid FeMFET-CMOS Analog Synapse Circuit for Neural Network Training and Inference" arXiv preprint arXiv:2004.00703
- [178] B Bhar, A Khanna, A Parihar, S Datta, A Raychowdhury "Stochastic Resonance in insulator-Metal-transition Systems" *Scientific Reports* 10 (1), 1-11
- [177] P Wang, W Shim, Z Wang, J Hur, S Datta, AI Khan, S Yu "Drain-Erase Scheme in Ferroelectric Field Effect Transistor--Part II: 3-D-NAND Architecture for In-Memory Computing" *IEEE Transactions on Electron Devices* Volume: 67 , Issue: 3 , Mar. 2020
- [176] P Wang, W Shim, Z Wang, J Hur, S Datta, AI Khan, S Yu "Drain-Erase Scheme in Ferroelectric Field-Effect Transistor--Part I: Device Characterization" *IEEE Transactions on Electron Devices* Volume: 67 , Issue: 3 , March 2020
- [175] Y Liang, Z Zhu, X Li, SK Gupta, S Datta, V Narayanan "Mismatch of Ferroelectric Film on Negative Capacitance FETs Performance" *IEEE Transactions on Electron Devices* Volume: 67 , Issue: 3 , Mar. 2020
- [174] E Corti, A Khanna, K Niang, J Robertson, K Moselund, B Gotsamnn, S Datta, S Karg "Time-delay encoded image recognition in a network of resistively-coupled VO₂ on Si oscillators" *IEEE Device Letters* Feb 2020 DOI: 10.1109/LED.2020.2972006
- [173] C Wu, H Ye, N Shaju, J Smith, B Grisafe, S Datta, and P Fay "Hf_{0.5}Zr_{0.5}O₂ Based Ferroelectric Gate HEMTs (FeHEMTs) with Large Threshold Voltage Tuning Range" *IEEE Electron Device Letters* Jan 9, 2020 DOI: 10.1109/LED.2020.2965330.
- [172] J Park, H Paik, K Nomoto, Ki Lee, B Park, B Grisafe, L Wang, S Salahuddin, S Datta, Y Kim, D Jena, HG Xing, and D Schlom "Fully transparent field-effect transistor with high drain current and on-off ratio" *APL Materials* Vol 8, 2020 Issue DOI: 10.1063/1.5133745
- [171] K Ni, X. Yin, AF Laguna, S Joshi, S Dunkel, M Trentzsch, J Muller, S Beyer, M Niemier, X S Hu, and S Datta "Ferroelectric ternary content-addressable memory for one-shot learning" *Nature Electronics* Nov 2019 2 (11), 521-529
- [170] X Chen, S Datta, XS Hu, M Jerry, AF Laguna, K Ni, M Niemier, D Reis, X Sun, P Wang, X Yin, and S Yu " The Impact of Ferroelectric FETs on Digital and Analog Circuits and Architectures" *IEEE Design & Test* Oct 2019 DOI: 10.1109/MDAT.2019.2944094
- [169] I Yoon, M Jerry, S Datta, and A Raychowdhury "Design Space Exploration of Ferroelectric FET based Processing -in-Memory" Aug 2019 arXiv preprint arXiv:1908.07942

- [168] V Misra, A Bozkurt, BH Calhoun, S Datta, M Dickey, M Kiani, J Lach, B Lee, J Jur, O Oralkan, M Ozturk, R. Rajagopalan, S Roundy, J Strohmaier, S Trolier-McKinstry, D Vashaee, D Wentzloff and D Werner "Optimizing the energy balance to achieve autonomous self-powering for vigilant health and IoT applications" *Journal of Physics: Conference Series* 1407 (1), 012001
- [167] Y Hu, J Hwang, Y Lee, P Conlin, DG Schlom, S Datta, and K Cho, "First principles calculations of intrinsic mobilities in tin-based oxide semiconductors SnO, SnO₂, and Ta₂SnO₆" *Journal of Applied Physics* 126 (18), 185701
- [166] C Wu, H Ye, B Grisafe, S Datta, and P Fay, "Ferroelectric Polarization Switching Behavior of Hf_{0.5}Zr_{0.5}O₂ Gate Dielectrics on Gallium Nitride High-Electron-Mobility-Transistor Heterostructures" *Physica Status Solidi A-Applications and Materials Science* 09 Nov. 2019 <https://doi.org/10.1002/pssa.201900717>
- [165] S Datta, S Dutta, B Grisafe, J Smith, S Srinivasa, and H Ye, "Back-End-of-Line Compatible Transistors for Monolithic 3-D Integration", *IEEE Micro*, vol 39 , Issue: 6 , 2019
- [164] Y Liang, Z Zhu, X Li, SK Gupta, S Datta, and V Narayanan, "Utilization of Negative-Capacitance FETs to Boost Analog Circuit Performances", *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol , no , pp 2019
- [163] B Grisafe, M Jerry, J A Smith, and S Datta, "Performance Enhancement of Ag/HfO₂ Metal Ion Threshold Switch Cross-point Selectors", *IEEE Electron Device Letters*, vol 40, no 10, pp 1602-1605, 2019
- [162] S Dutta, A Parihar, A Khanna, J Gomez, W Chakraborty, M Jerry, B Grisafe, A Raychowdhury, and S Datta, "Programmable coupled oscillators for synchronized locomotion" *Nature communications*, vol 10, no 1, pp 3299, 2019
- [161] D Reis, K Ni, W Chakraborty, X Yin, M Trentzsch, S D unkel, T Melde, J M uller, S Beyer, S Datta, M Niemier, and X S Hu, "Design and analysis of an ultra-dense, low-leakage and fast FeFET-based random access memory array", *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, July 2019
- [160] I Yoon, M Chang, K Ni, M Jerry, S Gangopadhyay, G H Smith, T Hamam, J Romberg, V Narayanan, A Khan, S Datta, and A Raychowdhury, "A FerroFET based in-memory processor for solving distributed and iterative optimizations via least-squares method", *IEEE Journal on Exploratory Solid-State Computational Devices and Circuits*, July 2019

- [159] AK Saha, K Ni, S Dutta, S Datta, and S Gupta, “Phase field modeling of domain dynamics and polarization accumulation in ferroelectric HZO”, *Applied Physics Letters* vol 114, no 20, pp 202903, 2019
- [158] Y Fang, J Gomez, Z Wang, S Datta, AI Khan, and A Raychowdhury, “Neuro-mimetic Dynamics of a Ferroelectric FET Based Spiking Neuron” *IEEE Electron Device Letters*, vol 40, no 7, pp 1213-1216, 2019
- [157] X Li, J Wu, K Ni, S George, K Ma, J Sampson, S K Gupta, Y Liu, H Yang, S Datta, and V Narayanan, “Design of 2T/cell and 3T/cell nonvolatile memories with emerging ferroelectric FETs”, *IEEE Design and Test*, vol 36, no 3, pp 39-45, 2019
- [156] Z Shen, S Srinivasa, A Aziz, S Datta, V Narayanan, and S K Gupta, “SRAMs and DRAMs With Separate Read–Write Ports Augmented by Phase Transition Materials”, *IEEE Transactions on Electron Devices*, vol 66, no 2, pp 929-937, 2019
- [155] Y Fang, Z Wang, J Gomez, S Datta, AI Khan, and A Raychowdhury, “A Swarm Optimization Solver Based on Ferroelectric Spiking Neural Networks” *Frontiers Neuroscience* 13, pp 855, 2019
- [154] R Zhao, B Grisafe, R Ghosh, K Wang, S Datta, and J Robinson, “Stabilizing the commensurate charge-density wave in 1T-tantalum disulfide at higher temperatures via potassium intercalation” *Nanoscale*, vol 11, no 13, pp 6016-6022, 2019
- [153] Zhesheng Shen, Srivatsa Srinivasa, Ahmedullah Aziz, Suman Datta, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, “SRAMs and DRAMs With Separate Read-Write Ports Augmented by Phase Transition Materials”, *IEEE Transactions on Electron Devices*, vol 66, no 2, pp. 929-937, 2019
- [152] Arijit Raychowdhury, Abhinav Parihar, Gus Henry Smith, Vijaykrishnan Narayanan, György Csaba, Matthew Jerry, Wolfgang Porod, Suman Datta, “Computing With Networks of Oscillatory Dynamical Systems”, *Proceedings of the IEEE*, vol 107, no 1, pp. 73-89, 2019
- [151] Xunzhao Yin, Kai Ni, Dayane Reis, Suman Datta, Michael Niemier, Xiaobo Sharon Hu, “An Ultra-dense 2FeFET TCAM Design based on a Multi-Domain FeFET Model”, *IEEE Transactions on Circuits and Systems*, 2018
- [150] Yuhua Liang, Xueqing Li, Sumeet Kumar Gupta, Suman Datta, Vijaykrishnan Narayanan, “Analysis of DIBL Effect and Negative Resistance Performance for NCFET Based on a Compact SPICE Model”, *IEEE Transactions on Electron Devices*, vol 65, no 12, pp. 5525-5529, 2018

- [149] Kai Ni, Xueqing Li, Jeffrey A Smith, Matthew Jerry, Suman Datta, “Write Disturb in Ferroelectric FETs and Its Implication for 1T-FeFET AND Memory Arrays”, *IEEE Electron Device Letters*, vol 39, no 11, pp 1656-1659, 2018
- [148] Xiaoming Chen, Kai Ni, Michael T Niemier, Yinhe Han, Suman Datta, Xiaobo Sharon Hu, “Power and Area Efficient FPGA Building Blocks Based on Ferroelectric FETs”, *IEEE Transactions on Circuits & Systems*, pp1-14, 2018
- [147] J-S Liu, M Clavel, R Pandey, S Datta, Y Xie, JJ Heremans, MK Hudait, “Heterogeneous integration of InAs/GaSb tunnel diode structure on silicon using 200 nm GaAsSb dislocation filtering buffer”, *AIP Advances*, vol 8, no 10, pp 105108, 2018
- [146] Benjamin Grisafe, Rui Zhao, Ram Krishna Ghosh, Joshua A Robinson, Suman Datta, “Electrically triggered insulator-to-metal phase transition in two-dimensional (2D) heterostructures”, *Applied Physics Letters*, vol 113, no 14, pp. 142101, 2018
- [145] S Datta, “Ten nanometer CMOS logic technology”, *Nature Electronics*, vol 1, no 9, pp-500-501, 2018
- [145] Yuhua Liang, Xueqing Li, Sumitha George, Srivatsa Srinivasa, Zhangming Zhu, Sumeet Kumar Gupta, Suman Datta, Vijaykrishnan Narayanan, “Influence of Body Effect on Sample-and-Hold Circuit Design Using Negative Capacitance FET”, *IEEE Transactions on Electron Devices*, vol 65, no 9, pp. 3909-3914, 2018
- [144] Matthew Jerry, Sourav Dutta, Arman Kazemi, Kai Ni, Jianchi Zhang, Pai-Yu Chen, Pankaj Sharma, Shimeng Yu, X Sharon Hu, Michael Niemier, Suman Datta, “A Ferroelectric field effect transistor based synaptic weight cell”, *Journal of Physics D: Applied Physics*, vol 51, no 43, pp 434001-, 2018
- [143] Sayeef Salahuddin, Kai Ni, Suman Datta, “The Era of Hyper-Scaling in Electronics”, *Nature Electronics*, vol 1, no 8, pp 442, 2018
- [142] Xueqing Li, Sumitha George, Yuhua Liang, Kaisheng Ma, Kai Ni, Ahmedullah Aziz, Sumeet Kumar Gupta, John Sampson, Meng-Fan Chang, Yongpan Liu, Huazhong Yang, Suman Datta, Vijaykrishnan Narayanan, “Lowering Area Overheads for FeFET-Based Energy-Efficient Nonvolatile Flip-Flops”, *IEEE Transactions on Electron Devices*, vol 65, no 6, pp. 2670-2674, 2018
- [141] Kai Ni, Pankaj Sharma, Jianchi Zhang, Matthew Jerry, Jeffery A Smith, Kandabara Tapily, Robert Clark, Souvik Mahapatra, Suman Datta, “Critical Role of Interlayer in Hf_{0.5}Zr_{0.5}O₂

Ferroelectric FET Nonvolatile Memory Performance”, *IEEE Transactions on Electron Devices*, vol 65, no 6, pp.2461-2469, 2018

[140] Abhinav Parihar, Matthew Jerry, Suman Datta, Arijit Raychowdhury, “Stochastic IMT (insulator-metal-transition) neurons: An interplay of thermal and threshold noise at bifurcation”, *Frontiers in Neuroscience*, vol 12, pp. 210, 2018

[139] Atanu K Saha, Suman Datta, Sumeet K Gupta, ““Negative capacitance” in resistor-ferroelectric and ferroelectric-dielectric networks: Apparent or intrinsic?”, *Journal of Applied Physics*, vol 123, no 10, pp 105102-, 2018

[138] Sushant Sonde, Bhaswar Chakrabarti, Yuzi Liu, Kiran Sasikumar, Jianqiang Lin, Liliana Stan, Ralu Divan, Leonidas E Ocola, Daniel Rosenmann, Pabitra Choudhury, Kai Ni, Subramanian KRS Sankaranarayanan, Suman Datta, Supratik Guha, “Silicon compatible Sn-based resistive switching memory”, *Nanoscale*, vol 10, non 20, pp 9441-9449, 2018

[137] R Zhao, B Grisafe, RK Ghosh, S Holoviak, B Wang, K Wang, N Briggs, A. Haque, S. Datta and J. Robinson, “Two-dimensional tantalum disulfide: controlling structure and properties via synthesis”, *2D Materials* 5 (2), 025001, 2018

[136] M. Jerry, K. Ni, A. Parihar, A. Raychowdhury and S. Datta, “Stochastic Insulator-to-Metal Phase Transition based True Random Number Generator”, *IEEE Electron Device Letters*, vol 39, pp 139-142, no 1, Jan 2018

[135] P Sharma, J Zhang, K Ni, S Datta, “Time-Resolved Measurement of Negative Capacitance”, *IEEE Electron Device Letters*, vol 39, no 2, pp 272-275, Dec 2017

[134] H Takeuchi, RJ Mears, RJ Stephenson, M Hytha, D Connelly, P Fastenko, R Burton, N Cody, D Weeks, D Choutov, N. Agrawal, S. Datta, “Punch-Through Stop Doping Profile Control via Interstitial Trapping by Oxygen-Insertion Silicon Channel”, *IEEE Journal of the Electron Devices Society*, Nov, 2017

[133] PJ Drummond, A Wali, MJ Barth, AM Diehm, S Datta, J Ruzyllo, “Photoconductance Decay Characterization of 3D Multi-Fin Silicon on SOI Substrates”, *IEEE Electron Device Letters* 38 (11), 1513-1515, Nov 2017

[132] X Li, S George, K Ma, WY Tsai, A Aziz, J Sampson, SK Gupta, MF Chang, Y. Liu, S. Datta, V. Narayanan, “Advancing nonvolatile computing with nonvolatile NCFET latches and flip-flops”, *IEEE Transactions on Circuits and Systems I: Regular Papers* 64 (11), 2907-2919, Nov 2017

- [131] C Schulte-Braucks, R Pandey, RN Sajjad, M Barth, RK Ghosh, B Grisafe, Pankaj Sharma, Nils von den Driesch, Anurag Vohra, Gilbert Bruce Rayner, Roger Loo, Siegfried Mantl, Dan Buca, Chih-Chieh Yeh, Cheng-Hsien Wu, Wilman Tsai, Dimitri A Antoniadis, Suman Datta Fabrication, Characterization, and Analysis of Ge/GeSn Heterojunction p-Type Tunnel Transistors”, *IEEE Transactions on Electron Devices* 64 (10), 4354-4362, Oct 2017
- [130] D Li, AA Sharma, N Shukla, H Paik, JM Goodwill, S Datta, DG Schlom, J. Bain, M. Skowronski,”ON-state evolution in lateral and vertical VO₂ threshold switching devices”, *Nanotechnology* 28 (40), 405201, Sept 2017
- [129] S. Gupta, M. Steiner, A. Aziz, V. Narayanan, S. Datta, S. Gupta, “Device-Circuit Analysis of Ferroelectric FETs for Low-Power Logic”, *IEEE Transactions on Electron Devices*, vol 64, no. 8, pp. 3092-3100, 2017
- [128] X. Li, J. Sampson, A.Khan, K, Ma, S. George, A. Aziz, S.Gupta, S.Salahuddin, M.F. Chang, S.Datta, V.Narayanan, “Enabling Energy-Efficient Nonvolatile Computing With Negative Capacitance FET”, *IEEE Transactions on Electron Devices*, vol 64, no. 8, pp. 3452-3458, 2017
- [127] X. Li, K.Ma, S. George, Win-San Khwa, J.Sampson, S.Gupta, Y. Liu, M.F. Chang, Suman Datta, V. Narayanan, “Design of Nonvolatile SRAM with Ferroelectric FETs for Energy-Efficient Backup and Restore”, *IEEE Transactions on Electron Devices*, vol 64, no. 7, pp. 3037-3040, 2017
- [126] X. Li, Sumitha George, Kaisheng Ma, Wei-Yu Tsai, Ahmedullah Aziz, John Sampson, Sumeet Kumar Gupta, Meng-Fan Chang, Yongpan Liu, Suman Datta, Vijaykrishnan Narayanan, “Advancing Nonvolatile Computing With Nonvolatile NCFET Latches and Flip-Flops”, *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol xx, pp yy, pp 1-13, 2017
- [125] Yuanxia Zheng, Sungwook Hong, George Psifogiannakis, G Bruce Rayner Jr, Suman Datta, Adri CT van Duin, Roman Engel-Herbert, “Modeling and in Situ Probing of Surface Reactions in Atomic Layer Deposition”, *ACS Applied Materials & Interfaces*, vol 9, no 18, pp 15848-15856, 2017
- [124] Quentin Smets, Anne Verhulst, Ji-Hong Kim, Jason P Campbell, David Nminibapiel, Dmitry Veksler, Pragya Shrestha, Rahul Pandey, Eddy Simoen, David Gundlach, Curt Richter, Kin P Cheung, Suman Datta, Anda Mocuta, Nadine Collaert, Aaron V-Y Thean, Marc M Heyns, “Pulsed IV on TFETs: Modeling and Measurements”, *IEEE Transactions on Electron Devices*, vol 64, no 4, pp 1489-1497, 2017

- [123] Abhinav Parihar, Nikhil Shukla, Matthew Jerry, Suman Datta, Arijit Raychowdhury, "Computing with dynamical systems based on insulator-metal-transition oscillators", *Nature Scientific Reports*, vol 6, no 3, pp 601-611, 2017
- [122] A. Aziz, N. Shukla, S. Datta, S. K. Gupta, "Steep Switching Hybrid Phase Transition FETs (Hyper-FET) for Low Power Applications: A Device-Circuit Co-design Perspective–Part I", *IEEE Trans. on Elec. Dev.*, vol. 64, no. 3, pp. 1350-1357, 2017.
- [121] M. Brahlek, L. Zhang, J. Lapano, H.-T. Zhang, R. Engel-Herbert, N. Shukla, S. Datta, H. Paik, D. G. Schlom, "Opportunities in vanadium-based strongly correlated electron systems", *MRS Communications*, pp. 1-26, Feb. 2017. doi: 10.1557/mrc.2017.2.
- [120] Y. J. Li, C. Y. Huang, C. C. Wu, Y. C. Chen, C. Y. Wang, S. Datta, V. Narayanan, "Dynamic Diagnosis for Defective Reconfigurable Single-Electron Transistor Arrays", *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, vol. , no.99, pp. 1-13, 2017.
- [119] S. Advani, P. Zientara, N. Shukla, I. Okafor, K. Irick, J. Sampson, S. Datta, V. Narayanan, "A Multitask Grocery Assist System for the Visually Impaired: Smart glasses, gloves, and shopping carts provide auditory and tactile feedback", *IEEE Consumer Electronics Magazine*, vol. 6, no. 1, pp. 73-81, 2017
- [118] J.-S. Liu, M. B. Clavel, R. Pandey, S. Datta, M. Meeker, G. A. Khodaparast, and M. K Hudait, "Growth and characterization of metamorphic InAs/GaSb tunnel heterojunction on GaAs by molecular beam epitaxy", *J. Appl. Phys.*, vol. 119, no. 24, pp. 244308, 2016.
- [117] M. Barth, A. Kumar, J. H. Warner, B. R. Bennett, B. Boos, C. D. Cress, N. J-H Roche, M. Raine, M. Gaillardin, P. Paillet, D. McMorro, K. Saraswat, S. Datta, "Single-Event Measurement and Analysis of Antimony-Based p-Channel Quantum-Well MOSFETs with High- κ Dielectric", *IEEE Trans. Nuc. Sci.*, no. 99, 2016.
- [116] A. Aziz, N. Jao, S. Datta, S. K. Gupta, "Analysis of Functional Oxide based Selectors for Cross-Point Memories", *IEEE Trans. on Circuits and Systems I: Regular Papers*, vol. 63, no. 12, pp. 2222-2235, 2016.
- [115] R. Pandey, S. Mookerjee, S. Datta, "Opportunities and Challenges of Tunnel FETs", *IEEE Trans. Circuits and Systems I: Regular Paper*, vol. 63 no. 12, pp. 2128-2138, 2016.
- [114] S. Srinivasa, A. Aziz, N. Shukla, X. Li, J. Sampson, S. Datta, J. P. Kulkarni, V. Narayanan, S. K. Gupta, "Correlated Material Enhanced SRAMs With Robust Low Power Operation", *IEEE Trans. Elec. Dev.* vol. 63, no. 12, pp. 4744-4752, 2016.

- [113] C.H. Ho, Y.C. Chen, C.Y. Wang, C.Y. Huang, S. Datta, V. Narayanan, "Area-Aware Decomposition for Single-Electron Transistor Arrays", *ACM Trans. on Design Automation of Electronic Systems (TODAES)*, vol. 21, no. 4, pp. 70, 2016.
- [112] H.-T. Zhang, L. Guo, G. Stone, L. Zhang, Y.-X. Zheng, E. Freeman, D. W. Keefer, S. Chaudhuri, H. Paik, J. A. Moyer, M. Barth, D. G. Schlom, J. V. Badding, S. Datta, V. Gopalan, R. Engel-Herbert, "Imprinting of Local Metallic States into VO₂ with Ultraviolet Light", *J. Adv. Func. Mat.*, vol. 26, no. 36, pp. 6612-6618, 2016.
- [111] Z. Y. A. Balushi, K. Wang, R. K. Ghosh, R. A. Vilá, S. M. Eichfeld, J. D. Caldwell, X. Qin, Y.-C. Lin, P. A. DeSario, G. Stone, S. Subramanian, D. F. Paul, R. M. Wallace, S. Datta, J. M. Redwing, J. A. Robinson, "Two-dimensional gallium nitride realized via graphene encapsulation", *Nature Materials*, vol. 15, pp. 1166–1171, 2016.
- [110] M.S. Kim, W. C.-Wissing, X. Li, J. Sampson, S. Datta, S.K. Gupta, V. Narayanan, "Comparative area and parasitics analysis in FinFET and heterojunction vertical TFET standard cells", *ACM J. on Emerging Tech. in Computing Systems (JETC)*, vol. 12, no. 4, pp. 38, 2016.
- [109] C.Y. Huang, Y.J. Li, C.W. Liu, C.Y. Wang, Y.C. Chen, S. Datta, V. Narayanan, "Diagnosis and Synthesis for Defective Reconfigurable Single-Electron Transistor Arrays", *IEEE Trans. on Very Large Scale Integration (VLSI) Systems*, vol. 24, no. 6, pp. 2321-2334, 2016.
- [108] JU Mehta, WA Borders, H Liu, R Pandey, S Datta, L Lunardi, "III–V tunnel FET model with closed-form analytical solution", *IEEE Trans. Elec. Dev.*, vol. 63 no. 5, pp. 2163-2168, 2016.
- [107] D Li, AA Sharma, DK Gala, N Shukla, H Paik, S Datta, DG Schlom, J. A Bain, M. Skowronski, "Joule Heating-Induced Metal–Insulator Transition in Epitaxial VO₂/TiO₂ Devices", *ACS Appl. materials & interfaces*, vol. 8 no. 20, pp. 12908-12914, 2016.
- [106] K Majumdar, S Datta, SP Rao, "Revisiting the theory of ferroelectric negative capacitance", *IEEE Trans. Elec. Dev.*, vol. 63 no. 5, pp. 2043-2049, 2016.
- [105] W.-Y. Tsai, X. Li, M. Jerry, B. Xie, N. Shukla, H. Liu, N. Chandramoorthy, M. Cotter, A. Raychowdhury, D. M Chiarulli, S. P Levitan, S. Datta, J. Sampson, N. Ranganathan, and V. Narayanan, "Enabling new computation paradigms with hyperFET-an emerging device", *IEEE Trans. on Multi-Scale Computing Sys.*, vol. 2, no. 1, pp. 30-48, 2016.
- [104] A. Aziz, S. Ghosh, S. Datta and S. K. Gupta, "Physics-Based Circuit-Compatible SPICE Model for Ferroelectric Transistors," *IEEE Elec. Dev. Lett.*, vol. 37, no. 6, pp. 805-808, June 2016.

- [103] M. S. Kim, X. Li, H. Liu, J. Sampson, S. Datta, and V. Narayanan, "Exploration of low-power high-SFDR current-steering D/A converter design using steep-slope Heterojunction Tunnel FETs," *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, VOL. 24, NO. 6, pp. 2299-2308, JUNE 2016.
- [102] W.-Y. Tsai, X. Li, M. Jerry, B. Xie, N. Shukla, H. Liu, N. Chandramoorthy, M. Cotter, A. Raychowdhury, D. M. Chiarulli, S. P. Levitan, S. Datta, J. Sampson, N. Ranganathan, and V. Narayanan, "Enabling new computation paradigms with Hyper-FET – an emerging device," *IEEE Transactions on Multi-Scale Computing Systems*, Vol 2, NO. 1, pp 30-48, Jan-Mar 2016.
- [101] M. S. Kim, W. Cane-Wissing, X. Li, and J. Sampson, S. Datta, S. K. Gupta and V. Narayanan, "Comparative Area and Parasitics Analysis in FinFET and Heterojunction Vertical TFET Standard Cells" *ACM Journal on Emerging Technologies in Computing Systems*, Vol. 12, No. 4, Article 38, May 2016.
- [100] K. Martens, J. W. Jeong, N. Aetukuri, C. Rettner, N. Shukla, E. Freeman, D. N. Esfahani, F. M. Peeters, T. Topuria, P. M. Rice, A. Volodin, B. Douhard, W. Vandervorst, M. G. Samant, S. Datta, and S. S. P. Parkin, "Field Effect and Strongly Localized Carriers in the Metal-Insulator Transition Material VO₂", *Physical Review Letters*, Nov 6, 2015.
- [99] H. Paik, J. A. Moyer, T. Spila, J.W. Tashman, J. A. Mundy, E. Freeman, N. Shukla, J.M. Lapano, R. Engel-Herbert, W. Zander, J. Schubert, D.A. Muller, S. Datta, P. Schiffer, and D. G. Schlom, "Transport properties of ultra-thin VO₂ films on (001) TiO₂ grown by reactive molecular-beam epitaxy" *Applied Physics Letters* 106, 163101, Oct 19, 2015.
- [98] Y. X. Zheng, A. Agrawal, G. B. Rayner, Jr., M. J. Barth, K. Ahmed, S. Datta, and R. Engel-Herbert "In Situ Process Control of Trilayer Gate-Stacks on p-Germanium With 0.85-nm EOT", *IEEE Electron Device Lett.*, vol. 36, no. 9, pp 881-883, Sep. 2015
- [97] J. U. Mehta, W. A. Borders, H. Liu, R. Pandey, S. Datta, and L. Lunardi, "III–V Tunnel FET Model With Closed-Form Analytical Solution", *IEEE Trans. Elec. Dev.*, vol. , no. , pp , Sept. 2015.
- [96] P. Maffezzoni, L. Daniel, N. Shukla, S. Datta, A. Raychowdhury, "Modeling and Simulation of Vanadium Dioxide Relaxation Oscillators", *IEEE Trans. Circuits and Systems*, vol. 62, no. 9, pp 2207-2215, Sept. 2015.
- [95] P. Maffezzoni, L. Daniel, N. Shukla, S. Datta, A. Raychowdhury and V. Narayanan, "Modelling hysteresis in vanadium dioxide oscillators", *IET Electron. Lett.*, vol. 51, pp 819-820, May 2015.

- [94] S. Dasgupta, A. Rajashekhar, K. Majumdar, N. Agrawal, A. Razavieh, S. Trolier-Mckinstry, and S. Datta, "Sub-kT/q Switching in Strong Inversion in PbZr_{0.52}Ti_{0.48}O₃ Gated Negative Capacitance FETs", *IEEE J. Exploratory Solid-State Comp. Dev. and Cir.*, vol. 1, pp 43-48, Aug. 2015.
- [93] N. Shukla, A. V. Thathachary, A. Agrawal, H. Paik, A. Aziz, D. G. Schlom, S. K. Gupta, R. Engel-Herbert and S. Datta, "A steep-slope transistor based on abrupt electronic phase transition", *Nature Comm.*, vol. 6, pp 7812, Jun. 2015.
- [92] Y.-C. Lin, R. K. Ghosh, R. Addou, N. Lu, S. M. Eichfeld, H. Zhu, M.-Y. Li, X. Peng, M. J. Kim, L.-J. Li, R. M. Wallace, S. Datta, and J. A. Robinson, "Atomically thin resonant tunnel diodes built from synthetic van der Waals heterostructures", *Nature Comm.*, vol. 6, pp 7311, Jun. 2015.
- [91] N. Agrawal, H. Liu, R. Arghavani, V. Narayanan, and S. Datta, "Impact of Variation in Nanoscale Silicon and Non-Silicon FinFETs and Tunnel FETs on Device and SRAM Performance", *IEEE Trans. Electron Devices*, vol. 62, no. 6, pp 1691-1697, Jun. 2015.
- [90] L. Liu, X. Li, V. Narayanan, and S. Datta, "A Reconfigurable Low-Power BDD Logic Architecture Using Ferroelectric Single-Electron Transistors", *IEEE Trans. Electron Devices*, vol. 62, no. 3, pp 1052-1057, Mar. 2015.
- [89] A. Parihar, N. Shukla, S. Datta, A. Raychowdhury, "Synchronization of pairwise-coupled, identical, relaxation oscillators based on metal-insulator phase transition devices: A Model Study", *J. Appl. Phys.* vol. 117, pp 054902 Feb. 2015
- [88] M. J. Hollander, Y. Liu, W.-J. Lu, L.-J. Li, Y.-P. Sun, J. A. Robinson, and S. Datta, "Electrically Driven Reversible Insulator–Metal Phase Transition in 1T-TaS₂", *NanoLetters* 15(3), pp 1861-1866, Jan. 2015
- [87] Himanshu Madan *, Matthew Jerry*, Alexej Pogrebnyakov , Theresa Mayer , and Suman Datta, "Quantitative Mapping of Phase Coexistence in Mott-Peierls Insulator during Electronic and Thermally Driven Phase Transition", *ACS Nano*, 9 (2), pp 2009–2017, January 2015 (First and second authors supervised by the candidate)
- [86] A. V. Thathachary*, G. Lavalley, M. Cantoro, K. K. Bhuiwarka, Y.C. Yeo, S. Maeda and S. Datta, "Impact of Sidewall Passivation and Channel Composition on In_xGa_{1-x}As FinFET Performance", *IEEE Electron Device Letters*, vol 36, no 2, pp 117, February 2015 (First author supervised by the candidate)

- [85] N. Agrawal*, A. V. Thathachary*, S. Mahapatra and S. Datta, "Impact of Varying Indium(x) Concentration and Quantum Confinement on PBTI Reliability in $\text{In}_x\text{Ga}_{1-x}\text{As}$ FinFET", *IEEE Electron Device Letters*, vol 36, no 2, pp 120, January 2015 (First and second authors supervised by the candidate)
- [84] Bijesh Rajamohanan*, Rahul Pandey*, Varistha Chobpattana, Canute Vaz, David Gundlach, Kin P. Cheung, John Suehle, Susanne Stemmer, and Suman Datta, "0.5 V Supply Voltage Operation of $\text{In}_{0.65}\text{Ga}_{0.35}\text{As}/\text{GaAs}_{0.4}\text{Sb}_{0.6}$ Tunnel FET", *IEEE Electron Device Letters*, vol 36, no 1, pp January 2015 (First and second authors supervised by the candidate)
- [83] M. Barth*, G. B. Rayner, S McDonnell, R.M. Wallace, B.R. Bennett, R. Engel-Herbert, and S. Datta "High quality $\text{HfO}_2/\text{p-GaSb}(001)$ metal-oxide-semiconductor capacitors with 0.8nm equivalent oxide thickness", *Applied Physics Letters*, 105, pp 222103, Dec 2, 2014 (First author supervised by the candidate)
- [82] Abhinav Parihar, Nikhil Shukla, Suman Datta, and Arijit Raychowdhury, "Exploiting Synchronization Properties of Correlated Electron Devices in a Non-Boolean Computing Fabric for Template Matching", *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol4, no 4, pp 400-411, December 2014 (Second author supervised by the candidate)
- [81] Yu-Chuan Lin, Chih-Yuan S. Chang, Ram Krishna Ghosh*, Jie Li,§ Hui Zhu, Rafik Addou, Bogdan Diaconescu, Taisuke Ohta, Xin Peng, Ning Lu, Moon J. Kim, Jeremy T. Robinson, Robert M Wallace, Theresa S. Mayer, Suman Datta, Lain-Jong Li, and Joshua A. Robinson, "Atomically Thin Heterostructures Based on Single-Layer Tungsten Diselenide and Graphene", *Nano Letters*, vol 14, pp 6936-6941 November 2014 (Third author supervised by the candidate)
- [80] W. Li, Q. Zhang, R. Bijesh*, O.A. Kirillov, Y. Liang, I. Levin, Lian-Mao Peng, C. A. Richter, X. Liang, S. Datta, D. J. Gundlach, and N. V. Nguyen "Electron and hole photoemission detection for band offset determination of tunnel field-effect transistor heterojunctions", *Applied Physics Letters* 105, 213501, November 10, 2014 (Third author supervised by the candidate)
- [79] M. S. Kim, H. Liu*, X. Li, S. Datta, and V. Narayanan, "A Steep-Slope Tunnel FET Based SAR Analog-to-Digital Converter", *IEEE Transactions on Electron Devices*, vol. 61, no.11, pp: 3661-3666, November 2014 (Second author supervised by the candidate)
- [78] X. Li, H. Liu*, S. Datta, R. Vaddi, V.Narayanan, K. Ma, "Tunnel FET RF Rectifier Design for Energy Harvesting Application", *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol 4, no 4, pp 400-411, October 2014 (Second author supervised by the candidate)

- [77] A. R. Trivedi, S. Datta, and S. Mukhopadhyay "Application of Silicon-Germanium Source Tunnel-FET to Enable Ultralow Power Cellular Neural Network-Based Associative Memory", *IEEE Transactions on Electron Devices*, vol. 61, no.11, pp 3707-3715, November 2014 (equal contribution by all authors)
- [76] M. Huefner, R. Ghosh*, E. Freeman*, N. Shulka*, H. Paik, D. G. Schlom, and S. Datta "Hubbard Gap Modulation in Vanadium Dioxide Nanoscale Tunnel Junctions", *Nano Letters*, vol 14, no 11, pp 6115-6120, September 2014 (Second, third and fourth authors supervised by the candidate)
- [75] A. Agrawal*, M. Barth*, H. Madan*, Yi-Jing Lee, You-Ru Lin, Cheng-Hsien Wu, Chih-Hsin Ko, C. H. Wann, D. Loubychev, A. Liu, J. Fastenau, J. Lindemuth, and S. Datta "Comparative analysis of hole transport in compressively strained InSb and Ge quantum well heterostructures", *Applied Physics Letters* 105, 052102, August 5, 2014 (First, second and third authors supervised by the candidate).
- [74] N. Shukla*, T. Joshi, S. Dasgupta*, P. Borisov, D. Lederman, and S. Datta "Electrically induced insulator to metal transition in epitaxial SmNiO₃ thin films", *Applied Physics Letters* 105, 012108, July 11, 2014 (First and third authors supervised by the candidate).
- [73] H. Liu*, M. Cotter*, S. Datta, and V. Narayanan, "Soft-Error Performance Evaluation on Emerging Low Power Devices", *IEEE Transactions on Device and Materials Reliability*, vol. 14, no. 2, June 3, 2014 (First author supervised by the candidate).
- [72] N. Shukla*, A. Parihar, E. Freeman*, H. Paik, G. Stone, V. Narayanan, H. Wen, Z. Cai, V. Gopalan, R. Engel-Herbert, D. G. Schlom, A. Raychowdhury, and S. Datta "Synchronized charge oscillations in correlated electron systems", *Nature Scientific Reports* 4:4964, May 14, 2014 (First and third authors supervised by the candidate)
- [71] A. Agrawal*, J. Lin, M. Barth*, R. White, B. Zheng, S. Chopra, S. Gupta, K. Wang, J. Gelatos, S. Mohny, and S. Datta "Fermi level depinning and contact resistivity reduction using a reduced titania interlayer in n-silicon metal-insulator-semiconductor ohmic contacts", *Applied Physics Letters* Vol.104, 112101, March 17, 2014 (First and third authors supervised by the candidate).
- [70] M. Hollander*, H. Madan*, N. Shukla*, D. Snyder, J. Robinson, and S. Datta, "Short-channel graphene nanoribbon transistors with enhanced symmetry between p- and n-branches", *Applied Physics Express*, 7, 055103, 2014 (First, second and third authors supervised by the candidate)

- [69] R. Pandey*, V. Saripalli*, J.P. Kulkarni, V. Narayanan, and S. Datta, "Impact of Single Trap Random Telegraph Noise on Heterojunction TFET SRAM Stability ", *IEEE Electron Device Letters* Vol. 35. NO. 3, March 2014 (First and second authors supervised by the candidate)
- [68] S Datta, H. Liu*, and V. Narayana "Tunnel FET technology: A reliability perspective", *Microelectronics Reliability Journal*, March 3, 2014 (Second author supervised by the candidate).
- [67] R. Pandey*, B. Rajamohanam*, H. Liu*, V. Narayanan, and S. Datta, "Electrical Noise in Heterojunction Interband Tunnel FETs", *IEEE Transactions on Electron Devices*, vol. 61, no.2, pp: 552-559, February 2014 (First, second and third authors supervised by the candidate).
- [66] A. V. Thathachary*, N. Agrawal*, L. Liu*, and S. Datta, "Electron Transport in Multigate InxGa1-x As Nanowire FETs: From Diffusive to Ballistic Regimes at Room Temperature", *Nano Letters* 14(2):626-33, Feb 2014 (First, second and third authors supervised by the candidate)
- [65] B. Rajamohanam*, D. Mohata*, Y. Zhu, M. Hudait, Z. Jiang, M. Hollander*, G. Klimeck, and S. Datta "Design, fabrication, and analysis of p-channel arsenide/antimonide hetero-junction tunnel transistors" *Journal of Applied Physics* 115, 044502. January 2014 (First, second and sixth authors supervised by the candidate)
- [64] E. Freeman*, G Stone, N. Shukla*, H. Paik, J. A. Moyer, Z. Cai, H. Wen, R. Engel-Herbert, D. G. Schlom V. Gopalan, and S. Datta "Nanoscale structural evolution of electrically driven insulator to metal transition in vanadium dioxide", *Applied Physics Letters* Vol.103, Issue 26 December 30, 2013 (First and second authors supervised by the candidate).
- [63] N. Agrawal*, Y. Kimura, R. Arghavani, and S. Datta, "Impact of Transistor Architecture (Bulk Planar, Trigate on Bulk, Ultrathin-Body Planar SOI) and Material (Silicon or III Semiconductor) on Variation for Logic and SRAM Applications", *IEEE Transactions on Electron Devices*, vol. 60, no.10, pp: 3298-3304, October 2013 (First author supervised by the candidate).
- [62] B. Rajamohanam*, D. Mohata*, D. Zhernokletov, B. Brennan, R. M. Wallace, R. Engel-Herbert, and S. Datta, "Low-Temperature Atomic-Layer-Deposited High-k Dielectric for p-Channel In_{0.7}Ga_{0.3}As/GaAs_{0.35}Sb_{0.65} Heterojunction Tunneling Field-Effect Transistor", *Applied Physics Express*, 6, 101201, 2013 (First and second authors supervised by the candidate)
- [61] C. Cress and S. Datta, "Nanoscale transistor – Just around the gate", *Science*, vol. 341, pp. 140-141, 2013 (contributed equally by all authors)
- [62] Lubyshev, Joel M. Fastenau, and Amy K. Liu "Structural, morphological, and defect properties of metamorphic In_{0.7}Ga_{0.3}As/GaAs_{0.35}Sb_{0.65} p-type tunnel field effect transistor structure grown by

molecular beam epitaxy" *Journal of Vacuum Science and Technology B* 31(4), pp 041203-1, Jul/Aug 2013 (Third author supervised by the candidate)

[61] R. Bijesh*, D. Mohata*, A. Ali*, and Suman Datta "Insight into the output characteristics of III-V tunneling field effect transistors" *Applied Physics Letters* 102, 092105 March 2013 (First, second and third authors supervised by the candidate)

[60] A Ali*, H. Madan*, M. Barth*, J. B. Boos, B. R. Bennett, and S. Datta "Effect of Interface States on the Performance of Antimonide nMOSFETs" *IEEE Electron Device Letters* Vol. 34. NO. 3, March 2013(first, second and third authors supervised by the candidate)

[59] YC Chen, S Eachempati, CY Wang, S Datta, Y Xie, V. Narayanan "A Synthesis Algorithm for Reconfigurable Single-Electron Transistor Arrays" *ACM Journal on Emerging Technologies in Computing Systems (JETC)* Volume 9 Issue 1, February 2013 (contributed equally by all authors)

[58] Ayan Kar*, Nikhil Shukla*, Eugene Freeman*, Hanjong Paik, Huichu Liu*, Roman Engel-Herbert, S. S. N. Bhardwaja, Darrell G. Schlom, and Suman Datta "Intrinsic electronic switching time in ultrathin epitaxial vanadium dioxide thin film" *Applied Physics Letters* 102, 072106 February 2013 (first, second, third and fifth authors supervised by the candidate)

[57] L. Liu*, V. Narayanan, and S.Datta "A programmable ferroelectric single electron transistor" *Appl. Phys. Lett.* 102, 053505 February 2013 (First author supervised by the candidate)

[56] Y. Zhu, N. Jain, S. Vijayaraghavan, D. K. Mohata*, S. Datta, D. Lubyshev, J. M. Fastenau, A. K. Liu, , and M. K. Hudait "Band offset determination of mixed As/Sb type-II staggered gap heterostructure for n-channel tunnel field effect transistor application" *J. Appl. Phys.* 113, 024319 January 2013 (Fourth author supervised by the candidate)

[55] S. K. Gupta, J. P Kulkarni, S. Datta and K. Roy, "Heterojunction Intra-band Tunneling (HIBT) FETs for Low Voltage SRAMs" *IEEE Transactions on Electron Devices*, vol. 59, no.12, pp: 3533-3542, December 2012 (contributed equally by all authors)

[54] Y. Zhu, N. Jain, S. Vijayaraghavan, D. K. Mohata*, S. Datta, D. Lubyshev, J. M. Fastenau, A. K. Liu, N. Monsegue, and M. K. Hudait "Defect assistant band alignment transition from staggered to broken gap in mixed As/Sb tunnel field effect transistor heterostructure" *J. Appl. Phys.* 122, 094312 October 2012 (Fourth author supervised by the candidate)

[53] H. Madan*, V. Saripalli*, H. Liu* , and S. Datta, "Asymmetric Tunnel Field-Effect Transistors as Frequency Multipliers" *IEEE Electron Device Letters* vol. 33, no. 11, pp. 1547-1549, November 2012 (First, second and third authors supervised by the candidate)

- [52] D. K. Mohata*, R. Bijesh*, T. Mayer, J. Fastenau, D. Lubyshev, A. W. K. Liu, and S. Datta, "Barrier Engineered Arsenide-Antimonide Hetero-junction Tunnel FETs with Enhanced Drive Current" *IEEE Electron Device Letters* vol. 33, no. 11, pp. 1568-1570, November 2012 (First and second authors supervised by the candidate)
- [51] J. D. Yearsley, J. C. Lin, E. Hwang*, S. Datta, and S. E. Mohnney "Ultra low-resistance palladium silicide Ohmic contacts to lightly doped n-InGaAs" *J. Appl. Phys.* 112, 054510, September 2012 (Third author supervised by the candidate)
- [50] Y. Zhu, N. Jain, D. K. Mohata*, S. Datta, D. Lubyshev, J. M. Fastenau, A. K. Liu, and M. K. Hudait "Structural properties and band offset determination of p-channel mixed As/Sb type-II staggered gap tunnel field-effect transistor structure" *Appl. Phys. Lett.* 101, 112106 September 2012 (Third author supervised by the candidate)
- [49] R. Bijesh*, I. Ok, S. Mujumdar, C. Hobbs, P. Majhi, R. Janmmmy, and S. Datta, "Correlated Flicker Noise and Hole Mobility Characteristics of (110)/<110> Uniaxially Strained SiGe FINFETs" *IEEE Electron Device Letters*, vol. 33, no. 09, pp. 1237-1239, September 2012. (First author supervised by the candidate)
- [48] F. Li*, R. Misra, Zhao Fang*, Y. Wu, P. Schiffer, Q. M. Zhang, S. Tadigadapa and S. Datta, "Magnetoelectric Flexural Gate Transistor with NanoTesla Sensitivity" *IEEE Journal of Microelectromechanical Systems (MEMS)*, vol 22, No 1, February 2013 (first and third authors supervised by the candidate)
- [47] Y. Zhu, N. Jain, S. Vijayaraghavan, D. K. Mohata*, S. Datta, D. Lubyshev, J. M. Fastenau, W. K. Liu, N. Monsegue, and M. K. Hudait, "Role of InAs and GaAs terminated heterointerfaces at source/channel on the mixed As-Sb staggered gap tunnel field effect transistor structures grown by molecular beam epitaxy" *Journal of Applied Physics*, 112, 024306 July 2012 (Fifth author supervised by the candidate).
- [46] A. Agrawal*, N. Shukla*, K. Ahmed, and S. Datta, "A Unified Model for Insulator Selection to Form Ultra-Low Resistivity Metal-Insulator-Semiconductor Contacts to n-Si, n-Ge and n-InGaAs" *Applied Physics Letters*, 101, 042108, July 2012 (first and second authors supervised by candidate)
- [45] L. Liu*, D. K. Mohata*, and S. Datta, "Scaling Length Theory of Double-Gate Interband Tunnel Field-Effect Transistors" *IEEE Transactions on Electron Devices*, 59(4), pp. 902-908, April 2012 (First and second authors supervised by candidate)

- [44] W. Cho, M. Luisier, D. K. Mohata*, S. Datta, D. Pawlik, S. L. Rommel, and G. Klimeck "Full band atomistic modeling of homo-junction InGaAs band-to-band tunneling diodes including band gap narrowing" *Applied Physics Letters*, 100(6), pp. 063504 - 063504-3, February 2012 (Third author supervised by the candidate)
- [43] S. Mujumdar*, K. Maitra, and S. Datta "Layout dependent strain optimization for p-channel Non-planar Tri-gate Transistors" *IEEE Transactions on Electron Devices*, vol 59, no. 1, pp. 72-78, January 2012. (First author supervised by the candidate)
- [42] A. Ali*, H. Madan*, A. Agrawal*, I. Ramirez, R. Misra, J. B. Boos, B. R. Bennett, J. Lindemuth and S. Datta, "Enhancement Mode Antimonide Quantum Well MOSFETs with High Electron Mobility and GHz Small-Signal Switching Performance," *IEEE Electron Device Letters*, vol 32, no 12, 1689 – 1691, December 2011 (First, second and third authors supervised by the candidate)
- [41] Matthew J Hollander*, Michael LaBella, Zachary R Hughes, Michael Zhu, Kathleen A Trumbull, Randal Cavalero, David W Snyder, Xiaojun Wang, Euichul Hwang, Suman Datta, and Joshua A Robinson, "Enhanced Transport and Transistor Performance with Oxide Seeded High-k Gate Dielectrics on Wafer-Scale Epitaxial Graphene," *Nano Letters*, p. 110804163939013, Aug. 2011. (First author supervised by the candidate)
- [40] Vinay Saripalli*, Guangyu Sun, Asit Mishra, Yuan Xie, Suman Datta and Vijaykrishnan Narayanan, "Exploiting Heterogeneity for Energy Efficiency in Chip Multiprocessors," *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, June 2011. (Invited paper) (First author supervised by the candidate)
- [39] E.Hwang*, S.Mookerjee*, M.K Hudait, S.Datta, "Investigation of scalability of In_{0.7}Ga_{0.3}As quantum well field effect transistor (QWFET) architecture for logic applications," *Solid-State Electronics*, vol. 62, pp. 82-89, August 2011. (First and second authors supervised by the candidate)
- [38] D. Mohata*, S. Mookerjee*, A. Agrawal*, Y. Li, T. Mayer, V. Narayanan, A. Liu and S. Datta, "Experimental Staggered-Source and N⁺ Pocket-Doped Channel III-V Tunnel Field-Effect Transistors and Their Scalabilities," *Applied Physics Express*, vol 4, pp. 024105, February 2011 (First, second and third authors supervised by the candidate)
- [37] A. Ali*, B. Bennett, B. Boos, H. Madan*, A. Agrawal*, P. Schiffer, R. Misra and S. Datta, , "Experimental Determination of Quantum and Centroid Capacitance in Arsenide-Antimonide Quantum-Well MOSFETs Incorporating Non-Parabolicity Effect," *IEEE Transactions on Electron*

Devices , vol. 58 , pp. 1397-1403, January 2011 (First and fourth authors supervised by the candidate)

[36] L. Liu* and S. Datta, "A Generalized Scaling Length Theory for Double Gate Inter-band Tunnel FETs," accepted for publication in *IEEE Transactions on Electron Devices*, number of pages: 6, Dec 2011 (first author supervised by the candidate)

[35] A. Vallett, S. Minassian, P. Kaszuba, S. Datta, J. M. Redwing and T.S. Mayer, "Fabrication and Characterization of Axially Doped Silicon Nanowire Tunnel Field-Effect Transistors," *NanoLetters*, vol. 10, pp. 4813-4818, November 2010.

[34] I. Geppert, M. Eizenberg, A. Ali* and S. Datta, "Band offsets determination and interfacial chemical properties of the Al₂O₃/GaSb system," *Applied Physics Letters*, vol. 97, pp. 162109, October 2010 (Third author supervised by the candidate)

[33] W.C. Kao*, A. Ali, E. Hwang*, S. Mookerjee* and S. Datta "Effect of interface states on sub-threshold response of III-V MOSFETs, MOS HEMTs and tunnel FETs", *Solid-State Electronics*, vol.54, pp. 16665-16668, August 2010. (First, second, third and fourth authors supervised by the candidate)

[32] A. Ali*, H. S. Madan*, A. P. Kirk, R. M. Wallace, D. A. Zhao, D. A. Mourey, M. K. Hudait, T. N. Jackson, B. R. Bennett, J. B. Boos, and S. Datta, "Fermi Level Unpinning of GaSb (100) using Plasma Enhanced Atomic Layer Deposition of Al₂O₃ Dielectric," submitted to *Applied Physics Letters*, number of pages: 3, July 2010 (First and second authors supervised by the candidate)

[31] V. Saripalli*, L. Liu*, S. Datta and V. Narayanan "Energy-Delay Analysis of Single Electron Transistor Based BDD Logic," *Journal of Low Power Electronics*, Vol, 6, No. 3, pp. , October 2010 (First and second authors supervised by the candidate)(This paper featured on the cover page of the journal)

[30] F. Li*, F. Zhao*, Q. M. Zhang, and S. Datta, "Low Frequency Voltage Mode Sensing of Magnetoelectric Sensor in Package," *Electronics Letters*, Vol. 46, No. 16, pp. August 2010 (First and second authors supervised by the candidate)

[29] S. Mookerjee*, D. Mohata*, T. Mayer, V. Narayanan, S. Datta, "Temperature-Dependent "Characteristics of a Vertical In_{0.53}Ga_{0.47}As Tunnel FET," *IEEE Electron Device Letters*, Vol. 31, No. 6, pp. 564-567, June 2010. (First and second authors supervised by the candidate)

- [28] A. Ali*, H. Madan*, S. Kovesnikov, S. Oktyabrsky, R. Kambhampati, T. Heeg, D. Schlom, and S. Datta, "Small signal response of inversion layers in high mobility In_{0.53}Ga_{0.47}As MOSFETs made with thin high-k dielectrics," *IEEE Transactions on Electron Devices*, Vol. 57, No. 4, p. 742-748, April 2010. (First and second authors supervised by the candidate)
- [27] B. Downey, S. Datta and S. Mohny, "Numerical study of reduced contact resistance via nanoscale topography at metal/semiconductor interfaces," *Semiconductor Science and Technology*, Vol. 25, No. 1, pp 1-4, January 2010. (Equal contributions by all authors)
- [26] F. Li*, S. H. Lee, Z. Fang*, P. Majhi, Q. Zhang, S. K. Banerjee, and S. Datta, "Flicker Noise Improvement in 100 nm Lg Si_{0.50}Ge_{0.50} Strained Quantum-Well Transistors using Ultra-Thin Si Cap Layer," *IEEE Electron Device Letters*, vol. 31, No. 1, pp. 47-49, January 2010. (First author supervised and third author co-supervised by the candidate)
- [25] S. Mookerjee*, R. Krishnan*, S. Datta, and V. Narayanan, "On Enhanced Miller Capacitance in Inter-Band Tunnel Transistors," *IEEE Electron Device Letters*, Vol. 30, No. 10, pp. 1102-1104, October 2009. (First author supervised and second author co-supervised by candidate)
- [24] A. Ali*, H. Madan*, S. Kovesnikov and S. Datta," Small Signal Response of Inversion Layers in High Mobility In_{0.53}Ga_{0.47}As MOSFETs Made with Thin High-k Dielectrics", *Electrochemical Society (ECS) Transactions*, Vol. 25, No. 6, pp. 271-284, Physics and Technology of High-k Gate Dielectrics, October 2009. (First and second authors supervised by the candidate)(Based on candidate's invited talk)
- [23] S. Mookerjee*, R. Krishnan*, S. Datta, and V. Narayanan, "Effective Capacitance and Drive Current for Tunnel-FET (TFET) CV/I Estimation," *IEEE Transactions on Electron Devices*, Vol. 56, No. 9, pp. 2092-2098, September 2009. (First author supervised and second author co-supervised by candidate)
- [22] Z. Fang*, S. G. Lu, F. Li*, S. Datta, and Q. M. Zhang, "Enhancing the Magnetoelectric Response of Metglas/Polyvinylidene fluoride Laminates by Exploiting the Flux Concentration Effect," *Applied Physics Letters*, Vol. 95, No. 11, pp. 112903_1-112903_3, September 2009. (First author co-supervised and third author supervised by the candidate)
- [21] S. Mookerjee*, R. Krishnan*, A. Vallett, T. Mayer and S. Datta, "Inter-band Tunnel Transistor Architecture using Narrow Gap Semiconductors", *ECS Transactions*, Vol 19, No. 5, Ge and III-V MOSFETs, pp. 287-292, May 2009. (First author co-supervised and third author supervised by the candidate) (Based on Invited Talk by the candidate)

- [20] D. Schlom, S. Guha, and S. Datta, "Gate Oxides Beyond SiO₂," *MRS Bulletin*, pp. 1017-1025, November 2008. (Equal contributions by all authors)
- [19] S. H. Lee, P. Majhi, J. Oh, B. Sassman, C. Young, A. Bowonder, W. Y. Loh, J. J. Choi, B. J. Cho, H. D. Lee, P. Kirsch, H. R. Harris, W. Tsai, S. Datta, H. H. Tseng, S. K. Banerjee, and R. Jammy, "Demonstration of Lg 55 nm pMOSFETs With Si Si_{0.25}Ge_{0.75} Si Channels, High Ion Ioff (5×10^4), and Controlled Short Channel Effects (SCEs)," *IEEE Electron Device Letters*, Vol. 29, No 9, pp. 1017-1020, September 2008. (Equal contribution by all authors)
- [18] C. I. Kuo; H. T. Hsu, E. Y Chang, C. Y. Chang; Y. Miyamoto, S. Datta; M. Radosavljevic, G-W. Huang, and C. T. Lee, "RF and Logic Performance Improvement of In_{0.7}Ga_{0.3}As/InAs/In_{0.7}Ga_{0.3}As Composite-Channel HEMT Using Gate-Sinking Technology," *IEEE Electron Device Letters*, Vol. 29, Issue 4, pp. 290-293, 2008. (Equal contribution by all authors)
- [17] R. Chau, B. Doyle, S. Datta, K. Kavalieros, and K. Zhang, "Integrated nanoelectronics for the future," *Nature Materials*, Vol. 6, pp. 810-812, 2007. (Equal contribution by all authors)
- [16] S. Datta, G. Dewey, J. M. Fastenau, M. K. Hudait, D. Loubychev, W. K. Liu, M. Radosavljevic, W. Rachmady, and R. Chau, "Ultrahigh-Speed 0.5 V Supply Voltage In_{0.7}Ga_{0.3}As Quantum-Well Transistors on Silicon Substrate," *IEEE Electron Device Letters*, Vol. 28, No. 8, pp. 685, 2007. (Nominated for the George E. Smith award for the best 2007 EDL paper by IEEE Electron Devices Society)
- [15] S. Datta, "III-V field-effect transistors for low power digital logic applications," *Journal of Microelectronic Engineering*, Vol. 84, No. 9-10, pp. 2133-2137, 2007. (Invited paper) (This paper is cited several times in the Emerging Research Devices section of the 2007 edition of the International Technology Roadmap for Semiconductors (ITRS). ITRS is a vital reference for the semiconductor industry and addresses the technology challenges and possible solutions for the industry over the next 15 years)
- [14] C. Y. Chang, H. T. Hsu, E. Y. Chang, C. I. Kuo, S. Datta, M. Radosavljevic, M. Miyamoto, and G. W. Y. Huang, "Investigation of Impact Ionization in InAs-Channel HEMT for High-Speed and Low-Power Applications," *IEEE Electron Device Letters*, Vol. 28, No. 10, pp. 856-858, 2007. (Equal contributions by all authors)
- [13] T. Ashley, L. Buckle, S. Datta, M.T. Emeny, D.G. Hayes, K.P. Hilton, R. Jefferies, T. Martin, T.J. Phillips, D.J. Wallis, P.J. Wilding, and R. Chau, "Heterogeneous InSb quantum well transistors

on silicon for ultra-high speed, low power logic applications,” *Electronics Letters*, Vol. 43, No. 14, 2007. (Equal contribution by all authors)

[12] R. Chau, S. Datta, M. Doczy, B. S. Doyle, B. Jin, J. Kavalieros, A. Majumdar, M. Metz, and M. Radosavljevic, “Benchmarking nanotechnology for high-performance and low-power logic transistor applications,” *IEEE Transactions on Nanotechnology*, Vol. 4, No. 2, pp. 153-158, 2005. (Equal contribution by all authors)

[11] R. Chau, J. Brask, S. Datta, G. Dewey, M. Doczy, B. Doyle, J. Kavalieros, B. Jin, M. Metz, A. Majumdar, and M. Radosavljevic, “Application of high-K gate dielectrics and metal gate electrodes to enable silicon and non-silicon logic nanotechnology,” *Journal of MicroElectronic Engineering*, Vol. 80, No. 17, pp. 1-6, 2005. (Equal contribution by all authors)

[10] R. Chau, S. Datta, M. Doczy, B. Doyle, J. Kavalieros, and M. Metz, “High-K/Metal-Gate Stack and its MOSFET Characteristics,” *IEEE Electron Device Letters*, Vol. 25, No. 6, pp. 408-410, 2004. (Equal contribution by all authors)

[9] R. Chau, B. Boyanov, B. Doyle, M. Doczy, S. Datta, S. Hareland, B. Jin, J. Kavalieros, and M. Metz, "Silicon Nano-transistors for Logic Applications," *Physica E, Low-Dimensional Systems and Nanostructures*, Vol. 19, No. 1-2, pp. 1-5, 2003. (Equal contribution by all authors)

[8] B. S. Doyle, S. Datta, M. Doczy, S. Hareland, B. Jin, J. Kavalieros, T. Linton, A. Murthy, R. Rios, and R. Chau, "High Performance Fully-Depleted Tri-Gate CMOS Transistors," *IEEE Electron Device Letters*, Vol. 24, No. 4, pp. 263-265, 2003. (Equal contribution by all authors)

[7] S. Datta, K. P. Roenker, M. M. Cahay, and L. M. Lunardi, “Analytical Modeling of Pnp InP/InGaAs Heterojunction Bipolar Transistors,” *Solid-State Electronics*, Vol. 44, No. 7, pp. 1331-1333, 2000.

[6] S. Datta, K. P. Roenker, and M. M. Cahay, “A Gummel-Poon Model for Pnp Heterojunction Bipolar Transistors with a Compositionally Graded Base,” *Solid-State Electronics*, Vol. 44, No. 6, pp. 991-1000, 2000.

[5] S. Datta, K. P. Roenker, and M. M. Cahay, “Emitter Series Resistance Effect of Multiple Heterojunction Contacts for Pnp Heterojunction Bipolar Transistors,” *Solid-State Electronics*, Vol. 43, No. 7, pp. 1299-1305, 1999.

[4] S. Datta, K. P. Roenker, and M. M. Cahay, “Hole Transport and Quasi-Fermi Level Splitting at the Emitter-Base Junction in Pnp Heterojunction Bipolar Transistors,” *Journal of Applied Physics*, Vol. 85, No. 3, pp. 1949-1955, 1999.

- [3] S. Datta, K. P. Roenker, and M. M. Cahay, “Implications of Hole versus Electron Transport Properties for High Speed Pnp Heterojunction Bipolar Transistors,” *Solid-State Electronics*, Vol. 43, No. 1, pp. 73-80, 1999.
- [2] S. Datta, S. Shi, K. P. Roenker, and M. M. Cahay and W. E. Stanchina, “Simulation and Design of InAlAs/InGaAs Pnp Heterojunction Bipolar Transistors,” *IEEE Transactions on Electron Devices*, Vol. 45, No. 8, pp. 1634-1643, 1998.
- [1] S. Datta, K. P. Roenker, and M. M. Cahay, “A Thermionic-Emission-Diffusion Model for a Graded Base Pnp Heterojunction Bipolar Transistors,” *Journal of Applied Physics*, Vol. 83, No. 12, pp. 8036-8045, 1998

Refereed Conference Proceedings Articles

- [263] J. Lee, C. Zhang, S. Park, H. Park, T.-H. Kim, L. J.-C. Liu, E. Ambrosi, M. Song, X. Bao, M.-F. Chang, Suman Datta, S. Yu, “Monolithic 3D integration of dual-gated ALD oxide-channel non-volatile capacitive memory on 40nm Si CMOS for digital compute-in-memory,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA, highlight paper.
- [262] O. Phadke, J. Lee, P.-K. Hsu, C. Zhang, A. I. Khan, Suman Datta, S. Yu, “Array-level demonstration of charge-domain in-memory search using back-end-of-line compatible ferroelectric nonvolatile capacitors,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [261] O. Phadke, M. Shon, S. Wodzro, H. Mulaosmanovic, S. Dunkel, S. Beyer, A. I. Khan, Suman Datta, S. Yu, “On the localized ferroelectric phase variation in scaled FeFET: Experiments and modeling,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [260] S. Deng, J. Sonawane, F. G. Waqar, O. Phadke, C. Zhang, M.-Y. Lee, S. Yu, Suman Datta, “Demonstration of 3T0C gain cells with self-aligned oxide transistors for parasitic-aware design at array-level,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [259] S. Deng, H.-C. Chou, C. Zhang, J. Kwak, A. D. Asokan, E. Quezada, E. Sarkar, J. Lee, J. Shin, D. Chakraborty, V. Sathe, S. Yu, Suman Datta, “High-efficiency BEOL-compatible on-chip switched capacitor DC-DC converter using planar and vertical oxide transistors with trench capacitors,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [258] H. Park, J. Shin, E. Quezada, H. J. Lee, E. Sarkar, S. Yu, A. I. Khan, S. Guha, W. Haensch, Suman Datta, “Fluid imprint: A reliability bottleneck in ferroelectric RAM read operation,” IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.

- [257] Y.-H. Kuo, C. Zhang, P. G. Ravikumar, S. Kang, T. Song, M. A. Villena, L. Larcher, H. Kim, M. Hong, P. Yun, G. Thareja, S. Yu, D. Ha, Suman Datta, J. E. Medvedeva, A. I. Khan, "Experiments and modeling of defect dynamics and BTI in doped In₂O₃ TFTs undergoing densification during 400 °C post-BEOL forming gas annealing (FGA)," IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [256] E. Sarkar, H. Park, M. A. Al Mamun, H. J. Lee, D. Chakraborty, E. Quezada, C. Zhang, S. Kirtania, M. Tian, A. I. Khan, S. Yu, K. Cho, H. Kim, C. Im, M. Hong, D. Ha, Suman Datta, "Overcoming threshold voltage-performance-stability tradeoff with double-gate gallium doped In₂O₃ MOSFETs using tri-layer HfO₂-ZrO₂-HfO₂ gate stack," IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [255] H. J. Lee, H. Park, C. Zhang, J. Shin, E. Sarkar, H. Kim, C. Im, M. Hong, D. Ha, S. Yu, A. I. Khan, Suman Datta, "DC and AC PBTI / NBTI analysis for BEOL oxide FETs via on-the-fly measurement and modeling," IEEE International Electron Devices Meeting (IEDM) 2025, San Francisco, USA.
- [254] P.-K. Hsu, S. Cho, J. Sharda, H. Park, Suman Datta, S. Yu, "Monolithic 3D stackable DRAM design with BEOL-compatible oxide channel access transistor," IEEE International Memory Workshop (IMW) 2025, Monterey, CA.
- [253] E. Sarkar, D. Chakraborty, W.-C. Wang, K. A. Aabrar, S. G. Kirtania, O. Phadke, C. Zhang, E. Quezada, J. Shin, S. Zhang, A. I. Khan, S. Yu, S. Mukhopadhyay, Suman Datta, "Analog in-memory-compute with multi-bit silicon ferro FinFET array for improved energy and area efficiency," IEEE International Memory Workshop (IMW) 2025, Monterey, CA.
- [252] L. Fernandes, P. V. Ravindran, M. Tian, Y.-H. Kuo, N. Afroze, S. Soliman, S. Kang, D. Chakraborty, T. Song, C. Zhang, K. Kim, K. Seo, K. Kim, W. Kim, D. Ha, S. Yu, Suman Datta, A. I. Khan, "Comparative study of channel materials for ferroelectric NAND applications," IEEE International Memory Workshop (IMW) 2025, Monterey, CA.
- [251] M. Shon, C. Park, P. V. Ravindran, L. Fernandes, J. Lee, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, Suman Datta, A. I. Khan, S. Yu, "A comprehensive modeling of gate stack interlayer engineering for ferroelectric vertical NAND," IEEE Device Research Conference (DRC) 2025, Durham, NC, best paper award.
- [250] D. Chakraborty, E. Sarkar, H. Park, H. J. Lee, E. Quezada, C. Zhang, M. Tian, A. Khan, Suman Datta, "Overcoming Mobility-Stability Trade-off by Gate Stack Engineering in BEOL

Compatible Ga Doped In₂O₃ MOSFETs," 2025 Device Research Conference (DRC) 2025, Durham, NC, USA, 2025, pp. 1-2

[249] E. Quezada, E. Sarkar, D. Chakraborty, H. Park, C. Zhang, M. Tian, A. Khan, Suman Datta, "Dipole Engineered Gate Stacks in AOS Channel FETs to Overcome the VT-ION Tradeoff," 2025 Device Research Conference (DRC) 2025, Durham, NC, USA, 2025, pp. 1-2

[248] J. Kwak, C. Zhang, G. Jeong, O. Phadke, S. G. Kirtania, J. Lee, Suman Datta, S. Yu, "Machine learning-assisted modeling of AC stress-induced bias temperature instability in oxide channel transistors for 2T gain cell eDRAM," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA.

[247] M. Shon, C. Park, P. V. Ravindran, L. Fernandes, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, Suman Datta, A. I. Khan, S. Yu, "Modeling dynamic interplay between charge traps and polarization for memory window enhancement in gate injection layers," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA.

[246] P. Venkatesan, A. Padovani, L. Fernandes, P. Ravikumar, C. Park, H. Tran, Z. Wang, T. Song, H. Chen, W. Chern, Z. Wang, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, Suman Datta, A. I. Khan, "Enhanced memory performance in ferroelectric NAND applications: The role of tunnel dielectric position for robust 10-year retention," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA, best paper award.

[245] P. G. Ravikumar, C. Park, P. Venkatesan, N. Afroze, M. Tian, W. Chern, Suman Datta, S. Yu, S. Mahapatra, A. I. Khan, "First write pulse-induced interface damage in ferroelectric field-effect transistors," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA.

[244] P. Ravikumar, A. Padovani, P. Venkatesan, C. Park, N. Afroze, M. Tian, Suman Datta, S. Yu, L. Larcher, G. Thareja, A. I. Khan, "Understanding correlation between memory window closure, leakage and read delay effects for FeFET reliability improvement: role of IL and FE traps," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA.

[243] S. G. Kirtania, F. G. Waqar, D. Chakraborty, J. Shin, E. Sarkar, J. Reiss, D. E. Wolfe, S. Yu, Suman Datta, "Radiation-resilient amorphous indium oxide FeFETs for embedded nonvolatile memory," IEEE International Reliability Physics Symposium (IRPS) 2025, Monterey, CA.

[242] H. Park, S. G. Kirtania, E. Sarkar, D. Chakraborty, C. Zhang, H.-J. Lee, J. Shin, S. Yu, A. I. Khan, H. Kim, C. Im, M. J. Hong, D. Ha, Suman Datta, "Demonstration of tungsten-doped

indium oxide MOSFETs with 3 Angstrom EOT, improved stability and high on-current,” IEEE Symposium on VLSI Technology and Circuits (VLSI) 2025, Kyoto, Japan.

[241] J. Sonawane, S. Deng, F. G. Waqar, O. Phadke, C. Zhang, M.-Y. Lee, Suman Datta, S. Yu, “Experimental demonstration of gain cells with self-aligned oxide transistors for reduced capacitive coupling,” IEEE Non-Volatile Memory Technology Symposium (NVMTS) 2025, Atlanta, GA.

[240] M. Vadlamani, D. Chakraborty, J. Jia, H. Mulaosmanovic, S. Duenkel, S. Beyer, Suman Datta, S. Yu, “Cryogenic characterization of ferroelectric non-volatile capacitors,” IEEE Non-Volatile Memory Technology Symposium (NVMTS) 2025, Atlanta, GA.

[239] H. Park, S. G. Kirtania, E. Sarkar, D. Chakraborty, J. Shin, A. I. Khan, S. Yu, Suman Datta, “A theoretical study of multi-domain ferroelectric switching in BEOL compatible ferroelectric field effect transistors,” IEEE International Conference on Nanotechnology (NANO) 2025, Washington DC.

[238] P.-K. Hsu, S. Cho, J. Sharda, H. Park, Suman Datta, S. Yu, “Monolithic 3D stackable DRAM design with BEOL-compatible oxide channel access transistor,” Conference on Future of Memory and Storage (FMS) 2025, Santa Clara, CA.

[237] J. Lee, C. Zhang, M. Shon, J. Read, S. Deng, O. Phadke, P. V. Ravindran, M. Tian, Y.-C. Luo, T.-H. Kim, A. I. Khan, S. Datta, S. Yu, “BEOL-compatible non-volatile capacitive synapse with ALD W-doped In₂O₃ semiconductor layer,” IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA.

[236] M. Choi, R. Dutta, P. Saha, M. Singh, S. Mukhopadhyay, S. Datta, S. Kumar, "Fast Prediction of Spatio-Temporal Temperature Profiles in FinFET Arrays via Numerical and Machine-Learning Approaches," IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, CA, USA

[235] J. Kwak, J. Shin, S. Deng, G. Jeong, J. Sharda, S. Datta, S. Yu, “Bias temperature instability analysis of oxide power transistors for BEOL on-chip voltage converter in thermally-constrained H3D systems,” IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA.

[234] S. Deng, J. Shin, C. Zhang, H. Park, O. Phadke, J. Kwak, S. Yu, S. Datta, “Boosted performance and enhanced reliability of BEOL-compatible dual-gate oxide power transistors for on-chip DC-DC voltage conversion,” IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA.

- [233] E. Sarkar, C. Zhang, D. Chakraborty, F. Waqar, S. Kirtania, K. A. Aabrar, H. Park, J. Shin, M. Tian, A. I. Khan, S. Yu, S. Datta, "First demonstration of W-doped In₂O₃ gate-all-around (GAA) nanosheet FET with improved performance and record threshold voltage stability," IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA.
- [232] S. G. Kirtania, O. Phadke, E. Sarker, K. A. Aabrar, D. Chakraborty, F. Waqar, S. Jaewon, T. Pantha, S. Dutta, A. I. Khan, S. Yu, S. Datta, "Amorphous indium oxide channel FeFETs with write voltage of 0.9V and endurance >10¹² for refresh-free 1T-1FeFET embedded memory," IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA.
- [231] N. Afroze, A. Padovani, J. Choi, P. G. Ravikumar, Y.-H. Kuo, C. Zhang, T. Song, M. Tian, E. Sarkar, M. Noor, P. V. Ravindran, K. A. Aabrar, B. Yildiz, S. Mahapatra, A. Kummel, K. Cho, S. Yu, S. Datta, J. H. Lee, L. Larcher, G. Thareja, A. I. Khan, "Self-healing ferroelectric capacitors with ~1000x endurance improvement at high temperatures (85-125°C)," IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, USA
- [230] T. H. Pantha, A. Khanna, H. Ye, S. Niazi, B. Chakraborty, E. G. Weinstock, N. Babu, S. Mukhopadhyay, S. Datta, K. Çamsari, S. Dutta, "Towards 3D CMOS+X Ising Machines: Addressing the Connectivity Problem with Back-end-of-line FeFETs," IEEE International Electron Devices Meeting (IEDM) 2024, San Francisco, CA, USA
- [229] T. H. Pantha, S. Kirtania, K. A. Aabrar, S. Deng, S. Datta and S. Dutta, "Design Space Exploration of Oxide Semiconductor based Monolithic 3D Gain Cell Memory," 2024 IEEE European Solid-State Electronics Research Conference (ESSERC) 2024 , Bruges, Belgium, , pp. 125-128
- [228] R. Saligram, A. Gaidhane, Y. Cao, S. Datta, A. Raychowdhury, "Cooling the Chaos: Mitigating the Effect of Threshold Voltage Variation in Cryogenic CMOS Memories", Proceedings of the 29th ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED) 2024 Newport Beach CA USA
- [227] L. A Shamieh, W. C. Wang, S. Zhang, R. Saligram, A. D Gaidhane, Y. Cao, A. Raychowdhury, S. Datta, S. Mukhopadhyay, "Cryogenic Operation of Computing-In-Memory based Spiking Neural Network", Proceedings of the 29th ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED) 2024 Newport Beach CA USA
- [226] S. Deng, J. Kwak, J. Lee, D. Chakraborty, J. Shin, O. Phadke, S. G. Kirtania, C. Zhang, K. A. Aabrar, S. Yu, S. Datta, "Demonstration of on-chip switched-capacitor DC-DC converters using

BEOL compatible oxide power transistors and superlattice MIM capacitors,” IEEE Symposium on VLSI Technology and Circuits (VLSI Symp) 2024, Hawaii, USA,

[225] S. Datta, S. Yu, A. I. Khan, S. Deng, K. A. Aabrar, J. Kwak, “Amorphous oxide semiconductors for monolithic 3D integrated circuits,” IEEE Symposium on VLSI Technology and Circuits (VLSI Symp) 2024, Hawaii, USA, invited.

[224] K. A. Aabrar, H. Park, S. G Kirtania, E. Sarkar, M . A Mamun, S. Deng, C. Zhang, G. B Rayner, K. Cho, S. Datta "On the Reliability of High-Performance Dual Gate (DG) W-Doped In₂O₃ FET," 2024 IEEE Symposium on VLSI Technology and Circuits (VLSI Symp) 2024, Honolulu, HI, USA,

[223] D. Das, L. Fernandes, P. V. Ravindran, T. Song, C. Park, N. Afroze, M. Tian, H. Chen, W. Chern, K. Kim, J. Woo, S. Lim, K. Kim, W. Kim, D. Ha, S. Yu, S. Datta, A. I. Khan, “Design framework for ferroelectric gate stack engineering of vertical NAND structures for efficient TLC and QLC operation,” IEEE International Memory Workshop (IMW) 2024, Seoul, South Korea.

[222] J. Lee, S. Deng, J. Kwak, M. Shon, S. Datta, S. Yu, “Optimization of backside of silicon-compatible high voltage superlattice capacitor for 12V-to-6V on-chip voltage conversion,” IEEE Device Research Conference (DRC) 2024, College Park, MD

[221] S. Deng, J. Kwak, J. Lee, S. Yu, S. Datta, “BEOL-compatible on-chip DC-DC converters,” IEEE International Interconnect Technology Conference (IITC) 2024, San Jose, CA

[220] P. G. Ravikumar, P. V. Ravindran, K. A. Aabrar, T. Song, S. G. Kirtania, D. Das, C. Park, N. Afroze, M. Tian, S. Yu, S. Datta, S. Mahapatra, A. I. Khan, “Comprehensive time-dependent dielectric breakdown (TDDB) characterization of ferroelectric capacitors under bipolar stress conditions,” IEEE International Reliability Physics Symposium (IRPS) 2024, Dallas, TX

[219] S. E. Wodzro, O. Phadke, S. Datta, S. Yu, “Modeling single-event-effects in silicon bulk FeFET,” IEEE Semiconductor Interface Specialists Conference (SISC) 2024, San Diego, CA

[218] D. Das, H. Park, Z. Wang, C. Zhang, P. V. Ravindran, C. Park, N. Afroze, P.-K. Hsu, M. Tian, H. Chen, W. Chern, S. Lim, K. Kim, K. Kim, W. Kim, D. Ha, S. Yu, S. Datta, A. I. Khan, “Ferroelectric Gate Stack Engineering with Tunnel Dielectric Insert for Achieving High Memory Window in FEFETs for NAND Applications,” IEEE Electron Devices Technology and Manufacturing Conference (EDTM) 2024, Bangalore, India

- [217] P. G. Ravikumar, C. Park, P. V. Ravindran, N. Afroze, M. Tian, W. Chern, S. Datta, S. Yu, S. Mahapatra, A. I. Khan, "First write pulse-induced interface damage in ferroelectric field-effect transistors," IEEE International Integrated Reliability Workshop (IIRW) 2024, Lake Tahoe, CA
- [216] R Saligram, S Datta, A Raychowdhury, Cryogenic CMOS as an Enabler for Low Power Dynamic Logic, *2023 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*
- [215] SG Kirtania, KA Aabrar, AI Khan, S Yu, S Datta, Cold-FeFET as Embedded Non-Volatile Memory with Unlimited Cycling Endurance, *2023 IEEE Symposium on VLSI Technology and Circuits (VLSI Symp)*
- [214] S Woo, G Choe, AI Khan, S Datta, S Yu, Design of Ferroelectric-Metal Field-Effect Transistor for Multi-Level-Cell 3D NAND Flash, *2023 IEEE International Memory Workshop (IMW)*
- [213] O Phadke, KA Aabrar, Y Luo, SG Kirtania, AI Khan, S Datta, S Yu, Low-Frequency Noise Characteristics of Ferroelectric Field-Effect Transistors, *2023 IEEE International Reliability Physics Symposium (IRPS)*, 1-4
- [212] Y Hu, KA Aabrar, A Palmieri, M Bergschneider, M Pešić, CD Young, On the Dopant, Defect States, and Mobility in W Doped Amorphous In₂O₃ for BEOL Transistors, *2023 7th IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*
- [211] D Durham, KA Aabrar, S Datta, N Zaluzec, S Guha, C Phatak, Direct imaging of grain and domain structure in nanocrystalline ferroelectric thin films, *2023 Bulletin of the American Physical Society (APS)*
- [210] M Passlack, N Tasneem, Z Wang, KA Aabrar, J Hur, H Chen, VDH Hou, Direct Quantitative Extraction of Internal Variables from Measured PUND Characteristics Providing New Key Insights into Physics and Performance of Silicon and Oxide Channel, *2022 International Electron Devices Meeting (IEDM)*, 32.4. 1-32.4. 4
- [209] KA Aabrar, J Read, SG Kirtania, S Stepanoff, DE Wolfe, S Yu, S Datta, Total Ionizing Dose Effect in Tri-gate Silicon Ferroelectric Transistor Memory, *2022 International Electron Devices Meeting (IEDM)*, 32.7. 1-32.7. 4
- [208] Y Hu, H Ye, KA Aabrar, SG Kirtania, W Chakraborty, S Datta, K Cho, Theoretical and Empirical Insight into Dopant, Mobility and Defect States in W Doped Amorphous In₂O₃ for High-Performance Enhancement Mode, *2022 International Electron Devices Meeting (IEDM)*

- [207] R Bhattacharya, A Khanna, B Bosworth, N Orloff, V Gambin, D Streit, ..., Thermally Resilient Microwave Switch and Power Limiter based on Insulator-Metal Transition of Lanthanum Cobalt Oxide, *2022 International Electron Devices Meeting (IEDM)*, 70-73
- [206] MM Hossain, P Pandey, A Aabrar, KG Serrano, T Moise, J Rodriguez, Pulsed Current-Voltage Protocol to Reveal Polarization-Continuation in Ferroelectric Memory: Implications for Partial State Storage, *2022 Device Research Conference (DRC)*
- [205] KA Aabrar, SG Kirtania, A Lu, A Khanna, W Chakraborty, M San Jose, A thousand state superlattice (SL) FEFET analog weight cell, *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI symp)*
- [204] A Khanna, H Ye, Y Luo, G Bajpai, M San Jose, W Chakraborty, S Yu, BEOL Compatible Ferroelectric Routers for Run-time Reconfigurable Compute-in-Memory Accelerators, *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Symp)*
- [203] W Chakraborty, P Shrestha, A Gupta, R Saligram, S Spetalnick, Multi-bit per-cell 1T SiGe Floating Body RAM for Cache Memory in Cryogenic Computing, *2022 IEEE Symposium on VLSI Technology and Circuits (VLSI Symp)*
- [202] YC Luo, H Ye, W Chakraborty, J Hur, PV Ravindran, AI Khan, S Datta, Low-Frequency Noise Characteristics of BEOL-Compatible IWO Transistor, *2022 IEEE Silicon Nanoelectronics Workshop (SNW)*, 1-2
- [201] Y Hu, W Chakraborty, H Ye, S Datta, K Cho "First-principles investigation of amorphous n-type In₂O₃ for BEOL transistor" *2021 International Conference on Simulation of Semiconductor Processes and Devices (SISPAD)*
- [200] W Chakraborty, KA Aabrar, J Gomez, R Saligram, A Raychowdhury, P Fay, S Datta "Cryogenic RF CMOS on 22nm FDSOI Platform with Record $f_T=495\text{GHz}$ and $f_{MAX}=497\text{GHz}$ " *2021 Symposium on VLSI Technology* (June 2021)
- [199] W Chakraborty, MS Jose, J Gomez, A Saha, KA Aabrar, P Fay, S Gupta, S Datta "Higher-k Zirconium Doped Hafnium Oxide (HZO) Trigate Transistors with Higher DC and RF Performance and Improved Reliability," *2021 Symposium on VLSI Technology* (June 2021)
- [198] P Wang, X Peng, W Chakraborty, A Khan, S Datta, and S Yu "Cryogenic Performance for Compute-in-Memory based Deep Neural Network Accelerator" *2021 IEEE International Symposium on Circuits and Systems (ISCAS)*

- [197] R Saligram, S Datta, and A Raychowdhury "CryoMem: A 4K-300K 1.3 GHz eDRAM Macro with Hybrid 2T-Gain-Cell in a 28nm Logic Process for Cryogenic Applications" *2021 IEEE Custom Integrated Circuits Conference (CICC)*
- [196] YC Luo, S Datta, and S Yu "Three-dimensional (3D) Non-volatile SRAM with IWO Transistor and HZO Ferroelectric Capacitor" *2021 International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA)*
- [195] Panni Wang, Xiaochen Peng, Wriddhi Chakraborty, Asif Islam Khan, Suman Datta, and Shimeng Yu, "Cryogenic Benchmarks of Embedded Memory Technologies for Recurrent Neural Network based Quantum Error Correction" *IEEE International Electron Devices Meeting (IEDM)*, 38.5.1-38.5.4 2020
- [194] Xiaochen Peng, Wriddhi Chakraborty, Ankit Kaul, Wonbo Shim, Muhannad S Bakir, Suman Datta, Shimeng Yu, "Benchmarking Monolithic 3D Integration for Compute-in-Memory Accelerators: Overcoming ADC Bottlenecks and Maintaining Scalability to 7nm or Beyond" *IEEE International Electron Devices Meeting (IEDM)*, 30.4.1-30.4.4 2020
- [193] Shan Deng, Zhouhang Jiang, Sourav Dutta, Huacheng Ye, Wriddhi Chakraborty, Santosh Kurinec, Suman Datta, and Kai Ni, "Examination of the Interplay Between Polarization Switching and Charge Trapping in Ferroelectric FET" *IEEE International Electron Devices Meeting (IEDM)*, 4.3.1-4.3.4 2020
- [192] A. K. Saha, M. Si, K. Ni, S. Datta, P. D. Ye, and S. K. Gupta, "Ferroelectric Thickness Dependent Domain Interactions in FEFETs for Memory and Logic: A Phase-field Model based Analysis" *IEEE International Electron Devices Meeting (IEDM)*, 4.3.1-4.3.4 2020
- [191] H. Ye, J. Gomez, W. Chakraborty, S. Spetalnick, S. Dutta, K. Ni, A. Raychowdhury, and S. Datta "Double-Gate W-Doped Amorphous Indium Oxide Transistors for Monolithic 3D Capacitorless Gain Cell eDRAM" *IEEE International Electron Devices Meeting (IEDM)*, 28.3.1-28.3.4 2020
- [190] H. Ye, J. Gomez, W. Chakraborty, S. Spetalnick, S. Dutta, K. Ni, A. Raychowdhury, and S. Datta "Double-Gate W-Doped Amorphous Indium Oxide Transistors for Monolithic 3D Capacitorless Gain Cell eDRAM" *IEEE International Electron Devices Meeting (IEDM)*, 28.3.1-28.3.4 2020
- [189] S. Dutta, H. Ye, W. Chakraborty, Y.-C. Luo, M. San Jose, B. Grisafe, A. Khanna, I. Lightcap, S. Shinde, S. Yu and S. Datta, "Monolithic 3D Integration of High Endurance Multi-Bit

Ferroelectric FET for Accelerating Compute-In-Memory”, *IEEE International Electron Devices Meeting (IEDM)*, 36.4.1-36.4.4 2020

[188] S Deng, G Yin, W Chakraborty, S Dutta, S Datta, X Li, and K Ni "A Comprehensive Model for Ferroelectric FET Capturing the Key Behaviors: Scalability, Variation, Stochasticity, and Accumulation" *2020 IEEE Symposium on VLSI Technology*, 1-2

[187] A Khanna, E Elmitwalli, S Dutta, S Deng, S Datta, S Köse, and K Ni "A Bias and Correlation Free True Random Number Generator Based on Quantized Oscillator Phase under Sub-Harmonic Injection Locking" *2020 IEEE Symposium on VLSI Technology*, 1-2

[186] W Chakraborty, B Grisafe, H Ye, I Lightcap, K Ni, and S Datta "BEOL Compatible Dual-Gate Ultra Thin-Body W-Doped Indium-Oxide Transistor with $SS = 73\text{mV/dec}$ and $\text{ratio} > 4 \times 10^9$ " *2020 IEEE Symposium on VLSI Technology*, 1-2 (**best paper Award**)

[185] W Chakraborty, U Sharma, S Datta, and S Mahapatra "Hot Carrier Degradation in Cryo-CMOS" *2020 IEEE International Reliability Physics Symposium (IRPS)* April 2020

[184] K Ni, S Dutta, S Datta, "Ferroelectrics: From Memory to Computing" *25th IEEE Asia and South Pacific Design Automation Conference (ASP-DAC)* 401-406. Jan 2020

[183] S Dutta, A Khanna, J Gomez, K Ni, Z Toroczka and S Datta, "Experimental Demonstration of Phase Transition Nano-Oscillator Based Ising Machine”, *IEEE International Electron Devices Meeting (IEDM)* 2019

[182] J. Gomez S. Dutta, K. Ni, J.A. Smith, B. Grisafe, A. Khan and S. Datta, "Hysteresis-free negative capacitance in the multi- domain scenario for logic applications”, *IEEE International Electron Devices Meeting (IEDM)* 2019

[181] K Ni, J Smith, H Ye, B Grisafe, G. B. Rayner, A Kummel, and S Datta, "A Novel Ferroelectric Superlattice Based Multi-Level Cell Non-Volatile Memory”, *IEEE International Electron Devices Meeting (IEDM)* 2019

[180] W Chakraborty, K Ni, J Smith, A Raychowdhury and S Datta, "An Empirically Validated Virtual Source FET Model for Deeply Scaled Cool CMOS”, *IEEE International Electron Devices Meeting (IEDM)* 2019

[179] K Ni, A Saha, W Chakraborty, H Ye, B Grisafe, J Smith, G B Rayner, S Gupta, S Datta, "Equivalent Oxide Thickness (EOT) Scaling With Hafnium Zirconium Oxide High- κ Dielectric Near Morphotropic Phase Boundary”, *IEEE International Electron Devices Meeting (IEDM)* 2019

- [178] AK Saha, B Grisafe, S Datta, SK Gupta, “Microscopic Crystal Phase Inspired Modeling of Zr Concentration Effects in $\text{Hf}_{1-x}\text{Zr}_x\text{O}_2$ Thin Films”, Symposium on VLSI Technology, T226-T227, June 2019
- [177] S Dutta, A Saha, P Panda, W Chakraborty, J Gomez, A Khanna, S Gupta, S Datta, “Biologically Plausible Ferroelectric Quasi-Leaky Integrate and Fire Neuron”, Symposium on VLSI Technology, T140-T141, June 2019
- [176] S Dutta, A Khanna, W Chakraborty, J Gomez, S Joshi, S Datta, “Spoken vowel classification using synchronization of phase transition nano-oscillators”, Symposium on VLSI Technology, T128-T129, June 2019
- [175] K Ni, W Chakraborty, J Smith, B Grisafe, S Datta, “Fundamental Understanding and Control of Device-to-Device Variation in Deeply Scaled Ferroelectric FETs”, Symposium on VLSI Technology, T40-T41, June 2019
- [174] S Dutta, W Chakraborty, J Gomez, K Ni, S Joshi, S Datta, “Energy-Efficient Edge Inference on Multi-Channel Streaming Data in 28nm HKMG FeFET Technology”, Symposium on VLSI Technology, T38-T39, June 2019
- [173] P Cadareanu, CG Almudever, A Khanna, A Raychowdhury, S Datta, “Rebooting Our Computing Models”, Design, Automation & Test in Europe Conference & Exhibition (DATE), 469-476, Mar 2019
- [172] J Gomez, S Dutta, K Ni, S Joshi, S Datta, “Steep Slope Ferroelectric Field Effect Transistor,” Electron Devices Technology and Manufacturing Conference (EDTM), 59-61, Mar 2019
- [171] K Ni, JA Smith, B Grisafe, T Rakshit, B Obradovic, JA Kittl, M Rodder, S Datta, “SoC Logic Compatible Multi-Bit FeMFET Weight Cell for Neuromorphic Applications”, IEEE International Electron Devices Meeting (IEDM), 13.2. 1-13.2. 4, Dec 2018
- [170] X Sun, P Wang, K Ni, S Datta, S Yu, “Exploiting Hybrid Precision for Training and Inference: A 2T-1FeFET Based Analog Synaptic Weight Cell”, IEEE International Electron Devices Meeting (IEDM), 3.1. 1-3.1. 4, Dec 2018
- [169] Z Wang, B Crafton, J Gomez, R Xu, A Luo, Z Krivokapic, L Martin, S Datta, A Khan, “Experimental Demonstration of Ferroelectric Spiking Neurons for Unsupervised Clustering”, IEEE International Electron Devices Meeting (IEDM), 13.3. 1-13.3. 4, Dec 2018

- [168] K Ni, B Grisafe, W Chakraborty, AK Saha, S Dutta, M Jerry, JA Smith, S Datta, “In-Memory Computing Primitive for Sensor Data Fusion in 28 nm HKMG FeFET Technology”, IEEE International Electron Devices Meeting (IEDM), 16.1. 1-16.1. 4, Dec 2018
- [167] K Ni, JA Smith, B Grisafe, T Rakshit, B Obradovic, JA Kittl, M Rodder, S Datta, “SoC Logic Compatible Multi-Bit FeMFET Weight Cell for Neuromorphic Applications”, 2018 IEEE International Electron Devices Meeting (IEDM), 13.2. 1-13.2. 4, Dec 2018
- [166] X Sun, P Wang, K Ni, S Datta, S Yu, “Exploiting Hybrid Precision for Training and Inference: A 2T-1FeFET Based Analog Synaptic Weight Cell”, 2018 IEEE International Electron Devices Meeting (IEDM), 3.1. 1-3.1. 4, Dec 2018
- [165] Z Wang, B Crafton, J Gomez, R Xu, A Luo, Z Krivokapic, L Martin, S Datta, A Khan, “Experimental Demonstration of Ferroelectric Spiking Neurons for Unsupervised Clustering”, 2018 IEEE International Electron Devices Meeting (IEDM), 13.3. 1-13.3. 4, Dec 2018
- [164] K Ni, B Grisafe, W Chakraborty, AK Saha, S Dutta, M Jerry, JA Smith, S Datta, “In-Memory Computing Primitive for Sensor Data Fusion in 28 nm HKMG FeFET Technology”, 2018 IEEE International Electron Devices Meeting (IEDM), 16.1. 1-16.1. 4, Dec 2018
- [163] Robert D Clark, Kandabara Tapily, Steven Consiglio, Cory S Wajda, Kai Ni, Sonal Dey, Vineetha Mukundan, K Beckman, Gert J Leusink, Nathaniel Cady, Alain C Diebold, S Datta, “Teaching a New Dog Old Tricks: Ferroelectric HfZrO Films and Devices”, The Electrochemical Society Meeting Abstracts, pp 696, July 2018
- [162] Abhinav Parihar, Anvesha Amravati, Matthew Jerry, Suman Datta, Arijit Raychowdhury, “Dynamics of Coupled Systems and their Computing Properties”, 2018 16th IEEE International New Circuits and Systems Conference (NEWCAS), pp 361-364, Jun 2018
- [161] M Jerry, JA Smith, K Ni, A Saha, S Gupta, S Datta, “Insights on the DC Characterization of Ferroelectric Field-Effect-Transistors”, 2018 76th Device Research Conference (DRC), 1-2, Jun 2018
- [160] A Aziz, N Shukla, A Seabaugh, S Datta, S Gupta, “Cockcroft-Walton Multiplier based on Unipolar Threshold Switch”, 2018 76th Device Research Conference (DRC), 1-2, June 2018
- [159] JA Smith, H Takeuchi, R Stephenson, YA Chen, M Hytha, RJ Mears, S Datta, “Experimental Investigation of N-Channel Oxygen-Inserted (OI) Silicon Channel MOSFETs with High-K/Metal Gate Stack”, 2018 76th Device Research Conference (DRC), 1-2, Jun 2018

- [158] B Grisafe, S Datta, “Investigation of Threshold Switch OFF-State Resistance on Performance Enhancement in 2D Mos2 Phase-FETs”, 2018 76th Device Research Conference (DRC), Jun 2018
- [157] I Yoon, M Chang, K Ni, M Jerry, S Gangopadhyay, G Smith, T Hamam, S Datta, “A FeFET Based Processing-In-Memory Architecture for Solving Distributed Least-Square Optimizations”, 76th Device Research Conference (DRC), 1-2, Jun 2018
- [156] Kai Ni, Matthew Jerry, Jeffrey A Smith, Suman Datta, “A circuit compatible accurate compact model for ferroelectric-FETs”, 2018 IEEE Symposium on VLSI Technology, pp 131-132, Jun 2018
- [155] M Jerry, A Aziz, K Ni, S Datta, Sumeet Kumar Gupta, N Shukla, “A Threshold Switch Augmented Hybrid-FeFET (H-FeFET) with Enhanced Read Distinguishability and Reduced Programming Voltage for Non-Volatile Memory Applications”, 2018 IEEE Symposium on VLSI Technology, pp129-130, Jun 2018
- [154] Gyorgy Csaba, Arijit Raychowdhury, Suman Datta, Wolfgang Porod, “Computing with Coupled Oscillators: Theory, Devices, and Applications”, Circuits and Systems (ISCAS), 2018 IEEE International Symposium on, pp 1-5, May 2018
- [153] Benjamin Grisafe, Rui Zhao, Matthew Jerry, Joshua A Robinson, Suman Datta, “Investigation of the abrupt phase transition in 1T-TaS2/MoS2heterostructures”, 2018 International Symposium on VLSI Technology, Systems and Application (VLSI-TSA), pp 1-2, Apr 2018
- [152] Ahmedullah Aziz, Evelyn T Breyer, An Chen, Xiaoming Chen, Suman Datta, Sumeet Kumar Gupta, Michael Hoffmann, Xiaobo Sharon Hu, Adrian Ionescu, Matthew Jerry, Thomas Mikolajick, Halid Mulaosmanovic, Kai Ni, Michael Niemier, Ian O'Connor, Atanu Saha, Stefan Slesazeck, Sandeep Krishna Thirumala, Xunzhao Yin, “Computing with ferroelectric FETs: Devices, models, systems, and applications”, Design, Automation & Test in Europe Conference & Exhibition (DATE), pp 1289-1298, Mar 2018
- [151] S Datta, A Kummel, “Technology Innovations in Selective ALD for Next-Generation Contacts and Vias”, 2018 IEEE 2nd Electron Devices Technology and Manufacturing Conference (EDTM), pp 102-103, Mar 2018
- [150] N Shukla, RK Ghosh, B Grisafe, S Datta, “Fundamental mechanism behind volatile and non-volatile switching in metallic conducting bridge RAM”, Electron Devices Meeting (IEDM), 2017 IEEE International, 4.3. 1-4.3. 4, Dec 2017

- [149] JA Smith, K Ni, RK Ghosh, J Xu, M Badaroglu, PRC Chidambaram, S Datta, “Investigation of electrically gate-all-around hexagonal nanowire FET (HexFET) architecture for 5 nm node logic and SRAM applications”, Solid-State Device Research Conference (ESSDERC), 2017 47th European, 188-191, 2017
- [148] M Badaroglu, J Xu, J Zhu, D Yang, J Bao, SC Song, P Feng, Ritzenthaler, Hans Mertens, Geert Eneman, Naoto Horiguchi, Jeffrey Smith, Suman Datta, David Kohen, Po-Wen Chan, Keagan Chen, PR Chidi Chidambaram, “PPAC scaling enablement for 5nm mobile SoC technology”, Solid-State Device Research Conference (ESSDERC), 2017 47th European, 240-24
- [147] A Aziz, N Jao, S Datta, V Narayanan, SK Gupta, “A computationally efficient compact model for leakage in cross-point array”, Simulation of Semiconductor Processes and Devices (SISPAD), 2017
- [146] S Datta, A Seabaugh, M Niemier, A Raychowdhury, D Schlom, D Jena, “In quest of the next information processing substrate”, Design Automation Conference (DAC), 2017 54th ACM/EDAC/IEEE, 1-6
- [145] P Sharma, J Zhang, AK Saha, S Gupta, S Datta, “Negative capacitance transients in metal-ferroelectric Hf 0.5 Zr 0.5 O 2-Insulator-Semiconductor (MFIS) capacitors,” Device Research Conference (DRC), 2017 75th Annual, 1-2
- [144] Ahmedullah Aziz, Xueqing Li, Nikhil Shukla, Suman Datta, Meng-Fan Chang, Vijaykrishnan Narayanan, Sumeet Kumar Gupta, “Low power current sense amplifier based on phase transition material” Device Research Conference(DRC), pp 1-2, June 2017
- [143] K Ni, JA Smith, M Barth, H Liu, JH Warner, K Saraswat, S Datta, “Soft error evaluation for InGaAs and Ge complementary FinFETs”, Device Research Conference (DRC), pp 1-2, June 2017
- [142] Jeffrey A Smith, Michael Barth, Kai Ni, Mirco Cantoro, Dong-Won Kim, Suman Datta, “Corrugated channel In 0.8 Ga 0.2 As quantum well transistors for low power logic applications” Device Research Conference (DRC), pp 1-2, June 2017
- [141] Matthew Jerry, Abhinav Parihar, Arijit Raychowdhury, Suman Datta, “A random number generator based on insulator-to-metal electronic phase transitions”, Device Research Conference (DRC), pp 1-2, June 2017

- [140] Suman Datta, Alan Seabaugh, Michael Niemier, Arijit Raychowdhury, Darrell Schlom, Debdeep Jena, Grace Xing, H-S Philip Wong, Eric Pop, Sayeef Salahuddin, Sumeet Gupta, Supratik Guha, "In Quest of the Next Information Processing Substrate", Proceedings of the 54th Annual Design Automation Conference (DAC), pp 17, 2017
- [139] P Sharma, K Tapily, AK Saha, J Zhang, A Shaughnessy, A Aziz, GL Snider, S Gupta, RD Clark, S Datta, "Impact of total and partial dipole switching on the switching slope of gate-last negative capacitance FETs with ferroelectric hafnium zirconium oxide gate stack", IEEE Symposium on VLSI Technology, Kyoto, June 2017
- [138] M Jerry, A Parihar, B Grisafe, A Raychowdhury, S Datta, "Ultra-low power probabilistic IMT neurons for stochastic sampling machines" IEEE Symposium on VLSI Technology, Kyoto, June 2017
- [137] Abhinav Parihar, Nikhil Shukla, Matthew Jerry, Suman Datta, Arijit Raychowdhury, "Computational paradigms using oscillatory networks based on state-transition devices", International Joint Conference on Neural Networks (IJCNN), 14-19 May, 2017
- [136] Benjamin Grisafe, Nikhil Shukla, Matthew Jerry, Suman Datta, "A steep slope Phase-FET based on 2D MoS₂ and the electronic phase transition in VO₂" VLSI Technology, Systems and Application (VLSI-TSA), 2017 International Symposium on, pp 1-2, 2017
- [135] Sumeet Kumar Gupta, Danni Wang, Sumitha George, Ahmedullah Aziz, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan, "Harnessing ferroelectrics for non-volatile memories and logic", Quality Electronic Design (ISQED), 2017 18th International Symposium on, pp 29-34, 2017
- [134] Robert J Mears, Hideki Takeuchi, Robert J Stephenson, Marek Hytha, Richard Burton, Nyles W Cody, Doran Weeks, Dmitri Choutov, Nidhi Agrawal, Suman Datta, "Punch-through stop doping profile control via interstitial trapping by oxygen-insertion silicon channel", Electron Devices Technology and Manufacturing Conference (EDTM), 2017 IEEE, pp 65-66, 2017
- [133] R. Pandey, C. Schulte-Braucks, R.N. Sajjad, M. Barth, R.K. Ghosh, B. Grisafe, P. Sharma, N. von den Driesch, A. Vohra, B. Rayner, R. Loo, S. Mantl, D. Buca, C.C. Yeh, C.H. Wu, W. Tsai, D. Antoniadis, S. Datta, "Performance benchmarking of p-type In_{0.65} Ga_{0.35} As/GaAs 0.4 Sb 0.6 and Ge/Ge_{0.93} Sn_{0.07} hetero-junction tunnel FETs", IEEE International Electron Devices Meeting (IEDM), 2016.
- [132] N. Shukla, B. Grisafe, R. K. Ghosh, N. Jao, A. Aziz, J. Frougier, M. Jerry, S. Sonde, S. Rouvimov, T. Orlova, S. Gupta, S. Datta, "Ag/HfO₂ based threshold switch with extreme non-

linearity for unipolar cross-point memory and steep-slope phase-FETs", IEEE International Electron Devices Meeting (IEDM), 2016.

[131] X. Yin, A. Aziz, J. Nahas, S. Datta, S. Gupta, M. Niemier, X.S. Hu, "Exploiting ferroelectric fets for low-power non-volatile logic-in-memory circuits", 35th International Conference on Computer-Aided Design, 2016.

[130] S. Datta, N. Shukla, "In quest of the next switch", International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), Nuremberg, Germany, 2016.

[129] R. K. Ghosh, S. Datta, "Orbitronics—Harnessing metal-insulator phase transition in 1T-MoSe₂", International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), Nuremberg, Germany, 2016.

[128] D Wang, S George, A Aziz, S Datta, V Narayanan, SK Gupta, "Ferroelectric Transistor based Non-Volatile Flip-Flop", International Symposium on Low Power Electronics and Design (ISLPED), 2016.

[127] A Parihar, N Shukla, S Datta, A Raychowdhury, "Computing with dynamical systems in the post-CMOS era", IEEE Photonics Society Summer Topical Meeting Series (SUM), 2016.

[126] S George, A Aziz, X Li, MS Kim, S Datta, J Sampson, S Gupta, V. Narayanan, "Device Circuit Co Design of FEFET Based Logic for Low Voltage Processors", IEEE Computer Society Annual Symposium on VLSI (ISVLSI), 2016.

[125] S Datta, "Transistor innovation in the 21st century—A lesson in serendipity", Device Research Conference (DRC), Delaware, USA, June 2016 (Plenary Talk)

[125] S Datta, N Shukla, A Parihar, A Raychowdhury, "Computing with coupled dynamical systems" IEEE Silicon Nanoelectronics Workshop (SNW), Honolulu, June 2016.

[124] S Datta, R Pandey, S Mookerjee, "Opportunities and challenges of tunnel FETs", IEEE International Symposium on Circuits and Systems (ISCAS), Canada, May 2016.

[123] S. K. Gupta, A. Aziz, N. Shukla, S. Datta, "On the potential of correlated materials in the design of spin-based cross-point memories", IEEE International Symposium on Circuits and Systems (ISCAS), Canada, May 2016.

[122] R. Pandey, R. K. Ghosh, S. Datta, "Band structure engineered Germanium-Tin (GeSn) p-channel tunnel transistors", 2016 International Symposium on VLSI Technology, Systems and Application (VLSI-TSA), Taiwan, Apr. 2016.

- [121] N. Shukla, S. Datta, A. Parihar, A. Raychowdhury, and V. Narayanan, "Computing with Coupled Dynamical Systems", 15th International Workshop on Cellular Nanoscale Networks and their Applications (CNNA), Dresden, Deutschland, Aug. 2016.
- [120] Nikhil Shukla, W.Y.- Tsai, M. Jerry, M. Barth, V. Narayanan, S. Datta, "Ultra-low powered coupled oscillators for computer vision applications" VLSI Symposium, Hawaii, June 2016
- [119] J. Frougier, N. Shukla, D. Deng, M. J. Jerry, A. Aziz, L. Liu, G. Lavallee, T. S. Mayer, S. K. Gupta and S. Datta, "Phase-transition-FET Exhibiting Steep Switching Slope of 8mV/Decade and 36% Enhanced ON Current", 2016 Symposia on VLSI Technology and Circuits, June 2016.
- [118] A. Aziz, S. Ghosh, S. K. Gupta, and S. Datta, "Polarization Charge and Coercive Field Dependent Performance of Negative Capacitance FETs", Device Research Conference (DRC) 2016, Delaware, USA.
- [117] M. Jerry, N. Shukla, H. Paik, D. G. Schlom and S. Datta, "Dynamics of electrically driven sub-nanosecond switching in Vanadium dioxide", Silicon Nanoelectronics Workshop (SNW), Honolulu HI, 2016.
- [116] M. Jerry, W. Tsai, B. Xie, X. Li, V. Narayanan, A. Raychodhury and S. Datta, "Phase Transition Oxide Neuron for Spiking Neural Networks" Device Research Conference (DRC), University of Delaware, 2016.
- [115] R Pandey, R. K. Ghosh, and S Datta, "Band structure engineered Germanium-Tin (GeSn) p-channel tunnel transistors", International Symposium on VLSI Technology, Systems and Application (VLSI-TSA), 2016.
- [114] M. J. Barth, A. Kumar, J. H. Warner, B. R. Bennett, J. B. Boos, C. D. Cress, N. J.-H. Roche, P. Paillet, M. Raine, M. Gaillardin, D. McMorro, K. Saraswat, S. Datta "Single-Event Measurement and Analysis of Antimony-Based P-Channel Quantum-Well MOSFETs with High-k Dielectric" 2016 NSREC Portland OR, July 2016.
- [113] S. George, K. Ma, A. Aziz, X. Li, A. Khan, S. Salahuddin, M.-F. Chang, S. Datta, J. Sampson, S. Gupta, V. Narayanan, "Nonvolatile Memory Design Based on Ferroelectric FETs", Digital Automation Conference (DAC '16), June 05-09, 2016, Austin, TX, USA. DOI: <http://dx.doi.org/10.1145/2897937.2898050>
- [112] D. Wang, S. George, A. Aziz, S. Datta, V. Narayanan and S. K. Gupta, "Ferroelectric Transistor Based Non-Volatile Flip-Flop", Accepted in International Symposium on Low Power Electronics and Design (ISLPED), 2016.

- [111] S. K Gupta, A. Aziz, N. Shukla and S. Datta, et al, "On the Potential of Correlated Materials in the Design of Spin-based Cross-point Memories (Invited)", to be presented in IEEE International Symposium on Circuits and Systems (ISCAS), 2016, Montreal, Canada.
- [110] S. George, A. Aziz, X. Li, M. S. Kim, J. Sampson, S. Datta, S. K. Gupta, V. Narayanan, "Device –Circuit Co Design of FEFET Based Logic for Low Voltage Processors", Accepted in IEEE Computer Society Annual Symposium on VLSI (ISVLSI) 2016
- [109] A. Aziz, N. Shukla, S. Datta and S. K Gupta, "Implication of Hysteretic Selector Device on the Biasing Scheme of a Cross-point Memory Array", International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), 2015, Washington DC, USA.
- [108] Ram Krishna Ghosh, Yu-Chuan Lin, Joshua A. Robinson and Suman Datta "Heterojunction resonant tunneling diode at the atomic limit", International Conference on Simulation of Semiconductor Processes and Devices (SISPAD), 2015, Washington DC, USA
- [107] A. Aziz, N. Shukla, S. Datta and S. K Gupta, "COAST: Correlated Material Assisted STT MRAMs for Optimized Read Operation", International Symposium on Low Power Electronics and Design (ISLPED), 2015, Rome, Italy
- [106] M. Barth, H. Liu, J. H. Warner, B. R. Bennett, D. McMorow, N. Roche, P. Paillet, M. Gaillardin, and S. Datta "Single Event Measurement and Analysis of Antimony based n-channel Quantum-Well MOSFET with High-k Dielectric" 2015 NSREC Boston MA, July 2015
- [105] M. Barth, G. Bruce Rayner, R. Engel-Herbert, and S. Datta "Preparation of high quality high-k/GaSb interfaces using in-situ spectroscopic ellipsometry and reflection high energy electron diffraction" 2015 AVS ALD Conference, Portland OR June 2015
- [104] R. Pandey, H. Madan, H. Liu, V. Chobpattana, M. Barth, B. Rajamohanam, M. J. Hollander, T. Clark, K. Wang, J- H. Kim, D. Gundlach, K. P. Cheung, J. Suehle, R. Engel-Herbert, S. Stemmer and S. Datta, "Demonstration of p-type $\text{In}_{0.7}\text{Ga}_{0.3}\text{As}/\text{GaAs}_{0.35}\text{Sb}_{0.65}$ and n-type $\text{GaAs}_{0.4}\text{Sb}_{0.6}/\text{In}_{0.65}\text{Ga}_{0.35}\text{As}$ Complimentary Heterojunction Vertical Tunnel FETs for Ultra-Low Power Logic", IEEE Symposia on VLSI Technology and Circuits, Kyoto, 2015.
- [103] A. V. Thathachary, N. Agrawal, K. K. Bhuwalka, M. Cantoro, Y-C Heo, G. Lavallee, S. Maeda, and S. Datta, "Indium Arsenide (InAs) Single and Dual Quantum-Well Heterostructure FinFETs", IEEE Symposia on VLSI Technology and Circuits, Kyoto, 2015.

- [102] A. Aziz, N. Shukla, S. Datta and S. K. Gupta, "Read Optimized MRAM with Separate Read-Write Paths based on Concerted Operation of Magnetic Tunnel Junction with Correlated Material", Device Research Conference (DRC), Ohio State University, 2015.
- [101] R. Pandey, N. Agrawal, R. Arghavani, and S. Datta, "Analysis of Local Interconnect Resistance at Scaled Process Nodes", Device Research Conference (DRC), Ohio State University, 2015.
- [100] N. Agrawal, A. Agrawal, S. Mukhopadhyay, S. Mahapatra, and S. Datta, "Electron Trapping Dominance in Strained Germanium Quantum Well Planar and FinFET devices with NBTI", Device Research Conference (DRC), Ohio State University, 2015
- [99] A. Agrawal*, M. Barth*, G. B. Rayner Jr., Arun V. T.*, C. Eichfeld, G. Lavallee, S-Y. Yu, X. Sang, S. Brookes, Y. Zheng, Y-J. Lee, Y-R. Lin, C-H. Wu, C-H. Ko, J. LeBeau, R. Engel-Herbert¹, S. E. Mohny, Y-C. Yeo and S. Datta "Enhancement Mode Strained (1.3%) Germanium Quantum Well FinFET (W_{Fin}=20nm) with High Mobility (Hole=700 cm²/Vs), Low EOT (~0.7nm) on Bulk Silicon Substrate", *IEEE International Electron Device Meeting (IEDM) Technical Digest*, pp. 414-417, December 2014 (First, second and fourth authors supervised by the candidate)
- [98] N. Shukla*, A. Parihar, M. Cotter*, M. Barth*, X. Li, N. Chandramoorthy, H. Paik, D. G. Schlom, V. Narayanan, A. Raychowdhury and S. Datta, "Pairwise Coupled Hybrid Vanadium Dioxide-MOSFET (HVFET) Oscillators for Non-Boolean Associative Computing", *IEEE International Electron Device Meeting (IEDM) Technical Digest*, pp. 673-676, December 2014 (First, third and fourth authors supervised by the candidate)
- [97] Moon Seok Kim, Huichu Liu*, Karthik Swaminathan, Xueqing Li, Suman Datta, Vijaykrishnan Narayanan "Enabling power-efficient designs with III-V Tunnel FETs" *IEEE Compound Semiconductor IC Symposium (CSICS)*, October, 2014 (Second author supervised by the candidate)
- [96] Matthew J. Hollander*, Himanshu Madan*, Gregory Pastir, Randal Cavalero, David Snyder, Joshua A. Robinson, and Suman Datta. "Enhanced Short-channel Performance and p-n Symmetry in Graphene Based Ambipolar Mixer Using Nano-ribbon Geometry" *4th International Symposium on Graphene Devices*, Sept. 2014 (First and second authors supervised by the candidate)
- [95] Huichu Liu*; Mahsa Shoaran; Xueqing Li; Suman Datta; Alexandre Schmid; Vijaykrishnan Narayanan, "Tunnel FET-Based Ultra-Low Power, Low-Noise Amplifier Design for Bio-signal

Acquisition”, *International Symposium on Low Power Electronics and Design (ISLPED)*, La Jolla, August 11-13, 2014 (First author supervised by the candidate)

[94] S. Datta, R. Pandey*, S. Gupta, R. Arghavani, “Impact of Contact and Local Interconnect Scaling on Logic Performance”, *IEEE Symposia on VLSI Technology and Circuits (VLSI)*, June 2014 (Second author supervised by the candidate) (invited talk)

[93] Arun. V Thathachary*, N. Agrawal*, G. Lavallee, M. Cantoro, S.-S. Kim, D.-W. Kim and S. Datta, “Investigation of InxGa1-xAs FinFET Architecture with varying Indium (x) concentration and quantum confinement”, *IEEE Symposia on VLSI Technology and Circuits (VLSI)*, June 2014 (First and second authors supervised by the candidate)

[92] S. Datta, N. Shukla*, M. Cotter, A. Parihar, A. Raychowdhury, “Neuro Inspired Computing with Coupled Relaxation Oscillators”, *ACM/EDAC/IEEE Annual Design Automation Conference (DAC)*, pp. 1-6, June 2014 (Second author supervised by the candidate) (invited talk)

[91] Vijaykrishnan Narayanan, Suman Datta, Gert Cauwenberghs, Don Chiarulli, Steve Levitan, and Philip Wong, "Video Analytics Using Beyond CMOS Devices", *Design, Automation and Test in Europe Conference and Exhibition (DATE)*, March 2014 (equal contribution by all authors) (invited talk)

[90] Chian-We Liu, Chang-En Chiang, Ching-Yi Huang, Chun-Yao Wang, Yung-Chih Chen, Suman Datta, Vijaykrishnan Narayanan, "Width Minimization in the Single-Electron Transistor Array Synthesis", *Design, Automation and Test in Europe Conference and Exhibition (DATE)*, 2014 March 2014 (equal contribution by all authors)

[89] R. Bijesh*, H. Liu*, H. Madan*, D. Mohata*, W. Li, N. V. Nguyen, D. Gundlach, C.A. Richter, J. Maier, K. Wang, T. Clarke, J. M. Fastenau, D. Loubychev, W. K. Liu, V. Narayanan and S. Datta, "Demonstration of InGaAs/GaAsSb Near Broken-gap Tunnel FET with $I_{on}=740\mu A/\mu m$, $G_m=700\mu S/\mu m$ and Gigahertz Switching Performance at $V_{DS}=0.5V$ ", *IEEE International Electron Device Meeting (IEDM) Technical Digest*, pp. 687-690, December 2013 (First, second, third and fourth authors supervised by the candidate)

[88] H. Liu*, S. Datta, and V. Narayanan "Steep Switching Tunnel FET: A Promise to Extend the Energy Efficient Roadmap for Post-CMOS Digital and Analog/RF Application" *International Symposium on Low Power Electronics and Design (ISLPED)* Beijing, China, September 4-6, 2013 (First author supervised by the candidate)

- [87] H. Liu*, R. Vaddi, S. Datta, and V. Narayanan "Tunnel FET based Ultra-Low Power, High Sensitivity UHF RFID Rectifier" *International Symposium on Low Power Electronics and Design (ISLPED)* Beijing, China, September 4-6, 2013 (First author supervised by the candidate)
- [86] M. Barth*, A. Agrawal*, A. Ali*, J. Fastenau, D. Loubychev, W.K. Liu and S. Datta "Compressively Strained InSb MOSFETs with High Hole Mobility for P-Channel Application" *IEEE Device Research Conference (DRC)*, University of Notre Dame, June 23-26, 2013 (First, second and third authors supervised by the candidate)
- [85] Arun V. Thathachary*, L. Liu* and S.Datta "Impact of fin width scaling on carrier transport in III-V FinFETs" *IEEE Device Research Conference (DRC)*, University of Notre Dame, June 23-26, 2013. (First and second authors supervised by the candidate)
- [84] Matthew J. Hollander*, Nikhil Shukla*, Nidhi Agrawal*, Himanshu Madan*, Joshua A. Robinson and Suman Datta "Reduction of Charge Transfer Region Using Graphene Nano-ribbon Geometry for Improved Complementary FET Performance at Sub-Micron Channel Length" *IEEE Device Research Conference (DRC)*, University of Notre Dame, June 23-26, 2013 (First, second, third and fourth authors supervised by the candidate)
- [83] H. Madan*, M. J. Hollander*, J. A. Robinson, and S. Datta "Analysis and Benchmarking of Graphene Based RF Low Noise Amplifiers" *IEEE Device Research Conference (DRC)*, University of Notre Dame, June 23-26, 2013 (First and second authors supervised by the candidate).
- [82] A. Agrawal*, J. Lin, B. Zheng, S. Sharma, S. Chopra, K. Wang, A. Gelatos, S. Mohnheyand, S. Datta "Barrier Height Reduction to 0.15eV and Contact Resistivity Reduction to $9.1 \times 10^{-9} \Omega\text{-cm}^2$ Using Ultrathin $\text{TiO}_2\text{-x}$ Interlayer between Metal and Silicon" *IEEE Symposia on VLSI Technology and Circuits (VLSI)*, Kyoto, Japan, June 11-14, 2013. (First author supervised by the candidate)(nominated for best paper award)
- [81] H. Madan*, M. J. Hollander*, J. A. Robinson and S. Datta "Graphene Transistors for Ambipolar Mixing at Microwave Frequencies" *223rd Electrochemical Society Meeting*, Toronto, Ontario, Canada May 14-18 2013 (First and second authors supervised by the candidate)
- [80] S. Datta, R. Bijesh*, H. Liu*, D. Mohata*, and V. Narayanan "Tunnel Transistors for Energy Efficient Computing" *IEEE International Reliability Physics Symposium (IRPS)*, Monterey, California, April 14- 18 2013 (Second, third and fourth authors supervised by the candidate) (invited paper)

- [79] K. Joshi, S. Hung, S. Mukhopadhyay, V. Chaudhary, N. Nanaware, B. Rajamohanan*, T. Sato, M. Bevan, A. Wei, A. Noori, B. McDougal, C. Ni, G. Saheli, C. Lazik, P. Liu, D. Chu, L. Date, S. Datta, A. Brand, J. Swenberg, and S. Mahapatra "HKMG Process Impact on N, P BTI: Role of Thermal IL Scaling, IL/HK Integration and Post HK Nitridation" *IEEE International Reliability Physics Symposium (IRPS)*, Monterey, California, April 14- 18 2013 (Sixth author supervised by the candidate)
- [78] H. Madan*, M.J. Hollander*, M. LaBella, R. Cavallero, D. Snyder, J. A. Robinson and S. Datta, "Record High Conversion Gain Ambipolar Graphene Mixer at 10 GHz Using Scaled Gate Oxide", *IEEE International Electron Device Meeting (IEDM) Technical Digest*, pp. 76-79, December 2012 (First and second authors supervised by the candidate)
- [77] Huichu Liu*, Matthew Cotter, Suman Datta and Vijay Narayanan, "Technology Assessment of Si and III-V FinFETs and III-V Tunnel FETs from Soft Error Rate Perspective", *IEEE International Electron Device Meeting (IEDM) Technical Digest*, pp. 577-580, December (2012) (First author supervised by the candidate)
- [76] E. Kultursay , K. Swaminathan , V. Saripalli*, S. Datta , V. Narayanan, and M. Kandemir, "Performance Enhancement under Power Constraints using Heterogeneous CMOS-TFET Multicores" *IEEE/ACM International Conference on Hardware/Software Codesign and System Synthesis*, Oct 7-12, 2012 (Third author supervised by the candidate)
- [75] R. Mukundrajan*, M. Cotter, V. Saripalli*, M. J. Irwin, S. Datta, and V. Narayanan, "Ultra Low Power Circuit Design using Tunnel FETs" *IEEE Computer Society Annual Symposium on VLSI (ISVLSI)*, Aug 19-21, Amherst, USA (First and third authors supervised by the candidate)
- [74] M. J. Hollander*, A. Agrawal*, M. S. Bresnehan, M. LaBella, K. A. Trumbull, R. Cavallero, S. Datta, and Joshua A. Robinson, "High Performance, Large Area Graphene Transistors on Quasi-Free- Standing Graphene Using Synthetic Hexagonal Boron Nitride Gate Dielectrics" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First and second authors supervised by the candidate) (won best paper award)
- [73] M. J. Hollander*, A. Agrawal*, M. S. Bresnehan, M. LaBella, K. A. Trumbull, R. Cavallero, S. Datta, and Joshua A. Robinson, "Effect of Transferred Hexagonal Boron Nitride Dielectrics on Quasi-Freestanding Epitaxial Graphene" *Electronic Materials Conference (EMC)*, Penn State University, June 18-20, 2012 (First and second authors supervised by the candidate)

- [72] H. Liu*, D. K. Mohata*, A. Nidhi*, V. Saripalli*, V. Narayanan and S. Datta, "Exploration of Vertical MOSFET and Tunnel FET Device Architecture for Sub 10nm Node Applications" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (first, second, third and fourth authors supervised by the candidate).
- [71] H. Madan*, M. J. Hollander*, J. A. Robinson, and S. Datta, "Extraction of Near Interface Trap Density in Top Gated Graphene Transistor Using High Frequency Current Voltage Characteristics" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First and second authors supervised by the candidate)
- [70] A. Agrawal*, J. Park, D. K. Mohata*, K. Ahmed, and S. Datta, "Experimental Demonstration of "Cold" Low Contact Resistivity Ohmic Contacts on Moderately Doped n-Ge with in-situ Atomic Hydrogen Clean" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First and third authors supervised by the candidate).
- [69] E. Freeman*, A. Kar*, N. Shukla*, R. Misra, R. Engel-Herbert, D. Schlom, V. Gopalan, K. Rabe, and S.Datta, "Characterization and Modeling of Metal-Insulator Transition (MIT) Based Tunnel Junctions" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First, second and third authors supervised by the candidate)
- [68] N. Agrawal*, V.Saripalli*, V.Narayanan, Y.Kimura, R.Arghavani, and S.Datta, "Will Strong Quantum Confinement Effect Limit Low Vcc Applications of III-V FinFETs?" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First and second authors supervised by the candidate)
- [67] R. Bijesh*, D. K. Mohata*, H. Liu*, and S. Datta, "Flicker Noise Characterization and Analytical Modeling of Homo and Hetero-Junction III-V Tunnel FETs" *IEEE Device Research Conference (DRC)*, Penn State University, June 18-20, 2012 (First, second and third authors supervised by the candidate)
- [66] A. Ali*, H. Madan*, M. J. Barth*, M. J. Hollander, J. B. Boos, B. R. Bennett, and S.Datta, "Antimonide NMOSFET with Source Side Injection Velocity of 2.7×10^7 cm/s for Low Power High Performance Logic Applications" *IEEE Symposia on VLSI Technology and Circuits (VLSI)*, Honolulu, June 12-15, 2012 (First, second and third authors supervised by the candidate).
- [65] D. K. Mohata*, R. Bijesh*, Y. Zhu, M. K. Hudait, R. Southwick, Z. Chbili, D. Gundlach, J. Suehle, J. M. Fastenau, D. Loubychev, A. K. Liu, T. S. Mayer, V. Narayanan and S. Datta, "Demonstration of Improved Heteroepitaxy, Scaled Gate Stack and Reduced Interface States

Enabling Heterojunction Tunnel FETs with High Drive Current and High On-Off Ratio" *IEEE Symposia on VLSI Technology and Circuits (VLSI)*, Honolulu, June 12-15, 2012 (First and second authors supervised by the candidate)

[64] K. Ahmed, S. Chopra, A. Agrawal* and S. Datta "Benchmarking of Novel Contact Architectures on Silicon and Germanium" *International Silicon-Germanium Technology and Device Meeting (ISTDM)*, June 4-6 2012 (Third author supervised by the candidate)

[63] F. Li*, R. Misra, Z. Fang, C. Curwen, Y. Wu, Q. M. Zhang, P. Schiffer, S. Tadigadapa, and S. Datta, "Magnetoelectric Resonant Gate Transistor" *Solid-State Sensors, Actuators, and Microsystems Workshop*, Hilton Head, June 3-7, 2012 (First author supervised by the candidate)

[62] K. Ahmed, A. Agrawal*, S. Chopra, and S. Datta, "Benchmarking of Novel Contact Architectures on Silicon and Germanium" *International Silicon-Germanium Technology and Device Meeting (ISTDM)*, June 4, 2012

[61] D. K. Mohata, R. Bijesh , S. Mujumdar, C. Eaton, R. Engel-Herbert, T. Mayer, V. Narayanan, J. Fastenau, D. Loubyshev, A. Liu and S. Datta, "Demonstration of MOSFET-Like On-Current Performance in Arsenide/Antimonide Tunnel FETs with Staggered Hetero-junctions for 300mV Logic Applications", *IEEE International Electron Devices Meeting Technical Digest*, pp. 33.5.1 - 33.5.4, Dec. 5-7, 2011 (First, second and third authors supervised by the candidate)

[60] L. Liu*, V. Saripalli*, V. Narayanan and S. Datta, "Device Circuit Co-Design Using Classical and Non-Classical III-V Multi-Gate Quantum-Well FETs (MuQFETs)", *IEEE International Electron Devices Meeting Technical Digest*, 4.5.1 - 4.5.4, Dec. 5-7, 2011 (First and second authors supervised by the candidate)

[59] V. Saripalli*, J. P. Kulkarni, N. Vijaykrishnan and S. Datta, "Variation-Tolerant Ultra Low-Power Heterojunction Tunnel FET SRAM Design", *IEEE/ACM International Symposium on Nanoscale Architectures (NANOARCH)*, pp. 45 – 52, July 2011(First author supervised by the candidate)

[58] A. Agrawal*, A. Ali*, R. Misra, P. E. Schiffer, J. B. Boos, B. R. Bennett and S. Datta, "Low Field Electron Transport in Mixed Arsenide Antimonide Quantum Well Heterostructures", *Electronic Materials Conference (EMC)*, Univ. of California, Santa Barbara, June 2011 (First and second authors supervised by the candidate)

[57] A. Agrawal*, A. Ali*, R. Misra, P. E. Schiffer, B. R. Bennett, J. B. Boos and S. Datta, "Experimental Determination of Dominant Scattering Mechanisms in Scaled InAsSb Quantum

Well", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 27-28, June 2011 (First and second authors supervised by the candidate)

[56] R. Bijesh*, I. OK, M. Baykan, C. Hobbs, P.Majhi, R.Jammy and S.Datta, "Hole Mobility Enhancement in Uniaxially Strained SiGe FINFETs: Analysis and Prospects", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 237-238, June 2011 (First author supervised by the candidate)

[55] D. K. Mohata*, R. Bijesh*, V. Saripalli*, T. Mayer and S. Datta," Self-aligned Gate NanoPillar In_{0.53}Ga_{0.47}As Vertical Tunnel Transistor", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 203-204, June 2011 (First, second and third authors supervised by the candidate)

[54] F. Li*, Z. Fang*, R. Misra, S. Tadigadapa, Q. Zhang and S. Datta, "Giant magnetoelectric effect in nanofabricated Pb(Zr_{0.52}Ti_{0.48})O₃-Fe₈₅B₅Si₁₀ Cantilevers and resonant gate transistors", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 237-238, pp. 69-70, June 2011 (First and second authors supervised by the candidate)

[53] L. Liu*, V. Saripalli*, V. Narayanan and S. Datta, "Experimental Investigation of Scalability and Transport in In_{0.7}Ga_{0.3}As Multi-Gate Quantum Well FET (MuQFET)", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 17-18, June 2011 (First and second authors supervised by the candidate)

[52] H. Madan*, D. Veksler, Y.T. Chen, J. Huang, N. Goel, G. Bersuker and S. Datta, "Interface States at high-k/InGaAs interface: H₂O vs. O₃ based ALD Dielectric", Device Research Conference (DRC), Univ. of California, Santa Barbara, pp. 117-118, June 2011 (First author supervised by the candidate)

[51] C. D. Young, M. Baykan, A. Agrawal*, H. Madan*, K. Akarvardar, C. Hobbs, I. OK, W. Taylor, C. E. Smith, M. M. Hussain, T. Nishida, S. Thompson, P. Majhi, P. Kirsch, S. Datta and R. Jammy, "Critical Discussion on (100) and (110) Orientation Dependent Transport : nMOS Planar and FinFET", Intl. Symposium on VLSI Technology (VLSI), Kyoto, Japan, June, 2011. (Third and fourth authors supervised by the candidate)

[50] L. Liu*, V. Saripalli*, E. Hwang*, V. Narayanan and S. Datta, "Multi-Gate Modulation Doped In_{0.7}Ga_{0.3}As Quantum Well FET for Ultra Low Power Digital Logic", accepted for publication in 219th Electro chemical Society (ECS) Meeting, Montreal, Canada, May 1-6, 2011. (First, second and third authors supervised by the candidate)

- [49] V. Saripalli*, A. Misra, S. Datta and V. Narayanan, "An Energy-Efficient Heterogeneous CMP based on Hybrid TFET-CMOS Cores," Design Automation Conference (DAC), San Diego, June 5-10, 2011. (First author supervised by the candidate)
- [48] Y.C. Chen, S. Soumya, G. Sun, Y. Xie, S. Datta and V. Narayanan, "Automated Mapping for Reconfigurable Single Electron Transistor Arrays," Design Automation Conference (DAC) , San Diego, June 5-10, 2011
- [47] Z. Fang*, F. Li*, N. Mokhariwale, S. Datta, and Q. M. Zhang, "Direct integration of magnetoelectric sensors with microelectronics—Improved field sensitivity, signal-to-noise ratio and frequency response," pp.15-16, *IEEE Sensors 2010 Conference*, pp. 614–619, Waikoloa, Hawaii, November 2010 (First and second authors supervised by the candidate)
- [46] A. Ali, H. Madan, R. Misra, E.Hwang, A. Agrawal, P. Schiffer, J. B. Boos, B. R. Bennett, I. Geppert, M. Eizenberg and S. Datta, "Advanced Composite High- κ Gate Stack for Mixed Anion Arsenide-Antimonide Quantum Well Transistors" *IEEE International Electron Devices Meeting (IEDM)*, San Francisco, December 2010 (First author supervised by the candidate)
- [45] S. Datta "Compound Semiconductor Based Tunnel Transistor Logic," *Lester Eastman Conference on High Performance Devices (LEC)*, pp.178-179, Troy, USA, August 2010. (Invited talk by the candidate)
- [44] S. Datta, A. Ali, S. Mookerjee, V. Saripalli, L. Liu, S. Eachempati, T. Mayer and V. Narayanan, "Non-silicon logic elements on silicon for extreme voltage scaling," *Proceedings of the Silicon Nanoelectronics Workshop (SNW)*, pp.15-16, Honolulu, Hawaii, June 2010. (Invited talk by the candidate)
- [43] A.Ali*, H. S. Madan*, A. P. Kirk, R.M. Wallace, D. A. Zhao, D. A. Mourey, M. Hudait, T. N. Jackson, B. R. Bennett, J. B. Boos, and S. Datta,"Fermi Level Unpinning of GaSb(100) using Plasma Enhanced ALD Al₂O₃ Dielectric," *IEEE Device Research Conference Digest*, pp. 27-28, South Bend, Indiana, June 2010. (First and second authors supervised by the candidate)
- [42] E. Hwang*, S. Mookerjee*, M. Hudait and S. Datta, "Scalability Study of In_{0.70}Ga_{0.30}As HEMTs for 22nm node and beyond Logic Applications," *IEEE Device Research Conference Digest*, pp. 61-62, South Bend, Indiana, June 2010. (First and second authors supervised by the candidate)
- [41] A.Vallett, S. Minassian, S. Datta, J. Redwing and T. Mayer,"Fabrication of Axially-Doped Silicon Nanowire Tunnel FETs and Characterization of Tunneling Current," *IEEE Device Research*

Conference Digest (DRC), pp. 273-274, South Bend, Indiana, June 2010. (Equal contribution by all authors)

[40] D. Pawlik, M. Barth, P. Thomas, S. Kurinec, S. Mookerjee*, D. Mohata*, S. Datta, S. Cohen, D. Ritter, S. Rommel, "Sub-Micron In_{0.53}Ga_{0.47}As Esaki Diodes With Record Current Density of 1MA/cm²," *IEEE Device Research Conference Digest (DRC)*, pp. 163-164, South Bend, Indiana, June 2010. (Fifth and sixth authors supervised by the candidate)

[39] D. K. Mohata*, D. Pawlik, L. Liu*, S. Mookerjee*, V. Saripalli*, S. Rommel and S. Datta, "Implications of Record Peak Current Density In_{0.53}Ga_{0.47}As Esaki Tunnel Diode on Tunnel FET Logic Applications," *IEEE Device Research Conference Digest (DRC)*, pp. 101-102, South Bend, Indiana, June 2010. (First, third, fourth, and fifth authors supervised by the candidate)

[38] L. Liu* and S. Datta, "Investigation of the Scalability of Ultra Thin Body Double Gate Tunnel FET using Physics based 2D Analytical Model," *IEEE Device Research Conference Digest (DRC)*, pp. 15-16, South Bend, Indiana, June 2010. (First author supervised by the candidate)

[37] V. Saripalli*, D. K. Mohata*, S. Mookerjee*, S. Datta and V. Narayanan, "Low Power Loadless 4T SRAM cell based on Degenerately Doped Source (DDS) In_{0.53}Ga_{0.47}As Tunnel FETs," *IEEE Device Research Conference Digest (DRC)*, pp. 103-104, South Bend, Indiana, June 2010. (First author supervised and second, third authors supervised by the candidate)

[36] S. Datta, S. Mookerjee, D. Mohata, L. Liu, V. Saripalli, V. Narayanan and T. Mayer "Compound Semiconductor Based Tunnel Transistor Logic," *IEEE CS MANTECH Conference*, pp. 203-204, Portland, Oregon, May 2010 (Invited talk by the candidate)

[32] S. Datta, "III-V compound MOSFET and TFET devices," *Proceedings of the IEEE 11th Ultimate Integration of Silicon (ULIS) Conference*, Glasgow, Scotland, March 2010. (Plenary Invited Talk by the candidate)

[35] J. Singh*, R. Krishnan*, S. Mookerjee*, S. Datta, V. Narayanan, "A Novel Si TFET Based SRAM design for Ultra Low-Power 0.3V VDD Applications," *Proceedings of 15th Asia South Pacific Design Automation Conference (ASPDAC)*, Yokohama, Japan, January 2010. (First and second author co-supervised and third author supervised by the candidate)

[34] V. Saripalli*, S. Datta and V. Narayanan "Analyzing Energy-Delay Behavior in Room Temperature Single Electron Transistors," *23rd International Conference on VLSI Design*, pp. 399-404, Bangalore, India, January 2010. (First author co-supervised by the candidate)

- [33] S. Mookerjee*, D. Mohata*, R. Krishnan*, J. Singh*, A. Vallett, A. Ali*, T. Mayer, V. Narayanan, D. Schlom, A. Liu and S. Datta, "Experimental Demonstration of 100nm Channel Length In_{0.53}Ga_{0.47}As-based Vertical Inter-band Tunnel Field Effect Transistors (TFETs) for Ultra Low-Power Logic and SRAM Applications," *IEEE International Electron Devices Meeting (IEDM) Technical Digest*, pp. 949-951, Baltimore, Maryland, December 2009. (First, second and sixth authors supervised, third and fourth authors co-supervised by the candidate) (This paper received worldwide press coverage by leading semiconductor news agencies)
- [32] Z. Fang*, S. Lu, F. Li*, N. Mokhariwale, S. Datta, and Q.M. Zhang, "Sensitivity enhancement of magnetic sensors based on Metglas/PVDF laminates using the flux concentration effect," *Nanoelectronic Devices for Defense and Security Conference (NANO DDS)*, Colorado Springs, Colorado, September 2009. (First author co-supervised and third author supervised by the candidate)
- [31] S. Mookerjee* and S. Datta, "Band-gap Engineered Hot Carrier Tunnel Transistors," *67th IEEE Device Research Conference (DRC)*, pp. 121-122, University Park, Pennsylvania, June 2009. (First author supervised by the candidate)
- [30] A. Ali*, S. Mookerjee*, E. Hwang*, S. Kovesnikov, S. Oktyabrsky, V. Tokranov, M. Yakimov, R. Kambhampati, W. Tsai, and S. Datta, "HfO₂ Gated, Self Aligned and Directly Contacted Indium Arsenide Quantum-well Transistors for Logic Applications – A Temperature and Bias Dependent Study," *67th IEEE Device Research Conference (DRC)*, pp. 55-56, University Park, Pennsylvania, June 2009. (First, second, and third authors supervised by the candidate)
- [29] D. J. Pawlik, P. Thomas, M. Barth, K. Johnson, S.L. Rommel, S. Mookerjee*, S. Datta, M. Luisier, G. Klimeck, Z.Cheng, J. Li, J.S. Park, J.M. Hydrick, J.G. Fiorenza, and A. Lochtefeld, "Indium Gallium Arsenide on Silicon Interband Tunnel Diodes for NDR-based memory and Steep Subthreshold Slope Transistor Applications," *67th IEEE Device Research Conference (DRC)*, pp. 69-70, University Park, Pennsylvania, June 2009. (Sixth author supervised by the candidate)
- [28] N. Goel, D. Heh, S. Kovesnikov, I. OK, S. Oktyabrsky, V. Tokranov, R. Kambhampati, M. Yakimov, Y. Sun, P. Pianetta, C. Gaspe, M. Santos, J. Lee, S. Datta, P. Majhi, and W. Tsai, "Addressing The Gate Stack Challenge For High Mobility InxGaAs Channels For NFETs," *IEEE International Electron Devices Meeting Technical Digest (IEDM)*, pp. 363-366, San Francisco, California, December 2008. (Equal contribution by all authors)

- [27] S. Datta, "Sub-Quarter Volt Supply Voltage III-V Tunnel Transistors for Green Nanoelectronics," *39th IEEE Semiconductor Interface Specialists Conference (SISC)*, San Diego, California, December 2008. (Invited Talk by the candidate)
- [26] V. Saripalli*, S. Mookerjee*, S. Datta, and V. Narayanan, "Ultra low power signal processing architectures," *IEEE Biomedical Circuits and Systems Conference (BioCAS)*, pp. 333–336, Baltimore, Maryland, November 2008. (First author co-supervised and second author supervised by the candidate)
- [25] S. Mookerjee* and S. Datta, "Comparative Study of Si, Ge and InAs Based Steep Subthreshold Slope Tunnel Transistors for 0.25V Supply Voltage Logic Applications," *IEEE Device Research Conference (DRC)*, pp. 47-48, Santa Barbara, California, June 2008. (First author supervised by the candidate)
- [24] S. Datta, "Compound Semiconductor as CMOS Channel Material - Deja vu or New Paradigm?", *IEEE Device Research Conference (DRC)*, pp 33-36, Santa Barbara, California, June 2008. (Invited Talk by the candidate)
- [23] S. Eachempati, V. Saripalli*, N. Vijaykrishnan, and S. Datta, "Reconfigurable BDD Based Quantum Circuits," *IEEE/ACM International Symposium on Nanoscale Architectures (NANOARCH)*, pp. 61-67, Anaheim, California, June 2008. (Second author co-supervised by the candidate)
- [22] S. Datta, "Enabling Green Transistors with Narrow Bandgap Ccompound Semiconductors," *32nd Workshop on Compound Semiconductor Devices and Integrated Circuits (WOCSDICE)*, Leuven, Belgium, May 2008. (Invited Talk)
- [21] M. K. Hudait, S. Datta, G. Dewey, J. M. Fastenau, J. Kavalieros, W. K. Liu, D. Lubyshev, R. Pillarisetty, M. Radosavljevic, and R. Chau, "Heterogeneous Integration of Enhancement Mode In_{0.7}Ga_{0.3}As Quantum Well Transistor on Silicon Substrate using Thin (<2 um) Composite Buffer Architecture for High-Speed and Low-voltage (0.5V) Logic Applications," *IEEE International Electron Devices Meeting (IEDM) Technical Digest*, pp. 625-628, Washington, D.C., December 10–12, 2007. (Equal contribution by all authors)
- [20] M. Chandhok, S. Datta, D. Lionberger, and S. Vesecky, "Impact of Line Width Roughness of Intel's 65 nm Process Devices," *Proceedings of SPIE*, pp. 6519, Orlando, Florida, 2007. (Equal contribution by all authors)

- [19] J. Kavalieros, B. S. Doyle, S. Datta, and G. Dewey, "Tri-Gate Transistor Architecture with High-k Gate Dielectrics, Metal Gates and Strain Engineering," *Digest of Technical Papers VLSI Technology Symposium*, pp. 62-63, Honolulu, Hawaii, June 15–17, 2006. (Equal contribution by all authors)
- [18] S. Datta, "Antimonide based Quantum Well Transistors for High Speed, Low Power Logic Applications," *Proceedings of the International Conference on Indium Phosphide and Related Materials (IPRM)*, pp. 174–176, Princeton, New Jersey, May 2006. (Invited talk by the candidate)
- [16] S. Datta, T. Ashley, J. Brask, L. Buckle, M. Doczy, M. Emeny, D. Hayes, K. Hilton, R. Jefferies, T. Martin, T. Phillips, D. Wallis, P. Wilding, and R. Chau, "85nm gate length enhancement and depletion mode InSb quantum well transistors for ultra high speed and very low power digital logic applications," *International Electron Devices Meeting (IEDM) Technical Digest*, pp. 763-766, Washington, D.C., December 5-7, 2005.
- [16] R. Chau, S. Datta, and A. Majumdar, "Opportunities and Challenges of III-V Nanoelectronics for Future High-speed, Low-power Logic Applications," *IEEE Compound Semiconductor Integrated Circuit Symposium (IEEE/CSICS) Technical Digest*, pp. 17-20, Palm Springs, California, November 2005. (Equal contribution by all authors)
- [15] S. Datta* and R. Chau, "Silicon and III-V nanoelectronics," *Proceedings on the International Conference on Indium Phosphide and Related Materials (IPRM)*, pp. 7-8, Glasgow, Scotland, May 8-12, 2005. (Invited talk by the candidate)
- [14] R. Chau, J. Brask, S. Datta, G. Dewey, M. Doczy, B. Doyle, J. Kavalieros, B. Jin, M. Metz, A. Majumdar, and M. Radosavljevic, "Emerging Silicon and Non-Silicon Nano-electronic Devices: Opportunities and Challenges for Future High-Performance and Low-Power Computational Applications," *Proceedings of Technical Papers, IEEE VLSI-TSA International Symposium on Very-large-scale integration (VLSI) Technology*, pp. 13-16, Hsinchu, Taiwan, April 2005. (Equal contribution by all authors)
- [13] T. Ashley, A. Bares, L. Buckle, S. Datta, A. Dean, M. Emeny, M. Fearn, D. Hayes, K. Hilton, R. Jefferies, T. Martin, K. Nash, T. Philips, W. Tang, P. Wilding, and R. Chau, "Novel InSb-based Quantum Well Transistors for Ultra-High Speed, Low Power Logic Applications," *Proceedings 7th International Conference on Solid-State and Integrated Circuits Technology (ICSICT)*, pp. 2253-2256, Beijing, China, October 18-21, 2004. (Equal contribution by all authors)

- [12] B. Jin, S. Datta, G. Dewey, M. Doczy, B. Doyle, K. Johnson, J. Kavalieros, M. Metz, U. Shah, N. Zelick, and R. Chau, "Mobility Enhancement in Compressively Strained SiGe Surface Channel pMOS(FET) with HfO₂/TiN Gate Stack," *Proceedings of the ECS 2004 Joint International Meeting, SiGe: Materials Processing and Devices*, pp. 111-122, Hawaii, October 2004. (Equal contribution by all authors)
- [11] S. Datta, "Advanced Si and SiGe Strained NMOS and PMOS Transistors with High-K/Metal-Gate Stack," *Proceedings of the IEEE Bipolar/BiCMOS Circuits and Technology Meetings (BCTM)*, pp. 194-197, Montreal, Canada, September 2004. (Invited talk)
- [10] S. Datta, G. Dewey, M. Doczy, B. Doyle, B. Jin, J. Kavalieros, M. Metz, N. Zelick, and R. Chau, "High mobility Si/SiGe strained channel MOS transistors with HfO₂/TiN gate stacks," *IEEE International Electron Devices Meeting (IEDM) Technical Digest*, pp. 28.1.1-28.1.4, Washington, D.C., 2003. (Equal contribution by all authors)
- [9] R. Chau, S. Datta, M. Doczy, J. Kavalieros, and M. Metz, "Gate Dielectric Scaling for High-Performance Complementary metal-oxide-semiconductor (CMOS): from SiO₂ to High-K," *Extended Abstracts of International Workshop on Gate Insulator (IWGI)*, pp. 124-126, Tokyo, Japan, November 2003. (Equal contribution by all authors)
- [8] R. Chau, B. Doyle, M. Doczy, S. Datta, S. Harelend, B. Jin, J. Kavalieros, and M. Metz, "Silicon Nano-Transistors and Breaking the 10nm Physical Gate Length Barrier," *Conference Digest of 61st IEEE Device Research Conference (DRC)*, pp. 123-126, Salt Lake City, Utah, June 23-25, 2003. (Equal contribution by all authors)
- [7] B. Doyle, B. Boyanov, S. Datta, M. Doczy, S. Harelend, B. Jin, J. Kavalieros, T. Linton, R. Rios, and R. Chau, "Tri-Gate Fully-Depleted CMOS Transistors: Fabrication, Design and Layout," *Digest of Technical Papers VLSI Technology Symposium*, pp. 133-134, Kyoto, Japan, June 10-12, 2003. (Equal contribution by all authors)
- [6] R. Chau, B. Doyle, J. Kavalieros, D. Barlage, A. Murthy, M. Doczy, R. Arghavani, and S. Datta, "Advanced Depleted-Substrate Transistors: Single-gate, Double-Gate and Tri-gate," *Extended Abstracts of the International Conference on Solid-State Devices and Materials (SSDM)*, pp. 68-69, Nagoya, Japan, September 17-19, 2002. (Equal contribution by all authors)
- [5] S. Datta, K. P. Roenker, R. E. Peddenpohl II, and M. M. Cahay, "Analysis of High Current Effects on the Performance of Pnp InP-Based Heterojunction Bipolar Transistors," *Proceedings of*

Twelfth International Conference on InP and Related Materials, pp. 134-137, Piscataway, New Jersey, May 18, 2000.

[4] S. Datta, K. P. Roenker, and M. M. Cahay, “Base Pushout and High Current Effects in InP-Based Pnp Heterojunction Bipolar Transistors,” *Proceedings of the State-of-the-Art Program on Compound Semiconductors XXXI, Electrochemical Society*, Vol. 99-17, Honolulu, Hawaii, October 17-19, 1999.

[3] S. Datta, K. P. Roenker, and M. M. Cahay, “High Current and Two Dimensional Effects in InP-Based Pnp Heterojunction Bipolar Transistors,” *Proceedings of the State-of-the-Art Program on Compound Semiconductors XXIX, Electrochemical Society*, Vol. 98-12, Boston, Massachusetts, November 1-6, 1998.

[2] S. Datta, S. Shi, K. P. Roenker, and M. M. Cahay, “Base Design for Pnp InAlAs/InGaAs Heterojunction Bipolar Transistors,” *Proceedings of the State-of-the-Art Program on Compound Semiconductors XXVI, Electrochemical Society*, Vol. 97-1, pp. 272-287, Montreal, Canada, May 4-9, 1997.

[1] S. Datta, S. Shi, K. P. Roenker, M. M. Cahay, and W. E. Stanchina, “Numerical Modeling and Design of Pnp InAlAs-InGaAs Heterojunction Bipolar Transistors,” *Proceedings of the Ninth International Conference on InP and Related Materials*, pp. 392-395, Piscataway, New Jersey, May 1997.

Recent Invited Talks (selected)

Universities

“Steep Slope Phase Transition FETs and their applications” École Polytechnique Fédérale de Lausanne EPFL, Lausanne, Switzerland, 1/2016

“Inter-band Tunnel Transistors: Opportunities and Challenges”, *NEEDS Seminar*, Purdue University, West Lafayette, IN, 10/2015

“Negative Capacitance Ferroelectric Transistors: A Promising Steep Slope Device Candidate?”, *NEEDS Seminar*, Purdue University, West Lafayette, IN, 10/2015

“Ultra Low Power Devices”, *IEEE EDS Distinguished lecture series in Workshop on Connected, Autonomously Powered Systems*, Columbia University, New York, NY, 04/2014

“Cool Device Strategies for Beyond CMOS Nanoelectronics”, *ECE Colloquium*, University of Texas, Dallas, TX, 02/2014

“Cool Device Strategies for Beyond CMOS Nanoelectronics”, *ECE Colloquium*, Cornell University, Ithaca, NY, 02/2012

“Nanoelectronics for Future Energy Efficient Information Processing”, *ECE Colloquium*, University of Illinois, Urbana Champagne, IL, 03/2012

“Tunnel Transistor Based Energy Efficient Logic”, *IEEE EDS Distinguished lecture series in Electronics/Photonics*, Ohio State University, Columbus, OH, 04/2010

“Energy Efficient Logic Transistors using Compound Semiconductors” Cornell University Electron Devices Society Lecture Series, Cornell University, Ithaca, NY, 04/2010

“Logic and Memory Design using Inter-band Tunnel Transistor” Nanoseminar Seminar Series, Arizona State University, 03/2010

“Compound Semiconductor based Logic Elements” IEEE Electron Devices Mini Colloquium, Indian Institute of Technology, Mumbai, 01/2010

“Ushering in the Green Transistor Era”, Rochester Institute of Technology, Rochester, New York, 5/2009

“Green Transistors to Green Architectures”, Institut für Materialien und Bauelemente der Elektronik, Leibniz Universität Hannover (University of Hannover), Hannover, Germany, 10/2009

“Green Nanoelectronic Computing Devices”, as part of the annual workshop on Emerging Trends in Photonic and Electronic Device Research held sponsored by The University of Illinois chapters of the Optical Society of America (OSA) and the IEEE Electron Devices Society (EDS) in conjunction with the Micro and Nanotechnology Laboratory (MNTL), University of Illinois, Urbana Champagne, Illinois, 09/2008

“Recent Advances in Silicon and Non-Silicon Nanoelectronic Devices for High-Performance, Energy Efficient Logic Applications”, Penn State University Computer Science and Engineering Colloquia Series, 11/2007

“Emerging Nanoelectronic Devices for High-Speed and Ultra-Low Power Applications”, sponsored by the University of Wisconsin, Madison, Materials Research Science and Engineering Center (MRSEC) in association with Electrical Engineering Department University of Wisconsin, Madison, Wisconsin, 12/2006

“Ultra Low Power Nanoelectronics for the Logic technology”, Taipei Local Chapter of IEEE Electron Devices Society (EDS), National Tshao-tung University (NCTU), Hsinshu, Taiwan:, 12/2006

“Emerging Nanoelectronic Devices for High-Speed and Ultra-Low Power Application”, Electrical Engineering Colloquium, University of Texas, Austin, 3/2006

“Silicon Nano-Transistors and Nanotechnology for High-Performance Logic Applications”, sponsored by the IEEE Phoenix Section Components, Packaging, and Manufacturing Technology Society Chapter, & Waves and Devices Chapter, Arizona State University, Tempe, Arizona, 11/2003

Government/Industry

“Steep Slope Transistors”, Speaker at the IEEE Rebooting Computing Workshop, Washington DC, 10/12/2015

“Steep Slope Transduction FETs”, Invited Speaker at Global Foundries, Malta, New York, 10/21/2015

“Strained Germanium Quantum Well FinFETs”, Invited Speaker at Global Foundries, Malta, New York, 09/01/2015

“Steep Slope Transistors”, Keynote Speaker at the NSF sponsored “The Workshop for Energy Efficient Computing” Arlington, VA, 04/14/2015

“Function Stacks for Logic and Memory Devices”, Invited Speaker at the SEMATECH's 7th International Symposium on Advanced Gate Stack Technology, in Albany, New York, 09/29/2010-10/01/ 2010

“Non silicon logic elements for extreme voltage scaling “, Invited Speaker at the IBM MRC Workshop on III/V Devices IBM Research, Zurich, Switzerland, 09/2010

“Binary Decision Diagram Logic for Single Electron Devices and Tunnel FETs”, Invited Speaker at the Nanoelectronics Research Initiative (NRI) sponsored Architecture & Device Benchmarking Workshop, University of Notre Dame, 08/09/2010

“Tunnel Transistors: From Circuits to Architecture”, Invited Speaker at the Nanoelectronics Research Initiative (NRI) sponsored Architecture & Device Benchmarking Workshop, University of Notre Dame, 08/09/2010

“High mobility channel MOSFETs: to include or not to include in the ITRS?”, Panelist at the Sematech/IMEC III-V Workshop for discussion on inclusion high mobility channel MOSFETs in the ITRS, Hilton Hawaiian Village, Honolulu, 06/2010

“Green Transistors to Green Architectures”, Tutorial at the 16th IEEE International Symposium on High-Performance Computer Architecture (HPCA), Bangalore, India, 01/2010

“High Mobility Channel MOSFETs”, Panelist at the Workshop for Future III-V Complementary Metal–Oxide–Semiconductor (CMOS) Technology, Washington DC , December 2009

“Heterojunction Tunnel Transistor Logic,” Intel on-site NRI sponsored PI’s Workshop, Intel Corporation, Portland, Oregon, 8/2009

“Tunnel Transistor Logic”, Intel Microprocessor Research Lab Seminar, Portland, Oregon, 10/09

"Looking Beyond Silicon - A Pipe Dream or the Inevitable Next Step?" Panelist on the IEDM sponsored evening panel called (This panel assembled internationally recognized panelists to discuss the future of Complementary Metal-Oxide Semiconductor (CMOS) and beyond CMOS for leading-edge advanced integrated circuit applications", 12/2007

"III-V Complementary Metal-Oxide Semiconductor (CMOS) on Si: Technical and Manufacturing Needs" Panelist on the Sematech and Aixtron sponsored workshop on readiness of III-V MOSFET Technology. This workshop received world-wide press coverage under the heading “III-V Compounds Emerging as Prime Materials for Future NMOS Channels, Technologists Indicate at SEMATECH & AIXTRON Workshop,” Washington, D.C., 12/2007

Inventions

[186] US Patent # 11532355 “Non-volatile multi-level cell memory using a ferroelectric superlattice and related systems”

[185] US Patent # 11031482 “Gate electrode having a capping layer”

[184] US Patent # 10937907 “Method for fabricating transistor with thinned channel ”

[183] US Patent # 10839880 “Low power sense amplifier based on phase transition material”

[182] US Patent # 10707319 “Gate electrode having a capping layer”

[181] US Patent # 10672475 “Nonvolatile digital computing with ferroelectric FET”

[184] US Patent # 10475514 “Nonvolatile digital computing with ferroelectric FET”

[183] US Patent # 10367093 “Method for fabricating transistor with thinned channel”

[182] US Patent # 10262714 “Low power sense amplifier based on phase transition material”

- [181] US Patent # 10141437 “Extremely high mobility CMOS logic”
- [180] US Patent # 10121897 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [179] US Patent # 9991172 “Forming arsenide-based complementary logic on a single substrate ”
- [178] US Patent # 9806195 “Method for fabricating transistor with thinned channel”
- [177] US Patent # 9800094 “Low power nanoelectronics”
- [176] US Patent # 9761724 “Semiconductor device structures and methods of forming semiconductor structures”
- [175] US Patent # 9748391 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [174] US Patent # 9691856 “Extremely high mobility CMOS logic”
- [173] US Patent # 9614083 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [172] US Patent # 9548363 “Extremely high mobility CMOS logic”
- [171] US Patent # 9425256 “Strain inducing semiconductor regions”
- [170] US Patent # 9391068 “Power rectifier using tunneling field effect transistor”
- [169] US Patent # 9385180 “Semiconductor device structures and methods of forming semiconductor structures”
- [168] US Patent # 9368583 “Gate electrode having a capping layer”
- [167] US Patent # 9287380 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [166] US Patent # 9337307 “Method for fabricating transistor with thinned channel ”
- [165] US Patent # 9287380 “Gate electrode having a capping layer”
- [164] US Patent # 9048314 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [163] US Patent # 9006707 “Forming arsenide-based complementary logic on a single substrate”
- [162] US Patent # 8841180 “Strain-inducing semiconductor regions”
- [161] US Patent # 8816394 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”

- [160] US Patent # 8803255 “Gate electrode having a capping layer ”
- [159] US Patent # 8802517 “Extreme high mobility CMOS logic ”
- [158] US Patent # 8664694 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [157] US Patent # 8638591 “TFET based 4T memory devices”
- [156] US Patent # 8581258 “Semiconductor device structures and methods of forming semiconductor structures”
- [155] US Patent # 8530884 “Strain inducing semiconductor regions”
- [154] US Patent # 8518768 “Extreme high mobility CMOS logic”
- [153] US Patent # 8421059 “Strain inducing semiconductor region”
- [152] US Patent # 8405164 “Tri-gate transistor device with stress incorporation layer and method of fabrication ”
- [151] US Patent # 8390082 “Gate electrode having a capping layer ”
- [150] US Patent # 8369134 “TFET based 6T SRAM cell ”
- [149] US Patent # 8368135 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication”
- [148] US Patent # 8294180 “CMOS devices with a single work function gate electrode and method of fabrication”
- [147] US Patent # 8288233 “Method to introduce uniaxial strain in multigate nanoscale transistors by self aligned SI to SIGE conversion processes and structures formed thereby ”
- [146] US Patent # 8273626 “Nonplanar semiconductor device with partially or fully wrapped around gate electrode and methods of fabrication”
- [145] US Patent # 8264004 “Mechanism for forming a remote delta doping layer of a quantum well structure”
- [144] US Patent # 8237234 “Transistor gate electrode having conductor material layer”
- [143] US Patent # 8232588 “Increasing the surface area of a memory cell capacitor”
- [142] US Patent # 8217383 “High hole mobility p-channel Ge transistor structure on Si substrate”
- [141] US Patent # 8193567 “Process for integrating planar and non-planar CMOS transistors on a bulk substrate and article made thereby ”

- [140] US Patent # 8183646 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication ”
- [139] US Patent # 8183556 “Extreme high mobility CMOS logic”
- [138] US Patent # 8169027 “Substrate band gap engineered multi-gate pMOS devices”
- [137] US Patent # 8148786 “Complementary metal oxide semiconductor integrated circuit using raised source drain and replacement metal gates”
- [136] US Patent # 8138042 “Capacitor, method of increasing a capacitance area of same, and system containing same”
- [135] US Patent # 8129795 “Inducing strain in the channels of metal gate transistors”
- [134] US Patent # 8124959 “High hole mobility semiconductor device”
- [133] US Patent # 8120065 “Tensile strained NMOS transistor using group III-N source/drain regions”
- [132] US Patent # 8119508 “Forming integrated circuits with replacement metal gate electrodes”
- [131] US Patent # 8084818 “High mobility tri-gate devices and methods of fabrication”
-
- [130] US Patent # 8071983 “Semiconductor device structures and methods of forming semiconductor structures,” J. Brask, J. Kavalieros, U. Shah, S. Datta, A. Majumdar, R. Chau, B. Doyle, Dec 06, 2011
- [129] US Patent # 79892807 “Dielectric interface for group III-V semiconductor device,” J. Brask, S. Datta, M. Doczy, J. Blackwell, M. Metz, J. Kavalieros, R. Chau, Aug 02, 2011
- [128] US Patent # 7968957 “Transistor gate electrode having conductor material layer,” A. Murthy, B. Boyanov, S. Datta, B. Doyle, B. Jin, S. Yu, R. Chau, June 28, 2011
- [127] US Patent # 7960794 “Non-planar pMOS structure with a strained channel region and an integrated strained CMOS flow,” B. Doyle, S. Datta, B. Jin, N. Zelik, R. Chau, June 14, 2011
- [126] US Patent # 7951673 “Forming abrupt source drain metal gate transistors,” N. Lindert, S. Datta, J. Kavalieros, M. Doczy, M. Metz, J. Brask, R. Chau, M. Bohr, A. Murthy, May 31, 2011
- [125] US Patent # 7915694 “Gate electrode having a capping layer,” G. Dewey, M. Doczy, S. Datta, J. Brask, M. Metz, Mar 29, 2011

- [124] US Patent # 7915167 “Fabrication of channel wraparound gate structure for field-effect transistor,” M. Radosavljevic, A. Majumdar, B. Doyle, J. Kavalieros, M. Doczy, J. Brask, U. Shah, S. Datta, R. Chau, Mar 29, 2011
- [124] US Patent # 7902058 “Inducing strain in the channels of metal gate transistors,” S. Datta, J. Kavalieros, M. Doczy, M. Metz, J. Brask, R. Chau, B. Doyle, Mar 08, 2011
- [123] US Patent # 7902014 “CMOS devices with a single work function gate electrode and method of fabrication,” B. Doyle, B. Jin, J. Kavalieros, S. Datta, J. Brask, R. Chau, Mar 08, 2011
- [122] US Patent # 7898041 “Block contact architectures for nanoscale channel transistors,” M. Radosavljevic, A. Majumdar, B. Doyle, J. Kavalieros, M. Doczy, J. Brask, U. Shah, S. Datta, R. Chau, Mar 01, 2011
- [121] US Patent # 7893506 “Field effect transistor with narrow bandgap source and drain regions and method of fabrication,” R. Chau, S. Datta, J. Kavalieros, J. Brask, M. Doczy, M. Metz, Feb 22, 2011
- [120] US Patent # 7888221 “Tunneling field effect transistor using angled implants for forming asymmetric source/drain regions,” J. Kavalieros, M. Metz, G. Dewey, B. Jin, J. Brask, S. Datta, R. Chau, Feb 15, 2011
- [119] US Patent # 7883951 “CMOS device with metal and silicide gate electrodes and a method for making it,” J. Brask, M. Doczy, J. Kavalieros, M. Metz, C. Barns, U. Shah, S. Datta, C. Thomas, R. Chau, Feb 08, 2011
- [118] US Patent # 7879739 “Thin transition layer between a group III-V substrate and a high-k gate dielectric layer,” W. Rachmady, J. Blackwell, S. Datta, J. Kavalieros, M. Hudait, Feb 01, 2011
- [117] US Patent # 7879675 “Field effect transistor with metal source/drain regions,” M. Radosavljevic, S. Datta, B. Doyle, J. Kavalieros, J. Brask, M. Doczy, A. Majumdar, R. Chau, Feb 01, 2011
- [116] US Patent # 7875937 “Semiconductor device with a high-k gate dielectric and a metal gate electrode,” M. Metz, M. Doczy, J. Brask, J. Kavalieros, R. Chau, Robert S, Jan 25, 2011
- [115] US Patent # 7871916 “Transistor gate electrode having conductor material layer,” A. Murthy, B. Boyanov, S. Datta, B. Doyle, B. Jin, S. Yu, R. Chau, Jan 18, 2011
- [114] US Patent # 7859081 “Capacitor, method of increasing a capacitance area of same, and system containing same,” B. Doyle, R. Chau, S. Datta, V. De, A. Keshavarzi, D. Somasekhar, Dec 28, 2010

- [113] US Patent # 7858481 "Method for fabricating transistor with thinned channel," J. Brask, R. Chau, S. Datta, M. Doczy, B. Doyle, J. Kavalieros, A. Majumdar, M. Metz, M. Radosavljevic, Dec 28, 2010
- [112] US Patent # 7825481 "Field effect transistor with narrow bandgap source and drain regions and method of fabrication," R. Chau, S. Datta, J. Kavalieros, J. Brask, M. Doczy, M. Metz, Nov 02, 2010
- [111] US Patent # 7825437 "Unity beta ratio tri-gate transistor static random access memory (SRAM)," R. Pillarisetty, S. Datta, J. Kavalieros, B. Doyle, U. Shah, Nov 02, 2010
- [110] US Patent # 7825400 "Strain-inducing semiconductor regions," S. Datta, J. Kavalieros, B. Jin, Nov 02, 2010
- [109] US Patent # 7820513 "Nonplanar semiconductor device with partially or fully wrapped around gate electrode and methods of fabrication," S. Hareland, R. Chau, B. Doyle, R. Rios, T. Linton, S. Datta, Oct 26, 2010
- [108] US Patent # 7820512 "Spacer patterned augmentation of tri-gate transistor gate length," R. Pillarisetty, S. Datta, J. Kavalieros, B. Doyle, U. Shah. Oct 26, 2010
- [107] US Patent # 7791063 "High hole mobility p-channel Ge transistor structure on Si substrate," M. Hudait, S. Datta, J. Kavalieros, P. Tolchinsky, Sept 07, 2010
- [106] US Patent # 7790536 8 "Dopant confinement in the delta doped layer using a dopant segregation barrier in quantum well structures" M. Hudait, A. Budrevich, D. Loubychev, J. Kavalieros, S. Datta, J. Fastenau, A. Liu Sept 07, 2010
- [105] US Patent # 7785958 "Method for making a semiconductor device having a high-k gate dielectric layer and a metal gate electrode," M. Doczy, J. Brask, J. Kavalieros, U. Shah, M. Metz, S. Datta, R. Nagisetty, R. Chau, Aug 31, 2010.
- [104] US Patent # 7776684 "Increasing the surface area of a memory cell capacitor," B. Doyle, R. Chau, V. De, D. Somasekhar, Aug 17, 2010.
- [103] US Patent # 7736956 "Lateral undercut of metal gate in SOI device," S. Datta, J. Brask, J. Kavalieros, B. Doyle, G. Dewey, M. Doczy, R. Chau, June 15, 2010.
- [102] US Patent # 7718479 "Forming integrated circuits with replacement metal gate electrodes," J. Kavalieros, J. Brask, M. Doczy, M. Metz, S. Datta, U. Shah, R. Chau, May 18, 2010.
- [101] US Patent # 7714397 "Tri-gate transistor device with stress incorporation layer and method of fabrication," S. Hareland, R. Chau, B. Doyle, S. Datta, B. Jin, May 11, 2010.

- [100] US Patent # 7713803 “Mechanism for forming a remote delta doping layer of a quantum well structure,” B. Jin, J. Kavalieros, S. Datta, A. Majumdar, R. Chau, B. Jin, May 11, 2010
- [99] US Patent # 7709909 “Method for making a semiconductor device having a high-k gate dielectric,” M. Doczy, G. Dewey, S. Datta, S. Pae, J. Brask, J. Kavalieros, M. Metz, A. Sherrill, M. Kuhn, R. Chau, May 04, 2010.
- [98] US Patent # 7704833 “Method of forming abrupt source drain metal gate transistor,” N. Lindert, S. Datta, J. Kavalieros, M. Doczy, M. Metz, J. Brask, R. Chau, M. Bohr, A. Murthy, April 27, 2010.
- [97] US Patent # 7671471 “Method for making a semiconductor device having a high-k dielectric layer and a metal gate electrode,” J. Brask, J. Kavalieros, M. Doczy, U. Shah, C. Barnes, M. Metz, S. Datta, A. Cappellani, R. Chau, March 03, 2010.
- [96] US Patent # 7642610 “Transistor gate electrode having conductor material layer,” A. Murthy, B. Boyanov, S. Datta, B. Doyle, B. Jin, S. Yu, R. Chau, January 05, 2010.
- [95] US Patent # 7642603 “Semiconductor device with reduced fringe capacitance,” S. Datta, T. Rakshit, J. Kavalieros, B. Doyle, January 05, 2010.
- [94] US Patent # 7638169 “Directing carbon nanotube growth,” M. Radosavljevic, J. Kavalieros, A. Majumdar, S. Datta, December 29, 2009.
- [93] US Patent # 7629643 “Independent n-tips for multi-gate transistors,” R. Pillarisetty, S. Datta, J. Kavalieros, B. Doyle, December 12, 2009
- [92] US Patent # 7615441 “Forming high-k dielectric layers on smooth substrates,” J. Brask, J. Kavalieros, M. Doczy, M. Metz, S. Datta, U. Shah, G. Dewey, R. Chau, November 10, 2009
- [91] US Patent # 7608883 “Transistor for non volatile memory devices having a carbon nanotube channel and electrically floating quantum dots in its gate dielectric” M. Radosavljevic, A. Majumdar, S. Datta, J. Brask, B. Doyle, R. Chau, October 27, 2009
- [90] US Patent # 7601980 “Dopant confinement in the delta doped layer using a dopant segregation barrier in quantum well structures” M. Hudait, A. Budrevich, D. Loubyshev, J. Kavalieros, S. Datta, J. Fastenau, A. Liu, October 13, 2009
- [89] US Patent # 7598560 “Hetero-bimos injection process for non-volatile flash memory,” J. Kavalieros, S. Datta, R. Chau, D. Kencke, October 6, 2009.
- [88] US Patent # 7592213 “Tensile strained NMOS transistor using group III-N source/drain regions,” S. Datta, J. Brask, B. Jin, J. Kavalieros, M. Hudait, September 22, 2009.

- [87] US Patent # 7575991 "Removing a high-k gate dielectric," M. Doczy, R. Norman, J. Brask, M. Metz, S. Datta, R. Chau, August 18, 2009.
- [86] US Patent # 7569857 "Dual crystal orientation circuit devices on the same substrate," P. Tolchinsky, J. Kavalieros, B. Doyle, S. Datta, August 4, 2009.
- [85] US Patent # 7569869 "Transistor having tensile strained channel and system including same," B. Jin, R. Chau, S. Datta, J. Kavalieros, M. Radosavljevic, August 4, 2009.
- [84] US Patent # 7566898 "Buffer architecture formed on a semiconductor wafer," M. Hudait, D. Loubychev, S. Datta, R. Chau, J. Fastenau, A. Liu, July 28, 2009.
- [83] US Patent # 7560756 "Tri-gate devices and methods of fabrication," R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, July 14, 2009.
- [82] US Patent # 7547637 "Methods for patterning a semiconductor film," J. Brask, J. Kavalieros, U. Shah, S. Datta, A. Majumdar, R. Chau, B. Doyle, June 16, 2009.
- [81] US Patent # 7531393 "Non-planar MOS structure with a strained channel region," B. Doyle, S. Datta, B. Jin, R. Chau, May 12, 2009.
- [80] US Patent # 7531404 "Semiconductor device having a metal gate electrode formed on an annealed high-k gate dielectric layer," S. Pae, J. Maiz, J. Brask, G. Dewey, J. Kavalieros, R. Chau, S. Datta, May 12, 2009.
- [79] US Patent # 7524727 "Gate electrode having a capping layer," G. Dewey, M. Doczy, S. Datta, J. Brask, M. Metz, April 28, 2009.
- [78] US Patent # 7525160 "Multigate device with recessed strain regions," J. Kavalieros, J. Brask, S. Datta, B. Doyle, R. Chau, April 28, 2009.
- [77] US Patent # 7514346 "Tri-gate devices and methods of fabrication," R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, April 14, 2009.
- [76] US Patent # 7518196 "Field effect transistor with narrow bandgap source and drain regions and method of fabrication," R. Chau, S. Datta, J. Kavalieros, J. Brask, M. Metz, April 14, 2009.
- [75] US Patent # 7514746 "Floating-body dynamic random access memory and method of fabrication in tri-gate technology," S. Tang, A. Keshavarzi, D. Somasekhar, F. Paillet, M. Khellah, Y. Ye, S. Lu, B. Doyle, S. Datta, V. De, April 7, 2009.
- [74] US Patent # 7504678 "Tri-gate devices and methods of fabrication," R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, March 17, 2009.

- [73] US Patent # 7501336 “Metal gate device with reduced oxidation of a high-k gate dielectric,” B. Doyle, J. Kavalieros, J. Brask, M. Metz, M. Doczy, S. Datta, R. Chau, March 10, 2009.
- [72] US Patent # 7569443 “Complementary metal oxide semiconductor integrated circuit using raised source drain and replacement metal gate,” J. Kavalieros, A. Cappellani, J. Brask, M. Doczy, S. Datta, C. Barnes, R. Chau, January 20, 2009.
- [71] US Patent # 7494862 “Methods for uniform doping of non-planar transistor structures,” B. Doyle, R. Chau, S. Datta, J. Kavalieros, February 24, 2009.
- [70] US Patent # 7485503 “Dielectric interface for group III-V semiconductor device,” J. Brask, S. Datta, M. Doczy, J. Blackwell, M. Metz, J. Kavalieros, R. Chau, February 3, 2009.
- [69] US Patent # 7479421 “Process for integrating planar and non-planar CMOS transistors on a bulk substrate and article made thereby,” J. Kavalieros, J. Brask, B. Doyle, U. Shah, S. Datta, M. Doczy, M. Metz, R. Chau, January 20, 2009.
- [68] US Patent # 7465976 “Tunneling field effect transistor using angled implants for forming asymmetric source/drain regions,” J. Kavalieros, M. Metz, G. Dewey, B. Jin, S. Datta, R. Chau, December 16, 2008.
- [67] US Patent # 7456476 “Nonplanar semiconductor device with partially or fully wrapped around gate electrode and methods of fabrication,” S. Hareland, R. Chau, B. Doyle, R. Rios, T. Linton, S. Datta, November 25, 2008.
- [64] US Patent # 7449373 “Method of ion implanting for tri-gate devices,” B. Doyle, S. Datta, J. Kavalieros, A. Majumdar, November 11, 2008.
- [63] US Patent # 7449756 “Semiconductor device with a high-k gate dielectric and a metal gate electrode or fully wrapped around gate electrode and methods of fabrication,” M. Metz, S. Datta, M. Doczy, J. Brask, J. Kavalieros, R. Chau, November 11, 2008.
- [62] US Patent # 7445980 “Method and apparatus for improving stability of a 6T CMOS SRAM cell,” S. Datta, B. Doyle, R. Chau, B. Jin, J. Kavalieros, B. Zheng, S. Hareland, November 4, 2008.
- [61] US Patent # 7442983 “Method for making a semiconductor device having a high-k gate dielectric,” M. Doczy, G. Dewey, S. Datta, S. Pae, J. Brask, J. Kavalieros, M. Metz, A. Sherrill, M. Kuhn, R. Chau, October 28, 2008.
- [60] US Patent # 7439113 “Forming dual metal complementary metal oxide semiconductor integrated circuits,” M. Doczy, M. Taylor, J. Brask, J. Kavalieros, S. Datta, M. Metz, R. Chau, J. Hwang, October 21, 2008.

- [59] US Patent # 7427794 “Tri-gate devices and methods of fabrication,” R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, S. Hareland, September 23, 2008.
- [58] US Patent # 7429747 “Sb-based CMOS devices,” M. Hudait, S. Datta, J. Kavalieros, M. Doczy, R. Chau, September 30, 2008.
- [57] US Patent # 7427541 “Carbon nanotube energy well (CNEW) field effect transistor,” S. Datta, M. Radosavljevic, B. Doyle, J. Kavalieros, J. Brask, A. Majumdar, R. Chau, September 23, 2008.
- [56] US Patent # 7425490 “Reducing reactions between polysilicon gate electrodes and high dielectric constant gate dielectrics,” J. Kavalieros, J. Brask, M. Doczy, U. Shah, M. Metz, S. Datta, R. Chau, September 16, 2008.
- [55] US Patent # 7425500 “Uniform silicide metal on epitaxially grown source and drain regions of three-dimensional transistors,” M. Metz, S. Datta, J. Kavalieros, M. Doczy, J. Brask, R. Chau, September 16, 2008.
- [54] US Patent # 7407847 “Stacked multi-gate transistor design and method of fabrication,” B. Doyle, T. Rakshit, R. Chau, S. Datta, J. Brask, U. Shah, August 5, 2008.
- [53] US Patent # 7402875 “Lateral undercut of metal gate in SOI device,” S. Datta, J. Brask, J. Kavalieros, B. Doyle, G. Dewey, M. Doczy, R. Chau, July 22, 2008.
- [52] US Patent # 7390947 “Forming field effect transistors from conductors,” A. Majumdar, J. Brask, M. Radosavljevic, S. Datta, B. Doyle, M. Doczy, J. Kavalieros, M. Metz, R. Chau, U. Shah, J. Blackwell, June 24, 2008.
- [51] US Patent # 7390709 “Method for making a semiconductor device having a high-k gate dielectric layer and a metal gate electrode,” M. Doczy, J. Brask, J. Kavalieros, U. Shah, M. Metz, S. Datta, R. Nagisetty, R. Chau, June 24, 2008.
- [50] US Patent # 7387927 “Reducing oxidation under a high K gate dielectric,” R. Turkot, J. Brask, J. Kavalieros, M. Doczy, M. Metz, U. Shah, S. Datta, R. Chau, June 17, 2008.
- [49] US Patent # 7384880 “Method for making a semiconductor device having a high-k gate dielectric,” J. Brask, J. Kavalieros, M. Doczy, S. Datta, R. Chau, June 10, 2008.
- [48] US Patent # 7381608 “Method for making a semiconductor device with a high-k gate dielectric and a metal gate electrode,” J. Brask, S. Pae, J. Kavalieros, M. Metz, M. Doczy, S. Datta, R. Chau, J. Maiz, June 03, 2008.
- [47] US Patent # 7358121 “Tri-gate devices and methods of fabrication,” R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, April 15, 2008.

- [46] US Patent # 7355281 "Method for making semiconductor device having a high-k gate dielectric layer and a metal gate electrode," J. Brask, J. Kavalieros, M. Doczy, U. Shah, C. Barnes, M. Metz, S. Datta, A. Cappellani, R. Chau, April 08, 2008.
- [45] US Patent # 7355254 "Pinning layer for low resistivity N-type source drain ohmic contacts," S. Datta, J. Kavalieros, M. Doczy, R. Chau, April 08, 2008.
- [44] US Patent # 7348284 "Non-planar pMOS structure with a strained channel region and an integrated strained CMOS flow," B. Doyle, S. Datta, B. Jin, N. Zelik, R. Chau, March 25, 2008.
- [43] US Patent # 7342277 "Transistor for non volatile memory devices having a carbon nanotube channel and electrically floating quantum dots in its gate dielectric," M. Radosavljevic, A. Majumdar, S. Datta, J. Brask, B. Doyle, R. Chau, March 11, 2008.
- [42] US Patent #7323423 "Forming high-k dielectric layers on smooth substrates," J. Brask, J. Kavalieros, M. Doczy, M. Metz, S. Datta, U. Shah, G. Dewey, R. Chau, January 29, 2008.
- [41] US Patent #7317231 "Method for making a semiconductor device having a high-K gate dielectric and a titanium carbide gate electrode," M. Metz, S. Datta, M. Doczy, J. Kavalieros, J. Brask, R. Chau, January 8, 2008.
- [40] US Patent #7279375 "Block contact architectures for nanoscale channel transistors," M. Radosavljevic, A. Majumdar, B. Doyle, J. Kavalieros, M. Doczy, J. Brask, U. Shah, S. Datta, R. Chau, October 9, 2007.
- [39] US Patent # 7268058 "Tri-gate transistors and methods to fabricate same," R. Chau, S. Datta, B. Doyle, B. Jin, September 11, 2007.
- [38] US Patent # 7241653 "Nonplanar device with stress incorporation layer and method of fabrication," S. Hareland, R. Chau, B. Doyle, S. Datta, B. Jin, July 10, 2007.
- [37] US Patent # 7235809 "Semiconductor channel on insulator structure," B. Jin, B. Doyle, S. Hareland, M. Doczy, M. Metz, B. Boyanov, S. Datta, J. Kavalieros, R. Chau, June 26, 2007.
- [36] US Patent # 7223679 "Transistor gate electrode having conductor material layer," A. Murthy, B. Boyanov, S. Datta, B. Doyle, B. Jin, S. Yu, R. Chau, May 29, 2007.
- [35] US Patent #7220635 "Method for making a semiconductor device with a metal gate electrode that is formed on an annealed high-k gate dielectric layer," J. Brask, M. Doczy, J. Kavalieros, U. Shah, M. Metz, C. Barnes, S. Datta, C. Thomas, R. Chau, May 22, 2007.
- [34] US Patent # 7193279 "Non-planar MOS structure with a strained channel region," B. Doyle, S. Datta, B. Jin, R. Chau, March 20, 2007.

- [33] US Patent # 7192890 “Depositing an oxide,” Y. Zhou, M. Metz, J. Brask, J. Burghard, M. Kuhn, S. Datta, R. Chau, March 20, 2007.
- [32] US Patent #719285 “Forming dual metal complementary metal oxide semiconductor integrated circuits,” M. Doczy, L. Wong, V. Dubin, J. Brask, J. Kavalieros, S. Datta, M. Metz, R. Chau, March 20, 2007.
- [31] US Patent #7176090 “Method for making a semiconductor device that includes a metal gate electrode,” J. Brask, J. Kavalieros, M. Doczy, M. Metz, S. Datta, U. Shah, B. Doyle, R. Chau, February 13, 2007.
- [30] US Patent #7170120 “Carbon nanotube energy well (CNEW) field effect transistor,” S. Datta, M. Radosavljevic, B. Doyle, J. Kavalieros, J. Brask, A. Majumdar, R. Chau, January 30, 2007.
- [29] US Patent #7160779 “Method for making a semiconductor device having a high-k gate dielectric,” M. Doczy, J. Kavalieros, J. Brask, M. Metz, S. Datta, B. Doyle, R. Chau, January 9, 2007.
- [28] US Patent #7157378 “Method for making a semiconductor device having a high-k gate dielectric layer and a metal gate electrode,” J. Brask, C. Barns, M. Doczy, U. Shah, J. Kavalieros, M. Metz, S. Datta, A. Miller, R. Chau, January 2, 2007.
- [27] US Patent #7153784 “Method for making a semiconductor device having a high-k gate dielectric layer and a metal gate electrode,” J. Brask, J. Kavalieros, M. Doczy, U. Shah, C. Barns, M. Metz, S. Datta, A. Cappellani, R. Chau, December 26, 2006.
- [26] US Patent #7153734 “CMOS device with metal and silicide gate electrodes and a method for making it,” Brask, Justin K. (Portland, OR, US) M. Doczy, J. Kavalieros, M. Metz, C. Barns, U. Shah, S. Datta, C. Thomas, R. Chau, December 26, 2006.
- [25] US Patent #7148548 “Semiconductor device with a high-k gate dielectric and a metal gate electrode,” M. Doczy, J. Kavalieros, M. Metz, J. Brask, S. Datta, R. Chau, December 12, 2006.
- [24] US Patent #7148099 “Reducing the dielectric constant of a portion of a gate dielectric,” S. Datta, J. Kavalieros, M. Doczy, M. Metz, J. Brask, R. Chau, December 12, 2006.
- [23] US Patent #7144783 “Reducing gate dielectric material to form a metal gate electrode extension,” S. Datta, J. Brask, J. Kavalieros, M. Doczy, M. Metz, R. Chau, December 5, 2006.
- [22] US Patent #7138323 “Planarizing a semiconductor structure to form replacement metal gates,” J. Kavalieros, J. Brask, M. Doczy, U. Shah, C. Barns, M. Metz, S. Datta, R. Chau, November 21, 2006.

- [21] US Patent #7138316 "Semiconductor channel on insulator structure," B. Jin, B. Doyle, S. Hareland, M. Doczy, M. Metz, Matthew, B. Boyanov, S. Datta, J. Kavalieros, R. Chau, November 21, 2006.
- [20] US Patent #7138305 "Method and apparatus for improving stability of a 6T CMOS SRAM cell," S. Datta, B. Doyle, R. Chau, J. Kavalieros, B. Zheng, S. Hareland, November 21, 2006.
- [19] US Patent #7126199 "Multilayer metal gate electrode," M. Doczy, J. Brask, J. Kavalieros, C. Barns, M. Metz, S. Datta, R. Chau, October 24, 2006.
- [18] US Patent #7125762 "Compensating the workfunction of a metal gate transistor for abstraction by the gate dielectric layer," J. Brask, J. Kavalieros, M. Doczy, M. Metz, S. Datta, U. Shah, R. Chau, October 24, 2006.
- [17] US Patent #7098507 "Floating-body dynamic random access memory and method of fabrication in tri-gate technology," S. Tang, A. Keshavarzi, D. Somasekhar, F. Paillet, M. Khellah, Y. Ye, S. Lu, B. Doyle, S. Datta, V. De, August 29, 2006.
- [16] US Patent #7087476 "Using different gate dielectrics with NMOS and PMOS transistors of a complementary metal oxide semiconductor integrated circuit," M. Metz, S. Datta, J. Kavalieros, M. Doczy, J. Brask, R. Chau, August 8, 2006.
- [15] US Patent #7084038 "Method for making a semiconductor device having a high-k gate dielectric," M. Doczy, G. Dewey, S. Datta, S. Pae, J. Brask, J. Kavalieros, M. Metz, A. Sherrill, M. Kuhn, R. Chau, August 1, 2006.
- [14] US Patent #7074680 "Method for making a semiconductor device having a high-k gate dielectric," M. Doczy, G. Dewey, S. Datta, S. Pae, J. Brask, J. Kavalieros, M. Metz, Matt, A. Sherrill, M. Kuhn, R. Chau, July 11, 2006.
- [13] US Patent #7064066 "Method for making a semiconductor device having a high-k gate dielectric and a titanium carbide gate electrode," M. Metz, S. Datta, M. Doczy, J. Kavalieros, J. Brask, R. Chau, June 20, 2006.
- [12] US Patent #7060568 "Using different gate dielectrics with NMOS and PMOS transistors of a complementary metal oxide semiconductor integrated circuit," M. Metz, S. Datta, J. Kavalieros, M. Doczy, J. Brask, R. Chau, June 13, 2006.
- [11] US Patent #7045428 "Method for making a semiconductor device with a high-k gate dielectric and a conductor that facilitates current flow across a P/N junction," J. Brask, J. Kavalieros, M. Doczy, M. Metz, U. Shah, C. Barns, S. Datta, R. Turkot, R. Chau, May 16, 2006.

- [10] US Patent #7042009 “High mobility tri-gate devices and methods of fabrication,” M. Shaheen, B. Doyle, S. Datta, R. Chau, P. Tolchinsky, May 9, 2006.
- [9] US Patent #7005366 “Tri-gate devices and methods of fabrication,” R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, S. Hareland, February 28, 2006.
- [8] US Patent #6974738 “Nonplanar device with stress incorporation layer and method of fabrication,” S. Hareland, R. Chau, B. Doyle, S. Datta, December 13, 2005.
- [7] US Patent #6970373 “Method and apparatus for improving stability of a 6T CMOS SRAM cell,” S. Datta, B. Doyle, R. Chau, J. Kavalieros, B. Zheng, S. Hareland, November 29, 2005.
- [6] US Patent #6787440 “Method for making a semiconductor device having an ultra-thin high-k gate dielectric,” C. Parker, M. Kuhn, Y. Zhou, S. Hareland, S. Datta, N. Lindert, R. Chau, T. Glassman, M. Metz, S. Tyagi, September 7, 2005.
- [5] US Patent #6914295 “Tri-gate devices and methods of fabrication,” R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, S. Hareland, July 5, 2005.
- [4] US Patent #6909151 “Nonplanar device with stress incorporation layer and method of fabrication,” S. Hareland, R. Chau, B. Doyle, S. Datta, B. Jin, June 21, 2005.
- [3] US Patent #6887800 “Method for making a semiconductor device with a high-k gate dielectric and metal layers that meet at a P/N junction,” M. Metz, S. Datta, J. Kavalieros, M. Doczy, J. Brask, U. Shah, R. Chau, May 3, 2005.
- [2] US Patent #6869889 “Etching metal carbide films,” J. Brask, J. Kavalieros, M. Doczy, M. Metz, S. Datta, U. Shah, T. Bacuita, R. Chau, March 22, 2005.
- [1] US Patent #6858478 “Tri-gate devices and methods of fabrication,” R. Chau, B. Doyle, J. Kavalieros, D. Barlage, S. Datta, S. Hareland, February 22, 2005.

Professional Activities

Associate Editor, IEEE Journal on Exploratory Solid-State Computational Devices and Circuits

Distinguished Lecturer, Electron Devices Society, IEEE

Fellow of IEEE; **Fellow** of National Academy of Inventors

General Chair, *IEEE International Electron Devices Meeting (IEDM '20)*, San Francisco, December 2020

Chair, *Technical Program, IEEE International Electron Devices Meeting (IEDM '19)*, San Francisco, December 2019

Vice Chair, *Technical Program, IEEE International Electron Devices Meeting (IEDM '18)*, San Francisco, December 2018

Chair, *Short Course, IEEE International Electron Devices Meeting (IEDM '17)*, San Francisco, December 2017

Vice Chair, *Short Course, IEEE International Electron Devices Meeting (IEDM '16)*, San Francisco, December 2016

Chair, *Publications, IEEE International Electron Devices Meeting (IEDM '15)*, Washington DC, December 2015

TPC Member, Focus Sessions Chair, VLSI Technology Symposium (*VLSI Symposium '23*), Honolulu, Hawaii

TPC Member, Focus Sessions Chair, VLSI Technology Symposium (*VLSI Symposium '22*), Kyoto, Japan

TPC Member, Forum Chair, VLSI Technology Symposium (*VLSI Symposium '21*), Honolulu, Hawaii

TPC Member, Arrangements Chair, VLSI Technology Symposium (*VLSI Symposium '20*), Kyoto, Japan

TPC Member, Arrangements Chair, VLSI Technology Symposium (*VLSI Symposium '19*), Honolulu, Hawaii

TPC Member, VLSI Technology Symposium (*VLSI Symposium '18*), Kyoto, Japan

TPC Member, Design Automation Conference (DAC '16, DAC '17)

TPC Member, International Conference on Simulation of Semiconductor Processes & Devices (SISPAD '16), Dresden, Germany

TPC Member, International Reliability Physics Symposium (IRPS '16, IRPS '17)

Chair, *Publicity, IEEE International Electron Devices Meeting (IEDM '14)*, San Francisco, California, December 2014

Chair, *Tutorials, IEEE International Electron Devices Meeting (IEDM '12)*, San Francisco, California, December 2012

General Chair, *IEEE Device Research Conference (DRC)*, Notre Dame University, University Park, PA, June 2013

Program Chair, *IEEE Device Research Conference (DRC)*, Pennsylvania State University, University Park, PA, June 2012

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Chair, *Emerging Technologies, IEEE International Electron Devices Meeting (IEDM '11)*, Washington DC, December 2011

TPC Vice-Chair, *IEEE Device Research Conference (DRC)*, University of California, Santa Barbara, CA, June 2011

Chair, *Quantum, Power, and Compound Semiconductor Devices Sub-Committee, IEEE International Electron Devices Meeting (IEDM '10)*, San Francisco, California, December 2010

Organizer and Chair, *Rump Session on Embedded Memory called "Looking for Extra Cache", IEEE Device Research Conference (DRC '10)*, Univ. of Notre Dame, June 2010

Organizer and Chair, *Rump Session on Steep Slope Transistors called "Steep Slope or Slippery Slope", IEEE Device Research Conference (DRC '10)*, Penn State University, June 2009

TPC Member, *IEEE Device Research Conference (DRC '10)*, Univ. of Notre Dame, June 2010

TPC Member, *Electronic Materials Conference (EMC '10)*, Univ. of Notre Dame, June 2010

TPC Member, *Semiconductor Interface Specialist Conference (SISC '10)*, San Diego, California, December 2010

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Seville, Spain, September, 2010

TPC Member, *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED '10)*, Austin, Texas, August 2009

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Athens, Greece, September, 2009

TPC Member, *Quantum, Power, and Compound Semiconductor Devices, IEEE International Electron Devices Meeting (IEDM '09)*, Baltimore, Maryland, December 7–9, 2009

TPC Member, *IEEE Device Research Conference (DRC '09)*, Penn State Univ., June 2009

TPC Member, *ACM/IEEE International Symposium on Low Power Electronics and Design (ISLPED '09)*, San Francisco, California, August 2009

TPC Member, *European Solid-State Device Research Conference (ESSDERC '09)*, Edinburgh, Scotland, September 2008

TPC Member, *Silicon Nanoelectronics Workshop (SNW '08)*, Honolulu, Hawaii, June 2008

TPC Member, *International Symposium on VLSI Technology, Systems, and Applications (2005 IEEE VLSI-TSA)*, Hinshu, Taiwan, April 25-27, 2005

Arrangements Chair, *IEEE Device Research Conference (DRC)*, Penn State University, June 2009

Panel Chair, *ACM/IEEE International Symposium Low Power Electronics and Design (ISLPED '09)*, San Francisco, California, August 2009

Session Chair, “*Non Volatile Memory*” at the *High-K Dielectric Materials and Gate Stack Symposium, Electrochemical Society Meeting (ECS '09)*, Vienna, Austria, October 2009

Session Chair, “*MOS Devices*” at the *IEEE Device Research Conference (DRC '09)*, June 2009

Session Chair, “*MOS Devices*” at the *IEEE Device Research Conference (DRC '09)*, June 2009

Session Co-Chair, “*III-V Logic Transistors with Advanced Gate Stack*” at the *IEEE International Electron Devices Meeting (IEDM '09)*, Baltimore, Maryland, December 8, 2009

Session Co-Chair, “*Heterostructure High-Speed Devices*” at the *IEEE International Electron Devices Meeting (IEDM '08)*, San Francisco, California, December 2008

Session Chair, “*Materials for Ge and Si devices*” at the *International Conference on Insulating Films on Semiconductors (INFOS '07)*, Athens, Greece, June 2007

Member, “*Front End Processes*” Tab and contributor to the 2003 Version of the *International Technology Roadmap for Semiconductors (ITRS)*

Member, *American Society for Engineering Education*, 2007 – 2015

NSF Review Panelist- In EPDT, EMT programs in years 2008-2017

Reviewer,

Science

Science Advances

NanoLetters

Nature Materials

Nature Communications

IEEE Transactions on Nanotechnology

IEEE Transactions on Electron Devices

IEEE Electron Device Letters

ACM Journal on Emerging Technologies in Computing Systems

Solid-State Electronics

Applied Physics Letters

Journal of Applied Physics

Journal of Vacuum Science and Technology B

Journal of Nanotechnology

Nanotechnology (Journal) from Institute of Physics (IOP)

Journal of Electronic Materials

ACS Nano

IEDM, DRC, ESSDERC, VLSI-TSA, IEEE ISLPED, SISC, EMC, IEEE SNW, VLSI Symposia
on Technology and Circuits

EXHIBIT F

Julia Rowe

Vice President

MA, Accountancy
DePaul University

BA, Business Administration
University of Michigan

Julia Rowe is a recognized expert in intellectual property licensing, the valuation of intellectual property assets, and economic analysis of commercialization investments. She holds dual professional credentials as a CPA and a Certified Licensing Professional (CLP).

Ms. Rowe's testimony experience spans multiple venues, including the U.S. International Trade Commission, Federal Court, State Court, and arbitration. Ms. Rowe has testified in several high-profile cases involving patent infringement, trademark infringement, copyright infringement, theft of trade secrets, breach of contract, unfair competition, and more. She has extensive experience analyzing international trade issues (including public interest, domestic industry and bonding), lost profits, reasonable royalties, and unjust enrichment damages.

In addition to litigation-related work, Ms. Rowe provides licensing-related consulting services, strategy consulting services related to intellectual property, and business and asset valuations.

Ms. Rowe's industry experience includes computer hardware and software, medical devices, telecommunications, and much more.

Prior to joining CRA, Ms. Rowe was a consultant at Kroll. She has a bachelor's degree in business administration from the University of Michigan and a master of accountancy degree from DePaul University.

Speeches, Presentations & Publications

"What Every IP Lawyer Needs to Know International Licensing," ACI's 5th Annual Summit for Women Leaders in IP Law, June 2025

"Valuing Intellectual Property and Intangible Assets." In *The Lawyer's Business Valuation Handbook* (chapter 28), August 2024 (co-author).

"Advanced Damages Techniques," 2023 ABA-IPL Section Annual Meeting, April 2023

"Strategies for Realizing Value from Patents Without Litigation," Continuing Legal Education Program for the Georgia Bar Patent Committee, January 2021

"ITC Investigations: Economic Issues" Continuing Legal Education Program for Faegre Drinker Biddle & Reath LLP, November 2020

"IP Value And the Benefits Of Cross-Department Collaboration," Duff & Phelps Webinar, November 2019

"IP Of the Future: Issues For Disruptive Technologies," Duff & Phelps 2018 IP Value Summit, November 2018

“The Growing Role Of Economics In IP-based Investigations At the ITC,” Annual Technology & Law Seminar, Kirkland Ellis LLP-Chicago, May 2018

“Hot Topics In Patent Licensing And Litigation,” Licensing Executives Society, Silicon Valley Chapter Meeting, March 2018

“Competitors And the Right To Exclude,” Duff & Phelps 2017 IP Value Summit, December 2017

“Connecting DC With Wall Street—How IP Investors View the Legal And Policy Landscape,” Intellectual Asset Management Patent Law and Policy Conference, November 2016

“Remedies In the Defend Trade Secrets Act,” Association of Corporate Counsel (Nationwide teleconference), July 2016

“Valuing Intellectual Property & Trade Secrets For Financial Reporting Purposes,” Michigan Association of Certified Public Accountants, May 2015

“Eastern District Of Texas (EDTX) General Order Regarding Alternative Procedure for Case Management,” Duff & Phelps IP Client Update, February 2014

“The Use Of Surveys For U.S. Patent Litigation,” Continuing Legal Education Program for Covington & Burling LLP (Nationwide teleconference), October 2013

“Recent Trends In IP Damages Law,” Licensing Executives Society, Inc. Chicago Chapter Meeting, June 2013

“Recent Trends In IP Damages Law and International Trade Commission Issues,” Continuing Legal Education Program For Duane Morris (Boston), October 2012

“Valuing Limited Partnership Interests,” Trusts & Estates, October 2010

Professional & Business History

2024–Present	<i>Vice President</i> , Charles River Associates, Chicago, IL
2011–May 2024	<i>Managing Director</i> , Kroll (formerly Duff & Phelps) <i>Managing Director</i> , 2021-2024 <i>Director</i> , 2013–2021 <i>Vice President</i> , 2011–2013 <i>Senior Associate</i> , 2011
2009–2011	<i>Associate</i> , CRA International, Chicago, IL <i>Analyst</i> , 2008–2009

Professional Associations and Certifications

Certified Public Accountant (CPA), Registered in Illinois

Certified Licensing Professional (CLP)

Member, Illinois CPA Society, Licensing Executives Society, Chicago Women in IP (“ChiWIP”)